The Localization Scale of the Block Anderson Model in One Dimension at Arbitrary Coupling and Volume

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Abstract

We study the one-dimensional block Anderson model with independent $W \times W$ Gaussian unitary ensemble potentials and deterministic nearest-neighbor hopping, with hopping-to-disorder ratio g. The system consists of L blocks, with total dimension $N = WL$. As $W \to \infty$, we identify the bulk localization length, up to an arbitrarily small power of W, with $\min\{N, \max\{W, W^2 g^2\}\}$, uniformly for deterministic couplings $g \geq 0$ and cycle lengths $L \geq 3$. For every fixed $D > 0$ and all sufficiently large W, the expected fraction of eigenvectors in the selected energy interval satisfying these bounds is at least $1 - W^{-D}$. In polynomial volume the bounds hold simultaneously for all eigenvectors in that interval with probability at least $1 - W^{-D}$. The localization length is the number of coordinates in the shortest cyclic interval containing at least half of an eigenvector's mass.

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1 Introduction

The localization length describes the region carrying a substantial fraction of an eigenvector's mass. Its dependence on disorder and on the number of interacting degrees of freedom is a central question in the theory of random Schrödinger operators. Anderson [4] introduced a random lattice Hamiltonian to explain the suppression of diffusion by disorder. Two fundamental approaches to rigorous localization are the multiscale analysis of Fröhlich–Spencer [13] and the fractional-moment method of Aizenman–Molchanov [1]. They establish localization at strong disorder or suitable spectral edges. Quantifying the localization length as the disorder weakens, or as the number of channels grows, requires additional estimates beyond the existence of localization.

Random band matrices provide a finite-dimensional setting for this question. Their independent entries are appreciable only within a band of width W, so the bandwidth controls the range of interaction. In one dimension, the predicted localization scale is W^2, and the corresponding transition for a matrix of size N occurs near $W = \sqrt{N}$; the supersymmetric analysis of Fyodorov–Mirlin [15] gives an early theoretical explanation of this scale. On the localization side, for each fixed small $\varepsilon > 0$, Schenker [28] proved localization for Gaussian band matrices when $W \leq N^{1/8-\varepsilon}$. Chen–Smart [7] and Cipolloni–Peled–Schenker–Shapiro [8] reached $W \leq N^{1/4-\varepsilon}$, by different refinements of fluctuation arguments. More recently, Drogin [9] proved an eigenfunction-correlator bound with exponential decay length W^2 and a polynomial prefactor in W, on any fixed compact energy interval, for a class of real band matrices with independent entries, including the real Wegner orbital model. On the delocalization side, Yau–Yin [33] proved bulk delocalization and universality for Gaussian band matrices when $W \geq N^{1/2+c}$, for any fixed $c > 0$. These results concern ensembles with random off-diagonal entries; the deterministic hopping in the present paper requires a different localization argument.

The dimension also changes the picture substantially. S. Dubova, K. Yang, H.-T. Yau and J. Yin [12] proved bulk delocalization and universality for two-dimensional Gaussian band matrices with $W \geq N^c$, for any fixed $c > 0$. S. Dubova, F. Yang, H.-T. Yau and J. Yin [11] established corresponding results in dimensions $d \geq 3$, and also treated block Anderson operators in a range of couplings. The stochastic analysis of resolvent loops developed in these works connects the quantum problem to a deterministic diffusion profile. The spatial estimates and coupling ranges depend on the dimension; the high-dimensional delocalization theorems do not themselves give the one-dimensional estimates considered here.

Block models retain a closer connection to the Anderson operator: each lattice site carries several internal coordinates, while hopping connects neighboring sites. This viewpoint goes back to Wegner's orbital model [32]. In the block Anderson model the onsite (i.e. within the same block) potential is random and the hopping is deterministic, whereas in the Wegner orbital model the hopping blocks are random as well. For normalized Gaussian blocks of size W and identity hopping of strength g, Peled–Schenker–Shamis–Sodin [24] proved exponential decay of fractional moments of the Green function for $g \leq cW^{-1/2}$ with c sufficiently small. This gives the upper bound on the scale of a single block in this regime, up to logarithmic factors. For small coupling above this threshold, Truong–Yang–Yin [31] proved the bulk lower bound on the scale $\min\{N, (Wg)^2\}$, up to arbitrarily small power losses, in polynomial finite volume for $W^{-1/2+\epsilon} \leq g \leq W^{-\epsilon}$ with fixed small $\epsilon > 0$. Their work also gives delocalization in two dimensions.

At order-one coupling, Drogin [10] proved the W^2 upper scale in coordinate distance for $g = 1$ and GOE onsite blocks. His bound concerns the limiting exponential decay rate of the Green function at each fixed energy and follows from a lower bound of order W^{-1} on the gaps between Lyapunov exponents.

Here we study a cycle of L blocks with independent normalized complex GUE potentials of size W, coupled by gI_W. In coordinate distance, our main result establishes matching lower and upper bounds on the scale $\min\{N, \max\{W, W^2g^2\}\}$, where $N = WL$, up to arbitrarily small

powers of W and uniformly over deterministic finite L and g.

For the lower bound we follow the loop hierarchy of Yau–Yin [33], adapted to block Schrödinger operators by Truong–Yang–Yin [31]. It propagates estimates for traces of cyclic products of Green functions to the spectral scale needed to bound the mass of an eigenvector on a single block. Our main addition is at large g, where transfer matrices give the initial estimates for this propagation and, for arbitrarily large g, reduce the spatial estimates to $W^{O(1)}$ sample points.

For the upper bound we use the fractional-moment method of Aizenman–Molchanov [1, 3]. Beyond the range covered by [24], we follow the boundary comparison of Macera–Sodin [23] to deduce Green-function decay from the weakest expansion of a transfer product on the subspace allowed by the boundary condition. At small g, we use Le Page's weighted-operator approach to random matrix products [21] and adapt Drogin's quantitative nonconcentration argument [10] to prove contraction. Our perturbation of the onsite blocks varies the expansion in one direction by matching the variation on the few directions that a short product strongly amplifies. At large g, we estimate the weakest expansion directly by perturbing around the free transfer. A correction to the logarithmic growth reveals a positive drift of order $(Wg^2)^{-1}$ per block. This argument covers the band center and, for $g \geq W$, the entire bulk.

1.1 The model

Let W, L be positive integers, and put $N \coloneqq WL$. We identify $\mathbb{Z}_N$ and $\mathbb{Z}_L$ with integers modulo N and L, respectively. Divide $\mathbb{Z}_N$ into the blocks

$$\mathcal{I}_a \coloneqq \{(a-1)W + 1, \dots, aW\}, \qquad a \in \mathbb{Z}_L, \tag{1.1}$$

where representatives $a = 1, \dots, L$ are used in this formula. For a coordinate $i \in \mathbb{Z}_N$, write $[i]$ for the block containing i. Indices $a, b, c, \dots$ denote blocks, and $i, j, \dots$ denote coordinates. We also identify $\mathbb{C}^N = \mathbb{C}^L \otimes \mathbb{C}^W$ in the order given by (1.1).

For each $a \in \mathbb{Z}_L$, let V_a be a normalized complex GUE matrix of size W. The matrices $(V_a)_{a \in \mathbb{Z}_L}$ are independent, and their covariance is

$$\mathbb{E}(V_a)_{\alpha\beta}(V_b)_{\gamma\delta} = W^{-1}\delta_{ab}\delta_{\alpha\delta}\delta_{\beta\gamma}, \qquad 1 \leq \alpha, \beta, \gamma, \delta \leq W. \tag{1.2}$$

Thus the diagonal entries are real Gaussians of variance W^{-1}; the real and imaginary parts of each upper off-diagonal entry are independent Gaussians of variance $(2W)^{-1}$.

Let A_L be the adjacency matrix of the cycle $\mathbb{Z}_L$ of L blocks, so $(A_L f)(a) = f(a-1) + f(a+1)$. For a coupling g, define

$$H = H_{g,W,L} \coloneqq \frac{gA_L \otimes I_W + V}{\max\{1, g\}}, \qquad V \coloneqq \operatorname{diag}(V_a : a \in \mathbb{Z}_L). \tag{1.3}$$

Equivalently, for a vector $\psi = (\psi_a)_{a \in \mathbb{Z}_L}$,

$$(H\psi)_a = \frac{V_a\psi_a + g\psi_{a-1} + g\psi_{a+1}}{\max\{1, g\}}, \qquad \psi_a \in \mathbb{C}^W. \tag{1.4}$$

The parameter g is the ratio of the hopping strength to the onsite disorder strength. The factor $\max\{1, g\}$ is included only for later convenience; it leaves the eigenvectors and their localization lengths unchanged. For $g \leq 1$, the model is $gA_L \otimes I_W + V$; for $g \geq 1$, it is $A_L \otimes I_W + V/g$. We use this normalization throughout. We write

$$\lambda_1 \leq \cdots \leq \lambda_N, \qquad H\psi_k = \lambda_k\psi_k, \qquad \|\psi_k\|_2 = 1, \tag{1.5}$$

for the eigenvalues and normalized eigenvectors. The spectrum is almost surely simple: the characteristic-polynomial discriminant is a polynomial in the real Gaussian coordinates, is nonzero for sufficiently separated diagonal onsite entries, and hence vanishes with probability zero.

Throughout the paper, $g \geq 0$ and $L \geq 3$ are finite deterministic parameters. We consider $W \to \infty$, allowing $g = g(W)$ and $L = L(W)$ to vary without further restriction. Probability estimates hold uniformly over the stated deterministic parameter range, with a separate event for each choice of g, L.

Notation. We write $\mathbb{C}_+ := \{z \in \mathbb{C} : \Im z > 0\}$ and $G(z) := (H - z)^{-1}$. For $\psi \in \mathbb{C}^N$,

$$\psi_a := (\psi(i))_{i \in \mathcal{I}_a}.$$

Matrix operator and entrywise maximum norms are denoted by $\|A\|$ and $\|A\|_{\max} := \max_{i,j} |A_{ij}|$, respectively. We use $\langle A \rangle := \operatorname{Tr} A$ for the unnormalized trace. For an $N \times N$ block matrix A, define its normalized block traces by

$$\langle A \rangle_{ab} := W^{-1} \operatorname{Tr} A_{ab}.$$

The trace identifies the orbital indices in blocks a and b. For block observables, we write

$$P_a := \operatorname{diag}(\mathbf{1}(i \in \mathcal{I}_a)), \qquad E_a := W^{-1} P_a$$

for the orthogonal projection onto block a and its normalized version.

Constants $C, c > 0$ may change from line to line and may depend on the fixed parameters in a statement, but not on W, L, g. The notation $X \lesssim Y$ means $X \le CY$, and $X \asymp Y$ means both $X \lesssim Y$ and $Y \lesssim X$.

Localization length.

A cyclic interval of length r is a set $I = \{x, x+1, \dots, x+r-1\} \subseteq \mathbb{Z}_N$, with $1 \le r \le N$, where addition is modulo N.

Definition 1.1 (Localization length)**.** *For a normalized vector $\psi \in \mathbb{C}^N$, define*

$$\ell(\psi) := \min \left\{ |I| : \ I \subseteq \mathbb{Z}_N \text{ is a cyclic interval}, \quad \sum_{i \in I} |\psi(i)|^2 \ge \frac{1}{2} \right\}. \tag{1.6}$$

In particular, $1 \le \ell(\psi) \le N$. The definition selects an interval containing half of the mass; the interval and its location may depend on ψ. Its length counts coordinates, rather than blocks. It measures finite-volume mass concentration and does not define an asymptotic exponential decay rate.

Introduce the deterministic coordinate scales

$$\ell_{W,g} := \max\{W, W^2 g^2\}, \qquad \ell_{W,g,L} := \min\{WL, \ell_{W,g}\}. \tag{1.7}$$

The uncapped scale is $\ell_{W,g}$, and the finite-volume scale is $\ell_{W,g,L}$. The corresponding block scales are $\ell_{W,g}/W$ and $\ell_{W,g,L}/W$. Equivalently,

$$\ell_{W,g,L} = \begin{cases} W, & Wg^2 \le 1, \\ W^2 g^2, & 1 < Wg^2 < L, \\ N, & Wg^2 \ge L. \end{cases} \tag{1.8}$$

The first regime is the single-block scale. In the second regime, the characteristic number of occupied blocks is Wg^2. In the third regime, the finite-volume scale is reached.

The deterministic bulk.

The deterministic band edges depend on g. We therefore define the bulk using the deterministic spectral distribution of the normalized model (1.3).

Let $m_g : \mathbb{C}_+ \to \mathbb{C}_+$ be the solution of the infinite-chain Dyson equation corresponding to (1.3):

$$m_g(z) = (2\pi)^{-1} \int_{-\pi}^{\pi} \left(\frac{2g \cos p}{\max\{1, g\}} - z - \frac{m_g(z)}{\max\{1, g\}^2} \right)^{-1} \mathrm{d}p, \qquad m_g(z) \sim -z^{-1} \quad (|z| \to \infty). \tag{1.9}$$

This is [31, (2.6)] with the hopping and onsite (i.e. within the same block) covariance of (1.3). Let ν_g be the probability measure with Stieltjes transform m_g, and define its distribution function by

$$F_g(x) := \nu_g((-\infty, x]) = \pi^{-1} \int_{-\infty}^{x} \Im m_g(E + i0)\, \mathrm{d}E. \tag{1.10}$$

Here $m_g(E + i0)$ denotes the boundary value from the upper half-plane. For every finite g, the measure ν_g is even and has a continuous density, positive inside a single spectral band.

As $g \to \infty$, ν_g tends to the arcsine law on $[-2, 2]$, with density

$$\rho_\infty(x) := \frac{1}{\pi\sqrt{4 - x^2}} \quad (|x| < 2), \qquad \rho_\infty(x) := 0 \quad (|x| \geq 2). \tag{1.11}$$

Lemma 1.2 (Deterministic spectral distribution and bulk windows)**.** *Let ν_g and F_g be the measure and distribution function defined above.*

(i) For every $g \geq 0$,

$$\operatorname{supp} \nu_g \subseteq [-4, 4]. \tag{1.12}$$

(ii) The measures are uniformly atomless:

$$\lim_{h \downarrow 0} \sup_{g \geq 0,\ x \in \mathbb{R}} \nu_g([x - h, x + h]) = 0. \tag{1.13}$$

(iii) For every $\kappa \in (0, 1/2)$, there are $\kappa_0 \in (0, 2)$ and $0 < c_\kappa < C_\kappa < \infty$, depending only on κ, such that

$$g \in [0, c_\kappa] \cup [C_\kappa, \infty), \qquad F_g(-|E|) \geq \kappa \quad \Longrightarrow \quad |E| \leq 2 - \kappa_0. \tag{1.14}$$

The proof, including the preceding distributional facts, is given in Appendix A.2.

The following rough density law is uniform in both parameters. It will determine the threshold for the cycle length L in our choice of energy interval.

Lemma 1.3 (A uniform rough density law)**.** *For every $\delta > 0$ there are $W_\delta, L_\delta, C_\delta, c_\delta > 0$ such that, for $W \geq W_\delta$, $L \geq L_\delta$, and every g,*

$$\mathbb{P}\left(\sup_{x \in \mathbb{R}} \left| N^{-1} \#\{k : \lambda_k \leq x\} - F_g(x) \right| > \delta \right) \leq C_\delta e^{-c_\delta W^2 L}. \tag{1.15}$$

Its proof is given in Appendix D.2. For $0 < \kappa < 1/2$, using the threshold $L_{\kappa/2}$ from Lemma 1.3, define

$$J_{g,\kappa,L} := \begin{cases} \mathbb{R}, & L < L_{\kappa/2}, \\ \{E : F_g(-|E|) \geq \kappa/2\}, & L \geq L_{\kappa/2}. \end{cases} \tag{1.16}$$

For $L \geq L_{\kappa/2}$, the interval excludes ν_g-mass $\kappa/2$ at each edge, so its total mass is $1 - \kappa$. The lemma, with $\delta = \kappa/2$, shows that it contains at least $(1 - 2\kappa)WL$ eigenvalues except on an event of probability $C_\kappa e^{-c_\kappa W^2 L}$, for all sufficiently large W.

1.2 Main results

Theorem 1.4 (Localization length)**.** *Fix $0 < \kappa < 1/2$ and $D, \tau > 0$. For all sufficiently large W, uniformly over L, g,*

$$\mathbb{E}\left[\frac{\#\{k : \lambda_k \in J_{g,\kappa,L},\ W^{-\tau} \ell_{W,g,L} \leq \ell(\psi_k) \leq W^{\tau} \ell_{W,g,L}\}}{\#\{k : \lambda_k \in J_{g,\kappa,L}\}} \right] \geq 1 - W^{-D}. \tag{1.17}$$

Here $\ell_{W,g,L}$ is defined in (1.7), and we set $0/0 = 0$.

It is easy to obtain the next corollary from Theorem 1.4 by Markov's inequality.

Corollary 1.5 (Simultaneous bounds in polynomial volume)**.** *Fix $B_L, D, \tau > 0$ and $0 < \kappa < 1/2$. For all sufficiently large W, uniformly over $L \le W^{B_L}$ and g,*

$$\mathbb{P}\left\{W^{-\tau}\ell_{W,g,L} \le \ell(\psi_k) \le W^{\tau}\ell_{W,g,L} \text{ for every } k \text{ with } \lambda_k \in J_{g,\kappa,L}\right\} \ge 1 - W^{-D}. \tag{1.18}$$

The power loss W^{τ} may be improvable. The unrestricted-volume statement controls the expected proportion of eigenvectors satisfying the bounds; it does not assert that every eigenvector in the selected interval satisfies them when L is arbitrarily large.

Organization of the paper. Sections 2 and 3 explain the lower- and upper-bound strategies, respectively, and state the component estimates. The proofs in Sections 2.5 and 3.7 assemble these estimates into Theorem 1.4. Section 4 proves the reduction from resolvent moments to spreading at arbitrary volume. Section 5 presents the loop hierarchy and the common propagation theorem, and completes Theorems 2.3 and 2.5 using the inputs proved later. Section 6 collects the deterministic Dyson, kernel and propagation-scale estimates. Section 7 proves the transfer estimates and spatial sampling, and verifies the stochastic inputs to the common propagation Theorem 5.12, uniformly for arbitrarily large coupling.

The upper-bound reduction is presented in Section 8. Section 9 proves the small-coupling vector-field estimates, Section 10 constructs Gaussian fields from the finite-product differential, and Section 11 proves the large-coupling transfer estimates.

Appendix A proves the deterministic inputs, Appendix B proves the loop identities and primitive-loop estimates, and Appendix C proves the bounded-coupling initialization and the common propagation theorem. Appendix D contains orbital spreading and the rough density law used by both bounds. Appendix E proves the fractional-moment conversion and the finite-volume criterion; Appendices F and G contain the circle-operator estimates and the auxiliary GUE estimates.

1.3 Version note and AI disclosure

Version note. This version gives the full proof of Theorem 1.4. The statement of the theorem is final, although the exposition may undergo substantial revision. The author will prepare a separate, shorter account for journal submission. The Lean formalization proves Theorem 1.4 assuming no mathematical results beyond mathlib. Differences between its arguments and those in the text are listed in `PAPER-VS-LEAN.md` in the repository (`https://github.com/JYin80/Lean-LSBA1d`). The author used Claude Opus 5.5 and Claude Sonnet 5.5 to write the Lean proofs.

AI disclosure. The author wrote the first draft of the main text. In subsequent work with OpenAI's ChatGPT 6 Astra, the author extended the results to the full range of couplings and volumes in Theorem 1.4. The model participated in mathematical discussions, explored proof strategies proposed by the author, and helped revise the manuscript.

2 Proof strategy for the lower bound

2.1 Spatial cutoff and the loop hierarchy

Spatial cutoff. To bound the localization length from below, we control the largest block mass $\max_a \|\psi_a\|^2$, where ψ_a is the ath block component of ψ, through the diagonal blocks of the Green function $G(z) = (H - z)^{-1}$. At the spectral heights used in the proof, Theorem 4.1 shows that $\Im G_{aa}$ on a cycle with $L \geq \ell_{W,g} = \max\{W, W^2 g^2\}$ is comparable to the same quantity on a local periodic restriction of at most $C\ell_{W,g}$ blocks. Thus it suffices to establish the resolvent estimates for $L \leq C\ell_{W,g}$, instead of arbitrary L. Theorem 2.2 combines this cutoff with the conversion from resolvent moments to spreading.

The basic strategy: loop hierarchy. The main tool for the spreading bound is the loop hierarchy developed in [33, Sections 2 and 5] and extended to block Schrödinger operators in [31, Sections 3–7]. We use a flow (H_t, z_t) in which the onsite blocks are independent Hermitian Brownian motions and the spectral height decreases along a deterministic characteristic. At $t = 1$, the matrix H_1 has the law of the target matrix H.

Resolvent loops $\mathcal{L}_k(t)$ are cyclic traces of products of $G_t = (H_t - z_t)^{-1}$, with normalized block projections inserted between the factors. Along the flow (5.1)–(5.3), Itô's formula expresses their drifts and quadratic variations through other loops, giving the *loop hierarchy*. The deterministic primitive loops $\mathcal{K}_k(t)$ retain the terms that split a loop into two, with the same initial data. Following the five-step bootstrap scheme of [33, 31], we control $D_k = \mathcal{L}_k - \mathcal{K}_k$ up to times close to 1 (Theorem 5.12). Matching the parameters at the terminal time to those of the target model then gives the required loop and resolvent bounds. The tree representation of primitive loops, their Ward identities, and their cancellation properties come from this earlier development. Fluctuation averaging uses the Dyson-residual approach of He–Knowles–Rosenthal [18], as developed for block operators in [31, Section 6].

2.2 Extremely small, small, and bounded coupling

Extremely small coupling. Invariance under the same unitary rotation in every block gives a localization length of at least cW, apart from an exponentially small expected proportion of eigenvectors (Lemma 2.1). Thus the spreading bound for $g \leq W^{-1/2+o(1)}$ follows immediately.

Small coupling. For fixed small $\epsilon > 0$ and polynomially small couplings $W^{-1/2+\epsilon} \leq g \leq W^{-\epsilon}$, the one-dimensional spreading result and the corresponding loop-hierarchy argument are already established in [31, Theorem 2.1] in polynomial finite volume. The corresponding resolvent moment bound here is included in Theorem 2.3.

Bounded coupling. In the range $W^{-\epsilon} \leq g \leq O(1)$, the required resolvent and stability estimates have a direct precedent in [11, Section 8], which studies block Anderson operators in dimensions $d \geq 3$. We adapt these inputs to the one-dimensional setting in Lemma 5.3; propagation then gives the resolvent moment bound of Theorem 2.3.

2.3 Large coupling: transfer matrices and initialization

For a long time, we did not see how to initialize the loop hierarchy at large coupling. The flow argument and the five-step bootstrap scheme could proceed once the initial estimates were available. The difficulty was obtaining these estimates: even at times bounded away from 1, the spectral height is already $\eta \asymp g^{-2}$. In the normalized model, large g means weak onsite randomness, and our usual resolvent methods did not give the bounds needed to start the flow.

The way forward was to reverse our perspective: weaker randomness also brings the model closer to deterministic free propagation, which is stable in the bulk. Taking inspiration from

the transfer-matrix methods used to prove localization, we therefore sought direct control of Green-function amplitudes on the block scale g^2. This is much shorter than the predicted localization scale of Wg^2 blocks. Thus the required initial bounds on Green-function block norms can already be obtained directly from transfer-matrix estimates.

For the normalized model with $g \geq 1$, fix a block b and $z = E + i\eta$ with $\eta > 0$, and write $G(z) = (H - z)^{-1}$. Away from the source block b, the resolvent equation gives

$$\begin{pmatrix} G_{j+1,b} \\ G_{jb} \end{pmatrix} = T_j(z) \begin{pmatrix} G_{jb} \\ G_{j-1,b} \end{pmatrix}, \qquad T_j(z) = \begin{pmatrix} zI_W - V_j/g & -I_W \\ I_W & 0 \end{pmatrix}.$$

Number the blocks consecutively along an arc of the cycle. For $j \leq j'$, provided none of the blocks $j, \dots, j'$ equals b modulo L, iteration gives

$$\begin{pmatrix} G_{j'+1,b} \\ G_{j'b} \end{pmatrix} = T_{j'}(z) \cdots T_j(z) \begin{pmatrix} G_{jb} \\ G_{j-1,b} \end{pmatrix}. \tag{2.1}$$

All Green blocks in this recurrence are evaluated at z. For $|E| \leq 2 - \kappa$, write $E = 2\sin\theta$ with $|\theta| < \pi/2$. The change of coordinates S_E from Definition 7.1 separates the free rotation from the random perturbation:

$$S_E^{-1} \cdot T_j(z) \cdot S_E = R_E + g^{-1} X_j + i\eta B_E, \qquad R_E = \operatorname{diag}(-ie^{i\theta}, ie^{-i\theta}) \otimes I_W.$$

Here X_j is a centered linear function of the onsite GUE matrix V_j, while B_E is deterministic and bounded. For fixed $p, C_0 > 0$ and all sufficiently large W, Lemma 7.2 gives

$$\sup_j \mathbb{E} \max_{j \leq j' < j+n} \left(\|T_{j'}(z) \cdots T_j(z)\|^p + \|(T_{j'}(z) \cdots T_j(z))^{-1}\|^p \right) \leq C_p,$$

for $n \leq C_0 g^2$ and $n\eta \leq C_0$. Thus the norms of the stacked block columns $\binom{G_{j+1,b}}{G_{jb}}$ remain comparable along at most n transfer steps avoiding b, with comparison factors having bounded moments. For sufficiently long cycles, summing the corresponding quadratic-form inequalities over $n \asymp g^2$ blocks and using the matrix Ward identity,

$$\eta \sum_j G_{jb}(z)^* G_{jb}(z) = \Im G_{bb}(z),$$

we obtain the initial bound of Lemma 5.5:

$$\mathbb{E}\|G_{ab}(E + i\eta)\|^p \leq C_p, \qquad \eta \asymp g^{-2},$$

uniformly in g, L. The same transfer and Ward estimates give spreading on the coordinate scale $W \min\{L, g^2\}$ (Lemma 2.4), handling short cycles directly.

The weighted Ward estimate of Lemma 7.3 adds exponential spatial decay to the initial resolvent-moment bound:

$$\mathbb{E}\|G_{ab}(z)\|^p \leq C_p e^{-pc\|a-b\|_L/g^2},$$

where $\|a - b\|_L$ is the block distance on the cycle. Together with unitary invariance, Gaussian fluctuation averaging, and Dyson stability, these bounds allow an early-time loop induction that yields Lemma 5.6, under its stated volume assumptions. Propagation then gives the resolvent moment bound of Theorem 2.5.

2.4 Extremely large coupling: spatial sampling

When $\log g \gg \log W$, even the reduced cycle can contain more than polynomially many blocks. A union bound over all block tuples would then destroy estimates whose failure probabilities are powers of W. On the other hand, in this regime, the transfer-matrix product $T_n(z) \cdots T_1(z)$ is

extremely stable when $n \lesssim g^2 W^{-B}$, for $B > 2$. More precisely, for $z = E + i\eta$, Lemma 7.7 gives, with arbitrarily high polynomial probability,

$$T_n(z) \cdots T_1(z) = T(E)^n + O(W^{-B/4}), \qquad T(E) = \begin{pmatrix} EI_W & -I_W \\ I_W & 0 \end{pmatrix}, \qquad n \lesssim g^2 W^{-B}, \quad 0 < \eta \lesssim g^{-2}.$$

For $j \le j'$ such that none of the blocks $j, \dots, j'$ equals b modulo L, write $n = j' - j + 1$. Using (2.1), we obtain

$$\begin{pmatrix} G_{j'+1,b} \\ G_{j'b} \end{pmatrix} \approx T(E)^n \begin{pmatrix} G_{jb} \\ G_{j-1,b} \end{pmatrix}, \qquad n \lesssim g^2 W^{-B}.$$

Thus there are scalar coefficients $c_{j',j}$ and $c_{j',j-1}$ such that

$$G_{j'b} \approx c_{j',j} G_{jb} + c_{j',j-1} G_{j-1,b}.$$

An analogous approximation holds in the index b. Recall that $L \lesssim g^2 W^2$ from the above spatial cutoff argument. Partition $\mathbb{Z}_L$ into $W^{O(1)}$ consecutive cells of length less than $g^2 W^{-B}$, and let $\mathcal{J}$ be the set of their first blocks. The set $\mathcal{J}$ is a $g^2 W^{-B}$-net in $\mathbb{Z}_L$. Then any G_{ab} can be approximated by the scalar source term $d_{ab} I_W$ plus a linear combination of the four blocks

$$G_{\alpha,\beta}, \quad G_{\alpha-1,\beta}, \quad G_{\alpha,\beta-1}, \quad G_{\alpha-1,\beta-1}, \quad \alpha,\ \beta \in \mathcal{J}$$

where α and a lie in the same cell, as do β and b. Proposition 7.4 interpolates every Green block, with error $\mathcal{E}_{ab}$ and scalar source term $d_{ab} I_W$, as follows:

$$G_{ab} = \sum_{\alpha,\beta} c_{a\alpha} c_{b\beta} G_{\alpha\beta} + d_{ab} I_W + \mathcal{E}_{ab},$$

where $\alpha, \beta \in \mathcal{J} \cup (\mathcal{J} - 1)$. By Lemma 7.5, both the resolvent loops $\mathcal{L}_k$ in (5.5) and the primitive loops $\mathcal{K}_k$ in (5.10) admit similar approximations. By the maximum-volume argument of Lemma 7.6, Lemma 5.8 gives deterministic sets $\mathcal{S}_k$ with $|\mathcal{S}_k| \le W^C$ such that, with probability at least $1 - W^{-D}$,

$$\max_{\boldsymbol{a}} |D_k(\boldsymbol{a}, \boldsymbol{\sigma})| \le C_k \max_{\boldsymbol{a} \in \mathcal{S}_k} |D_k(\boldsymbol{a}, \boldsymbol{\sigma})| + W^{-D}, \qquad D_k = \mathcal{L}_k - \mathcal{K}_k.$$

We can therefore restrict the estimates to these sampling points and apply the same five-step scheme of [33, 31] as in the other coupling regimes to prove Theorem 2.5 also in this regime.

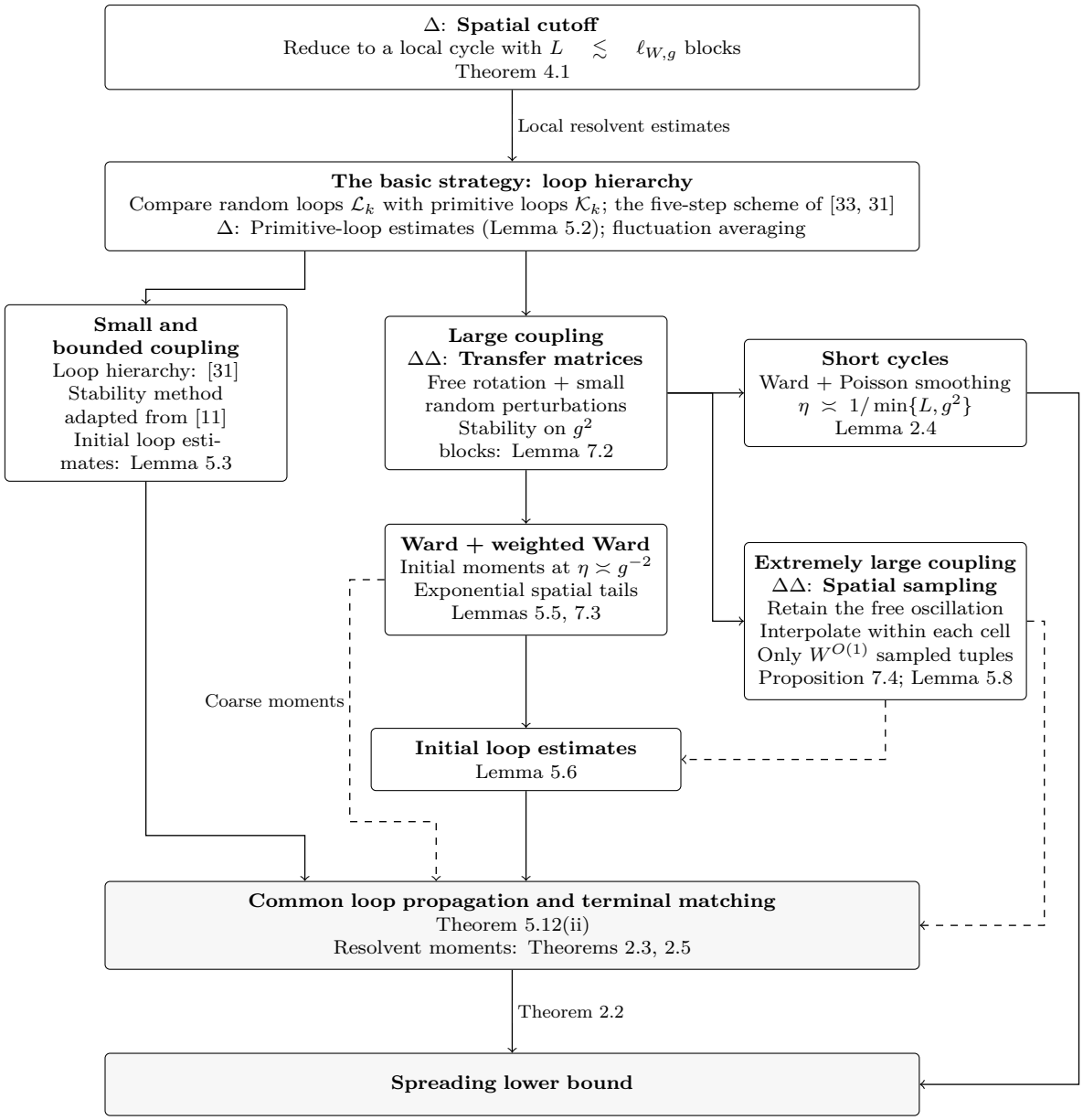


Figure 1: Lower-bound methods in the order of the preceding explanation. Δ: main contributions of this work; ΔΔ: the two central contributions.

2.5 Component estimates and proof of the lower bound

Lemma 2.1 handles the single-block scale, and Lemma 2.4 handles short cycles at large coupling. For the remaining parameters, Theorem 2.2 reduces spreading to a resolvent moment bound on a local cycle of $L_c < \ell_{W,g}$ blocks, with g unchanged. Theorems 2.3 and 2.5 supply this bound in the respective coupling ranges. Figures 1 and 2 summarize the methods and coupling regimes in the order of the preceding explanation.

Lemma 2.1 (Spreading from unitary invariance)**.** *For the model (1.3), there are absolute constants $c_0, c > 0$ such that, for all sufficiently large W and all L, g,*

$$\frac{1}{WL}\mathbb{E}\#\{k : \ell(\psi_k) < c_0 W\} \le 4We^{-cW}. \tag{2.2}$$

There is no energy restriction. Moreover, for every $D > 0$ there is $c_D > 0$ such that, with

Coupling	Main idea	Corresponding results
Extremely small $g \le W^{-1/2+o(1)}$	Common unitary invariance already gives the single-block scale.	Lemma 2.1
Small $W^{-1/2+\epsilon} \le g \le W^{-\epsilon}$	The loop-hierarchy method of [31].	Theorem 2.3
Bounded $W^{-\epsilon} \le g \le C$	Adapt the resolvent and stability method of [11], then use the same loop hierarchy.	Lemma 5.3; Theorem 2.3
Large $g \ge C_\kappa$	Transfer and Ward estimates give short-cycle spreading directly; the remaining cycles use initialization and loop propagation.	Short cycles: Lemma 2.4. Remaining cycles: Theorem 2.5, using Lemmas 5.5–5.6.
Extremely large $\log g \gg \log W$	For the remaining cycles, retain free oscillations and interpolate within cells: only $W^{O(1)}$ sampled tuples are needed for loop propagation.	Short cycles: Lemma 2.4. Remaining cycles: Theorem 2.5, using Proposition 7.4 and Lemma 5.8.

Figure 2: Qualitative coupling regimes in the order of the preceding explanation, with fixed $0 < \epsilon < 1/4$ and $C < \infty$. The regimes may overlap; the last row extends the large-coupling argument. The cited results give the precise energy and volume assumptions.

probability at least $1 - N^{-D}$*, simultaneously for all eigenvectors,*

$$\ell(\psi_k) \ge c_D (\log N)^{-1} W. \tag{2.3}$$

The proof is in Appendix D.1. Lemma 2.1 gives the required lower bound in the region

$$\ell_{W,g,L} \le c_0 W^{1+\tau}. \tag{2.4}$$

From resolvent moments to spreading at arbitrary volume.

Theorem 2.2 (Resolvent moments imply spreading, with volume extension)**.** *Fix* $E_0, D > 0$, $0 < \tau < 1$, *and* $0 < \nu < \tau/2$. *Let* $J \subset [-E_0, E_0]$ *be a deterministic interval. Put*

$$\eta = W^\nu / \ell_{W,g}, \qquad R = \lceil 2/\eta \rceil. \tag{2.5}$$

Choose the cycle length and moment height

$$L_c = \min\{L, 2R+1\}, \qquad \eta_c = \frac{W^\nu}{\ell_{W,g,L_c}}, \qquad J_{\eta_c} = J + [-\eta_c, \eta_c].$$

Let $G_{L_c}(z) = (H_{g,W,L_c} - z)^{-1}$ *be the resolvent of the normalized periodic model* (1.3) *on this cycle. Suppose that, for some* $p > 1 + (D+1)/(\tau - \nu)$ *and a constant* C_p*,*

$$\sup_{E \in J_{\eta_c}} \max_{1 \le x \le W L_c} \mathbb{E} \left(\Im (G_{L_c}(E + i\eta_c))_{xx} \right)^p \le C_p. \tag{2.6}$$

Then, for all sufficiently large W*,*

$$\frac{1}{WL} \mathbb{E} \# \left\{ k : \lambda_k \in J,\ \ell(\psi_k) < W^{-\tau} \ell_{W,g,L} \right\} \le W^{-D}. \tag{2.7}$$

The width threshold depends only on $E_0, D, \tau, \nu, p, C_p$ *and the threshold in the moment assumption. It is independent of* L, g *whenever that assumption holds uniformly in these parameters.*

The proof is in Section 4. The assumption requires only moments at deterministic energies; in applications we keep the original interval $J = J_{g,\kappa,L}$ on the cycle of L_c blocks.

Bounded coupling.

Theorem 2.3 (Bounded-coupling resolvent moments)**.** *Fix $C, p > 0$, $0 < \kappa < 1/2$, and $0 < \nu < 1$. Suppose that the deterministic coupling $g = g(W)$ and integer cycle length $L = L(W)$ satisfy*

$$W^{-1/2} \le g \le C, \qquad \log W \ll L \le W^2. \tag{2.8}$$

For all sufficiently large W,

$$\sup_{E:\, F_g(-|E|)\ge\kappa} \max_{1\le x\le WL} \mathbb{E}(\Im G_{xx}(E+i\eta_L))^p \le C_p, \qquad \eta_L = \frac{W^\nu}{\ell_{W,g,L}}. \tag{2.9}$$

The moment constant depends only on C, p, κ, ν. For every fixed positive function $\omega(W) \to \infty$, the width threshold may additionally depend on ω, and is uniform over deterministic parameters satisfying $W^{-1/2} \le g \le C$ and $\omega(W) \log W \le L \le W^2$.

The proof in Section 5.5 follows the loop hierarchy method of [31]; we prove the bounded-coupling inputs here by adapting the resolvent and stability method of [11, Section 8 and Appendix B.1].

Large coupling.

Lemma 2.4 (Spreading on the scale g^2)**.** *Fix $0 < \kappa < 1/2$. There are constants $W_0, C_\kappa, c_\kappa > 0$ such that, for every $W \ge W_0$, all L, and all $g \ge C_\kappa$,*

$$\frac{1}{WL}\mathbb{E}\#\left\{k : \lambda_k \in J_{g,\kappa,L},\ \ell(\psi_k) < c_\kappa W \min\{L, g^2\}\right\} \le C_\kappa e^{-c_\kappa W}. \tag{2.10}$$

The constants are independent of L, g.

The proof in Section 7.2 uses the transfer estimates and Ward's identity, followed by Poisson smoothing.

Theorem 2.5 (Large-coupling resolvent moments)**.** *Fix $0 < \kappa < 2$, $0 < \nu < 1$, and $p > 0$. There are constants $C_\kappa > 1$ and $C < \infty$ such that, for the periodic model (1.3), if deterministic $g = g(W)$ and $L = L(W)$ satisfy*

$$g \ge C_\kappa, \qquad L \gg g^2 \log W, \qquad L \le g^2 W^2,$$

then, for all sufficiently large W,

$$\sup_{|E|\le 2-\kappa} \max_{1\le x\le WL} \mathbb{E}(\Im G_{xx}(E+i\eta_L))^p \le C, \qquad \eta_L = \frac{W^\nu}{\ell_{W,g,L}}. \tag{2.11}$$

Here $L \gg g^2 \log W$ means $L/(g^2 \log W) \to \infty$ as $W \to \infty$. The moment constant C depends only on κ, ν, p, not on g, L. For every fixed positive function $\omega(W) \to \infty$, the width threshold may additionally depend on ω, and is uniform over deterministic $g \ge C_\kappa$ and $\omega(W) g^2 \log W \le L \le g^2 W^2$.

The proof of Theorem 2.5 is completed in Section 5.5.

Proof of the lower bound in Theorem 1.4. For both bounds, the case $\tau \ge 1$ follows from $\tau = 1/2$ by monotonicity. Fix $0 < \tau < 1$, $D > 0$, and $0 < \nu < \tau/2$, and choose p as in Theorem 2.2. Define

$$B_- = \#\{k : \lambda_k \in J_{g,\kappa,L},\ \ell(\psi_k) < W^{-\tau}\ell_{W,g,L}\}.$$

Lemma 2.1 covers (2.4). Otherwise $\min\{WL, W^2g^2\} = \ell_{W,g,L} > c_0 W^{1+\tau}$ for large W, so $L > c_0 W^\tau$ and $g \ge W^{-1/2}$. Set $J := J_{g,\kappa,L}$. By (1.16) and Lemma 1.2(i), $J \subseteq [-4, 4]$ and $F_g(-|E|) \ge \kappa/2$ on J. Fix a sufficiently large C_* depending only on κ, ν, p.

If $g \geq C_*$ and $L \leq W^{\tau/2} g^2$, then $\ell_{W,g,L} = WL$ and $W^{-\tau}\ell_{W,g,L} \leq W^{-\tau/2} W \min\{L, g^2\}$. Lemma 2.4 therefore gives $(WL)^{-1}\mathbb{E}B_- \leq C_\kappa e^{-c_\kappa W} \leq W^{-D}$ for sufficiently large W.

For the remaining parameters, select L_c, η_c as in Theorem 2.2 with $E_0 = 4$. The exclusions above and (2.5) give

$$\log W \ll L_c, \qquad \begin{cases} L_c \leq W^2, & g < C_*, \\ g^2 W^{\tau/2} \leq L_c \leq g^2 W^2, & g \geq C_*. \end{cases}$$

For $g \geq C_*$, we thus have $L_c/(g^2 \log W) \geq W^{\tau/2}/\log W \to \infty$ uniformly. Lemma 1.2(ii) gives

$$\begin{aligned} \eta_c &\leq W^{\nu-1} = o(1), \\ F_g(-|E|) &\geq \frac{\kappa}{2} - \sup_{x\in\mathbb{R}} \nu_g([x-\eta_c, x+\eta_c]) \geq \frac{\kappa}{4}, \qquad E \in J_{\eta_c}, \end{aligned}$$

uniformly in g, L. Apply Theorem 2.3 for $g < C_*$ and Theorem 2.5 for $g \geq C_*$ at this reduced margin, using Lemma 1.2(iii) in the latter case. This supplies (2.6), and Theorem 2.2 yields

$$(WL)^{-1}\mathbb{E}B_- \leq W^{-D}. \tag{2.12}$$

□

3 Proof strategy for the upper bound

Our goal is to place at least half the mass of most eigenvectors in the selected energy interval within $\max\{W, W^2 g^2\}$ consecutive coordinates, up to arbitrarily small powers of W. The total-volume cap is WL, so we may assume $L \geq W g^2$.

3.1 From fractional moments to transfer growth

The starting point: Fractional moment method. The classical fractional-moment method [1, 3] reduces the upper bound to decay of Green-function moments. For the Hamiltonian H_Λ on a finite path or cycle Λ of L blocks, write $G_\Lambda(x, y; E)$ for the (x, y) block of $(H_\Lambda - E)^{-1}$ and $\|x - y\|_L$ for the block distance. The Gaussian regularization estimates of [2] control the local Green-function factors used in this reduction. For $g < W$, we use the fixed-energy input (8.2) on the selected energy window J:

$$\sup_{E\in J} \mathbb{E}\|G_\Lambda(x, y; E)\|^q \leq W^C e^{-c\|x-y\|_L/\mathfrak{b}}, \qquad 0 < q < 1,$$

with $\mathfrak{b} = \max\{1, Wg^2\}$ up to arbitrarily small powers of W. Lemma 8.1(i) converts this into concentration within $C\mathfrak{b}\log W$ blocks.

The fixed-energy argument applies for $g \leq W^B$ for any fixed $B > 0$, since the factor g from single-block spectral averaging can then be absorbed into a power of W. The two methods overlap. We switch at $g = W$ because the negative-moment estimate of Theorem 11.1 covers the entire bulk, including $E = 0$, uniformly for $g \geq W$. In this range we use the energy-averaged version (8.3) on finite Dirichlet paths,

$$\int_J \mathbb{E}\|G_\Lambda(x, y; E)\|^q \, dE \leq W^C e^{-c\|x-y\|_L/(Wg^2)},$$

where J is a fixed bulk window. To remove the factor g for arbitrarily large couplings, we average over groups of $m \asymp g^2$ blocks. Transfer products and their inverses remain stable on this scale (Lemma 7.2); normalization reduces the coupling-dependent cost to $g/\sqrt{m} \asymp 1$. Lemma 8.1(ii) then gives concentration within $CWg^2 \log W$ blocks.

For $g \leq c_* W^{-1/2}$, the required fractional-moment decay is already supplied by [24, Theorem 1], yielding Theorem 3.1. In the remaining ranges, we obtain it from transfer growth, as described next.

Reduction to transfer growth estimates. We follow the transfer-to-Green-function comparison of Macera–Sodin [23, Section 2.2]. To illustrate the passage from transfer growth to Green-function decay, take $g \geq 1$ and a local interval $\Lambda = [0, n]$. Let L_0, R_0 be Hermitian endpoint shifts and let ι_j insert a vector into block j. As in (8.5), set

$$H_\Lambda^{L_0,R_0} := H_\Lambda + \iota_0 L_0 \iota_0^* + \iota_n R_0 \iota_n^*.$$

This is the standard boundary reduction: the shifts vanish for Dirichlet truncation and encode the eliminated exterior of a larger path. At a real energy E away from poles, write

$$\widetilde{G} = (H_\Lambda^{L_0,R_0} - E)^{-1},$$

and denote its blocks by $\widetilde{G}_{ab}$. Using the transfer matrices $T_j(E)$ from Section 2, iteration from n to 1, away from the source at 0, gives

$$\begin{pmatrix} \widetilde{G}_{00} \\ \widetilde{G}_{10} \end{pmatrix} = T_1(E) \cdots T_n(E) \begin{pmatrix} \widetilde{G}_{n0} \\ \widetilde{G}_{n+1,0} \end{pmatrix}, \quad \widetilde{G}_{n+1,0} := R_0 \widetilde{G}_{n0}.$$

Consequently,

$$\|\widetilde{G}_{n0}\| \leq \left(\inf_{\|v\|=1} \left\| T_1(E) \cdots T_n(E) \begin{pmatrix} v \\ R_0 v \end{pmatrix} \right\| \right)^{-1} (\|\widetilde{G}_{00}\| + \|\widetilde{G}_{10}\|).$$

The infimum measures the least growth allowed by the right boundary. Use the fixed unitary change of coordinates S from Definition 7.1, and let $\mathcal{F}_W$ denote the set of orthonormal boundary frames in these coordinates:

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \otimes I_W, \qquad \mathcal{F}_W = \left\{ \Phi \in \mathbb{C}^{2W \times W} : \Phi^* \Phi = I_W, \quad \Phi^* \operatorname{diag}(I_W, -I_W) \Phi = 0 \right\}.$$

For every Hermitian R_0, there exists $\Phi_{R_0} \in \mathcal{F}_W$ such that $S\Phi_{R_0}$ is an orthonormal frame of $\{(v, R_0 v) : v \in \mathbb{C}^W\}$ (Definition 8.3). Normalizing the boundary data gives

$$\|\widetilde{G}_{n0}\| \leq \left(\inf_{\|v\|=1} \|T_1(E) \cdots T_n(E) S \Phi_{R_0} v\| \right)^{-1} (\|\widetilde{G}_{00}\| + \|\widetilde{G}_{10}\|).$$

To make $\widetilde{G}_{n0}$ small, we seek expansion by a power of W with high probability at a suitable deterministic block length. The bound must be uniform over deterministic R_0, but need not hold simultaneously for all R_0. Define the amplitude matrix as in (8.10):

$$D_n(\Phi; E) = \sqrt{2} \begin{pmatrix} I_W & 0 \end{pmatrix} S^* T_n(E) \cdots T_1(E) S \Phi.$$

Here we switch the product order for convenience. The smallest singular value of $D_n(\Phi; E)$ measures the weakest expansion:

$$\|D_n(\Phi; E)^{-1}\|^{-1} = \inf_{\|v\|=1} \|T_n(E) \cdots T_1(E) S \Phi v\|.$$

For general $g > 0$, we henceforth use the normalized transfers and D_n from Definition 8.3. For $g < W$, the near-source Green-function factors have fractional moments bounded by powers of W. Sufficiently strong high-probability transfer growth absorbs this cost and yields the local resolvent target (8.6) (Proposition 8.4(i)). For fixed $D > 0$, the high-probability input has the form below, with $\mathfrak{b}$ supplied by the estimates for each coupling range:

$$\sup_{E \in J} \sup_{\Phi \in \mathcal{F}_W} \mathbb{P}\Big(\|D_{n_0}(\Phi; E)^{-1}\| > W^{-D} \Big) \leq W^{-D}, \qquad n_0 \lesssim \mathfrak{b} \log W. \tag{3.1}$$

For $g \geq W$, we instead use the negative-moment bound from Section 3.6: for some fixed $s \in (0, 1)$ and a fixed bulk interval J,

$$\sup_{E \in J} \sup_{\Phi \in \mathcal{F}_W} \mathbb{E} \|D_n(\Phi; E)^{-1}\|^s \leq C W e^{-cn/(W g^2)}, \qquad n \geq 1. \tag{3.2}$$

3.2 Reduction to operator contraction

In this subsection, we discuss the proof of (3.1) for $c_* W^{-1/2} \le g < W^{\epsilon}$, and away from zero energy for $W^{\epsilon} \le g < W$, with any fixed $0 < \epsilon < 1$. Using the projective and flag geometry of [14, 22], and adapting Drogin's framework [10, Sections 2 and 4], we reduce growth in the weakest direction of a long transfer product to contraction of $P_{1/2}$. Our target $\|D_n^{-1}\|^{-1}$ is the smallest singular value of D_n. The total volume expansion records the product of all W singular values; dividing by the expansion on a $(W-1)$-dimensional subspace isolates one transverse direction. For $\mathcal{T}_n = S^* T_n \cdots T_1 S$, $Y = \operatorname{ran} \Phi$ with $\Phi \in \mathcal{F}_W$, and $U \subset Y$ of dimension $W-1$, define

$$c(\mathcal{T}_n; U, Y) = \log \frac{\operatorname{vol}(\mathcal{T}_n|_Y)}{\operatorname{vol}(\mathcal{T}_n|_U)}, \qquad \|D_n(\Phi)^{-1}\|^{-1} = \inf_{U \subset Y,\, \dim U = W-1} e^{c(\mathcal{T}_n; U, Y)}, \quad Y = \operatorname{ran} \Phi$$

Here $\operatorname{vol}(\mathcal{T}_n|_Z)$ denotes the volume expansion on Z. Since the exponent D in (3.1) is arbitrary, polynomial factors in W can be absorbed. We can therefore use the W row norms of D_n^{-1} instead of locating the random weakest direction. For $0 < s \le 1$, the row identity in (8.23) and $\|D_n^{-1}\|^s \le \sum_{j=1}^{W} \|e_j^* D_n^{-1}\|^s$ give

$$\mathbb{E}\|D_n^{-1}\|^s \le W \sup_{\substack{U \subset Y \\ \dim U = W-1}} \mathbb{E} e^{-s \cdot c(\mathcal{T}_n; U, Y)}, \quad Y = \operatorname{ran} \Phi$$

We group a suitable fixed number $q = O(1)$ of blocks into one step $\mathcal{T}_q = S^* T_q \cdots T_1 S$ and define

$$(P_s \circ h)(U, Y) = \mathbb{E}\big[e^{-s c(\mathcal{T}_q; U, Y)} h(\mathcal{T}_q U, \mathcal{T}_q Y)\big].$$

Here h is a bounded function of the pair. The advantage is that estimating P_s requires analyzing only $q = O(1)$ transfer matrices, while D_n contains a long product of n such matrices. Since c adds under successive transfers, iteration over independent batches gives, as in Lemma 8.6,

$$\sup_{\Phi \in \mathcal{F}_W} \mathbb{E}\|D_{qk}(\Phi; E)^{-1}\|^s \le W \|(P_s)^k \circ 1\|_\infty, \qquad k \ge 1,$$

where the supremum norm ranges over the pairs (U, Y) above. The ideal input would be a one-step supremum-norm contraction

$$\|P_s\|_{\infty \to \infty} \le 1 - c/\mathfrak{b},$$

which would immediately give exponential decay of the iterates on scale $\mathfrak{b}$.

The proof uses an angular L^4 bound. For fixed U, the admissible spaces $Y = \operatorname{ran} \Phi$ containing U, with $\Phi \in \mathcal{F}_W$, form a circle $\mathcal{C}_U$ with normalized angular measure m_U. Write $\|h\|_{\mathrm{ang},4} := \sup_U \|h(U, \cdot)\|_{L^4(\mathcal{C}_U, m_U)}$ (Definition 8.8). The angular Jacobian satisfies

$$\frac{dm_{\mathcal{T}_q U}(\mathcal{T}_q Y)}{dm_U(Y)} = e^{-2c(\mathcal{T}_q; U, Y)} = \big(e^{-s c(\mathcal{T}_q; U, Y)}\big)^{2/s}.$$

Take $s = 1/2$. For each fixed U, Jensen's inequality and then the change of variables $Y' = \mathcal{T}_q Y$ give

$$\begin{aligned} \int_{\mathcal{C}_U} |(P_{1/2} \circ h)(U, Y)|^4 \, dm_U(Y) &\le \mathbb{E} \int_{\mathcal{C}_U} e^{-2c(\mathcal{T}_q; U, Y)} |h(\mathcal{T}_q U, \mathcal{T}_q Y)|^4 \, dm_U(Y) \\ &= \mathbb{E} \int_{\mathcal{C}_{\mathcal{T}_q U}} |h(\mathcal{T}_q U, Y')|^4 \, dm_{\mathcal{T}_q U}(Y') \le \|h\|_{\mathrm{ang},4}^4. \end{aligned} \tag{3.3}$$

The last step uses the supremum over all possible first subspaces in the angular norm. Then the supremum over U yields

$$\|P_{1/2} \circ h\|_{\mathrm{ang},4} \le \|h\|_{\mathrm{ang},4}.$$

The constant 1 comes from the exact match between the fourth power of the weight $e^{-c/2}$ and the Jacobian e^{-2c}. This bound holds for every bounded measurable h. The same calculation uses $L^{2/s}$ for other small positive s; we choose $s = 1/2$ for convenience. To apply Lemma 8.10, we need the strict improvement

$$\|P_{1/2} \circ h\|_{\mathrm{ang},4} \le (1 - c/\mathfrak{b})\|h\|_{\mathrm{ang},4}$$

for bounded nonnegative h invariant under the same unitary rotation in both W-dimensional components, as in (8.25). The functions $h = (P_{1/2})^r \circ 1$ have this invariance, so iteration gives $\|(P_{1/2})^k \circ 1\|_{\mathrm{ang},4} \le (1 - c/\mathfrak{b})^k$.

This angular estimate still allows narrow peaks, so we use smoothing to obtain control at each fixed boundary pair. Following Drogin's regularization of the transfer-product law [10, Section 4], we split $P_{1/2} = G_{1/2} + B_{1/2}$ using input weights χ_P and $1 - \chi_P$. The first part maps angular L^4 to L^∞, while the second contracts in L^∞ (Lemma 8.9). Repeated angular contraction, followed by one smoothing step and a geometric summation of the remainder terms, gives

$$\|(P_{1/2})^k \circ 1\|_\infty \le W^C e^{-ck/\mathfrak{b}}$$

by Lemma 8.10. The smoothing cost W^C is paid only once. Lemma 8.6 then gives (3.1) at $n_0 = qk \asymp \mathfrak{b} \log W$ with a sufficiently large constant.

3.3 A criterion for contraction of $P_{1/2}$

We adapt Drogin's quantitative nonconcentration criterion [10, Proposition 2.6 and Claims 2.7–2.8], using Gaussian integration by parts; see [17, Section 3] for this mechanism. Recall that equality in the Jensen step of (3.3) requires, for fixed (U, Y), the quantity being averaged to be almost surely constant in the Gaussian input $\omega = (V_1, \dots, V_q)$. Denote this quantity by

$$Z_h := e^{-c(\mathcal{T}_q;U,Y)/2} h(\mathcal{T}_q U, \mathcal{T}_q Y), \qquad (P_{1/2} \circ h)(U, Y) = \mathbb{E} Z_h.$$

To obtain strict contraction, we therefore need to rule out near constancy of Z_h in ω quantitatively.

We vary $(V_1, \dots, V_q)$ in a direction $X = (X_1, \dots, X_q)$, with each X_i Hermitian. Write D_X for differentiation along this perturbation. We aim to choose X so that

$$D_X[c(\mathcal{T}_q; U, Y)] = 1, \qquad D_X[h(\mathcal{T}_q U, \mathcal{T}_q Y)] = 0.$$

To ensure the second identity, we require the first-order changes of the two terminal subspaces $\mathcal{T}_q U, \mathcal{T}_q Y$ induced by X to be a common unitary rotation acting identically on both W-dimensional components, as in (8.34). Every invariant h then has zero derivative. This condition controls the motion of the subspaces while allowing stretching within them. The task therefore has two parts:

1. Achieve $D_X[c(\mathcal{T}_q; U, Y)] = 1$ under this geometric condition.
2. Control the loss of Gaussian probability weight along X.

The choice of X may depend on (U, Y) but works for all such h.

The construction needs only work on a sufficiently large region of the Gaussian inputs. A smooth cutoff $0 \le \chi \le 1$ restricts it to that region, where the two derivative identities hold, and makes χX a regular field on the full input space. Its Gaussian divergence is defined by

$$\delta_{\mathrm{G}}(\chi X): \quad \mathbb{E} D_{\chi X} \varphi = \mathbb{E}[\varphi\, \delta_{\mathrm{G}}(\chi X)]$$

for smooth test functions φ. This measures the cost of integration by parts, including the variation of the Gaussian density and the derivatives of the field and cutoff. The required bounds are

$$\mathbb{E}\chi \ge \tfrac{1}{2}, \qquad \mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\mathfrak{b}.$$

The first prevents us from placing the entire perturbation on a rare event. The second determines the contraction scale. To see the connection, test integration by parts with a bounded primitive of the indicator of $[a-t, a+t]$, evaluated at $\log Z_h$. The primitive has absolute value at most t, while $D_{\chi X} \log Z_h = -\chi/2$. Cauchy–Schwarz therefore gives

$$\mathbb{E}\Big[\chi\, \mathbf{1}_{\{|\log Z_h - a| \le t\}}\Big] \le Ct\big(\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2\big)^{1/2} \le Ct\sqrt{\mathfrak{b}}.$$

Taking $t = c_0 \mathfrak{b}^{-1/2}$ with c_0 sufficiently small, we see that $\log Z_h$ cannot concentrate within distance t of any constant. This quantitative nonconcentration yields the strict contraction of Lemma 8.11:

$$\|P_{1/2} \circ h\|_{\mathrm{ang},4} \le (1 - c/\mathfrak{b})\|h\|_{\mathrm{ang},4}.$$

The next two subsections explain how to construct X and control its Gaussian cost in the different coupling ranges.

3.4 Small coupling: exploiting amplified directions

We first examine how the transfer matrices act on U and Y, using T_j and the fixed unitary matrix S from Definition 8.3. Set $U_0 = SU$, $Y_0 = SY$, and write $U_j = T_j \cdots T_1 U_0$, $Y_j = T_j \cdots T_1 Y_0$. For each fixed incoming flag, almost surely for every $j \ge 1$, we represent Y_j by a Hermitian matrix R_j, and U_j by R_j and a nonzero vector v_j, as

$$Y_j = \left\{ \begin{pmatrix} x \\ -gR_j x \end{pmatrix} : x \in \mathbb{C}^W \right\}, \qquad U_j = \left\{ \begin{pmatrix} x \\ -gR_j x \end{pmatrix} : x \in \mathbb{C}^W,\ v_j^* x = 0 \right\}.$$

Choose a nonzero representative v_1 and fix the subsequent representatives by the recurrence

$$R_j = (V_j - EI_W - g^2 R_{j-1})^{-1}, \qquad v_j = R_j v_{j-1}, \qquad j \ge 2.$$

First consider a simultaneous unitary rotation. Fix $K^* = -K$, set $\widehat{K} = \mathrm{diag}(K, K)$, and let

$$V_j(s) = e^{sK} V_j e^{-sK}, \qquad 1 \le j \le q, \qquad U_0(s) = e^{s\widehat{K}} U_0, \qquad Y_0(s) = e^{s\widehat{K}} Y_0.$$

Every propagated pair then undergoes that same unitary rotation: $U_j(s) = e^{s\widehat{K}} U_j$, $Y_j(s) = e^{s\widehat{K}} Y_j$. The normal representatives may be chosen as $v_j(s) = e^{sK} v_j$. Take this simultaneous variation of the matrices and the initial flag as $X^{[0]}$, and write $D_{\mathrm{rot}} = D_{X^{[0]}}$. Then

$$D_{\mathrm{rot}} V_j = [K, V_j], \quad D_{\mathrm{rot}} R_j = [K, R_j], \qquad D_{\mathrm{rot}} v_j = K v_j, \qquad D_{\mathrm{rot}} c = D_{\mathrm{rot}} h = 0.$$

Keeping the initial subspaces and the preparation blocks fixed. We now keep U_0, Y_0 fixed, as required in the contraction criterion. The first four blocks give the regularity and nondegeneracy estimates for R_4, v_4 in Lemma 9.1. We freeze these blocks as well. Thus R_4, v_4 are fixed. Our first modification is the uncorrected field $X^{[1]}$, which rotates only the later matrices:

$$X_j^{[1]} = 0 \quad (1 \le j \le 4), \qquad X_j^{[1]} = [K, V_j] \quad (5 \le j \le q).$$

At the first active block this gives

$$D_{X^{[1]}} R_5 = [K, R_5] - g^2 R_5 [K, R_4] R_5, \qquad D_{X^{[1]}} v_5 = K v_5 - R_5 K v_4 - g^2 R_5 [K, R_4] v_5.$$

The terms involving $[K, R_4]$ arise because R_4 no longer rotates; even Y_5 therefore need not follow a pure unitary rotation. The term $-R_5 K v_4$ arises because the incoming normal v_4 is fixed.

Correcting the first active block. To remove the terms involving $[K, R_4]$, define $X^{[2]}$ by changing only the fifth component of $X^{[1]}$:

$$X_5^{[2]} = [K, V_5] - g^2 [K, R_4], \qquad X_j^{[2]} = X_j^{[1]} \quad (j \ne 5).$$

Propagating this correction gives, for $5 \le j \le q$,

$$D_{X^{[2]}} R_j = [K, R_j], \qquad D_{X^{[2]}} v_j \;\; = Kv_j - (R_j \cdots R_5) K v_4.$$

Thus each Y_j for $5 \le j \le q$ follows the unitary rotation. The variation of v_j, which specifies U_j through $v_j^* x = 0$, retains the extra term caused by fixing v_4.

The changes in c and h under $X^{[2]}$. Write

$$\xi_q = (R_q \cdots R_5) K v_4, \qquad M_q = (I_W + g^2 R_q^2)^{-1}.$$

The common unitary rotation contributes nothing to c or h. After removing it, R_q is fixed and the infinitesimal change of v_q is $-\xi_q$. With the first four blocks fixed, (9.14) expresses e^{-2c} as a fixed multiple of $v_q^* M_q v_q$. Including the rotation term in the normal variation, we obtain

$$D_{X^{[2]}} c = \frac{\Re(v_q^* M_q \xi_q)}{v_q^* M_q v_q}, \qquad D_{X^{[2]}} v_q = K v_q - \xi_q.$$

The ideal change $\xi_q = v_q$ would give $D_{X^{[2]}} c = 1$ and $D_{X^{[2]}} h = 0$.

Choosing K at the penultimate block. Exact matching is impossible. We instead seek an approximate match at $q-1$, choosing K using only $V_1, \ldots, V_{q-1}$ and reserving the last block for correcting the terminal motion. Here

$$D_{X^{[2]}} v_{q-1} = K v_{q-1} - \xi_{q-1}, \qquad \xi_{q-1} = (R_{q-1} \cdots R_5) K v_4.$$

We therefore seek a small skew-Hermitian K such that

$$(R_{q-1} \cdots R_5) K v_4 \approx (R_{q-1} \cdots R_5) v_4 = v_{q-1}.$$

The error is measured relative to $\|v_{q-1}\|$. A small Kv_4 can reproduce this output if it matches v_4 on the few directions that contribute substantially after propagation.

Correcting the terminal motion. A scalar normalization of K makes $\Re(v_{q-1}^* \xi_{q-1}) = \|v_{q-1}\|^2$. The remaining matching error has no real radial component, so we can choose a small skew-Hermitian L satisfying

$$L v_{q-1} = \xi_{q-1} - v_{q-1}.$$

Define $X^{[3]}$ by changing only the last component:

$$X_q^{[3]} = X_q^{[2]} - [L, V_q - g^2 R_{q-1}], \qquad X_j^{[3]} = X_j^{[2]} \quad (j < q).$$

The terminal variations are

$$D_{X^{[3]}} R_q = [K - L, R_q], \qquad D_{X^{[3]}} v_q = (K - L) v_q - v_q.$$

We take $X = X^{[3]}$. The unit radial contraction gives $D_X c = 1$. Since the radial term $-v_q$ does not change the hyperplane $v_q^* x = 0$, both terminal subspaces undergo only the common unitary rotation generated by $K - L$. Invariance of h then gives $D_X h = 0$ (Lemma 9.2).

It remains to achieve the match at $q-1$ with as small a skew-Hermitian generator K as possible.

3.4.1 Constructing K

Write $\mathcal{R} = R_{q-1} \cdots R_5$. We first look for a small vector y such that $\mathcal{R} y \approx \mathcal{R} v_4$, and then realize y as the image of v_4 under a small skew-Hermitian generator. Skew-Hermiticity requires $\Re(v_4^* y) = 0$. We will impose the stronger condition $v_4^* y = 0$.

Isolating V_5 in the product. For two blocks we have

$$R_6R_5 = (V_6 - EI_W)^{-1}\big[R_5^{-1} - g^2(V_6 - EI_W)^{-1}\big]^{-1}.$$

The first factor depends only on V_6. To isolate V_5 in the full product, define

$$\overleftarrow{R}_{q-1} = (V_{q-1} - EI_W)^{-1}, \quad \overleftarrow{R}_j = (V_j - EI_W - g^2\overleftarrow{R}_{j+1})^{-1}, \qquad 6 \le j < q-1.$$

The same elimination gives

$$\mathcal{R} = R_{q-1}\cdots R_5 = \underbrace{\overleftarrow{R}_{q-1}\cdots\overleftarrow{R}_6}_{A}\underbrace{(V_5 - EI_W - g^2R_4 - g^2\overleftarrow{R}_6)^{-1}}_{\widetilde{R}}. \tag{3.4}$$

The two shifts in $\widetilde{R}$ come from the blocks to the left and right of block 5, respectively (Section 9.2). Conditioning on all inputs except V_5 fixes both shifts, A, and v_4, while V_5 retains its GUE law.

First take $W^{-\beta} \le g \le W^{-\alpha}$ with fixed $0 < \alpha < \beta < 1/2$, so g^2 exceeds the bulk level spacing $1/W$ by a power of W. In this range, up to small powers of W, we need to retain at most Wg^2 right singular directions of A. For each such unit vector u,

$$u^*\widetilde{R}v_4 = (\widetilde{R}u)^*v_4.$$

For the fixed vector v_4, averaging over V_5 with the cutoff weight χ^2 bounds its squared projection onto this span by $O(g^2)\|v_4\|^2$, up to small powers of W. By choosing y with the same projection, we can construct a small skew-Hermitian K such that $\mathcal{R}Kv_4 \approx \mathcal{R}v_4$ and $\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 \lesssim W^\epsilon g^2$.

Why Wg^2 directions suffice in one block. The factors of A obey the same recursion from the right as R_j does from the left. For two consecutive blocks we have

$$R_j = \Gamma_j^{-1}, \qquad \Gamma_j = V_j - EI_W - g^2R_{j-1}, \qquad \Gamma_{j+1} = V_{j+1} - EI_W - g^2R_j.$$

If $\Gamma_j e = \lambda e$ and $\|e\| = 1$, then $R_j e = \lambda^{-1}e$. The next block therefore sees a shift $-g^2/\lambda$ along this direction. This shift reaches the order-one scale of the GUE block when $|\lambda| \asymp g^2$. Conditional on the preceding blocks, Γ_j is a GUE matrix with a fixed Hermitian shift, so the one-block estimate (9.58) gives

$$\mathbb{E}_{V_j}\#\{i : |\lambda_i(\Gamma_j)| \le g^2\} \lesssim Wg^2.$$

Block j amplifies the squared norm in this direction by $1/\lambda^2$. In block $j+1$, the shift $-g^2/\lambda$ along e gives a spectral density at zero, smoothed on scale g^2, of order $[1 + (g^2/\lambda)^2]^{-1}$ by (G.2) on the regular chain. Multiplying this density by $O(g^2)$ bounds the spectral mass of e in the band $[-g^2, g^2]$ of the next block. The squared amplification in block j times the next-block density is of order

$$\frac{1}{\lambda^2}\cdot\frac{1}{1 + (g^2/\lambda)^2} = \frac{1}{\lambda^2 + g^4}.$$

Thus for $|\lambda| \lesssim g^2$ the weight per direction is of order g^{-4}. For $k \ge 0$ with $2^k g^2 \ll 1$, the band $|\lambda| \asymp 2^k g^2$ has about $2^k Wg^2$ directions. Its weight per direction is of order 2^{-2k} relative to the band at g^2, so its total contribution is only of order 2^{-k} times that of the band at g^2.

The directions in the band $|\lambda| \asymp g^2$ change from block to block. A product term passing through this band has rank at most the rank of its spectral projection, hence at most Wg^2 up to small powers of W on the regular chain. For a vector fixed by the preceding blocks, the band in the next block captures at most an $O(g^2)$ fraction of its squared norm on average, up to the same power loss, by (G.7). Inversion then amplifies this part by order g^{-4} in squared norm, giving a one-step gain in mean squared norm of at most order $g^2 \cdot g^{-4} = g^{-2}$. The product moment estimate combines the contributions from all spectral bands to bound $\operatorname{Tr}|A|^p$.

A first choice of K. Recall that our target is $A\widetilde{R}Kv_4 \approx A\widetilde{R}v_4$. We retain the right singular directions of A with singular values above t_{cut}, and let Q project orthogonally onto their span after applying $\widetilde{R}$. To count these directions we use

$$\operatorname{rank} Q = \#\{i : \sigma_i(A) > t_{\mathrm{cut}}\} \le t_{\mathrm{cut}}^{-p} \operatorname{Tr}|A|^p.$$

With p as in (9.54), the moment bound (9.53) gives the count (9.55). If $Qx = 0$, then $\widetilde{R}x$ lies entirely in the discarded right singular subspace of A. We choose t_{cut} to make the discarded output small relative to $v_{q-1} = A\widetilde{R}v_4$. On the cutoff support the choice in (9.41) gives

$$\|A\widetilde{R}x\| \le t_{\mathrm{cut}}\|\widetilde{R}\|\,\|v_4\| \lesssim W^{-4}\|v_{q-1}\| \qquad (Qx = 0,\ \|x\| \le \|v_4\|).$$

Write $v_4 = a + b$, with $a = Qv_4$ and $b = (I_W - Q)v_4$. The overlap estimate (9.65) controls $\|a\|^2/\|v_4\|^2$. We might try to set $Kv_4 = a$. This is impossible for a skew-Hermitian K when $a \neq 0$, since $v_4^* a = \|a\|^2$ whereas $\Re(v_4^* K v_4) = 0$.

When $\|a\| \ll \|v_4\|$, we have $\|b\| \asymp \|v_4\|$. We can use this large component b to produce a. As a zeroth choice, take the rank-one map $K^{[0]} = ab^*/\|b\|^2$, so $K^{[0]}v_4 = a$. Subtracting its adjoint gives our first skew-Hermitian choice,

$$K^{[1]} = K^{[0]} - (K^{[0]})^* = \frac{ab^* - ba^*}{\|b\|^2}, \qquad K^{[1]}v_4 = a - \frac{\|a\|^2}{\|b\|^2}\,b.$$

The correction lies along b, so $QK^{[1]}v_4 = a$ exactly. Since $a \perp b$, we have $\|K^{[1]}\|_{\mathrm{HS}}^2 = 2\|a\|^2/\|b\|^2$. Thus the small overlap gives a small generator, and (9.71) controls the relative error from the discarded output.

A spectral cutoff for g^2R_4. The Gaussian divergence at block 5 contains the term $-W\operatorname{Tr}(V_5[\chi K, g^2R_4])$ in (3.5). To control the commutator, take $K^{[2]} = PK^{[1]}P$, where P is the spectral cutoff for g^2R_4 in (9.4). Here Q selects the component we must match, whereas P restricts the directions on which K may act. By construction,

$$K^{[1]}b = a, \qquad K^{[1]}a = -\frac{\|a\|^2}{\|b\|^2}\,b.$$

Thus the component a in $QK^{[1]}v_4$ comes entirely from $K^{[1]}b$. Suppose, for example, that $Pa = a$ and $Pb = 0$. Applying $K^{[2]}$ then gives

$$v_4 = a + b \ \xrightarrow{P}\ a \ \xrightarrow{K^{[1]}}\ -\frac{\|a\|^2}{\|b\|^2}\,b \ \xrightarrow{P}\ 0.$$

The cutoff preserves the target a but removes the input b that the first generator needs to produce it.

For the third choice, $K^{[3]}$, we first choose its image $y = K^{[3]}v_4$ inside $\operatorname{ran} P$ so that it still matches $a = Qv_4$. In the range $W^{-\beta} \le g \le W^{-\alpha}$, up to small powers of W, Q retains at most Wg^2 input directions selected by the large singular values of A, while P discards at most Wg^2 directions corresponding to large eigenvalues of $|g^2R_4|$. In the critical window the same construction takes $P = I_W$ and uses the rank and overlap bounds in (9.65). In Section 9.2 we construct y by (9.16). On the cutoff support $Pv_4 \neq 0$ and

$$Py = y, \qquad v_4^* y = 0, \qquad Qy = Qv_4, \qquad \|y\| \le 2\|Qv_4\|.$$

As for $K^{[1]}$, take a rank-one map minus its adjoint,

$$K^{[3]} = \|Pv_4\|^{-2}\big(y(Pv_4)^* - (Pv_4)y^*\big).$$

Both y and Pv_4 lie in $\operatorname{ran} P$, so $PK^{[3]}P = K^{[3]}$. The generator sends v_4 to y and has squared norm

$$\|K^{[3]}\|_{\mathrm{HS}}^2 = 2\frac{\|y\|^2}{\|Pv_4\|^2} \lesssim \frac{\|Qv_4\|^2}{\|Pv_4\|^2}.$$

Since $Q(y-v_4) = 0$ and $\|y-v_4\| \lesssim \|v_4\|$, the cutoff bound above gives $\|A\widetilde{R}(y-v_4)\| \lesssim W^{-4}\|v_{q-1}\|$. Section 9.2 denotes $K^{[3]}$ by K_0. To match the radial component at $q-1$ exactly, we use (9.18),

$$\xi = \frac{\Re(v_{q-1}^* \mathcal{R} K^{[3]} v_4)}{\|v_{q-1}\|^2} = 1 + O(W^{-4}), \qquad K = K^{[3]}/\xi.$$

The terminal correction L then absorbs the remaining error.

Property of K. Recall the recursion for the block normals and the factorization (3.4),

$$R_j = (V_j - EI_W - g^2 R_{j-1})^{-1}, \quad v_j = R_j v_{j-1} \quad (j \geq 2), \qquad \mathcal{R} = R_{q-1}\cdots R_5 = A\widetilde{R}.$$

At block 4, $P = \mathbf{1}_{\{|g^2 R_4| \leq W^{\varepsilon/800}\}}$ selects the directions on which K may act, with $P = I_W$ in the critical window. The input directions seen by the retained output form the range of Q, the orthogonal projection onto the image under $\widetilde{R}$ of the right singular subspace of A above t_{cut}. For fixed incoming data, $K = K(V_1, \ldots, V_{q-1})$. The smooth cutoff χ from Section 3.3 selects inputs for which $Pv_4 \neq 0$ and the discarded output is small. On its support, with $\xi_{q-1} = \mathcal{R}Kv_4$ and $v_{q-1} = \mathcal{R}v_4$, we have

$$\begin{aligned}
\text{Generator} \quad & K^* = -K, \qquad K = PKP, \qquad \operatorname{rank} K \leq 2, \\
\text{Size} \quad & \|K\|_{\mathrm{HS}} \lesssim \|Qv_4\|/\|Pv_4\|, \\
\text{Block 5} \quad & \|g^2[K, R_4]\|_{\mathrm{HS}} \leq 2\|g^2 P R_4 P\|\,\|K\|_{\mathrm{HS}}, \\
\text{Matching error} \quad & \|\xi_{q-1} - v_{q-1}\| \lesssim W^{-4}\|v_{q-1}\|, \\
\text{Radial component} \quad & \Re(v_{q-1}^* \xi_{q-1}) = \|v_{q-1}\|^2.
\end{aligned}$$

With the terminal correction $Lv_{q-1} = \xi_{q-1} - v_{q-1}$, the field satisfies $D_X c = 1$ and $D_X h = 0$ by Lemma 9.2.

3.4.2 Controlling the Gaussian cost

To estimate $\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2$, write $\delta_{\mathrm{G},j}$ for the Gaussian divergence in V_j. The four preparation blocks do not move, so

$$\delta_{\mathrm{G}}(\chi X) = \sum_{j=5}^{q} \delta_{\mathrm{G},j}(\chi X_j).$$

Condition on the preparation blocks, so R_4, v_4, P are fixed. Put $\widetilde{V}_5 = V_5 - g^2 R_4$ and $\widetilde{V}_j = V_j$ for $6 \leq j < q$. For $5 \leq j < q$ we have $\chi X_j = [\chi K, \widetilde{V}_j]$, and the Gaussian divergence formula (8.33) reads

$$\delta_{\mathrm{G},j}(\chi X_j) = W \operatorname{Tr}(V_j[\chi K, \widetilde{V}_j]) - \operatorname{div}_{V_j}(\chi X_j).$$

Varying V_j in a Hermitian direction B gives

$$D_{V_j}(\chi X_j)[B] = [D_{V_j}(\chi K)[B], \widetilde{V}_j] + [\chi K, B].$$

To take the ordinary divergence, put $B = E_\alpha$ and sum the E_α components over a real orthonormal basis of Hermitian matrices, using $\langle A, B\rangle_{\mathrm{HS}} = \Re \operatorname{Tr}(A^* B)$. The second commutator drops out since $\langle E_\alpha, [\chi K, E_\alpha]\rangle_{\mathrm{HS}} = 0$. In the first, expand $D_{V_j}(\chi K)[E_\alpha]$ in a real orthonormal basis (H_k) of the skew-Hermitian matrices supported on $\operatorname{ran} P$. The sum over α then reconstructs the direction $\sum_\alpha \langle E_\alpha, [H_k, \widetilde{V}_j]\rangle_{\mathrm{HS}} E_\alpha = [H_k, \widetilde{V}_j]$. Since $\operatorname{Tr}(V_j[\chi K, V_j]) = 0$, this gives

$$\delta_{\mathrm{G},j}(\chi X_j) = -\mathbf{1}_{\{j=5\}} W g^2 \operatorname{Tr}(V_5[\chi K, R_4]) - \sum_k \langle H_k, D_{V_j}(\chi K)[[H_k, \widetilde{V}_j]]\rangle_{\mathrm{HS}}. \tag{3.5}$$

The shift at block 5. The coefficient $[\chi K, g^2 R_4]$ in the trace in (3.5) depends on V_5. Let M_0 bound $\|g^2 P R_4 P\|$ on the cutoff support. Since P commutes with R_4 and $\chi K = P(\chi K)P$, we have

$$W\|[\chi K, g^2 R_4]\|_{\mathrm{HS}}^2 \le 4W M_0^2 \|\chi K\|_{\mathrm{HS}}^2.$$

Write $\delta_{\mathrm{act}} = \sum_{j=5}^{q-1} \delta_{\mathrm{G},j}(\chi X_j)$. Two Gaussian integrations by parts for $\mathbb{E}|\delta_{\mathrm{act}}|^2$ give (9.32). There the shift costs at most $CW(1+M_0^2)\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 + CWM_0 e^{-cW}$, as in (9.34).

Summing the blocks for each rotation direction. For the derivative contribution to δ_{act}, exchange the j and k sums in (3.5) using

$$\sum_{j=5}^{q-1} \sum_k \langle H_k, D_{V_j}(\chi K)[[H_k, \widetilde{V}_j]]\rangle_{\mathrm{HS}} = \sum_k \left\langle H_k, \sum_{j=5}^{q-1} D_{V_j}(\chi K)[[H_k, \widetilde{V}_j]] \right\rangle_{\mathrm{HS}}.$$

The inner sum is the derivative $D_{\mathrm{mat}}^{H_k}(\chi K)$ obtained by conjugating every $\widetilde{V}_j$ by $\mathcal{U}_t = e^{tH_k}$ while holding v_4 fixed.

Another key property of K. Under this rotation the matching target becomes $Q_t v_4$, where $Q_t = \mathcal{U}_t Q \mathcal{U}_t^*$. If we also rotate v_4, as in (9.24), the rotated vector $\mathcal{U}_t y$ matches the new target since

$$Q_t(\mathcal{U}_t y) = \mathcal{U}_t Q v_4 = Q_t(\mathcal{U}_t v_4).$$

Both y and Pv_4 in the rank-two formula rotate by $\mathcal{U}_t$, while ξ and χ stay fixed. Thus χK becomes $\mathcal{U}_t(\chi K)\mathcal{U}_t^*$. Differentiating at $t=0$ and subtracting the contribution from v_4 gives the derivative in the inner sum,

$$\underbrace{D_{\mathrm{mat}}^{H_k}(\chi K)}_{\text{matrices only}} = \underbrace{[H_k, \chi K]}_{\text{matrices and vector}} - \underbrace{D_{v_4}(\chi K)[H_k v_4]}_{\text{vector only}}.$$

Here D_{v_4} varies only the vector argument in the construction of χK, with all matrix inputs fixed. In the scalar divergence, $\langle H_k, [H_k, \chi K]\rangle_{\mathrm{HS}} = 0$ by cyclicity. Substitution therefore gives the exact identity

$$\delta_{\mathrm{act}} = -Wg^2 \operatorname{Tr}(V_5[\chi K, R_4]) + \sum_k \langle H_k, D_{v_4}(\chi K)[H_k v_4]\rangle_{\mathrm{HS}}.$$

In the second-moment bound (9.34), substitute the same derivative identity before estimating the squared norms. By (9.33), the squared commutators sum to at most $2W\|\chi K\|_{\mathrm{HS}}^2$, which is absorbed into the size term below. For the vector derivatives, $H_k v_4 = H_k(Pv_4)$. In coordinates with Pv_4 along the first axis, only the first column of H_k moves the vector. Summing over k as in (9.35) therefore gives

$$\mathbb{E}|\delta_{\mathrm{act}}|^2 \lesssim \mathbb{E}\left[\|Pv_4\|^2 \sum_a \|D_{v_4}(\chi K)[e_a]\|_{\mathrm{HS}}^2\right] + W(1+M_0^2)\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 + WM_0 e^{-cW},$$

where (e_a) is a real orthonormal basis of the tangent directions $\{w \in \operatorname{ran} P : \Re(v_4^* w) = 0\}$. Only the first term requires a derivative.

Variations of v_4 that change or preserve Qv_4. Vary v_4 in a unit tangent direction w, keeping the matrices fixed. Writing $\dot{y} = D_{v_4} y[w]$ and differentiating $Qy = Qv_4$ and $v_4^* y = 0$ gives

$$Q\dot{y} = Qw, \qquad v_4^* \dot{y} = -w^* y.$$

If Qw is of order one, y must follow the moving target at order-one speed, however small y itself is. If $Qw = 0$, the target stays put. The adjustment needed to keep y orthogonal to v_4 is driven by $w^* y$, which already carries the small factor $\|y\|$. For our rank-two generator this gives $\|Pv_4\|\|D_{v_4}K[w]\|_{\mathrm{HS}} \lesssim \|K\|_{\mathrm{HS}}$ when both $Qw = 0$ and $D_{v_4}\xi[w] = 0$, by (9.27).

Even when $Qw = 0$, the scalar normalization ξ can change. Choose the tangent basis adapted to $Qw = 0$ and $D_{v_4}\xi[w] = 0$. The first condition imposes at most $2 \operatorname{rank} Q$ real constraints, and the second adds at most one. Thus at most $2 \operatorname{rank} Q + 1$ basis directions need the order-one bound. On the others the mean squared cost is bounded by $C\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2$. Differentiating χ adds only $O(1)$ in total by (9.30). For $W^{-\beta} \le g \le W^{-\alpha}$, suppressing small powers of W, the construction gives $\operatorname{rank} Q \lesssim Wg^2$ on the cutoff support and $\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 \lesssim g^2$. Thus the derivative bound in (9.38) comes from

$$\underbrace{O(Wg^2) \times O(1)}_{\text{target or normalization changes}} + \underbrace{O(W) \times O(g^2)}_{\text{remaining directions, cost controlled by } \mathbb{E}\|\chi K\|_{\mathrm{HS}}^2} = O(Wg^2).$$

The resulting contraction scale. Conditional on $V_1, \dots, V_{q-1}$, the last block contributes a centered Gaussian with variance $Wg^4\|[\chi L, R_{q-1}]\|_{\mathrm{HS}}^2$ by (9.36). Its cross term with δ_{act} vanishes. The W^{-4} matching error gives $\|L\|_{\mathrm{HS}} \lesssim W^{-4}$ on the cutoff support, where also $\|R_{q-1}\| \lesssim W^3$, so this block costs only $O(W^{-1})$.

Averaging over the preparation blocks and restoring the small powers of W, Lemma 8.12 gives $\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \lesssim \mathfrak{b}$ with

$$\mathfrak{b} = \begin{cases} W^{\varepsilon/2} W g^2, & W^{-\beta} \le g \le W^{-\alpha}, \\ W^{18\theta}, & c_* W^{-1/2} \le g \le W^{-1/2+\theta/2}. \end{cases}$$

Here $0 < \alpha < \beta < 1/2$, $\varepsilon, c_* > 0$ and $0 < \theta \le 1/100$ are fixed. Since $\mathbb{E}\chi \ge \frac{1}{2}$, the criterion in Section 3.3 gives the angular L^4 contraction of $P_{1/2}$ by $1 - c/\mathfrak{b}$ on bounded nonnegative functions invariant under simultaneous unitary rotations, uniformly over incoming flags and energies in a fixed bulk interval. Iteration and smoothing as in Section 3.2 then give (3.1) at length $n_0 \lesssim \mathfrak{b} \log W$.

3.5 Perturbations from a fixed matrix direction

The range $g = W^{o(1)}$ lies between two regimes in which stability has different origins. At small g, the product strongly amplifies directions selected by the random blocks, but averaging over V_5 controls their overlap with the incoming normal. At large g, each transfer is a small perturbation of the free transfer, so a fixed-length product stays close to the free evolution.

It may be possible to obtain the main bound using just these two approaches. Extending Section 9 to $g = W^{o(1)}$ would require controlling the matching error and the final correction even when $g \asymp 1$. Extending Section 11 to the full bulk for $W^{\epsilon} \le g < C_\kappa \sqrt{W}$ would require a drift estimate uniform in the energies outside its near-zero window. For simplicity, we treat both ranges by adapting Drogin's construction for the GOE model [10, Section 4] to our GUE blocks. The Gaussian cost of realizing the prescribed stretch is at most $W^{1+o(1)}$ for $g = W^{o(1)}$. Away from zero energy we bound the cost by $O(Wg^2)$ for $W^{\epsilon} \le g < W$, with fixed $0 < \epsilon < 1$.

We switch from the notation of Section 9 to that of Section 10, in which P and R_i denote blocks of test matrices. We fix U pointwise and stretch the unit direction $e \in Y \cap U^{\perp}$ by e^t before applying the transfers. This leaves both terminal subspaces fixed and adds t to their logarithmic volume ratio.

Write $F = T_1 \cdots T_q$ in the product order of Section 10. In the coordinates of F, the stretch is $e^{\sqrt{2}tM}$ with

$$M = \tfrac{1}{\sqrt{2}}\, S(ee^* - (J_{\mathrm{sig}}e)(J_{\mathrm{sig}}e)^*)S^*, \qquad J_{\mathrm{sig}} = \operatorname{diag}(I_W, -I_W).$$

We seek Hermitian perturbations $X = (X_1, \dots, X_q)$ of the GUE blocks that realize the stretch to first order, so we require

$$\frac{d}{dt} F(V_1 + tX_1, \dots, V_q + tX_q)\Big|_{t=0} = \sqrt{2}\, FM.$$

Multiplying on the left by F^{-1} and using the product rule gives the equivalent equation

$$\mathcal{A}(X) := F^{-1}D_X F = \frac{1}{g}\sum_{i=1}^{q} Q_i^{-1}\begin{pmatrix} 0 & 0 \\ X_i & 0 \end{pmatrix} Q_i = \sqrt{2}M, \qquad Q_i = T_{i+1}\cdots T_q, \quad Q_q = I.$$

3.5.1 The general perturbation and its Gaussian cost

Taking the real Hilbert–Schmidt inner product of both sides with a $2W \times 2W$ test matrix Z in the transfer Lie algebra, we need

$$\langle X, \mathcal{A}^* Z\rangle = \langle \sqrt{2}M, Z\rangle \qquad \text{for every } Z.$$

If we can recover $Z = \mathcal{K}_0\mathbf{R}$ from the data $\mathbf{R} = (R_1, \dots, R_q) = g\mathcal{A}^*Z$, then the right side equals $\langle \sqrt{2}\, g\mathcal{K}_0^* M, \mathcal{A}^* Z\rangle$. Thus $X = \sqrt{2}\, g\mathcal{K}_0^* M$ solves the equation.

To bound its size, Section 10 uses Drogin's block elimination [10, Proposition 4.4] on the support of the smoothing cutoff χ_P for $G_{1/2}$ from Section 3.2. Subtracting two block equations gives the coefficient $(V_{k_j} - V_{i_j})/g$. Normalizing this difference costs a factor g. The GUE bound (10.14), obtained using [6], then controls the inverse uniformly in W. For $g \geq 1$ this gives

$$\|\mathcal{K}_0\| \lesssim g, \qquad \|X\| \leq \sqrt{2}\, g\|\mathcal{K}_0\| \lesssim g^2,$$

since $\|M\|_{\mathrm{HS}} = 1$.

The Gaussian cost. The divergence also depends on how X varies with the onsite blocks. For any fixed $B > 0$, Section 10 obtains throughout $W^{-B} \leq g \leq W^B$

$$\|X\| \lesssim \max\{g^{-1}, g^2\}, \qquad \|\nabla_V X\|_{\mathrm{HS}} \lesssim \sqrt{W}\max\{g^{-1}, g^2\} \quad \text{on } \operatorname{supp}\chi_P.$$

Here $\|X\|^2 = \sum_i \|X_i\|_{\mathrm{HS}}^2$. The factor $\sqrt{W}$ comes from summing squared derivatives over a Hilbert–Schmidt orthonormal basis of Hermitian matrices, as in (10.19). Substituting the bounds for X and $\nabla_V X$ in the Gaussian integration-by-parts estimate gives

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \leq W\mathbb{E}\|\chi_P X\|^2 + \mathbb{E}\|\nabla_V(\chi_P X)\|_{\mathrm{HS}}^2 \lesssim W\max\{g^{-2}, g^4\}.$$

This is the divergence bound of Lemma 8.13(i). Since $\mathbb{E}\chi_P \geq 1/2$ and the stretch fixes the terminal flag with $D_X c = 1$, the criterion of Section 3.3 (Lemma 8.11) gives contraction by $1 - c/\mathfrak{b}$ with $\mathfrak{b} = W\max\{g^{-2}, g^4\}$. For intermediate coupling $g = W^{o(1)}$, the ratio of this scale to the target Wg^2 is $\max\{g^{-4}, g^2\} = W^{o(1)}$. This loss fits within the arbitrarily small power of W allowed in the upper bound.

3.5.2 Saving a factor of g away from zero energy

For $W^\epsilon \leq g < W$, the general perturbation costs Wg^4 and we need Wg^2. Retain one block equation (10.13) before subtraction. With E the energy and P the upper-left block of Z, it reads

$$E(P + P^*) = F_i(\mathbf{R}) + g^{-1}(V_i P + P^* V_i).$$

On the cutoff, $\|F_i(\mathbf{R})\|_{\mathrm{HS}} \lesssim \|\mathbf{R}\|$ and $\|V_i\| \lesssim 1$. The factor g^{-1} therefore lets us use the coarse bound $\|P\|_{\mathrm{HS}} \lesssim g\|\mathbf{R}\|$ to obtain

$$|E|\,\|P + P^*\|_{\mathrm{HS}} \lesssim \|\mathbf{R}\| + g^{-1}\|P\|_{\mathrm{HS}} \lesssim \|\mathbf{R}\|.$$

For $|E| \geq \kappa_1 > 0$, division by $|E|$ saves a factor of g on $P + P^*$. This is exactly the diagonal combination tested by the Hermitian target M in $\langle M, Z\rangle$. Section 10 bounds the off-diagonal blocks by back substitution and estimates the derivative of the minimum-norm perturbation. With $\mathbf{R} = g\mathcal{A}^*Z$, the resulting size and Gaussian cost are

$$\|X\| \lesssim_{\kappa_1} g, \qquad \mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \lesssim_{\kappa_1} Wg^2.$$

This gives the divergence input of Lemma 8.13(ii) at scale $\mathfrak{b} = Wg^2$, hence (3.1) by contraction, iteration and smoothing. Near zero, Section 3.6 estimates transfer growth directly.

3.6 Direct transfer growth

We need a new method for two reasons. First, near $E = 0$, the estimate on $E(P + P^*)$ in Section 3.5 gives no uniform control of $P + P^*$. The general perturbation then costs Wg^4, too large for the target scale Wg^2. Second, the smoothing estimate in Section 3.2 assumes $g \le W^B$ for a fixed B, whereas our upper bound must also cover superpolynomial g.

In both ranges we treat the random term V_j/g as a perturbation of the free rotation, which preserves lengths. The larger g is, the closer each step is to this rotation and the more accurate the expansion in g^{-1} becomes. As g decreases, the drift we seek grows like $(Wg^2)^{-1}$, but the second-order error bound $O(g^{-4})$ grows faster. Its size relative to the drift is $O(W/g^2)$. Thus for $g \ge W$ second order suffices even at $E = 0$. The harder case is near zero energy when $W^\epsilon \le g \ll \sqrt{W}$. There the error can mask the drift, so we must expand further.

Following one direction through the product. For each fixed energy and initial frame, our targets are the two bounds on $\|D_n^{-1}\|$ recalled below, uniform in both parameters.

$$\begin{aligned} \mathbb{P}(\|D_{n_0}^{-1}\| > W^{-D}) &\le W^{-D} && \text{at } n_0 \asymp Wg^2 \log W, \\ \mathbb{E}\|D_n^{-1}\|^s &\le CWe^{-cn/(Wg^2)} && \text{for every } n \ge 1. \end{aligned}$$

The first is (3.1) near zero energy for $W^\epsilon \le g < W$ and fixed $0 < \epsilon < 1$. The second is (3.2) for a fixed $s \in (0, 1)$ and applies throughout the bulk for $g \ge W$, including at $E = 0$. Roughly speaking, we estimate the weakest growth of D_n directly by finding a gain of order $(Wg^2)^{-1}$ per block in the logarithmic volume ratio.

How one block moves the normal. As in Section 3.4, we follow a hyperplane $U \subset Y = \operatorname{ran}\Phi$ through the transfer product, with $\Phi \in \mathcal{F}_W$ and $\dim U = W - 1$. In the free coordinates of Definition 7.1, the two spaces after j blocks are

$$Y_j = S_E^{-1}T_j(E)\cdots T_1(E)SY, \qquad U_j = S_E^{-1}T_j(E)\cdots T_1(E)SU,$$

with the empty product at $j = 0$. Section 3.4 used a Hermitian graph matrix R_j and an unnormalized normal. Here C_j is unitary, and v_j is a unit normal in the upper coordinate:

$$\begin{aligned} Y_j &= \{(u, C_j u) : u \in \mathbb{C}^W\}, \qquad C_j^* C_j = I_W, \\ U_j &= \{(u, C_j u) : u \in \mathbb{C}^W,\ v_j^* u = 0\}, \qquad \|v_j\| = 1. \end{aligned}$$

Write $E = 2\sin\theta$, $h_\theta = (2\cos\theta)^{-1}$, and $\lambda = -ie^{i\theta}$. The onsite block V_j updates the pair (C_{j-1}, v_{j-1}) by

$$\begin{aligned} C_j &= -e^{-2i\theta}[C_{j-1} + \tfrac{ih_\theta}{g}V_j(I_W + C_{j-1})]F_j^{-1}, \\ v_j &= \lambda\frac{F_j^{-*}v_{j-1}}{\|F_j^{-*}v_{j-1}\|}, \qquad F_j = I_W - \tfrac{ih_\theta}{g}V_j(I_W + C_{j-1}), \end{aligned}$$

where $F_j^{-*} = (F_j^{-1})^*$. The upper coordinate evolves by $u' = \lambda F_j u$. The normal transforms by the inverse adjoint because

$$v_{j-1}^* u = 0 \iff v_{j-1}^*(\lambda F_j)^{-1}u' = 0 \iff (\lambda F_j^{-*}v_{j-1})^* u' = 0.$$

For $g \gg 1$, the term involving V_j/g perturbs the free rotation $u \mapsto \lambda u$. After n blocks the transfer on the upper coordinate is $\lambda^n F_n \cdots F_1$. Since $|\lambda| = 1$, the coordinate comparison (11.6) gives

$$\|D_n(\Phi; E)^{-1}\| \asymp_\kappa \|(F_n \cdots F_1)^{-*}\|.$$

The one-step logarithmic volume ratio in (8.17) is $c_j := -\log\|F_j^{-*}v_{j-1}\|$. Starting from a unit normal $v_0 = w$ and multiplying the normalization factors gives

$$\ell_n(C_0, w) := \sum_{j=1}^{n} c_j\big|_{v_0=w} = -\log\|(F_n \cdots F_1)^{-*}w\|.$$

The weakest growth is the minimum of this sum over all initial unit normals. Thus, for each realization,

$$\log \|D_n(\Phi; E)^{-1}\| = -\min_{\|w\|=1} \sum_{j=1}^{n} c_j\big|_{v_0=w} + O_\kappa(1).$$

Martingalization of c_i. Write $\mathbb{E}_{i-1}$ for expectation conditional on $V_1, \dots, V_{i-1}$. Centering each increment gives

$$\sum_{i=1}^{n} c_i = \sum_{i=1}^{n} \mathbb{E}_{i-1} c_i + \underbrace{\sum_{i=1}^{n} (c_i - \mathbb{E}_{i-1} c_i)}_{\text{martingale}}.$$

The martingale enters the final estimates as an error term (Lemmas 11.4 and 11.6).

Where the positive drift is hidden. For the incoming state, write

$$(C, v) = (C_{i-1}, v_{i-1}), \qquad m_1 := W^{-1} \operatorname{Tr} C, \qquad f_1 := v^* C v.$$

The logarithmic gain from block i is

$$c_i = -\tfrac{1}{2} \log\big[v^* (F_i^* F_i)^{-1} v\big], \qquad F_i = I_W - \tfrac{i h_\theta}{g} V_i (I_W + C).$$

The linear term is $(h_\theta/g)\Im(v^* V_i C v)$. It has mean zero, but expanding the logarithm to second order brings in its square. The GUE covariance gives

$$\mathbb{E}_{i-1}\left[\left(\tfrac{h_\theta}{g}\Im(v^* V_i C v)\right)^2\right] = \frac{h_\theta^2}{2W g^2}(1 - \Re(f_1^2)).$$

The constant part of this variance is the gain we seek. The full expansion (11.12) gives, uniformly in the incoming state,

$$\mathbb{E}_{i-1} c_i = \underbrace{\frac{h_\theta^2}{2W g^2}}_{\text{desired gain}} + \frac{h_\theta^2}{g^2}\left[\Re f_1 + \Re(m_1 f_1) - \frac{\Re(f_1^2)}{2W}\right] + O(g^{-4}).$$

At a single block, the first term carries a factor $1/W$, while the terms involving $\Re f_1$ and $\Re(m_1 f_1)$ can be of order g^{-2}. Over n blocks the first term has the same positive value each time, so it adds up to $n h_\theta^2/(2W g^2)$. The other terms change with the state. For example, at $E = 0$ the free rotation sends f_1 to $-f_1$, so consecutive linear terms cancel in the free evolution. The correction below uses this cancellation and controls the quadratic terms to recover a positive drift on the scale $n/(W g^2)$.

Correcting the drift at a bounded endpoint cost. Choose a bounded function Q of the state. Before taking conditional expectations, replace each increment c_i by

$$Y_i := c_i - (Q_i - Q_{i-1}), \qquad Q_i := Q(C_i, v_i).$$

The differences $Q_i - Q_{i-1}$ telescope. Applying the martingale decomposition to Y_i therefore gives

$$\sum_{i=1}^{n} c_i = \underbrace{Q_n - Q_0}_{\text{endpoint difference}} + \underbrace{\sum_{i=1}^{n} \mathbb{E}_{i-1} Y_i}_{\text{corrected drift}} + \underbrace{\sum_{i=1}^{n} (Y_i - \mathbb{E}_{i-1} Y_i)}_{\text{martingale}}.$$

The endpoint difference costs at most $2\|Q\|_\infty$ for any n. For the choices of Q below, the martingale again contributes an error term. To see how Q changes the drift, write

$$\mathbb{E}_{i-1} Y_i = \mathbb{E}_{i-1} c_i - \mathbb{E}_{i-1} Q(C_i, v_i) + Q(C_{i-1}, v_{i-1}).$$

For a bounded test function F of the state, define the transition operator P by

$$(P \circ F)(C_{i-1}, v_{i-1}) = \mathbb{E}_{i-1} F(C_i, v_i).$$

The same operator applies at every step because the update rule is fixed and the onsite blocks V_i are independent and identically distributed. Thus

$$\mathbb{E}_{i-1} Y_i = \mathbb{E}_{i-1} c_i - [(P - I) \circ Q](C_{i-1}, v_{i-1}).$$

We choose Q so that its expected change absorbs the state-dependent terms in $\mathbb{E}_{i-1} c_i$ and reveals the positive drift. This use of a correction follows Schulz-Baldes [29, Lemma 1 and Section 6.1] and Sadel–Schulz-Baldes [27, Sections 3–5]. In both ranges, we seek a bounded correction Q such that $\mathbb{E}_{i-1} Y_i \geq c_\kappa/(Wg^2)$ for most steps i. Near zero energy, for $W^\epsilon \leq g < W$, this bound holds with high probability simultaneously for $n_* < i \leq n_0$, where the initial interval has length $n_* \asymp g^2 \log W$ and the target length is $n_0 \asymp W g^2 \log W$.

3.6.1 A quadratic drift correction for $g \geq W$

Suppose $g \geq W$ and fix $\kappa \in (0, 2)$. Throughout the bulk $|E| \leq 2 - \kappa$, the drift bound holds for every incoming state from the first block onward. Together with the fluctuation estimate, this gives the negative-moment bound (3.2) at every length $n \geq 1$. In this range the $O(g^{-4})$ remainder in the drift expansion is already smaller than the target scale $(Wg^2)^{-1}$.

Using the free rotation to cancel the linear term. The random block enters the update only through V_i/g. For the smooth functions F used here, we can therefore compute $P \circ F$ to the required order by expanding about the free step and averaging over V_i. The term linear in V_i has mean zero. For example, the linear term in the conditional drift (11.12) is $h_\theta^2 g^{-2} \Re f_1$. At energy $E = 2 \sin\theta$ the free step multiplies f_1 by $-e^{-2i\theta}$, so its change is $-(1 + e^{-2i\theta}) f_1$. This gives the first correction

$$Q^{[1]}(C, v) = -h_\theta^2 g^{-2} \Re[f_1/(1 + e^{-2i\theta})], \qquad (P - I) \circ Q^{[1]} = h_\theta^2 g^{-2} \Re f_1 + O(g^{-4}).$$

Using damping to absorb the two quadratic terms. For simplicity we describe the quadratic correction at $E = 0$, where $h_0 = 1/2$. Section 11.2 gives the calculation for general bulk energies with h_θ and the free multiplier $-e^{-2i\theta}$. After the linear correction, the state-dependent part at order g^{-2} is

$$(4g^2)^{-1}[\Re(m_1 f_1) - (2W)^{-1} \Re(f_1^2)].$$

At $V_i = 0$ the free step gives

$$(C_i, v_i) = (-C_{i-1}, -i v_{i-1}), \qquad (m_1, f_1) \longmapsto (-m_1, -f_1).$$

Thus $m_1 f_1 \mapsto (-m_1)(-f_1) = m_1 f_1$ and $f_1^2 \mapsto (-f_1)^2 = f_1^2$. For the random step, the terms linear in V_i/g average to zero, so

$$\begin{aligned} P \circ m_1 &= -m_1 + O(g^{-2}), & P \circ f_1 &= -f_1 + O(g^{-2}), \\ P \circ (m_1 f_1) &= m_1 f_1 + O(g^{-2}), & P \circ (f_1^2) &= f_1^2 + O(g^{-2}). \end{aligned}$$

For f_1, dividing by the change in the free multiplier, $-1 - 1 = -2$, gave $Q^{[1]}$. For the two quadratic terms this divisor is $1 - 1 = 0$. Since we seek a lower bound on the drift, it suffices to offset their negative contribution. For this we use the corrected moments (11.16)–(11.17),

$$\begin{aligned} M_1(C) &= m_1 - (8g^2)^{-1}(1 + m_2 + 2m_1^2), \quad m_2 = W^{-1} \operatorname{Tr} C^2, \\ F_1(C, v) &= f_1 - (8g^2)^{-1}(4m_1 f_1 + 2f_2 - f_1^2 - |f_1|^2), \quad f_2 = v^* C^2 v. \end{aligned}$$

For the original moments, the $O(g^{-2})$ errors in the one-step averages are as large as the damping we seek. The corrections $M_1 = m_1 + O(g^{-2})$ and $F_1 = f_1 + O(g^{-2})$ give the more precise bounds (11.18)–(11.19),

$$-(P-I)\circ|M_1|^2 \geq g^{-2}\big[c_0|M_1|^2 - C(W^{-2}+g^{-4})\big],$$
$$-(P-I)\circ|F_1|^2 \geq g^{-2}\big[c_0|F_1|^2 - C|M_1|^2 - C(W^{-1}+g^{-4})\big].$$

To apply these bounds to the quadratic drift, replace m_1, f_1 by M_1, F_1. Since all four moments are uniformly bounded,

$$m_1 f_1 - M_1 F_1 = (m_1 - M_1)f_1 + M_1(f_1 - F_1) = O(g^{-2}), \qquad f_1^2 - F_1^2 = O(g^{-2}).$$

The prefactor $(4g^2)^{-1}$ makes the resulting drift error $O(g^{-4})$. Take

$$Q = Q^{[1]} + A|M_1|^2 + B|F_1|^2$$

First choose $B > 0$ small enough that the error $CB/(Wg^2)$ uses at most half of the gain $1/(8Wg^2)$. By Young's inequality the product $|M_1F_1|$ is bounded by $\delta|F_1|^2 + (4\delta)^{-1}|M_1|^2$. Taking δ small compared with B and then A large, the two damping bounds absorb the quadratic loss at scale g^{-2} and leave

$$\mathbb{E}_{i-1}Y_i \geq c_1/(Wg^2) - Cg^{-4} - C/(W^2g^2) \geq c/(Wg^2).$$

Since $g \geq W$, we have $W/g^2 \leq 1/W$, so both errors are at most C/W times $(Wg^2)^{-1}$. For large W the lower bound therefore holds for every incoming state from the first block onward.

Lemma 11.4 combines this drift with the fluctuation bound to give $\mathbb{E}_{i-1}e^{-sY_i} \leq e^{-c/(Wg^2)}$ for some fixed $s \in (0,1)$. Iterating the conditional bound and using the bounded endpoint difference and the basis estimate (11.9) gives $\mathbb{E}\|D_n(\Phi;E)^{-1}\|^s \leq CWe^{-cn/(Wg^2)}$ for every $n \geq 1$, as required in (3.2).

3.6.2 A higher-order correction near zero energy

Fix $0 < \epsilon < 1$ and suppose $W^\epsilon \leq g < W$. We explain the argument at $E = 0$, where $h_0 = 1/2$ and the free step sends C to $-C$. Lemma 11.5 proves the bound uniformly on a fixed interval around zero. Here, for each fixed $D > 0$, the target is

$$\sup_{\Phi\in\mathcal{F}_W} \mathbb{P}(\|D_{n_0}(\Phi;0)^{-1}\| > W^{-D}) \leq W^{-D}, \qquad n_0 = \lceil C_1 W g^2 \log W\rceil,$$

for sufficiently large fixed C_1 and all large W.

The initial interval. For $g \geq W$, the corrected drift bound in the preceding subsection holds for every incoming state from the first block onward. Here the early increments c_i can have either sign, and losses can accumulate. In the moment estimate (11.60), the contribution of the initial state is bounded by Ce^{-ci/g^2}. This bound is still of order one at $i \asymp g^2$ and falls to a negative power of W at $n_* = \lceil C_* g^2 \log W\rceil$. We therefore split the total growth as

$$\sum_{i=1}^{n_0} c_i = \underbrace{\sum_{i=1}^{n_*} c_i}_{\ell_{n_*}} + \sum_{i=n_*+1}^{n_0} c_i.$$

For the first sum, the rough bound $|\mathbb{E}_{i-1}c_i| \leq Cg^{-2}$ and the martingale estimate give $|\ell_{n_*}| \leq C_D \log W$ with probability at least $1 - W^{-D}$ by (11.29). Once the moments have stabilized, the corrected conditional drift is of order $(Wg^2)^{-1}$ per block, or W^{-1} over g^2 blocks. Over the remaining $n_0 - n_*$ blocks it therefore contributes order $C_1 \log W$. Choosing C_1 large makes this gain exceed the possible initial loss $C_D \log W$.

Further expansion. We first estimate the trace and directional moments that occur in the conditional drift. The first two moments already show why an even directional term needs a different correction from $Q^{[1]}$.

Smallness of the first trace and directional moments. For $n_* \le i < n_0$ we estimate c_{i+1} at the incoming state (C_i, v_i). To illustrate the first two corrections, take $g \asymp W^\epsilon$ with $0 < \epsilon < 1/4$, so $W^{-1} = o(g^{-4})$. Start with the first corrections to $m_1 = W^{-1} \operatorname{Tr} C_i$ and $f_1 = v_i^* C_i v_i$ from the preceding subsection, denoted here by $M_1^{[1]}$ and $F_1^{[1]}$. The damping estimates (11.18)–(11.19) show why we bound the trace moment first. Write $u_i = \mathbb{E}|M_1^{[1]}(C_i)|^2$ and $w_i = \mathbb{E}|F_1^{[1]}(C_i, v_i)|^2$. Taking expectations gives

$$u_{i+1} \le (1 - cg^{-2})u_i + Cg^{-6},$$
$$w_{i+1} \le (1 - cg^{-2})w_i + Cg^{-2}u_i + Cg^{-6}.$$

In the first line the geometric sum of the damping factors is $O(g^2)$. Thus $u_i \lesssim e^{-ci/g^2} + g^{-4}$. Inserting this in the second line and summing again gives $w_i \lesssim e^{-c'i/g^2} + g^{-4}$ for some $c' > 0$. For $i \ge n_*$, choosing C_* sufficiently large gives

$$u_i \lesssim g^{-4}, \qquad w_i \lesssim g^{-4}.$$

For every fixed $1 \le p < \infty$, iterating (11.36)–(11.37) gives $\|M_1^{[1]}(C_i)\|_{L^p} + \|F_1^{[1]}(C_i, v_i)\|_{L^p} \lesssim_p g^{-2}$ for $i \ge n_*$, with C_* chosen for p.

Estimating $M_2^{[1]}$. At $E = 0$ the free step $C \mapsto -C$ leaves $m_2 = W^{-1} \operatorname{Tr} C^2$ unchanged. Squaring the one-step update (11.3) and averaging its normalized trace gives

$$(P - I) \circ m_2 = -(2g)^{-2}\big[(1 + 2m_1 + m_2)^2 + 2(1 + m_1)(m_1 + 2m_2 + m_3)\big] + O(g^{-4}).$$

The odd part of the bracket is $2(3m_1 + m_3 + 4m_1m_2)$. It changes sign under the free step, so the correction $M_2^{[1]} = m_2 - (2g)^{-2}(3m_1 + m_3 + 4m_1m_2)$ cancels it. The even part is $1 + 6m_2 + m_2^2 + 6m_1^2 + 2m_1m_3$, as in (11.52). For $i \ge n_*$, the first-moment bound on m_1 and $|m_3| \le 1$ give, in each fixed L^p,

$$(P - I) \circ M_2^{[1]} = -(2g)^{-2}(1 + 6m_2 + m_2^2) + O(g^{-4}).$$

The unique zero of this leading polynomial in the unit disk is $a_2^{(0)} = -3 + \sqrt{8}$. Set $X_i = M_2^{[1]}(C_i) - a_2^{(0)}$. Factoring the polynomial as $(m_2 - a_2^{(0)})(6 + m_2 + a_2^{(0)})$ and using $m_2(C_i) - a_2^{(0)} = X_i + O(g^{-2})$ gives

$$\mathbb{E}_i X_{i+1} = \big[1 - (2g)^{-2}(6 + m_2(C_i) + a_2^{(0)})\big] X_i + O(g^{-4}).$$

Since $|m_2(C_i)| \le 1$ and $\Re(6 + m_2(C_i) + a_2^{(0)}) > 4$, the multiplier has modulus at most $1 - cg^{-2}$ for large g. The same moment iteration as for $M_1^{[1]}$ gives

$$\|M_2^{[1]}(C_i) - a_2^{(0)}\|_{L^p} \lesssim_p g^{-2}, \qquad n_* \le i < n_0.$$

Since $m_2 - M_2^{[1]} = O(g^{-2})$, the same estimate holds for m_2.

Estimating $F_2^{[1]}$. The free step also fixes $f_2 = v^* C^2 v$. Each term in the bracket below changes sign under the free step. Thus the odd terms in (11.38) cancel in the drift of the first correction (11.39),

$$F_2^{[1]} = f_2 - (8g^2)^{-1}\big[(3 + 4m_2)f_1 + 3f_3 + 8m_1 f_2 - (f_1 + \overline{f_1})f_2\big].$$

In the even part of the order-$(2g)^{-2}$ coefficient, every term except $-6f_2 - 2m_2 f_2$ contains m_1 or f_1. Both first moments are $O(g^{-2})$ for $i \ge n_*$. Using $f_2 = F_2^{[1]} + O(g^{-2})$ in the remaining term gives, in each fixed L^p,

$$(P - I) \circ F_2^{[1]} = -(2g^2)^{-1}(3 + m_2)F_2^{[1]} + O(g^{-4}).$$

Since $\Re(3+m_2) \geq 2$, the random step damps $F_2^{[1]}$ toward zero at rate g^{-2}. Using the bounds above for $M_1^{[1]}, F_1^{[1]}$ in the moment recursion (11.41) gives

$$\|F_2^{[1]}(C_i, v_i)\|_{L^p} \lesssim_p g^{-2}, \qquad n_* \leq i < n_0.$$

The same bound holds for f_2.

An even term in the corrected drift. Substitute these estimates into the drift after $Q^{[1]} = -(8g^2)^{-1}\Re f_1$. By (11.12),

$$\mathbb{E}_i c_{i+1} - [(P-I)\circ Q^{[1]}](C_i, v_i) = (8Wg^2)^{-1} + (4g^2)^{-1}\left[\Re(m_1 f_1) - (2W)^{-1}\Re(f_1^2)\right] + O(g^{-4}).$$

The two products are $O(g^{-4})$ in each fixed L^p by the first-moment bounds. As we expand further, a term linear in an odd f_j with a fixed coefficient can be cancelled just as for f_1, since its free increment is $-2f_j$. Consider instead a fixed multiple of $\Re f_2$. Its free increment is zero.

To see where such a term enters, keep only the coefficient of f_2, taking real parts and omitting terms with an explicit W^{-1}. At fourth order the logarithm in c_{i+1} and the update of $Q^{[1]}$ give $(2g)^{-4}[-(1+m_1)^2 + (1+m_1)]$. The constant parts cancel. Substituting the first trace correction $m_1 = M_1^{[1]} + (8g^2)^{-1}(1 + m_2 + 2m_1^2)$ leaves a contribution $-\frac{1}{2}(1 + a_2^{(0)})$ to the coefficient of $(2g)^{-6}\Re f_2$ when the traces are set to their centers. The next Wick terms from the logarithm and the update of $Q^{[1]}$ contribute $(4 + 2a_2^{(0)}) - (5 + 2a_2^{(0)}) = -1$. Thus, by (11.43), the even term to be matched is

$$-\tfrac{1}{2}(2g)^{-6}(3 + a_2^{(0)})\Re f_2.$$

The $F_2^{[1]}$ calculation above explains how to match it. At the trace centers, a unit coefficient of f_2 produces an f_2 coefficient $-(2g^2)^{-1}(3 + a_2^{(0)})$ in $(P-I)\circ f_2$ by (11.38). To match the displayed term, we therefore include $\frac{1}{4}(2g)^{-4}\Re f_2$ in $Q^{[2]} - Q^{[1]}$. The factor g^2 in this division is the cost of the even mode.

Correcting the moments to higher order. For the full range $W^\epsilon \leq g < W$ we carry out the moment corrections first. Write $M_j^{[r]}, F_j^{[r]}$ for moment j after r rounds. Each round expands their one-step updates one order further in g^{-2}. A monomial of total index d acquires the free sign $(-1)^d$. If its parity differs from that of j, the two free multipliers differ by ± 2, so its coefficient can be cancelled as in the first corrections above. The terms of the same parity remain in the update. Consider the two even terms $m_2 f_2$ and $m_1 f_3$ in the example above. The first combines with $3f_2$ to give the damping coefficient $3 + m_2$, whose real part is at least 2. For the second, the bound on m_1 already suffices. Since $|f_3| \leq 1$, its contribution to the one-step drift is $O(g^{-4})$ in each fixed L^p.

For general j, we first center the trace moments at a_j, as we did for the second trace. The second-order drift formula for the directional moments (11.51) gives the terms of the same parity as j. Its diagonal term gives damping at order g^{-2}, and the terms involving lower directional moments are controlled inductively. The only higher directional moment at this order is f_{j+1}, which appears multiplied by m_1. Thus $|f_{j+1}| \leq 1$ suffices at this step. Higher-order couplings carry at least g^{-4}, so even after summing over the $O(g^2)$ damping time they retain a factor g^{-2}. Iterating the moment recursion (11.59) uses this gain to reach the required accuracy.

Wick pairing restricts the indices at each order. For instance, at order $g^{-2r}W^0$ in the conditional drift, each index has modulus at most r and the sum of their moduli is at most $2r$ (see the proof of Lemma 11.7). Choose the expansion depth R with $2\epsilon R > 1$. A remaining one-step error $O(g^{-2R-2})$ accumulates over the damping time $O(g^2)$ to $O(g^{-2R}) = o(W^{-1})$. Writing M_j, F_j for the resulting moments, (11.60) gives, for fixed small $\eta > 0$, with high probability simultaneously at the required indices and times $n_* \leq i < n_0$,

$$|M_j(C_i) - a_j| \leq W^{-1+\eta}, \qquad |F_j(C_i, v_i)| \leq W^{-1/2+\eta}.$$

Choosing the drift correction. Now expand $\mathbb{E}_i c_{i+1}$ in these corrected moments. Set the traces to their centers and collect the terms linear in F_j. The preceding bounds control the products and the errors from replacing the traces. At each round we add a linear combination of the F_j's to $Q^{[s]}$ so that $(P-I)\circ(Q^{[s+1]}-Q^{[s]})$ matches the linear coefficients still present at that order. The divisor found for f_2 persists for every even index. For $j>0$, the two diagonal terms in the second-order drift formula for the directional moments (11.51) are $-3jF_j$ and $-ja_2F_j$ at the trace centers. Including the free sign, the coefficient of F_j in $(P-I)\circ F_j$ through order g^{-2} at $E=0$ is

$$(-1)^j\bigl[1-j(2g)^{-2}(3+a_2)\bigr]-1.$$

It is of order one for odd j and of order g^{-2} for even j, since $3+a_2=3+a_2^{(0)}+O(g^{-2})=\sqrt{8}+O(g^{-2})$ by (11.56). The other linear terms at this order have lower indices of the same parity, so we can match coefficients by triangular inversion. The higher-order part of the update is $O(g^{-4})$, so even after the $O(g^2)$ inverse each successive correction gains a factor $O(g^{-2})$. Section 11.3.4 carries out this finite iteration in (11.67), retaining the free phases for energies near zero. At $E=0$ the state-independent term $(8Wg^2)^{-1}$ from (11.12) remains, and the resulting bounded correction Q_g gives $\mathbb{E}_i Y_{i+1}=(8Wg^2)^{-1}+o((Wg^2)^{-1})$ by (11.70).

Returning to the two sums split at n_*, the second contributes corrected drift of order $C_1\log W$ at $n_0=\lceil C_1Wg^2\log W\rceil$. By Lemma 11.6, the martingale costs at most $C_D\sqrt{C_1}\log W$ and the initial interval at most $C_D\log W$ with high probability. Including the bounded endpoint difference, (11.73) yields

$$\ell_{n_0}(C_0,w)\geq\bigl(cC_1-C_D\sqrt{C_1}-C_D\bigr)\log W$$

for each fixed initial normal w. Taking C_1 sufficiently large and applying (11.9) to a fixed orthonormal basis with one extra power of W in the failure bound gives the claimed estimate for $\|D_{n_0}(\Phi;0)^{-1}\|$.

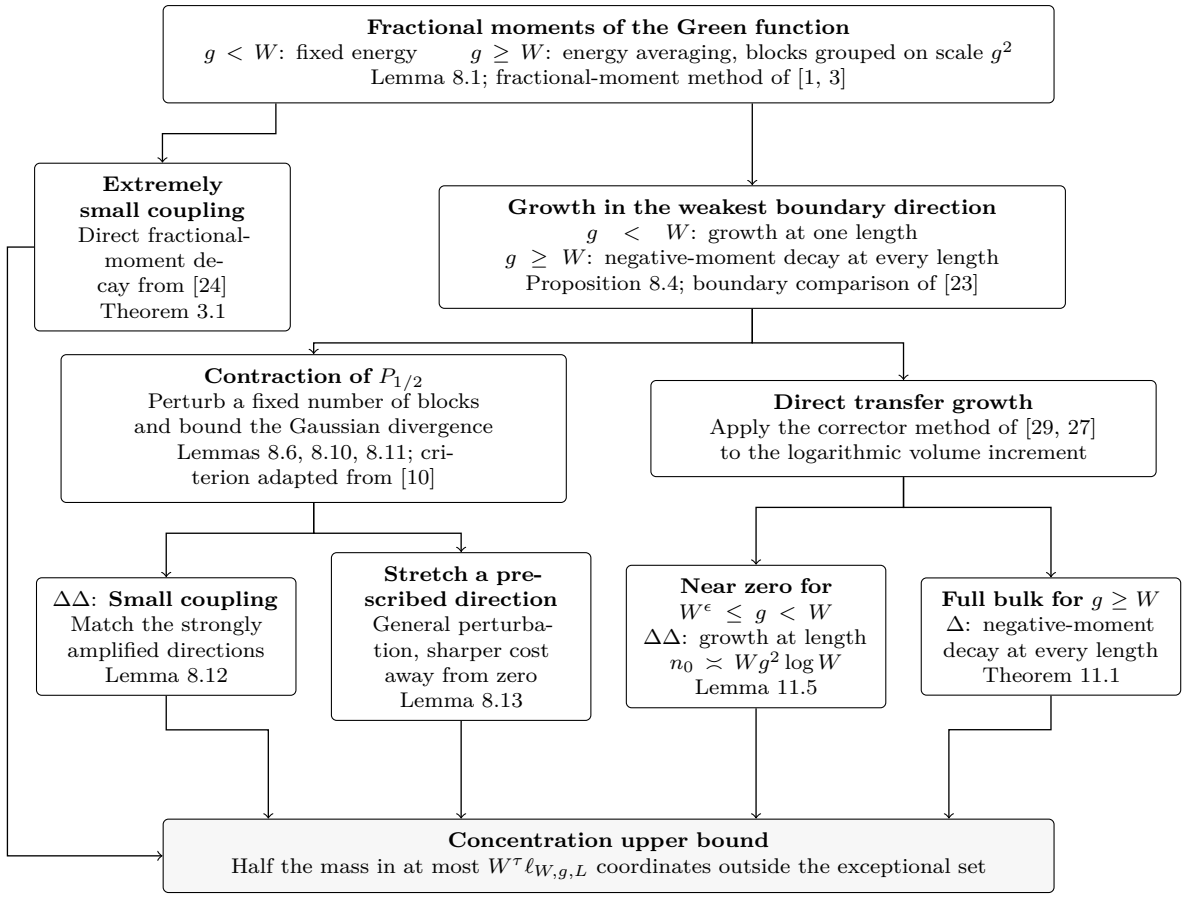


Figure 3: Upper-bound methods. $\Delta\Delta$ marks the small-coupling construction and the near-zero growth estimate. Δ marks the full-bulk estimate for $g \geq W$. Both direct transfer estimates use a corrected drift on the scale $(Wg^2)^{-1}$. Figure 4 gives the coupling and energy ranges.

3.7 Component estimates and proof of the upper bound

The deterministic bound $\ell(\psi_k) \leq WL$ suffices whenever $L \leq W^\tau \max\{1, Wg^2\}$. For longer cycles we use the four concentration estimates below. The first uses [24] directly. The other three use transfer growth, with fixed-energy fractional moments for $g < W$ and energy-averaged fractional moments for $g \geq W$. Figures 3 and 4 give the methods and their coupling ranges, respectively.

Coupling	Main idea	Corresponding results
Extremely small $g \le c_* W^{-1/2}$	The fractional-moment estimate of [24] gives concentration on the single-block scale.	Theorem 3.1
Small $c_* W^{-1/2} \le g \le W^{-\epsilon}$	Match the amplified directions to obtain contraction at the block scale $\max\{1, Wg^2\}$ up to a small power of W.	Lemma 8.12; Theorem 3.2
Intermediate $W^{-\epsilon} \le g \le W^{\epsilon}$	A fixed matrix direction gives the block scale $W \max\{g^{-2}, g^4\}$, sufficient for small ϵ.	Lemma 8.13(i); Theorem 3.3
Large $W^{\epsilon} \le g < W$	Away from zero, sharpen the contraction of $P_{1/2}$. Near zero, estimate transfer growth directly. Both give growth at one length.	Lemmas 8.13(ii), 11.5; Theorem 3.4
Largest couplings $g \ge W$	Direct negative-moment decay at every length gives the energy-averaged bound throughout the bulk.	Theorem 11.1; Lemma 8.1(ii); Theorem 3.4

Figure 4: Upper-bound regimes for fixed $0 < \epsilon < 1/4$. For $W^{\epsilon} \le g < W$ the two energy windows overlap. The component theorems give the precise bounds.

The four theorems specify an energy window J and a block radius R for which, given $D > 0$,

$$\frac{1}{WL}\mathbb{E}\#\left\{k : \lambda_k \in J, \quad \min_{a \in \mathbb{Z}_L} \sum_{\|b-a\|_L > R} \|\psi_{k,b}\|_2^2 > W^{-D}\right\} \le W^{-D}. \tag{3.6}$$

For every remaining eigenvector with $\lambda_k \in J$, a block ball of radius R contains at least half the mass, so $\ell(\psi_k) \le \min\{N, W(2\lceil R\rceil + 1)\}$ for large W.

Theorem 3.1 (Strong-disorder concentration; finite-volume corollary of [24, Theorem 1])**.** *There exists a sufficiently small constant $c_* \in (0,1)$ such that, for every fixed $D > 0$, all sufficiently large W, every deterministic $g \le c_* W^{-1/2}$, and every L, (3.6) holds with*

$$R := \lceil (\log W)^2 \rceil, \qquad J = \mathbb{R}. \tag{3.7}$$

The width threshold is independent of L, g.

The proof is given at the end of Section 8.1.

Theorem 3.2 (Sharp concentration at small coupling)**.** *Fix $0 < \alpha < 1/2$, $c_*, D, \varepsilon > 0$, and $\kappa \in (0,2)$. There is W_0 such that, for every $W \ge W_0$, every L, and every deterministic*

$$c_* W^{-1/2} \le g \le W^{-\alpha},$$

(3.6) holds with

$$R \le W^{\varepsilon} \max\{1, Wg^2\}, \qquad J = [-2+\kappa, 2-\kappa]. \tag{3.8}$$

The width threshold is independent of L, g in the stated range.

The proof in Section 8.5 uses Lemma 8.12 from Section 9.

Theorem 3.3 (Concentration from the general lift)**.** *Fix $B, E_0, D > 0$. For all sufficiently large W, every deterministic $W^{-B} \le g \le W^B$, and every L, (3.6) holds with*

$$R \le W \max\{g^{-2}, g^4\}(\log W)^2, \qquad J = [-E_0, E_0]. \tag{3.9}$$

The width threshold depends only on D, B, E_0 and is uniform in g, L.

The proof in Section 8.5 uses the general lift in Section 10.

Theorem 3.4 (Concentration at large coupling). *Fix $\kappa \in (0,2)$ and $D, \epsilon > 0$. There is W_0 such that, for every $W \geq W_0$, every deterministic finite $g \geq W^{\epsilon}$, and every L, (3.6) holds with*

$$R \leq Wg^2(\log W)^2, \qquad J = [-2+\kappa, 2-\kappa]. \tag{3.10}$$

The width threshold depends only on κ, D, ϵ, and not on L, g.

The proof concludes Section 11.

Proof of the upper bound and completion of Theorem 1.4. Fix $0 < \tau < 1$ and define

$$B_+ = \#\{k : \lambda_k \in J_{g,\kappa,L},\ \ell(\psi_k) > W^{\tau}\ell_{W,g,L}\}.$$

If $L \leq W^{\tau}\max\{1, Wg^2\}$, then $B_+ = 0$ by the cap $\ell(\psi_k) \leq WL$. Otherwise $\ell_{W,g,L} = \ell_{W,g}$ and $L > W^{\tau} \geq L_{\kappa/2}$ for large W, so $F_g(-|E|) \geq \kappa/2$ on $J_{g,\kappa,L}$ by (1.16). Split the remaining couplings at $c_*W^{-1/2}$, $W^{-\tau/8}$ and $W^{\tau/4}$. In increasing order use Theorem 3.1, Theorem 3.2 with $(\alpha, \varepsilon) = (\tau/8, \tau/2)$, Theorem 3.3 with $(B, E_0) = (\tau/4, 4)$, and Theorem 3.4 with $\epsilon = \tau/4$. The first theorem holds on $\mathbb{R}$. For the small- and large-coupling theorems, (1.14) at $\kappa/2$ gives a common bulk margin $\kappa_0 > 0$. For the intermediate range, (1.12) gives $J_{g,\kappa,L} \subseteq [-4, 4]$. For Theorem 3.3, the radius ratio to $\max\{1, Wg^2\}$ costs at most $W^{\tau/2}(\log W)^2$, since the extra factors are g^{-4} for $g \leq 1$ and g^2 for $g \geq 1$. The radii in all four theorems therefore satisfy $W(2\lceil R\rceil + 1) \leq W^{\tau}\ell_{W,g}$ for large W. Thus (3.6) in the applicable range gives

$$(WL)^{-1}\mathbb{E}B_+ \leq W^{-D}. \tag{3.11}$$

By (1.16) and Lemma 1.3, the expected fraction in (1.17), with $0/0 = 0$, is at least

$$1 - (1-2\kappa)^{-1}(WL)^{-1}\mathbb{E}(B_- + B_+) - C_{\kappa}e^{-c_{\kappa}W^2L} \geq 1 - W^{-D}, \tag{3.12}$$

using (2.12) and (3.11) with exponent $D+2$. □

4 Spreading from local resolvent moments

We prove Theorem 2.2 by counting large spectral atoms. For large volumes, a comparison with a local periodic resolvent first transfers the moment bound to the full cycle.

Theorem 4.1 (Local resolvent comparison). *Fix a central block a and an integer $R \geq 1$ with $2R+1 < L$. Let $I = [a-R, a+R]$, and let H_I^{per} be the restriction of H to I with periodic boundary conditions and the same onsite blocks and hopping. For every $z = E + i\eta$ with $\eta R \geq 2$, we have, in the ordering of positive semidefinite $W \times W$ matrices,*

$$C^{-1}\Im G_{I,aa}^{\mathrm{per}}(z) \ \leq\ \Im G_{L,aa}(z) \ \leq C\Im G_{I,aa}^{\mathrm{per}}(z), \tag{4.1}$$

where C is an absolute constant. The estimate is deterministic.

Proof. Define a rectangular multiplication operator from the full cycle of L blocks to I by $\chi(b) = \max\{0, 1 - |b-a|/R\}$. Onsite (i.e. within the same block) terms commute with χ. The geometric resolvent identity reads

$$G_I^{\mathrm{per}}KG_L = \chi G_L - G_I^{\mathrm{per}}\chi, \quad K := H_I^{\mathrm{per}}\chi - \chi H_L$$

For $u \in \mathbb{C}^W$, let u_L, u_I denote its embeddings at block a in the full cycle and in I, respectively. Since $\chi(a) = 1$, $(\chi G_L)_{aa} = (G_L)_{aa}$ and $(G_I^{\mathrm{per}}\chi)_{aa} = (G_I^{\mathrm{per}})_{aa}$. Cauchy–Schwarz and Ward's identity give

$$\begin{aligned}
\left|u^*(G_{L,aa} - G_{I,aa}^{\mathrm{per}})u\right| &= |u_I^*G_I^{\mathrm{per}}KG_Lu_L| \\
&\leq \|K\|\ \|(G_I^{\mathrm{per}})^*u_I\|\ \|G_Lu_L\| \leq \|K\|\ \frac{1}{\eta}\,(u^*\Im G_{L,aa}u)^{1/2}\left(u^*\Im G_{I,aa}^{\mathrm{per}}u\right)^{1/2} \\
&\leq \|K\|\ \frac{1}{2\eta}\left(u^*\Im G_{L,aa}u + u^*\Im G_{I,aa}^{\mathrm{per}}u\right)
\end{aligned}$$

The only potentially nonzero blocks of the $(2R+1)\times L$ block matrix K are

$$K_{bc} = \frac{g}{\max\{1,g\}}(\chi(c)-\chi(b))I_W, \qquad b,c\in I, \quad |b-c|=1.$$

The Schur bound gives $\|K\| \le 2g/(\max\{1,g\}R) \le \eta$. Thus (4.1) holds with $C=3$. □

In the application to Theorem 2.2, it suffices to bound the resolvent moments on the local cycle of L_c blocks.

The comparison (4.1) transfers local moments to the full cycle. We now convert these moments into a bound on the number of eigenvectors with short half-mass length.

Lemma 4.2 (Poisson smoothing of heavy spectral atoms)**.** *Let $\mu = \sum_k p_k\delta_{\lambda_k}$ be a finite positive measure, $\eta, h > 0$, and*

$$f(E) = \int \frac{\eta}{(\lambda-E)^2+\eta^2}\,d\mu(\lambda), \qquad J_\eta = J + [-\eta,\eta].$$

Then

$$\sum_{\lambda_k\in J} p_k \mathbf{1}_{\{p_k>h\}} \le \frac{2}{\pi}\int_{J_\eta} f(E)\mathbf{1}_{\{f(E)>h/(2\eta)\}}\,dE. \tag{4.2}$$

There is also a matrix version. If $\boldsymbol{\mu} = \sum_k v_k v_k^*\delta_{\lambda_k}$, $m_k = \|v_k\|^2$, and $F(E) = \int \eta((\lambda-E)^2+\eta^2)^{-1}\,d\boldsymbol{\mu}(\lambda)$, then

$$\sum_{\lambda_k\in J} m_k \mathbf{1}_{\{m_k>h\}} \le \frac{2}{\pi}\int_{J_\eta} \operatorname{Tr} F(E)\mathbf{1}_{\{\|F(E)\|>h/(2\eta)\}}\,dE. \tag{4.3}$$

Proof. If $p_k > h$ and $|E-\lambda_k| \le \eta$, positivity gives $f(E) \ge p_k/(2\eta) > h/(2\eta)$. Also $\int_{\lambda_k-\eta}^{\lambda_k+\eta} \eta((\lambda_k - E)^2+\eta^2)^{-1}\,dE = \pi/2$. On the union of these intervals, f is at least the sum of the corresponding Poisson contributions, each restricted to its own interval. Tonelli's theorem proves (4.2), even when the intervals overlap. In the matrix case, the rank-one atom forces $\|F(E)\| \ge m_k/(2\eta)$ on the same interval, and the trace contains its scalar contribution. The identical argument proves (4.3). □

Proof of Theorem 2.2. Put $\eta_L = W^\nu/\ell_{W,g,L}$. If $L_c < L$, then $WL_c > \ell_{W,g}$, $\eta_c = \eta_L = \eta$, and $\eta R \ge 2$. Theorem 4.1 and the law of the local periodic restriction transfer (2.6) to the original cycle. For $L_c = L$, apply the assumption directly. Thus, with C_p enlarged as needed,

$$\sup_{E\in J_{\eta_L}} \max_{1\le x\le WL} \mathbb{E}(\Im G_{xx}(E+i\eta_L))^p \le C_p.$$

Set $r = W^{-\tau}\ell_{W,g,L}$, and let $B_-(r)$ count eigenvectors with eigenvalue in J and half-mass length at most r. Using (4.2) at each coordinate, we obtain

$$\begin{aligned} B_-(r) &\le 4\sum_{k:\lambda_k\in J}\sum_x |\psi_k(x)|^2\mathbf{1}_{\{|\psi_k(x)|^2>1/(4r)\}} \\ &\le \frac{8}{\pi}\sum_x \int_{J_{\eta_L}} \Im G_{xx}(E+i\eta_L)\mathbf{1}_{\{\Im G_{xx}(E+i\eta_L)>1/(8r\eta_L)\}}\,dE. \end{aligned}$$

The moment bound above and $|J_{\eta_L}| \le 2E_0+2$ then give

$$\frac{\mathbb{E}B_-(r)}{WL} \le C_p W^{-(p-1)(\tau-\nu)}.$$

Since $(p-1)(\tau-\nu) > D+1$, this is at most W^{-D} for sufficiently large W, proving (2.7), uniformly in g, L under the stated assumptions. □

5 The loop hierarchy and propagation

Basic strategy: Following [33, 31], we obtain resolvent bounds through the resolvent loops defined in (5.5). In particular, the mixed two-loop controls the Hilbert–Schmidt norm of a Green block,

$$\mathcal{L}_2(t; a, b; +, -) = W^{-2}\|(G_t)_{ba}\|_{\mathrm{HS}}^2,$$

which, together with unitary invariance and fluctuation averaging, yields entrywise and averaged resolvent bounds. Under the characteristic stochastic flow (5.1)–(5.3), these loops satisfy the hierarchy (5.8): both the drift and the quadratic variations of the martingales are expressed in terms of loops. The main task is to show that the random loops $\mathcal{L}_k(t)$ remain close to the deterministic primitive loops $\mathcal{K}_k(t)$, defined by (5.10), as the spectral height decreases to η_L. For bounded coupling, we follow the loop hierarchy method and prove its inputs here, adapting the stability analysis of [11, Section 8 and Appendix B.1].

The main new difficulty is to make this argument uniform for arbitrarily large g. Here we use transfer matrices, a tool more commonly associated with localization proofs. Their role is natural: after normalization, the onsite randomness has size g^{-1}, while the free transfer matrix in the bulk is conjugate to a unitary matrix. We exploit this deterministic structure and treat the onsite randomness as a small centered perturbation, a viewpoint that also underlies our transfer-matrix analysis of the upper bound.

Transfer estimates provide three inputs to the loop hierarchy. Combined with Ward's identity, they give initial resolvent moments at $\eta \asymp g^{-2}$, as well as coarse moment bounds throughout the subsequent propagation. They also yield spatial interpolation of Green functions within cells much shorter than g^2. This interpolation, extended to both random and primitive loops, reduces the spatial estimates to $W^{O(1)}$ sampled block tuples, even when the cycle length is superpolynomial in W. These inputs allow the loop hierarchy to propagate the sharp estimates down to η_L, with constants uniform in g.

5.1 The characteristic stochastic flow

For each coupling g, define

$$\begin{aligned} A_g &\coloneqq \sqrt{\gamma_g}\, g A_L, \quad \gamma_g \coloneqq \max\{1, g\}^{-2}. \\ H_t &\coloneqq H_t(g) = A_g \otimes I_W + \mathcal{B}_t^{(g)}, \qquad 0 \le t \le 1. \end{aligned} \tag{5.1}$$

The process $\mathcal{B}_t^{(g)}$ is block diagonal, with independent Hermitian Brownian blocks started at zero and covariance

$$\mathbb{E}(d\mathcal{B}_t^{(g)})_{ai,aj}(d\mathcal{B}_t^{(g)})_{bk,b\ell} = W^{-1}\gamma_g \delta_{ab}\delta_{i\ell}\delta_{jk}\, dt.$$

At $t = 1$, $H_1(g)$ has the same law as the normalized Hamiltonian H in (1.3) at coupling g.

For $z \in \mathbb{C}_+$, define the finite-cycle Dyson solution by

$$m_{g,L}(z) = L^{-1} \sum_{p \in (2\pi/L)\mathbb{Z}_L} \frac{1}{2\sqrt{\gamma_g}\, g \cos p - z - \gamma_g m_{g,L}(z)}. \tag{5.2}$$

We take the Herglotz solution with asymptotic $-1/z$ and use its boundary values at real energies. Equation (5.2) specializes [31, (3.2)–(3.4)] to our hopping and covariance.

Fix a real energy E with $\Im m_{g,L}(E) > 0$, and define

$$\begin{aligned} m &\coloneqq m(g,E) = m_{g,L}(E), \\ M_B &\coloneqq M_B(g,E) = (A_g - E - \gamma_g m(g,E))^{-1}, \\ M &\coloneqq M(g,E) = M_B(g,E) \otimes I_W, \\ z_t &\coloneqq z_t(g,E) = E + (1-t)\gamma_g m(g,E), \\ \eta_t &\coloneqq \eta_t(g,E) = \Im z_t(g,E) = (1-t)\gamma_g \Im m(g,E), \\ G_t &\coloneqq G_t(g,E) = (H_t(g) - z_t(g,E))^{-1}, \qquad 0 \le t < 1. \\ \widetilde{G}_t &\coloneqq \widetilde{G}_t(g,E) = G_t(g,E) - M(g,E) \end{aligned} \tag{5.3}$$

By definition, $\widetilde{G}_0 = 0$. We will propagate the approximation $G_t(g,E) \approx M(g,E)$ along the characteristic, with the deterministic matrix $M(g,E)$ independent of t. These characteristics are those of [31, Definitions 3.1–3.2, (3.5)–(3.8)] with variance rate γ_g; see also [33, Definition 2.7]. We suppress (g,E) within a fixed flow, except when comparing flows.

For an $N \times N$ block matrix $A = (A_{ab})_{a,b \in \mathbb{Z}_L}$ with $W \times W$ blocks, define

$$\langle A \rangle_{ab} \coloneqq W^{-1} \operatorname{Tr} A_{ab} = W^{-1} \sum_{1 \le i \le W} A_{ai,bi}.$$

Since $M_{aa} = m I_W$, we have $\langle \widetilde{G}_t \rangle_{aa} = \langle G_t \rangle_{aa} - m$. Itô's formula gives

$$dG_t = -G_t \, d\mathcal{B}_t^{(g)} \, G_t + \gamma_g G_t \operatorname{diag}(\langle \widetilde{G}_t \rangle_{aa} I_W : a \in \mathbb{Z}_L) G_t \, dt. \tag{5.4}$$

Equation (5.4) is [31, (3.9)–(3.10)] with covariance operator $X \mapsto \gamma_g \operatorname{diag}(\langle X \rangle_{aa} I_W : a \in \mathbb{Z}_L)$; see also the scalar-matrix equation following [33, Definition 2.7].

5.2 The loop hierarchy and primitive loops

Recall $E_a = P_a / W$, where P_a is the projection onto block a. Write $G_t^+ \coloneqq G_t$, $G_t^- \coloneqq G_t^*$, and use the same convention for $\widetilde{G}_t$, M, and scalars. For signs $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_k)$ and block labels $\boldsymbol{a} = (a_1, \dots, a_k)$, define

$$\mathcal{L}_k(t; \boldsymbol{a}, \boldsymbol{\sigma}) \coloneqq \mathcal{L}_k(t; g, E; \boldsymbol{a}, \boldsymbol{\sigma}) \coloneqq \operatorname{Tr}(G_t^{\sigma_1} E_{a_1} \cdots G_t^{\sigma_k} E_{a_k}). \tag{5.5}$$

This is [31, Definition 3.4, (3.16)–(3.19)], equivalently [33, Definition 2.9], with block size W. The trace is unnormalized. In particular,

$$\mathcal{L}_1(t; a, +) = \langle G_t \rangle_{aa}, \qquad \mathcal{L}_2(t; a, b; +, -) = W^{-2} \|(G_t)_{ba}\|_{\mathrm{HS}}^2.$$

The mixed two-loop measures the resolvent mass between two blocks. Longer loops arise when this quantity is differentiated or its martingale fluctuations are estimated.

We describe the cut operations explicitly, following [31, Definition 3.5, (3.20)–(3.22)] and [33, Definition 2.10]. Inserting E_c into the i-th resolvent factor means replacing $G_t^{\sigma_i}$ by $G_t^{\sigma_i} E_c G_t^{\sigma_i}$. Denote the resulting $(k+1)$-loop by $\operatorname{Cut}_i^c \mathcal{L}_k(t)$. For $i < j$, the two cut words have indices

$$\begin{aligned} \operatorname{Cut}_{L,i,j}^c : \quad & \text{signs } (\sigma_1, \dots, \sigma_i, \sigma_j, \dots, \sigma_k), \\ & \text{labels } (a_1, \dots, a_{i-1}, c, a_j, \dots, a_k), \\ \operatorname{Cut}_{R,i,j}^c : \quad & \text{signs } (\sigma_i, \dots, \sigma_j), \\ & \text{labels } (a_i, \dots, a_{j-1}, c). \end{aligned} \tag{5.6}$$

Thus the two loops share the label c, and their lengths are $k + i - j + 1$ and $j - i + 1$. Their sum is $k + 2$, and each length lies between two and k. These operations act on the indices of any family of loops, including the primitive loops defined by (5.10).

Let $\mathcal{M}_k = \mathcal{M}_k(t; g, E; \boldsymbol{a}, \boldsymbol{\sigma})$ be the martingale part of $\mathcal{L}_k(t)$, with $\mathcal{M}_k(0) = 0$, and define the centered insertion term by

$$\mathcal{W}_k \coloneqq \mathcal{W}_k(t; g, E; \boldsymbol{a}, \boldsymbol{\sigma}) \coloneqq \gamma_g W \sum_{i=1}^{k} \sum_{c} \langle \widetilde{G}_t^{\sigma_i} \rangle_{cc} \operatorname{Cut}_i^c \mathcal{L}_k(t). \tag{5.7}$$

This is the centered insertion in [31, (3.26)], or [33, (2.47)], with the projected covariance $\gamma_g I_L$. Equation (5.4) and the contraction of two differentiated resolvent factors yield the *loop hierarchy*:

$$d\mathcal{L}_k(t) = d\mathcal{M}_k + \mathcal{W}_k \, dt + \gamma_g W \sum_{i<j} \sum_{c} (\operatorname{Cut}_{L,i,j}^c \mathcal{L}_k(t))(\operatorname{Cut}_{R,i,j}^c \mathcal{L}_k(t)) \, dt. \tag{5.8}$$

This is [31, Lemma 3.6, (3.24)], corresponding to [33, Lemma 2.11, (2.45)], with our covariance and block normalization. The coefficient is $\gamma_g W$: Gaussian contraction contributes γ_g / W, and each of the two new normalized block projectors contributes a factor W. For two loops $\mathcal{L}_k, \mathcal{L}'_k$ with martingale parts $\mathcal{M}_k, \mathcal{M}'_k$, put

$$B_i = G_t^{\sigma_i} E_{a_i} \cdots G_t^{\sigma_{i+k-1}} E_{a_{i+k-1}} G_t^{\sigma_i}, \qquad 1 \le i \le k,$$

with indices read modulo k. Define B'_j in the same way for the second loop $\mathcal{L}'_k$. Their exact martingale contraction is

$$\frac{d}{dt} \langle \mathcal{M}_k, \overline{\mathcal{M}'_k} \rangle = W^{-1} \gamma_g \sum_{c} \sum_{i,j=1}^{k} \operatorname{Tr}(P_c B_i P_c B_j'^{*}). \tag{5.9}$$

The proof is in Appendix B.1. Replacing P_c by $W E_c$ makes the right side $\gamma_g W$ times joined $(2k+2)$-loops with a summed block label. Consequently (5.8) cannot be closed at a single length: the drift contains a $(k+1)$-loop and the bracket contains still longer loops.

The primitive loops keep the quadratic part of (5.8) and its exact initial data. With $a_0 \coloneqq a_k$, set $\mathcal{K}_1(t; a, \sigma) \coloneqq \mathcal{K}_1(t; g, E; a, \sigma) = m(g, E)^{\sigma}$. For $k \ge 2$, write $\mathcal{K}_k(t; \boldsymbol{a}, \boldsymbol{\sigma}) = \mathcal{K}_k(t; g, E; \boldsymbol{a}, \boldsymbol{\sigma})$ and define

$$\begin{aligned} \partial_t \mathcal{K}_k(t) &= \gamma_g W \sum_{i<j} \sum_{c} (\operatorname{Cut}_{L,i,j}^c \mathcal{K}_k(t))(\operatorname{Cut}_{R,i,j}^c \mathcal{K}_k(t)), \\ \mathcal{K}_k(0; \boldsymbol{a}, \boldsymbol{\sigma}) &= \operatorname{Tr} \prod_{j=1}^{k} (M^{\sigma_j} E_{a_j}) = W^{1-k} \prod_{j=1}^{k} (M_B^{\sigma_j})_{a_{j-1}, a_j}. \end{aligned} \tag{5.10}$$

The initial value is an M-loop. Unlike G-loops, primitive loops form a triangular deterministic system once the two-loops have been solved. Equation (5.10) is [31, Definition 3.7, (3.27)–(3.29)] with the cut coefficient $\gamma_g W$. The corresponding band-matrix definition is [33, Definition 2.12, (2.48)], with a scalar initial comparison matrix.

Lemma 5.1 (Mixed Ward identity)**.** *Let $\mathcal{X} \in \{\mathcal{L}, \mathcal{K}\}$ be defined by (5.5) or (5.10). For $k \ge 2$, $0 \le t < 1$ with $\eta_t > 0$, and every choice of the remaining signs and labels,*

$$\begin{aligned} &\sum_{c \in \mathbb{Z}_L} \mathcal{X}_k(t; a_1, \dots, a_{k-1}, c; +, \sigma_2, \dots, \sigma_{k-1}, -) \\ &\quad = (2i)^{-1} W^{-1} \eta_t^{-1} [\mathcal{X}_{k-1}(t; a_1, \dots, a_{k-1}; +, \sigma_2, \dots, \sigma_{k-1}) \\ &\qquad\qquad - \mathcal{X}_{k-1}(t; a_1, \dots, a_{k-1}; -, \sigma_2, \dots, \sigma_{k-1})]. \end{aligned} \tag{5.11}$$

Cyclic permutation gives the identity at any vertex between opposite signs.

The proof, including the block normalization, is in Appendix B.1.

For the two-loop equation, define matrices on block labels by

$$P_{ab} \coloneqq P_{ab}(g,E) = \gamma_g |(M_B(g,E))_{ab}|^2, \qquad Q_{ab} \coloneqq Q_{ab}(g,E) = \gamma_g (M_B(g,E))_{ab}^2. \tag{5.12}$$

The matrices P/γ_g and Q/γ_g are the mixed-sign and same-sign initial pair matrices in [31, Definition 3.8, (3.31)]. The symmetry of the hopping matrix A_g in (5.1) gives the two-loop formula [31, Example 3.12, (3.47)] in the form

$$\begin{aligned} \mathcal{K}_2(t;a,b;+,-) &= \gamma_g^{-1} W^{-1} [P(I-tP)^{-1}]_{ab}, \\ \mathcal{K}_2(t;a,b;+,+) &= \gamma_g^{-1} W^{-1} [Q(I-tQ)^{-1}]_{ab}. \end{aligned} \tag{5.13}$$

For example, $\partial_t \mathcal{K}_2^{+-}(t) = \gamma_g W (\mathcal{K}_2^{+-}(t))^2$ and $\mathcal{K}_2^{+-}(0) = P/(\gamma_g W)$, which proves the first formula. The other signs are obtained by conjugation. The Dyson equation implies $\sum_b P_{ab} = 1$, hence $\sum_b [P(I-tP)^{-1}]_{ab} = (1-t)^{-1}$. The same-sign inverse stays stable at the regular energies under consideration. Thus only mixed signs produce the growing propagation length.

For $k \geq 3$, adjacent cuts, including the cyclic pair $(1,k)$, produce one two-loop and one k-loop. The other cuts produce two loops of lengths strictly less than k. Following [33, Definition 5.2], denote the resulting linear operator on block tensors by $\Theta_{t,\boldsymbol{\sigma}} = \Theta_{t,\boldsymbol{\sigma}}(g,E)$, with $\sigma_{k+1} \coloneqq \sigma_1$. This is the generator in [31, Definition 7.3, (7.18)] with our covariance:

$$\begin{aligned} (\Theta_{t,\boldsymbol{\sigma}} F)(\boldsymbol{a}) \coloneqq \gamma_g W \sum_{j=1}^{k} \sum_{c} \mathcal{K}_2(t; a_j, c; \sigma_j, \sigma_{j+1}) \\ \cdot F(a_1, \dots, a_{j-1}, c, a_{j+1}, \dots, a_k). \end{aligned} \tag{5.14}$$

Then

$$\partial_t \mathcal{K}_k(t) = \Theta_{t,\boldsymbol{\sigma}} \mathcal{K}_k(t) + \gamma_g W \sum_{i<j \text{ nonadjacent}} \sum_{c} (\mathrm{Cut}^c_{L,i,j} \mathcal{K}_k(t))(\mathrm{Cut}^c_{R,i,j} \mathcal{K}_k(t)).$$

The source on the right only uses shorter primitive loops. The linear evolution acts separately on each block label. Its one-label kernels $U^{\sigma\sigma'}_{s,t} = U^{\sigma\sigma'}_{s,t}(g,E)$, as in [31, (7.19)] and [33, (5.17)], are

$$\begin{aligned} U^{+-}_{s,t} = U^{-+}_{s,t} &\coloneqq (I-sP)(I-tP)^{-1}, \\ U^{++}_{s,t} \coloneqq (I-sQ)(I-tQ)^{-1}, &\qquad U^{--}_{s,t} \coloneqq \overline{U^{++}_{s,t}}. \end{aligned} \tag{5.15}$$

Here $P_{ab} = \gamma_g |(M_B)_{ab}|^2$ and $Q_{ab} = \gamma_g (M_B)_{ab}^2$, as in (5.12). For instance, $U^{+-}_{s,t} = I + (t-s)P(I-tP)^{-1}$ and $\partial_t U^{+-}_{s,t} = P(I-tP)^{-1} U^{+-}_{s,t}$. With $\mathcal{U}^{\boldsymbol{\sigma}}_{s,t} = \mathcal{U}^{\boldsymbol{\sigma}}_{s,t}(g,E) \coloneqq \bigotimes_{j=1}^{k} U^{\sigma_j \sigma_{j+1}}_{s,t}$, Duhamel's principle gives the primitive-loop counterpart of [31, (7.21)], with the deterministic forcing formed by the nonadjacent cuts; see [33, Lemma 5.3] for the general variation-of-constants formula:

$$\begin{aligned} \mathcal{K}_k(t) = \mathcal{U}^{\boldsymbol{\sigma}}_{s,t} \mathcal{K}_k(s) \\ + \gamma_g W \int_s^t \mathcal{U}^{\boldsymbol{\sigma}}_{u,t} \sum_{i<j \text{ nonadjacent}} \sum_{c} (\mathrm{Cut}^c_{L,i,j} \mathcal{K}_k(u))(\mathrm{Cut}^c_{R,i,j} \mathcal{K}_k(u))\, du. \end{aligned} \tag{5.16}$$

For $k = 3$ the source is empty. Iterating (5.16) expresses every fixed-order primitive in terms of its initial M-loops and the two-loop propagators. The tree representation in [31, Lemma 4.16], extending [33, Lemma 3.4] to initial M-loops with M not proportional to I_N, organizes this expansion. Its size cannot in general be estimated by taking absolute values before the cancellations between its terms.

Write $D_k = D_k(t; g, E; \boldsymbol{a}, \boldsymbol{\sigma}) := \mathcal{L}_k(t) - \mathcal{K}_k(t)$. Subtraction gives the exact error equation [31, (7.14)–(7.15)], corresponding to [33, (5.12)–(5.13)], with cut coefficient $\gamma_g W$:

$$\begin{aligned} dD_k = d\mathcal{M}_k + \mathcal{W}_k\, dt \\ + \gamma_g W \sum_{i<j} \sum_{c} [(\mathrm{Cut}^c_{L,i,j}\, \mathcal{K}_k(t))(\mathrm{Cut}^c_{R,i,j}\, D_k) \\ + (\mathrm{Cut}^c_{L,i,j}\, D_k)(\mathrm{Cut}^c_{R,i,j}\, \mathcal{K}_k(t)) \\ + (\mathrm{Cut}^c_{L,i,j}\, D_k)(\mathrm{Cut}^c_{R,i,j}\, D_k)]\, dt. \end{aligned} \tag{5.17}$$

For $k \geq 3$, its adjacent linear terms are again generated by $\Theta_{t,\boldsymbol{\sigma}}$. The remaining terms consist of shorter linear errors, quadratic errors, centered insertions, and the martingale. This decomposition is the basis of the time induction in Appendix C.4.

The spatial scale of the mixed kernel depends on the coupling. The bounded-coupling scale has the form of [31, (3.42)]; the large-coupling scale follows from the kernel estimates in Lemma A.10. For the two extensions considered here we use

$$\ell_t := \ell_t(g) = \min\left\{L, 1 + \frac{g}{\sqrt{\gamma_g(1-t)}}\right\}, \qquad \zeta_t := \zeta_t(g) = \frac{1}{W\ell_t(g)(1-t)\gamma_g}. \tag{5.18}$$

The length ℓ_t counts blocks, whereas $\ell_{W,g,L}$ counts coordinates. In the parameter regimes of Lemma 5.2 below, $\Im m \asymp 1$, so $\zeta_t \asymp (W\ell_t\eta_t)^{-1}$. Directly from the definition, $\zeta_t \geq c/W$ and $0 \leq \partial_t \log \zeta_t \leq (1-t)^{-1}$ almost everywhere, including the capped scale; hence $\zeta_s \leq \zeta_t \leq (1-s)(1-t)^{-1}\zeta_s$ for $0 \leq s \leq t < 1$. Under the volume assumptions of Theorem 5.12, $\ell_t \asymp \max\{1, g^2\}$ and $\zeta_t \asymp W^{-1}$ for $0 \leq t \leq 1/2$.

Lemma 5.2 (Size of the primitive loops)**.** *Fix an integer $k \geq 2$. Consider either of the following parameter regimes for deterministic $g = g(W)$ and cycle length $L = L(W)$.*

(a) Bounded coupling. *Fix $C, c > 0$ and assume $|E| \leq C$ and $\Im m_{g,L}(E) \geq c$ and*

$$0 < g \leq C, \qquad \log W \ll L \leq W^2.$$

(b) Large coupling. *Fix $\kappa \in (0,2)$. There exist $C_\kappa > 1$, independent of k, and $C_* = C_*(\kappa, k) > 0$ such that the size estimates below hold when $|E| \leq 2 - \kappa$ and*

$$g \geq C_\kappa, \qquad L \geq C_* g^2 \log W. \tag{5.19}$$

For sufficiently large W, uniformly on $0 \leq t < 1$, $1 - t \geq W^{-3}$, the primitive loops of length $k \geq 3$ satisfy

$$|\mathcal{K}_k(t)| \leq C' \zeta_t^{k-1}, \tag{5.20}$$

for all block labels and sign words. In case (a)*, C' depends only on C, c, k; in case* (b)*, only on κ, k, with no upper bound on g or L. At length two,*

$$\max_{a,b,\sigma_1,\sigma_2} |\mathcal{K}_2(t; a, b; \sigma_1, \sigma_2)| \leq C' \zeta_t.$$

For every fixed $\epsilon, Q > 0$, under case (a)*, or under case* (b) *with the stronger volume condition $L \gg g^2 \log W$,*

$$\sup_{0 \leq t \leq 1 - W^{-3}} \max_{\boldsymbol{\sigma} \in \{+,-\}^k} \sup_{\mathrm{diam}(\boldsymbol{a}) > W^\epsilon \ell_t} |\mathcal{K}_k(t; \boldsymbol{a}, \boldsymbol{\sigma})| \leq W^{-Q}. \tag{5.21}$$

In case (b)*, the constants are independent of g, L; the width threshold for* (5.21) *may depend on the divergence rate of $L/(g^2 \log W)$, uniformly over families with a common rate.*

The lower bound $L \geq C_* g^2 \log W$ controls the finite-cycle comparison at large coupling. Lemma 2.4 covers the complementary short cycles in the proof of the lower bound in Theorem 1.4. Primitive-loop estimates are a central ingredient in Yau–Yin [33, Section 3 and Lemma 2.17] and Truong–Yang–Yin [31, Section 4 and Lemma 3.15]. Their method combines a tree representation, the primitive Ward identity and sum-zero cancellation; estimates for individual molecule cores then close under tree gluing. Our main task here is to extend these estimates uniformly to arbitrarily large g. This requires additional ideas to handle the long decay scale of M without a $\log g$ loss, but the proof follows the same underlying method of YY and TYY. The size estimates in cases (a) and (b) are proved in Appendices B.3 and B.5, respectively; (5.21) is proved in Appendix B.6.

Definition (Stochastic domination). For random variables X and $Y \geq 0$ depending on W, we write $X \prec Y$ if for every $\epsilon, D > 0$,

$$\mathbb{P}(|X| > W^{\epsilon} Y) \leq W^{-D}$$

for sufficiently large W. Unless otherwise stated, the estimate is uniform over the deterministic parameters in the statement.

The basic estimates to be propagated along the flow are the following, written at a deterministic time $0 \leq s < 1$, with fixed $\theta \in (0, 1/2]$ and $c_{\mathrm{p}} > 0$:

$$\begin{aligned} &\max_{\boldsymbol{a},\boldsymbol{\sigma}} \frac{|\mathcal{L}_k(s)|}{\zeta_s^{k-1}} \prec 1, \qquad \max_{\boldsymbol{a},\boldsymbol{\sigma}} \frac{|D_k(s)|}{\zeta_s^{k}} \prec 1, \qquad \max_{a,b} \frac{|\langle \widetilde{G}_s \rangle_{ab}|}{\zeta_s} \prec 1, \qquad \frac{\|\widetilde{G}_s\|_{\max}}{\sqrt{\zeta_s}} \prec 1, \\ &\max_{a,b} \frac{|D_2(s;a,b;+,-)|}{\zeta_s^2 (e^{-c_{\mathrm{p}}(\|a-b\|_L/\ell_s)^{\theta}} + W^{-F})} \prec 1, \\ &\max_{\boldsymbol{\sigma}\in\{+,-\}^k} \max_{\boldsymbol{a}:\, \mathrm{diam}(\boldsymbol{a}) > W^{\epsilon}\ell_s} (|\mathcal{L}_k(s)| + |D_k(s)|) \prec W^{-A}, \qquad k \geq 2. \end{aligned} \tag{5.22}$$

They are required for every fixed loop length k and $\epsilon, A, F > 0$, uniformly over the stated deterministic parameters.

At bounded coupling, $\eta_u \asymp 1$ uniformly for $0 \leq u \leq 1/2$. We prove the following initial input by adapting the resolvent and stability method of [11, Section 8 and Appendix B.1].

Lemma 5.3 (Initial loop estimates at bounded coupling)**.** *Assume the bounded-coupling conditions of Lemma 5.2(a). There exists $c_{\mathrm{p}} > 0$ such that*

$$(5.22)|_{s=u,\ \theta=1/2}, \quad 0 \leq u \leq 1/2, \qquad \sup_{0\leq u\leq 1/2} \max_{a,b} \|(G_u)_{ab}\|_{\mathrm{op}} \prec 1. \tag{5.23}$$

The time suprema are inside the probability, and all estimates are uniform over the stated deterministic parameters.

The proof is in Appendix C.2. Starting from this input at $s = 1/2$, the flow propagates (5.22) to the terminal spectral height η_L. For large g, already the initial height is of order g^{-2}, and transfer matrices are needed. The main new inputs for this regime are presented in Section 5.3.

For spatial interpolation of primitives, we use an auxiliary flow with orbital dimension $\widetilde{W}$; the parameters of the original flow remain fixed.

Lemma 5.4 (Auxiliary orbital limit)**.** *Fix $g > 0$, $L, W \in \mathbb{N}$, $E \in \mathbb{R}$ and $0 \leq T < 1$. Assume that the Dyson solution $m = m_{g,L}(E)$ in (5.3) satisfies $\Im m > 0$. For each $\widetilde{W} \in \mathbb{N}$, let $\mathcal{B}_t^{(g,\widetilde{W})}$ be block diagonal, with independent Hermitian Brownian blocks starting at zero and covariance*

$$\mathbb{E}\, d(\mathcal{B}_t^{(g,\widetilde{W})})_{ai,aj}\, d(\mathcal{B}_t^{(g,\widetilde{W})})_{bk,b\ell} = \frac{\gamma_g}{\widetilde{W}} \delta_{ab}\delta_{i\ell}\delta_{jk}\, dt, \qquad a, b \in \mathbb{Z}_L, \quad i, j, k, \ell \in \{1, \dots, \widetilde{W}\}.$$

Using A_g and γ_g from (5.1), define

$$H_t^{(\widetilde{W})} = A_g \otimes I_{\widetilde{W}} + \mathcal{B}_t^{(g,\widetilde{W})}, \qquad G_t^{(\widetilde{W})} = (H_t^{(\widetilde{W})} - z_t I_{L\widetilde{W}})^{-1}, \qquad z_t = E + (1-t)\gamma_g m.$$

Let $P_a^{(\widetilde{W})}$ project onto block a and set $E_a^{(\widetilde{W})} = P_a^{(\widetilde{W})}/\widetilde{W}$, $G_t^{(\widetilde{W}),+} = G_t^{(\widetilde{W})}$ and $G_t^{(\widetilde{W}),-} = (G_t^{(\widetilde{W})})^*$. With the unnormalized trace, put

$$\mathcal{L}_k^{(\widetilde{W})}(t;\boldsymbol{a},\boldsymbol{\sigma}) = \operatorname{Tr}(G_t^{(\widetilde{W}),\sigma_1} E_{a_1}^{(\widetilde{W})} \cdots G_t^{(\widetilde{W}),\sigma_k} E_{a_k}^{(\widetilde{W})}).$$

For every fixed $k \geq 1$, with $\mathcal{K}_k$ from (5.10),

$$\lim_{\widetilde{W}\to\infty} \sup_{0\leq t\leq T} \max_{\boldsymbol{a}\in\mathbb{Z}_L^k,\, \boldsymbol{\sigma}\in\{+,-\}^k} \left| \widetilde{W}^{k-1}\mathbb{E}\mathcal{L}_k^{(\widetilde{W})}(t;\boldsymbol{a},\boldsymbol{\sigma}) - W^{k-1}\mathcal{K}_k(t;\boldsymbol{a},\boldsymbol{\sigma}) \right| = 0. \tag{5.24}$$

The parameters W, g, L, E, T, and hence m and z_t, are fixed in this limit; no convergence rate uniform in g, L is asserted.

The proof is in Appendix B.

5.3 Applications of transfer matrices to the loop hierarchy

This subsection presents the main new inputs to the lower-bound argument: transfer-matrix resolvent moments and initialization (Lemmas 5.5 and 5.6), and spatial sampling of random and primitive loops (Lemma 5.8). We also state the residual and continuity estimates needed to use these inputs in the loop hierarchy. Their large-coupling proofs are given in Section 7, independently of the propagation theorem below.

Lemma 5.5 (Resolvent moments at large coupling)**.** *For every fixed $0 < \kappa < 2$ and $p, C_0 > 0$, if $g \geq C_\kappa$, $L \geq 4g^2$, $|E| \leq 2-\kappa$, and $0 < \eta \leq C_0/g^2$, then*

$$\sup_{0\leq t\leq 1} \max_{a,b} \mathbb{E} \left\| [(H_t(g) - E - i\eta)^{-1}]_{ab} \right\|_{\mathrm{op}}^p \leq C_p (g^2\eta)^{-p}. \tag{5.25}$$

The constants and width threshold depend only on the fixed parameters; there is no upper bound on g.

The proof in Section 7.2 combines transfer estimates with Ward's identity. These moments supply both the initialization below and the coarse control used during propagation.

Lemma 5.6 (Initial loop estimates at large coupling)**.** *Fix $0 < \kappa < 2$ and $C \geq 2$. For a sufficiently large C_κ, assume*

$$g \geq C_\kappa, \qquad L \gg g^2 \log W, \qquad L \leq C g^2 W^2, \qquad |E| \leq 2-\kappa.$$

Uniformity is over deterministic parameter families for which $L/(g^2 \log W)$ diverges uniformly; the width threshold may depend on its rate. There exists $c_{\mathrm{p}} > 0$ such that (5.23) holds. The time suprema are inside the probability, and there is no upper bound on g.

The proof is in Section 7.6.

Definition 5.7 (Cell partitions $\mathcal{P}_B$)**.** *A deterministic partition of the block cycle $\mathbb{Z}_L$ into nonempty consecutive intervals is a* cell partition*, and its intervals are called* cells*. For fixed g, L, W and $B > 0$, set $n = \max\{1, \lfloor g^2 W^{-B} \rfloor\}$ and fix a deterministic partition $\mathcal{P}_B(g, L, W)$ into $\lceil L/n \rceil$ cells whose lengths differ by at most one. Each cell then has length at most n.*

Lemma 5.8 (Spatial sampling of loop errors)**.** *Fix $0 < \kappa < 2$, $C > 1$, $D > 0$, and $K \in \mathbb{N}$. Choose a fixed $B > 2$ sufficiently large depending on D and the other fixed parameters. The following holds for all sufficiently large W. Assume that g, L, E are deterministic and*

$$4g^2 \leq L \leq g^2 W^C,$$
$$|E| \leq 2-\kappa, \qquad |m_{g,L}(E)| \leq C, \quad \Im m_{g,L}(E) \geq C^{-1}.$$

Use the partition $\mathcal{P}_B$ from Definition 5.7; it satisfies $|\mathcal{P}_B| \le 3W^{B+C}$. For each $1 \le k \le K$, choose at most $d_k := 6^k$ deterministic points from each product $I_1 \times \cdots \times I_k$ of cells $I_1, \dots, I_k \in \mathcal{P}_B$, and let $\mathcal{S}_k$ be their union. The points can be chosen independently of time and of the sign word so that, with probability at least $1 - W^{-D}$,

$$\max_{\boldsymbol{a} \in \mathbb{Z}_L^k} |D_k(t; \boldsymbol{a}, \boldsymbol{\sigma})| \le C_k \max_{\boldsymbol{a} \in \mathcal{S}_k} |D_k(t; \boldsymbol{a}, \boldsymbol{\sigma})| + W^{-D}, \qquad D_k = \mathcal{L}_k - \mathcal{K}_k, \tag{5.26}$$

simultaneously for $1 \le k \le K$, all sign words $\boldsymbol{\sigma}$, and $0 \le t \le 1 - W^{-3}$. The same assertion holds with D_k replaced by $\mathcal{L}_k$. Here C_k depends only on k; all other constants and the width threshold depend only on the fixed parameters, not on g, L, E.

If $g^2 < W^B$, all cells are singletons and $L \le W^{C+B}$, so taking $\mathcal{S}_k = \mathbb{Z}_L^k$ gives exact sampling. The proof in Section 7.3 therefore only needs interpolation when $g^2 \ge W^B$. Green-block interpolation, together with an auxiliary orbital limit, places random and primitive loops in a common approximate finite-dimensional space on each product of cells. Sampling points for this space are chosen within that product. The argument does not use the sharp local law being proved.

For the flow (5.1), the Dyson quantities at time $0 \le u < 1$ and spectral point $z \in \mathbb{C}_+$, together with their values on the characteristic (5.3), are

$$\begin{aligned} M_{B,u}(z; g) &:= (A_g - z - u\gamma_g m_u(z; g))^{-1}, \quad m_u(z; g) := L^{-1} \operatorname{Tr} M_{B,u}(z; g), \\ M_u(z; g) &:= M_{B,u}(z; g) \otimes I_W, \qquad \Im m_u(z; g) > 0, \\ m_u(z_u; g) &= m(g, E), \qquad M_{B,u}(z_u; g) = M_B(g, E). \end{aligned} \tag{5.27}$$

For the next two lemmas and the accompanying discussion, suppress g in these Dyson quantities and write $G = (H_u(g) - z)^{-1}$ and $\widetilde{G} = G - M_u(z)$. Following [31, definition preceding (6.37)], define the residual by

$$\begin{aligned} \Pi(G) &:= I_N + (z - A_g \otimes I_W)G + u\gamma_g \operatorname{diag}(\langle G \rangle_{aa} I_W)G \\ &= \mathcal{B}_u^{(g)} G + u\gamma_g \operatorname{diag}(\langle G \rangle_{aa} I_W)G, \qquad R_{ab} := \langle M_u(z)\Pi(G) \rangle_{ab}. \end{aligned} \tag{5.28}$$

The first expression gives $\Pi(M_u(z)) = 0$; the second follows from $(H_u - z)G = I_N$. For $a = b$, R_{aa} is the averaged residual in [31, (6.50)].

The next two lemmas organize the uniform coarse moments of Lemma 5.5 and the spatial interpolation underlying Lemma 5.8 into propagation inputs. For arbitrarily large g, the coarse moments control exceptional-event contributions in fluctuation averaging, while the interpolation reduces residual maxima to polynomially many sampled pairs.

Lemma 5.9 (Fluctuation averaging)**.** *Use the parameter ranges of Lemma 5.3 or Lemma 5.6, with fixed energy and density margins. Fix deterministic $0 \le u \le 1 - W^{-3}$ and take $z = z_u(g, E)$. For deterministic $1/2 \le s \le u$, also allow any deterministic point*

$$z = \Re z_u + iy, \qquad \eta_u \le y \le \xi^{-1}\eta_s(g', E'), \qquad \xi = \sqrt{\frac{s\gamma_{g'}}{u\gamma_g}},$$

where (g', E') is supplied by Lemma 5.11.

(i) Averaged residual. *For deterministic polynomially bounded $K \ge 1$ and $\chi \ge W^{-1/2}$,*

$$\max_{a,b} |\langle G \rangle_{ab}| \prec K, \quad \max_{ai,bj} |G_{ai,bj} - \delta_{ij}\langle G \rangle_{ab}| \prec K\chi \quad \Longrightarrow \quad \max_{a,b} |R_{ab}| \prec K^2\chi^2. \tag{5.29}$$

(ii) Entrywise estimate. *For each pair a, b and deterministic $\Psi > 0$,*

$$W^{-2}\|G_{ab}\|_{\mathrm{HS}}^2 \prec \Psi^2 \quad \Longrightarrow \quad \max_{i,j} |G_{ai,bj} - \delta_{ij}\langle G \rangle_{ab}| \prec \Psi. \tag{5.30}$$

(iii) Residual moments. *For each fixed even $p \geq 2$ and $C > 0$, there is C_p such that, for deterministic $\chi \geq W^{-1/2}$,*

$$\max_{ai,bj} \|\widetilde{G}_{ai,bj}\|_{L^{2p}} \leq C\chi \quad \Longrightarrow \quad \max_{a,b} \mathbb{E}|R_{ab}|^p \leq C_p \chi^{2p}. \tag{5.31}$$

All estimates are uniform in the allowed deterministic parameters. Probabilities and moments use the original Gaussian measure, without conditioning on a stopping event.

On the characteristic $z = z_u$, resolvent subtraction gives, with Q from (5.12),

$$\begin{aligned} \langle \widetilde{G} \rangle_{ab} &= -R_{ab} + u\gamma_g \sum_c M_{B,ac} \langle \widetilde{G} \rangle_{cc} \langle G \rangle_{cb}, \\ (I - uQ)(\langle \widetilde{G} \rangle_{aa})_a &= -(R_{aa})_a + u\gamma_g \left(\sum_c M_{B,ac} \langle \widetilde{G} \rangle_{cc} \langle \widetilde{G} \rangle_{ca} \right)_a . \end{aligned} \tag{5.32}$$

These identities convert residual estimates into the averaged local law.

The proof in Section 7.5 uses the Gaussian calculations in Appendix C.1, following [31, (6.40), (6.50)] and the averaged improvement in [33, proof of (4.5), (4.12)]. For the random centering $\langle G \rangle_{ab} I_W$ in part (i), that appendix gives the block and orbital index counts, pairing the covariance factors $u\gamma_g$ with absolute or squared Dyson row sums. Part (ii) uses common Haar conjugation.

For the next lemma, the same coarse moments yield time-increment bounds through Itô's formula and BDG. Unweighted sampling of loops and loop errors follows directly from Lemma 5.8. Weighted loop sampling and sampling of centered block traces and traceless entries use the same interpolation mechanism; residual sampling requires the additional proof in Proposition 7.10.

Lemma 5.10 (Sampling and continuity for propagation)**.** *Use the coupling, volume and regularity assumptions of Lemma 5.9. Fix a finite set of loop lengths and $D, D' > 0$. The following estimates are uniform in the deterministic parameters, for $0 \leq u, v \leq 1 - W^{-3}$.*

(a) Coarse moments and time increments. *Along the fixed characteristic (g, E), let $X(u)$ be any of*

$$\begin{gathered} (G_u)_{ab}, \qquad \langle \widetilde{G}_u \rangle_{ab}, \qquad (G_u)_{ai,bj} - \delta_{ij} \langle G_u \rangle_{ab}, \\ \mathcal{L}_k(u; \boldsymbol{a}, \boldsymbol{\sigma}), \qquad D_k(u; \boldsymbol{a}, \boldsymbol{\sigma}), \qquad R_{ab}(u). \end{gathered}$$

The block indices $a, b, \boldsymbol{a}$, orbital indices i, j, and signs $\boldsymbol{\sigma}$ are fixed when comparing u and v. Here $R_{ab}(u)$ is the residual in (5.28) evaluated at $z = z_u(g, E)$. Use operator norm for $(G_u)_{ab}$ and absolute value for the other five quantities. There is a fixed exponent C such that, for every fixed $p \geq 2$,

$$\max_{a,b} \mathbb{E}\|(G_u)_{ab}\|_{\mathrm{op}}^p \leq C_p W^{Cp}, \qquad \mathbb{E}\|X(v) - X(u)\|^p \leq C_p W^{Cp} |v - u|^{p/2}. \tag{5.33}$$

The exponent may depend on the fixed set of loop lengths and regularity margins, but not on p, g, L in the arbitrary-coupling range.

(b) Spatial suprema. *For fixed signs and orbital indices, choose one of the scalar fields*

$$\begin{aligned} X(u; \boldsymbol{a}) &\in \{\mathcal{L}_k(u; \boldsymbol{a}, \boldsymbol{\sigma}), D_k(u; \boldsymbol{a}, \boldsymbol{\sigma})\}, && \boldsymbol{a} \in \mathbb{Z}_L^k, \\ X(u; \boldsymbol{a}) &\in \{\langle \widetilde{G}_u \rangle_{ab}, (G_u)_{ai,bj} - \delta_{ij} \langle G_u \rangle_{ab}, R_{ab}(u)\}, && \boldsymbol{a} = (a, b) \in \mathbb{Z}_L^2. \end{aligned}$$

Take $w_u = 1$; for loops and loop errors one may instead fix $c, F > 0$, $0 < \theta \leq 1/2$ and take $w_u(\boldsymbol{a}) = e^{-c(\mathrm{diam}(\boldsymbol{a})/\ell_u)^\theta} + W^{-F}$. There is a deterministic set $\mathcal{S}_X$ of at most W^C block tuples, independent of time and allowed to depend on the fixed weight parameters, D, D', such that, with probability at least $1 - W^{-D'}$, simultaneously for all $0 \leq u \leq 1 - W^{-3}$,

$$\max_{\boldsymbol{a}} \frac{|X(u; \boldsymbol{a})|}{w_u(\boldsymbol{a})} \leq C \max_{\boldsymbol{a} \in \mathcal{S}_X} \frac{|X(u; \boldsymbol{a})|}{w_u(\boldsymbol{a})} + W^{-D}. \tag{5.34}$$

For joint-tail restrictions, sampling is cellwise and enlarges the diameter boundary by at most twice the cell length, which is $o(W^{\epsilon}\ell_u)$ for every fixed $\epsilon > 0$. For entries there is an additional union over the W^2 orbital indices. In polynomial volume one may take all block tuples. In the arbitrary-coupling range, the constants are independent of g, L; uniformity of the width threshold is as in Lemma 5.6.

At each fixed deterministic (u, z) on the comparison segments in Lemma 5.9, the first bound in (5.33) holds for $G = (H_u(g) - z)^{-1}$. At this point, (5.34) with $w = 1$ holds for $\langle G - M_u(z)\rangle_{ab}$, $G_{ai,bj} - \delta_{ij}\langle G\rangle_{ab}$ and R_{ab}; the deterministic sample sets may depend on (u, z).

The proof is in Section 7.4. At each fixed time, the derivative bound (7.24) is used to extend estimates from a discrete height mesh to the full vertical comparison segment. In propagation, spatial sampling is applied to the actual fields after adapted stopping and moment estimates, not to their martingale differentials.

5.4 Propagation of the loop estimates

We propagate the estimates along a finite sequence of times $1/2 = s_0 < s_1 < \cdots < 1$, using the estimates obtained at each step as the input for the next. The rough bounds at the start of a step for $G_{s_{i+1}}(g, E)$ come from the earlier resolvent $G_{s_i}(g', E')$, with adjusted coupling and energy. Lemma 5.11 gives the relation between g' and g and controls E', so that the new characteristic retains the required bulk regularity. In particular, E' remains a bulk anchor for coupling g', with $\Im m_{g',L}(E')$ bounded below by a positive constant.

Lemma 5.11 (Resolvent comparison at two times)**.** *Fix deterministic $1/2 \le s \le u \le 1 - W^{-3}$. Use (5.1), (5.3) and (5.18), and set*

$$g' := g\sqrt{s/u}, \quad R := (1-s)/(1-u).$$

(a) Assume the coupling and volume bounds of Lemma 5.2(a), and suppose $|E| \le C'$ and $\Im m_{g,L}(E) \ge c'$ for fixed $C', c' > 0$. There is a constant $C'' > 0$, depending only on the fixed coupling and regularity bounds, and a deterministic $E' \in \mathbb{R}$ such that

$$\begin{aligned} \Im G_u(g,E) &\overset{\mathrm{d}}{\preceq} C''R\,\Im G_s(g',E'), \\ \max_{a,b}\|(G_u(g,E))_{ab}\|_{\mathrm{op}} &\overset{\mathrm{d}}{\le} C''R\max_{a,b}\|(G_s(g',E'))_{ab}\|_{\mathrm{op}}, \\ \max_{a,b}|\langle G_u(g,E)\rangle_{ab}| &\overset{\mathrm{d}}{\le} C''R\max_{a,b}|\langle G_s(g',E')\rangle_{ab}|, \\ \zeta_s(g') \asymp \zeta_s(g), \qquad \zeta_s(g') \le \zeta_u(g), \qquad |E'| \le C'', \qquad \Im m_{g',L}(E') &\ge (C'')^{-1}. \end{aligned} \tag{5.35}$$

Here d *means that a coupling exists under which the three comparisons hold simultaneously, and $\preceq$ denotes the positive-semidefinite order.*

(b) Assume the coupling and volume bounds of Lemma 5.2(b), with $|E| \le 2 - \kappa$, where C_κ is sufficiently large and C_* is chosen after the prescribed finite orders and accuracy exponents. There are a constant $C'' > 0$ and a deterministic $E' \in \mathbb{R}$ such that the three resolvent comparisons and the scale bounds in (5.35) hold, together with

$$|E'| \le 2 - \kappa/2, \qquad |E' - E| \le C''((g')^{-2} - g^{-2}). \tag{5.36}$$

The proof is given in Appendix A.9. Following the two-time comparison of [33, Lemma 2.8] and [31, Lemma 3.3], we first use Gaussian scaling to identify the two matrix laws up to a bounded factor. We then choose E' so that the corresponding spectral points have the same

real part and comparable imaginary parts, with loss at most CR. Spectral calculus and Ward's identity yield the three resolvent comparisons.

On a time step $[s,t]$, Lemma 5.11 supplies rough bounds with losses in powers of $R = (1-s)/(1-t)$. Theorem 5.12(i) uses the loop hierarchy to obtain sharp bounds that serve as the input for the next step; part (ii) iterates these bounds to the target time.

Theorem 5.12 (Propagation of the loop estimates)**.** *Use the flow, loops and scales in* (5.1)–(5.18)*, in the parameter ranges of Lemma 5.3 or Lemma 5.6. Below, case* (a) *denotes bounded coupling and case* (b) *denotes large coupling, including arbitrarily large g. Assume the primitive-loop estimates of Lemma 5.2 and the following inputs from Section 5.3, in their respective parameter ranges:*

- *the transfer-matrix resolvent moment bounds of Lemma 5.5, at large coupling;*
- *the spatial sampling estimates for random loops and loop errors in Lemma 5.8, at large coupling;*
- *the fluctuation-averaging and entrywise estimates of Lemma 5.9;*
- *the coarse moments, weighted spatial sampling and time-continuity estimates of Lemma 5.10, including its estimates at the comparison spectral points.*

All inputs are required uniformly over the deterministic parameters and comparison points used below, with fixed regularity margins. There is no separate polynomial upper bound on g or L.
(i) One time step. *Fix $c, c_{\mathrm{p}} > 0$, $\theta \in (0, 1/2]$ and deterministic times $1/2 \le s < t < 1$. Take a deterministic pair (g, E) in the range*

$$\text{(a)} \quad |E| \le C', \quad \Im m_{g,L}(E) \ge c'; \qquad \text{(b)} \quad |E| \le 2 - \kappa/2, \tag{5.37}$$

where $C' > 0$ and $c' \in (0,1)$ are fixed in case (a)*. Assume* (5.22) *at (g, E) for every fixed k and $\epsilon, A, F > 0$. The times satisfy*

$$1 - t \ge W^{-3}, \qquad \left(\frac{1-s}{1-t}\right)^{20} \zeta_t \le W^{-c}. \tag{5.38}$$

We also assume

$$(5.22)|_{g\to g',\, E\to E'} \quad \text{uniformly for } s \le u \le t,$$

where Lemma 5.11 supplies (g', E') for the given g, E, s, u. Then there exists a constant $c'_{\mathrm{p}} > 0$ such that

$$(5.22)|_{s\to u,\, \theta\to\theta/2,\, c_{\mathrm{p}}\to c'_{\mathrm{p}}}, \qquad u \in [s,t]. \tag{5.39}$$

The supremum over $u \in [s,t]$ of each normalized left-hand side is taken inside the probability; the supremum over the allowed deterministic (g, E) is outside.

(ii) Iteration. *Assume in addition the initial loop estimates of Lemma 5.3 or Lemma 5.6, uniformly at the required comparison pairs. The initial pair (g, E) satisfies*

$$\text{(a)} \quad |E| \le C', \quad \Im m_{g,L}(E) \ge c'; \qquad \text{(b)} \quad |E| \le 2 - 3\kappa/4. \tag{5.40}$$

Fix $c_0 > 0$. There exist constants $\theta_* = \theta_*(c_0) > 0$ and $c^*_{\mathrm{p}} > 0$ such that for every deterministic t_* satisfying

$$\frac{1}{2} \le t_* \le 1 - W^{-3}, \qquad \zeta_{t_*} \le W^{-c_0}, \tag{5.41}$$

the conclusion is

$$(5.22)|_{s\to u,\, \theta\to\theta_*,\, c_{\mathrm{p}}\to c^*_{\mathrm{p}}}, \qquad u \in [1/2, t_*], \tag{5.42}$$

in the same probability sense as in part (i).

The deterministic inputs are the propagator and Dyson bounds of Lemmas 6.1 and 6.4, and the spatial budgets of Lemma 6.2. Lemma 6.3(b) controls the finite comparison chains in part (ii): it preserves bounded energies and a fixed positive density margin at bounded coupling, and carries the initial margin $|E| \le 2 - 3\kappa/4$ into $|E'| \le 2 - \kappa/2$ at large coupling, with at most a factor two in the upper-volume constant.

The proof in Appendix C follows the five-step propagation and finite induction of [33, Theorem 2.21, (2.68)–(2.72), Section 5] and [31, Theorem 5.4, (5.6)–(5.10), Section 7]. The main new work is the transfer-matrix and sampling inputs proved in Section 7; Appendix C.3 checks their compatibility with spatial sums, stopping and time suprema.

5.5 Completion of the resolvent moment bounds

We complete Theorems 2.3 and 2.5.

Lemma 5.13 (Deterministic bounds for the terminal comparison)**.** (a) *Fix $C > 0$ and $0 < \kappa < 1/2$. For $g \le C$ and all sufficiently large L, the condition $F_g(-|E|) \ge \kappa$ implies*

$$|E| \le C', \qquad \Im m_{g,L}(E) \ge c' > 0. \tag{5.43}$$

Also $|m_{g,L}(e)| \le \max\{1, g\}$ for every real e. The constants and the volume threshold depend only on C, κ.

(b) *Fix $0 < \kappa < 2$. There is $C_\kappa > 1$ such that, if $g \ge C_\kappa$ and $L \ge C_\kappa g^2$, then for every $|E| \le 2 - \kappa$ and $0 \le \delta \le 1$ there exists a deterministic $E_0 \in [-2 + 3\kappa/4, 2 - 3\kappa/4]$ satisfying*

$$E = E_0 + \delta g^{-2} \Re m_{g,L}(E_0), \qquad c \le \Im m_{g,L}(E_0) \le |m_{g,L}(E_0)| \le C. \tag{5.44}$$

The constants are independent of g, L, E, δ.

Parts (a) and (b) are proved in Appendices A.3 and A.4.1, respectively.

Lemma 5.14 (Terminal matching at bounded coupling)**.** *Fix $C > 0$ and $0 < \kappa < 1/2$. Let g, L, E, η be deterministic parameters, with L an integer, satisfying*

$$0 < g \le C, \qquad \log W \ll L \le W^2, \qquad F_g(-|E|) \ge \kappa, \qquad 0 < \eta = o(1).$$

Set $m := m_{g,L}(E + i\eta)$ and define

$$t_* := \frac{\gamma_g \Im m}{\gamma_g \Im m + \eta}, \qquad g_* := \sqrt{t_*}\, g, \qquad c_* := \sqrt{\frac{t_* \gamma_{g_*}}{\gamma_g}}, \qquad E_* := c_* \left(E - \eta \frac{\Re m}{\Im m} \right). \tag{5.45}$$

For sufficiently large W, these parameters satisfy

$$\tfrac{1}{2} \le t_* < 1, \qquad 1 - t_* \asymp \eta, \qquad |E_*| \le C', \qquad \Im m_{g_*,L}(E_*) \ge c', \tag{5.46}$$

and

$$G(E + i\eta) \overset{d}{=} c_* G_{t_*}(g_*, E_*). \tag{5.47}$$

Here $C', c' > 0$ and the comparison constants depend only on C, κ. The threshold in W may depend on the rates in the asymptotic assumptions and is uniform over families with uniform rates.

The proof is given under *Terminal matching* in Appendix A.

Proof of Theorem 2.3. By (1.7), $\ell_{W,g,L} \ge W$. Set

$$\eta := W^\nu / \ell_{W,g,L} \le W^{\nu-1} = o(1), \qquad c_0 := \nu/3. \tag{5.48}$$

For each deterministic E with $F_g(-|E|) \geq \kappa$, define t_*, g_*, c_*, E_* by (5.45). Lemma 5.14 gives (5.46) and (5.47). In particular, $0 < g_* \leq g \leq C$ and $\gamma_{g_*} \asymp 1$. Together with the energy and density bounds, this gives the bounded-coupling input range (5.40)(a); it allows the comparison coupling g_* to lie below $W^{-1/2}$. Here $\ell_{W,g,L} = \min\{WL, W^2g^2\}$, so $\ell_{W,g,L} \leq WL$ and $\ell_{W,g,L} \leq W^2g^2 \leq C^2W^2$. Using (5.18) at coupling g_* and $1 - t_* \asymp \eta$, we obtain

$$\begin{aligned} &\zeta_{t_*}(g_*) \leq C'\left(\frac{1}{WL\eta} + \frac{1}{Wg\sqrt{\eta}}\right) \quad \leq C'(W^{-\nu} + W^{-\nu/2}) \leq W^{-c_0}, \\ &1 - t_* \asymp \eta \geq cW^{\nu-2} \gg W^{-3}. \end{aligned} \tag{5.49}$$

Lemmas 5.2 and 5.3 supply the primitive-loop and initial estimates, respectively, while Lemmas 5.9 and 5.10 supply the residual and continuity inputs. These inputs are uniform for fixed energy and density margins, including the comparison pairs controlled by Lemma 6.3(b). Theorem 5.12(ii), with (5.49), therefore gives

$$\|\widetilde{G}_{t_*}(g_*, E_*)\|_{\max} \prec \sqrt{\zeta_{t_*}(g_*)} \leq W^{-c_0/2}.$$

Since $M_{xx}(g_*, E_*) = m_{g_*,L}(E_*)$ and Lemma 5.13(a) gives $|m_{g_*,L}(E_*)| \leq \max\{1, g_*\} \leq \max\{1, C\}$, the decomposition $G_{t_*} = M + \widetilde{G}_{t_*}$ bounds the diagonal imaginary parts of $G_{t_*}(g_*, E_*)$ by a fixed constant except on an event of probability at most W^{-A}, for every fixed $A > 0$. Combining (5.47) with $0 < c_* \leq 1$, which follows from (5.45) and (5.46), yields

$$\max_x \Im G_{xx}(E + i\eta) \leq C' \tag{5.50}$$

with probability at least $1 - W^{-A}$, uniformly over the stated deterministic parameters. Since $\Im G_{xx} \leq \eta^{-1} \leq C'W^2$, choosing $A > 2p + 1$ bounds the exceptional-event contribution to the pth moment by C_pW^{2p-A} and proves (2.9). □

Proof of Theorem 2.5. Fix a deterministic $E \in [-2+\kappa, 2-\kappa]$. Put $q = L/g^2$, so $\log W \ll q \leq W^2$, and set $\delta_* = g^2\eta_L = W^\nu/[W \min(q, W)] \leq W^{\nu-1} = o(1)$, $t_* = 1 - \delta_*$ and $\widetilde{g} = \sqrt{t_*}\, g$. For sufficiently large W, $t_* \geq 1/2$ and $L \geq C_*g^2 \log W$. Lemma A.18, with $g_* = \widetilde{g}$, therefore gives $E_0 \in [-2 + 3\kappa/4, 2 - 3\kappa/4]$ such that, with

$$\eta(E) := \Im z_{t_*}(\widetilde{g}, E_0) = \eta_L t_*^{-1} \Im m_{\widetilde{g},L}(E_0) \asymp \eta_L,$$

we have $z_{t_*}(\widetilde{g}, E_0) = E + i\eta(E)$ and $G_{t_*}(\widetilde{g}, E_0) \overset{d}{=} G(E + i\eta(E))$ by (A.76).

We have $1 - t_* = \delta_* \geq W^{\nu-2} \gg W^{-3}$, and (5.18) gives

$$\zeta_{t_*}(\widetilde{g}) \leq C\left(\frac{1}{Wq\delta_*} + \frac{1}{W\sqrt{\delta_*}}\right) \leq C(W^{-\nu} + W^{-\nu/2}) \leq W^{-\nu/3}.$$

We apply Theorem 5.12(ii) with bulk parameter $\kappa/2$ and $c_0 = \nu/3$. The comparison-parameter calculation (6.11) in its proof gives $g/2 \leq g' \leq g$ and $|E'| \leq 2 - \kappa/4$. Consequently,

$$\frac{L}{(g')^2 \log W} \geq \frac{L}{g^2 \log W} \longrightarrow \infty, \qquad L \leq 4(g')^2W^2.$$

Lemma 5.13(b), with $\delta = 0$ and bulk margin $\kappa/4$, bounds $|m_{g',L}(E')|$ above and $\Im m_{g',L}(E')$ below uniformly. After enlarging the coupling threshold, Lemmas 5.2(b), 5.5 and 5.6 supply the primitive, resolvent-moment and initial estimates at every required pair; use upper-volume constant 4 in the initial lemma. The displayed volume bounds and the verified regularity bounds also give the hypotheses of Lemma 5.8 with a fixed $C \geq 3$ chosen also to cover the regularity bounds, which supplies spatial sampling without any additional coupling threshold. Lemmas 5.9

and 5.10 supply the remaining stochastic inputs. All conditions of Theorem 5.12(ii) are therefore satisfied, and (5.42) gives

$$\|\widetilde{G}_{t_*}(\widetilde{g}, E_0)\|_{\max} \prec \sqrt{\zeta_{t_*}(\widetilde{g})} \le W^{-\nu/6}. \tag{5.51}$$

Since $M_{ii}(\widetilde{g}, E_0) = m_{\widetilde{g},L}(E_0)$ is uniformly bounded by (5.44), the matrix identity in law above yields $\max_i \Im G_{ii}(E + i\eta(E)) \le C$ outside an event of probability at most W^{-D}, for every fixed $D > 0$.

To obtain unconditional moments, take $\eta_0 = g^{-2}$. Since $\eta(E) \asymp \delta_* \eta_0$ and $\delta_* = o(1)$, comparison of Poisson kernels gives $\Im G_{ii}(E + i\eta(E)) \le C\delta_*^{-1} \Im G_{ii}(E + i\eta_0)$. Lemma 5.5, with $t = 1$, $C_0 = 1$ and moment order $2p$, bounds the seed moment uniformly. Cauchy–Schwarz therefore bounds the exceptional-event contribution to the pth moment by $C\delta_*^{-p} W^{-D/2} \le CW^{(2-\nu)p - D/2} = o(1)$ when $D > 4p$. Finally, $\eta(E) \asymp \eta_L$ gives pointwise comparable Poisson kernels, so the same moment bound holds at η_L, uniformly over the stated E, g, L and all coordinates, proving (2.11). □

6 Dyson solutions, pair kernels, and propagation scales

We state the deterministic inputs used by the main-text proofs, in proof-dependency order. Complete proofs and appendix-only refinements are in Appendix A. Use the characteristic, Dyson matrices, pair kernels and scales defined in (5.3), (5.27), (5.12) and (5.18).

Lemma 6.1 (Scales and deterministic propagators)**.** *Use $\gamma = \gamma_g$ and the flow, scales, pair kernels and propagators in (5.3), (5.18), (5.12) and (5.15). Assume either*

$$\begin{aligned} &g \le C, \quad |E| \le B_0, \quad \Im m_{g,L}(E) \ge c', \quad \log W \ll L \le W^{B_L}; \\ &g \ge C_\kappa, \quad |E| \le 2 - \kappa, \quad L \gg g^2 \log W. \end{aligned} \tag{6.1}$$

The constants C, B_0, c', B_L, κ are fixed. Uniformly for $1/2 \le s \le u \le t < 1$, $1 - t \ge W^{-3}$, we have

$$\begin{aligned} c \le \Im m \le C', \qquad \max_{a,b} |M_{B,ab}| \le C', \\ \gamma \max_a \sum_b (|M_{B,ab}| + |M_{B,ab}|^2) \le C', \qquad \max_a \sum_b |[(I - uQ)^{-1}]_{ab}| \le C'. \end{aligned} \tag{6.2}$$

For every prescribed $A > 0$, the following bounds hold with c, C' independent of A; C_A and the width threshold may depend on A:

$$0 \le [P(I - uP)^{-1}]_{ab} \le \frac{C'}{(1-u)\ell_u} e^{-c\|a-b\|_L/\ell_u} + C_A \ell_u^{-1} W^{-A}. \tag{6.3}$$

Set $\theta_u = (1-u)P(I - uP)^{-1}$. Since $P\mathbf{1} = \mathbf{1}$, this is a probability profile, with $\|\theta_u\|_\infty \le C/\ell_u$ by (6.3).

For a scalar function h on the cycle and $n \ge 2$, define the row-restoration commutator by

$$C_v(h)(\boldsymbol{a}) := \sum_c [P(I - vP)^{-1}]_{a_1 c} h(c) \cdot \left[\prod_{j=2}^n \theta_v(c - a_j) - \prod_{j=2}^n \theta_v(a_1 - a_j) \right]. \tag{6.4}$$

Then, with $\operatorname{row} A(a_1) = \sum_{a_2,\dots,a_n} A(\boldsymbol{a})$,

$$\operatorname{row} C_v(h) = 0, \qquad \|C_v(h)\|_\infty \le \frac{C_n \|h\|_\infty}{(1-v)\ell_v^{n-1}}. \tag{6.5}$$

Constants are independent of g, L; the width threshold may depend on the rate of divergence in the volume lower bound.

The proof is in Appendix A.7.

Lemma 6.2 (Spatial summation and weights)**.** *Use the parameter ranges of Theorem 5.12, with fixed regularity margins, and $0 \le u \le 1 - W^{-3}$. For a fixed C, the scales in (5.18) satisfy*

$$\gamma_g L + L/\ell_u \le W^C, \qquad \gamma_g W \ell_u \zeta_u = (1-u)^{-1}. \tag{6.6}$$

For fixed $c_1, F > 0$ and $0 < \theta \le 1/2$, define $w_u(r) = e^{-c_1 (r/\ell_u)^\theta} + W^{-F}$. Then

$$e^{-C(n/\ell_u)^\theta} \le w_u(r')/w_u(r) \le e^{C(n/\ell_u)^\theta} \quad (|r - r'| \le 2n,\ n \ge 0). \tag{6.7}$$

In particular the weight ratio is at most two on cells of length $n \le g^2 W^{-B}$ in the large-coupling range. If $LW^{-F/2} \le \ell_u$, then

$$\sum_b w_u(d(a,b)) w_u(d(b,c)) \le C_\theta \ell_u w_u(d(a,c)), \qquad \sum_b \sqrt{w_u(d(a,b)) w_u(d(b,c))} \le C_\theta \ell_u \sqrt{w_u(d(a,c))}. \tag{6.8}$$

All constants and exponents are independent of g, L in the stated ranges.

The proof is in Appendix A.8.

Lemma 6.3 (Comparison parameters through finite iteration)**.** (a) *Use the parameter hypotheses of Lemma 5.11, and fix $1/2 \le s \le u < 1$. Set $g' = g\sqrt{s/u}$ and $\xi = (s\gamma_{g'}/(u\gamma_g))^{1/2}$. There are deterministic E', w, with the energy, density and scale bounds of Lemma 5.11, such that*

$$\Re w = \Re z_u(g, E), \qquad 1 \le \frac{\Im w}{\eta_u(g,E)} \le C \frac{1-s}{1-u}, \qquad w = \xi^{-1} z_s(g', E'). \tag{6.9}$$

At large coupling, $\xi = 1$. The corresponding density and energy bounds hold along the vertical segment from $z_u(g,E)$ to w.

(b) *Start from (g_0, E_0) in the initial parameter range of Theorem 5.12(ii). Consider a chain $(g_j, E_j)_{j=0}^m$ of $m \le J$ comparisons from part* (a)*, where J is fixed, interspersed with movement to earlier times at fixed coupling, all times in $[1/2, 1)$. At bounded coupling, all resulting energies are bounded and all $\Im m_{g,L}(E)$ have a positive lower bound depending only on the initial bounds and J; also $c_J g_0 \le g \le C$. At large coupling the domain*

$$u\gamma_g \le g_0^{-2}, \qquad s\gamma_{g'} = u\gamma_g \tag{6.10}$$

is preserved. For sufficiently large C_κ, the chain satisfies

$$g_0/\sqrt{2} \le g \le g_0, \qquad |E| \le 2 - \kappa/2. \tag{6.11}$$

The volume bounds remain valid, with at most a factor two in the large-coupling upper-volume constant. Earlier comparisons preserve both $1 - u \ge W^{-3}$ and an upper bound $\zeta_u(g) \le W^{-c_0}$.

Parts (a) and (b) are proved in Appendices A.9 and A.9.1, respectively.

Lemma 6.4 (Dyson bounds on comparison segments)**.** *The scalar equation in (5.27) has a unique Herglotz solution; at the characteristic it has the values given there. In either parameter range of Lemma 5.11, fix $1/2 \le s \le u < 1$. Let the deterministic comparison point be $w = \xi^{-1} z_s(g', E')$, where $g' = g\sqrt{s/u}$, $\xi = (s\gamma_{g'}/(u\gamma_g))^{1/2}$ and E' is the comparison energy in Lemma 5.11. Let $z = \Re z_u(g,E) + iy$, $\eta_u(g,E) \le y \le \Im w$. Use the Dyson solution and matrix in (5.27), and put $\gamma = \gamma_g$. Uniformly on this segment,*

$$\begin{aligned} &c \le \Im m_u(z) \le C', \qquad \max_{a,b} |M_{B,u}(z)_{ab}| \le C', \\ &\gamma \max_a \sum_b (|M_{B,u}(z)_{ab}| + |M_{B,u}(z)_{ab}|^2) \le C', \\ &\max_a \sum_b \left| \left[(I - u\gamma (M_{B,u}(z)^2_{ab})_{a,b})^{-1} \right]_{ab} \right| \le C'. \end{aligned} \tag{6.12}$$

At large coupling, these bounds also hold directly for $|x| \le 2-\kappa/2$, $z = x+iy$, $0 < y \le C/g^2$ *and* $1/2 \le u \le 1$, *provided* $g \ge C_\kappa$ *and* $L \gg g^2 \log W$. *At the characteristic for every* $0 \le u < 1$, *and on the comparison segments above, one also has*

$$\max_{a,c} |M_{B,ac}| \le C, \qquad u\gamma_g \max_a \sum_c (|M_{B,ac}| + |M_{B,ac}|^2) \le C. \tag{6.13}$$

Here $M_B = M_{B,u}(z)$. *At the upper comparison point,* $M_{B,u}(w) = \xi M_B(g', E')$. *For* $r = \Im z/\gamma_g$, *with* $u, g, \Re z$ *held fixed, one has*

$$\max_{a,b} |\partial_r M_{B,u}(z)_{ab}| + \gamma_g \max_a \sum_b |\partial_r M_{B,u}(z)_{ab}| \le C. \tag{6.14}$$

Constants are independent of g, L.

The proof is in Appendix A.10.

Lemma 6.5 (Free rows and interval interpolation)**.** *Fix* $|E| \le 2-\kappa$, $g \ge C_\kappa$, $L \ge 4g^2$, *and* $m = m_{g,L}(E)$ *with* $c \le \Im m \le |m| \le C$. *For* $M_B = (A_L - E - g^{-2}m)^{-1}$,

$$\max_{b,c} |M_{B,bc}| \le C, \qquad g^{-2} \max_b \sum_c |M_{B,bc}| \le C. \tag{6.15}$$

Let I *be a deterministic consecutive interval with* $|I| \le cg^2$ *and first block* b_0; *block indices are taken modulo* L. *Write* $E = 2\sin\theta$, $-\pi/2 < \theta < \pi/2$, *and lift* I *to an integer interval starting at* b_0. *For* $a \in I$ *and any block* b, *define*

$$\begin{aligned} c_{a,b_0} &= \frac{\sin((a-b_0+1)(\pi/2-\theta))}{\cos\theta}, \\ c_{a,b_0-1} &= -\frac{\sin((a-b_0)(\pi/2-\theta))}{\cos\theta}, \\ q_{ab} &= \mathbf{1}_{\{b_0 \le b < a\}} \frac{\sin((a-b)(\pi/2-\theta))}{\cos\theta}. \end{aligned} \tag{6.16}$$

Set $c_{a\alpha} = 0$ *for* $\alpha \notin \{b_0, b_0-1\}$. *In the indicator, lift* b *to* $[b_0, b_0+L)$. *The coefficients* c, q *are real and bounded by* C, *independent of* g, *time and the random matrices once* E *and* I *are fixed. Then*

$$M_{B,ab} = \sum_\alpha c_{a\alpha} M_{B,\alpha b} + q_{ab} + e^M_{ab}, \qquad |e^M_{ab}| \le C|I|/g^2. \tag{6.17}$$

These are the coefficients of the real free forced recurrence, so the same c, q *apply when that recurrence is perturbed by onsite matrices. The same interpolation bound holds for the large-coupling comparison matrices of Lemma 6.4. Interpolation in both coordinates follows by applying* (6.17) *separately in each coordinate on its respective interval, using symmetry.*

The proof is in Appendix A.10.1.

For the upper-bound argument in Section 9, we also need the following one-block Dyson profile.

Lemma 6.6 (Dyson profile for a regular inverse)**.** *Fix* $\kappa_0 \in (0,2)$ *and* $C_* > 0$. *Let* $g = o(1)$ *and let* R *be Hermitian with* $\operatorname{rank} \mathbf{1}_{\{|g^2R| > g\}} \le Wg^{1/2}$. *For* $|E| \le 2-\kappa_0$, *define the Herglotz solution* $M(i\eta) = (-EI - g^2R - i\eta - m(i\eta))^{-1}$ *and* $m(i\eta) = W^{-1}\operatorname{Tr} M(i\eta)$. *Then, for sufficiently large* W,

$$\Im m(i\eta) \asymp 1, \qquad \Im M(i\eta) \asymp (I + g^4R^2)^{-1}, \qquad 0 < \eta \le C_*. \tag{6.18}$$

The matrix comparison is in the sense of quadratic forms, with constants depending only on κ_0, C_*.

The proof is in Appendix A.16.

7 Transfer estimates, spatial sampling, and loop initialization

We prove the large-coupling inputs stated in Section 5.3, which allow the common Theorem 5.12 to be used without an upper bound on g. The transfer estimate of Lemma 7.2 gives the resolvent seed through Ward's identity and also supplies the coarse moments used during propagation. The same estimates prove the short-cycle spreading bound of Lemma 2.4 in Section 7.2. The weighted Ward estimate in Section 7.2 adds exponential spatial decay. Section 7.3 then proves the spatial sampling of Lemma 5.8: short-cell transfer control gives Green-function interpolation; the auxiliary orbital limit transfers this interpolation to primitive loops; and a common finite-dimensional space reduces the required spatial maxima to polynomially many ordinary tuples. Section 7.4 establishes the continuity, weighted sampling and residual interpolation needed for propagation. Sections 7.5 and 7.6 prove fluctuation averaging and the complete initial estimates, respectively, uniformly in g. The five-step propagation is proved in Appendix C.3.

Definition 7.1 (Transfer matrices and free coordinates)**.** *For $g > 0$, $z \in \mathbb{C}$, and Hermitian onsite blocks V_j, define*

$$T_j(z) := \begin{pmatrix} zI_W - V_j/g & -I_W \\ I_W & 0 \end{pmatrix}, \qquad T(z) := \begin{pmatrix} z & -1 \\ 1 & 0 \end{pmatrix} \otimes I_W.$$

For time-dependent Hermitian onsite blocks $V_j(t)$, define

$$T_j(z,t) := \begin{pmatrix} zI_W - V_j(t)/g & -I_W \\ I_W & 0 \end{pmatrix}.$$

The free transfer matrix $T(z)$ corresponds to zero onsite blocks; for fixed z, it is independent of j and t. For $E \in (-2,2)$, write $E = 2\sin\theta$ with $-\pi/2 < \theta < \pi/2$ and set

$$S_E := (2\cos\theta)^{-1/2} \begin{pmatrix} 1 & 1 \\ ie^{-i\theta} & -ie^{i\theta} \end{pmatrix} \otimes I_W,$$
$$R_E := S_E^{-1} T(E) S_E = \operatorname{diag}(-ie^{i\theta}, ie^{-i\theta}) \otimes I_W.$$

The matrix R_E is unitary, and $\|S_E\| + \|S_E^{-1}\| \le C_\kappa$ for $|E| \le 2-\kappa$. Write $S := S_E|_{E=0}$, which is also unitary.

7.1 Transfer moments on the g^2 scale

The products in Lemma 7.2 are controlled at each deterministic energy, with constants independent of the coupling and the number of blocks in a window of length at most $C_0 g^2$.

Lemma 7.2 (Maximal transfer moments on scattering-length windows)**.** *Fix $\kappa > 0$ and $C_0, C_1 < \infty$. There is $C_\kappa < \infty$ such that, for $g \ge C_\kappa$, $|E| \le 2-\kappa$, $n \le C_0 g^2$, and $0 \le \eta \le 1$ with $n\eta \le C_1$, the matrices $T_j(z)$ from Definition 7.1, with $z = E + i\eta$, satisfy, for every fixed $p > 0$ and all sufficiently large W,*

$$\mathbb{E} \max_{0 \le j \le n} \left(\|T_j(z) \cdots T_1(z)\|^p + \left\| (T_j(z) \cdots T_1(z))^{-1} \right\|^p \right) \le C. \tag{7.1}$$

Products at $j = 0$ are identities. The constants and width threshold are independent of g, n, E, η in the displayed range. The same bound holds when the onsite variance is any number in $[0,1]$.

Proof. We use the real-energy normal form of Sadel–Schulz-Baldes [27, Section 5.1, (22)]. In the coordinates S_E of Definition 7.1, substituting $z = E + i\eta$ gives

$$M_j = S_E^{-1} T_j(z) S_E = R_E + i\eta B_E + g^{-1} X_j,$$

where $\|B_E\| \le C$ and X_j is a centered real-linear function of the GUE block. For every deterministic unit vector v, the real Gaussian vector $\xi = X_j v \in \mathbb{R}^{4W}$ has covariance Σ_v satisfying

$$\operatorname{Tr}\Sigma_v \le C, \qquad \|\Sigma_v\| \le C/W.$$

Indeed, for normalized GUE, $\mathbb{E}\,\|Vw\|^2 = \|w\|^2$ and the variance of every real linear functional of Vw is at most $C\,\|w\|^2/W$. The bounded deterministic conjugations preserve these estimates.

We record the needed Gaussian calculation. Put $w = R_E v + i\eta B_E v$, so $\|w\|^2 \le 1 + C\eta$. For $q \ge 2$, $x^{q/2} \le e^{q(x-1)/2}$ gives

$$\mathbb{E}\,\|M_j v\|^q \le e^{Cq\eta}\mathbb{E}\exp\Big\{\frac{q}{g}\Re\langle w,\xi\rangle + \frac{q}{2g^2}\,\|\xi\|^2\Big\}.$$

The Gaussian quadratic-moment formula, diagonalizing Σ_v, bounds the Gaussian expectation by

$$\exp\Big\{Cq/g^2 + Cq^2/(Wg^2)\Big\}, \qquad q/(Wg^2) \le c.$$

To check the constants directly, the logarithmic determinant term is at most $C(q/g^2)\operatorname{Tr}\Sigma_v$, and the linear term is at most $C(q^2/g^2)\,\|\Sigma_v\|\,\|w\|^2$. Choose $q = W$ and increase C_κ if necessary. Homogeneity then gives, for every past-measurable vector v,

$$\mathbb{E}[\|M_j v\|^q \mid V_1,\dots,V_{j-1}] \le e^a\,\|v\|^q, \qquad a = Cq(g^{-2}+\eta). \tag{7.2}$$

For deterministic unit v, $e^{-aj}\,\|M_j\cdots M_1 v\|^q$ is a nonnegative supermartingale. The maximal inequality therefore yields

$$\mathbb{P}\{\max_{j\le n}\|M_j\cdots M_1 v\| > u\} \le e^{an}u^{-q}.$$

A $1/2$-net of the unit sphere of $\mathbb{C}^{2W}$ has at most 5^{4W} elements, and the operator norm is bounded by twice the maximum over that net. Since $an \le Cq(C_0+C_1)$ and $q = W$,

$$\mathbb{P}\{\max_{j\le n}\|M_j\cdots M_1\| > u\} \le \min\{1,(C/u)^W\}. \tag{7.3}$$

For $W \ge 2p$, integrating this tail gives a bound C independent of g, n. The conjugation by S_E changes only constants.

For the inverse prefixes, apply the identical argument to the fresh left-multiplying factors M_j^{-*}:

$$(M_j\cdots M_1)^{-*} = M_j^{-*}\cdots M_1^{-*}.$$

They have the same structure as above. In fact

$$T_j(z)^{-1} = \begin{pmatrix} 0 & I_W \\ -I_W & zI_W - V_j/g \end{pmatrix}$$

is affine in the same fresh Gaussian block, and the free inverse-adjoint conjugated matrix is $R_E^{-*} = R_E$. Thus the same covariance and deterministic-perturbation bounds hold. This proves (7.1). □

7.2 Resolvent moments and exponential spatial decay

Proof of Lemma 5.5: (5.25). We first prove a pathwise bound and coarse moments for general L. Fix $g \ge C_\kappa$, $|E| \le 2-\kappa$, and $t \in [0,1]$, and set

$$n = \min\Big\{\Big\lfloor\frac{L-1}{2}\Big\rfloor, \lfloor g^2\rfloor\Big\}. \tag{7.4}$$

Write $G = (H_t(g) - E - i\eta)^{-1}$ and let $f_x = G_{xb}u$ for a deterministic vector u. At every block except b, f obeys the homogeneous transfer recurrence. For each a, select a deterministic direction so that the arc from a through $a+n$ has no b in its interior; such a direction exists because $2n < L$. Let K_a be the maximum inverse-transfer norm along this arc, including the identity. The transfer recurrence and summation over $0 \le j < n$ give

$$\|f_a\|^2 \le K_a^2(\|f_{a+j}\|^2 + \|f_{a+j+1}\|^2), \qquad \|f_a\|^2 \le \frac{2K_a^2}{n}\|f\|^2,$$

since each block occurs at most twice in the sum. This applies also at $a = b$, because the source is an endpoint and its inhomogeneous equation is not used. Ward's identity gives $\eta\|f\|^2 = \Im\langle u, f_b\rangle \le \|u\|\,\|f_b\|$. Combining these inequalities first at b and then at a yields

$$\left\|\left[(H_t(g) - E - i\eta)^{-1}\right]_{ab}\right\|_{\mathrm{op}} \le \frac{2K_aK_b}{n\eta}. \tag{7.5}$$

For $0 < \eta \le \min\{1, C_0/n\}$, we have $n \le g^2$ and $n\eta \le C_0$, so Lemma 7.2 bounds $\mathbb{E}K_a^{2p}$ and $\mathbb{E}K_b^{2p}$ uniformly in the onsite variance $t \in [0,1]$. Cauchy–Schwarz, without any independence requirement, gives

$$\sup_{0\le t\le 1}\max_{a,b}\mathbb{E}\left\|\left[(H_t(g) - E - i\eta)^{-1}\right]_{ab}\right\|_{\mathrm{op}}^p \le C_p(n\eta)^{-p}, \qquad 0 < \eta \le \min\{1, C_0/n\}. \tag{7.6}$$

At the seed height $\eta = c_0/n$, with fixed $0 < c_0 \le \min\{1, C_0\}$, this gives the bounded resolvent moments needed for initialization, without an upper bound on g or L.

Finally, if $L \ge 4g^2$, then $n = \lfloor g^2 \rfloor \asymp g^2$; thus $0 < \eta \le \min\{1, C_0/g^2\}$ implies $n\eta \le C_0$, and (7.6) gives (5.25). If $1 < \eta \le C_0/g^2$, then $g^2 < C_0$ and the deterministic bound $\|(H_t(g) - E - i\eta)^{-1}\| \le \eta^{-1} \le C_0(g^2\eta)^{-1}$ proves the remaining case. □

Proof of Lemma 2.4. For $L < L_{\kappa/2}$, Lemma 2.1 proves the claim after reducing c_κ, since $\min\{L, g^2\} \le L_{\kappa/2}$. Assume henceforth $L \ge L_{\kappa/2}$ and write $J = J_{g,\kappa,L}$. By Lemma 1.2(iii) with margin $\kappa/2$, choosing C_κ large enough puts J in a fixed free-bulk interval. Take n from (7.4) and $\eta = c_0/n$, decreasing c_0 so that $J_\eta = J + [-\eta, \eta]$ remains in the free bulk. All constants below depend only on κ.

First we prove the seed tail estimates (7.7). At $t = 1$, the inverse-product version of (7.3) and (7.5), with $a = b$, give $\mathbb{P}(K_a > v) \le \min\{1, (C/v)^W\}$ and $X_a(E) := \|G_{aa}(E + i\eta)\| \le 2K_a^2/c_0$. Consequently, uniformly for $E \in J_\eta$ and every block a,

$$\begin{aligned} \mathbb{P}(X_a(E) > v) &\le \min\{1, (C/v)^{W/2}\}, \qquad v > 0, \\ \mathbb{E}[X_a(E)\mathbf{1}_{\{X_a(E)>A\}}] &\le CA(C/A)^{W/2}, \qquad A > C, \quad W \ge 4. \end{aligned} \tag{7.7}$$

The second inequality follows from $\mathbb{E}[X\mathbf{1}_{\{X>A\}}] = A\mathbb{P}(X > A) + \int_A^\infty \mathbb{P}(X > v)\,dv$.

Next we obtain the heavy-block mass estimate (7.8). Since $n \ge \frac{1}{4}\min\{L, g^2\}$, $\frac{C'_\kappa}{2\eta\min\{L,g^2\}} \ge \frac{C'_\kappa}{8c_0}$; choose C'_κ so that the last constant exceeds $2C$ in (7.7). Apply (4.3) to each block spectral measure, using $\|\Im G_{aa}\| \le X_a(E)$ and $\operatorname{Tr}\Im G_{aa} \le WX_a(E)$. Summing over the blocks gives

$$\begin{aligned} &\frac{1}{WL}\mathbb{E}\sum_{k:\lambda_k\in J}\sum_a \|\psi_{k,a}\|^2\mathbf{1}_{\{\|\psi_{k,a}\|^2 > C'_\kappa/\min\{L,g^2\}\}} \\ &\quad\le \frac{C}{L}\sum_a\int_{J_\eta}\mathbb{E}[X_a(E)\mathbf{1}_{\{X_a(E)>C'_\kappa/(8c_0)\}}]\,dE \le Ce^{-cW}, \end{aligned} \tag{7.8}$$

where the last inequality uses (7.7) and the bounded length of J_η.

Finally we deduce (2.10). By (7.8), the expected number of eigenvectors with $\lambda_k \in J$ whose total mass on these heavy blocks exceeds $1/8$ is at most $8CWLe^{-cW}$. For every remaining eigenvector, a cyclic coordinate interval I meets at most $|I|/W + 2$ blocks, so

$$\sum_{x\in I} |\psi_k(x)|^2 \le \frac{1}{8} + \left(\frac{|I|}{W} + 2\right) \frac{C'_\kappa}{\min\{L, g^2\}} < \frac{1}{2}$$

whenever $|I| \le c_\kappa W \min\{L, g^2\}$, c_κ is sufficiently small, and $\min\{L, g^2\}$ is sufficiently large. For bounded $\min\{L, g^2\}$, reduce c_κ so that the target length is at most the single-block scale in (2.2); that estimate covers the remaining case. Absorbing its factor W into the exponential proves (2.10), uniformly in L, g. □

Lemma 7.3 (Weighted Ward estimate)**.** *For $g \ge 1$ and $0 \le t \le 1$, let $G(z) = (H_t(g) - z)^{-1}$, and let $\iota_b : \mathbb{C}^W \to \mathbb{C}^{WL}$ insert a vector into block b. For $z = E + i\eta$, $0 < \eta \le 1$, and $u \in \mathbb{C}^W$, put*

$$f = G(z)\iota_b u, \qquad f_c := G_{cb}(z)u \in \mathbb{C}^W, \qquad w_c = \exp(\alpha d(c, b)), \qquad c \in \mathbb{Z}_L,$$

where $0 \le \alpha \le \eta/8$. Then

$$\eta \sum_c w_c^2 \|f_c\|^2 \le 2\Im\langle u, G_{bb}(z)u\rangle.$$

The estimate holds pathwise, with the same constant in the auxiliary orbital model.

Proof. We use a Combes–Thomas-type exponential-weight argument; see [16, Appendix A.2, p. 827] for the classical exponential-conjugation method. The calculation below retains the local imaginary part on the right-hand side.

Fix a realization and use $\langle x, y\rangle = x^* y$. Since $g \ge 1$, the normalization in (5.1) gives $\sqrt{\gamma_g}\, g = 1$, so the hopping blocks are I_W. Writing $B_c = B_c^*$ for the onsite blocks, the resolvent equation is

$$(B_c - E)f_c + f_{c-1} + f_{c+1} - i\eta f_c = \mathbf{1}_{\{c=b\}} u,$$

with all block indices taken modulo L. Set $S := \sum_c w_c^2 \|f_c\|^2$. Multiply the equation by $w_c^2 f_c^*$ from the left and sum over c. The onsite contribution $\sum_c w_c^2 \langle f_c, (B_c - E)f_c\rangle$ is real. Reindexing the term involving f_{c-1} writes the hopping contribution as

$$\sum_c \left[w_c^2 \langle f_c, f_{c+1}\rangle + w_{c+1}^2 \langle f_{c+1}, f_c\rangle \right].$$

Since $\langle f_{c+1}, f_c\rangle = \overline{\langle f_c, f_{c+1}\rangle}$, its imaginary part is $\sum_c (w_c^2 - w_{c+1}^2)\Im\langle f_c, f_{c+1}\rangle$. On the right, $w_b = 1$ and $f_b = G_{bb}(z)u$ give

$$\Im\langle f_b, u\rangle = -\Im\langle u, G_{bb}(z)u\rangle.$$

Taking imaginary parts and rearranging therefore yields the exact identity

$$\eta S = \Im\langle u, G_{bb}(z)u\rangle + \mathcal{E}, \qquad \mathcal{E} := \sum_c (w_c^2 - w_{c+1}^2)\Im\langle f_c, f_{c+1}\rangle.$$

The distance satisfies $|d(c+1, b) - d(c, b)| \le 1$. Hence

$$|\mathcal{E}| \le 2\sinh\alpha \sum_c w_c w_{c+1} \|f_c\| \|f_{c+1}\| \le \sinh\alpha \sum_c (w_c^2 \|f_c\|^2 + w_{c+1}^2 \|f_{c+1}\|^2) = 2\sinh\alpha\, S.$$

Since $\alpha \le \eta/8 \le 1$, we have $2\sinh\alpha \le 4\alpha \le \eta/2$. Consequently

$$\eta S \le \Im\langle u, G_{bb}(z)u\rangle + \frac{\eta}{2} S, \qquad \eta \sum_c w_c^2 \|f_c\|^2 \le 2\Im\langle u, G_{bb}(z)u\rangle.$$

Only Hermiticity of the onsite blocks and the normalized nearest-neighbor hopping were used. The calculation is pathwise and independent of the orbital dimension, so the same constant applies to the auxiliary model. □

Exponential spatial decay. For $L \geq 4g^2$, $g \geq C_\kappa$, $|E| \leq 2-\kappa$ and fixed $0 < c_0 \leq C_0$, the normalized model satisfies

$$\mathbb{E}\|G_{ab}(E+i\eta)\|_{\mathrm{op}}^p \leq C_p e^{-pc\, d(a,b)/g^2}, \qquad c_0/g^2 \leq \eta \leq C_0/g^2, \tag{7.9}$$

for every fixed $p > 0$, with $G(z) = (H-z)^{-1}$. The same bound holds with $G(z) = (H_t(g)-z)^{-1}$ at each fixed $0 \leq t \leq 1/2$. Constants are uniform in g, L, t, and c is independent of p; no polynomial volume upper bound is needed here.

Proof of (7.9). For $\eta > 1$, the weighted identity in the proof of Lemma 7.3 with fixed small $\alpha > 0$ and $\|G_{bb}\| \leq \eta^{-1}$ gives $\|G_{ab}\| \leq \sqrt{2}\,\eta^{-1} e^{-\alpha d(a,b)}$, which suffices since $g \geq 1$. Assume henceforth $\eta \leq 1$. For $u \in \mathbb{C}^W$, put $f = G(E+i\eta)\iota_b u$. Choose from a a source-avoiding arc of $n \asymp g^2$ sites, as in the derivation of (7.5), and let K_a be the maximal inverse-transfer norm there. The homogeneous recurrence implies

$$\|f_a\|^2 \leq \frac{2K_a^2}{n} \sum_{c \text{ on arc}} \|f_c\|^2.$$

On this arc $d(c,b) \geq d(a,b) - n - 1$. Apply Lemma 7.3 with $\alpha = c\eta$, observing $\alpha n = O(1)$:

$$\|G_{ab}u\|^2 \leq \frac{CK_a^2}{n\eta} e^{-2\alpha d(a,b)} \|G_{bb}\|\, \|u\|^2.$$

Here $n\eta \asymp 1$. Lemmas 7.2 and 5.5 bound every fixed moment of K_a and G_{bb}, since $g^2\eta \geq c_0$; no independence between them is required. Taking the supremum over unit vectors u and applying Hölder proves (7.9), since $\alpha \geq c/g^2$. The Brownian-time version uses the same one-step covariance bounds with variance no larger than one. □

7.3 Spatial sampling

Sampling controls maxima over all block tuples using only polynomially many sampling points in W, avoiding a union bound over all L^k tuples. We deduce Lemma 5.8 from scalar Green-function interpolation with a uniform coarse envelope, a common finite-dimensional space for random and primitive loops, and a classical sampling inequality.

Throughout this subsection, unless otherwise stated, the volume and regularity assumptions of Lemma 5.8 are in force, with $g \geq C_\kappa$ sufficiently large for the auxiliary transfer estimates. In the sampling proof these estimates are used only when $g^2 \geq W^B$, which ensures this fixed lower bound for sufficiently large W. Recall the loop definition (5.5). Estimate constants are denoted by C'' to distinguish them from the fixed input bound C and may change between estimates; they may depend on the fixed parameters of Lemma 5.8.

Green-function interpolation and its uniform envelope.

Proposition 7.4 (Scalar Green-block interpolation)**.** *Fix $0 < \kappa < 2$, $C > 1$, $D > 0$ and $B > 2$. For all sufficiently large W, assume that g, L, E are deterministic and*

$$g^2 \geq W^B, \qquad 4g^2 \leq L \leq g^2 W^C,$$
$$|E| \leq 2-\kappa, \qquad |m_{g,L}(E)| \leq C, \quad \Im m_{g,L}(E) \geq C^{-1}.$$

For the flow (5.1), set $G_t = (H_t(g) - z_t)^{-1}$ along the characteristic (5.3), $0 \leq t \leq 1 - W^{-3}$. All constants and the width threshold depend only on the fixed parameters κ, C, B, D, not on g, L, E. For $\mathcal{P} = \mathcal{P}_B(g, L, W) = \{I_j\}$ from Definition 5.7, let b_j be the first block of I_j, with block indices taken modulo L. Write

$$\mathcal{G}_t(\mathcal{P}) := \max_{\substack{I_i, I_j \in \mathcal{P} \\ \alpha \in \{b_i, b_i-1\},\ \beta \in \{b_j, b_j-1\}}} \|(G_t)_{\alpha\beta}\|_{\mathrm{op}}.$$

For $a \in I_j$ and any block b, define $c_{a\alpha}$ and q_{ab} by (6.16) with $I = I_j$ and $b_0 = b_j$, and set $c_{a\alpha} = 0$ for $\alpha \notin \{b_j, b_j - 1\}$. Thus the first index of c or q determines the cell used in its definition. Set

$$d_{ab} = q_{ab} + \sum_{\alpha \in \{b_j, b_j - 1\}} c_{a\alpha} q_{b\alpha},$$

where $q_{b\alpha}$ is defined using the cell containing b. All coefficients are real and bounded by C''. Once the partition is fixed, they depend only on E. With probability at least $1 - W^{-D}$, simultaneously for all a, b and $0 \le t \le 1 - W^{-3}$,

$$(G_t)_{ab} = \sum_{\alpha,\beta} c_{a\alpha} c_{b\beta} (G_t)_{\alpha\beta} + d_{ab} I_W + \mathcal{E}_{ab}(t), \qquad \|\mathcal{E}_{ab}(t)\|_{\mathrm{op}} \le C'' W^{-B/4} (1 + \mathcal{G}_t(\mathcal{P})). \tag{7.10}$$

On the same event,

$$\sup_{0 \le t \le 1 - W^{-3}} \mathcal{G}_t(\mathcal{P}) \le W^{C+10}. \tag{7.11}$$

The envelope exponent is independent of B, D, g, L.

For a fixed deterministic comparison $1/2 \le s \le u \le 1 - W^{-3}$ as in Lemma 6.3(a), with upper comparison point w, the interpolation and error bound in (7.10) hold on the same event and with the same coefficients for $G = (H_u(g) - z)^{-1}$, uniformly over $z = \Re z_u + iy$, $\eta_u \le y \le \Im w$, with G replacing G_t also in the maximum defining $\mathcal{G}_t(\mathcal{P})$. For the adjoint resolvents, the same bound holds on the same event, with the same homogeneous coefficients and with d_{ab} replaced by d_{ba}.

A common interpolation space for loops and primitives.

Lemma 7.5 (A common scalar interpolation space)**.** *Under the assumptions of Proposition 7.4, use its partition and envelope. Fix $1 \le k \le K$. Fix cells $I_1, \dots, I_k \in \mathcal{P}$. There is a deterministic space $\mathcal{V}$ of scalar functions on $X = I_1 \times \cdots \times I_k$, with $1 \le \dim \mathcal{V} \le d_k = 6^k$. For $0 \le t \le 1 - W^{-3}$, the second bound below holds deterministically, and the first holds on the event of Proposition 7.4:*

$$\begin{aligned} \operatorname{dist}_\infty(W^{k-1} \mathcal{L}_k(t), \mathcal{V}) &\le W^{-B/8} (1 + \mathcal{G}_t(\mathcal{P}))^k, \\ \operatorname{dist}_\infty(W^{k-1} \mathcal{K}_k(t), \mathcal{V}) &\le W^{-B/4} (1 - t)^{-C''}. \end{aligned} \tag{7.12}$$

$$\operatorname{dist}_\infty(f, \mathcal{V}) := \inf_{v \in \mathcal{V}} \max_{\boldsymbol{x} \in X} |f(\boldsymbol{x}) - v(\boldsymbol{x})|.$$

The exponent C'' depends only on k and the fixed volume and regularity parameters, not on B, g, L; the width threshold may also depend on B, K. The same space works for every time and every sign tuple of length k.

Finite-dimensional sampling.

Lemma 7.6 (Finite maximum-volume sampling)**.** *If $\mathcal{V}$ is a d_0-dimensional space, $d_0 \ge 1$, of complex functions on a finite set X, there exist d_0 points $\xi_1, \dots, \xi_{d_0}$ such that, for all $Z, F : X \to \mathbb{C}$,*

$$\begin{aligned} \sup_X |Z - F| \le d_0 \max_{1 \le j \le d_0} |Z(\xi_j) - F(\xi_j)| \\ + (d_0 + 1)(\operatorname{dist}_\infty(Z, \mathcal{V}) + \operatorname{dist}_\infty(F, \mathcal{V})). \end{aligned} \tag{7.13}$$

Proof. This is the classical maximum-volume (Fekete-point) argument. Choose a basis and evaluation points maximizing the absolute determinant of the evaluation matrix, which is nonzero by linear independence. Cramer's rule expresses any evaluation as a combination of the selected ones with coefficients of modulus at most one: each numerator replaces one selected row and cannot exceed the maximal determinant. Apply this representation to the difference of best approximations in $\mathcal{V}$ and add their errors to obtain (7.13). □

Proof of Lemma 5.8. If $g^2 < W^B$, take $\mathcal{S}_k = \mathbb{Z}_L^k$ and sampling is exact. Otherwise, work on the event of Proposition 7.4, of probability at least $1 - W^{-D}$. By (7.11)–(7.12), the sum of the two approximation errors in Lemma 7.5 is at most

$$2^K W^{-B/8+K(C+10)} + W^{-B/4+3C''} \le W^{-D-1},$$

uniformly over all cell products, $k \le K$, signs, and times, after choosing B sufficiently large; both exponents $C + 10$ and C'' are independent of B. On each product, take the at most d_k sampling points supplied by Lemma 7.6 for its common space, and let $\mathcal{S}_k$ be their union. These sets are deterministic and independent of time and signs. Applying (7.13) and dividing by W^{k-1} proves (5.26) with $C_k = d_k$, since the error is at most $(d_k + 1)W^{-D-k} \le W^{-D}$ for large W. Taking $F = 0$ in the same inequality proves the assertion for $\mathcal{L}_k$. □

7.3.1 Green-block interpolation and uniform envelope

We prove Proposition 7.4. The short-cell transfer estimates below give (7.10); the subsequent maximal-increment estimate gives (7.11). Use the normalized Brownian blocks $B_j(t) := g(\mathcal{B}_t^{(g)})_{jj}$ from (5.1). Fix $|E| \le 2 - \kappa$ and use $T_j(z,t)$ from Definition 7.1 with $V_j(t) = B_j(t)$. For prefix products on a cell or interval, number its blocks from 1 in consecutive order, starting at its first block, and put

$$P_j(z,t) := T_j(z,t) \cdots T_1(z,t).$$

Here $1 \le j \le |I|$ on the chosen interval I, and $P_0(z,t) = I_{2W}$.

Lemma 7.7 (Polynomially many cells)**.** *Fix $B > 2$ and $D > 0$, and assume $g^2 \ge W^B$. With probability at least $1 - W^{-D}$, simultaneously for every cell $I \in \mathcal{P}_B$, every $1 \le j \le |I| \le \lfloor g^2 W^{-B} \rfloor$, all $0 \le t \le 1$, and all $z = E + i\eta$ with $0 \le \eta \le g^{-2}$,*

$$T_j(z,t) \cdots T_1(z,t) = T(E)^j + O(W^{-B/4}).$$

The error is in operator norm. The assertion holds for sufficiently large W depending on the fixed parameters, uniformly in g, L.

Proof. Set $Q_j(t) := R_E^{-j} S_E^{-1} P_j(E,t) S_E$, with $Q_0(t) = I_{2W}$. For every fixed $p \ge 2$, in any orbital dimension W and on any interval of length n with $n/g^2 \le c_p$, we first prove

$$\mathbb{E} \sup_{0 \le t \le 1} \max_{1 \le j \le n} \|Q_j(t) - I\|_{\mathrm{HS}}^p \le C_p W^{p/2} (n/g^2)^{p/2}, \tag{7.14}$$

where $c_p, C_p > 0$ are independent of g, n, W. At time 1, the sequence $Q_j(1)$ is a spatial martingale: conditional expectation over the new block replaces $T_j(E,1)$ by $T(E)$. The conditional Gaussian moment bound gives

$$\mathbb{E}[\|Q_j - Q_{j-1}\|_{\mathrm{HS}}^p \mid \mathcal{F}_{j-1}] \le C_p g^{-p} \|Q_{j-1}\|_{\mathrm{HS}}^p .$$

The Hilbert-space martingale maximal inequality and Hölder imply

$$X := \mathbb{E} \max_{1 \le j \le n} \|Q_j - I\|_{\mathrm{HS}}^p \le C_p (n/g^2)^{p/2} \mathbb{E} \max_{0 \le j \le n} \|Q_j\|_{\mathrm{HS}}^p \le C_p (n/g^2)^{p/2} (W^{p/2} + X).$$

For n/g^2 sufficiently small, absorb the last term.

For every fixed prefix j, $Q_j(t)$ is also a matrix martingale with respect to the joint Brownian filtration of $B_1, \dots, B_n$. Indeed each matrix product contains each independent block Brownian motion at most once, and cross quadratic variations between different blocks vanish. The finite maximum $\max_{1 \le j \le n} \|Q_j(t) - I\|_{\mathrm{HS}}$ is a nonnegative submartingale by convexity. Doob's L^p inequality in time bounds its supremum moment by the terminal spatial maximum just estimated, proving (7.14).

Now each cell of $\mathcal{P}_B$ has length at most $\lfloor g^2 W^{-B} \rfloor$, and there are at most $3W^{C+B}$ cells. Apply (7.14), Markov's inequality, and a union over these cells. The failure bound is $C_p W^{C+B+p/2-Bp/4}$; since $B > 2$, choose the fixed moment order p sufficiently large after $C + B, D$. The identity $P_j(E,t) - T(E)^j = S_E R_E^j (Q_j(t) - I) S_E^{-1}$, with R_E unitary and S_E, S_E^{-1} uniformly bounded in the bulk, gives $\|P_j(E,t) - T(E)^j\| = O(W^{-B/4})$ on this event.

Fix a cell I of length $n = |I|$ and a Brownian time $u \in [0,1]$, and suppress this time in $T_j(z,u)$ and $P_j(z,u)$. The real-energy prefixes and inverse prefixes have bounded norm. Discrete variation of constants gives

$$P_j(z) = P_j(E) + \sum_{r=1}^{j} P_j(E) P_r(E)^{-1} [T_r(z) - T_r(E)] P_{r-1}(z). \tag{7.15}$$

The inserted difference has norm $|z-E|$. Since $n \le g^2 W^{-B}$, discrete Gronwall and the real-energy estimate give

$$\|P_j(z) - T(E)^j\| \le C'' W^{-B/4} + C'' n |z-E| e^{C'' n |z-E|} = O(W^{-B/4}),$$

uniformly over all prefixes, cells and times whenever $|z-E| \le C''/g^2$, for any fixed C''. This proves the assertion on the same event. □

Proof of Proposition 7.4. Work on the event of Lemma 7.7, fix a cell I of length $n = |I|$ and a Brownian time $u \in [0,1]$, and suppress this time in $T_j(z,u)$ and $P_j(z,u)$. The lemma's proof gives $P_j(z) = T(E)^j + O(W^{-B/4})$ throughout $|z-E| \le C''/g^2$. Since the real free powers and their inverses are uniformly bounded, the inverse identity and multiplication show that $P_j(z)$, $P_j(z)^{-1}$ and $P_j(z)P_r(z)^{-1}$ are within $O(W^{-B/4})$ in norm of the free products $T(E)^j$, $T(E)^{-j}$ and $T(E)^{j-r}$, respectively, uniformly over $0 \le r \le j \le n$, all cells, $u \in [0,1]$, and $|z-E| \le C''/g^2$.

Along the characteristic (5.3), $|m| \le C$ gives $|z_t - E| \le C/g^2$. We now prove (7.10) on this event for $0 \le t \le 1 - W^{-3}$, writing $G = G_t$. In the original block indices, the Green column obeys the forced recurrence

$$\begin{pmatrix} G_{r+1,b} \\ G_{r,b} \end{pmatrix} = T_r(z_t, t) \begin{pmatrix} G_{r,b} \\ G_{r-1,b} \end{pmatrix} + \begin{pmatrix} \mathbf{1}_{r=b} I_W \\ 0 \end{pmatrix}.$$

For $a \in I_j$, iterate in the first coordinate from $(G_{b_j,b}, G_{b_j-1,b})$ to G_{ab}. There is at most one source term, at b, and the homogeneous and source coefficients differ from their scalar free counterparts by $O(W^{-B/4})$. Thus

$$G_{ab} = \sum_{\alpha} c_{a\alpha} G_{\alpha b} + q_{ab} I_W + \mathcal{R}_{ab}, \qquad \|\mathcal{R}_{ab}\| \le C'' W^{-B/4} (1 + \|G_{b_j,b}\| + \|G_{b_j-1,b}\|).$$

For $b \in I_l$, apply the same first-coordinate recurrence on I_l to $(G^*)_{b\alpha}$ at $\overline{z_t}$ and take its adjoint. The real free coefficients give second-coordinate coefficients $c_{b\beta}$, $\beta \in \{b_l, b_l - 1\}$, and source term $q_{b\alpha} I_W$. In particular,

$$\max_{\substack{I_i \in \mathcal{P} \\ \alpha \in \{b_i, b_i - 1\}}} \|G_{\alpha b}\| \le C''(1 + \mathcal{G}_t(\mathcal{P})).$$

This bounds $\mathcal{R}_{ab}$ by $C'' W^{-B/4}(1 + \mathcal{G}_t(\mathcal{P}))$; substituting the second-coordinate interpolation gives (7.10) with the stated d_{ab}.

For $G = (H_u(g) - z)^{-1}$ on the comparison segment in Proposition 7.4, (6.9) and its parameter bounds give $|z-E| \le |\Re z_u - E| + \Im w \le C''/g^2$ throughout the comparison segment. The transfer bound obtained from (7.15) and the Green recurrence give (7.10) for these resolvents and, at conjugate spectral parameters, for their adjoints.

To prove (7.11) and the increments used in Lemma 5.10, we establish (7.17). For this increment estimate, assume only $L\gamma_g \le C W^{C_L}$ for fixed $C_L > 0$ and $|m(g,E)| \le C$. Take

$0 < T < 1$ and use G_s on the characteristic; then $|z_s'| = \gamma_g |m(g, E)| \le C\gamma_g$. Assume that, for every fixed $q > 0$,

$$\sup_{a,b,\, 0\le s\le T} \mathbb{E}\|(G_s)_{ab}\|^q \le C_q W^{3q}, \tag{7.16}$$

where C_q is independent of W, g, L. For every fixed $p \ge 2$ and block pair a, b, we claim that

$$\begin{aligned} &\mathbb{E} \sup_{0\le s\le T} \|(G_s)_{ab}\|^p \le C_p W^{(C_L+9)p}, \\ &\mathbb{E}\|(G_v - G_u)_{ab}\|^p \le C_p W^{(C_L+9)p}(v-u)^{p/2}, \qquad 0 \le u \le v \le T. \end{aligned} \tag{7.17}$$

Itô's formula for the flow (5.1) gives $d(G_s)_{ab} = A_{ab}(s)\, ds + d\mathcal{M}_{ab}(s)$, where

$$\begin{aligned} A_{ab}(s) &= z_s' \sum_c (G_s)_{ac}(G_s)_{cb} + \gamma_g \sum_c (G_s)_{ac}\langle G_s\rangle_{cc}(G_s)_{cb}, \\ d\mathcal{M}_{ab}(s) &= -\sum_c (G_s)_{ac}\, d(\mathcal{B}_s^{(g)})_{cc}\, (G_s)_{cb}. \end{aligned}$$

Define the scalar quadratic variation $[\mathcal{M}_{ab}]_{\mathrm{HS}}(s) := \sum_{\mu,\nu=1}^{W} [(\mathcal{M}_{ab})_{\mu\nu}, \overline{(\mathcal{M}_{ab})_{\mu\nu}}]_s$. Independence of the Brownian blocks and the GUE covariance give

$$\frac{d}{ds}[\mathcal{M}_{ab}]_{\mathrm{HS}}(s) = \frac{\gamma_g}{W} \sum_c \|(G_s)_{ac}\|_{\mathrm{HS}}^2 \|(G_s)_{cb}\|_{\mathrm{HS}}^2.$$

Using (7.16), $\|X\|_{\mathrm{HS}} \le \sqrt{W}\|X\|$ and $|z_s'| \le C\gamma_g$, Minkowski and Hölder give

$$\big\| \|A_{ab}(s)\| \big\|_{L^p} \le C_p(L\gamma_g)W^9, \qquad \left\| \frac{d}{ds}[\mathcal{M}_{ab}]_{\mathrm{HS}}(s) \right\|_{L^{p/2}} \le C_p(L\gamma_g)W^{13}.$$

Hilbert-space BDG and Hölder in time, with $L\gamma_g \le CW^{C_L}$ and $T < 1$, prove (7.17), using (7.16) at $s = 0$ for the first bound.

We return to the large-coupling assumptions and take $T = 1 - W^{-3}$. Here $|\Re z_t| \le 2 - \kappa/2$ and $0 < \eta_t \le C/g^2$ for large W. Since $g^2\eta_t = (1-t)\Im m \asymp 1 - t$ and $1 - t \ge W^{-3}$, (5.25) with margin $\kappa/2$ verifies (7.16) uniformly in g, L; $L\gamma_g = L/g^2 \le W^C$ allows $C_L = C$ in the increment estimate.

Apply the first estimate in (7.17) and Markov's inequality to the $O(W^{2(C+B)})$ endpoint pairs in $\mathcal{G}_t(\mathcal{P})$. A union bound gives

$$\mathbb{P}\left(\sup_{t\le 1-W^{-3}} \mathcal{G}_t(\mathcal{P}) > W^{C+10} \right) \le C_p W^{2(C+B)-p}.$$

Choose $p > 2(C+B) + D + 1$ and use failure exponent $D+1$ in Lemma 7.7. For sufficiently large W, the combined failure probability is at most W^{-D}, proving (7.11) together with (7.10). □

7.3.2 A common interpolation space

Proof of Lemma 7.5. Set $n = \lfloor g^2 W^{-B} \rfloor$ and let b_j be the first block of I_j, $1 \le j \le k$, with cyclic subscripts. On $I_j \times I_{j+1}$, let $\mathcal{F}_j$ consist of the four functions $c_{x\alpha}c_{y\beta}$, with $\alpha \in \{b_j, b_j - 1\}$ and $\beta \in \{b_{j+1}, b_{j+1} - 1\}$, and the two functions d_{xy}, d_{yx} from Proposition 7.4. For $x_{k+1} = x_1$, set

$$\mathcal{V} = \operatorname{span}\left\{ \prod_{j=1}^{k} f_j(x_j, x_{j+1}) : f_j \in \mathcal{F}_j \right\}.$$

One homogeneous product takes value one at $(b_1, \dots, b_k)$, so $1 \le \dim \mathcal{V} \le 6^k$. The coefficients are deterministic and, for fixed cells and E, independent of time, signs and orbital dimension.

Expanding (7.10) in the cyclic trace word, using d_{yx} for an adjoint block, places its interpolant in $\mathcal{V}$. On the event of Proposition 7.4, telescoping the product in (7.10) and using $W^{-1}|\operatorname{Tr} X| \le \|X\|$ prove the random bound in (7.12) after absorbing the fixed prefactor into $W^{B/8}$ for sufficiently large W.

Uniform interpolation error before taking the limit. For the primitive bound in (7.12), use the auxiliary model of Lemma 5.4, keeping the original cells fixed. At fixed $0 \le t \le 1 - W^{-3}$, use the transfer matrices of Definition 7.1 and the cellwise prefix notation of Section 7.3.1 at width $\widetilde{W}$, with $1 \le j \le |I| \le n$. Put $\|Y\|_{2,\widetilde{W}} = \widetilde{W}^{-1/2}\|Y\|_{\mathrm{HS}}$. Estimate (7.14) bounds the normalized Hilbert–Schmidt L^p norm of $P_j(E,t) - T(E)^j$ by $C_p\sqrt{n/g^2}$, uniformly in $\widetilde{W}, g, L$. Since $\Re z_t = E + O(g^{-2})$ remains in the buffered free bulk after increasing C_κ, (7.15) and the prefix moments of Lemma 7.2, with Hölder and Minkowski, bound the same norm of $P_j(z_t,t) - P_j(E,t)$ by $C_p n|z_t - E| \le C_p n/g^2$. The inverse-prefix moments and Hölder then give a $C_p\sqrt{n/g^2}$ bound in this norm for $P_j(z_t,t)P_r(z_t,t)^{-1} - T(E)^{j-r}$, $0 \le r \le j \le |I|$. Applying (5.25) at width $\widetilde{W}$ with bulk margin $\kappa/2$ and $g^2\eta_t = (1-t)\Im m \asymp 1 - t$ gives

$$\sup_{b,c} \mathbb{E}\|(G_t^{(\widetilde{W})})_{bc}\|^p \le C_p(1-t)^{-p} \tag{7.18}$$

for every fixed p and all $\widetilde{W} \ge \widetilde{W}_0(p,\kappa,C)$, uniformly in g, L, E in the present parameter range. Let $\mathcal{E}_{ab}^{(\widetilde{W})}(t)$ be the algebraic remainder in (7.10) for the auxiliary model, with the same coefficients and summation ranges. Iterate the forced Green recurrence from the proof of Proposition 7.4 in each coordinate. The preceding prefix bounds, (7.18), and Hölder give

$$\sup_{a,b} \big\| \|\mathcal{E}_{ab}^{(\widetilde{W})}(t)\|_{2,\widetilde{W}} \big\|_{L^p} \le C_p\sqrt{n/g^2}(1-t)^{-1}.$$

Replacing each Green block by its scalar interpolation in the cyclic word defines $\widetilde{W}^{k-1}\widetilde{\mathcal{L}}_k^{(\widetilde{W})} \in \mathcal{V}$. Telescoping this product, using Hölder and $|\widetilde{W}^{-1}\operatorname{Tr} AB| \le \|A\|_{2,\widetilde{W}}\|B\|_{2,\widetilde{W}}$, gives

$$\sup_{\boldsymbol{x}\in X} \mathbb{E}|\widetilde{W}^{k-1}\mathcal{L}_k^{(\widetilde{W})}(t;\boldsymbol{x}) - \widetilde{W}^{k-1}\widetilde{\mathcal{L}}_k^{(\widetilde{W})}(t;\boldsymbol{x})| \le C''\sqrt{n/g^2}(1-t)^{-C''}.$$

The constants and width threshold remain independent of g, L. This is a supremum of pointwise expected errors, not an expectation of a spatial supremum; no microscopic union bound is used. Since $\widetilde{W}^{k-1}\mathbb{E}\widetilde{\mathcal{L}}_k^{(\widetilde{W})} \in \mathcal{V}$, the same bound holds for $\operatorname{dist}_\infty(\widetilde{W}^{k-1}\mathbb{E}\mathcal{L}_k^{(\widetilde{W})}, \mathcal{V})$.

Passage to the primitive at fixed parameters. At fixed W, g, L, E, cells and $t < 1$, Lemma 5.4 gives convergence in $\ell^\infty(X)$ because X is finite; since $\mathcal{V}$ is closed and independent of $\widetilde{W}$,

$$\operatorname{dist}_{\ell^\infty(X)}(W^{k-1}\mathcal{K}_k, \mathcal{V}) = \lim_{\widetilde{W}\to\infty} \operatorname{dist}_{\ell^\infty(X)}(\widetilde{W}^{k-1}\mathbb{E}\mathcal{L}_k^{(\widetilde{W})}, \mathcal{V}) \le C''\sqrt{n/g^2}(1-t)^{-C''}.$$

Since $n/g^2 \le W^{-B}$, absorbing the fixed prefactor into $W^{B/4}$ for sufficiently large W proves the primitive bound in (7.12). Uniformity in g, L comes from the prelimit bound, without requiring a uniform convergence rate. □

7.4 Sampling and continuity for propagation

We prove Lemma 5.10. We first extend the Green-block increments proved in Section 7.3 to the fields used in propagation. We then treat spatial weights, centered traces and entries, and the averaged residual, before proving the extension from height meshes to full vertical comparison segments at fixed times.

7.4.1 Time increments and weighted sampling

We prove the time-increment bounds (5.33) in Lemma 5.10(a) and the weighted sampling bound (7.23) in Corollary 7.8. Together, they reduce the weighted suprema used in spatial stopping, to any prescribed polynomial accuracy, to ordinary spatial samples and a polynomial-size time net.

Proof of Lemma 5.10(a): (5.33). Under the volume and regularity assumptions of Lemma 5.8, with $g \geq C_\kappa$ sufficiently large, the block moments verified in the proof of Proposition 7.4 give $\sup_{a,b,\,0\leq s\leq 1-W^{-3}} \mathbb{E}\|(G_s)_{ab}\|_{\mathrm{op}}^q \leq C_q W^{3q}$. The second estimate in (7.17), with $C_L = C$, therefore supplies the Green-block increments in this range. We next prove the normalized-loop bounds

$$\mathbb{E}|X(v) - X(u)|^p \leq C_p W^{C_* p}|v-u|^{p/2}, \qquad X \in \{W^{k-1}\mathcal{L}_k, W^{k-1}(\mathcal{L}_k - \mathcal{K}_k)\}, \tag{7.19}$$

for $p \geq 2$, fixed block and sign labels, and $0 \leq u \leq v \leq 1 - W^{-3}$. The constant C_* depends only on C, the fixed loop lengths and regularity margins, not on p, g, L.

For $W^{k-1}\mathcal{L}_k$, telescope the difference of its k Green-block factors. Hölder, (7.17), and the coarse block moments bound each term: its L^p norm is at most $C_p W^{C+9+3(k-1)}|v-u|^{1/2}$. This proves (7.19) for $X = W^{k-1}\mathcal{L}_k$. For the primitive, the following deterministic argument uses only $\max_{a,b}|M_{B,ab}| \leq C_M$, the exact Ward identity and (5.10); its constants depend only on k, C_M. Set $F_k = W^{k-1}\mathcal{K}_k$ and

$$R_k(t) := \max_{1\leq j\leq k} \max_{\boldsymbol{a}_{\neq j},\boldsymbol{\sigma}} \sum_{a_j\in\mathbb{Z}_L} |F_k(t;\boldsymbol{a},\boldsymbol{\sigma})|.$$

Here $\|F_k\|_\infty$ denotes the maximum over all labels and signs. The entry bound and $\gamma_g \sum_b |M_{B,ab}|^2 = 1$ give $\|F_k(0)\|_\infty + \gamma_g R_k(0) \leq C_k$ for $k \geq 2$: for the row, apply Cauchy–Schwarz to the two edges incident to a_j. The pair formula (5.13), $|Q| \leq P$ and $P\mathbf{1} = \mathbf{1}$ give $\|F_2(t)\|_\infty + \gamma_g R_2(t) \leq C(1-t)^{-1}$. By (5.15), the absolute row and column sums of each one-label propagator are at most $(1-u)/(1-t)$. Thus the k-label propagator has norm at most $((1-u)/(1-t))^k$ for both $\|\cdot\|_\infty$ and R_k.

For a nonadjacent cut in (5.6), let $k_L, k_R \geq 3$ be the two child lengths, with $k_L + k_R = k+2$. The normalized source is $\gamma_g \sum_c F_{k_L} F_{k_R}$. Its supremum is bounded by $\gamma_g \|F_{k_L}\|_\infty R_{k_R}$, and its one-label row by $\gamma_g R_{k_L} R_{k_R}$: the free external label belongs to one child, whereas c belongs to both. Induction in (5.16) therefore gives

$$\|F_k(t)\|_\infty + \gamma_g R_k(t) \leq C_k(1-t)^{-(2k-3)}, \qquad k \geq 2, \quad 0 \leq t < 1. \tag{7.20}$$

Indeed, $k = 3$ has no source; for $k \geq 4$ the Duhamel bound is $C_k(1-t)^{-k}[1 + \int_0^t (1-u)^{2-k}\,du] \leq C_k(1-t)^{-(2k-3)}$. For $k = 2$, the pair Riccati equation gives the derivative bound below. For $k \geq 3$, the generator $\Theta_{t,\boldsymbol{\sigma}}$ in (5.14) satisfies $\|\Theta_{t,\boldsymbol{\sigma}}\|_{\ell^\infty\to\ell^\infty} \leq k/(1-t)$. Together with the nonadjacent-source supremum estimate and (7.20), this proves

$$\|\partial_t(W^{k-1}\mathcal{K}_k(t))\|_\infty \leq C_k(1-t)^{-(2k-2)}, \qquad k \geq 2. \tag{7.21}$$

Consequently the normalized primitive increments are bounded by $C_k W^{6k-6}|v-u|$ on $[0, 1-W^{-3}]$; $\mathcal{K}_1 = m$ is constant. The triangle inequality and the already proved case $X = W^{k-1}\mathcal{L}_k$ of (7.19) give the case $X = W^{k-1}(\mathcal{L}_k - \mathcal{K}_k)$, after increasing C_* if necessary.

For Lemma 5.10(a), both parameter ranges satisfy $L\gamma_g \leq CW^2$. At bounded coupling, $\eta_s^{-1} \leq C''W^3$ gives the same block moment input, while $|z_s'| = \gamma_g|m| \leq C\gamma_g$ supplies the characteristic derivative bound. Thus (7.17), with $C_L = 2$, gives $\mathbb{E}\|(G_v - G_u)_{ab}\|_{\mathrm{op}}^p \leq C_p W^{11p}|v-u|^{p/2}$ in both cases. Normalized traces and entries inherit the block increment bound; centering does not change the trace increment because $M(g,E)$ is independent of time. Dividing (7.19) by $W^{(k-1)p}$ gives the bounds for $\mathcal{L}_k$ and D_k at large coupling. At bounded coupling, $\max_{a,b}|M_{B,ab}| \leq (\gamma_g \Im m)^{-1} \leq C'$,

so the same product telescoping and the primitive Lipschitz bound (7.21) give these two bounds, including the constant length-one primitive.

For the residual, solve the first identity in (5.32) for R_{ab} and subtract it at v, u. In the quadratic term, separate the increment of the time prefactor and telescope the two trace factors. Using $\gamma_g \sum_c |M_{B,ac}| \le C$, Minkowski and Hölder with (7.16) and the second estimate in (7.17), with $C_L = 2$, give $\|R_{ab}(v) - R_{ab}(u)\|_{L^p} \le C_p W^C |v-u|^{1/2}$, with C independent of p, g, L in the arbitrary-coupling range. This proves the residual case of (5.33).

Finally, fix $D > 0$ and a deterministic collection $\mathcal{S}$ of at most W^{C_S} tuples. Choose $p > 4$ sufficiently large after D, C_S. Markov's inequality on successive dyadic increments, followed by summation of their geometric probabilities and a union over $\mathcal{S}$, gives a polynomial 1/4-Hölder seminorm: for a fixed C_H,

$$\mathbb{P}\left(\max_{\boldsymbol{a}\in\mathcal{S}} \sup_{0\le u<v\le 1-W^{-3}} \frac{\|X(v;\boldsymbol{a}) - X(u;\boldsymbol{a})\|}{|v-u|^{1/4}} > W^{C_H} \right) \le W^{-D}. \tag{7.22}$$

This holds for each field X in Lemma 5.10(a); for Green blocks, normalized block traces, $W^{k-1}\mathcal{L}_k$ and $W^{k-1}(\mathcal{L}_k - \mathcal{K}_k)$, it holds under the assumptions of Lemma 5.8 with $g \ge C_\kappa$ sufficiently large. The exponent is independent of g, L. No energy or coupling union is used. □

Corollary 7.8 (Weighted spatial sampling)**.** *Fix $D, D', F, c_1 > 0$ and $0 < \theta \le 1/2$. Under the assumptions of Lemma 5.8, with $K \ge 2$ and $g \ge C_\kappa$ sufficiently large, take ℓ_t from (5.18) and set*

$$w_t(r) = \exp[-c_1 (r/\ell_t)^\theta] + W^{-F}.$$

For B sufficiently large depending on these fixed parameters, the deterministic sample sets $\mathcal{S}_k$ from that lemma have polynomial cardinality and are independent of time and signs. With probability at least $1 - W^{-D'}$,

$$\max_{a,b\in\mathbb{Z}_L} \frac{|D_2(t;a,b,\boldsymbol{\sigma})|}{w_t(d(a,b))} \le 2d_2 \max_{(a,b)\in\mathcal{S}_2} \frac{|D_2(t;a,b,\boldsymbol{\sigma})|}{w_t(d(a,b))} + W^{-D}, \tag{7.23}$$

simultaneously for $0 \le t \le 1 - W^{-3}$ and all sign pairs $\boldsymbol{\sigma}$, with D_2 as in (5.26). For $1 \le k \le K$, the same bound holds for D_k or $\mathcal{L}_k$ on order-k tuples, with d_k, $\mathcal{S}_k$ and the weight $w_t(\mathrm{diam}(\boldsymbol{a}))$, simultaneously for all sign words. Sample tuples move any diameter by at most $2n$, so joint-tail conditions transfer with this explicit enlargement of their boundary. Together with (7.22), any fixed polynomial accuracy in such time-dependent suprema is determined by a polynomial-size time net and ordinary sample set.

Proof. If $g^2 < W^B$, all cells are singletons and spatial sampling is exact. Otherwise, within each product of cells, (6.7) gives a weight ratio at most $\exp(C(n/\ell_t)^\theta) \le 2$ for large W, since each distance or diameter changes by at most $2n$. Apply Lemma 5.8 with its exponent at least $\max\{D + F, D'\} + 1$, using Lemma 7.6 on each product of cells. Dividing the remainder by the floor W^{-F} leaves an error at most W^{-D}, proving (7.23) and its higher-order version with failure probability at most $W^{-D'-1}$. Direct differentiation of w_u, using $w_u \ge W^{-F}$ and $1 - u \ge W^{-3}$, gives polynomial time moduli for w_u and w_u^{-1}, uniformly in g, L. Use (7.22) with failure exponent $D' + 1$ and combine the two events. This proves the last assertion on a mesh W^{-M} with $M/4$ larger than the fixed accuracy and modulus exponents. □

Remark 7.9 (Trace and entry envelopes). The scalar traces $\langle \widetilde{G} \rangle_{ab}$ obey the Green-block interpolation of Proposition 7.4 directly: the particular scalar term cancels against the same particular term in the deterministic M_B recurrence. Thus their supremum reduces to polynomially many pairs formed from the first blocks of the cells and their predecessors, plus an arbitrarily small remainder. The same statement holds for individual orbital entries after subtracting the scalar diagonal part, with an additional polynomial union over orbital indices. This observation concerns norming and continuity only. It does not replace the residual or fluctuation-averaging estimate at a fixed pair.

7.4.2 Residual sampling and continuity in spectral height

We prove residual sampling in Proposition 7.10 and the spectral-height derivative bound (7.25). We combine these results with Corollary 7.8 and the preceding trace and entry envelopes to prove Lemma 5.10(b).

Proposition 7.10 (Compression of the averaged Dyson residual)**.** *Define the exact averaged residual*

$$R_{ab} = -\langle \widetilde{G} \rangle_{ab} + \frac{t}{g^2} \sum_c M_{B,ac} \langle \widetilde{G} \rangle_{cc} \langle G \rangle_{cb}.$$

Under the assumptions of Lemma 5.8, with $g \geq C_\kappa$ sufficiently large, for every prescribed $D, D' > 0$ the cell exponent B can be chosen so that, with probability at least $1 - W^{-D'}$, simultaneously for every microscopic pair and $0 \leq t \leq 1 - W^{-3}$,

$$R_{ab} = \sum_{\alpha,\beta} c_{a\alpha} c_{b\beta} R_{\alpha\beta} + O(W^{-D}).$$

The coefficients are those of Proposition 7.4: for $a \in I_j$ and $b \in I_l$, the sums use $\alpha \in \{b_j, b_j - 1\}$ and $\beta \in \{b_l, b_l - 1\}$, where b_j and b_l are the first blocks of the respective cells. Consequently a pointwise residual estimate on polynomially many pairs of these blocks yields its full microscopic supremum. The assertion requires only a crude polynomial Green envelope and does not assume the small-branch local law which the residual is later used to prove.

Proof. If $g^2 < W^B$, retain every block and take $c_{a\alpha} = \mathbf{1}_{a=\alpha}$; the assertion is then exact and $L \leq W^{C+B}$. Otherwise, on the prefix event, Proposition 7.4 and the coarse time envelope give $\max_{a,b,t} \|G_{ab}(t)\| \leq W^{C''}$ for a fixed exponent C'' independent of B, g, L. Thus $|\langle \widetilde{G} \rangle_{cc}| + |\langle G \rangle_{cb}| \leq C'' W^{C''}$. Use (6.15) and apply (6.17) on the cell containing a. The hypotheses of Lemma 6.5 follow from those of Lemma 5.8. Use second-coordinate interpolation on the scalar trace, with $c_{b\beta}$ and q_{bc} defined by the cell containing b:

$$\langle G \rangle_{cb} = \sum_\beta c_{b\beta} \langle G \rangle_{c\beta} + q_{bc} + e^{\mathrm{tr}}_{cb}, \qquad |e^{\mathrm{tr}}_{cb}| \leq C'' W^{-B/4} (1 + W^{C''}).$$

Again q_{bc} is bounded and supported on at most n values of c. Inserting these two formulas in $K_{ab} = tg^{-2} \sum_c M_{B,ac} \langle \widetilde{G} \rangle_{cc} \langle G \rangle_{cb}$ gives its homogeneous part $\sum_{\alpha,\beta} c_{a\alpha} c_{b\beta} K_{\alpha\beta}$. Terms containing a particular solution cost at most $C''(n/g^2) W^{2C''}$. Terms containing e^{tr} are bounded with the absolute row of M_B, and cost at most $C'' W^{-B/4} W^{2C''}$. Terms containing e^M may be bounded crudely with $L/g^2 \leq W^C$, and cost at most $C''(n/g^2) W^{C+2C''}$. Products of errors or particular terms satisfy the same larger bound. Hence

$$\left| K_{ab} - \sum_{\alpha,\beta} c_{a\alpha} c_{b\beta} K_{\alpha\beta} \right| \leq C'' W^{C+2C''} (W^{-B/4} + n/g^2).$$

Applying (6.17) in both coordinates gives the same particular term d_{ab} as in (7.10), so these terms cancel in $M_{B,ab} - \langle G \rangle_{ab}$. Their remaining error is at most $C'' W^{-B/4}(1 + W^{C''}) + C'' n/g^2$. Choosing B after C, C'', D, D' proves the proposition. As in the proof of Proposition 7.4, the exponent C'' of the crude envelope can be fixed before B; the moment order is chosen after the polynomial number of cells. □

Proof of Lemma 5.10(b) and the comparison-height estimates. **Proof of (5.34).** At bounded coupling, or at large coupling with $g^2 < W^B$, the volume is at most CW^{B+2}. Retain every block tuple, so spatial norming is an identity. For the remaining range $g^2 \geq W^B$, Lemma 5.8 treats $\mathcal{L}_k, D_k$; their weighted versions follow from (7.23) and its higher-order form in Corollary 7.8, which also gives the diameter enlargement. The Green-block interpolation of Proposition 7.4 also

applies to normalized traces and orbital entries. The particular scalar terms in the random and deterministic recurrences agree and cancel in $\langle \widetilde{G} \rangle$, and those in an orbital entry cancel against its block trace. Thus the same polynomial collection of pairs formed from cell first blocks and their predecessors controls both centered fields. Proposition 7.10 proves the residual case. For a weight with floor W^{-F}, choose the unified exponent in Lemma 5.8 at least $\max\{D+F, D'\}$, and increase it further by the fixed normalization and mesh costs. This gives precisely (5.34).

For vertical comparison in the bounded-coupling range, $y \geq \eta_u \geq cW^{-3}$ gives $\|(H_u - z)^{-1}\| \leq CW^3$, and retaining all block tuples gives exact norming in polynomial volume. In the arbitrary-coupling range, fix an allowed deterministic u, z. The real part remains in the buffered bulk, and $W^{-3} \lesssim \Im z/\gamma_g \lesssim 1$. Moreover, $|z - E| \leq Cg^{-2}$ and $|z + u\gamma_g m_u(z) - E| \leq Cg^{-2}$, by (6.9) and (6.12). The comparison-point clause of Proposition 7.4 supplies the random Green interpolation at (u, z). The comparison-matrix clause of Lemma 6.5 uses the same real free energy E and the same scalar source. Equation (6.12) bounds the γ_g-weighted row sums of $M_{B,u}(z)$ needed for interpolation. The coefficient in the residual identity is $u\gamma_g$. The proof of Proposition 7.10 consequently applies with $M_{B,u}(z)$. For a coarse Green envelope W^{C_G}, terms retaining $M_{B,u}(z)$ use its bounded γ_g-weighted absolute row. Particular-source terms have support on at most n labels and cost $C(n/g^2)W^{2C_G}$. The comparison-matrix clause of Lemma 6.5, applied on the cell containing a, gives $|e^M_{ac}| \leq Cn/g^2$, so terms where this error replaces M cost at most

$$C(n/g^2)(L/g^2)W^{2C_G} \leq C(n/g^2)W^{2+2C_G}.$$

Choose the cell exponent B after the fixed coarse-envelope and accuracy exponents. If $g^2 < W^B$, retain every block: norming is exact and $L \leq CW^{B+2}$. Otherwise choose $n \leq g^2 W^{-B}$ and apply the displayed error budgets. The scalar particular terms in $M_{B,u}(z)_{ab} - \langle G \rangle_{ab}$ cancel as before. Lemma 5.5 supplies the fixed-height block moments. Minkowski and Hölder in the exact residual identity give, with a fixed C_0 independent of p, g, L,

$$\begin{aligned}\|R_{ab}(u,z)\|_{L^p} &\leq \|\langle G - M_u(z)\rangle_{ab}\|_{L^p} + u\gamma_g \sum_c |M_{B,u}(z)_{ac}| \|\langle G - M_u(z)\rangle_{cc}\|_{L^{2p}} \|\langle G\rangle_{cb}\|_{L^{2p}} \\ &\leq C_p W^{2C_0}.\end{aligned}$$

This proves fixed-height norming and coarse moments without a small residual estimate.

Interpolation on comparison-height segments. We prove (7.25) to extend residual estimates from discrete height meshes to full vertical comparison segments at fixed times, before applying Dyson stability. Fix any polynomial collection of allowed deterministic times and their comparison segments. For each such time u, write $z = x + i\gamma_g r$ and $G_u(z) = (H_u - z)^{-1}$, with u, x, g fixed and $cW^{-3} \leq r \leq C$. Thus r is the normalized spectral height. Since $\partial_r G_u(z) = i\gamma_g G_u(z)^2$, the two block Ward identities and Cauchy–Schwarz give

$$\|\partial_r G_u(z)_{ab}\|_{\mathrm{op}} \leq r^{-1} \sqrt{\|\Im G_u(z)_{aa}\|_{\mathrm{op}} \|\Im G_u(z)_{bb}\|_{\mathrm{op}}}. \tag{7.24}$$

Choose a fixed coarse envelope exponent $C_G > C_0 + 2$ before choosing any sampling accuracy or moment order. At deterministic height mesh points the preceding fixed-height moments, trace and traceless-entry samples, and the union over W^2 orbital indices give $\max_{a,b} \|G_u(z)_{ab}\|_{\mathrm{op}} \leq W^{C_G}$ with arbitrarily high probability. Here entries are reconstructed by adding the centered trace and the bounded Dyson entry, and $\|A\|_{\mathrm{op}} \leq W \max_{i,j} |A_{ij}|$ accounts for the extra factor W. The moment order is chosen after the polynomial number of samples and mesh points, so C_G is independent of their number. By (7.24), while the envelope is at most $2W^{C_G}$ every block derivative is at most CW^{C_G+3}. A mesh of spacing $o(W^{-3})$ therefore excludes a first exit from $2W^{C_G}$ and gives this derivative bound throughout each segment.

In the notation of Lemma 5.9, the residual identity is

$$R_{ab}(u,z) = -\langle \widetilde{G}_u(z)\rangle_{ab} + u\gamma_g \sum_c M_{B,u}(z)_{ac} \langle \widetilde{G}_u(z)\rangle_{cc} \langle G_u(z)\rangle_{cb}.$$

Differentiating at fixed u and applying the covariance-weighted row bounds (6.12) and (6.14) for $M_{B,u}(z)$ and its derivative, together with the preceding Green-block and derivative bounds, gives $|\partial_r R_{ab}(u,z)| \le CW^{2C_G+3}$, with no extra factor L or g. Consequently, for a fixed $C_H > 2C_G + 4$ and arbitrary $D' > 0$,

$$\mathbb{P}\Big(\max_{u} \sup_{r} \max_{a,b} [\|G_u(z)_{ab}\|_{\mathrm{op}} + \|\partial_r G_u(z)_{ab}\|_{\mathrm{op}} + |\partial_r M_{B,u}(z)_{ab}| + |\partial_r R_{ab}(u,z)|] > W^{C_H} \Big) \le W^{-D'}. \tag{7.25}$$

The maximum over u is over the prescribed deterministic collection; each supremum is over its comparison segment. Given $D > 0$, choose a height mesh of spacing W^{-q_h} with $q_h > C_H + D + 1$, and then choose the sampling and pointwise moment failure exponents after the mesh cardinality. Outside the exceptional event in (7.25), $G_u(z), M_u(z), R(u,z)$ differ from their nearest mesh values by at most W^{-D}. This is a height assertion at fixed times; it does not use or assert an off-characteristic time increment estimate.

For the time suprema, use (7.22) and a mesh of spacing W^{-M}, with $M/4$ exceeding the fixed modulus, weight and accuracy exponents. All assertions are uniform in fixed deterministic parameters; no energy or coupling union and no stopped Gaussian integration by parts is used. □

7.5 Fluctuation averaging and uniform entrywise estimates

Proof of Lemma 5.9. Fix deterministic u, z in either parameter range of the lemma, including its permitted comparison points. All moments below use the original Gaussian law.

(i) and (iii): residual estimates. Set $Z_{ab} := G_{ab} - \langle G \rangle_{ab} I_W$. For each fixed even integer $p \ge 2$, fixed $C > 0$, and deterministic K, χ as in part (i), we use the moment implication

$$\left.\begin{aligned} \max_{a,b} \|\langle G \rangle_{ab}\|_{L^{2p}} &\le CK, \\ \max_{a,b,i,j} \|(Z_{ab})_{ij}\|_{L^{2p}} &\le CK\chi \end{aligned}\right\} \quad \Longrightarrow \quad \max_{a,b} \mathbb{E}|R_{ab}|^p \le C_p (K^2\chi^2)^p. \tag{7.26}$$

The constant C_p depends only on p, C and the fixed regularity margins, uniformly in the allowed parameters. Appendix C.1 proves (7.26) and (5.31) by the same Gaussian expansion, using the covariance-weighted Dyson rows (6.13). This establishes part (iii).

To deduce part (i), fix p and $\varepsilon > 0$. The stochastic-domination hypotheses, together with Hölder and the higher coarse moments (5.33) on their exceptional events, give the unconditional moment assumptions in (7.26) with K replaced by $W^{\varepsilon}K$. The failure exponent is chosen after the required moment order and accuracy. Apply (7.26) and Markov with sufficiently large fixed p at the polynomial residual sample set from (5.34); its norming inequality gives (5.29). In polynomial volume this set can be all pairs. No sampling or stopping event is inserted into Gaussian integration by parts.

(ii): entrywise estimate. For an independent Haar unitary $U \in U(W)$, common conjugation preserves the deterministic hopping and the joint law of the GUE blocks. Thus

$$(G_{ab})_{a,b} \overset{d}{=} (UG_{ab}U^*)_{a,b}, \qquad \operatorname{Tr}(UG_{ab}U^*) = \operatorname{Tr} G_{ab}.$$

For every fixed integer $p \ge 1$, deterministic $A \in \mathbb{C}^{W \times W}$ and $1 \le i, j \le W$, Haar integration gives

$$\mathbb{E}_U \left| (UAU^*)_{ij} - \delta_{ij} W^{-1} \operatorname{Tr} A \right|^{2p} \le CW^{-2p} \|A\|_{\mathrm{HS}}^{2p}. \tag{7.27}$$

The constant depends only on p; the proof is in Appendix C.1. Apply (7.27) conditionally on G_{ab}, then use Markov and a union bound over i, j to obtain (5.30). The same auxiliary U acts on all blocks; independence between resolvent blocks is not required. □

We next combine (7.27) with the sampling and time increments of Section 7.4. Fix a characteristic and a deterministic interval $I \subset [0, 1 - W^{-3}]$. Let Ψ_u be a positive deterministic scale satisfying $W^{-C_*} \le \Psi_u \le W^{C_*}$ and $|\partial_u \log \Psi_u| \le W^{C_*}$ almost everywhere, with Ψ absolutely continuous. If the fixed-pair Hilbert–Schmidt hypothesis holds uniformly in deterministic u, a, b, then

$$\Big[W^{-2}\|(G_u)_{ab}\|_{\mathrm{HS}}^2 \prec \Psi_u^2 \text{ uniformly in } u, a, b\Big] \quad \Longrightarrow \quad \sup_{u\in I} \max_{ai,bj} \frac{|(G_u)_{ai,bj} - \delta_{ij}\langle G_u\rangle_{ab}|}{\Psi_u} \prec 1. \tag{7.28}$$

Here and in this implication the deterministic characteristic parameters are fixed, and the constants are uniform over their allowed range.

Proof of (7.28). Fix $\epsilon, D, Q > 0$ and put $Z_{ai,bj}(u) = (G_u)_{ai,bj} - \delta_{ij}\langle G_u\rangle_{ab}$. The conditional Haar bound (7.27) and removal of exceptional events with (5.33) give

$$\mathbb{E}|Z_{ai,bj}(u)/\Psi_u|^{2p} \le C_{p,\delta} W^{2p\delta}$$

for every fixed p and $\delta > 0$. Choose deterministic traceless-entry samples from (5.34) with failure exponent $D+3$ and accuracy $Q' > C_* + Q + 2$. Their cardinality is at most $W^{C_{\mathrm{S}}}$ for each orbital pair, with C_{S} now fixed. The coarse time increments and the scale assumptions give a fixed C_Z, independent of the moment order, such that

$$\|Z_{ai,bj}(v)/\Psi_v - Z_{ai,bj}(u)/\Psi_u\|_{L^p} \le C_p W^{C_Z}|v-u|^{1/2}.$$

Indeed, split the difference into increments of Z and Ψ^{-1}, and use $|\partial_u \Psi_u^{-1}| \le W^{2C_*}$. For a mesh of spacing $h = W^{-q}$, Markov at the mesh points and within-interval dyadic subdivision (with $p > 2$) bound the probabilities of a sampled normalized value exceeding W^{ϵ} and of a normalized increment exceeding W^{-Q}, respectively, by

$$C_{p,\delta} W^{C_{\mathrm{S}}+q+2-2p(\epsilon-\delta)}, \qquad C_p W^{C_{\mathrm{S}}+q+2+p(C_Z+Q-q/2)}.$$

Here the exponent 2 accounts for the orbital pairs, and the dyadic maximum has L^p norm at most $C_p W^{C_Z} h^{1/2}$. First choose $q > 2(C_Z + Q + 1)$, then $\delta < \epsilon$, and then the moment orders so that both probabilities are at most W^{-D-1}. The sample failure probability is at most W^{-D-1} after the orbital union; the additive norming error divided by Ψ_u is $O(W^{-Q-2})$. Spatial norming proves (7.28). All moments use the original Gaussian law, and interpolation applies only to the actual entry field. □

7.6 Initial loop estimates at large coupling

We prove Lemma 5.6 on $0 \le t \le 1/2$, using the spatial decay (7.9) and the residual and Haar estimates of Lemma 5.9. The spatial sums below use (6.6), uniformly even when g, L are superpolynomial. Choose each tail-floor exponent after the fixed polynomial costs and required accuracy.

Proof of Lemma 5.6. Fix g, L, E as in Lemma 5.6. Use the flow $H_t(g)$ and characteristic $z_t = z_t(g, E)$ from (5.1) and (5.3), with all Dyson quantities and scales evaluated at (g, E). The estimates are uniform in these deterministic parameters, with spatial maxima and time suprema inside probability. The scale definition (5.18) gives $\ell_t \asymp g^2$ and $\zeta_t \asymp W^{-1}$ on $0 \le t \le 1/2$; hence (5.23) requires actual loops of size W^{1-k}, errors of size W^{-k}, and the pair profile and joint tails stated in (5.22). For the block operator bound, at every deterministic time $t \le 1/2$, (5.25) applies at z_t, since $g^2 \Im z_t \asymp 1$ and its real part stays in the buffered bulk. Thus $\mathbb{E}\|G_{ab}(t)\|^p \le C_p$ for every fixed p. For sufficiently large W, we have $4 \le L/g^2 \le CW^2 \le W^3$, so the spatial sampling estimates apply with volume exponent three on $0 \le t \le 1/2$. Enlarge the fixed constant $C \ge 3$ to cover the volume and regularity bounds, and take $\mathcal{P} = \mathcal{P}_B(g, L, W)$ from Definition 5.7.

Let b_j be the first block of each cell $I_j \in \mathcal{P}$. If $g^2 < W^B$, the cells are singletons and the b_j run over all blocks; otherwise Proposition 7.4 applies. For the polynomial collection of pairs with $\alpha \in \{b_i, b_i - 1\}$ and $\beta \in \{b_j, b_j - 1\}$, (7.22) gives a simultaneous 1/4-Hölder seminorm at most $W^{C''}$, with arbitrary polynomial probability. Choose a time mesh of spacing W^{-M} with $M > 4(C'' + 1)$. For each $\epsilon, D > 0$, Markov's inequality at the mesh points gives

$$\mathbb{P}\left(\max_{t \text{ on the mesh}} \max_{I_i, I_j \in \mathcal{P}} \max_{\alpha \in \{b_i, b_i - 1\}} \max_{\beta \in \{b_j, b_j - 1\}} \|G_{\alpha\beta}(t)\| > W^{\epsilon} \right) \le C_p W^{2(C+B)+M-p\epsilon} \le W^{-D},$$

by choosing p after the cell and mesh exponents. Hölder continuity then gives $\sup_{t \le 1/2} \mathcal{G}_t(\mathcal{P}) \prec 1$. Finally (7.10) has four homogeneous terms and a bounded particular term (and no interpolation is needed for singleton cells), so

$$\sup_{t \le 1/2} \max_{a,b} \|G_{ab}(t)\| \le C''(1 + \sup_{t \le 1/2} \mathcal{G}_t(\mathcal{P})) \prec 1.$$

If the cells are singletons, the same mesh argument already includes all blocks. No interpolation of a matrix norm by ordinary loops is used. The operator bound gives $W^{-2}\|G_{ab}(t)\|_{\mathrm{HS}}^2 \le W^{-1}\|G_{ab}(t)\|_{\mathrm{op}}^2 \prec W^{-1}$. Apply (7.28) with the constant scale $\Psi_t = W^{-1/2}$ on $[0, 1/2]$ to obtain the simultaneous traceless-entry bound. The residual estimate (5.29) gives $R \prec W^{-1}$. The scalar equations and stable inverse imply, for every fixed sufficiently small $\nu > 0$, $U \le CW^{-1+\nu} + CW^{\nu}U^2$ with arbitrarily high probability, where $U = \max_a |\langle \widetilde{G} \rangle_{aa}|$. At time zero $G = M$; continuity selects the small branch, giving $\max_{a,b} |\langle \widetilde{G} \rangle_{ab}| \prec W^{-1}$. Residual compression and the polynomial time modulus justify this simultaneous continuity statement without a microscopic union.

For a word with one freely summed label c, the two incident Green edges, Cauchy–Schwarz, and exact Ward give absolute row $\prec g^2 W^{1-k}$. In fact the highest-length linear coefficient uses a pair row bounded by Cg^2/W with a fixed constant, because $\max_a \Im\langle G \rangle_{aa}$ is now bounded. For primitives, including $k = 2$, (7.20) at coupling g and $t \le 1/2$ gives the deterministic row bound $C_k g^2 W^{1-k}$. The assumption $L/(g^2 \log W) \to \infty$ ensures that the primitive size and tail bounds of Lemma 5.2(b) apply on this time interval. Subtract the primitive equation as $\mathcal{L}_{k_L} D_{k_R} + D_{k_L} \mathcal{K}_{k_R}$, where k_L, k_R are the child lengths and $k_L + k_R = k + 2$. The interior cuts in (5.17), the light insertion (5.7), and the bracket (5.9) have respective budgets

$$\frac{W}{g^2}(g^2 W^{1-k_L}) W^{-k_R} = W^{-k}, \qquad \frac{W}{g^2} W^{-1}(g^2 W^{-k}) \quad = W^{-k}, \qquad \frac{W}{g^2}(g^2 W^{-2k-1}) = W^{-2k}.$$

The first uses the shorter-error induction, the second the centered one-loop bound, and the third only raw loops of length $2k + 2$. Predictably stop only on already established crude and shorter-word envelopes. BDG at the polynomial ordinary sample tuples and the norming theorem control the full supremum of D_k. Pathwise Gronwall has a fixed coefficient C, so induction gives $D_k \prec W^{-k}$. No interpolation property of the martingale itself is assumed.

The exponential moments (7.9), spatial sampling and (7.22) give uniform microscopic tails with arbitrary polynomial floors on $0 \le t \le 1/2$. In particular, the raw pair bound $W^{-1} e^{-cd(a,b)/g^2}$ and the sharp centered pair bound W^{-2} combine into $W^{-2+o(1)} e^{-c\sqrt{d(a,b)/g^2}}$, with any prescribed polynomial floor; their crossover is at $d(a,b)/g^2 = O(\log W)$. The tail bound has no prefactor $\eta^{-1} \asymp g^2$, while the sharp centered bound comes from the residual estimate and early-time loop induction above. A long edge controls every longer word, and (5.21) supplies the primitive tails. Together with Lemma 5.2(b), these estimates prove (5.23). □

The residual and initial estimates proved in Sections 7.5 and 7.6 supply the corresponding inputs to Theorem 5.12(ii), whose five-step proof and finite time iteration are in Appendices C.3 and C.4.

8 Transfer estimates and eigenvector concentration

We first convert fractional-moment decay of the resolvent to concentration, then obtain the required resolvent estimates from growth of D_n. The growth estimates are proved through contraction of $P_{1/2}$ or directly through the corrected scalar increments of Section 11, according to the coupling and energy.

8.1 Reduction to fractional-moment bounds

Following the fractional-moment method of Aizenman–Molchanov [1], we give two sufficient resolvent inputs for (3.6): fixed-energy decay with polynomially bounded parameters, and energy-averaged decay with constants uniform in the coupling. We use the split at $g = W$ explained in Section 3.1. For the first route, the finite-volume criterion below reduces decay to contraction at one deterministic length.

For a finite path or cycle Λ, let H_Λ denote the corresponding Hamiltonian, with Dirichlet boundary conditions on a path, and let ι_x denote the inclusion of block x. Write $L = |\Lambda|$; $\|x - y\|_L$ denotes cyclic block distance on a cycle and distance along the path otherwise. Set

$$G_\Lambda(E) \coloneqq (H_\Lambda - E)^{-1}, \qquad G_\Lambda(x, y; E) \coloneqq \iota_x^* G_\Lambda(E) \iota_y. \tag{8.1}$$

The following conversion uses classical spectral averaging. On bounded energy sets, the argument of Aizenman–Schenker–Friedrich–Hundertmark already accepts energy-averaged fractional-moment bounds; see [3, Appendix A, proof of Theorem A.1, (A.10)–(A.12)]. For the energy-averaged input, we average over cells of about g^2 blocks. Transfer moments remain bounded on this scale, and normalization by the cell dimension removes the coupling loss in single-block spectral averaging. The proof retains the transfer-dependent weight to obtain constants uniform in g.

Lemma 8.1 (From fractional-moment bounds to concentration)**.** *Consider the model (1.3). Fix $E_0 > 0$, $q \in (0, 1)$ and $C, c > 0$, and set $J = [-E_0, E_0]$. Assume one of the following bounds.*
(i) Fixed-energy bounds. *For a fixed $C_0 \geq 1$, suppose $0 \leq g \leq W^{C_0}$, $1 \leq \mathfrak{b} \leq W^{C_0}$, and*

$$\sup_{E \in J} \mathbb{E}\|G_\Lambda(x, y; E)\|^q \leq W^C e^{-c\|x-y\|_L/\mathfrak{b}}. \tag{8.2}$$

The bound is uniform over finite Dirichlet paths or cycles Λ and blocks $x, y \in \Lambda$.
(ii) Energy-averaged bounds. *Suppose $E_0 < 2$, $g \geq W$, $\mathfrak{b} = Wg^2$, and*

$$\int_J \mathbb{E}\|G_\Lambda(x, y; E)\|^q \, dE \leq W^C e^{-c\|x-y\|_L/\mathfrak{b}}. \tag{8.3}$$

The bound is uniform over finite Dirichlet paths Λ and blocks $x, y \in \Lambda$.

Then, for every fixed $D > 0$ and all sufficiently large W, (3.6) holds on every finite cycle with a block radius

$$R \leq C_D \mathfrak{b} \log W. \tag{8.4}$$

In particular, the expected number of eigenvectors in J with $\ell > C_D W \mathfrak{b} \log W$, divided by WL, is at most W^{-D}. The output constants and width threshold are independent of $g, L, \mathfrak{b}$ within the stated ranges.

The proof is given in Appendix E.1.

Remark. For $g \leq c_* W^{-1/2}$, the required fractional-moment estimate is due to Peled–Schenker–Shamis–Sodin [24, Theorem 1]; its application is given at the end of this subsection. At coupling of order one, the coordinate scale W^2 was obtained by Drogin [10, Theorem 1.2 and Corollary 1.4] for GOE blocks and $g = 1$, in the form of Lyapunov gaps and asymptotic Green-function decay. Our small- and intermediate-coupling proofs use his quantitative nonconcentration criterion [10,

Proposition 2.6 and Claims 2.7–2.8]; the general lift adapts [10, Proposition 4.4]. We work with GUE blocks, retain finite-length and coupling-uniform estimates, and supply the small-coupling construction in Section 9.

For polynomially bounded coupling, the next criterion reduces fractional-moment decay to contraction at one deterministic length.

For a finite block interval $\Lambda = [a, b]$ and deterministic Hermitian $W \times W$ endpoint shifts L_0, R_0, define

$$H_\Lambda^{L_0,R_0} := H_\Lambda + \iota_a L_0 \iota_a^* + \iota_b R_0 \iota_b^*, \qquad G_\Lambda^{L_0,R_0}(E) := (H_\Lambda^{L_0,R_0} - E)^{-1}. \tag{8.5}$$

Write $G_\Lambda^{L_0,R_0}(x, y; E) := \iota_x^* G_\Lambda^{L_0,R_0}(E)\iota_y$.

The following criterion is a block version of the finite-volume iteration in [3, proof of Theorem 1.1, (2.23)–(2.24)].

Proposition 8.2 (Finite-volume criterion for fractional-moment decay)**.** *Take the model and $C_0, E_0, g, \mathfrak{b}$ as in Lemma 8.1*(i)*, with $g > 0$. Fix $s \in (0, 1/2)$ and $C_1 \geq 1$, and allow J to be any deterministic subinterval of $[-E_0, E_0]$. In place of (8.2), assume that one deterministic integer $1 \leq n \leq C_1 \mathfrak{b} \log W$ satisfies*

$$\sup_{E\in J} \sup_{L_0=L_0^*,\, R_0=R_0^*} \mathbb{E}\|G_{[0,n]}^{L_0,R_0}(0, n; E)\|^s \leq W^{-2}. \tag{8.6}$$

The endpoint shifts have no norm restriction; the expectation is over the independent GUE blocks in $[0, n]$, and the same n works for all E, L_0, R_0.

Then, for all sufficiently large W, (8.2) holds with $q = s/2$ on every finite Dirichlet path and cycle, with the same scale $\mathfrak{b}$. The output constants and width threshold are independent of $L, g, \mathfrak{b}, J$. When $J = [-E_0, E_0]$, Lemma 8.1(i) *converts this bound to (3.6).*

The proof is given in Appendix E.2.

We record the a priori bound used in that proof and in the transfer comparison below. For each fixed $0 < p < 1$, every finite block interval Λ and $x, y \in \Lambda$,

$$\sup_{E\in\mathbb{R}} \sup_{L_0=L_0^*,\, R_0=R_0^*} \mathbb{E}\|G_\Lambda^{L_0,R_0}(x, y; E)\|^p \leq C \max\{1, g\}^p W^{2p}. \tag{8.7}$$

The constant depends only on p; the same bound holds for unshifted resolvents on finite cycles. To obtain it, write $V_x = V_x^\circ + \xi_x I_W$, where $\xi_x := W^{-1} \operatorname{Tr} V_x \sim N(0, W^{-2})$ is independent of V_x°. For $x = y$, conditioning on the remaining variables gives the diagonal Schur complement $((\xi_x/\max\{1, g\})I_W - A)^{-1}$ with A Hermitian. For $x \neq y$, condition on the traceless parts, all other blocks, and $\xi_x - \xi_y$, leaving the common scalar $(\xi_x + \xi_y)/2$ in the two-block compression. After scaling by $\max\{1, g\}^{-1}$, the scalar density is bounded by $CW \max\{1, g\}$. A union over at most $2W$ eigenvalues therefore gives the tail $\min\{1, CW^2 \max\{1, g\}/h\}$ at level h. Integrating it proves (8.7), uniformly in the endpoint shifts and the volume, and also on cycles.

Proof of Theorem 3.1. For sufficiently small c_*, [24, Theorem 1], with W orbitals and exponent $1/2$, followed by summation over the W coordinate vectors, gives

$$\sup_{E\in\mathbb{R}} \mathbb{E}\|G_\Lambda(a, b; E)\|^{1/2} \leq W^C e^{-c|a-b|}$$

on every finite Dirichlet path, uniformly for $g \leq c_* W^{-1/2}$. On cycles, the Schur identity (E.15), (8.7) and Cauchy–Schwarz give the same decay with exponent $q = 1/4$; Jensen's inequality gives this exponent on paths as well. Thus (8.2) holds with $\mathfrak{b} = 1$.

Set $J = [-3, 3]$. Since the hopping has norm $2g \leq 1/2$ for large W, Weyl's inequalities and the one-block GUE norm tail give

$$\frac{1}{WL}\mathbb{E}\#\{k : \lambda_k \notin J\} \leq \frac{1}{L}\sum_a \mathbb{P}(\|V_a\| > 5/2) \leq Ce^{-cW},$$

uniformly in L, g. Applying Lemma 8.1(i) on J with exponent $D+1$ gives an exceptional proportion at most W^{-D-1}, with mass threshold W^{-D-1} and radius $C_{D+1}\log W$. For large W, this radius is at most $\lceil(\log W)^2\rceil$; adding the spectral-tail count proves (3.6) on $\mathbb{R}$, since $W^{-D-1}+Ce^{-cW}\le W^{-D}$. □

8.2 From transfer growth to fractional resolvent moments

The basic reduction is explained in Section 3. For the transfer-growth route with $g < W$, high-probability growth at one length gives the local bound (8.6). Proposition 8.2 then gives the fixed-energy input of Lemma 8.1(i). For $g \ge W$, negative-moment decay gives the energy-averaged input of part (ii) directly.

Definition 8.3 (Boundary frames and transfer amplitudes)**.** *Recall the fixed unitary matrix*

$$S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \otimes I_W$$

from Definition 7.1, and define

$$\mathcal{F}_W := \left\{ \Phi \in \mathbb{C}^{2W\times W} : \Phi^*\Phi = I_W, \quad \Phi^* \operatorname{diag}(I_W, -I_W)\Phi = 0 \right\}. \tag{8.8}$$

Take an i.i.d. sequence of GUE blocks with covariance (1.2), without cyclic identification. For $g > 0$ and a real energy E of the normalized model (1.3), use Definition 7.1 and abbreviate

$$T_i := T_i\left(\frac{\max\{1,g\}E}{g}\right). \tag{8.9}$$

For every deterministic $\Phi \in \mathcal{F}_W$ and $n \ge 1$, define

$$\mathcal{T}_n := S^*(T_n \cdots T_1)S, \qquad D_n(\Phi; E) := \sqrt{2}\begin{pmatrix} I_W & 0 \end{pmatrix}\mathcal{T}_n\Phi. \tag{8.10}$$

Proposition 8.4 (From transfer growth to fractional resolvent moments)**.** *Consider the model (1.3) and D_n defined in (8.10).*
(i) High-probability growth. *Fix $C_0 \ge 1$, $E_0 > 0$ and $t \in (0, 1/2)$, and let $g, \mathfrak{b}$ and the energy interval J be deterministic, with*

$$0 < g \le W^{C_0}, \qquad 1 \le \mathfrak{b} \le W^{C_0}, \qquad J \subset [-E_0, E_0].$$

For a sufficiently large fixed D', suppose that there are constants C, W_0, independent of $W, g, \mathfrak{b}, J$, such that for $W \ge W_0$ some

$$2\log W \le n_0 = n_0(W, g, \mathfrak{b}, J, D') \le C\mathfrak{b}\log W$$

satisfies

$$\sup_{E\in J}\sup_{\Phi\in\mathcal{F}_W} \mathbb{P}(\|D_{n_0}(\Phi; E)^{-1}\| > W^{-D'}) \le W^{-D'}. \tag{8.11}$$

The same deterministic length n_0 works for all displayed E, Φ, with both suprema outside the probability. Then (8.6) holds with $s = t$ and $n = n_0$, with C_1 in Proposition 8.2 chosen to cover the length constant after D' is fixed. This bound is uniform in $W, g, \mathfrak{b}, J$ and the Hermitian endpoint shifts; n_0 is independent of E, L_0, R_0.
(ii) Negative-moment decay. *Take $E_0, g, \mathfrak{b}, J$ as in Lemma 8.1(ii). Suppose that, for fixed $s \in (0, 1)$ and $C, c > 0$,*

$$\sup_{E\in J}\sup_{\Phi\in\mathcal{F}_W} \mathbb{E}\|D_n(\Phi; E)^{-1}\|^s \le CWe^{-cn/(Wg^2)}, \qquad n \ge 1, \tag{8.12}$$

uniformly in W, g, n. Then (8.3) holds with $q = s/4$, uniformly on finite Dirichlet paths. The output constants and width threshold are independent of g and the path length.

Proof of Proposition 8.4(i). Fix $0 < t < 1/2$ and select the input exponent D' below. For $G = G^{L_0,R_0}_{[0,n_0]}(E)$ and $u \in \mathbb{C}^W$, set $f_i := G(i,0)u$, $0 \le i \le n_0$. The right boundary condition is $f_{n_0+1} = \max\{1,g\}R_0 f_{n_0}/g$. Choose a deterministic $\Psi = \Psi(g,R_0) \in \mathcal{F}_W$ so that $S\Psi$ is an orthonormal frame of the boundary subspace

$$\left\{ \begin{pmatrix} v \\ \max\{1,g\}R_0 v/g \end{pmatrix} : v \in \mathbb{C}^W \right\}.$$

This choice depends only on g, R_0, not on the onsite blocks or L_0. Direct multiplication gives

$$(S^*T_iS)^* \operatorname{diag}(I_W, -I_W)(S^*T_iS) = \operatorname{diag}(I_W, -I_W).$$

Hence, for $\Phi \in \mathcal{F}_W$ and $D_n = D_n(\Phi;E)$, there is a unitary U_n such that

$$\mathcal{T}_n\Phi = 2^{-1/2}\begin{pmatrix} D_n \\ U_nD_n \end{pmatrix}, \qquad \|(T_n\cdots T_1)S\Phi v\|_2 = \|D_nv\|_2. \tag{8.13}$$

Thus $\|D_n^{-1}\|$ is the reciprocal of the least expansion on $S\operatorname{ran}\Phi$. Applying the boundary comparison of [23, Claim 2.2(1) and proof of Claim 2.4] with the fixed right boundary frame Ψ and the reversed product $T_1\cdots T_{n_0}$ gives

$$\|G(n_0,0)\| \le \|D_{n_0}(\Psi;E)^{-1}\|(\|G(0,0)\| + \|G(1,0)\|). \tag{8.14}$$

Here $D_{n_0}(\Psi;E)$ is formed with the reversed product, which has the same law as $T_{n_0}\cdots T_1$ for this deterministic frame. The boundary-pair norm dominates $\|f_{n_0}\|_2$, so no $\|R_0\|$ factor occurs; L_0 affects only the source equation. On $\{\|D_{n_0}(\Psi;E)^{-1}\| \le W^{-D'}\}$ use this comparison. The complement has probability at most $W^{-D'}$ by (8.11); there, use Cauchy–Schwarz and (8.7) with exponent $2t < 1$. Thus

$$\mathbb{E}\|G(n_0,0)\|^t \le C\max\{1,g\}^t W^{2t}(W^{-D't} + W^{-D'/2}).$$

Taking D' sufficiently large in terms of t, C_0 makes the right-hand side at most W^{-2}. Adjunction gives (8.6) with $s = t$ and $n = n_0$. The same deterministic n_0 works for every energy and both endpoint shifts. □

Proof of Proposition 8.4(ii). On a finite Dirichlet path Λ, take $x < y$ and $n = y - x$. Applying (8.14) to the Schur complement onto $[x,y]$ gives

$$\|G_\Lambda(y,x;E)\| \le Z(E)(\|G_\Lambda(x,x;E)\| + \|G_\Lambda(x+1,x;E)\|). \tag{8.15}$$

Here $Z(E) := \|D_n(\Phi;E)^{-1}\|$ is formed from the reversed segment product $T_{x+1}\cdots T_y$ and the boundary frame $\Phi \in \mathcal{F}_W$ determined by the right exterior. Conditioning on the right exterior at each fixed energy, (8.12) and reversal of the independent blocks give $\mathbb{E}Z(E)^s \le CWe^{-cn/\mathfrak{b}}$.

To control the resolvent factor after Cauchy–Schwarz in energy, we use Boole's identity [25, (1.5)]. For each realization and vector $v \in \mathbb{C}^{W|\Lambda|}$, its spectral measure for H_Λ has mass $\|v\|^2$, so

$$\operatorname{Leb}\{E \in \mathbb{R} : |\langle v, G_\Lambda(E)v\rangle| > h\} = \frac{2\|v\|^2}{h}, \qquad h > 0.$$

Integrating the tail truncated by $|J|$ gives $\int_J |\langle v, G_\Lambda(E)v\rangle|^s\,dE \le C_s|J|^{1-s}\|v\|^{2s}$. Polarization, subadditivity for $s < 1$, and summation over the W^2 entries of a block therefore give

$$\int_J \|G_\Lambda(a,b;E)\|^s\,dE \le C_sW^2|J|^{1-s}, \qquad a,b \in \Lambda, \tag{8.16}$$

uniformly in the volume.

Take $p = s/2$. Cauchy–Schwarz on $dE\,d\mathbb{P}$, using (8.15) and (8.16), now gives

$$\int_J \mathbb{E}\,\|G_\Lambda(y,x;E)\|^p\,dE \le CW^{3/2}e^{-cn/(2\mathfrak{b})}.$$

Use adjunction for the other direction and (8.16) for the diagonal. Hölder on the finite measure $J \times \Omega$ lowers the exponent to $q = p/2 = s/4$, proving (8.3) on paths. □

8.3 From operator $P_{1/2}$ contraction to transfer growth

The reduction is explained in Section 3.2. We give the definitions, results and proofs needed for (8.11).

Definition 8.5 (Transfer batches and the operator $P_{1/2}$)**.** *Let $\Phi \in \mathcal{F}_W$ from (8.8), $Y = \operatorname{ran}\Phi$, and $U \subset Y$ with $\dim U = W - 1$. For an invertible matrix $\mathcal{T}$ and either subspace $Z = U, Y$, choose an orthonormal frame Φ_Z whose columns span Z and define*

$$\operatorname{vol}(\mathcal{T}|_Z) := \det(\Phi_Z^* \mathcal{T}^* \mathcal{T} \Phi_Z)^{1/2}, \qquad \operatorname{vol}(\mathcal{T}|_{\{0\}}) := 1.$$

Define the logarithmic volume ratio by

$$c(\mathcal{T}; U, Y) := \log \frac{\operatorname{vol}(\mathcal{T}|_Y)}{\operatorname{vol}(\mathcal{T}|_U)}. \tag{8.17}$$

Let q be a fixed positive integer, chosen separately in the lemmas below, and set

$$F_\ell := S^* T_{\ell q} \cdots T_{(\ell-1)q+1} S. \tag{8.18}$$

These batches are independent and identically distributed, with $F_1 = \mathcal{T}_q$ and $F_k \cdots F_1 = \mathcal{T}_{qk}$. For a bounded measurable function h of the pair (U, Y), define

$$(P_{1/2} h)(U, Y) := \mathbb{E}\big[e^{-c(\mathcal{T}_q; U, Y)/2} h(\mathcal{T}_q U, \mathcal{T}_q Y)\big]. \tag{8.19}$$

Write $\|h\|_\infty := \sup_{U,Y} |h(U, Y)|$.

Lemma 8.6 (From $\|P_{1/2}^k 1\|_\infty$ to the transfer input)**.** *Use Definition 8.5 with a fixed batch length q, under the model and parameter assumptions of Proposition 8.4(i), with the additional lower bound $g \geq W^{-C_0}$. For each deterministic $\Phi \in \mathcal{F}_W$, let $Y = \operatorname{ran}\Phi$ and let U_j be the span of its columns other than the jth. Use identity matrices for empty products; then $D_0 = \sqrt{2}[I_W \ 0]\Phi$ is unitary by (8.8). Thus*

$$\begin{aligned} (P_{1/2}^k 1)(U, Y) &= \mathbb{E} e^{-c(F_k \cdots F_1; U, Y)/2}, \\ \mathbb{E}\|D_{qk}(\Phi; E)^{-1}\|^{1/2} &\leq \sum_{j=1}^{W} (P_{1/2}^k 1)(U_j, Y), \qquad k \geq 0. \end{aligned} \tag{8.20}$$

Suppose that constants $c, C > 0$, uniform in the stated deterministic $g, \mathfrak{b}, E$, satisfy

$$\|P_{1/2}^k 1\|_\infty \leq W^C e^{-ck/\mathfrak{b}} \quad (k \geq 0). \tag{8.21}$$

For every fixed $D' > 0$, a sufficiently large fixed $C_{D'}$ gives

$$n_0 := q \left\lceil C_{D'} \mathfrak{b} \log W / q \right\rceil, \qquad 2 \log W \leq n_0 \leq 2 C_{D'} \mathfrak{b} \log W, \tag{8.22}$$

at which (8.11) holds for all sufficiently large W. The same deterministic n_0 works for all displayed energies and frames, with the suprema outside the probability.

Proof of Lemma 8.6. The Gram matrix of the image frame in (8.13) is $D_n^* D_n$. Its determinant and the jth principal minor are the squared volumes on Y and U_j. Their ratio, by the diagonal cofactor formula for $(D_n^* D_n)^{-1}$, gives (see the proof in Appendix F)

$$\|e_j^* D_n^{-1}\| = e^{-c(\mathcal{T}_n; U_j, Y)}, \quad \|D_n^{-1}\|^{1/2} \leq \sum_{j=1}^{W} e^{-c(\mathcal{T}_n; U_j, Y)/2}. \tag{8.23}$$

For transfer products $\mathcal{T}, \mathcal{T}'$, multiplicativity of restricted volumes gives

$$c(\mathcal{T}'\mathcal{T}; U, Y) = c(\mathcal{T}; U, Y) + c(\mathcal{T}'; \mathcal{T}U, \mathcal{T}Y).$$

This identity and successive conditioning on the independent batches give the first line of (8.20); (8.23) then gives the second line. The product order is $F_k \cdots F_1$: F_1 acts first.

At the multiple of q in (8.22), (8.20) and (8.21) give

$$\sup_{E\in J}\sup_{\Phi\in\mathcal{F}_W} \mathbb{E}\|D_{n_0}(\Phi;E)^{-1}\|^{1/2} \le W^{C+1-cC_{D'}/q}.$$

Markov's inequality therefore gives

$$\sup_{E\in J}\sup_{\Phi\in\mathcal{F}_W} \mathbb{P}(\|D_{n_0}(\Phi;E)^{-1}\| > W^{-D'}) \le W^{C+1+D'/2-cC_{D'}/q} \le W^{-D'}$$

for a sufficiently large fixed $C_{D'}$ and then large W. This is exactly (8.11). □

Definition 8.7 (Isotropic flags and the transfer group)**.** (i) *Set* $J_{\mathrm{sig}} := \mathrm{diag}(I_W, -I_W)$.

(ii) *A subspace* Y *is* J_{sig}-isotropic *if* $x^* J_{\mathrm{sig}} y = 0$ *for all* $x, y \in Y$.

(iii) *A* J_{sig}*-isotropic subspace* Y *is* maximal isotropic *if* $\dim Y = W$. *Thus the maximal isotropic subspaces are precisely* $Y = \operatorname{ran}\Phi$ *with* $\Phi \in \mathcal{F}_W$.

(iv) *A pair* (U, Y) *from Definition 8.5 is called an* isotropic flag.

(v) *Set*

$$\mathcal{G} := \{\mathcal{T} \in \mathbb{C}^{2W\times 2W} : \mathcal{T}^* J_{\mathrm{sig}} \mathcal{T} = J_{\mathrm{sig}},\ \det \mathcal{T} = 1\}. \tag{8.24}$$

The conjugated transfers $S^* T_i S$ belong to $\mathcal{G}$: they preserve J_{sig}, as shown in the proof of Proposition 8.4(i), and $\det T_i = 1$.

Definition 8.8 (Angular L^4 norm)**.** *For each* U *as in Definition 8.5, let* $\mathcal{C}_U$ *be the circle of its maximal isotropic extensions* Y. *If* e_+, e_- *are unit eigenvectors of* J_{sig} *with eigenvalues* $+1, -1$, *respectively, in* $(U + J_{\mathrm{sig}}U)^\perp$, *its normalized angular measure is*

$$Y_\theta = U \oplus \mathbb{C}(e_+ + e^{i\theta} e_-)/\sqrt{2}, \qquad dm_U = d\theta/(2\pi).$$

For a bounded measurable function h *of the pair* (U, Y), *define*

$$\|h\|_{\mathrm{ang},4} := \sup_U \|h(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)} = \sup_U \left(\int_{\mathcal{C}_U} |h(U,Y)|^4 \, dm_U(Y)\right)^{1/4}.$$

Definition (Simultaneous unitary invariance). A scalar function h on these pairs (U, Y) is called *simultaneously unitarily invariant* if

$$h(\mathrm{diag}(\widetilde{V}, \widetilde{V})U, \mathrm{diag}(\widetilde{V}, \widetilde{V})Y) = h(U, Y) \tag{8.25}$$

for every such flag (U, Y) and every $W \times W$ unitary matrix $\widetilde{V}$.

For the density decomposition below, let $d\mathcal{T}$ denote Haar measure on $\mathcal{G}$.

Lemma 8.9 (Good and bad components and their operator bounds)**.** *For the model (1.3), fix* $B, E_0 > 0$ *and an integer* $q \ge 1000$ *divisible by 1000. Let* μ *be the law of* $F_1 = \mathcal{T}_q$ *from Definition 8.5. Uniformly for deterministic* $W^{-B} \le g \le W^B$ *and* $|E| \le E_0$, μ *has, for all sufficiently large* W, *a decomposition with a weight* $\widetilde{\pi} \in [1/2, 1]$*:*

$$\begin{aligned} &\mu = \widetilde{\pi} f \, d\mathcal{T} + \mu_{\mathrm{bad}}, \qquad \mu_{\mathrm{bad}} \ge 0, \qquad \mu_{\mathrm{bad}}(\mathcal{G}) = 1 - \widetilde{\pi}, \\ &f \in C_c^\infty(\mathcal{G}), \qquad f \ge 0, \qquad \int f \, d\mathcal{T} = 1. \end{aligned} \tag{8.26}$$

The two measures on the right of (8.26) are the pushforwards of the input law weighted by χ_P and $1-\chi_P$, respectively, under the transfer-product map. Here χ_P is a smooth cutoff. The cutoff is independent of the incoming flag, and

$$\widetilde{\pi} \ge 1 - Ce^{-cW} \ge \tfrac{1}{2}, \qquad \sup_{\mathcal{T}\in\operatorname{supp} f} \max\{\|\mathcal{T}\|, \|\mathcal{T}^{-1}\|\} \le W^C, \\ \int \|\mathcal{T}\|^{1/2}\, d\mu_{\mathrm{bad}}(\mathcal{T}) \le \tfrac{1}{2}. \tag{8.27}$$

With

$$(G_{1/2}h)(U,Y) := \widetilde{\pi} \int e^{-c(\mathcal{T};U,Y)/2} h(\mathcal{T}U, \mathcal{T}Y) f(\mathcal{T})\, d\mathcal{T}, \\ (B_{1/2}h)(U,Y) := \int e^{-c(\mathcal{T};U,Y)/2} h(\mathcal{T}U, \mathcal{T}Y)\, d\mu_{\mathrm{bad}}(\mathcal{T}), \qquad P_{1/2} = G_{1/2} + B_{1/2}, \tag{8.28}$$

every bounded measurable h satisfies

$$\|G_{1/2}h\|_\infty \le W^C \sup_U \|h(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)}, \qquad \|B_{1/2}h\|_\infty \le \tfrac{1}{2}\|h\|_\infty. \tag{8.29}$$

All constants and width thresholds depend only on B, E_0, q.

The proof is given in Appendix G.5. It follows the product-differential and Gaussian integration-by-parts arguments of [10, Proposition 4.1, Lemma 4.2 and Section 4.3], with the GUE coercivity estimate of Appendix G.4 and the $G_{1/2}, B_{1/2}$ estimates of Appendix F.3.

Lemma 8.10 (From L^4 contraction to decay of $\|P_{1/2}^k 1\|_\infty$)**.** *Use the model and deterministic parameter range of Proposition 8.4(i), with $g \ge W^{-C_0}$, and suppose that Lemma 8.9 applies to the chosen fixed q. Assume that, for every bounded nonnegative measurable function h that is simultaneously unitarily invariant in the sense of (8.25), one has, uniformly in the stated deterministic parameters,*

$$\sup_U \|(P_{1/2}h)(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)} \le (1 - c/\mathfrak{b}) \sup_U \|h(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)}, \qquad 0 < c \le \tfrac{1}{4}. \tag{8.30}$$

Then (8.21) holds.

Proof of Lemma 8.10. Simultaneous conjugation $V_i \mapsto \widetilde{V} V_i \widetilde{V}^*$ preserves the joint GUE law, so (8.19) implies that $P_{1/2}^r 1$ satisfies (8.25) for every $r \ge 0$. Set $\rho := 1 - c/\mathfrak{b} \ge 3/4$. Applying (8.30) repeatedly to $P_{1/2}^r 1$ gives angular norm at most ρ^r. In the exact expansion

$$P_{1/2}^k = B_{1/2}^k + \sum_{j=0}^{k-1} B_{1/2}^j G_{1/2} P_{1/2}^{k-j-1}, \tag{8.31}$$

apply the angular bound to the rightmost factor, the smoothing bound once, and then the pointwise bound for $B_{1/2}$. After increasing C, this yields

$$\|P_{1/2}^k 1\|_\infty \le 2^{-k} + W^C \sum_{j=0}^{k-1} 2^{-j} \rho^{k-j-1} \le \left(1 + \frac{W^C}{\rho - 1/2}\right) \rho^k \le W^C e^{-ck/\mathfrak{b}}. \tag{8.32}$$

Only the rightmost powers applied to 1 need be simultaneously unitarily invariant in the sense of (8.25). □

8.4 A criterion for contraction of $P_{1/2}$

The criterion is explained in Section 3.3. For a field $X = (X_1, \dots, X_q)$ on $\mathrm{Herm}(W)^q$, each X_i is Hermitian and may depend on all input blocks $V = (V_1, \dots, V_q)$. For a smooth scalar function $\varphi : \mathrm{Herm}(W)^q \to \mathbb{R}$, write $D_X\varphi = \sum_i \langle \nabla_{V_i}\varphi, X_i\rangle$, with real Hilbert–Schmidt inner products. Since each real orthonormal Hermitian coordinate of V_i has variance W^{-1}, the Gaussian divergence is

$$\delta_{\mathrm{G}} X := W \sum_{i=1}^{q} \langle V_i, X_i\rangle - \operatorname{div} X, \qquad \mathbb{E} D_X\varphi = \mathbb{E}[\varphi\, \delta_{\mathrm{G}} X], \quad \varphi \in C_c^\infty(\mathrm{Herm}(W)^q). \tag{8.33}$$

Here $\operatorname{div} X$ is the ordinary Euclidean divergence in these real coordinates.

Lemma 8.11 (From Gaussian divergence to contraction)**.** *Work with the model (1.3). Fix $B_g, B_b, E_0 > 0$ and a positive integer q, and let $g, \mathfrak{b}, J$ be deterministic with*

$$W^{-B_g} \le g \le W^{B_g}, \qquad 1 \le \mathfrak{b} \le W^{B_b}, \qquad J \subset [-E_0, E_0].$$

Use Definition 8.5 with this q, writing $\mathcal{T}_q = \mathcal{T}_q(V; E)$. Suppose that for every deterministic $E \in J$ and incoming flag (U, Y) there exist a cutoff and a vector field

$$\chi : \mathrm{Herm}(W)^q \longrightarrow [0,1], \qquad X = (X_1, \dots, X_q) : \mathrm{Herm}(W)^q \longrightarrow \mathrm{Herm}(W)^q.$$

These are deterministic functions of V, allowed to depend on W, g, E, U, Y. Input derivatives and expectations are taken with these parameters fixed.

Require that, for every $V \in \operatorname{supp}\chi$, there exists a matrix $K_{\mathrm{out}} = K_{\mathrm{out}}(V; W, g, E, U, Y) \in \mathbb{C}^{W\times W}$ such that

$$D_X c(\mathcal{T}_q; U, Y) = 1,$$
$$D_X(\mathcal{T}_q U, \mathcal{T}_q Y) = \frac{d}{dt}\Big|_{t=0} (\mathrm{diag}(e^{tK_{\mathrm{out}}}, e^{tK_{\mathrm{out}}})\mathcal{T}_q U, \mathrm{diag}(e^{tK_{\mathrm{out}}}, e^{tK_{\mathrm{out}}})\mathcal{T}_q Y), \qquad K_{\mathrm{out}}^* = -K_{\mathrm{out}}. \tag{8.34}$$

In the right-hand derivative, $\mathcal{T}_q, U, Y$ and the value of K_{out} at the current input are held fixed. Assume that χX is smooth in its active input coordinates, extends by zero across the cutoff boundary, and is integrable together with the derivatives needed for Gaussian integration by parts. These regularity conditions may be checked after conditioning on coordinates whose field components vanish identically. With the original, unconditional Gaussian law, suppose that

$$\mathbb{E}\chi \ge \tfrac{1}{2}, \qquad \mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\mathfrak{b}, \tag{8.35}$$

with δ_{G} as in (8.33).

Then the contraction estimate (8.30) holds for every bounded nonnegative measurable function h that is simultaneously unitarily invariant in the sense of (8.25). If the fields can be chosen with $K_{\mathrm{out}} = 0$ throughout $\operatorname{supp}\chi$, the estimate holds for every bounded measurable h, without the invariance assumption. The contraction constant and width threshold are uniform over the deterministic parameters whenever the field bounds are uniform.

Proof of Lemma 8.11. We adapt the circle nonconcentration argument of [10, Proposition 2.6 and Claims 2.7–2.8], using Gaussian integration by parts in the input variables. By positivity and homogeneity, take a simultaneously unitarily invariant $h \ge 0$ in the sense of (8.25), with angular L^4 norm at most one. Initially take h strictly positive. Fix U and a positive probability density $\rho : \mathcal{C}_U \to (0, \infty)$. We test the angular L^4 norm against $\rho^{3/4}$, whose $L^{4/3}(dm_U)$ norm is one, and define

$$R_{\mathcal{T}_q}(Y) := \rho(Y)^{-1} e^{-2c(\mathcal{T}_q; U, Y)} h(\mathcal{T}_q U, \mathcal{T}_q Y)^4.$$

Keep $U, Y, \rho(Y)$ fixed and vary only the GUE blocks. Along the local flow of the smooth cut field χX, every smooth invariant function of the terminal flag has zero derivative by (8.34). Such functions separate the orbits of the compact unitary group, so the terminal flag stays in one orbit; no regularity of K_{out} is needed. Thus even a measurable invariant h is constant along this flow. Together with the unit cocycle derivative, this gives, in the flow sense,

$$D_{\chi X} \log R_{\mathcal{T}_q}(Y) = -2\chi.$$

Let φ_t be a primitive of $\mathbf{1}_{[-t,t]}$ with $\|\varphi_t\|_\infty \le t$, first smoothed if necessary. Changing variables along the local flow justifies Gaussian integration by parts for the bounded function $\varphi_t(\log R_{\mathcal{T}_q}(Y))$ using its flow derivative. Equation (8.33) gives

$$2\mathbb{E}[\chi \mathbf{1}_{\{|\log R_{\mathcal{T}_q}(Y)| \le t\}}] = -\mathbb{E}[\varphi_t(\log R_{\mathcal{T}_q}(Y))\, \delta_{\mathrm{G}}(\chi X)] \le t\, \mathbb{E}|\delta_{\mathrm{G}}(\chi X)| \le Ct\sqrt{\mathfrak{b}}.$$

Taking $t = c_0/\sqrt{\mathfrak{b}} \le 1$ with sufficiently small $c_0 > 0$ and using $\mathbb{E}\chi \ge 1/2$ yields, for every fixed Y,

$$\mathbb{P}\{|\log R_{\mathcal{T}_q}(Y)| > t\} \ge \tfrac{1}{4}.$$

The scalar deficit satisfies

$$\frac{3}{4} + \frac{r}{4} - r^{1/4} \ge 0, \qquad \inf_{|\log r| > t} \left(\frac{3}{4} + \frac{r}{4} - r^{1/4} \right) \ge ct^2 \qquad (0 < t \le 1).$$

For $\mathcal{T} \in \mathcal{G}$, the map $Y \mapsto \mathcal{T}Y$ has the exact angular Jacobian

$$\frac{dm_{\mathcal{T}U}(\mathcal{T}Y)}{dm_U(Y)} = e^{-2c(\mathcal{T};U,Y)}. \tag{8.36}$$

Its proof is given under **Proof of** (8.36) in Appendix F. Applying (8.36) with $\mathcal{T} = \mathcal{T}_q$ gives

$$\int_{\mathcal{C}_U} \rho(Y) R_{\mathcal{T}_q}(Y)\, dm_U(Y) = \int_{\mathcal{C}_{\mathcal{T}_q U}} h(\mathcal{T}_q U, Y')^4\, dm_{\mathcal{T}_q U}(Y') \le 1.$$

Thus the deficit and the pointwise nonconcentration bound imply

$$\begin{aligned} \int \rho(Y)^{3/4} (P_{1/2}h)(U,Y)\, dm_U(Y) &= \int \rho(Y) \mathbb{E} R_{\mathcal{T}_q}(Y)^{1/4}\, dm_U(Y) \\ &\le 1 - c/\mathfrak{b}. \end{aligned}$$

Adding a positive constant and normalizing, then using dominated convergence, removes the strict positivity assumptions on h and ρ. Taking the supremum over the dual tests $\rho^{3/4}$ and then over U proves (8.30). If $K_{\mathrm{out}} = 0$, the endpoint flag is constant along the field, so the same argument requires no invariance of h. In this case, $|P_{1/2}h| \le P_{1/2}|h|$ extends the estimate to arbitrary bounded measurable h. □

Lemma 8.12 (Gaussian divergence bounds at small coupling)**.** *Use the model* (1.3)*, fix* $\kappa_0 \in (0,2)$*, and let* $J = [-2+\kappa_0, 2-\kappa_0]$*. Consider the following two ranges.*
(i) *Fix* $0 < \alpha < \beta < 1/2$ *and* $\varepsilon > 0$*. For each deterministic* $a \in [\alpha, \beta]$*, set* $g = W^{-a}$*.*
(ii) *Fix* $c_* > 0$ *and* $0 < \theta \le 1/100$*, and let* g *be any deterministic coupling in the critical window*

$$c_* W^{-1/2} \le g \le W^{-1/2+\theta/2}. \tag{8.37}$$

In each range there is a fixed local length q *such that, for all sufficiently large* W*, every deterministic* $E \in J$ *and incoming flag admit* χ, X *satisfying the geometric and regularity assumptions of Lemma* 8.11*, with* $0 \le \chi \le 1$*,* $\mathbb{E}\chi \ge 1/2$*, and*

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\mathfrak{b}, \qquad \mathfrak{b} := \begin{cases} W^{\varepsilon/2} W g^2, & \text{in (i)}, \\ W^{18\theta}, & \text{in (ii)}. \end{cases} \tag{8.38}$$

The local length, constant C *and width threshold depend only on* $\alpha, \beta, \varepsilon, \kappa_0$ *in (i), and on* c_*, θ, κ_0 *in (ii). They are uniform over the displayed deterministic couplings, energies and flags.*

The proof of Lemma 8.12 is given in Section 9.3.

Lemma 8.13 (Gaussian divergence bounds from the product differential)**.** *Use the model (1.3), fix $B, E_0 > 0$, and take $q = 1000$. For all sufficiently large W and every deterministic $W^{-B} \le g \le W^B$, $|E| \le E_0$ and incoming flag, there are χ_P, X satisfying the geometric and regularity assumptions of Lemma 8.11, with $K_{\mathrm{out}} = 0$, $0 \le \chi_P \le 1$ and $\mathbb{E}\chi_P \ge 1/2$, and the following estimates.*
(i) *Throughout this range,*

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le C\mathfrak{b}, \qquad \mathfrak{b} := W \max\{g^{-2}, g^4\}. \tag{8.39}$$

The constant and width threshold depend only on B, E_0.
(ii) *If $g \ge 1$ and $\kappa_1 \le |E| \le E_0$, where $0 < \kappa_1 < E_0$ is fixed, one can choose a field with*

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le C_{\kappa_1} W g^2.$$

The constant and width threshold may additionally depend on κ_1. The same cutoff is used in both estimates and is independent of the incoming flag; the fields may differ.

The proof is given in Section 10.

8.5 Concentration at small and intermediate coupling

The constructions in Sections 9 and 10 provide the transfer input (8.11). Proposition 8.4(i) then gives the local resolvent bound (8.6); the finite-volume criterion and Lemma 8.1(i) give concentration.

Proof of Theorems 3.2 and 3.3. For either construction below, apply Lemma 8.11 to a field of local length q_0. For $Q = \mathrm{lcm}(q_0, 1000)$, independence and the cocycle identity imply that the operator of a Q-block batch is $P_{1/2}^{Q/q_0}$, where $P_{1/2}$ denotes the local q_0-block operator. Since $P_{1/2}$ preserves simultaneous unitary invariance (8.25), the contraction factor of $P_{1/2}^{Q/q_0}$ is at most $(1 - c/\mathfrak{b})^{Q/q_0} \le 1 - c/\mathfrak{b}$. At this fixed length Q, Lemmas 8.9 and 8.10 therefore apply. This uses operator iteration and requires no extension of the local field to the longer product.

For Theorem 3.2, set

$$\theta = \min\left\{\frac{\varepsilon}{200}, \frac{1-2\alpha}{4}, \frac{1}{100}\right\}, \qquad \beta = \frac{1}{2} - \frac{\theta}{2} > \alpha.$$

If $c_* W^{-1/2} \le g \le W^{-\beta}$, use Lemma 8.12(ii) with $\kappa_0 = \kappa$; its scale is $W^{18\theta} \le W^{\varepsilon/2}$. If $W^{-\beta} \le g \le W^{-\alpha}$, use part (i) of that lemma with loss parameter ε; its scale is $W^{\varepsilon/2} W g^2$. In both cases, Lemmas 8.11, 8.10 and 8.6 give (8.11) on $J = [-2 + \kappa, 2 - \kappa]$ with a scale at most $\mathfrak{b} = W^{\varepsilon/2} \max\{1, Wg^2\}$. Fix $0 < q < 1/4$. Proposition 8.4(i), with $t = 2q$, gives (8.6). Proposition 8.2 now proves (8.2) on paths and cycles. In each application, first choose C_0 to cover the polynomial ranges of $g, \mathfrak{b}$, then D' as required by Proposition 8.4(i), and finally C_1 in Proposition 8.2 to cover the resulting length constant. All parameters are fixed before W, g, L; the two ranges meet at $g = W^{-\beta}$, so taking the larger constants gives uniformity on their union. Lemma 8.1(i) gives (3.6) with

$$R \le C_D W^{\varepsilon/2} \max\{1, Wg^2\} \log W.$$

For large W, absorbing $C_D \log W$ into the remaining power proves (3.8) and the coordinate bound $\min\{WL, W^{\varepsilon} \max\{W, W^2 g^2\}\}$.

For Theorem 3.3, use Lemma 8.13(i) with batch length 1000 and $J = [-E_0, E_0]$. The same chain gives (8.11), then (8.6) and (8.2), with $\mathfrak{b} = W \max\{g^{-2}, g^4\} \le W^{1+4B}$. Lemma 8.1(i) gives $R \le C_D W \max\{g^{-2}, g^4\} \log W$; absorbing C_D into $\log W$ proves (3.9). Both bounds hold for every deterministic g, L in the stated ranges, with uniform constants and the finite-volume cap WL. □

9 Vector-field construction and estimates at small coupling

The construction is explained in Section 3.4. Here we prove Lemma 8.12; the passage to contraction is given by Lemma 8.11, following [10, Proposition 2.6 and Claims 2.7–2.8].

9.1 Constructing $X \in (\mathbb{C}^{W\times W})^q$ from $K \in \mathbb{C}^{W\times W}$

Fix $0 < g \le 1$, $E \in \mathbb{R}$, an isotropic flag (U, Y), and an integer $q \ge 7$. For the original input variables and transfer product, write

$$V = (V_1, \dots, V_q) \in \mathrm{Herm}(W)^q, \qquad \mathcal{T}_q = \mathcal{T}_q(V; E) = S^* T_q \cdots T_1 S.$$

Here T_i is defined in (8.9).

Data used to define the field. We apply the original transfers T_j to SU, SY, using the fixed unitary matrix S from Definition 7.1. For each fixed incoming flag, the first-block projections of $T_j \cdots T_1 SY$, $1 \le j \le q$, are almost surely invertible. Define the Hermitian matrices R_j and the normal lines $\mathbb{C}v_j$ by

$$T_j \cdots T_1 SY = \mathrm{ran}\begin{pmatrix} I_W \\ -gR_j \end{pmatrix}, \quad T_j \cdots T_1 SU = \left\{ \begin{pmatrix} x \\ -gR_j x \end{pmatrix} : x \in \mathbb{C}^W,\ v_j^* x = 0 \right\}. \tag{9.1}$$

Choose a nonzero representative v_1. The subsequent representatives can be chosen so that

$$R_j = (V_j - EI_W - g^2 R_{j-1})^{-1}, \qquad v_j = R_j v_{j-1}, \qquad 2 \le j \le q. \tag{9.2}$$

Call an inverse state R regular if

$$\|R\| \le W^3, \qquad \mathrm{rank}\, \mathbf{1}_{\{|g^2 R| > g\}} \le W g^{1/2}. \tag{9.3}$$

For the two regimes of Lemma 8.12 set

$$P := \begin{cases} \mathbf{1}_{\{|g^2 R_4| \le W^{\varepsilon/800}\}}, & \text{in regime (i)}, \\ I_W, & \text{in regime (ii)}. \end{cases} \tag{9.4}$$

Lemma 9.1 (Estimates after the first four blocks)**.** *Fix a deterministic incoming flag. In either regime, with probability $1 - o(1)$ uniformly over the flag and the stated energies and couplings, R_4 is regular and the following bounds hold. In regime (i),*

$$\|R_4\| \le W^3, \qquad \|Pv_4\| \ge W^{-10}\|v_4\|, \qquad b_4 = v_4^*(I + g^4 R_4^2)^{-1} v_4 \le C\|Pv_4\|^2, \tag{9.5}$$

and

$$\mathrm{rank}\, \mathbf{1}_{\{|g^2 R_4| > W^{-\varepsilon/800}\}} \le W g^2 W^{\varepsilon/400}. \tag{9.6}$$

In regime (ii),

$$\|R_4\| \le W^{1+\theta}, \qquad \mathrm{rank}\, \mathbf{1}_{\{|g^2 R_4| > W^{-\theta}\}} \le W^{3\theta}, \qquad g^2 \|R_4\| \le W^{2\theta}. \tag{9.7}$$

The proof, including first-step invertibility, is in Appendix G.1.1.

We hold the chosen representative v_4 fixed when varying $V_5, \dots, V_q$. For fixed E, g, U, Y, let $\chi \in [0, 1]$ and $K \in \mathbb{C}^{W\times W}$ depend only on $V_1, \dots, V_{q-1}$. Write $R_{[q-1,5]} := R_{q-1} \cdots R_5$ and assume

$$K^* = -K, \qquad \mathrm{Re}(v_{q-1}^* R_{[q-1,5]} K v_4) = \|v_{q-1}\|^2 \quad \text{on } \mathrm{supp}\, \chi. \tag{9.8}$$

The formula for X. For $V \in \operatorname{supp}\chi$, first form the correction vector and matrix

$$\begin{aligned} v_{\mathrm{corr}} &\coloneqq R_{[q-1,5]} K v_4 - v_{q-1}, \\ L &\coloneqq \|v_{q-1}\|^{-2}(v_{\mathrm{corr}} v_{q-1}^* - v_{q-1} v_{\mathrm{corr}}^*) - \|v_{q-1}\|^{-4} v_{q-1}^* v_{\mathrm{corr}}\, v_{q-1} v_{q-1}^*. \end{aligned} \tag{9.9}$$

With $[A,B] = AB - BA$, define X on $\operatorname{supp}\chi$ by the following formula and set $X = 0$ outside $\operatorname{supp}\chi$:

$$\boxed{X_i(V) \coloneqq \begin{cases} 0, & 1 \le i \le 4, \\ [K, V_5] - g^2[K, R_4], & i = 5, \\ [K, V_i], & 6 \le i \le q-1, \\ [K, V_q] - [L, V_q - g^2 R_{q-1}], & i = q. \end{cases}} \tag{9.10}$$

Lemma 9.2 (The constructed field satisfies the geometric conditions)**.** *Let χ, K be as above, satisfying (9.8), and define L, X by (9.9) and (9.10). Then every X_i is Hermitian and, for each $V \in \operatorname{supp}\chi$, we have*

$$\begin{aligned} D_X c(\mathcal{T}_q; U, Y) &= 1, \\ D_X(\mathcal{T}_q U, \mathcal{T}_q Y) &= \frac{d}{dt}\Big|_{t=0} (\operatorname{diag}(e^{t(K-L)}, e^{t(K-L)})\mathcal{T}_q U, \operatorname{diag}(e^{t(K-L)}, e^{t(K-L)})\mathcal{T}_q Y). \end{aligned} \tag{9.11}$$

Proof. Equations (9.8) and (9.9) give

$$\Re(v_{q-1}^* v_{\mathrm{corr}}) = 0, \qquad L^* = -L, \qquad L v_{q-1} = v_{\mathrm{corr}}. \tag{9.12}$$

Hermiticity follows from (9.10) and the skew-Hermitian properties in (9.8) and (9.12). Since the first four components vanish, $D_X R_4 = D_X v_4 = 0$. Put $M_q \coloneqq V_q - E I_W - g^2 R_{q-1}$. Differentiating (9.2), using (9.10) and (9.12), gives

$$\begin{aligned} D_X R_j &= -R_j(X_j - g^2 D_X R_{j-1}) R_j = [K, R_j], \qquad 5 \le j \le q-1, \\ D_X v_{q-1} &= [K, R_{[q-1,5]}] v_4 = (K-L) v_{q-1} - v_{q-1}, \\ D_X M_q &= [K - L, M_q]. \end{aligned} \tag{9.13}$$

The terminal spaces have the parametrizations

$$\begin{aligned} S\mathcal{T}_q Y &= \{(-g^{-1} M_q x, x) : x \in \mathbb{C}^W\}, \\ S\mathcal{T}_q U &= \{(-g^{-1} M_q x, x) : v_{q-1}^* x = 0\}, \end{aligned}$$

which do not require M_q to be invertible. The base-times-height formula and cocycle additivity give

$$e^{-2c(\mathcal{T}_q; U, Y)} = e^{-2c(\mathcal{T}_4; U, Y)} \cdot g^{2(q-4)} \cdot \frac{v_{q-1}^*(g^2 I_W + M_q^2)^{-1} v_{q-1}}{v_4^*(I_W + g^2 R_4^2)^{-1} v_4} \tag{9.14}$$

Where R_q exists, the numerator equals $v_q^*(I_W + g^2 R_q^2)^{-1} v_q$; the displayed form is defined for every V_q. Only this terminal quadratic form varies in (9.14). The terminal identities in (9.13) cancel the commutator terms and give

$$D_X\big(v_{q-1}^*(g^2 I_W + M_q^2)^{-1} v_{q-1}\big) = -2 v_{q-1}^*(g^2 I_W + M_q^2)^{-1} v_{q-1}.$$

Taking a logarithmic derivative of (9.14) yields $D_X c(\mathcal{T}_q; U, Y) = 1$.

At the current input V, freeze the values $K(V), L(V)$. The last two equations in (9.13) are the infinitesimal variation of

$$(M_q, v_{q-1}) \longmapsto (e^{t(K-L)} M_q e^{-t(K-L)},\ e^{-t} e^{t(K-L)} v_{q-1}).$$

In the terminal parametrizations above, this sends x to $e^{t(K-L)}x$; the scalar factor e^{-t} does not change the normal's kernel. Thus $S\mathcal{T}_q U, S\mathcal{T}_q Y$ undergo the same block-diagonal unitary rotation. Since this rotation commutes with S, the second conclusion in (9.11) follows. □

9.2 Construction of K

Fix either regime of Lemma 8.12 and condition on the first four blocks. Use the factorization $R_{[q-1,5]} = A\widetilde{R}$ from Section 3.4.1.

Proof of (3.4). Consider the block tridiagonal matrix on $\{5, \dots, q-1\}$ with first diagonal block $V_5 - EI_W - g^2 R_4$, remaining diagonal blocks $V_j - EI_W$, and adjacent off-diagonal blocks gI_W. Whenever the inverses exist, successive Schur complements from the left and from the right express its $(q-1,5)$ inverse block as $(-g)^{q-6}R_{[q-1,5]}$ and $(-g)^{q-6}A\widetilde{R}$, respectively. Cancelling the common factor proves (3.4). □

At the deterministic threshold $t_{\mathrm{cut}} > 0$ in (9.41), set

$$A_> \coloneqq A \cdot \mathbf{1}_{(t_{\mathrm{cut}}^2,\infty)}(A^*A), \qquad A_< \coloneqq A - A_>.$$

Let Q be the orthogonal projection with

$$\operatorname{ran} Q = \widetilde{R}(\operatorname{ran}(A_>^*)). \tag{9.15}$$

Thus $A_>\widetilde{R}(I-Q) = 0$ since $\widetilde{R}$ is Hermitian, and $\|A_<\| \le t_{\mathrm{cut}}$ by the singular-value cutoff. Use P from (9.4). In (i) it suffices to take $\varepsilon > 0$ small in terms of α, β, since decreasing ε strengthens the bounds. For $Pv_4 \neq 0$ let Π project onto $\operatorname{ran} P \cap v_4^{\perp}$ and set

$$\Pi \coloneqq P - \|Pv_4\|^{-2}(Pv_4)\cdot(Pv_4)^*, \qquad B_Q \coloneqq Q\Pi Q\big|_{\operatorname{ran} Q}.$$

If $B_Q > 0$, define

$$y \coloneqq \Pi Q B_Q^{-1} Q v_4, \qquad K_0 \coloneqq \|Pv_4\|^{-2}(y\cdot(Pv_4)^* - Pv_4\cdot y^*). \tag{9.16}$$

Then $Py = y$, $v_4^* y = 0$ and $Qy = Qv_4$, since $Q\Pi Q = B_Q$ on $\operatorname{ran} Q$. Hence

$$K_0^* = -K_0, \qquad K_0 = PK_0P, \qquad \operatorname{rank} K_0 \le 2, \qquad K_0 v_4 = y.$$

Since $A_>\widetilde{R}(y - v_4) = 0$, the matching error is

$$e_{q-1} \coloneqq A_<\widetilde{R}(y - v_4) = R_{[q-1,5]}(y - v_4). \tag{9.17}$$

For the normalization (9.8), set

$$\xi \coloneqq \|v_{q-1}\|^{-2}\Re(v_{q-1}^* R_{[q-1,5]} y), \qquad K \coloneqq K_0/\xi. \tag{9.18}$$

These formulas are used on the cutoff support, where

$$\|Pv_4\| > 0, \qquad B_Q \ge \tfrac{1}{2} I_{\operatorname{ran} Q}, \qquad \|y\| \le \|v_4\|, \qquad \|A_<\widetilde{R}\|\,\|v_4\| \le \tfrac{1}{4}\|v_{q-1}\|. \tag{9.19}$$

Under these conditions, $\|B_Q^{-1}\| \le 2$, and the two orthogonal rank-one terms in (9.16) give

$$\|y\| \le 2\|Qv_4\|, \qquad \|K_0\|_{\mathrm{HS}}^2 = 2\|Pv_4\|^{-2}\|y\|^2 \le 8\|Pv_4\|^{-2}\|Qv_4\|^2. \tag{9.20}$$

By (9.17) and $\|y\| \le \|v_4\|$,

$$\xi = 1 + \|v_{q-1}\|^{-2}\Re(v_{q-1}^* e_{q-1}), \qquad \|e_{q-1}\| \le 2\|A_<\widetilde{R}\|\,\|v_4\|, \qquad \tfrac{1}{2} \le \xi \le \tfrac{3}{2}.$$

Combining these bounds with (9.18) and the definition of L in (9.9) yields

$$\begin{aligned} &|\xi - 1| \le 2\|v_{q-1}\|^{-1}\|A_<\widetilde{R}\|\,\|v_4\|, \qquad \|K\|_{\mathrm{HS}}^2 \le 32\|Pv_4\|^{-2}\|Qv_4\|^2, \\ &v_{\mathrm{corr}} = \xi^{-1}(e_{q-1} - (\xi-1)v_{q-1}), \qquad \|v_{\mathrm{corr}}\| \le 8\|A_<\widetilde{R}\|\,\|v_4\|, \qquad \|L\|_{\mathrm{HS}} \le 24\|v_{q-1}\|^{-1}\|A_<\widetilde{R}\|\,\|v_4\|. \end{aligned} \tag{9.21}$$

Lemma 9.3 (Bounds and equivariance of K)**.** *In each regime of Lemma 8.12 one can choose a fixed length q and a cutoff χ so that (9.19) holds and K in (9.18) has the following properties. Pointwise bounds hold on* $\operatorname{supp}\chi$*. Expectations include the preparation blocks.*

(i) Support and normalization.

$$K^* = -K, \qquad K = PKP, \qquad \Re(v_{q-1}^* R_{[q-1,5]} K v_4) = \|v_{q-1}\|^2. \tag{9.22}$$

(ii) Source dependence and unitary symmetry. *In the construction replace v_4 by a variable $\widehat{v}_4 \neq 0$, keeping E, g, R_4, P and the preparation indicator fixed. This gives a family smooth near the support of the source cutoff $\widehat{\chi}$ constructed below, with*

$$K = \widehat{K}(V_5, \dots, V_{q-1}; v_4; R_4). \tag{9.23}$$

The family is invariant under $\widehat{v}_4 \mapsto a\widehat{v}_4$ for $a \in \mathbb{C} \setminus \{0\}$. For a unitary $\mathcal{U}$ acting identically on $\ker P$*, the transformation*

$$V_5 - g^2 R_4 \longmapsto \mathcal{U} \cdot (V_5 - g^2 R_4) \cdot \mathcal{U}^*, \quad V_i \longmapsto \mathcal{U} \cdot V_i \cdot \mathcal{U}^*, \quad 6 \le i \le q-1, \quad \widehat{v}_4 \longmapsto \mathcal{U}\widehat{v}_4 \tag{9.24}$$

sends $\widehat{K}$ to $\mathcal{U}\widehat{K}\mathcal{U}^*$.

(iii) Size and matching error. *We have*

$$\|K\|_{\mathrm{HS}}^2 \le 32, \qquad \mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 \le \begin{cases} C g^2 W^{\varepsilon/4}, & \text{in regime (i)}, \\ C W^{-1+14\theta}, & \text{in regime (ii)}, \end{cases} \tag{9.25}$$

and

$$\|R_{[q-1,5]} K v_4 - v_{q-1}\| \le C W^{-4} \|v_{q-1}\|. \tag{9.26}$$

(iv) Derivative bounds in v_4. *Differentiate the vector argument with all matrices fixed, in the real tangent space*

$$\mathcal{H}_{v_4} := \{w \in \operatorname{ran} P : \Re(v_4^* w) = 0\} = \{A v_4 : A^* = -A,\ A = PAP\}.$$

There is a real subspace $\mathcal{E}_{v_4} \subseteq \mathcal{H}_{v_4}$ such that

$$\begin{aligned} &\|Pv_4\| \cdot \|(w \cdot \nabla_{v_4})\widehat{K}\|_{\mathrm{HS}} \le C\|w\|, && w \in \mathcal{H}_{v_4}, \\ &\|Pv_4\| \cdot \|(w \cdot \nabla_{v_4})\widehat{K}\|_{\mathrm{HS}} \le C\|K\|_{\mathrm{HS}}\|w\|, && w \in \mathcal{H}_{v_4} \cap \mathcal{E}_{v_4}^{\perp}, \end{aligned} \tag{9.27}$$

where orthogonality is for the real Euclidean inner product and

$$\dim_{\mathbb{R}} \mathcal{E}_{v_4} \le \begin{cases} C W g^2 W^{\varepsilon/4}, & \text{in regime (i)}, \\ C W^{14\theta}, & \text{in regime (ii)}. \end{cases} \tag{9.28}$$

Summing over an adapted real orthonormal basis (e_a) of $\mathcal{H}_{v_4}$ and using $\dim_{\mathbb{R}} \mathcal{H}_{v_4} \le 2W$ gives

$$\|Pv_4\|^2 \cdot \sum_a \|(e_a \cdot \nabla_{v_4})\widehat{K}\|_{\mathrm{HS}}^2 \le C \dim_{\mathbb{R}} \mathcal{E}_{v_4} + CW\|K\|_{\mathrm{HS}}^2. \tag{9.29}$$

(v) Cutoff conditions. *The cutoff family $\widehat{\chi}$ has the parameter dependence and invariances in part (ii), with*

$$0 \le \widehat{\chi} \le 1, \qquad \chi = \widehat{\chi}|_{\widehat{v}_4 = v_4}, \qquad \mathbb{E}\chi \ge \tfrac{1}{2}.$$

On its support,

$$|R_{q-1}|_{\mathrm{op}} \le C W^3, \qquad g^2 |R_4|_{\mathrm{op}} \le C W^{2\theta} \quad \text{in regime (ii)}.$$

Its derivative satisfies

$$\mathbb{E}\left[\|Pv_4\|^2 \|K\|_{\mathrm{HS}}^2 \|\nabla_{v_4}\widehat{\chi}\|^2\right] \le C. \tag{9.30}$$

The gradient is taken along $\mathcal{H}_{v_4}$. The local length, constants, and width threshold have the dependence and uniformity specified in Lemma 8.12.

The proof is given in Section 9.4.

9.3 The Gaussian divergence bound

Lemma 9.4 (Gaussian divergence bound for the constructed field)**.** *Take P, K, χ from Lemma 9.3 and define L, X by (9.9) and (9.10). Conditional on $V_1, \dots, V_4$, write $\mathbb{E}$ for expectation over $V_5, \dots, V_q$. If $g^2\|PR_4P\| \le M_0$ on $\operatorname{supp}\chi$ for a deterministic $M_0 > 0$, then*

$$\begin{aligned}\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\mathbb{E}\left[\|Pv_4\|^2 \sum_a \left\|(e_a \cdot \nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2\right] \\ + CW(1+M_0^2)\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 + Wg^4\mathbb{E}\|[\chi L, R_{q-1}]\|_{\mathrm{HS}}^2 + CWM_0e^{-cW}.\end{aligned} \tag{9.31}$$

Here $C, c > 0$ are absolute, (e_a) is a real orthonormal basis of $\mathcal{H}_{v_4}$, and the derivative is evaluated at the prepared v_4.

Proof. To bound the divergence from blocks $5, \dots, q-1$ in (9.34), expand $\chi K = \sum_l a_l \mathsf{K}_l$ in a real Hilbert–Schmidt orthonormal basis of $\{H : H^* = -H,\ H = PHP\}$. The matrix field multiplying a_l in (9.10) differentiates along

$$Z_l \coloneqq [\mathsf{K}_l, V_5 - g^2R_4] \cdot \nabla_{V_5} + \sum_{i=6}^{q-1} [\mathsf{K}_l, V_i] \cdot \nabla_{V_i}.$$

The commutators have zero Euclidean divergence, and $\langle V_i, [\mathsf{K}_l, V_i]\rangle = 0$ for $\langle A, B\rangle = \Re\operatorname{Tr}(A^*B)$. Thus only the shift at block 5 contributes to the Gaussian adjoint,

$$Z_l^* = b_l - Z_l, \qquad b_l \coloneqq Wg^2\langle R_4, [\mathsf{K}_l, V_5 - g^2R_4]\rangle.$$

Writing $\delta_{\mathrm{act}} = \sum_l Z_l^* a_l$, two integrations by parts give

$$\mathbb{E}\delta_{\mathrm{act}}^2 = \mathbb{E}\sum_{l,j}(Z_j a_l)(Z_l a_j) + \mathbb{E}\sum_{l,j} a_l a_j Z_l b_j + \mathbb{E}\sum_{l,j} a_l [Z_j, Z_l] a_j. \tag{9.32}$$

By Cauchy–Schwarz and orthonormality, the first term is bounded in absolute value by $\mathbb{E}\sum_l \|Z_l(\chi K)\|_{\mathrm{HS}}^2$. The second equals

$$Wg^2\mathbb{E}\Re\operatorname{Tr}\Big(R_4[\chi K, [\chi K, V_5 - g^2R_4]]\Big).$$

Using $PR_4 = R_4P$, $\chi K = P(\chi K)P$, and $g^2\|PR_4P\| \le M_0$, we obtain

$$\begin{aligned}\left|\mathbb{E}\sum_{l,j} a_l a_j Z_l b_j\right| &\le CWM_0\mathbb{E}[(\|V_5\| + M_0)\|\chi K\|_{\mathrm{HS}}^2] \\ &\le CW(1+M_0^2)\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 + CWM_0e^{-cW}.\end{aligned}$$

The last inequality uses the GUE norm tail and the uniform bound on $\|\chi K\|_{\mathrm{HS}}$ in (9.25). For the third term, summing in a matrix-unit basis gives

$$\sum_l \|[\mathsf{K}_l, \chi K]\|_{\mathrm{HS}}^2 \le 2W\|\chi K\|_{\mathrm{HS}}^2. \tag{9.33}$$

Since $[Z_j, Z_l] = Z_{[\mathsf{K}_l, \mathsf{K}_j]}$, with Z_H defined by replacing K_l by H, the invariance of the inner product gives

$$\begin{aligned}\left|\mathbb{E}\sum_{l,j} a_l [Z_j, Z_l] a_j\right| &= \left|\mathbb{E}\sum_l \langle [\mathsf{K}_l, \chi K], Z_l(\chi K)\rangle\right| \\ &\le \tfrac{1}{2}\mathbb{E}\sum_l \|Z_l(\chi K)\|_{\mathrm{HS}}^2 + W\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2.\end{aligned}$$

Combining the three estimates yields

$$\mathbb{E}\delta_{\mathrm{act}}^2 \le C\mathbb{E}\sum_l \|Z_l(\chi K)\|_{\mathrm{HS}}^2 + CW(1+M_0^2)\mathbb{E}\|\chi K\|_{\mathrm{HS}}^2 + CWM_0e^{-cW}. \tag{9.34}$$

To bound the derivatives in (9.34), differentiate (9.24) with $\mathcal{U} = e^{t\mathsf{K}_l}$. By Lemma 9.3(ii),(v),

$$Z_l(\chi K) = [\mathsf{K}_l, \chi K] - ((\mathsf{K}_l v_4)\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K}).$$

In coordinates on $\operatorname{ran} P$ with Pv_4 in the first coordinate direction, the real covariance of $(\mathsf{K}_l v_4)_l$ has eigenvalue $\|Pv_4\|^2$ in the phase direction iPv_4 and $\|Pv_4\|^2/2$ in the remaining tangent directions. Applying this to each real coordinate of the derivative yields

$$\sum_l \left\|((\mathsf{K}_l v_4)\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2 \le \|Pv_4\|^2 \sum_a \left\|(e_a\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2.$$

Thus the chain rule and (9.33) give

$$\sum_l \|Z_l(\chi K)\|_{\mathrm{HS}}^2 \le C\|Pv_4\|^2 \sum_a \left\|(e_a\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2 + CW\|\chi K\|_{\mathrm{HS}}^2. \tag{9.35}$$

Finally condition on $V_1,\dots,V_{q-1}$, which fixes χ, K, L, R_{q-1}. The last component of (9.10) is

$$\chi X_q = [\chi K - \chi L, V_q] + g^2[\chi L, R_{q-1}].$$

The commutator field has zero Gaussian divergence, while the constant field contributes

$$\delta_{\mathrm{last}} := Wg^2\langle V_q, [\chi L, R_{q-1}]\rangle, \qquad \delta_{\mathrm{G}}(\chi X) = \delta_{\mathrm{act}} + \delta_{\mathrm{last}}.$$

The conditional GUE mean and variance give

$$\mathbb{E}[\delta_{\mathrm{last}} \mid V_1,\dots,V_{q-1}] = 0, \qquad \mathbb{E}[\delta_{\mathrm{last}}^2 \mid V_1,\dots,V_{q-1}] = Wg^4\|[\chi L, R_{q-1}]\|_{\mathrm{HS}}^2. \tag{9.36}$$

Since δ_{act} is fixed under this last conditional expectation, the cross term vanishes. Combining (9.35), (9.34), and (9.36) proves (9.31). □

Proof of Lemma 8.12. For the field in (9.10), (9.22) and Lemma 9.2 give (8.34). The first four components vanish, so integration by parts is conditional on preparation. The smooth cutoff gives the required regularity in the active blocks, and truncation is valid in V_q since the field is affine in V_q with the other inputs fixed.

We now average (9.31) over preparation. For its derivative term the product rule and (9.29) give

$$\begin{aligned}\|Pv_4\|^2 \sum_a \left\|(e_a\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2 &\le C\chi^2(\dim_{\mathbb{R}}\mathcal{E}_{v_4} + W\|K\|_{\mathrm{HS}}^2)\\ &\quad + C\|Pv_4\|^2\|K\|_{\mathrm{HS}}^2\|\nabla_{v_4}\widehat{\chi}\|^2.\end{aligned} \tag{9.37}$$

Taking expectations and using the size, dimension, and cutoff estimates in (9.25), (9.28), and (9.30) gives

$$\mathbb{E}\left[\|Pv_4\|^2 \sum_a \left\|(e_a\cdot\nabla_{v_4})(\widehat{\chi}\,\widehat{K})\right\|_{\mathrm{HS}}^2\right] \le \begin{cases} CWg^2W^{\varepsilon/4}, & \text{in regime (i)},\\ CW^{14\theta}, & \text{in regime (ii)}.\end{cases} \tag{9.38}$$

Here $Wg^2 \ge W^{1-2\beta}$ in (i) and $W^{14\theta} \ge 1$ in (ii) absorb the constant cutoff cost. For the last block, (9.9) and (9.26) give $\|L\|_{\mathrm{HS}} \le CW^{-4}$ on $\operatorname{supp}\chi$. There $\|R_{q-1}\| \le CW^3$, so

$$Wg^4\mathbb{E}\|[\chi L, R_{q-1}]\|_{\mathrm{HS}}^2 \le CWg^4W^6W^{-8} \le CW^{-1},$$

since $g \le 1$ in both parameter regimes for large W.

In (i), (9.4) permits $M_0 = W^{\varepsilon/8}$. Substituting (9.25) and (9.38) into (9.31) gives

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\left(Wg^2W^{\varepsilon/4} + W^{1+\varepsilon/4}g^2W^{\varepsilon/4} + W^{-1}\right) \le CWg^2W^{\varepsilon/2}. \tag{9.39}$$

In (ii), Lemma 9.3(v) permits $M_0 = CW^{2\theta}$, and the same substitution gives

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi X)|^2 \le C\left(W^{14\theta} + W^{1+4\theta}W^{-1+14\theta} + W^{-1}\right) \le CW^{18\theta}. \tag{9.40}$$

In both cases $WM_0e^{-cW} = O(W^{-1})$. The cutoff mass $\mathbb{E}\chi \ge 1/2$, the fixed length q and the uniformity of the bounds follow from Lemma 9.3. □

9.4 Proof of Lemma 9.3

9.4.1 Step 1. Retained directions and overlap

In the two regimes of Lemma 8.12, choose the fixed length q by (9.54) and (9.62), respectively, and set

$$t_{\mathrm{cut}} := \begin{cases} W^{-40}g^{-q}, & \text{in regime (i)},\\ W^{-30+(1-2\theta)(q-6)/2}, & \text{in regime (ii)}. \end{cases} \tag{9.41}$$

The goal is to bound the rank of Q in (9.15) and its overlap with the prepared vector v_4, as in (9.65). We first establish the two inverse-image estimates used in regimes (i) and (ii).

Lemma 9.5 (Relative range)**.** *Fix $\kappa_0 \in (0,2)$ and $\delta > 0$. Let $|E| \le 2-\kappa_0$ and let V be normalized complex GUE of size W. The data E, B_0, B_1, Q_0, U, v below are deterministic. Let $B_0 = B_{\mathrm{bad}} + B_{\mathrm{small}}$ be Hermitian, with a spectral projection Q_0 of B_0 and $B_{\mathrm{bad}} := Q_0B_0Q_0$, with $\|B_{\mathrm{small}}\| = o(1)$. Let B_1 be Hermitian of norm $o(1)$, U a fixed subspace, and v a fixed vector. Set*

$$R := (V - EI - B_0 - B_1)^{-1}.$$

If $W^{\delta} \le d = o(W)$, $d \ge \dim U + 2\operatorname{rank} Q_0 + 1$, and $\|Q_0P_U\| \le C\sqrt{d/W}$, then, with failure $W^{-D} + Ce^{-cd}$ for each prescribed D,

$$\|P_{RU}v\|^2 \le C(d/W)(\log W)^2 v^*(I + B_0^2)^{-1}v. \tag{9.42}$$

Here P_{RU} is the orthogonal projection onto RU, and the probability event may depend on the fixed vector v. The constants do not depend on $\|B_{\mathrm{bad}}\|$.

In Lemma 9.5, the fresh cross block is used both to form the inverse-image subspace and to test its overlap. We first prove the Gaussian estimate that controls this dependence.

Lemma 9.6 (Dependent Gaussian projection)**.** *Fix $\delta > 0$ and constants $C, C_1 > 0$. Let $d \ge W^{\delta}$, $d = o(W)$, $\eta := Ad/W$ with fixed sufficiently large A, and let R be a fixed Hermitian inverse on a space of dimension at most W. Suppose*

$$C_0 d \le \operatorname{rank} \mathbf{1}_{|R|\ge\eta^{-1}} \le C_1W\eta, \quad \operatorname{Tr}|R_<| \le CW\log W, \quad \operatorname{Tr} R_<^2 \le CW/\eta, \quad \|R_<\| \le \eta^{-1}, \tag{9.43}$$

where $R_< := R\mathbf{1}_{|R|<\eta^{-1}}$ and C_0 is sufficiently large. Let F be a deterministic subspace of the space on which R acts, with dimension at most d, and let Y have independent centered complex Gaussian entries of variance $1/W$, one row per dimension of this space, and at most d columns. All projections in this lemma are orthogonal. Then, with failure Ce^{-cd},

$$\left\|Y^*P_{F+\mathrm{ran}(RY)}\right\| \le C\sqrt{\eta}\log W. \tag{9.44}$$

Proof. The dependence appears in the quadratic form $Y^*(I-P_F)RY$. We split the spectrum so that the large inverse directions provide a lower bound for the generated vectors, while the remaining quadratic form has only a logarithmic cost. Set

$$Y_\perp := (I-P_F)RY, \qquad E_> := \operatorname{ran} \mathbf{1}_{\{|R|\geq \eta^{-1}\}}.$$

Set $H = E_> \cap F^\perp$, so $\dim H \geq (C_0-1)d$ and $P_H Y_\perp = P_H RY$. Each column has covariance $W^{-1}P_H R^2 P_H \geq (W\eta^2)^{-1}P_H$. The rectangular Gaussian lower bound therefore gives

$$\|Y_\perp b\| \geq c\eta^{-1/2}\|b\|. \tag{9.45}$$

The Gaussian norm and quadratic-form bounds also give

$$\|Y^*P_F\| + \left\|Y^*P_{(I-P_F)E_>}\right\| \leq C\sqrt{\eta}, \qquad \|(I-P_F)R_<Y\| \leq C\eta^{-1/2}, \tag{9.46}$$

and

$$\|Y^*(I-P_F)R_<Y\| \leq C\log W. \tag{9.47}$$

These three estimates hold simultaneously with failure Ce^{-cd}; their elementary verifications are in Appendix G.2.1.

Write $Y_\perp = Y_{\perp,>} + Y_{\perp,<}$ according to the split of R. Equations (9.45)–(9.47) imply

$$\|Y^*Y_\perp b\| \leq C\sqrt{\eta}\,\|Y_{\perp,>}b\| + C\log W\,\|b\| \leq C\sqrt{\eta}\log W\,\|Y_\perp b\|.$$

Here $\|Y_{\perp,<}b\| \leq C\eta^{-1/2}\|b\|$ is absorbed using (9.45). Since F and $\operatorname{ran} Y_\perp$ are orthogonal and span $F + \operatorname{ran}(RY)$, (9.44) follows. □

Proof of Lemma 9.5. Write $\Gamma = R^{-1} = V - EI - B_0 - B_1$. Choose a d-dimensional space Q containing $\operatorname{ran} Q_0, U, \operatorname{ran}(B_1Q_0)$ and $(I-Q_0)v$. Decompose $Q = \operatorname{ran} Q_0 \oplus Q_1$ and then $Q \oplus Q^\perp$. The complementary matrix Γ_c has only a small-norm deformation. For every prescribed $D>0$, its spectral input is

$$(9.43)|_{R=\Gamma_c^{-1},\,\eta=Ad/W} \quad \text{with probability at least } 1-W^{-D}. \tag{9.48}$$

Here A is a sufficiently large fixed constant. The counting-law and dyadic-summation proof is given under **Proof of** (9.48) in Appendix G.2.2. Write the cross block as (Y_0, Y_1) according to $\operatorname{ran} Q_0, Q_1$. Because $\operatorname{ran}(B_1Q_0) \subset Q$, Y_0 has zero mean and is independent of both Γ_c and Y_1. The latter block can have a deterministic mean. Condition on these two objects and put $F := \operatorname{ran}(\Gamma_c^{-1}Y_1)$. Lemma 9.6 gives

$$\left\|Y_0^* P_{F+\operatorname{ran}(\Gamma_c^{-1}Y_0)}\right\| \leq C\sqrt{\eta}\log W, \qquad \eta = Ad/W. \tag{9.49}$$

There is also the following uniform inverse-image compression:

$$\|x_Q\| \leq C\sqrt{\eta}\,\|x\| \quad \text{whenever } \Gamma x \in Q. \tag{9.50}$$

Indeed $x_c = -\Gamma_c^{-1}Yx_Q$. Let $\mathcal{E}$ be the spectral subspace of Γ_c for $[-\eta,\eta]$, of dimension at least C_0d. Conditional on Γ_c, write $\overline{Y} := \mathbb{E}[Y \mid \Gamma_c]$ and let $\mathcal{N} \subset \mathcal{E}$ be the kernel of $(P_{\mathcal{E}}\overline{Y})^*$ restricted to $\mathcal{E}$. Then $\dim\mathcal{N} \geq \dim\mathcal{E} - d$. For every column vector u, the inverse spectral bound is applied before the row projection:

$$\left\|\Gamma_c^{-1}Yu\right\| \geq \eta^{-1}\|P_{\mathcal{E}}Yu\| \geq \eta^{-1}\|P_{\mathcal{N}}P_{\mathcal{E}}Yu\|.$$

The remaining centered Gaussian block $P_{\mathcal{N}}P_{\mathcal{E}}Y$ has least singular value at least $c\sqrt{d/W}$, in orthonormal coordinates on $\mathcal{N}$. Taking $u = x_Q$ gives $\|x_c\| \geq c\eta^{-1}\sqrt{d/W}\,\|x_Q\|$. This event and (9.49) can be intersected without any independence assertion between them.

For $x \in RU$, let $u := \Gamma x \in U$. Transversality and the usual GUE norm bound imply

$$\|u\| \le 2\,\|(I-Q_0)u\| \le 2\,\|V - EI - B_{\mathrm{small}} - B_1\|\,\|x\| \le C\,\|x\|\,, \qquad \|Q_0 u\| \le C\sqrt{\eta}\,\|x\|\,.$$

Writing $x_0 = Q_0 x$, the Q_0 component of $\Gamma x = u$ is

$$B_{\mathrm{bad}} x_0 = Q_0(V - EI - B_{\mathrm{small}} - B_1)x_Q + Y_0^* x_c - Q_0 u.$$

The deterministic Q_0-to-complement block vanishes by the choice of Q; its Gaussian part is exactly Y_0^*. Since $x_c = -\Gamma_c^{-1} Y x_Q \in F + \mathrm{ran}(\Gamma_c^{-1} Y_0)$, (9.49) bounds $\|Y_0^* x_c\|$ by $C\sqrt{\eta}\log W \|x_c\|$. The bounded norm of $V - EI - B_{\mathrm{small}} - B_1$, (9.50), and the bound on $Q_0 u$ above now give

$$\|B_{\mathrm{bad}} x_0\| + \|x_Q\| \le C\sqrt{\eta}\log W\,\|x\|\,. \tag{9.51}$$

Finally split v into its good and bad components and use weighted Cauchy–Schwarz:

$$\begin{aligned} |v^* x| &\le \|v_{\mathrm{good}}\|\,\|x_Q\| + \left\|(I + B_{\mathrm{bad}}^2)^{-1/2} v_{\mathrm{bad}}\right\| \left\|(I + B_{\mathrm{bad}}^2)^{1/2} x_0\right\| \\ &\le C\sqrt{\eta}\log W \sqrt{v^*(I + B_0^2)^{-1} v}\,\|x\|\,. \end{aligned}$$

Take the supremum over unit vectors in RU. □

When the deformation profile is not needed, the compression step in Lemma 9.5 gives the following consequence.

Corollary 9.7 (Fresh-matrix compression)**.** *Fix $\kappa_0 \in (0,2)$ and $\delta > 0$. All matrices and vectors below act on $\mathbb{C}^W$. Let $B = B_{\mathrm{small}} + B_{\mathrm{bad}}$ be deterministic Hermitian, $\|B_{\mathrm{small}}\| = o(1)$ and $\operatorname{rank} B_{\mathrm{bad}} \le r_0$. Let U be a deterministic subspace of dimension at most m, and let v be a deterministic unit vector. Put $R := (V - EI - B)^{-1}$, where V is fresh normalized complex GUE and $|E| \le 2 - \kappa_0$. If $W^\delta \le d = o(W)$ and $d \ge m + r_0 + 1$, then, for each fixed $D' > 0$,*

$$\|P_{RU} v\|^2 \le Cd/W \tag{9.52}$$

with failure at most $W^{-D'} + Ce^{-cd}$, where P_{RU} is the orthogonal projection onto RU. The constants are independent of $\|B_{\mathrm{bad}}\|$ and of any relative orientation of U, v and $\operatorname{ran} B_{\mathrm{bad}}$.

Proof. Choose a deterministic d-dimensional space $\mathcal{Q}$ containing U, v, and $\operatorname{ran} B_{\mathrm{bad}}$. The complementary block of $V - EI - B$ has only a small-norm deformation. The argument proving (9.50) applies without the transversality hypothesis: at $\eta = Ad/W$, its near-zero spectral space has dimension at least a large constant times d. After removing the deterministic cross-block mean inside that space, the remaining Gaussian matrix has least singular value at least $c\sqrt{d/W}$. The complementary row equation consequently gives $\|x_{\mathcal{Q}}\| \le C\sqrt{d/W}\|x\|$ for every $x \in RU$. Since $v \in \mathcal{Q}$, take the supremum of $|v^* x|$ over unit vectors in RU. The local-law and Gaussian failures are respectively $W^{-D'}$ and Ce^{-cd}, as in the proof of (9.50). □

Rank and overlap for the retained space. On the event in Lemma 9.1, fix $V_1, \dots, V_4$. A singular-value moment bounds the retained dimension; the fresh matrix V_5 then controls its overlap with v_4. The cutoff in Section 9.4.4 will enforce the resulting good events.

The right product A is the adjoint of a forward product in reversed order with initial state zero. The product moment bound (G.6) in Appendix G.3, with $h = q - 6$, gives

$$\operatorname{rank} Q = \operatorname{rank} A_> \le t_{\mathrm{cut}}^{-p} \operatorname{Tr}|A|^p \le t_{\mathrm{cut}}^{-p} W^{1+e}, \qquad 0 < p < \frac{2}{q-5}, \tag{9.53}$$

on an invariant event of probability $1 - o(1)$, for each fixed $e > 0$.

Regime (i). For $g = W^{-a}$ with $a \in [\alpha, \beta] \subset (0, 1/2)$, choose

$$q = 5 + \left\lceil 25600\varepsilon^{-1}(40+\beta) \right\rceil, \qquad p = \frac{2}{q-5}(1 - \varepsilon/6400). \tag{9.54}$$

Using the threshold (9.41) and $e = \varepsilon/3200$ in (9.53) yields

$$\operatorname{rank} Q \le W^{1+\varepsilon/3200+40p} g^{qp} < W g^2 W^{\varepsilon/800}. \tag{9.55}$$

The last inequality follows by substitution of (9.54), uniformly in a; in particular q, p do not depend on a, W.

Small overlap and compatibility with P. To control $\|Qv_4\|/\|Pv_4\|$, apply Lemma 9.5 with incoming deformation $g^2 R_4$ and spectral projection

$$Q_0 := \mathbf{1}_{\{|g^2 R_4| > W^{-\varepsilon/800}\}}.$$

The rank of Q_0 is bounded by (9.6). We first check the required transversality and remove the large right self-energy from the deformation.

Use the retained space $\operatorname{ran}(A_>^*)$ from (9.55), and split $g^2 \overleftarrow{R}_6 = B_{1,>} + B_{1,<}$ at $W^{-\varepsilon/800}$. Define

$$U := \operatorname{ran}(A_>^*) + \operatorname{ran} B_{1,>}.$$

The joint right-cavity law is invariant under common unitary conjugation, independently of the incoming data. Therefore U is Haar conditional on its dimension and invariant events. The Gaussian representation of a Haar subspace gives

$$\|Q_0 P_U\| \le C\sqrt{(\dim U + \operatorname{rank} Q_0 + W^{(1-2\beta)/4})/W} \tag{9.56}$$

with exponential failure in the numerator's dimension scale. This uses only common orbital invariance, not independent cavity matrices. Since $\operatorname{ran} Q = \widetilde{R} \operatorname{ran}(A_>^*)$, the enlarged space satisfies

$$\operatorname{ran} Q \subset \widetilde{R} U = (V_5 - EI - g^2 R_4 - B_{1,<})^{-1} U. \tag{9.57}$$

Both inverse images are the set of x for which the corresponding matrix times x belongs to U; their difference has range in U. We apply the range estimates to this enlarged space and then restrict them to $\operatorname{ran} Q$. For each new block, condition on its Hermitian shift and write $\Gamma = V - EI_W - g^2 R$ with V normalized complex GUE. For an interval J and $\delta > 0$, the single-block Gaussian regularization estimates of [2, Theorem 1] (see also [24, Theorem 2, (1.10)]) give

$$\mathbb{E} N(\Gamma, J) \le CW|J|, \qquad \mathbb{P}\left(\left\|\Gamma^{-1}\right\| > W^{1+\delta}\right) \le CW^{-\delta}. \tag{9.58}$$

Applying the count bound to the right recursion gives $\operatorname{rank} B_{1,>} \le W g^2 W^{\varepsilon/400}$ with failure $O(W^{-\varepsilon/800})$. For the dimension parameter in Lemma 9.5, with $\delta = (1-2\beta)/4$, choose

$$d := \lceil 8 W g^2 W^{\varepsilon/200} \rceil. \tag{9.59}$$

Equations (9.55) and (9.6) give $d \ge \dim U + 2\operatorname{rank} Q_0 + 1$, while

$$d \ge W^{1-2\beta}, \qquad d/W \le CW^{-2\alpha+\varepsilon/200} = o(1).$$

Thus d dominates the numerator in (9.56), and the spectral scale d/W is mesoscopic. Also $\operatorname{ran}(I-P) \subset \operatorname{ran} Q_0$. Use the compression bound (9.50) in the proof of Lemma 9.5, and apply (9.42) separately to the two fixed vectors v_4, Pv_4. Put $b_4 = v_4^*(I + g^4 R_4^2)^{-1} v_4$; preparation in (9.5) gives $b_4 \le C\|Pv_4\|^2$. For the projection Q in (9.15), this gives

$$\|(I-P)Q\|^2 \le Cd/W = o(1), \qquad \|Qv_4\|^2 \le g^2 W^{\varepsilon/100} b_4, \qquad \|QPv_4\|^2 \le g^2 W^{\varepsilon/100} \|Pv_4\|^2. \tag{9.60}$$

The fixed constants and $(\log W)^2$ are absorbed into $W^{\varepsilon/200}$ with a common threshold.

Regime (ii). Here $P = I_W$, and preparation gives (9.7). To control $\|R_{[q-1,5]}v_4\|$, the physical scale g^2 may be too small for the local law. Introduce the larger spectral scale

$$\upsilon = W^{-1+2\theta}, \qquad c_* W^{-1/2} \le g \le W^{-1/2+\theta/2}. \tag{9.61}$$

Then $g^2 \le \upsilon$ and $W\upsilon = W^{2\theta}$, so Lemma G.1 applies at $\eta = \upsilon$. We use this scale in Step 2 to bound $\|v_{q-1}\|$ from below.

Choose

$$q := 5 + \lceil 10/\theta \rceil, \qquad p := \frac{2}{q-5}(1-\theta/4). \tag{9.62}$$

Using the threshold (9.41) and $e = \theta/4$ in (9.53) gives

$$\operatorname{rank} Q \le W^{1+\theta/4+30p}\upsilon^{(q-6)p/2} \le W^{9\theta}, \tag{9.63}$$

by (9.62). Again q is fixed independently of g, W.

Small overlap. Since $P = I_W$, the absolute compression bound in Corollary 9.7 suffices for the overlap. Split each of g^2R_4 and $g^2\overleftarrow{R}_6$ spectrally at $W^{-\theta}$. The two large pieces have ranks at most $W^{3\theta}$ with failure $O(W^{-\theta})$: their expected counts are at most $CWg^2W^{\theta} \le CW^{2\theta}$. The small pieces have total norm at most $2W^{-\theta}$. Conditional on all right-cavity data, apply Corollary 9.7 to

$$B = g^2R_4 + g^2\overleftarrow{R}_6, \qquad U = \operatorname{ran} A^*_{>}, \qquad d := \lceil W^{12\theta} \rceil.$$

Indeed $d \ge W^{9\theta} + 2W^{3\theta} + 1$ for large W and $d = o(W)$. The source vector v_4 and all these data are fixed before V_5. Applying Corollary 9.7 to $v_4/\|v_4\|$ and using $\operatorname{ran} Q = \widetilde{R}U$ yields

$$\|v_4\|^{-2}\|Qv_4\|^2 \le CW^{-1+12\theta} \le W^{-1+13\theta} \tag{9.64}$$

with probability $1 - o(1)$, uniformly in (8.37).

Common conclusion. In both regimes the overlap estimates give, on $\operatorname{ran} Q$,

$$B_Q \ge \left(1 - \|(I-P)Q\|^2 - \|Pv_4\|^{-2}\|QPv_4\|^2\right) I_{\operatorname{ran} Q} \ge \tfrac{1}{2} I_{\operatorname{ran} Q}, \qquad \|Pv_4\|^{-2}\|Qv_4\|^2 = o(1).$$

Here the first loss vanishes when $P = I_W$. By (9.20), $\|y\| = o(\|v_4\|)$, so the first three conditions in (9.19) hold with room for the cutoff margins. Combining (9.55), (9.60), (9.63) and (9.64) gives, for any $0 \le \chi \le 1$ supported on these events,

$$\operatorname{rank} Q \le \begin{cases} CWg^2W^{\varepsilon/4}, & \text{in (i)}, \\ CW^{14\theta}, & \text{in (ii)}, \end{cases} \qquad \mathbb{E}\left[\chi^2\|Pv_4\|^{-2}\|Qv_4\|^2\right] \le \begin{cases} Cg^2W^{\varepsilon/4}, & \text{in (i)}, \\ CW^{-1+14\theta}, & \text{in (ii)}. \end{cases} \tag{9.65}$$

The rank bound is pointwise on the support, where $\|Qv_4\| \le \|Pv_4\|$. The expectation includes the preparation blocks.

9.4.2 Step 2. The matching error

We prove the relative error bound (9.71). Since $\|A_{<}\| \le t_{\mathrm{cut}}$, the numerator is at most $t_{\mathrm{cut}}\|\widetilde{R}\|\,\|v_4\|$. We therefore need an upper bound on $\|\widetilde{R}\|$ and a lower bound on the transported vector $\|v_{q-1}\|$. The inverse bound is common to both regimes.

The active-chain matrix has diagonal blocks $V_5 - EI - g^2R_4, V_6 - EI, \dots, V_{q-1} - EI$ and adjacent off-diagonal blocks gI. Impose the bound W^3 on its inverse. It has failure $C(q-5)^{3/2}/W$, uniformly over the prepared boundary: the common scalar Gaussian shift has variance $1/((q-5)W^2)$, independently of the remaining coordinates, and a union bound covers the $(q-5)W$ eigenvalues. The exact Schur identities give

$$\left\|\widetilde{R}\right\|, \|R_{q-1}\| \le W^3, \qquad \left\|R_{[q-1,5]}\right\| \le W^3 g^{-(q-6)}. \tag{9.66}$$

For the output cutoff, set

$$v_{\min} := \begin{cases} g^{-(q-5)}\|Pv_4\|W^{-3\varepsilon/800}, & \text{in (i)}, \\ v^{-(q-5)/2}\|v_4\|W^{-4\theta}, & \text{in (ii)}. \end{cases} \tag{9.67}$$

For every fixed $D > 0$ and sufficiently large W, the probability that the chain is regular and $\|v_{q-1}\| < 2v_{\min}$ is at most W^{-D}.

Regime (i). To bound the denominator $\|v_{q-1}\|$ from below, define

$$F_j := (I + g^4 R_j^2)^{-1}, \qquad 4 \le j \le q-1.$$

Write $R_{[j,5]} = R_j \cdots R_5$ for $5 \le j \le q-1$. For a regular past, u fixed before V_j, and $\eta = g^2$, the relative local law (G.2) in Appendix G.1 gives

$$\Im\, u^*(R_j^{-1} - ig^2 I_W)^{-1}u \asymp u^*(I_W + g^4 R_{j-1}^2)^{-1}u$$

with failure probability at most W^{-D} for every fixed $D > 0$ and all sufficiently large W. The probability that the chain is regular and the following comparison fails for some $5 \le j \le q-1$ is also at most W^{-D}:

$$v_4^* R_{[j,5]}^* F_j R_{[j,5]} v_4 \asymp_{j-4} g^{-2(j-4)} v_4^* F_4 v_4, \qquad 5 \le j \le q-1. \tag{9.68}$$

Indeed, $R_j F_j R_j = g^{-2}\Im(R_j^{-1} - ig^2)^{-1}$, and the Dyson profile at the next step is comparable to F_{j-1}. Normalize the conditional vector at each step and sum the failure probabilities over the fixed number of steps, taking the one-step exponent larger than D. Regularity itself has probability $1 - o(1)$.

Since $b_4 \ge (1 + W^{\varepsilon/400})^{-1}\|Pv_4\|^2$ by the definition of P, (9.68) gives

$$\|v_{q-1}\| \ge c_{q-5} g^{-(q-5)}\|Pv_4\|W^{-\varepsilon/800}.$$

The comparison constants are uniform in a and q is fixed, so this exceeds $2v_{\min}$ for large W.

Regime (ii). Condition on the prepared R_4, v_4. By (9.7), $\|R_4\| \le W^{1+\theta}$ and R_4 is regular on the preparation event. For this fixed v_4, the lower bound proved in Appendix G.3.1 is

$$\left\|R_{[q-1,5]}v_4\right\| \ge c_{q-5} v^{-(q-5)/2}\|v_4\|W^{-3\theta}. \tag{9.69}$$

The probability that this bound fails and the entire chain is regular is at most $W^{-D'}$ for every fixed $D' > 0$. It gives $\|v_{q-1}\| \ge 2v_{\min}$ since $c_{q-5}W^{\theta} > 2$ for large W. The estimate uses forward conditioning, whereas (9.64) uses right-cavity conditioning; we combine their events by a union bound, without conditioning either estimate on the other's data.

Common error bounds. On $\|v_{q-1}\| \ge v_{\min}$, substitute (9.41) into $t_{\mathrm{cut}}W^3\|v_4\|/v_{\min}$ and, in regime (i), use $\|v_4\| \le W^{10}\|Pv_4\|$ from (9.5). This gives

$$\|v_{q-1}\|^{-1}\|A_< \widetilde{R}\|\,\|v_4\| \le \begin{cases} g^{-5}W^{-27+3\varepsilon/800}, & \text{in (i)}, \\ v^{1/2}W^{-27+4\theta}, & \text{in (ii)}. \end{cases} \tag{9.70}$$

Using $g^{-5} \le W^{5\beta}$ and $\beta < 1/2$ in (i), we obtain in both regimes

$$\|v_{q-1}\|^{-1}\|A_< \widetilde{R}\|\,\|v_4\| \le CW^{-4}. \tag{9.71}$$

This verifies the last condition in (9.19) and controls $|\xi - 1|$ by (9.21). For the normalization and cutoff derivatives, (9.66) gives the common ratio bound

$$v_{\min}^{-2}\|Pv_4\|^2\|R_{[q-1,5]}\|^2 \le \begin{cases} g^2 W^{6+3\varepsilon/400}, & \text{in (i)}, \\ C_{c_*,q} W^{5+8\theta+2\theta(q-5)} \le C_{c_*,q} W^{25+10\theta}, & \text{in (ii)}. \end{cases} \tag{9.72}$$

In the second line we used $g \ge c_* W^{-1/2}$ and $2\theta(q-5) \le 20 + 2\theta$. All restrictions above have common probability $1 - o(1)$ and will be imposed by the cutoff. In particular, (9.21) gives the matching estimate (9.26) and, together with Step 1, both size estimates in (9.25). The real normalization (9.18) gives (9.22).

9.4.3 Step 3. Derivatives in v_4

Keep the matrices fixed. Both thresholds in (9.41) are independent of v_4, so P, Q are fixed as v_4 varies. We prove a common directional estimate, then use the error bounds from Section 9.4.2 to control the scalar normalization.

We first isolate the directions in which the derivative can be large. Define

$$u := QB_Q^{-1}Qv_4.$$

Since $u \in \operatorname{ran} Q$ and $y = \Pi u$, we have

$$\|y\|^2 = u^* B_Q u \ge \tfrac{1}{2}\|u\|^2.$$

For $w \in \mathcal{H}_{v_4}$, the directional derivatives of P, Q and $\|Pv_4\|$ vanish. Differentiation therefore yields

$$\begin{aligned} (w \cdot \nabla_{v_4})\Pi &= -\|Pv_4\|^{-2}(w \cdot (Pv_4)^* + Pv_4 \cdot w^*), \\ (w \cdot \nabla_{v_4})y &= (I - \Pi QB_Q^{-1}Q) \cdot ((w \cdot \nabla_{v_4})\Pi) \cdot u + \Pi QB_Q^{-1}Qw. \end{aligned}$$

Using $\|B_Q^{-1}\| \le 2$ and $\|u\| \le \sqrt{2}\|y\|$ gives

$$\|(w \cdot \nabla_{v_4})y\| \le C\|Qw\| + C\|Pv_4\|^{-1}\|y\|\|w\|. \tag{9.73}$$

For the source derivative of the unnormalized generator, define

$$\widehat{K_0}(V_5, \dots, V_{q-1}; \widehat{v}_4; R_4) := K_0|_{v_4 = \widehat{v}_4}.$$

The product rule in (9.16), together with (9.73) and (9.20), gives

$$\|Pv_4\|\,\|(w \cdot \nabla_{v_4})\widehat{K_0}\|_{\mathrm{HS}} \le C\|Qw\| + C\|K_0\|_{\mathrm{HS}}\|w\|. \tag{9.74}$$

In particular, $\|Qv_4\| \le \|Pv_4\|$ implies

$$\|(w \cdot \nabla_{v_4})y - w\| \le C\|w\|. \tag{9.75}$$

It remains to control differentiation of the scalar normalization. With the matrices fixed,

$$(w \cdot \nabla_{v_4})v_{q-1} = R_{[q-1,5]}w, \qquad (w \cdot \nabla_{v_4})e_{q-1} = A_< \widetilde{R}((w \cdot \nabla_{v_4})y - w).$$

Apply the quotient rule to $\xi = 1 + \|v_{q-1}\|^{-2}\Re(v_{q-1}^* e_{q-1})$. Using (9.75) and $\|e_{q-1}\| \le 2\|A_<\widetilde{R}\|\,\|v_4\|$, then taking the supremum over unit $w \in \mathcal{H}_{v_4}$, gives

$$\|Pv_4\|^2\|\nabla_{v_4}\xi\|^2 \le C\|v_{q-1}\|^{-2}\|A_<\widetilde{R}\|^2\|v_4\|^2\left(1 + \|v_{q-1}\|^{-2}\|Pv_4\|^2\|R_{[q-1,5]}\|^2\right). \tag{9.76}$$

Substituting (9.70) and (9.72) into (9.76) yields

$$\|Pv_4\|^2\|\nabla_{v_4}\xi\|^2 \le \begin{cases} C(g^{-10}W^{-54+3\varepsilon/400} + g^{-8}W^{-48+3\varepsilon/200}), & \text{in (i)},\\ CW^{-30+20\theta}, & \text{in (ii)}. \end{cases} \tag{9.77}$$

These are uniformly bounded: in (i), use $g \ge W^{-\beta}$ and $\beta < 1/2$, and in (ii), use the stipulated small θ. With the normalization derivative controlled in both regimes, define the exceptional real subspace by

$$\mathcal{E}_{v_4} := \{w \in \mathcal{H}_{v_4} : Qw = 0,\ (w\cdot\nabla_{v_4})\xi = 0\}^{\perp_{\mathcal{H}_{v_4}}}.$$

Thus the preceding bound on $\nabla_{v_4}\xi$ and the real codimension of this common kernel give

$$\|Pv_4\|^2\|\nabla_{v_4}\xi\|^2 \le C, \qquad \dim_{\mathbb{R}}\mathcal{E}_{v_4} \le 2\operatorname{rank}Q + 1. \tag{9.78}$$

For $w \in \mathcal{E}_{v_4}^{\perp}\cap\mathcal{H}_{v_4}$, both Qw and $(w\cdot\nabla_{v_4})\xi$ vanish. The identity

$$(w\cdot\nabla_{v_4})\widehat{K} = \xi^{-1}(w\cdot\nabla_{v_4})\widehat{K}_0 - \xi^{-2}K_0(w\cdot\nabla_{v_4})\xi$$

then proves the second estimate in (9.27) by (9.74). For an arbitrary $w \in \mathcal{H}_{v_4}$, the same identity, $1/2 \le \xi \le 3/2$, $\|Qv_4\| \le \|Pv_4\|$, and (9.77) prove the first estimate in (9.27). Step 1 now gives (9.28). Summing (9.27) over an adapted orthonormal basis gives (9.29).

9.4.4 Step 4. The cutoff and its cost

Use nested smooth cutoffs with support restrictions

$$\|Pv_4\|^{-2}\|Qv_4\|^2 \le Cg^2W^{3\varepsilon/200}, \qquad \|Pv_4\|^{-2}\|QPv_4\|^2 \le \tfrac{1}{8}, \qquad \|v_{q-1}\| \ge v_{\min} \quad \text{in (i)}, \tag{9.79}$$

and

$$\|v_4\|^{-2}\|Qv_4\|^2 \le 2W^{-1+14\theta}, \qquad \|v_{q-1}\| \ge v_{\min} \quad \text{in (ii)}. \tag{9.80}$$

Take fixed relative transition widths and the output plateau $\|v_{q-1}\| \ge 2v_{\min}$. By (9.60) and (9.64), the overlap plateaux have probability $1 - o(1)$. By (9.68) and (9.69), the probability that the output lies in the transition region and the chain is regular is at most W^{-D} for every fixed $D > 0$ and sufficiently large W. The QPv_4 restriction is unnecessary when $P = I_W$.

For the source family, multiply these cutoffs by

$$\varphi\Big(W^{20}\|P\widehat{v}_4\|^2/\|\widehat{v}_4\|^2\Big), \qquad \widehat{v}_4 \ne 0,$$

where $\varphi \in C^\infty([0,\infty);[0,1])$ vanishes on $[0,1/4]$ and equals one on $[1,\infty)$. This factor equals one at every prepared v_4 by (9.5); in regime (ii), $P = I_W$. On the resulting support, $\|\widehat{v}_4\| \le 2W^{10}\|P\widehat{v}_4\|$, so (9.70) holds for the source family with at most an extra factor two. Thus $1/2 \le \xi \le 3/2$ there for large W. Define the source cutoff to be zero where $P\widehat{v}_4 = 0$; the new factor vanishes in a neighborhood of this set. It is homogeneous of degree zero, invariant under (9.24), and has zero derivative along $\mathcal{H}_{\widehat{v}_4}$, since both squared norms have zero derivative in these directions. It therefore adds no source derivative cost.

For $w \in \mathcal{H}_{v_4}$, the directional derivatives of P, Q and the norms $\|Pv_4\|, \|v_4\|$ vanish, so

$$\|Pv_4\|^2\|\nabla_{v_4}(\|Pv_4\|^{-2}\|Qv_4\|^2)\|^2 \le 4\|Pv_4\|^{-2}\|Qv_4\|^2.$$

For the overlap cutoff $\widehat{\chi}_Q$, the fixed relative transition width and chain rule therefore give, on its transition,

$$\|Pv_4\|^2\|\nabla_{v_4}\widehat{\chi}_Q\|^2 \le C\|Qv_4\|^{-2}\|Pv_4\|^2.$$

Together with $\|K\|_{\mathrm{HS}}^2 \le 32\|Qv_4\|^2/\|Pv_4\|^2$, this gives

$$\|Pv_4\|^2\|K\|_{\mathrm{HS}}^2\|\nabla_{v_4}\widehat{\chi}_Q\|^2 \le C. \tag{9.81}$$

The same calculation with QPv_4 in place of Qv_4 shows that the fixed-level cutoff for $\|QPv_4\|^2/\|Pv_4\|^2$ costs at most $C\|K\|_{\mathrm{HS}}^2$.

For the lower cutoff $\widehat{\chi}_{q-1}$, take $v_{\min}$ with $\nabla_{v_4} v_{\min} = 0$ and require

$$\widehat{\chi}_{q-1} = 0 \quad (\|v_{q-1}\| \le v_{\min}), \qquad \widehat{\chi}_{q-1} = 1 \quad (\|v_{q-1}\| \ge 2v_{\min}).$$

Differentiating a smooth function of $\|v_{q-1}\|^2/v_{\min}^2$ gives

$$\|Pv_4\|^2 \|\nabla_{v_4} \widehat{\chi}_{q-1}\|^2 \le C v_{\min}^{-2} \|Pv_4\|^2 \|R_{[q-1,5]}\|^2 \mathbf{1}_{\{v_{\min} \le \|v_{q-1}\| \le 2v_{\min}\}}. \tag{9.82}$$

The ratio in (9.82) is polynomially bounded by (9.72). Since (9.21) bounds $\|K\|_{\mathrm{HS}}^2$ on the support, the arbitrarily small polynomial transition probability makes this expected weighted cost bounded in both regimes.

Impose regularity, (9.66), and the spectral moment and rank restrictions of Section 9.4.1, with nested margins. Also impose the matrix support restriction

$$\|(I-P)Q\|^2 \le \tfrac{1}{8}. \tag{9.83}$$

Its plateau has probability $1 - o(1)$ by (9.60); it is automatic when $P = I_W$. Together with the source cutoffs (9.79)–(9.80), it gives $B_Q \ge \frac{3}{4} I_{\operatorname{ran} Q}$ throughout the support for large W. This restriction is invariant under the allowed unitaries and independent of v_4, so it adds no source derivative cost. Before compact exhaustion, remove the inverse-pole sets and the set where a singular value of A equals t_{cut}. The threshold is deterministic, and the latter set is null: clearing inverse denominators gives a nonzero real polynomial, since sufficiently large scalar right onsite matrices make all singular values smaller than the threshold. On this invariant open set, the retained projection and Q are smooth.

Choose a compact subset losing at most W^{-1} mass and saturate it under the allowed compact unitary group. Average a smooth cutoff equal to one on this saturation and supported in the invariant open set. The average is still one on the saturation, so the mass loss remains at most W^{-1} without any invariance assumption on the Gaussian law. Choose the exhaustion measurably in R_4, P and independently of v_4, so this additional cutoff has zero derivative in v_4. Multiply by the source cutoffs above and set the result to zero on failed preparation. This last indicator is evaluated at the actual prepared vector and held fixed throughout the source family. Only the active source cutoffs are varied under (9.24); they are invariant because the source norms are preserved and Q and the output transform jointly. No preparation derivatives enter because $X_1 = \cdots = X_4 = 0$.

All plateaux have common probability $1 - o(1)$. The product rule and (9.81)–(9.82) give

$$0 \le \chi \le 1, \qquad \mathbb{E}\chi \ge \tfrac{1}{2}, \qquad \mathbb{E}\left[\|Pv_4\|^2 \|K\|_{\mathrm{HS}}^2 \|\nabla_{v_4} \widehat{\chi}\|^2\right] \le C. \tag{9.84}$$

This is (9.30) together with the required cutoff mass. Near the support of $\widehat{\chi}$, the inverses and the retained spectral projection are smooth, and ξ is bounded away from zero. Under (9.24), P is fixed, $A, \widetilde{R}, Q, \Pi$ are conjugated by $\mathcal{U}$, and y becomes $\mathcal{U}y$. Thus K_0 is conjugated and ξ is unchanged. Rescaling $\widehat{v}_4$ by $a \ne 0$ scales y by a and leaves K_0, ξ unchanged. This gives the smooth family (9.23) and its equivariance. The support bounds follow from (9.66) and (9.7). The lengths in (9.54) and (9.62) are fixed before W grows, and all estimates are uniform in the energy, incoming flag, and coupling ranges of Lemma 8.12. □

10 Vector fields X from a fixed matrix direction

We prove Lemma 8.13 by the construction explained in Section 3.5, adapting Drogin's construction [10, Proposition 4.1, Lemma 4.2 and Proposition 4.4] to the GUE inputs. Use the transfers (8.9) and set $q = 1000$, $F = T_1 \cdots T_q$ and $\mathcal{T}_q^{\mathrm{rev}} = S^* F S$. The Lie algebra of the group (8.24) is

$$\mathfrak{g} := \{M \in \mathbb{C}^{2W \times 2W} : M^* J_{\mathrm{sig}} + J_{\mathrm{sig}} M = 0,\ \operatorname{Tr} M = 0\}. \tag{10.1}$$

For a fixed incoming flag (U, Y) and any unit $e \in Y \cap U^{\perp}$, set

$$M = \tfrac{1}{\sqrt{2}}\, S(ee^* - (J_{\mathrm{sig}}e)(J_{\mathrm{sig}}e)^*)S^*.$$

Since $J_{\mathrm{sig}}e \perp Y$, $M = M^*$, $\|M\|_{\mathrm{HS}} = 1$ and $S^*MS \in \mathfrak{g}$. The flow of S^*MS fixes U pointwise and multiplies e by $e^{t/\sqrt{2}}$. As $Y = U \oplus \mathbb{C}e$, this gives

$$e^{tS^*MS}U = U, \qquad e^{tS^*MS}Y = Y, \qquad c(e^{tS^*MS}; U, Y) = t/\sqrt{2}. \tag{10.2}$$

A general target M with $S^*MS \in \mathfrak{g}$ has the block form

$$M = \begin{pmatrix} A & B_1 \\ B_2 & -A^* \end{pmatrix}, \qquad B_1 = B_1^*, \quad B_2 = B_2^*, \quad \Im \operatorname{Tr} A = 0. \tag{10.3}$$

We seek a Hermitian input field X with $D_X F = \sqrt{2}\, FM$. Use the real Hilbert–Schmidt inner product $\langle A, B\rangle = \Re \operatorname{Tr}(A^*B)$ for matrix targets and its sum over components for input arrays. Linear maps carry the induced operator norm and Jacobians the Hilbert–Schmidt norm. For a smooth compactly supported field $\chi_P X$, Gaussian integration by parts with covariance I/W gives

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le W\mathbb{E}\|\chi_P X\|^2 + \mathbb{E}\|\nabla_V(\chi_P X)\|_{\mathrm{HS}}^2. \tag{10.4}$$

See the proof of (10.4) in Appendix G.5.

Lemma 10.1 (Gaussian divergence of a lift of a Lie direction)**.** *In the model (1.3), fix $B, E_0 > 0$ and take q, F as above. There is a smooth compactly supported cutoff χ_P in the GUE inputs with*

$$\begin{gathered} 0 \le \chi_P \le 1, \qquad \mathbb{P}(\chi_P = 1) \ge 1 - Ce^{-cW}, \qquad \|\nabla_V \chi_P\| \le C, \\ \operatorname{supp}\chi_P \subset \{\max_i \|V_i\|_{\mathrm{op}} \le C\}. \end{gathered} \tag{10.5}$$

Uniformly for deterministic $W^{-B} \le g \le W^B$, $|E| \le E_0$, and every fixed M of the form (10.3) with $\|M\|_{\mathrm{HS}} = 1$, there is a field X in the original GUE variables such that

$$D_X F = \sqrt{2}\, FM \quad \text{on a neighborhood of } \operatorname{supp}\chi_P,$$

and, for all sufficiently large W,

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le CW \max\{g^{-2}, g^4\}. \tag{10.6}$$

If $g \ge 1$, $\kappa_1 \le |E| \le E_0$, and $M = M^*$, where $0 < \kappa_1 < E_0$ is fixed, a lift using the same cutoff satisfies

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le C_{\kappa_1} W g^2. \tag{10.7}$$

The cut fields satisfy the integration-by-parts regularity conditions of Lemma 8.11. Constants and width thresholds depend only on B, E_0 and, for the second estimate, κ_1.

Set $Q_i := T_{i+1} \cdots T_q$ for $1 \le i < q$ and $Q_q = I$. For a Hermitian input array X, the product rule at fixed E, g gives

$$\mathcal{A}(X) := F^{-1} D_X F = g^{-1} \sum_{i=1}^{q} Q_i^{-1} \begin{pmatrix} 0 & 0 \\ X_i & 0 \end{pmatrix} Q_i. \tag{10.8}$$

Thus both fields solve $\mathcal{A}(X) = \sqrt{2}\, M$. For the cutoff (10.5), the product rule and (10.4) give

$$\mathbb{E}|\delta_{\mathrm{G}}(\chi_P X)|^2 \le C\mathbb{E}\Big[\mathbf{1}_{\operatorname{supp}\chi_P}\left(W\|X\|^2 + \|\nabla_V X\|_{\mathrm{HS}}^2\right)\Big]. \tag{10.9}$$

10.1 The general perturbation and its Gaussian cost

Proof of (10.6) *in Lemma 10.1.* On a neighborhood of $\operatorname{supp}\chi_P$, the reconstruction (G.18)–(G.20) in Appendix G.5 gives a real linear map $\mathcal{K}_0$ on Hermitian input arrays such that, for every test matrix Z of the form (10.3),

$$\mathcal{K}_0(g\mathcal{A}^*Z) = Z. \tag{10.10}$$

Taking adjoints gives $\mathcal{A}(g\mathcal{K}_0^*) = I$, so $X = \sqrt{2}\,g\mathcal{K}_0^*M$ solves $\mathcal{A}(X) = \sqrt{2}M$. We use the following block formulas from the appendix to bound $\mathcal{K}_0$ on arbitrary Hermitian data arrays $\mathbf{R} = (R_1, \dots, R_q)$. Set

$$C_i \coloneqq (\max\{1, g\}EI_W - V_i)/g. \tag{10.11}$$

Define $E_q = 0$ and $E_{i-1} = -R_iC_i - E_i^*$ for $2 \le i \le q$. For odd $3 \le i \le q-1$, set

$$F_i(\mathbf{R}) \coloneqq E_iC_i + C_iE_i^* + C_iR_iC_i + R_{i+1} - R_{i-1}. \tag{10.12}$$

When $\mathbf{R} = g\mathcal{A}^*Z$, let P be the upper-left block of Z. The odd-index identity (G.19) gives

$$C_iP + P^*C_i = F_i(\mathbf{R}). \tag{10.13}$$

For the disjoint odd-index pairs $(i_j, k_j) = (4j+1, 4j-1)$, set

$$\widehat{V}_j \coloneqq 2^{-1/2}g(C_{i_j} - C_{k_j}) = 2^{-1/2}(V_{k_j} - V_{i_j}), \qquad 1 \le j \le 128.$$

Define

$$\mathcal{F}\mathbf{R} \coloneqq (F_{i_j}(\mathbf{R}) - F_{k_j}(\mathbf{R}))_{j=1}^{128}, \qquad \mathcal{L}(P) \coloneqq (\widehat{V}_jP + P^*\widehat{V}_j)_{j=1}^{128},$$

so that the paired equations give $\mathcal{L}P = 2^{-1/2}g\mathcal{F}\mathbf{R}$. The normalized differences $\widehat{V}_j$ are independent GUE matrices. Appendix G.4, (G.11)–(G.12), constructs the cutoff (10.5) with the additional bound

$$s_{\min}(\mathcal{L}|_{\{\Im\operatorname{Tr}P=0\}}) \ge c \quad \text{on a neighborhood of } \operatorname{supp}\chi_P. \tag{10.14}$$

Taking the left inverse on $\{\Im\operatorname{Tr}P = 0\}$ gives

$$\mathcal{J} \coloneqq (\mathcal{L}^*\mathcal{L})^{-1}\mathcal{L}^*, \qquad \mathcal{K}_P \coloneqq 2^{-1/2}g\mathcal{J}\mathcal{F}, \qquad P = \mathcal{K}_P\mathbf{R}.$$

Back substitution in (G.20) gives

$$\mathcal{K}_0\mathbf{R} = \begin{pmatrix} P & PC_q + C_qP^* + C_qR_qC_q - R_{q-1} \\ R_q & -P^* \end{pmatrix}, \qquad P = \mathcal{K}_P\mathbf{R}. \tag{10.15}$$

This formula defines $\mathcal{K}_0$ on all data arrays. The left-inverse identity (10.10) applies when $\mathbf{R} = g\mathcal{A}^*Z$.

Size of the perturbation. On $\operatorname{supp}\chi_P$, $\|C_i\|_{\mathrm{op}} \le C\max\{1, g^{-1}\}$. Each term of $F_i(\mathbf{R})$ contains one data block and at most two C factors, by the recursion for E_i. Since q is fixed,

$$\|F_i(\mathbf{R})\|_{\mathrm{HS}} \le C\max\{1, g^{-2}\}\|\mathbf{R}\|. \tag{10.16}$$

Taking the fixed number of differences defining $\mathcal{F}$ gives

$$|\mathcal{F}|_{\mathrm{op}} \le C\max\{1, g^{-2}\}. \tag{10.17}$$

The lower bound on $s_{\min}(\mathcal{L})$ and (10.17) give

$$\|\mathcal{J}\|,\ \|(\mathcal{L}^*\mathcal{L})^{-1}\| \le C, \qquad \|\mathcal{K}_P\| \le Cg\|\mathcal{F}\| \le C\max\{g^{-1}, g\}.$$

Bounding the four blocks in (10.15) for an arbitrary input array gives

$$\|\mathcal{K}_0\| \le C\big[(1+\|C_q\|_{\mathrm{op}})\|\mathcal{K}_P\| + 1 + \|C_q\|_{\mathrm{op}}^2\big].$$

Substituting the preceding bounds and using $X = \sqrt{2}\, g\mathcal{K}_0^* M$ yields

$$\|\mathcal{K}_0\| \le C\max\{g^{-2}, g\}, \qquad \|X\| \le C\max\{g^{-1}, g^2\}\|M\|. \tag{10.18}$$

Input derivatives and divergence. Let $(H_\alpha)_{\alpha=1}^{W^2}$ be a trace-orthonormal basis of Hermitian matrices and write $D_{i,\alpha} = H_\alpha \cdot \nabla_{V_i}$. For any $W \times W$ matrix B we have

$$\sum_{\alpha=1}^{W^2} H_\alpha^2 = W I_W, \qquad \sum_{\alpha=1}^{W^2} \|H_\alpha B\|_{\mathrm{HS}}^2 = \sum_{\alpha=1}^{W^2} \|B H_\alpha\|_{\mathrm{HS}}^2 = W\|B\|_{\mathrm{HS}}^2. \tag{10.19}$$

Test the adjoints of $\mathcal{F}$ and $\mathcal{L}$ on a Hermitian array $\mathbf{Y} = (Y_j)_{j=1}^{128}$. For example, differentiating the left factor of $C_a R_k C_b$ in direction $D_{a,\alpha}$ gives, in the adjoint, the Hermitian projection of $-g^{-1}H_\alpha Y_j C_b$. Its squared sum over α is bounded by $Wg^{-2}\|Y_j C_b\|_{\mathrm{HS}}^2$ by (10.19). The same bound with at most one remaining C factor applies to each differentiated term of F_i, giving

$$\sum_{i=1}^{q}\sum_{\alpha=1}^{W^2} \|(D_{i,\alpha}\mathcal{F})^*\mathbf{Y}\|^2 \le CW\max\{g^{-2}, g^{-4}\}\|\mathbf{Y}\|^2. \tag{10.20}$$

For $\mathcal{L}$, if $i \in \{i_j, k_j\}$, then V_i occurs only in $\widehat{V}_j$, with $D_{i,\alpha}\widehat{V}_j = \pm 2^{-1/2}H_\alpha$. Its contribution is

$$((D_{i,\alpha}\mathcal{L})P)_j = \pm 2^{-1/2}(H_\alpha P + P^* H_\alpha), \qquad (D_{i,\alpha}\mathcal{L})^*\mathbf{Y} = \pm\sqrt{2}\, H_\alpha Y_j.$$

All unlisted components vanish. The adjoint lies in $\{P : \Im \operatorname{Tr} P = 0\}$ since H_α, Y_j are Hermitian, and (10.19) gives

$$\sum_{i=1}^{q}\sum_{\alpha=1}^{W^2} \|(D_{i,\alpha}\mathcal{L})P\|^2 \le CW\|P\|^2, \qquad \sum_{i=1}^{q}\sum_{\alpha=1}^{W^2} \|(D_{i,\alpha}\mathcal{L})^*\mathbf{Y}\|^2 \le CW\|\mathbf{Y}\|^2. \tag{10.21}$$

Differentiating $\mathcal{J}^* = \mathcal{L}(\mathcal{L}^*\mathcal{L})^{-1}$ gives

$$D_{i,\alpha}\mathcal{J}^* = (I - \mathcal{L}\mathcal{J})(D_{i,\alpha}\mathcal{L})(\mathcal{L}^*\mathcal{L})^{-1} - \mathcal{J}^*(D_{i,\alpha}\mathcal{L})^*\mathcal{J}^*. \tag{10.22}$$

Here $I - \mathcal{L}\mathcal{J}$ is an orthogonal projection. Applying (10.21) to each of the two terms, with its argument fixed at the current input, yields

$$\sum_{i=1}^{q}\sum_{\alpha=1}^{W^2} \|(D_{i,\alpha}\mathcal{J}^*)P\|^2 \le CW\big(\|(\mathcal{L}^*\mathcal{L})^{-1}P\|^2 + \|\mathcal{J}^*P\|^2\big) \le CW\|P\|^2. \tag{10.23}$$

For $\mathcal{K}_P^* = 2^{-1/2}g\mathcal{F}^*\mathcal{J}^*$, the product rule gives one term with $D_{i,\alpha}\mathcal{F}^*$ and one with $D_{i,\alpha}\mathcal{J}^*$. Applying (10.20) to $\mathcal{J}^*P$ in the first, and (10.17) and (10.23) in the second, gives

$$\sum_{i,\alpha} \|(D_{i,\alpha}\mathcal{K}_P)^*P\|^2 \le CWg^2\big(\max\{g^{-2}, g^{-4}\} + \max\{1, g^{-4}\}\big)\|P\|^2 \le CW\max\{g^{-2}, g^2\}\|P\|^2. \tag{10.24}$$

To differentiate $\mathcal{K}_0^* M$ with M fixed, take the adjoint of (10.15). With ι_i denoting insertion in data slot i, cyclicity of the trace gives

$$\mathcal{K}_0^* M = \mathcal{K}_P^*\big(2(A + B_1 C_q)\big) + \iota_q(C_q B_1 C_q + B_2) - \iota_{q-1}B_1. \tag{10.25}$$

Here $\Im \operatorname{Tr}(A + B_1 C_q) = 0$. When the derivative hits $\mathcal{K}_P^*$, apply (10.24) to $2(A + B_1 C_q)$, whose norm is at most $C \max\{1, g^{-1}\}\|M\|$. When it hits C_q, the identity $D_{i,\alpha} C_q = -g^{-1}\delta_{iq} H_\alpha$ and (10.19) bound their contribution to the squared sum by $CWg^{-2}(\|\mathcal{K}_P\|^2 + \|C_q\|_{\mathrm{op}}^2)\|B_1\|^2$. Thus

$$\sum_{i,\alpha} \|D_{i,\alpha}(\mathcal{K}_0^* M)\|^2 \le CW \max\{g^{-4}, g^2\}\|M\|^2.$$

Since $X = \sqrt{2}\, g\mathcal{K}_0^* M$, it follows that

$$\|\nabla_V X\|_{\mathrm{HS}} \le C\sqrt{W} \max\{g^{-1}, g^2\}\|M\|. \tag{10.26}$$

For $\|M\|_{\mathrm{HS}} = 1$, inserting (10.18) and (10.26) into (10.9) proves (10.6). □

10.2 The Hermitian perturbation away from zero energy

Proof of (10.7) *in Lemma 10.1.* Assume $g \ge 1$, $\kappa_1 \le |E| \le E_0$, and $M = M^*$. For a test matrix Z write (P, U, R_q) for the blocks (A, B_1, B_2) in (10.3), and let $\mathbf{R} = g\mathcal{A}^* Z$. Since $C_i = EI_W - g^{-1}V_i$, at an interior odd index (10.13) reads

$$E(P + P^*) = F_i(\mathbf{R}) + g^{-1}(V_i P + P^* V_i).$$

By (10.18), $\|P\|_{\mathrm{HS}} \le \|Z\|_{\mathrm{HS}} = \|\mathcal{K}_0 \mathbf{R}\| \le Cg\|\mathbf{R}\|$. Thus (10.16) and $\|V_i\|_{\mathrm{op}} \le C$ give

$$\kappa_1 \|P + P^*\|_{\mathrm{HS}} \le C\|\mathbf{R}\| + Cg^{-1}\|P\|_{\mathrm{HS}} \le C\|\mathbf{R}\|.$$

By (10.15),

$$U = E(P + P^*) - g^{-1}(PV_q + V_q P^*) + C_q R_q C_q - R_{q-1},$$

so $\|U\|_{\mathrm{HS}} \le E_0\|P + P^*\|_{\mathrm{HS}} + Cg^{-1}\|P\|_{\mathrm{HS}} + C\|\mathbf{R}\|$. Together with $\|R_q\|_{\mathrm{HS}} \le \|\mathbf{R}\|$ this gives

$$\|P + P^*\|_{\mathrm{HS}} + \|U\|_{\mathrm{HS}} + \|R_q\|_{\mathrm{HS}} \le C_{\kappa_1} g\|\mathcal{A}^* Z\|. \tag{10.27}$$

For the Hermitian target M in (10.3), we have $A = A^*$ and $B_1 = B_2 \coloneqq B$. The pairing is

$$\langle M, Z\rangle = \Re \operatorname{Tr}[A(P + P^*) + B(U + R_q)].$$

Cauchy–Schwarz, with $\|M\|_{\mathrm{HS}} = 1$, gives the dual estimate

$$|\langle M, Z\rangle| \le C_{\kappa_1} g\|\mathcal{A}^* Z\|. \tag{10.28}$$

Set $\Gamma = \mathcal{A}\mathcal{A}^*$. Since $\|Z\| \le Cg\|\mathbf{R}\| = Cg^2\|\mathcal{A}^* Z\|$, we have

$$\Gamma \ge cg^{-4} I. \tag{10.29}$$

Take the minimum-norm solution

$$X = \sqrt{2}\, \mathcal{A}^* \Gamma^{-1} M, \qquad \mathcal{A}(X) = \sqrt{2} M, \qquad X \perp \ker \mathcal{A}. \tag{10.30}$$

Applying (10.28) to $Z = \Gamma^{-1} M$ yields

$$\|\mathcal{A}^* \Gamma^{-1} M\|^2 = \langle M, \Gamma^{-1} M\rangle \le C_{\kappa_1} g\|\mathcal{A}^* \Gamma^{-1} M\|$$

and hence $\langle M, \Gamma^{-1} M\rangle \le C_{\kappa_1} g^2$. Using $\|\Gamma^{-1} M\|^2 \le \|\Gamma^{-1}\|\langle M, \Gamma^{-1} M\rangle$ and (10.29), we obtain

$$\|X\|^2 = 2\langle M, \Gamma^{-1} M\rangle \le C_{\kappa_1} g^2, \qquad \|\Gamma^{-1} M\| \le C_{\kappa_1} g^3. \tag{10.31}$$

Input derivatives and divergence. By (8.9), differentiating a factor T_ℓ or T_ℓ^{-1} inserts H_α with a factor g^{-1}. The map in (10.8) and its adjoint already have a prefactor g^{-1}, while all undifferentiated factors are bounded on the cutoff neighborhood. Thus a typical term in $(D_{i,\alpha}\mathcal{A}^*)Z$ has the

form $g^{-2}B_0H_\alpha B_1Z_{ab}B_2$, with bounded B_j, or the same form with H_α to the right of Z_{ab}. By (10.19), its square sum over α is at most $CWg^{-4}\|Z_{ab}\|_{\mathrm{HS}}^2$. For a Hermitian input array H', the same calculation applies to $(D_{i,\alpha}\mathcal{A})H'$ with H'_k in place of Z_{ab}. Summing over the fixed number of terms, blocks, and input slots gives

$$\sum_{i=1}^{q}\sum_{\alpha=1}^{W^2}\|(D_{i,\alpha}\mathcal{A}^*)Z\|^2 \le CWg^{-4}\|Z\|^2, \qquad \sum_{i=1}^{q}\sum_{\alpha=1}^{W^2}\|(D_{i,\alpha}\mathcal{A})H'\|^2 \le CWg^{-4}\|H'\|^2. \tag{10.32}$$

The inverse differentiation identity (10.22), with $\mathcal{L}=\mathcal{A}^*$, applied to (10.30) gives

$$D_{i,\alpha}X = \sqrt{2}\,(I-\mathcal{A}^*\Gamma^{-1}\mathcal{A})(D_{i,\alpha}\mathcal{A}^*)\Gamma^{-1}M - \mathcal{A}^*\Gamma^{-1}(D_{i,\alpha}\mathcal{A})X.$$

The first prefactor in parentheses is an orthogonal projection, and $\|\mathcal{A}^*\Gamma^{-1}\|^2 = \|\Gamma^{-1}\| \le Cg^4$ by (10.29). Applying (10.32) to the two terms and then (10.31) gives

$$\begin{aligned}\|\nabla_V X\|_{\mathrm{HS}}^2 &\le CWg^{-4}\|\Gamma^{-1}M\|^2 + CWg^{-4}\|\mathcal{A}^*\Gamma^{-1}\|^2\|X\|^2\\ &\le CW(g^{-4}\|\Gamma^{-1}M\|^2 + \|X\|^2) \le C_{\kappa_1}Wg^2.\end{aligned} \tag{10.33}$$

Inserting (10.31) and (10.33) into (10.9) proves (10.7). For both fields the inverses in their construction are smooth on a neighborhood of $\operatorname{supp}\chi_P$ by (10.14) and (10.29), respectively. Multiplication by χ_P and extension by zero give the required smooth compactly supported fields. □

Proof of Lemma 8.13. Take the unit Hermitian target M of (10.2). For either field of Lemma 10.1, $D_X\mathcal{T}_q^{\mathrm{rev}} = \sqrt{2}\,\mathcal{T}_q^{\mathrm{rev}}S^*MS$ on $\operatorname{supp}\chi_P$. Equation (10.2) and the cocycle identity give $D_Xc = 1$ and zero variation of both terminal subspaces, as required in (8.34) with $K_{\mathrm{out}} = 0$. The cutoff bound (10.5) gives $\mathbb{E}\chi_P \ge 1/2$ for large W, and (10.6) and (10.7) give the two divergence bounds. Reversing the inputs, field and cutoff restores the order of Definition 8.5 and preserves their bounds by the iid law. □

11 Large-coupling transfer growth

The direct transfer argument is explained in Section 3.6.

11.1 One-step logarithmic growth and its conditional mean

Fix $g \ge 1$, $|E| \le E_0 < 2$ and $\Phi \in \mathcal{F}_W$. To estimate $D_n(\Phi;E)^{-1}$ from (8.10), use the free coordinates S_E, rotation R_E, and angle $E = 2\sin\theta$ of Definition 7.1. Set

$$h_\theta := (2\cos\theta)^{-1}. \tag{11.1}$$

Set

$$J_{\mathrm{ph}} := \begin{pmatrix} 0 & -iI_W \\ iI_W & 0 \end{pmatrix}, \qquad S^*J_{\mathrm{ph}}S = S_E^*J_{\mathrm{ph}}S_E = J_{\mathrm{sig}}.$$

These identities show that $S_E^{-1}S$ preserves J_{sig}-isotropy. For each transfer T_i in (8.9), multiplication gives

$$S_E^{-1}T_iS_E = R_E\left[I - ih_\theta g^{-1}\begin{pmatrix} V_i & V_i \\ -V_i & -V_i\end{pmatrix}\right].$$

Every maximal J_{sig}-isotropic subspace has the graph representation

$$Y_C := \left\{\begin{pmatrix} u \\ Cu\end{pmatrix} : u \in \mathbb{C}^W\right\}, \qquad C \in U(W). \tag{11.2}$$

Since $\Phi \in \mathcal{F}_W$, the transformed initial frame $S_E^{-1}S\Phi$ is also isotropic. Write

$$S_E^{-1}S\Phi = \begin{pmatrix} I_W \\ C_0 \end{pmatrix} B, \quad C_0 \in U(W).$$

Thus the initial subspace in these coordinates is Y_{C_0}. Conditional on $V_1, \dots, V_{i-1}$, the graph $Y_{C_{i-1}}$ is fixed. The independent GUE block V_i gives the next graph Y_{C_i} by

$$\begin{gathered} F(C,V) \coloneqq I - ih_\theta g^{-1} V(I+C), \qquad C_i \coloneqq -e^{-2i\theta}[C_{i-1} + ih_\theta g^{-1} V_i(I + C_{i-1})]F(C_{i-1}, V_i)^{-1}, \\ S_E^{-1} T_i S_E \begin{pmatrix} I_W \\ C_{i-1} \end{pmatrix} = -ie^{i\theta} \begin{pmatrix} I_W \\ C_i \end{pmatrix} F(C_{i-1}, V_i). \end{gathered} \tag{11.3}$$

In particular C_i is unitary by isotropy. For a general initial subspace Y_{C_0}, define the normal-coordinate version of D_n by

$$D_n(C_0; E) \coloneqq \begin{pmatrix} I_W & 0 \end{pmatrix} S_E^{-1}(T_n \cdots T_1) S_E \begin{pmatrix} I_W \\ C_0 \end{pmatrix}. \tag{11.4}$$

Iterating (11.3) gives the factorization

$$D_n(C_0; E) = (-i)^n e^{in\theta} F(C_{n-1}, V_n) \cdots F(C_0, V_1). \tag{11.5}$$

Returning to (8.10) gives

$$D_n(\Phi; E) = \sqrt{2} \begin{pmatrix} I_W & 0 \end{pmatrix} S^* S_E \begin{pmatrix} I_W \\ C_n \end{pmatrix} D_n(C_0; E) B.$$

Isotropy and the uniform bounds on the coordinate changes give

$$2 \left\| \begin{pmatrix} I & 0 \end{pmatrix} S^* S_E \begin{pmatrix} I \\ C_n \end{pmatrix} x \right\|^2 = \left\| S^* S_E \begin{pmatrix} I \\ C_n \end{pmatrix} x \right\|^2 \asymp \|x\|^2.$$

Similarly, $2\|Bx\|^2 = \|S_E^{-1} S\Phi x\|^2 \asymp \|x\|^2$ because $\Phi \in \mathcal{F}_W$. Thus both exterior factors are uniformly invertible on $|E| \le E_0 < 2$, and

$$\|D_n(\Phi; E)^{-1}\| \asymp \|D_n(C_0; E)^{-1}\|. \tag{11.6}$$

We use $A^{-*} \coloneqq (A^{-1})^* = (A^*)^{-1}$. For fixed C_0 and a fixed unit vector v_0, (11.5) therefore gives

$$D_i(C_0; E)^{-*} v_0 = -ie^{i\theta} F(C_{i-1}, V_i)^{-*} D_{i-1}(C_0; E)^{-*} v_0.$$

Normalize after each step. The incoming direction v_{i-1} then depends only on $V_1, \dots, V_{i-1}$ and is independent of V_i. Define

$$v_i \coloneqq -ie^{i\theta} \frac{F(C_{i-1}, V_i)^{-*} v_{i-1}}{\|F(C_{i-1}, V_i)^{-*} v_{i-1}\|}, \quad c_i \coloneqq -\log \|F(C_{i-1}, V_i)^{-*} v_{i-1}\|. \tag{11.7}$$

Multiplication of the normalization factors gives

$$\ell_n(C_0, v_0) \coloneqq -\log \|D_n(C_0; E)^{-*} v_0\| = \sum_{i=1}^{n} c_i. \tag{11.8}$$

Geometrically, v_{i-1} specifies the subspace $\{(u, C_{i-1}u) : v_{i-1}^* u = 0\}$ inside $Y_{C_{i-1}}$, so c_i is also its one-step logarithmic volume ratio (8.17).

Taking v_0 in an orthonormal basis $(e_a)_{a=1}^W$ and using (11.6) bounds the matrix inverse by

$$\|D_n(\Phi;E)^{-1}\|^2 \lesssim \sum_{a=1}^{W} e^{-2\ell_n(C_0,e_a)} \le W \max_{1\le a\le W} e^{-2\ell_n(C_0,e_a)}. \tag{11.9}$$

Write $\mathbb{E}_{i-1}$ for expectation conditional on $V_1,\dots,V_{i-1}$. The trace and directional moments at the incoming state are

$$f_k(C_{i-1},v_{i-1}) \coloneqq v_{i-1}^* C_{i-1}^k v_{i-1}, \qquad m_k(C_{i-1}) \coloneqq W^{-1}\operatorname{Tr} C_{i-1}^k, \qquad f_0 = m_0 = 1. \tag{11.10}$$

For $k\in\mathbb{Z}$, unitarity gives $|m_k|,|f_k|\le 1$.

For functions of the state, write

$$(P_g f)(C_{i-1},v_{i-1}) \coloneqq \mathbb{E}_{i-1} f(C_i,v_i). \tag{11.11}$$

In the calculations below, m_k, f_k without an argument are evaluated at the incoming state, with $m_{-k}=\overline{m_k}$ and $f_{-k}=\overline{f_k}$.

The second-order conditional mean. Uniformly in the incoming state, W, and $|E|\le E_0<2$, for all sufficiently large g we have

$$\mathbb{E}_{i-1}c_i = h_\theta^2 g^{-2}\left[\tfrac12 W^{-1} + \Re f_1 + \Re(m_1 f_1) - \tfrac12 W^{-1}\Re(f_1^2)\right] + O(g^{-4}). \tag{11.12}$$

Proof of (11.12). By (11.7),

$$c_i = -\tfrac12 \log\left[v_{i-1}^*\left(F(C_{i-1},V_i)^* F(C_{i-1},V_i)\right)^{-1} v_{i-1}\right].$$

Multiplication gives

$$F(C_{i-1},V_i)^*F(C_{i-1},V_i) = I + h_\theta g^{-1}A + h_\theta^2 g^{-2}B,$$

where the linear and quadratic terms in V_i are

$$A \coloneqq -i(V_i C_{i-1} - C_{i-1}^* V_i), \qquad B \coloneqq (I+C_{i-1}^*)V_i^2(I+C_{i-1}).$$

Taking the quadratic form against v_{i-1} and then expanding the scalar logarithm gives

$$\mathbb{E}_{i-1}c_i = h_\theta^2 g^{-2}\Big[\tfrac12\mathbb{E}_{i-1}(v_{i-1}^*Bv_{i-1}) - \tfrac12\mathbb{E}_{i-1}(v_{i-1}^*A^2v_{i-1}) + \tfrac14\mathbb{E}_{i-1}\big[(v_{i-1}^*Av_{i-1})^2\big]\Big] + O(g^{-4}).$$

The GUE covariance (1.2) gives

$$\begin{aligned}
\mathbb{E}_{i-1}(v_{i-1}^*Bv_{i-1}) &= v_{i-1}^*(2I + C_{i-1} + C_{i-1}^*)v_{i-1} = 2 + 2\Re f_1,\\
\mathbb{E}_{i-1}(v_{i-1}^*A^2v_{i-1}) &= v_{i-1}^*(2I - m_1C_{i-1} - \overline{m_1}C_{i-1}^*)v_{i-1} = 2 - 2\Re(m_1f_1).\\
\mathbb{E}_{i-1}\big[(v_{i-1}^*Av_{i-1})^2\big] &= 2\mathbb{E}_{i-1}|v_{i-1}^*V_iC_{i-1}v_{i-1}|^2 - 2\Re\mathbb{E}_{i-1}\big[(v_{i-1}^*V_iC_{i-1}v_{i-1})^2\big]\\
&= 2W^{-1}(1-\Re(f_1^2)).
\end{aligned}$$

Substituting these expressions gives the coefficient in (11.12). For the remainder, (11.3) gives the exact identity

$$F(C_{i-1},V_i)^{-1} = I + ih_\theta g^{-1}V_i(I - e^{2i\theta}C_i).$$

Since C_{i-1}, C_i are unitary, it follows that

$$\|F(C_{i-1},V_i)\|, \left\|F(C_{i-1},V_i)^{-1}\right\| \le 1 + Cg^{-1}\|V_i\|, \qquad |c_i| \le \log(1+Cg^{-1}\|V_i\|). \tag{11.13}$$

To justify the expected remainder, expand through cubic order on $\{\|V_i\|\le c_0 g\}$, with a sufficiently small fixed $c_0>0$. Odd terms vanish under $V_i\mapsto -V_i$, and the fourth-order remainder is $Cg^{-4}\|V_i\|^4$. A sphere net and scalar Gaussian tails give

$$\mathbb{P}(\|V_i\|>R) \le 2\exp(2W\log 9 - WR^2/8), \qquad \mathbb{E}\|V_i\|^p \le C_p \tag{11.14}$$

for every fixed p. On the complement of this cutoff, (11.13), boundedness of m_k, f_k, and sufficiently high fixed moments give another $O(g^{-4})$ error. □

The one-step fluctuations. For the fluctuation bounds we differentiate with the incoming state fixed. For a Hermitian variation $\dot V_i$, differentiating (11.3) and (11.7) gives

$$\begin{aligned} dc_i &= \Re[ih_\theta g^{-1} v_i^* F(C_{i-1},V_i)^{-*}(I+C_{i-1}^*)\dot V_i v_i],\\ dC_i &= ih_\theta g^{-1}(C_i - e^{-2i\theta}I)\dot V_i (I+C_{i-1})F(C_{i-1},V_i)^{-1}. \end{aligned} \tag{11.15}$$

The derivative of v_i is the derivative of $F^{-*}v_{i-1}$ followed by differentiation of its normalization in (11.7).

For a bounded real Q put $Q_i = Q(C_i, v_i)$ and $Y_i = c_i - Q_i + Q_{i-1}$ as in [29, Lemma 1], [27, Section 3]. Then

$$\ell_n = Q_n - Q_0 + \sum_{i=1}^n \mathbb{E}_{i-1} Y_i + \sum_{i=1}^n (1-\mathbb{E}_{i-1})Y_i.$$

11.2 The quadratic correction for $g \ge C_\kappa\sqrt{W}$

Fix $0 < E_0 < 2$. For $E_0 = 2-\kappa$, denote the coupling threshold constant below by C_κ.

Theorem 11.1 (Negative transfer moments)**.** *There exist $C_*, W_0, C, c > 0$ and $s \in (0,1)$ such that, for every $W \ge W_0$, finite $g \ge C_*\sqrt{W}$, integer $n \ge 0$, deterministic $|E| \le E_0$, and deterministic $\Phi \in \mathcal{F}_W$,*

$$\mathbb{E}\left\| D_n(\Phi;E)^{-1}\right\|^s \le CW\exp\{-cn/(Wg^2)\}.$$

The initial frame may depend on random variables independent of $V_1,\dots,V_n$. The exponent and width threshold are independent of g, n.

The linear and quadratic corrections. Write $h = h_\theta$, $X = (C, v)$ for the incoming state, and $X' = (C', v')$ for the next state. Denote the increment c_i by $c(X, V)$ and its conditional mean by $P_g c$. On the bulk interval, h, h^{-1} and $(1+e^{-2i\theta})^{-1}$ are uniformly bounded.

Under the free step, f_1 acquires the phase $-e^{-2i\theta}$. To cancel $h^2 g^{-2}\Re f_1$ in (11.12), set

$$Q = -g^{-2}\Re\left[\frac{h^2}{1+e^{-2i\theta}} f_1\right].$$

Define the corrected moments

$$M = m_1 - h^2 g^{-2}\frac{e^{-2i\theta} + m_2 + 2m_1^2}{1+e^{-2i\theta}}, \tag{11.16}$$

$$F = f_1 - h^2 g^{-2}\frac{4m_1 f_1 + 2f_2 - f_1^2 - e^{-2i\theta} f_1 f_{-1}}{1+e^{-2i\theta}}. \tag{11.17}$$

Thus $M = m_1 + O(g^{-2})$, $F = f_1 + O(g^{-2})$.

Lemma 11.2 (Second-moment drifts)**.** *For constants $a, C > 0$, uniformly in the state,*

$$(P_g - I)|M|^2 \le -ag^{-2}|M|^2 + Cg^{-2}(W^{-2} + g^{-4}), \tag{11.18}$$

$$(P_g - I)|F|^2 \le -ag^{-2}|F|^2 + Cg^{-2}|M|^2 + Cg^{-2}(W^{-1} + g^{-4}). \tag{11.19}$$

Proof of Lemma 11.2. Write $A = I - ihg^{-1}V(I+C)$ in the graph update. Expanding with the cutoff used for (11.12) gives, uniformly in the incoming state,

$$P_g m_1 = -e^{-2i\theta}m_1 + e^{-2i\theta}h^2 g^{-2}(1+m_1)(1+2m_1+m_2) + O(g^{-4}), \tag{11.20}$$

$$P_g f_1 = -e^{-2i\theta}f_1 + e^{-2i\theta}h^2 g^{-2}(B_o + B_e) + O(g^{-4} + g^{-2}/W), \tag{11.21}$$

where

$$B_o = (3+m_2)f_1 + 2m_1 f_2 - f_1(m_1 f_1 + m_{-1} f_{-1}),$$
$$B_e = 4m_1 f_1 + 2f_2 - f_1^2 - f_1 f_{-1}.$$

To prove (11.20)–(11.21), expand the graph update and use $\mathbb{E}VBV = W^{-1}\operatorname{Tr}(B)I$. To see the directional term, set $Y = v^* A^{-1} A^{-*} v$ and $N = -e^{2i\theta} v^* A^{-1} C' A^{-*} v$. With $a = h/g$, write $Y = 1 + a y_1 + a^2 y_2 + \cdots$ and $N = f_1 + a n_1 + a^2 n_2 + \cdots$. The contractions give

$$\mathbb{E}y_2 = -[(1+m_1)f_1 + (1+m_{-1})f_{-1}],$$
$$\mathbb{E}n_2 = -(3+4m_1+m_2)f_1 - 2(1+m_1)f_2,$$
$$\mathbb{E}(n_1 y_1) - f_1 \mathbb{E}y_1^2 = O(W^{-1}).$$

Inserting these identities in $\mathbb{E}(N/Y) = f_1 + a^2 \mathbb{E}[n_2 - n_1 y_1 + f_1(y_1^2 - y_2)] + O(g^{-4})$ gives (11.21). The normalized trace of the graph expansion gives (11.20).

The zero-noise step multiplies m_k, f_k by $(-e^{-2i\theta})^k$. Applying it to the correction polynomials yields

$$P_g M = \alpha M + O(g^{-4}), \qquad P_g F = \alpha F + O(g^{-2}(|M| + g^{-2} + W^{-1})),$$
$$\alpha = -e^{-2i\theta}[1 - h^2 g^{-2}(3+m_2)]. \tag{11.22}$$

Since $\Re(3+m_2) \geq 2$, $|\alpha|^2 \leq 1 - a_1 g^{-2}$ for some fixed $a_1 > 0$, when g^{-2} is small.

The Gaussian Poincare inequality in real Hilbert–Schmidt coordinates of variance W^{-1} gives $\mathbb{E}|Z - \mathbb{E}Z|^2 \leq W^{-1}\mathbb{E}\,\|\nabla Z\|_{\mathrm{HS}}^2$. By (11.15) and the normalization in (11.7), the vector derivative has norm at most $Cg^{-1}(1 + g^{-1}\|V\|)^r \|dV\|_{\mathrm{HS}}$ for a fixed r. Product differentiation then gives

$$\|\nabla M(C')\|_{\mathrm{HS}} \leq Cg^{-1}W^{-1/2}(1 + g^{-1}\|V\|)^r, \quad \|\nabla F(C', v')\|_{\mathrm{HS}} \leq Cg^{-1}(1 + g^{-1}\|V\|)^r.$$

The normalized trace supplies $W^{-1/2}$. Thus conditional innovation variances are at most Cg^{-2}/W^2 and Cg^{-2}/W, respectively. In $P_g|Z|^2 = |P_g Z|^2 + \mathbb{E}|Z - P_g Z|^2$, insert (11.22). Young's inequality absorbs small multiples of $g^{-2}|M|^2, g^{-2}|F|^2$. The Cg^{-4} mean remainder costs Cg^{-6}; the other costs $Cg^{-2}(|M|^2 + g^{-4} + W^{-2})$. This proves both bounds. □

Lemma 11.3 (Uniform positive corrected drift)**.** *There are fixed $A, B, a_0, C_*, W_0 > 0$ such that for $W \geq W_0$, $g \geq C_*\sqrt{W}$, the bounded real function*

$$\mathcal{Q} = Q + R, \qquad R = A|M|^2 + B|F|^2$$

satisfies

$$P_g c - (P_g - I)\mathcal{Q} \geq a_0 g^{-2}/W. \tag{11.23}$$

Its supremum norm is independent of W, g. For a fixed sufficiently small $c_0 > 0$, on $\{\|V\| \leq c_0 g\}$, the derivative in V of $c - \mathcal{Q}(C', v')$ is at most C/g in Hilbert–Schmidt norm.

Proof of Lemma 11.3. By (11.12) and (11.21), the linear correction leaves

$$d := P_g c - (P_g - I)Q = \frac{h^2 g^{-2}}{2W} + h^2 g^{-2}\left[\Re(m_1 f_1) - \frac{\Re(f_1^2)}{2W}\right] + O(g^{-4}).$$

Let $h^2/2 \geq a_* > 0$. For any fixed $\varepsilon > 0$, Young's inequality and sufficiently large W give

$$h^2|m_1 f_1| + \frac{h^2}{2W}|f_1|^2 \leq K_\varepsilon |M|^2 + \varepsilon |F|^2 + K_\varepsilon g^{-4}.$$

Choose $B > 0$ so small that the noise term CB/W from (11.19) is at most $a_*/(8W)$. Take $\varepsilon = aB/2$, then choose A so large that $aA - CB \geq K_\varepsilon + 1$. The coefficients of $|M|^2, |F|^2$ in $d - (P_g - I)R$ are nonnegative. The remaining lower bound is

$$g^{-2}\left[\frac{a_*}{W} - C\frac{A}{W^2} - C\frac{B}{W} - Cg^{-2} - C(A + B + K_\varepsilon)g^{-4}\right].$$

Increase C_* to make $Cg^{-2} \le a_*/(8W)$ for $g^{-2} \le C_*^{-2}W^{-1}$, then W_0 to absorb the W^{-2} terms. This proves (11.23), with $a_0 = a_*/2$. Boundedness and the derivative bound follow from the fixed polynomials, the preceding derivative estimates, and (11.13) on the cutoff. □

Set $Y = c(X, V) - \mathcal{Q}(X') + \mathcal{Q}(X)$.

Lemma 11.4 (One-step exponential contraction)**.** *There exist fixed $s, c > 0$ such that, uniformly in the state and in the range of Lemma 11.3, $\mathbb{E}(e^{-sY} \mid X) \le \exp(-cg^{-2}/W)$.*

Proof of Lemma 11.4. Fix the incoming state X.

On the convex set $\mathcal{K}_g = \{\|V\| \le c_0 g\}$, Y is C/g-Lipschitz with the same C for every state. Set

$$\widetilde{Y}(V) \coloneqq \inf_{H \in \mathcal{K}_g} \{Y(H) + (C/g)\|V - H\|_{\mathrm{HS}}\}.$$

This extension has the same Lipschitz constant and is measurable in the incoming state. At zero noise the uncorrected squared moduli do not change, so $Y(0) = O(g^{-2})$, whence

$$|\widetilde{Y}(V)| \le Cg^{-2} + C\,\|V\|_{\mathrm{HS}}/g.$$

Gaussian concentration gives $\mathbb{E}e^{z(\widetilde{Y} - \mathbb{E}\widetilde{Y})} \le \exp(Cz^2 g^{-2}/W)$.

By (11.14), $\mathbb{P}(\mathcal{K}_g^c) \le 2e^{-cWg^2}$, while $|Y| \le C + \log(1 + C\,\|V\|/g)$. For every fixed $|z| \le 1$, the exponential moments of $|Y|, |\widetilde{Y}|$ of order $4|z|$ are uniformly bounded. For the extension use $\mathbb{E}\,\|V\|_{\mathrm{HS}} \le \sqrt{W}$, $g \ge C_*\sqrt{W}$, and Gaussian concentration of the norm. For Y use its logarithmic bound and fixed norm moments. Cauchy–Schwarz on the exceptional set yields

$$|\mathbb{E}Y - \mathbb{E}\widetilde{Y}| \le Ce^{-cWg^2}, \qquad \mathbb{E}e^{z(Y - \mathbb{E}Y)} \le \exp(Cz^2 g^{-2}/W) + Ce^{-cWg^2}.$$

Reducing c absorbs the centering error. Since $\mathbb{E}Y \ge a_0 g^{-2}/W$, choose fixed $s > 0$ with $Cs^2 \le a_0 s/4$. Increase W_0 so that $Ce^{-cWg^2} \le a_0 s\, g^{-2}/(4W)$ uniformly over $g \ge C_*\sqrt{W}$. This uses the decrease of ue^{-cu} for large $u = Wg^2$. Combining the inequalities proves the claim. □

Proof of Theorem 11.1. For $\ell_n = \sum_{i=1}^n c_i$, iterate Lemma 11.4 and telescope the bounded corrector:

$$\mathbb{E}e^{-s\ell_n} \le Ce^{-cn/(Wg^2)}. \tag{11.24}$$

For $0 < s < 1$, (11.9) and subadditivity give $\|D_n(\Phi; E)^{-1}\|^s \le C\sum_{a=1}^W e^{-s\ell_n(C_0, e_a)}$. Applying (11.24) to each term proves the bound for all n. For an independent random frame, condition on the frame.

□

11.3 Higher-order correction near zero energy

Fix $0 < \epsilon < 1$ and $W^\epsilon \le g < W$.

Lemma 11.5 (Transfer growth near zero)**.** *Fix $0 < \epsilon < 1$, $\kappa_0 \in (0, 2)$, and $D' > 0$. There exist $0 < \kappa_1 < (2 - \kappa_0)/2$ and constants C_1, W_0 such that, for every deterministic $W^\epsilon \le g < W$ and $W \ge W_0$, (8.11) holds with*

$$J = [-2\kappa_1, 2\kappa_1], \qquad \mathfrak{b} = Wg^2, \qquad n_0 = \lceil C_1 Wg^2 \log W \rceil.$$

One may take any sufficiently large fixed C_1. The window and the lower threshold for C_1 depend only on ϵ, κ_0, D'; the width threshold may also depend on C_1. The length is independent of E, Φ and the potentials, and the suprema in (8.11) remain outside the probability.

Fix $D > 0$ and a polynomial horizon W^{B_1} with $B_1 > 3$. Put

$$n_*(g, W) := \lceil C_* g^2 \log W \rceil, \tag{11.25}$$

where C_* is a fixed constant. For $n \geq n_*$,

$$\ell_n - \ell_{n_*} = \sum_{i=n_*+1}^{n} \mathbb{E}_{i-1} c_i + \sum_{i=n_*+1}^{n} (c_i - \mathbb{E}_{i-1} c_i). \tag{11.26}$$

For a deterministic function $Q_g(C, v)$, telescoping its increments gives

$$\begin{aligned} \ell_n - \ell_{n_*} = Q_g(C_n, v_n) - Q_g(C_{n_*}, v_{n_*}) + \sum_{i=n_*+1}^{n} \mathbb{E}_{i-1}[c_i - Q_g(C_i, v_i) + Q_g(C_{i-1}, v_{i-1})] \\ + \sum_{i=n_*+1}^{n} (1 - \mathbb{E}_{i-1})[c_i - Q_g(C_i, v_i)]. \end{aligned} \tag{11.27}$$

Lemma 11.6 (Initial interval and martingale fluctuations)**.** *Fix $0 < \epsilon < 1$, $D > 0$, $B_1 > 3$, and a deterministic family of real functions $Q_g = Q_{g,\theta,W}$ on $U(W) \times \{v : \|v\| = 1\}$ satisfying, with a common constant and for real tangent directions,*

$$\begin{aligned} \|Q_g\|_\infty &\leq C g^{-2}, \\ |(\dot{C} \cdot \nabla_C + \dot{v} \cdot \nabla_v) Q_g(C, v)| &\leq C g^{-2} (\|\dot{C}\|_{\mathrm{op}} + \|\dot{v}\|_2). \end{aligned} \tag{11.28}$$

For all sufficiently large W the bounds below hold uniformly for $W^\epsilon \leq g < W$, $|\theta| \leq \delta$ with $\delta > 0$ fixed sufficiently small, and all deterministic initial states.

1. *For any fixed $C_* > 0$ and n_* as in (11.25),*

$$\mathbb{P}\{|\ell_{n_*}| > C \log W\} \leq W^{-D}. \tag{11.29}$$

2. *For each deterministic $n_* \leq n \leq W^{B_1}$, with probability at least $1 - W^{-D}$,*

$$\max_{n_* \leq j \leq n} \left| \sum_{i=n_*+1}^{j} (1 - \mathbb{E}_{i-1})[c_i - Q_g(C_i, v_i)] \right| \leq C \sqrt{n g^{-2} W^{-1} \log W}. \tag{11.30}$$

For a fixed constant in (11.28), the constants are independent of n and the choice of Q_g.

Proof of Lemma 11.6. The expansion (11.12), together with $|m_1|, |f_1| \leq 1$ and bounded h_θ, gives, uniformly in the incoming state,

$$|\mathbb{E}_{i-1} c_i| \leq C g^{-2}. \tag{11.31}$$

The martingale and initial interval. Fix the incoming state. On the cutoff $\|V_i\| \leq K$, the derivative formulas (11.15) and the inverse bound (11.13) give

$$\|\nabla_{V_i} c_i\|_{\mathrm{HS}} \leq C g^{-1}, \qquad \|dC_i\|_{\mathrm{op}} + \|dv_i\|_2 \leq C g^{-1} \|\dot{V}_i\|_{\mathrm{HS}}.$$

The chain rule and (11.28) give $\|\nabla_{V_i} Q_g(C_i, v_i)\|_{\mathrm{HS}} \leq C g^{-3}$ on the same cutoff, and hence $\|\nabla_{V_i}(c_i - Q_g(C_i, v_i))\|_{\mathrm{HS}} \leq C g^{-1}$. Extend this function from the cutoff set with the same Lipschitz constant. After conditional centering, Gaussian concentration gives exponential moment bounds with variance parameter $C g^{-2}/W$, since the covariance of the normalized GUE is W^{-1}. Summing the extended increments from $i = n_* + 1$ and applying the exponential supermartingale inequality at threshold $C\sqrt{n g^{-2} W^{-1} \log W}$ gives failure probability at most $\frac{1}{2} W^{-D}$, with C chosen depending on D.

To return to the true increments, the cutoff complement has probability Ce^{-cW}, while $|c_i| \le \log(1+2h_\theta g^{-1}\|V_i\|)$ and $\|Q_g\|_\infty \le Cg^{-2}$. The extension is bounded by $Cg^{-2}+C\|V_i\|_{\mathrm{HS}}/g$, so its conditional L^2 norm is at most $C(g^{-2}+\sqrt{W}/g)$. Cauchy–Schwarz bounds each conditional-mean error by this polynomial factor times an exponential tail, hence by Ce^{-cW} after decreasing c. Summing the conditional-mean errors and taking a union bound over at most n inputs bounds each contribution by Cne^{-cW}. Since $n \le W^{B_1}$ and $g < W$, these errors are absorbed into the bound in (11.30) and the remaining $\frac{1}{2}W^{-D}$ failure probability for large W. This proves (11.30).

For the initial interval up to n_* in (11.25), combine (11.31) with the same concentration argument applied to $\sum_{i=1}^{n_*}(1-\mathbb{E}_{i-1})c_i$. Using $\ell_{n_*} = \sum_{i=1}^{n_*} c_i$ from (11.8) gives

$$|\ell_{n_*}| \le Cg^{-2}n_* + C\sqrt{n_* g^{-2}W^{-1}\log W} \le C\log W$$

with probability at least $1-W^{-D}$, proving (11.29). □

11.3.1 First-order moment corrections at zero energy

Set $E=0$ and $t=(2g)^{-1}$. Here M_j, F_j denote the first corrections $M_j^{[1]}, F_j^{[1]}$ of Section 3.6. The free update and Gaussian symmetry give, for each fixed k,

$$P_g m_k = (-1)^k m_k + O(g^{-2}), \qquad P_g f_k = (-1)^k f_k + O(g^{-2}). \tag{11.32}$$

First trace and directional moments. Take M_1, F_1 from (11.16)–(11.17) at $\theta=0$. Equations (11.20) and (11.22) give

$$P_g m_1 = -m_1 + t^2(1+m_1)(1+2m_1+m_2) + O(t^4), \tag{11.33}$$

$$P_g M_1 = -[1-t^2(3+m_2)]M_1 + O(t^4). \tag{11.34}$$

For M_1 the order-t^2 coupling to other moments is zero; for F_1 it is $O(t^2|M_1|)$ by (11.22). For each of the fixed trace or directional polynomials Z below, (11.15) bounds its gradient in the new GUE block by $Cg^{-1}(1+g^{-1}\|V\|)^C$, with an additional $W^{-1/2}$ for a normalized trace. Gaussian L^{2p} Poincaré and (11.14) give, conditionally on the incoming state, for fixed integers $p \ge 1$,

$$\|(1-\mathbb{E}_{i-1})Z(C_i,v_i)\|_{2p,i-1} \le C_p g^{-1}\begin{cases} W^{-1}, & Z \text{ involves only normalized traces,} \\ W^{-1/2}, & Z \text{ also involves directional moments.}\end{cases} \tag{11.35}$$

Expanding the $2p$th power around the conditional mean eliminates the linear fluctuation. Young's inequality absorbs a fraction of the $cg^{-2}|M_1|^{2p}$ damping from (11.34), leaving

$$P_g|M_1|^{2p} \le (1-c_p g^{-2})|M_1|^{2p} + C_p g^{-2}(W^{-2p}+g^{-4p}). \tag{11.36}$$

For F_1 the same expansion gives

$$P_g|F_1|^{2p} \le (1-c_p g^{-2})|F_1|^{2p} + C_p g^{-2}(|M_1|^{2p}+W^{-p}+g^{-4p}). \tag{11.37}$$

Iterating first for M_1 and then for F_1 gives, for $n_* \le i \le W^{B_1}$,

$$\|M_1(C_i)\|_{2p} \le C_p(W^{-1}+g^{-2}), \qquad \|F_1(C_i,v_i)\|_{2p} \le C_p(W^{-1/2}+g^{-2}).$$

Here and below C_* in (11.25) is chosen for p: the initial terms are bounded by $C_p(1+i/g^2)^2 e^{-c_p i/g^2}$ before taking the $2p$th root.

Second trace moment. Write $H = I + C$ and $\mathrm{tr}_W(B) = W^{-1}\operatorname{Tr} B$. In (11.3), $C' = -C - itHVH + t^2HVHVH + O(t^3\|V\|^3)$ on the Taylor cutoff. The square of the linear term and the two cross terms with $-C$ give

$$(P_g - I)m_2 = -t^2[\mathrm{tr}_W(H^2)^2 + 2\,\mathrm{tr}_W(H)\,\mathrm{tr}_W(CH^2)] + O(t^4)$$

by $\mathbb{E}VBV = \mathrm{tr}_W(B)I$ and Gaussian symmetry. The odd part of the bracket is $2(3m_1 + m_3 + 4m_1m_2)$. Set $M_2 = m_2 - t^2(3m_1 + m_3 + 4m_1m_2)$ and $a = -3 + \sqrt{8}$. Since $1 + 6a + a^2 = 0$, (11.32) yields

$$P_g(M_2 - a) = [1 - t^2(6 + m_2 + a)](M_2 - a) + O(t^2|M_1| + t^4).$$

Here the remaining terms $6m_1^2 + 2m_1m_3$ are $O(|M_1| + t^2)$. The multiplier has modulus at most $1 - cg^{-2}$. Using (11.35) and iterating as for M_1 gives $\|M_2(C_i) - a\|_{2p} \le C_p(W^{-1} + g^{-2})$ for $i \ge n_*$.

Second directional moment. Let $A = (I - itVH)^{-1}$. Expand $v^*A(C')^2A^*v/(v^*AA^*v)$, contracting each pair by $\mathbb{E}VBV = \mathrm{tr}_W(B)I$. At order t^2, contractions between two separate quadratic forms carry W^{-1}. This gives

$$\begin{aligned} P_g f_2 = f_2 + \tfrac{1}{4}g^{-2}\big[&-2(3 + m_2)f_2 - (m_3 + 6m_1)f_1 - 3m_1f_3 \\ &+ f_2(m_1f_1 + m_{-1}f_{-1}) - (3 + 4m_2)f_1 - 3f_3 \\ &- 8m_1f_2 + (f_1 + f_{-1})f_2\big] + O(g^{-4} + W^{-1}g^{-2}). \end{aligned} \tag{11.38}$$

The free update changes the sign of each term in the bracket defining

$$F_2 := f_2 - \tfrac{1}{2}t^2[(3 + 4m_2)f_1 + 3f_3 + 8m_1f_2 - (f_1 + f_{-1})f_2]. \tag{11.39}$$

Their increments cancel the odd terms in (11.38), leaving

$$P_gF_2 = [1 - 2t^2(3 + m_2)]F_2 + O(g^{-2}(|M_1| + |F_1|) + g^{-4} + W^{-1}g^{-2}). \tag{11.40}$$

Using (11.35) as for F_1 gives

$$P_g|F_2|^{2p} \le (1 - c_pg^{-2})|F_2|^{2p} + C_pg^{-2}(|M_1|^{2p} + |F_1|^{2p} + W^{-p} + g^{-4p}). \tag{11.41}$$

Iteration gives $\|F_2(C_i, v_i)\|_{2p} \le C_p(W^{-1/2} + g^{-2})$ for $i \ge n_*$. In particular all four first corrections have the precision used in Section 3.6 when $g \asymp W^\epsilon$, $0 < \epsilon < 1/4$.

11.3.2 Wick expansion and the first even-mode correction

Lemma 11.7 (Uniform polynomial expansions)**.** *For fixed $0 < \epsilon < 1$, $W^\epsilon \le g < W$, and $|\theta| \le \delta$ with $\delta > 0$ fixed sufficiently small, let $p = p(m, f, W^{-1})$ be a fixed polynomial in finitely many m_k, f_k and in nonnegative powers of W^{-1}. For every fixed integer $s \ge 0$,*

$$P_gp = \sum_{r=0}^{s} g^{-2r} \sum_{\ell \ge 0} W^{-\ell} p_{r,\ell}(m, f; \theta) + O_{p,s}(g^{-2s-2}).$$

Each $p_{r,\ell}$ is a polynomial, and only finitely many are nonzero. Their coefficients and the remainder bound are uniform in W, the energy, and the incoming state $C \in U(W)$, $\|v\|_2 = 1$. For $0 \le r \le s$, the coefficients have the following properties.

(i) Let $\mathscr{J} := (W^{-1}, f_k : k \ne 0)$ be the ideal generated by W^{-1} and the f_k with $k \ne 0$. Then

$$p \in \mathscr{J} \quad \Longrightarrow \quad \sum_{\ell \ge 0} W^{-\ell} p_{r,\ell} \in \mathscr{J}.$$

In particular, $p_{r,0}(m, 0; \theta) = 0$ for $p \in \mathscr{J}$: each term with no explicit W^{-1} retains at least one nonzero-index f_k factor.

(ii) For any two such polynomials $p, \widetilde{p}$,

$$(p\widetilde{p})_{r,0} = \sum_{a+b=r} p_{a,0}\, \widetilde{p}_{b,0}.$$

Here $(p\widetilde{p})_{r,0}$ denotes the coefficient of $g^{-2r}W^0$ in $P_g(p\widetilde{p})$, with the analogous notation for the two factors. Thus the formal W^0 expansion preserves multiplication.

(iii) If p depends only on the m_k's, then

$$p_{r,\ell} = p_{r,\ell}(m;\theta), \qquad p_{r,1} = 0.$$

Thus these expansions contain only traces, and their first possible finite-width correction has a factor W^{-2}.

Proof. Isotropy and the shear/inverse-shear bounds give $\|F(C_{i-1},V_i)^{-1}\| \le 1 + 2h_\theta g^{-1}\|V_i\|$. On $h_\theta g^{-1}\|V_i\| \le c_0$, expand the inverses and normalized vector in (11.7). Fixed-order derivatives are bounded by $C\|V_i\|^a$, uniformly in the state and width; each coefficient at g^{-2r} increases degree by at most $2r$. Gaussian symmetry eliminates the odd powers of V_i after expectation. A sphere-net estimate for GUE quadratic forms gives uniform operator-norm moments of every fixed order. On the complement of the Taylor event, these moments and the boundedness of the observables give any requested power of g^{-1}. This proves the remainder estimate uniformly in the incoming state.

To verify the assertions about the f_k factors, write $\alpha,\beta,\gamma,\delta_0$ for the blocks of the unrotated shear at step i, so $F(C_{i-1},V_i) = \alpha + \beta C_{i-1}$. For an auxiliary w, set $\alpha_w := \alpha - w\gamma$, $\beta_w := \beta - w\delta_0$, $K_w := \alpha_w^{-1}\beta_w$, $\mathcal{S}_w := (I + K_w C_{i-1})^{-1}$. The J_{sig}-unitary block identities give exactly

$$[F(C_{i-1},V_i) - w(\gamma + \delta_0 C_{i-1})]^{-1} F(C_{i-1},V_i)^{-*} = \mathcal{S}_w + \mathcal{S}_0^* - I. \tag{11.42}$$

Indeed, $\alpha_w\alpha^* - \beta_w\beta^* = I$ and multiplication reduces the claim to $\mathcal{S}_w(I - K_wK_0^*)\mathcal{S}_0^* = \mathcal{S}_w + \mathcal{S}_0^* - I$. If $Y := \|F(C_{i-1},V_i)^{-*}v_{i-1}\|^2$, then $Y = v_{i-1}^*(\mathcal{S}_0 + \mathcal{S}_0^* - I)v_{i-1}$ and

$$v_i^*(I + we^{2i\theta}C_i)^{-1}v_i = 1 + v_{i-1}^*(\mathcal{S}_w - \mathcal{S}_0)v_{i-1}/Y.$$

Expanding $\mathcal{S}_w$ shows that every initially marked word has a nonzero same-sign C_{i-1}-power adjacent to its rank-one factor $v_{i-1}v_{i-1}^*$; negative powers are obtained by adjunction.

Contract each Gaussian pair using $\mathbb{E}(V_i)_{ab}(V_i)_{cd} = W^{-1}\delta_{ad}\delta_{bc}$. With r pairs, t initially normalized unmarked cycles and u final unmarked cycles, the coefficient carries $W^{-(r+t-u)}$. Each split increases u by at most one, and a join decreases it. Thus $r + t - u \ge 0$. At exponent zero only splits creating unmarked cycles are possible. Their cuts occur at V_i-letters, so the nonzero same-sign C_{i-1}-power adjacent to $v_{i-1}v_{i-1}^*$ stays in its marked cycle. With no joins, opposite-sign powers from other marked cycles cannot cancel it, and a factor f_k with $k \ne 0$ survives. Starting from an input containing a nonzero-index f_j, a term with no remaining f_k factor therefore carries W^{-1}. When the input contains only normalized traces, $r + t - u$ is twice the number of joins, giving W^{-2} as the first width correction. Joins of two marked cycles cost W^{-1}, so in the coefficient of W^0, contractions act separately on distinct initial factors. This proves properties (i)–(iii); only finitely many words and Gaussian pairings occur at each fixed order. □

The linear f_2 coefficient. At $E = 0$ set $t = (2g)^{-1}$ and $Q^{[1]} = -\frac{1}{2}t^2\Re f_1$. We compute the W^0 coefficient linear in f_2 in $P_g c - (P_g - I)Q^{[1]}$, in the convention of Section 3.6. Let w be a formal variable and use $\mathcal{S}_w$ from (11.42). Write $\mathbb{E}_0$ for the W^0 part of the Wick expansion, $H = I + C$ and $H_w = H(I - wC)^{-1}$. Through any fixed Taylor order,

$$\mathbb{E}_0\mathcal{S}_w = (I + d_wC)^{-1}, \qquad d_w = (s_w - w)/(1 + s_w), \qquad s_w = t^2(1+w)^2\,\mathrm{tr}_W\left(H_w(I + s_wH_w)^{-1}\right).$$

Indeed, putting $u = t(1 + w)$ gives $\mathcal{S}_w = (I - wC)^{-1}(I - iuVH_w)^{-1}(I - iuV)$. For $K = \mathbb{E}_0(I - iuVH_w)^{-1}$, pairing the first V with its partner gives $K = I - u^2\,\mathrm{tr}_W(H_wK)H_wK$. Only noncrossing pairs contribute by the splitting rule in Lemma 11.7. Applying the same pairing to the factor $I - iuV$ on the right gives $\mathbb{E}_0\mathcal{S}_w = (I - wC)^{-1}K(1 + s_w)$, as claimed.

Set $d = d_0$ and $d' = \partial_w d_w|_{w=0}$. In $Y = v^*(\mathcal{S}_0 + \mathcal{S}_0^* - I)v$ the constant term in the directional moments is one. Products factor at W^0, so the coefficient of $\Re f_2$ in $-\frac{1}{2}\mathbb{E}_0 \log Y$ at real trace

centers is $-d^2$. For $P_g f_1$, extract $-[w]$ in the generating function following (11.42). The coefficient of f_2 is $-2dd'$ because the numerator vanishes at $w = 0$ and has no constant directional term. Thus the two contributions in the corrected drift sum to $-d(d + t^2 d')$. Writing $a = -3 + \sqrt{8}$, the trace substitution needed to this order is $m_2 = a + O(t^2)$ and $m_1 = \frac{1}{2}t^2(1 + a) + O(t^4)$: the corrected odd trace has center zero and $M_1 = m_1 - \frac{1}{2}t^2(1 + m_2 + 2m_1^2)$ through its first correction. From the equation for s_w,

$$d = t^2(1 + m_1) - t^4[\mathrm{tr}_W(H)\,\mathrm{tr}_W(H^2) + \mathrm{tr}_W(H)^2] + O(t^6) = t^2 - \tfrac{1}{2}(3 + a)t^4 + O(t^6),$$
$$d' = -1 + (3 + a)t^2 + O(t^4).$$

In particular the fourth-order constant cancels, and the sixth-order coefficient and the corresponding term of the correction are

$$-d(d + t^2 d') = -\tfrac{1}{2}(3 + a)t^6 + O(t^8), \qquad \frac{-\frac{1}{2}(3 + a)t^6}{-2(3 + a)t^2} = \tfrac{1}{4}t^4. \tag{11.43}$$

The divisor is the f_2 coefficient in (11.38) at the trace centers. This verifies the $\Re f_2$ term of $Q^{[2]} - Q^{[1]}$.

11.3.3 Higher-order moment estimates

Lemma 11.8 (Stability of M_k and F_k)**.** *Fix $0 < \epsilon < 1$, $D, \eta > 0$, $B_1 > 3$, an integer $R \geq 2$ with $2\epsilon R > 1$, and finitely many nonzero indices of the moments (11.10). There exist $\delta, C_* > 0$ such that, for all sufficiently large W and $W^\epsilon \leq g < W$, the moments admit polynomial corrections of the form*

$$M_k = m_k + \sum_{r=1}^{R+2} g^{-2r}\mathcal{M}_{k,r}(m;\theta), \quad F_k = f_k + \sum_{r=1}^{R+2} g^{-2r}\mathcal{F}_{k,r}(m, f;\theta).$$

Here $\mathcal{M}_{k,r}$ is a polynomial in finitely many m_j's, and $\mathcal{F}_{k,r}$ is a polynomial in finitely many m_j's and f_j's. Every monomial of $\mathcal{F}_{k,r}$ contains some f_j with $j \neq 0$. The order R, polynomial degrees and index sets are independent of W. The coefficient bounds are uniform on $|\theta| \leq \delta$, with δ chosen for the fixed indices and order. In particular, $M_k = m_k + O(g^{-2})$ and $F_k = f_k + O(g^{-2})$ uniformly in the state. There are also bounded deterministic centers $a_k = a_k(g, \theta)$ such that, for each deterministic initial state and $|\theta| \leq \delta$, for all sufficiently large W, with probability at least $1 - W^{-D}$,

$$|M_k(C_n) - a_k| \leq W^{\eta-1}, \qquad |F_k(C_n, v_n)| \leq W^{\eta-1/2}, \qquad n_* \leq n \leq W^{B_1}, \tag{11.44}$$

simultaneously for these indices and times. Here n_* is given by (11.25). The centers satisfy

$$a_{2j+1} = 0, \qquad |a_2| < 1/4 + O(g^{-2}), \qquad a_{-k} = \overline{a_k}.$$

Proof of Lemma 11.8. **Polynomial corrections.** For this fixed R, starting from the required indices, include those needed by the polynomial corrections, inverse substitutions, center corrections, and fixed-depth moment iterations below, and by the finite Neumann construction and its final residual in Section 11.3.4. Close each index set under lower nonzero indices. Only finitely many indices are added at each step; the triangular inverses use only lower indices. The resulting finite set is independent of W.

Write $[H]_{\leq s,0}$ for the terms of orders $1, g^{-2}, \ldots, g^{-2s}$ in H with no explicit factor W^{-1}, expressed in the coordinates used at that step. For a monomial, set

$$p = \prod_j M_j^{\alpha_j} F_j^{\beta_j}, \qquad \deg p = \sum_j j(\alpha_j + \beta_j), \qquad \alpha_j, \beta_j \in \mathbb{Z}_{\geq 0}.$$

We require

$$\begin{aligned}[P_g M_k]_{\le R+2,0} &= (-1)^k e^{-2ik\theta} M_k + \sum_{r=1}^{R+2} g^{-2r} \sum_{\deg p \equiv k \ (2)} c^M_{k,r,p}\, p(M),\\ [P_g F_k]_{\le R+2,0} &= (-1)^k e^{-2ik\theta} F_k + \sum_{r=1}^{R+2} g^{-2r} \sum_{\deg p \equiv k \ (2)} c^F_{k,r,p}\, p(M,F).\end{aligned} \tag{11.45}$$

Here $p(M)$ contains only trace moments.

Compute the coefficients by Lemma 11.7, keeping the phases at the specified energy. Initialize the recursion by

$$M_k^{(0)} = m_k, \qquad F_k^{(0)} = f_k \qquad (k \neq 0), \qquad M_0^{(s)} = F_0^{(s)} = 1.$$

The superscript s denotes the correction step. At step s, $1 \le s \le R+2$, the already constructed functions have the form

$$M^{(s-1)} = m + \Delta^M(m), \qquad F^{(s-1)} = f + \Delta^F(m,f), \qquad \Delta^M, \Delta^F = O(g^{-2}).$$

Here $m, f, M^{(s-1)}, F^{(s-1)}$ denote the collections of coordinates, and Δ^M, Δ^F are the known correction polynomials at this step. To express m, f in the current coordinates, start with $m^{[0]} = M^{(s-1)}$, $f^{[0]} = F^{(s-1)}$ and substitute recursively:

$$\begin{aligned} m^{[r+1]} &= M^{(s-1)} - \Delta^M(m^{[r]}),\\ f^{[r+1]} &= F^{(s-1)} - \Delta^F(m^{[r]}, f^{[r]}) \qquad (0 \le r < R+2).\end{aligned} \tag{11.46}$$

Truncate after g^{-2R-4} at each substitution. Since the correction polynomials start at g^{-2}, each substitution determines one more order: $m = m^{[r]} + O(g^{-2r-2})$ and $f = f^{[r]} + O(g^{-2r-2})$. Thus $r = R+2$ gives the required inverse substitution.

Now compute the conditional updates in m, f by the expansion rule above, substitute $m^{[R+2]}, f^{[R+2]}$, and collect powers of g^{-2}. For this fixed step s, this defines the polynomials $B^M_{k,r}, B^F_{k,r}$ by

$$\begin{aligned}[P_g M_k^{(s-1)}]_{\le R+2,0} &= (-1)^k e^{-2ik\theta} M_k^{(s-1)} + \sum_{r=1}^{R+2} g^{-2r} B^M_{k,r}(M^{(s-1)}),\\ [P_g F_k^{(s-1)}]_{\le R+2,0} &= (-1)^k e^{-2ik\theta} F_k^{(s-1)} + \sum_{r=1}^{R+2} g^{-2r} B^F_{k,r}(M^{(s-1)}, F^{(s-1)}).\end{aligned} \tag{11.47}$$

At order g^0, the conditional update in the current coordinates multiplies a monomial p of degree d by $(-1)^d e^{-2id\theta}$. Corrections to this rule start at g^{-2}; after multiplication by g^{-2s}, they affect only orders g^{-2s-2} and higher. For each monomial p, take its coefficients in these two polynomials:

$$b^M_{k,s,p} \coloneqq [p] B^M_{k,s}, \qquad b^F_{k,s,p} \coloneqq [p] B^F_{k,s},$$

where $[p]$ denotes the coefficient of p in the current coordinates. For the finitely many opposite-parity pairs (k,d) occurring in these corrections, choose a fixed $\delta > 0$ such that $|k-d|\delta \le \pi/3$ for every pair. Then

$$\inf_{|\theta| \le \delta} \left| (-1)^k e^{-2ik\theta} - (-1)^d e^{-2id\theta} \right| = \inf_{|\theta| \le \delta} 2|\cos((k-d)\theta)| \ge 1. \tag{11.48}$$

Both the window and this lower bound are independent of W. For every monomial whose degree has parity opposite to k, make the simultaneous updates

$$\begin{aligned} M_k^{(s)} &= M_k^{(s-1)} + g^{-2s} \sum_{\deg p \not\equiv k \ (2)} \frac{b^M_{k,s,p}\, p}{(-1)^k e^{-2ik\theta} - (-1)^{\deg p} e^{-2i(\deg p)\theta}},\\ F_k^{(s)} &= F_k^{(s-1)} + g^{-2s} \sum_{\deg p \not\equiv k \ (2)} \frac{b^F_{k,s,p}\, p}{(-1)^k e^{-2ik\theta} - (-1)^{\deg p} e^{-2i(\deg p)\theta}}.\end{aligned} \tag{11.49}$$

Here p is evaluated at the coordinates from step $s-1$; each sum contains only monomials with nonzero coefficients in that update. Expand back in the original m, f and discard powers beyond g^{-2R-4} after each step. Make no correction to same-parity terms. Define negative-index corrections by conjugation. Finally, the exact definitions are

$$M_k \coloneqq M_k^{(R+2)}, \qquad F_k \coloneqq F_k^{(R+2)}. \tag{11.50}$$

For a term $b\,p$ with $\deg p = d$ of parity opposite to k, the increment of the added polynomial in (11.49) multiplies its coefficient by $(-1)^d e^{-2id\theta} - (-1)^k e^{-2ik\theta}$, cancelling b at order g^{-2s}.

By Lemma 11.7, the part of the transition with no explicit W^{-1} preserves multiplication, polynomials in the traces, and the ideal generated by the nonzero-index f_k's. The successive substitutions preserve these properties, and induction in s gives (11.45). Applying (11.46) to these final corrections gives the inverse polynomials $m_k^{[R+2]}(M)$ and $f_k^{[R+2]}(M,F)$, with uniform remainder $O(g^{-2R-6})$. Each inverse substitution preserves the property that every monomial in $f_k^{[R+2]}$ contains a nonzero-index F_j.

The leading terms of the corrected transition. We now compute the order-g^{-2} terms that preserve parity. They determine both the trace centers and the contraction used below. For this coefficient calculation, set $\mathcal{F}(w) \coloneqq \sum_{k\ge 0} f_k w^k$, $\mathcal{M}(w) \coloneqq \sum_{k\ge 0} m_{k+1} w^k$. Let $h_\theta^2 \mathcal{D}_1$ denote the coefficient of $g^{-2}W^0$ in the expected unrotated transition. The shear gives $K_w = -wI - ih_\theta g^{-1}(1+w)^2 V_i + h_\theta^2 g^{-2}(1+w)^3 V_i^2 + O(g^{-3})$. Expanding (11.42) yields, for $k \ge 1$,

$$\mathcal{D}_1 f_k = -[w^k](1+w)^3[1+(1+w)\mathcal{M}(w)]\mathcal{F}'(w) + [(1+m_1)f_1 + (1+m_{-1})f_{-1}]f_k.$$

Retain only monomials whose degree has the same parity as k, and denote the resulting operator by $\mathcal{L}$. For $k > 0$ this gives the *second-order drift formula for the directional moments*,

$$\begin{aligned}\mathcal{L}f_k = \; & -3kf_k - \mathbf{1}_{k\ge 3}(k-2)f_{k-2} - \sum_{n=1}^{k+1} n f_n m_{k+2-n} - 6\sum_{n=1}^{k-1} n f_n m_{k-n} - \sum_{n=1}^{k-3} n f_n m_{k-2-n} \\ & + f_k(m_1 f_1 + m_{-1} f_{-1}).\end{aligned} \tag{11.51}$$

The first two coefficients of the update from C_{i-1} before rotation, in powers of $h_\theta g^{-1}$, are $\delta C_1 \coloneqq i(I+C_{i-1})V_i(I+C_{i-1})$, $\delta C_2 \coloneqq -(I+C_{i-1})V_i(I+C_{i-1})V_i(I+C_{i-1})$. Contracting their contributions to $W^{-1}\operatorname{Tr} C_{i-1}^k$ and retaining monomials of parity k gives

$$\mathcal{L}m_k = -3k\, m_k - 3k\sum_{a=1}^{k-1} m_a m_{k-a} - \frac{k}{2}\sum_{a=0}^{k-2} m_a m_{k-2-a} - \frac{k}{2}\sum_{a=1}^{k+1} m_a m_{k+2-a}. \tag{11.52}$$

Negative indices are conjugates. Empty sums vanish. The only leading higher index is $k+1$, multiplied by m_1. In particular, $\mathcal{L}m_2 = -1-6m_2-m_2^2-6m_1^2-2m_1m_3$, and $\mathcal{L}f_1 = -(3+m_2)f_1 - 2m_1 f_2 + f_1(m_1 f_1 + m_{-1}f_{-1})$. The leading retained map is $y_k \mapsto (-1)^k e^{-2ik\theta}[y_k + h_\theta^2 g^{-2}\mathcal{L}y_k]$; the phase is the input phase, not separate phases of output monomials.

Constructing the centers a_k. The updates of the M_k's involve only the M_j's. Define their truncated expected increment by

$$\mathcal{E}_k(M) \coloneqq [(P_g - I)M_k]_{\le R+2,0}, \qquad \mathcal{E}_k^{\mathrm{lead}}(M) \coloneqq [(P_g - I)M_k]_{\le 1,0},$$

where both polynomials are expressed in the final coordinates. We choose a as an approximate zero of this expected increment. The concentration estimate for $M_k - a_k$ will follow from the contraction of these updates.

First solve the leading equations $\mathcal{E}_{2j}^{\mathrm{lead}}(a^{(0)}) = 0$ in increasing j, with $a_0^{(0)} = 1$, $a_{2j+1}^{(0)} = 0$, and negative indices given by conjugation. The first equation is

$$(a_2^{(0)})^2 + d a_2^{(0)} + 1 = 0, \quad d \coloneqq 6 - g^2 h_\theta^{-2}(1 - e^{4i\theta}), \qquad |a_2^{(0)}| < 1/4. \tag{11.53}$$

The small root specifies $a_2^{(0)}$ uniquely. For $j \geq 2$, the $2j$th equation is linear in $a_{2j}^{(0)}$, with coefficient

$$e^{-4ij\theta}[1 - 2jh_\theta^2 g^{-2}(3 + a_2^{(0)})] - 1;$$

all its other nonzero terms involve already determined even indices. Thus these equations are a recursive definition of every required $a_{2j}^{(0)}$. At $E = 0$ they give $a_{2j}^{(0)} = (-3 + \sqrt{8})^j$.

To include the higher orders, let J be the Jacobian of $\mathcal{E}^{\text{lead}}$ at $a^{(0)}$ on the even coordinates. Starting from $a^{(0)}$, perform R corrections:

$$a^{(s+1)} = a^{(s)} - \mathsf{J}^{-1}\mathcal{E}(a^{(s)}) \quad (0 \leq s < R), \qquad a \coloneqq a^{(R)}. \tag{11.54}$$

Throughout the iteration, keep $a_0 = 1$, set the odd coordinates to zero, and define the negative coordinates by conjugation.

Bounds for the trace centers and the inverse matrices. The trace formula (11.52) gives (11.53). For sufficiently large g, $|d|^2 = 36 + (2g^4 h_\theta^{-4} - 12g^2 h_\theta^{-2})(1 - \cos 4\theta) \geq 36$. The contraction $a \mapsto -1/(d + a)$ on $|a| \leq 1/4$ proves existence and uniqueness of $a_2^{(0)}$ uniformly in $g^2\theta$. At $\theta = 0$, $a_{2p}^{(0)} = (-3 + \sqrt{8})^p$, $a_{2p+1}^{(0)} = 0$ solves all leading rows; substitution in (11.52) gives $-p^2(a_2^{(0)})^{p-1}[1 + 6a_2^{(0)} + (a_2^{(0)})^2] = 0$. The higher even equations are triangular. The diagonal coefficients used in these equations have the form

$$d_k \coloneqq (-1)^k e^{-2ik\theta}[1 - h_\theta^2 g^{-2} k(3 + a_2)] - 1, \qquad |d_k| \geq c_k g^{-2}. \tag{11.55}$$

For the leading trace equations use $a_2^{(0)}$ in place of a_2. For even k, $e^{-2ik\theta} - 1$ competes with g^{-2}, so here $\theta = o(1)$ does not justify setting $\theta = 0$. The numerators in the higher-center equations have a factor g^{-2}; hence the leading centers remain bounded at each fixed index. The second-row Jacobian is $e^{-4i\theta}[1 - h_\theta^2 g^{-2}(6 + 2a_2^{(0)})] - 1$; the higher rows are lower triangular with the diagonals just described. All off-diagonal entries are $O(g^{-2})$, so successive substitution on each fixed finite set gives $\|\mathsf{J}^{-1}\| = O(g^2)$.

The initial residual $\mathcal{E}(a^{(0)})$ is $O(g^{-4})$. In (11.54), a residual $O(g^{-2(s+2)})$ gives a correction $O(g^{-2(s+1)})$ and a new residual $O(g^{-2(s+3)})$: the Jacobian differs from J by $O(g^{-4})$ and the leading second derivative is $O(g^{-2})$. The finite iteration therefore gives

$$a = a^{(0)} + O(g^{-2}), \quad \mathcal{E}(a) = O(g^{-2R-2}), \quad a_{2k+1} = 0, \quad |a_2| < 1/4 + C_R g^{-2}. \tag{11.56}$$

The construction also gives $|M_k|, |F_k| \leq C_{k,R}$ and $|M_2| \leq 1 + C_R g^{-2}$. Their conditional means satisfy the corrected updates in (11.45), with remainders

$$O(g^{-2}W^{-2} + g^{-2R-2}) \quad \text{for the } M_k \text{ updates}, \qquad O(g^{-2}W^{-1} + g^{-2R-2}) \quad \text{for the } F_k \text{ updates}. \tag{11.57}$$

The powers of W^{-1} follow from Lemma 11.7(i), (iii) after the truncated inverse substitution. The same lemma gives a uniform bound on the Taylor remainder. In the moment estimates, $M_{k,n} = M_k(C_n)$ and $F_{k,n} = F_k(C_n, v_n)$ denote evaluation at time n.

The one-step fluctuations. The goal is the conditional innovation bound (11.58). Differentiating $M_k(C_i)$ and $F_k(C_i, v_i)$ by the product rule, using (11.15) and the derivative of the normalized vector v_i, gives

$$\|\nabla_{V_i} M_k(C_i)\|_{\text{HS}} \leq C g^{-1} W^{-1/2}(1 + g^{-1}\|V_i\|)^C,$$
$$\|\nabla_{V_i} F_k(C_i, v_i)\|_{\text{HS}} \leq C g^{-1}(1 + g^{-1}\|V_i\|)^C.$$

The Gaussian L^{2p} Poincaré inequality therefore gives, conditionally on the current state,

$$\|M_{k,n+1} - \mathbb{E}_n M_{k,n+1}\|_{2p,n} \leq C_{k,p} g^{-1} W^{-1}, \quad \|F_{k,n+1} - \mathbb{E}_n F_{k,n+1}\|_{2p,n} \leq C_{k,p} g^{-1} W^{-1/2}. \tag{11.58}$$

The dimension-independent L^{2p} inequality follows from the gradient commutation formula for the Gaussian Ornstein–Uhlenbeck semigroup and Hölder's inequality; its constant depends only

on p. The covariance contributes $W^{-1/2}$ and uniform GUE norm moments bound the polynomial factors.

Contraction of the conditional means. The goal is the moment recursion (11.59). The factors $(-1)^k e^{-2ik\theta}$ have modulus one and therefore do not affect these contraction estimates. For the first trace row and the higher M_k, F_k rows, the principal multiplier has modulus $|1 - h_\theta^2 g^{-2} k(3 + M_2)|$; for the centered second trace it is $|1 - h_\theta^2 g^{-2}(6 + M_2 + a_2)|$. Their squares are $\le 1 - c_k g^{-2}$, since $|M_2| \le 1 + O(g^{-2})$ and $|a_2| < 1/4 + O(g^{-2})$. The remaining leading trace terms involve deviations at lower indices or the product $M_1 M_{k+1}$, which is bounded by $C_k|M_1|$ on the bounded state space. Terms in the F_k update contain lower-index F_j's or are bounded by $C_k|M_1|$, including the nonlinear term in (11.51). Choose positive weights in increasing order of the indices to absorb these finitely many terms into the corresponding diagonal contractions.

For a finite index set J containing, with each k, every j with $0 < |j| \le |k|$, let

$$A_{J,p}(n) := \sum_{k \in J} u_k \mathbb{E}|M_{k,n} - a_k|^{2p}, \qquad B_{J,p}(n) := \sum_{k \in J} v_k \mathbb{E}|F_{k,n}|^{2p},$$

where the positive weights u_k, v_k are chosen as just described and are fixed independently of W. Expansion around the conditional means, using the same conditional-moment inequality as in Section 11.3.1, gives

$$\begin{aligned} A_{J,p}(n+1) &\le (1 - cg^{-2}) A_{J,p}(n) + Cg^{-2}[W^{-2p} + g^{-4p} A_{J',p}(n) + g^{-4pR}], \\ B_{J,p}(n+1) &\le (1 - cg^{-2}) B_{J,p}(n) + Cg^{-2}[W^{-p} + A_{J',p}(n) + g^{-4p} B_{J',p}(n) + g^{-4pR}]. \end{aligned} \tag{11.59}$$

Here J' is a larger fixed set, also closed under lower nonzero indices, containing the indices appearing in the higher-order update terms for J. To justify the powers, use

$$g^{-4}|Y|^{2p-1}|U| \le cg^{-2}|Y|^{2p} + Cg^{-2(2p+1)}|U|^{2p}.$$

Centered innovations cancel the linear term. Their variance term $g^{-2}W^{-2}|Y|^{2p-2}$ leaves $Cg^{-2}W^{-2p}$; for the F_k's replace W^{-2} by W^{-1}. Higher innovation terms from (11.58) are no larger after Young's inequality. The residual g^{-2R-2} leaves g^{-4pR} after dividing by the contraction rate. The finite-width errors in the conditional means are smaller than the variance contributions.

Iteration and the probability bound. We now prove (11.60) and deduce the simultaneous probability bound (11.44). First enlarge the trace index sets to include every index needed in the F_k estimates. Iterate each contracting convolution at least R times, retaining a bounded terminal moment. The normalized geometric kernel has total mass at most one; the coupling to the larger index set gains g^{-4p} at each iteration. Initial-state terms are bounded by $C(1 + g^{-2}n)^K e^{-cg^{-2}n}$ for a fixed finite K, hence by $Ce^{-c'g^{-2}n}$. It follows that

$$A_{J,p}(n) \le C(e^{-c'g^{-2}n} + W^{-2p} + g^{-4pR}), \quad B_{J,p}(n) \le C(e^{-c'g^{-2}n} + W^{-p} + g^{-4pR}). \tag{11.60}$$

Choose R with $2\epsilon R > 1$ and choose p large enough to take a union bound over W^{B_1} times with total failure probability at most W^{-D}. Then choose C_* in (11.25) so that the exponential term is also sufficiently small for every $n \ge n_*$. Markov's inequality at thresholds $W^{\eta-1}$ and $W^{\eta-1/2}$ proves (11.44) simultaneously for the finite set of indices used in the calculation. The sets of indices and the moment order are fixed before the union bound.

This completes the proof of Lemma 11.8. □

11.3.4 Construction of the scalar correction and its drift

Lemma 11.9 (A bounded corrector and positive drift)**.** *Under the coupling assumptions of Lemma 11.5, fix $D > 0$ and a polynomial horizon W^{B_1} with $B_1 > 3$. There are constants $\delta, C_*, C > 0$ such that, for all sufficiently large W and each deterministic $|\theta| \le \delta$, $W^\epsilon \le g < W$, there is a deterministic smooth real-valued function Q_g on $U(W) \times \{v : \|v\|_2 = 1\}$ with the following uniform bounds. The notation Q_g suppresses its dependence on θ, W.*

1. *The size and derivative bounds* (11.28) *hold.*

2. *Let* n_* *be as in* (11.25)*. For each deterministic initial state, with probability at least* $1 - W^{-D}$*, simultaneously for* $n_* \le i - 1 < W^{B_1}$*,*

$$\left|\mathbb{E}_{i-1}[c_i - Q_g(C_i, v_i) + Q_g(C_{i-1}, v_{i-1})] - \tfrac{1}{2}h_\theta^2 g^{-2}W^{-1}\right| \le \tfrac{1}{6}h_\theta^2 g^{-2}W^{-1}. \tag{11.61}$$

Proof of Lemma 11.9. **The coefficient equation.** Use the corrected coordinates M_k, F_k in (11.50) and the centers a_k in (11.54), constructed in Lemma 11.8 through order g^{-2R-4}, with $2\epsilon R > 1$.

Lemma 11.7 and the polynomial substitutions give the expansion of $(P_g - I)F_r$. For $\mathbb{E}_{i-1}c_i$, continue the inverse and logarithm expansions used to prove (11.12). The proof of Lemma 11.7 shows that every positive-order expected coefficient of the squared norm contains a nonzero-index f_k factor or an explicit W^{-1}. The logarithm expansion and the inverse substitution from m, f to M, F preserve this property, with F_k in place of f_k. The same cutoff argument gives a uniform expected remainder at the chosen finite order.

Express both expansions in M, F. In the terms without an explicit W^{-1}, set $M = a$ and extract the terms linear in F:

$$\begin{aligned} \mathbb{E}_{i-1}c_i &= \tfrac{1}{2}h_\theta^2 g^{-2}W^{-1} + \sum_{k\neq 0} b_k F_k + \mathcal{R}_0, \\ (P_g - I)F_r &= \sum_{k\neq 0} \mathsf{A}_{kr}F_k + \mathcal{R}_r. \end{aligned} \tag{11.62}$$

With $[F_k]$ denoting the coefficient of exactly one factor F_k and no other F_j factor, the coefficients are

$$\begin{aligned} b_k &\coloneqq [F_k]\ [\mathbb{E}_{i-1}c_i]_{\le R+2,0}\Big|_{M=a}, \\ \mathsf{A}_{kr} &\coloneqq [F_k]\ [(P_g - I)F_r]_{\le R+2,0}\Big|_{M=a}. \end{aligned} \tag{11.63}$$

Here the brackets have the truncation convention of Section 11.3.3, applied after the polynomial change of variables and before substituting $M = a$. Positive and negative indices are separate formal variables. The remainders collect terms containing a trace deviation $M_j - a_j$ together with an F factor, at least two nonzero-index F_j factors, an explicit W^{-1}, or a discarded Taylor order. The constant positive drift is kept separate from b.

For $Q_g = \sum_r q_r F_r$, we obtain

$$\mathbb{E}_{i-1}c_i - (P_g - I)Q_g = \tfrac{1}{2}h_\theta^2 g^{-2}W^{-1} + \sum_k [b_k - (\mathsf{A}q)_k]F_k + \mathcal{R}_0 - \sum_r q_r \mathcal{R}_r. \tag{11.64}$$

(1) Bounding the remainders. The goal is (11.65). Apply (11.44) with $\eta/2$ in place of η, where $0 < \eta < \min(\epsilon, 1/4)$. For the fixed finite set of indices used below, it gives

$$|\mathcal{R}_0| = o(W^{-1}g^{-2}), \qquad \max_r |\mathcal{R}_r| \le C_R W^{-1+\eta}g^{-2}. \tag{11.65}$$

Indeed, the terms in $\mathcal{R}_r$ start at g^{-2} and contain $(M_j - a_j)F_k$, two F factors, or an explicit W^{-1}; (11.44) bounds each of these by $O(W^{-1+\eta})$. For $\mathcal{R}_0$, (11.12) and $a_1 = 0$ show that its order-g^{-2} terms are bounded by constant multiples of $g^{-2}|(M_1 - a_1)F_1|$ and $g^{-2}W^{-1}|F_1|^2$, both $o(W^{-1}g^{-2})$ by (11.44). The remaining terms have the extra factor g^{-2} and total size $O(g^{-4}W^{-1+\eta}) = o(W^{-1}g^{-2})$. The Taylor and inverse-substitution errors $O(g^{-2R-2})$ are also negligible since $2\epsilon R > 1$. Thus, keeping the coefficient factor explicit,

$$\left|\mathcal{R}_0 - \sum_r q_r \mathcal{R}_r\right| \le o(W^{-1}g^{-2}) + C_R W^{-1+\eta}g^{-2}\sum_r |q_r|. \tag{11.66}$$

(2) Solving the coefficient equation. The goal is the coefficient and residual bound (11.69). By (11.45), every retained monomial in the update of F_r has degree congruent to r modulo two. At $M = a$, a monomial containing an odd-index M_j vanishes, since $a_{2j+1} = 0$. A surviving term linear in F_k therefore has $k \equiv r \pmod 2$. Thus

$$\mathsf{A}_{kr} = 0 \quad (k \not\equiv r \pmod 2), \qquad \mathsf{A} = \begin{pmatrix} \mathsf{A}_{\mathrm{odd}} & 0 \\ 0 & \mathsf{A}_{\mathrm{even}} \end{pmatrix}.$$

Let A_0 consist of the terms of orders 1 and g^{-2} in A:

$$(\mathsf{A}_0)_{kr} := [F_k]\,\big[(P_g - I)F_r\big]_{\le 1,0}\Big|_{M=a}.$$

In (11.51), set $m = a$ and $f = F$. Since $a_1 = 0$, the only higher-index term $-(k+1)a_1F_{k+1}$ vanishes; the vanishing of all odd a_j leaves only F_k and lower indices of the same parity, so each parity block of A_0 is triangular. The diagonal contribution is $-3kF_k - ka_2F_k = -k(3+a_2)F_k$, where $-ka_2F_k$ comes from $n = k$ in the first sum. Including the phase and subtracting I gives the diagonal entry, for $k > 0$,

$$(-1)^k e^{-2ik\theta}[1 - kh_\theta^2 g^{-2}(3 + a_2)] - 1,$$

and the negative-index diagonals are the conjugates. Shrink δ so that $|k|\delta \le \pi/3$ for every retained odd index. Then $|1 + e^{-2ik\theta}| \ge 1$, so the odd diagonals remain bounded away from zero after the $O(g^{-2})$ correction. The even diagonals have modulus at least $c_k g^{-2}$ by (11.55). Off-diagonal entries are $O(g^{-2})$, so triangular inversion gives $O(1)$ on the odd block and $O(g^2)$ on the even block. In particular,

$$\|\mathsf{A} - \mathsf{A}_0\| \le C_R g^{-4}, \qquad \|\mathsf{A}_0^{-1}(\mathsf{A} - \mathsf{A}_0)\| \le C_R g^{-2}.$$

The second bound uses the exact energy dependence of the even diagonals; it does not replace θ by zero.

Since $m_1 = M_1 + O(g^{-2})$, $f_1 = F_1 + O(g^{-2})$, and $a_1 = 0$, (11.12) gives

$$\sum_k b_k F_k = h_\theta^2 g^{-2} \Re F_1 + \sum_k O(g^{-4}) F_k.$$

Thus the odd coefficients of b start at g^{-2}, whereas the even coefficients start at g^{-4}. The extra factor g^{-2} in the even right-hand side compensates for its $O(g^2)$ inverse:

$$\|(\mathsf{A}_0^{-1}b)_{\mathrm{odd}}\|_1 \le C_R g^{-2}, \qquad \|(\mathsf{A}_0^{-1}b)_{\mathrm{even}}\|_1 \le C_R g^2 g^{-4} = C_R g^{-2}.$$

Define the successive coefficient corrections by

$$q^{(0)} := \mathsf{A}_0^{-1} b, \qquad q^{(s+1)} := -\mathsf{A}_0^{-1}(\mathsf{A} - \mathsf{A}_0)q^{(s)}, \quad 0 \le s < R, \tag{11.67}$$

and set

$$q := \sum_{s=0}^{R} q^{(s)}, \qquad Q_g(C_{i-1}, v_{i-1}) := \sum_{k \ne 0} q_k F_k(C_{i-1}, v_{i-1}). \tag{11.68}$$

Each correction gains a factor g^{-2}: $\|q^{(s)}\|_1 \le C_R g^{-2s-2}$. Telescoping the coefficient equations gives $\mathsf{A}q - b = (\mathsf{A} - \mathsf{A}_0)q^{(R)}$, and hence

$$\sum_k |q_k| \le C_R g^{-2}, \qquad \|\mathsf{A}q - b\|_1 \le C_R g^{-2R-6}. \tag{11.69}$$

The recursion acts on the retained polynomial coefficients.

The finite-index convention in Section 11.3.3 includes the support of b, the indices in all R corrections and the final residual, and the trace indices in their coefficients and centers. Apply Lemma 11.8 to this entire list; no term is dropped because its index lies outside an initial list.
(3) The norm and derivative bounds for Q_g. The goal is (11.28). Conjugation of (11.67) gives $q_{-k} = \overline{q_k}$, so Q_g is real. For real tangent directions $\dot{C}, \dot{v}$,

$$|(\dot{C} \cdot \nabla_C) m_j| \le |j| \|\dot{C}\|_{\mathrm{op}},$$
$$|(\dot{C} \cdot \nabla_C + \dot{v} \cdot \nabla_v) f_j| \le |j| \|\dot{C}\|_{\mathrm{op}} + 2\|\dot{v}\|_2.$$

The finite polynomial definitions of F_k, their bounded coefficients, and $|m_j|, |f_j| \le 1$ therefore give

$$|F_k| \le C_R, \qquad |(\dot{C} \cdot \nabla_C + \dot{v} \cdot \nabla_v) F_k| \le C_R(\|\dot{C}\|_{\mathrm{op}} + \|\dot{v}\|_2).$$

Combining these bounds with (11.69) proves both bounds in (11.28).
Completion of the proof. By (11.69) and the bound $|F_k| \le C_R$, the linear residual in (11.64) has modulus at most $C_R g^{-2R-6}$. Substituting $\sum_r |q_r| \le C_R g^{-2}$ into (11.66) and using (11.65) therefore gives

$$\begin{aligned} &\left| \mathbb{E}_{i-1} c_i - (P_g - I) Q_g - \tfrac{1}{2} h_\theta^2 g^{-2} W^{-1} \right| \\ &\quad \le o(W^{-1} g^{-2}) + C_R W^{-1+\eta} g^{-4} + C_R g^{-2R-6} = o(W^{-1} g^{-2}), \end{aligned} \tag{11.70}$$

since $g \ge W^\epsilon$, $\eta < \epsilon$, and $2\epsilon R > 1$. For sufficiently large W, the error in the corrected drift is at most $h_\theta^2/(6Wg^2)$. On the event (11.44), this holds simultaneously for $n_* \le i - 1 < W^{B_1}$, with probability at least $1 - W^{-D}$. This proves (11.61). □

11.3.5 From the corrected drift to transfer growth

Lemma 11.10 (Scalar growth)**.** *Under the coupling assumptions of Lemma 11.5, let ℓ_n be defined by (11.8). For every $D > 0$, there are $\delta > 0$ and $C_D > 0$ such that, for every fixed $C_1 \ge C_D$ and all sufficiently large W, the deterministic length $n = \lceil C_1 W g^2 \log W \rceil$ satisfies*

$$\sup_{|\theta| \le \delta} \sup_{\substack{C_0 \in U(W) \\ \|v_0\|_2 = 1}} \mathbb{P}\{\ell_n(C_0, v_0) < D \log W\} \le W^{-D}, \tag{11.71}$$

uniformly for $W^\epsilon \le g < W$. The constants δ, C_D depend only on ϵ, D; the width threshold may also depend on C_1.

Proof of Lemma 11.10. For $n \ge n_*$, take absolute values of the error terms in (11.27) to obtain

$$\begin{aligned} \ell_n \ge \frac{h_\theta^2 g^{-2}(n - n_*)}{2W} - |\ell_{n_*}| - 2\|Q_g\|_\infty - \left| \sum_{i=n_*+1}^{n} (1 - \mathbb{E}_{i-1})[c_i - Q_g(C_i, v_i)] \right| \\ - \sum_{i=n_*+1}^{n} \left| \mathbb{E}_{i-1}[c_i - Q_g(C_i, v_i) + Q_g(C_{i-1}, v_{i-1})] - \tfrac{1}{2} h_\theta^2 g^{-2} W^{-1} \right|. \end{aligned} \tag{11.72}$$

Fix $D > 0$ and $B_1 > 3$. Apply Lemmas 11.9 and 11.6 with probability exponent $D + 3$. For their common n_* from (11.25) and each deterministic $n_* \le n \le W^{B_1}$, the four bounds in those lemmas imply

$$\ell_n \ge h_\theta^2 g^{-2}(n - n_*)/(3W) - C \log W - C\sqrt{n g^{-2} W^{-1} \log W}, \tag{11.73}$$

with failure probability at most $3W^{-D-3}$. Here the martingale sum is bounded directly by (11.30), and the endpoint cost Cg^{-2} is absorbed into $C \log W$.

Since $h_\theta^2 \geq \frac{1}{4}$, at $n = \lceil C_1 W g^2 \log W \rceil$ equation (11.73) gives

$$\ell_n \geq (C_1/12 - C\sqrt{C_1} - C) \log W,$$

absorbing the initial-length contribution $O(C_* \log W/W)$. Thus $\ell_n \geq D \log W$ for every fixed $C_1 \geq C_D$, with failure probability at most W^{-D} for large W. The choice of B_1 puts n inside the stated horizon. Uniformity in the deterministic initial state and energy gives (11.71). □

Proof of Lemma 11.5. Apply Lemma 11.10 with $D > D' + 2$ and choose $0 < \kappa_1 < \min\{\sin\delta, (2 - \kappa_0)/2\}$. Applying (11.9) and taking a union bound over the W basis vectors gives (8.11) with $J = [-2\kappa_1, 2\kappa_1]$, $\mathfrak{b} = Wg^2$, and the length $n_0 = n$ from Lemma 11.10, uniformly in the deterministic energy and initial frame. □

Completion of the upper bound

Proof of Theorem 3.4. Set $J = [-2 + \kappa, 2 - \kappa]$ and $\mathfrak{b} = Wg^2$.
The range $g \geq W$. For sufficiently large W, we have $g \geq W \geq C_\kappa\sqrt{W}$, so Theorem 11.1 gives (8.12), uniformly in g, n. Proposition 8.4(ii) converts this to (8.3) with $q = s/4$, and Lemma 8.1(ii) gives (3.6) with $R \leq C_D \mathfrak{b} \log W$. The constants and width threshold are independent of g, L.
The range $W^\epsilon \leq g < W$. This range is empty if $\epsilon \geq 1$. For $0 < \epsilon < 1$, we have $\mathfrak{b} < W^3$. Fix $0 < q < 1/4$ and the exponent D' required by Proposition 8.4(i) with $t = 2q$ and polynomial-range parameter $C_0 = 3$. Lemma 11.5 gives (8.11) on $[-2\kappa_1, 2\kappa_1]$, with scale $\mathfrak{b} = Wg^2$. The near-zero window is fixed after D'; neither depends on the exceptional-count exponent D. On $[\kappa_1, 2 - \kappa]$ and $[-2 + \kappa, -\kappa_1]$, Lemma 8.13(ii), followed by Lemmas 8.11, 8.10 and 8.6, gives the same transfer input with batch length 1000.

On each interval, Proposition 8.4(i) gives (8.6), and Proposition 8.2 gives (8.2) with exponent q and scale $\mathfrak{b} = Wg^2$. Choose the length parameter C_1 in the finite-volume criterion to cover the resulting length constants, keeping $C_0 = 3$. The lengths may differ between the three intervals; taking the larger prefactor exponent and smaller decay rate gives (8.2) on their union J. All energy and frame suprema in the transfer inputs remain outside the probability; there is no union over couplings or volumes. Lemma 8.1(i) gives (3.6) with $R \leq C_D \mathfrak{b} \log W$, with constants and width threshold uniform in this range and in L.

Taking the larger constants for the two ranges gives the same bound uniformly for all $g \geq W^\epsilon$. For sufficiently large W, $C_D \log W \leq (\log W)^2$, proving (3.10) with a width threshold independent of g, L. When the radius fills the cycle, the tail is empty. □

A Dyson solutions and deterministic kernel estimates

This appendix proves the deterministic results of Section 6, together with the scalar comparison facts used in Sections 5 and 7. Each auxiliary result is proved before its use.

A.1 Free Green functions and scalar identities

To compare onsite variances at fixed hopping in Lemma 6.3(a), allow a general variance $\gamma > 0$ and put $A_{g,\gamma} := \sqrt{\gamma}\, g A_L$. The choice $\gamma = \gamma_g$ gives $A_{g,\gamma_g} = A_g$ and recovers the notation of (5.2)–(5.3). For $w \in \mathbb{C}_+$, define

$$\begin{aligned} f_g(w;\gamma) &:= \int_{-\pi}^{\pi} \frac{\mathrm{d}p/(2\pi)}{2\sqrt{\gamma}\, g\cos p - w} = -\frac{1}{\sqrt{w^2 - 4\gamma g^2}}, \\ f_{g,L}(w;\gamma) &:= L^{-1} \sum_{p \in (2\pi/L)\mathbb{Z}_L} \frac{1}{2\sqrt{\gamma}\, g\cos p - w}. \end{aligned} \tag{A.1}$$

The square root is asymptotic to w at infinity and has positive imaginary part on $\mathbb{C}_+$. At $g = 0$, $f_0(w;\gamma) = -1/w$. The scalar Dyson solutions and block matrix are

$$\begin{aligned} m_g(z;\gamma) &= f_g(z+\gamma m_g(z;\gamma);\gamma), \\ m_{g,L}(z;\gamma) &= f_{g,L}(z+\gamma m_{g,L}(z;\gamma);\gamma), \\ M_B(z;\gamma) &\coloneqq (A_{g,\gamma} - z - \gamma m_{g,L}(z;\gamma))^{-1}, \qquad M(z;\gamma) \coloneqq M_B(z;\gamma)\otimes I_W. \end{aligned} \tag{A.2}$$

We retain the Herglotz normalization and regular boundary values, write $m \coloneqq m_{g,L}(z;\gamma)$, and suppress fixed variance arguments. Let $\nu_g(\cdot;\gamma), F_g(\cdot;\gamma)$ be the measure and distribution function of $m_g(\cdot;\gamma)$. At $\gamma = \gamma_g$, $m_{g,L}(z;\gamma_g) = m_{g,L}(z)$ and $M_B(E;\gamma_g) = M_B(g,E)$, as in (5.2)–(5.3).

At a real regular energy E, take $m = m_{g,L}(E;\gamma)$ and $M = M(E;\gamma)$. Translation invariance and (A.2) give $M_{aa} = mI_W$. With $\Lambda = A_{g,\gamma}\otimes I_W$, $z_t = E + (1-t)\gamma m$ and $\mathcal{S}[R] = \gamma W^{-1}\operatorname{diag}(\operatorname{Tr} R_{aa})_a \otimes I_W$, direct substitution gives

$$\mathcal{S}[M] = \gamma m I_{WL}, \qquad \Lambda - z_t - t\mathcal{S}[M] = \Lambda - E - \gamma m = M^{-1}. \tag{A.3}$$

The spatial decay factor ρ of the free Green function will be used for finite-volume comparisons and the calculation of propagation scales. For $w \in \mathbb{C}_+$ put

$$\rho(w) \coloneqq \frac{2\sqrt{\gamma}\, g}{w + \sqrt{w^2 - 4\gamma g^2}}, \qquad |\rho(w)| < 1. \tag{A.4}$$

Lemma A.1 (Free Green functions)**.** *Write $R_{g,\infty}(x;w)$ and $R_{g,L}(x;w)$ for the $(0,x)$ entries of the free resolvents on $\mathbb{Z}$ and $\mathbb{Z}_L$, respectively, both with hopping $\sqrt{\gamma}\, g$. Thus $R_{g,L}(x;w) = ((A_{g,\gamma} - w)^{-1})_{0x}$. For $0 \le x < L$, these entries satisfy*

$$R_{g,\infty}(x;w) = f_g(w)\rho(w)^{|x|}, \qquad R_{g,L}(x;w) = f_g(w)\frac{\rho(w)^x + \rho(w)^{L-x}}{1-\rho(w)^L}. \tag{A.5}$$

At $x = 0$ the numerator is $1+\rho^L$. At $g = 0$ the limiting row is $-w^{-1}\mathbf{1}_{x=0}$. In particular, writing $M_B(x) = (M_B)_{0x}$, we have $M_B(x) = R_{g,L}(x; z+\gamma m)$.

Proof of Lemma A.1 and (A.1). For $g > 0$, apply the free Jacobi Green function formula [30, Section 1.3, (1.121)] at spectral parameter $w/(2\sqrt{\gamma}\, g)$ and multiply by $(2\sqrt{\gamma}\, g)^{-1}$. This gives the line formula, including $f_g(w)$ at $x = 0$; Fourier diagonalization gives the finite sum in (A.1). Periodization and geometric summation yield

$$R_{g,L}(x;w) = \sum_{n\in\mathbb{Z}} R_{g,\infty}(x+nL;w) = f_g(w)\frac{\rho(w)^x + \rho(w)^{L-x}}{1-\rho(w)^L}.$$

The $g = 0$ formula is immediate.

Proof of (A.6). For the following identities, take $m = m_g(z;\gamma)$ with normalized Fourier integration on the line, or $m = m_{g,L}(z;\gamma)$ with normalized averaging over the finite grid, and denote the corresponding average by $\langle\cdot\rangle$. Write $R(p) \coloneqq (2\sqrt{\gamma}\, g\cos p - z - \gamma m)^{-1}$ and $z = E + i\eta$. Taking imaginary parts in the Dyson equation gives

$$\langle |R|^2\rangle = \frac{\Im m}{\eta + \gamma \Im m}, \qquad |m| \le \gamma^{-1/2}, \qquad \Re(1 - \gamma\langle R^2\rangle) \ge 2\gamma(\Im m)^2. \tag{A.6}$$

The middle bound follows from $|m|^2 \le \langle |R|^2\rangle$. For the last inequality, use $\Re R^2 = |R|^2 - 2(\Im R)^2$ and Jensen's inequality. For real E with $\Im m_{g,L}(E;\gamma) > 0$, Parseval gives the exact identity $\gamma\sum_x |M_B(x)|^2 = 1$. For $\eta > 0$, the difference of two Dyson equations has coefficient of modulus less than one by Cauchy–Schwarz and the first identity; this proves uniqueness. The same identities control regular boundary values by continuity.

A.2 Bulk regularity and the deterministic distribution

We first prove the scalar regularity used in the distribution and finite-cycle statements.

Lemma A.2 (Bulk regularity). *Fix $0 < C < \infty$, $\kappa \in (0, 1/2)$, and $\max\{1, C\}^{-2} \le \gamma \le 1$. Use the Dyson functions and their probability measures in (A.2), suppressing the fixed variance argument γ throughout. There are $b_\kappa, a_{\mathrm{strip}}, c, C' > 0$, depending only on C, κ, such that, whenever*

$$g \le C, \qquad \operatorname{dist}(x, \{E : F_g(-|E|; \gamma) \ge \kappa\}) \le b_\kappa, \qquad z = x + i\eta, \qquad 0 \le \eta \le 1. \tag{A.7}$$

we have $|x| \le 2C + 2$ and

$$|m_g(z)| \le \gamma^{-1/2}, \qquad \Im m_g(z) \ge a_{\mathrm{strip}}, \qquad \Re(1 - \gamma f_g'(z + \gamma m_g(z))) \ge 2\gamma(\Im m_g(z))^2. \tag{A.8}$$

Every fixed mixed derivative of m_g in g and z is uniformly bounded on this set. The free row defined in (A.5) satisfies

$$\sum_{n \in \mathbb{Z}} e^{c|n|} |R_{g,\infty}(n; z + \gamma m_g(z))| \le C'.$$

The same conclusions hold with the bulk interval replaced by interval $\sqrt{\gamma}[-2 + \kappa_0, 2 - \kappa_0]$, $\kappa_0 \in (0, 2)$, with constants depending on C, κ_0.

Proof of Lemma A.2. First take $\gamma = 1$. For the law τ_g of $2g \cos p$, the boundary description in [5, Lemmas 2–4, Proposition 2 and Corollary 3, pp. 709–711] uses the negative of our Stieltjes transform. Since $\int (u-y)^{-2} \tau_g(dy)$ is infinite on $[-2g, 2g]$ and strictly decreases to zero for $u > 2g$, its set U_1 is $(-b_g, b_g)$; at $g = 0$, $b_0 = 1$. The boundary homeomorphism and density formula therefore give one positive-density band $(-e_g, e_g)$, with

$$\int (b_g - y)^{-2} \tau_g(dy) = 1, \quad b_g \le 2g + 1, \quad 2 \le b_g + b_g^{-1} \le e_g = b_g + \int (b_g - y)^{-1} \tau_g(dy) \le 2g + 2.$$

Here Jensen gives the lower bound and Cauchy–Schwarz the upper bound. The density is continuous and at most $1/\pi$ by (A.6), so each interior quantile has distance at least $\pi\kappa$ from its band edge. Weak continuity in g, interior quantile continuity, and implicit stability in (A.6) give, by compactness, fixed $b \le \pi\kappa/4$ and $c' > 0$ with $\Im m_g \ge c'$ on the $2b$-neighborhood of the quantile interval, uniformly for $g \le C$, including $g = 0$. For x in its b-neighborhood and any fixed H, Poisson integration on $[x - b, x + b]$ gives

$$\Im m_g(x + i\eta) \ge \frac{2c'}{\pi} \arctan(b/\eta) \ge \frac{2c'}{\pi} \arctan(b/H) > 0 \qquad (0 < \eta \le H).$$

The stability denominator in (A.6) is thus uniformly positive. Implicit differentiation gives all fixed mixed derivatives; shifting the free Fourier contour by $\pm 2ic$, with $2C \sinh(2c)$ smaller than half the imaginary gap, gives $|R_{g,\infty}(n)| \le C' e^{-2c|n|}$ and the weighted row. The fixed interval $[-2 + \kappa_0, 2 - \kappa_0]$ is handled in the same way, since $e_g \ge 2$.

For general γ, use

$$m_g(z; \gamma) = \gamma^{-1/2} m_g(z/\sqrt{\gamma}; 1), \qquad R_{g,\infty}(n; w; \gamma) = \gamma^{-1/2} R_{g,\infty}(n; w/\sqrt{\gamma}; 1).$$

Take $H = \max\{1, C\}$ and the physical buffer at most b/H. These identities cover $\eta \le 1$, rescale the quantiles and fixed interval by $\sqrt{\gamma}$, and transfer the derivative and row bounds uniformly over the stated variance interval; derivatives hold γ fixed. □

Proof of Lemma 1.2. In a fixed free realization, ν_g is the law of $(g/\max\{1, g\})A + \max\{1, g\}^{-1} S$, where A is arcsine and S is semicircular of variance one. Symmetry and the norm bound give (1.12); continuity of the coefficients on $[0, \infty]$ gives weak continuity and the two endpoint laws. Lemma A.2 supplies the single band and continuous density for finite g.

Every law in this weakly continuous compact family is atomless, including the arcsine endpoint. Compactness and the Portmanteau theorem therefore give (1.13).

For (1.14), write F_0, F_∞ for the semicircle and arcsine distribution functions. In the same free realization, $\|gA\| \le 2g$ and $\|S/g\| \le 2/g$, so spectral comparison gives

$$F_g(x) \le F_0(x+2g) \quad (g \le 1), \qquad F_g(x) \le F_\infty(x+2/g) \quad (g \ge 1).$$

Take $\kappa_0 = c_\kappa = \kappa^2/4$ and $C_\kappa = 8/\kappa^2$. For $g \le c_\kappa$ the first bound at $x = -2+\kappa_0$ is at most $F_0(-2+3\kappa^2/4) < \kappa$; for $g \ge C_\kappa$ the second is at most $F_\infty(-2+\kappa^2/2) < \kappa$, by the explicit endpoint densities. Monotonicity of F_g proves (1.14). □

A.3 Finite-cycle regularity at bounded coupling

Lemma A.3 (Finite-cycle Dyson regularity)**.** *Fix $C < \infty$, $0 < \kappa < 1/2$, and $\max\{1, C\}^{-2} \le \gamma \le 1$. There are $b, c, C' > 0$ such that, for $g \le C$ and all sufficiently large L, uniformly on*

$$z = x + i\eta, \qquad 0 \le \eta \le 1, \qquad \operatorname{dist}\big(x, \{E : F_g(-|E|;\gamma) \ge \kappa\}\big) \le b, \tag{A.9}$$

the line and finite-cycle solutions satisfy

$$\begin{gathered} |x| + |m_g(z)| + |m_{g,L}(z)| \le C', \qquad \Im m_g(z), \Im m_{g,L}(z) \ge c, \\ |\partial_g^j \partial_z^r (m_{g,L} - m_g)(z)| \le C_{j,r}(1+L)^{C_{j,r}} e^{-cL}, \qquad j, r \ge 0 \text{ fixed}, \\ \sum_{x \in \mathbb{Z}_L} e^{c\|x\|_L} |M_B(z)_{0x}| \le C'. \end{gathered} \tag{A.10}$$

The fixed derivatives of each scalar solution are bounded. The same conclusions hold on a fixed neighborhood of $\sqrt{\gamma}[-2+\kappa_0, 2-\kappa_0]$, $0 < \kappa_0 < 2$. Alternatively, for fixed $B_0, c' > 0$, they hold locally at each (g, E) with $g \le C$, $|E| \le B_0$ and $\Im m_{g,L}(E;\gamma) \ge c'$, with constants depending only on C, B_0, c' and the fixed variance interval. All derivatives hold γ fixed.

Proof of Lemma A.3. Use the free-row formula (A.5). On $\Im w \ge c > 0$ and $g \le C$, the identity $w = \sqrt{\gamma}\, g(\rho + \rho^{-1})$ gives $-\log|\rho| \ge \operatorname{arsinh}(c/(2C))$ for $g > 0$. Also $\rho = O(g)$ near zero.

Uniform exponential decay yields, at every fixed derivative order,

$$R_{g,L}(0; w) - f_g(w) = O((1+L)^{C'} e^{-cL}). \tag{A.11}$$

The quantitative implicit function theorem, using (A.8), gives the finite solution $m_{g,L}(z) = R_{g,L}(0; z + \gamma m_{g,L}(z))$ near the line solution, with the same error and all fixed derivatives. At real regular points it has positive imaginary part, and its local upper continuation is the unique finite Herglotz solution by the same Ward uniqueness argument. No contraction $|f_g'| < 1 - c$ is assumed. Uniformly on (A.7) and $0 \le \eta \le 1$, for $L \ge L_0$,

$$\Im m_{g,L}(x + i\eta) \ge a_{\text{strip}}/2. \tag{A.12}$$

The finite implicit comparison is also valid uniformly for $g \le C$, $|E| \le B_0$ and $\Im m_{g,L}(E;\gamma) \ge c'$, starting with its finite branch and finding the nearby line branch. Free arguments there have imaginary part at least $\gamma c'$ and bounded modulus; their stable denominator is at least $2\gamma(c')^2$.

The derivative bounds and the weighted row in (A.10) follow by differentiating the same implicit equation and summing (A.5) with a smaller exponential weight. Lemma A.2 supplies the stated energy buffer and the alternative fixed interval. □

Proof of Lemma 5.13(a). Apply Lemma A.2 at $\gamma = \gamma_g$ and use (A.12). The energy bound in (A.7) and the positive lower bound in (A.12) give (5.43). The modulus bound in (A.6), passed to real boundary values, gives $|m_{g,L}(e)| \le \gamma_g^{-1/2} = \max\{1, g\}$. □

An internal small-coupling expansion. We prove (A.13), which supplies the uniform semicircle bulk margin used in the bounded-coupling transport calculation. For $g \le C$, $\max\{1, C\}^{-2} \le \gamma \le 1$ and sufficiently large L as in Lemma A.3, assume $|E| \le B_0$ and $\Im m_{g,L}(E;\gamma) \ge c'$, and write $m := m_{g,L}(E;\gamma)$. Ward's identity gives $|m_{g,L}| \le \gamma^{-1/2}$. The argument $E + \gamma m_{g,L}$ of the free resolvent has imaginary part at least $\gamma c'$, so its exponential rows and implicit derivatives are uniformly bounded. For sufficiently small g, the Neumann expansion in $A_{g,\gamma}/(E+\gamma m)$ has no linear term since $\operatorname{Tr} A_{g,\gamma} = 0$. Thus

$$m = -\frac{1}{E+\gamma m} + O\left(\frac{g^2}{|E+\gamma m|^3}\right), \qquad \gamma m^2 + Em + 1 = O_{c',C}(g^2),$$

where $|E+\gamma m| \ge \gamma c'$ controls the remainder uniformly. Taking imaginary parts and dividing by $\Im m \ge c'$ gives the first estimate below; substituting it into the real part gives the second:

$$\sqrt{\gamma}\,\Re m_{g,L} = -2^{-1}\gamma^{-1/2}E + O_{c',C}(g^2), \qquad \gamma(\Im m_{g,L})^2 = 1 - 4^{-1}\gamma^{-1}E^2 + O_{c',C}(g^2). \tag{A.13}$$

Thus, for such (g, E) with g sufficiently small, $|E| \le 2\sqrt{\gamma}\sqrt{1-\gamma(c')^2/2} < 2\sqrt{\gamma}$. In particular, expansions around zero hopping have a uniform semicircle bulk margin. For g bounded away from zero, the positive imaginary part and the explicit free row provide the corresponding uniform bounds. These facts give uniform bounds under the stated energy and density conditions.

A.4 Uniform finite-cycle comparison at large coupling

Lemma A.4 (Uniform periodic comparison)**.** *Fix $0 < \kappa < 2$ and $|E| \le 2-\kappa$. There exist $C_\kappa \ge 1$ and $R_\kappa, c, C > 0$ with the following property. Define*

$$f(w) = -\frac{1}{\sqrt{w^2-4}}, \qquad \rho(w) = \frac{2}{w+\sqrt{w^2-4}}, \qquad w \in \mathbb{C}_+,$$

where the square root is asymptotic to w at infinity. For every $g \ge C_\kappa$ and $L/g^2 \ge R_\kappa$, the physical Dyson solutions on the line and the cycle of L blocks at variance g^{-2} satisfy

$$\Im m_g(E), \Im m_{g,L}(E) \asymp 1, \qquad |m_{g,L}(E) - m_g(E)| \le Ce^{-cL/g^2}.$$

Regard P, Q from (5.12) as convolution kernels on $\mathbb{Z}_L$, and let P_0 be the periodization of $g^{-2}|f(E+g^{-2}m_g(E))\rho(E+g^{-2}m_g(E))^{|x|}|^2$. Then P, P_0 are even nonnegative probability kernels and

$$\|P - P_0\|_1 + g^2\,\|P - P_0\|_\infty \le C(1+L/g^2)^C e^{-cL/g^2} =: \varepsilon_L. \tag{A.14}$$

The first norm in (A.14) also admits the weight $e^{c'd(x,0)/g^2}$ for sufficiently small $c' > 0$, where d is cyclic distance. The actual free row obeys

$$|(M_B)_{0x}| \le Ce^{-cd(x,0)/g^2}, \qquad \sum_x e^{c'd(x,0)/g^2}|(M_B)_{0x}| \le Cg^2 \quad (0 < c' < c). \tag{A.15}$$

Every same-sign inverse $V = (I - tQ)^{-1}$ or its conjugate satisfies

$$\sum_x e^{cd(x,0)/g^2}|V(x)| \le C, \qquad 0 \le t \le 1. \tag{A.16}$$

All constants are independent of g, L.

Proof of Lemma A.4. We prove the comparison also at positive height for later use. Keep hopping A_L (or $A_\mathbb{Z}$), allow covariance u/g^2, and, for fixed $C_0 > 0$, assume

$$|x| \le 2-\kappa, \quad \tfrac{1}{2} \le u \le 1, \quad z = x + iy, \quad 0 \le y \le C_0 g^{-2}, \quad g \ge C_{\kappa,C_0}, \quad L/g^2 \ge R_{\kappa,C_0}. \tag{A.17}$$

Write $m_{u,\infty}(z)$ and $m_u(z)$ for the line and cycle solutions, and $M_{B,u}(z) = (A_L - z - ug^{-2}m_u(z))^{-1}$. For μ in a fixed small disk about $i/\sqrt{4-x^2}$, put $w = z + ug^{-2}\mu$. The functions f, ρ in the statement satisfy

$$f(w) = \frac{i}{\sqrt{4-x^2}} + O(g^{-2}), \qquad |ug^{-2}f'(w)| \le Cg^{-2}, \qquad \Im w \asymp -\log|\rho(w)| \asymp g^{-2}.$$

These estimates follow from analyticity on the upper branch over the buffered bulk. The exact finite trace $f_L(w) = f(w)(1+\rho(w)^L)/(1-\rho(w)^L)$ gives

$$|(f_L - f)(w)| \le Ce^{-cL/g^2}, \qquad |\partial_\mu(f_L - f)(w)| \le C(1+L/g^2)e^{-cL/g^2}. \tag{A.18}$$

Increasing the two thresholds in (A.17) makes both Dyson maps contractions on the disk. Their fixed points are the Herglotz solutions by (A.6), with boundary values obtained as $y \downarrow 0$. Subtracting the fixed-point equations gives $|m_u - m_{u,\infty}| \le Ce^{-cL/g^2}$ and $\Im m_u, \Im m_{u,\infty} \asymp 1$.

The exact periodic row is $M_{B,u}(z)_{0r} = \sum_{n\in\mathbb{Z}} f(w_L)\rho(w_L)^{|r+nL|}$, where $w_L = z + ug^{-2}m_u(z)$. Its line summand has μ-derivative bounded by $Cg^{-2}(1+|r+nL|)e^{-c|r+nL|/g^2}$. Replacing m_u by $m_{u,\infty}$ therefore costs at most $C(1+L/g^2)^C e^{-cL/g^2}$ pointwise and g^2 times this in exponentially weighted absolute row norm. Geometric summation and Ward give, for a fixed sufficiently small $c' > 0$,

$$\begin{gathered}\Im m_u(z) \asymp 1, \qquad |M_{B,u}(z)_{0r}| \le Ce^{-cd(r,0)/g^2}, \qquad \sum_r e^{c'd(r,0)/g^2}|M_{B,u}(z)_{0r}| \le Cg^2, \\ \frac{u}{g^2}\sum_r |M_{B,u}(z)_{0r}|^2 = \frac{ug^{-2}\Im m_u(z)}{y + ug^{-2}\Im m_u(z)} \le 1.\end{gathered} \tag{A.19}$$

When squaring the rows, the coefficient u/g^2 cancels their row length g^2. The difference between the square of the periodized row and the periodization of its square consists of distinct-image products, for which $|r+nL| + |r+mL| \ge L$ if $n \ne m$. Their pointwise sum, after multiplication by u/g^2, is at most $Cg^{-2}(1+L/g^2)^C e^{-cL/g^2}$, and their weighted absolute row is at most $C(1+L/g^2)^C e^{-cL/g^2}$. These bounds apply to both modulus squares and complex squares. At $u = 1, y = 0$, Ward gives row one for the mixed kernels; this proves (A.14) and (A.15).

For same-sign stability let $w_\infty = z + ug^{-2}m_{u,\infty}(z)$, $R(p) = (2\cos p - w_\infty)^{-1}$, and $Q_\infty(r) = ug^{-2}f(w_\infty)^2\rho(w_\infty)^{2|r|}$. Then $\widehat{Q}_\infty(k) = ug^{-2}\int R(p)R(k-p)\,dp/(2\pi)$. Since $\Im R(p) > 0$,

$$\begin{aligned}\Re\widehat{Q}_\infty(k) &\le \frac{u}{g^2}\int (\Re R(p))^2\,\frac{dp}{2\pi} = \frac{ug^{-2}\int |R(p)|^2\,dp/(2\pi) + \Re\widehat{Q}_\infty(0)}{2} \le \tfrac{1}{2} + O(g^{-2}), \\ \widehat{Q}_\infty(0) &= ug^{-2}f'(w_\infty) = O(g^{-2}).\end{aligned}$$

Here the first inequality uses $2ab \le a^2 + b^2$, and the last uses the line Ward mass, which is at most one as in (A.19). Thus $\Re(1-\tau\widehat{Q}_\infty) \ge c > 0$ for $0 \le \tau \le 1$. The profile $|Q_\infty(r)| \le Cg^{-2}e^{-c|r|/g^2}$ preserves this gap in $|\Im k| \le c'/g^2$. Parseval on the edges of a narrower strip gives

$$\sum_r e^{2c''|r|/g^2}|[Q_\infty(I-\tau Q_\infty)^{-1}](r)|^2 \le C/g^2.$$

Weighted Cauchy–Schwarz gives a bounded exponential absolute row, since $\sum_r e^{-c''|r|/g^2} = O(g^2)$. The inverse $I + \tau Q_\infty(I-\tau Q_\infty)^{-1}$ has the same bound, as does its periodization. For the actual kernel $Q_L(r) = ug^{-2}M_{B,u}(z)_{0r}^2$, the image calculation above bounds its difference from the periodized Q_∞ by $C(1+L/g^2)^C e^{-cL/g^2}$ in weighted row norm. The resolvent identity in the weighted convolution algebra is therefore a convergent perturbation series for sufficiently large R_{κ,C_0}, giving

$$\sum_r e^{cd(r,0)/g^2}|[(I-\tau Q_L)^{-1}](r)| \le C, \qquad 0 \le \tau \le 1. \tag{A.20}$$

Taking $u = 1, y = 0$ proves (A.16); conjugation gives the other sign. □

A.4.1 Deterministic terminal bounds

Proof of Lemma 5.13(b). Apply Lemma A.4 on the buffered interval $|e| \le 2 - 3\kappa/4$. It gives $|m_{g,L}(e)| \le C$ and $\Im m_{g,L}(e) \ge c$. At $u = 1, y = 0$, the fixed-point derivative estimate (A.18), together with $g^{-2}|f'(w)| \le Cg^{-2}$, bounds $g^{-2}|f_L'(w)|$, where $w = e + g^{-2}m_{g,L}(e)$. Differentiating the Dyson equation gives, after increasing both the coupling and L/g^2 thresholds,

$$g^{-2}|\partial_e m_{g,L}(e)| \le \frac{g^{-2}|f_L'(w)|}{1 - g^{-2}|f_L'(w)|} \le C[g^{-2} + (1 + L/g^2)e^{-cL/g^2}] \le \tfrac{1}{2}. \tag{A.21}$$

Thus $e \mapsto e + \delta g^{-2}\Re m_{g,L}(e)$ has derivative in $[1/2, 3/2]$ and displacement at most Cg^{-2}, uniformly for $0 \le \delta \le 1$. Increasing C_κ so that $Cg^{-2} \le \kappa/4$, the endpoint images cover $[-2 + \kappa, 2 - \kappa]$. The intermediate value theorem gives E_0 in the claimed interval; the bounds on $m_{g,L}(E_0)$ give (5.44). □

A.5 Line kernels and bounded-coupling transport

Lemma A.5 (Actual finite transport and same-sign bounds)**.** *Fix $C, B_0, c' > 0$, and assume L is sufficiently large depending only on these constants. Suppose*

$$g \le C, \qquad |E| \le B_0, \qquad \Im m_{g,L}(E) \ge c'. \tag{A.22}$$

Use the finite Dyson matrix (5.3) and the pair kernels (5.12). Uniformly for $0 \le t < 1$, with

$$\ell_t = \min\left\{L, 1 + \frac{g}{\sqrt{\gamma_g(1-t)}}\right\}, \quad \zeta_t = (\gamma_g W(1-t)\ell_t)^{-1}, \tag{A.23}$$

the actual mixed kernel satisfies exact row mass $(1-t)^{-1}$, value comparable to $((1-t)\ell_t)^{-1}$ at zero, and upper exponential profile $C'((1-t)\ell_t)^{-1}e^{-c\|x\|_L/\ell_t}$. The free row, mixed kernel, and same-sign inverse $(I - tQ)^{-1}$ satisfy

$$\begin{aligned} \sum_{x\in\mathbb{Z}_L} e^{c\|x\|_L}|M_B(0,x)| &\le C', \\ \sum_{x\in\mathbb{Z}_L} e^{c\|x\|_L/\ell_t}|[P(I-tP)^{-1}]_{0x}| &\le \frac{C'}{1-t}, \\ \sum_{x\in\mathbb{Z}_L} e^{c\|x\|_L}|[(I-tQ)^{-1}]_{0x}| &\le C'. \end{aligned} \tag{A.24}$$

The actual Fourier multipliers also satisfy

$$1 - \widehat{P}(k) \asymp g^2(1 - \cos k), \qquad U_t(k) := (1 - t\widehat{P}(k))^{-1} \asymp ((1-t) + g^2|k|_{\mathbb{T}}^2)^{-1}. \tag{A.25}$$

With normalized averaging over the Fourier grid,

$$H := \langle U_t \rangle = 1 + t[P(I-tP)^{-1}]_{00} \ge 1, \qquad H \asymp ((1-t)\ell_t)^{-1}. \tag{A.26}$$

A common line kernel. On the line, at a real regular energy, write $m := m_g(E;\gamma)$ and $M_\infty(x) := [(\sqrt{\gamma}\, gA_{\mathbb{Z}} - E - \gamma m)^{-1}]_{0x}$, and set

$$P_\infty(x) := \gamma|M_\infty(x)|^2, \qquad Q_\infty(x) := \gamma M_\infty(x)^2.$$

On the cycle of L blocks, use P, Q from (5.12) with γ_g replaced by γ and M_B by $M_B(E;\gamma)$ from (A.2). Products of kernels are convolutions, and all Fourier averages are normalized. Write $\rho = e^{-q-i\theta}$, $q > 0$. Define the diffusivity, suppressing its dependence on E, γ, by

$$D_g := \frac{1}{2(\cosh 2q - 1)}. \tag{A.27}$$

Lemma A.6 (The line mixed kernel)**.** *For every* $0 \le t < 1$,

$$\begin{aligned} \widehat{P}_\infty(k) &= \frac{1}{1+2D_g(1-\cos k)}, \qquad \frac{\widehat{P}_\infty(k)}{1-t\widehat{P}_\infty(k)} = \frac{1}{1-t+2D_g(1-\cos k)}, \\ [P_\infty(I-tP_\infty)^{-1}](x) &= \frac{e^{-\lambda|x|}}{\sqrt{(1-t)(1-t+4D_g)}}, \qquad \cosh\lambda = 1+2^{-1}D_g^{-1}(1-t). \end{aligned} \tag{A.28}$$

At $g=0$, $P_\infty(I-tP_\infty)^{-1} = (1-t)^{-1}\mathbf{1}_0$. *Periodizing* (A.28) *gives a comparison kernel with row mass* $(1-t)^{-1}$ *and block scale*

$$\min\{L, 1+\sqrt{D_g/(1-t)}\}.$$

At bounded coupling in the regularity range of Lemma A.3, $D_g \asymp g^2$. *At normalized large coupling in the range of Lemma A.4,* $D_g \asymp g^4$.

Proof of Lemma A.6. Write $\rho = e^{-q-i\theta}$ in (A.5). Exact Ward gives $\gamma|m|^2 = \tanh q$. Summing $\sum_{x\in\mathbb{Z}} e^{-2q|x|}e^{-ikx}$ proves the first Fourier identity in (A.28); division by $1-t\widehat{P}_\infty(k)$ proves the second. Inverse Fourier summation gives the displayed exponential kernel and its periodization, with the stated limiting formula at $g=0$. For $g>0$, put $w = E+\gamma m = 2\sqrt{\gamma}\,g\cosh(q+i\theta)$. Since $D_g = (4\sinh^2 q)^{-1}$,

$$(\Im w)^2 \le 4\gamma g^2\sinh^2 q \le |w|^2, \qquad \frac{\gamma g^2}{|w|^2} \le D_g \le \frac{\gamma g^2}{(\Im w)^2}.$$

At bounded coupling, Lemma A.3 transfers the finite density margin to $\Im m_g(E;\gamma) \ge c'/2$ for sufficiently large L. Together with $|E| \le B_0$, $|m| \le \gamma^{-1/2}$ from (A.6), and the fixed positive variance interval, this gives $|w| \le C'$, $\Im w \ge c$ and hence $D_g \asymp g^2$. The uncapped block scale is therefore comparable to that in (5.18), since $\gamma_g \asymp 1$. At large coupling, the proof of Lemma A.4 already gives $q \asymp g^{-2}$; substituting this into (A.27) proves $D_g \asymp g^4$.

Lemma A.7 (Auxiliary increment and spatial-moment bounds)**.** *Use the parameters, kernels and scales of Lemma A.5. For* $1-t \ge W^{-3}$, *a fixed polynomial upper bound on* L, *and every accuracy* A, $L \ge C_A \log W$ *gives, for* $x, y \in \mathbb{Z}_L$,

$$\begin{aligned} &|[P(I-tP)^{-1}]_{0,x+y} - [P(I-tP)^{-1}]_{0,x}| \\ &\quad\le \frac{C'}{(1-t)\ell_t}\min\{\|y\|_L/\ell_t, 1\}\left(e^{-c\|x\|_L/\ell_t} + e^{-c\|x+y\|_L/\ell_t}\right) + W^{-A}. \end{aligned} \tag{A.29}$$

For every fixed integer $j \ge 0$, *the difference of the* j*th time derivatives satisfies* (A.29) *with* $C'/((1-t)\ell_t)$ *replaced by* $C'_j/((1-t)^{j+1}\ell_t)$ *and the same remainder* W^{-A}, *choosing* C_A *also depending on* j. *For every fixed* $r>0$,

$$\sum_{x\ne 0} \|x\|_L^r |[(I-tQ)^{-1}]_{0x}| \le C'_r g^2. \tag{A.30}$$

Proof of Lemmas A.5 and A.7. We prove the estimates for every fixed $\max\{1,C\}^{-2} \le \gamma \le 1$, with $g \le C$, $|E| \le B_0$ and $\Im m_{g,L}(E;\gamma) \ge c'$. Use the finite kernels defined above and the scales (A.23), replacing γ_g by γ in ζ_t; the constants are uniform over this variance interval. The choice $\gamma = \gamma_g$ gives the two stated lemmas. The free-row bound in (A.24) is (A.10) under these conditions. Write $m := m_{g,L}(E;\gamma)$ and set $Q(x) = \gamma M_B(x)^2$, $Q' := Q - \gamma m^2\mathbf{1}_0$. Exact Ward gives $\|Q'\|_1 = 1-\gamma|m|^2$. For $\Im m \ge a > 0$,

$$|1-t\gamma m^2|^2 = (1-t\gamma|m|^2)^2 + 4t\gamma(\Im m)^2, \quad |1-t\gamma m^2| - t(1-\gamma|m|^2) \ge \gamma a^2/2. \tag{A.31}$$

For $t \le 1/2$ use the triangle inequality and a gap at least $1/2$; for $t \ge 1/2$ rationalize the square-root difference in (A.31). Thus the complete, convergent expansion

$$(I - tQ)^{-1} = (1 - t\gamma m^2)^{-1} \sum_{j \ge 0} \left(\frac{tQ'}{1 - t\gamma m^2} \right)^{*j} \tag{A.32}$$

has absolute row $O((\gamma a^2)^{-1})$. A sufficiently small exponential weight changes the gap by less than half. At small g, the off-diagonal weighted row of Q is $O(g^2)$, so the nonconstant part of the full inverse is $O(g^2)$ too. The remaining compact range bounded away from zero uses this proved weighted bound with an enlarged fixed constant. This proves (A.30). No fixed-depth truncation is used.

On the actual torus, for $L \ge L_0$,

$$P(1) + P(-1) \ge cg^2, \quad \sum_{x \ne 0} e^{c\|x\|_L} P(x) \le C' g^2, \quad \inf_k \widehat{P}(k) \ge c. \tag{A.33}$$

For small g, the free Neumann and implicit expansions give $m = m_0 + O(g^2)$, $M_B = m_0 \mathbf{1}_0 - \sqrt{\gamma}\, g m_0^2 A_L(0,\cdot) + O_{\ell^1_c}(g^2)$, $|m_0| = \gamma^{-1/2}$. The zero adjacency diagonal removes the linear scalar term, and nearest-neighbor squares are at least $g^2/4$. For $g \ge c > 0$, line comparison and the explicit geometric nearest-neighbor coefficient and positive symbol prove (A.33) by choosing a fixed larger L_0. This separation avoids dividing an absolute comparison error by an arbitrarily small g. Evenness and row one now give exactly on the Fourier grid (A.25). Use nearest neighbors for the lower bound and $1 - \cos(kx) \le x^2(1 - \cos k)$ for the upper bound. In particular the zero mode is exact.

Lift P to an even probability on minimal integer representatives, splitting the Nyquist mass for even L. Its symbol $p^\sharp$ obeys $\Re p^\sharp(k + iy) - p^\sharp(k) \le C' g^2 y^2$. With $|y| = c/(1 + g/\sqrt{1-t})$ the denominator remains at least $c((1-t) + g^2|k|^2)$ in real part. Contour shifting and $\int dk/((1-t) + g^2k^2) \asymp [(1-t)((1-t)+g^2)]^{-1/2}$ give an exponential line-lift profile. Its convergent nonnegative convolution series periodizes exactly to the actual $P(I - tP)^{-1}$; summing translates proves the capped upper profile and weighted row. The positive grid symbol in (A.33) gives

$$[P(I - tP)^{-1}]_{00} \asymp L^{-1} \sum_k ((1-t) + g^2|k|^2)^{-1}$$
$$\asymp [(1-t)((1-t)+g^2)]^{-1/2} + (L(1-t))^{-1} \asymp ((1-t)\ell_t)^{-1}.$$

Fourier inversion gives $\langle U_t \rangle = 1 + t[P(I - tP)^{-1}]_{00}$. For $t \ge 1/2$ the displayed diagonal estimate proves (A.26); for $t \le 1/2$, both H and $((1-t)\ell_t)^{-1}$ are comparable to one, since g is bounded.

Proof of (A.29) and its time derivatives. Compare with the periodized geometric kernel of (A.28). Its discrete difference is $C'((1-t)\ell_t)^{-1} \min\{\|h\|_L/\ell_t, 1\}$ times the two endpoint profiles. This also holds when $\ell_t \asymp 1$, by applying the pointwise bound at both endpoints. The probability-kernel resolvent identity bounds the actual/comparison row difference by $(1-t)^{-2}\|P - \widetilde{P}_L\|_1$. Here $\widetilde{P}_L$ is the periodized line square kernel, not the square of the periodized line resolvent. Lemma A.3 bounds their difference by $C'(1+L)^{C'} e^{-cL}$. Choose C_A after all polynomial cutoffs and derivative orders. Finally $\partial_t^j [P(I-tP)^{-1}] = j![P(I-tP)^{-1}]^{j+1}$ has row $j!(1-t)^{-j-1}$. Apply (A.29), with accuracy $A + 3j + 1$, to one convolution factor and the mixed weighted row bound in (A.24) to the other j factors. Their weighted row is at most $(C'/(1-t))^j$, which gives the stated derivative prefactor. The error is at most $j!(1-t)^{-j} W^{-A-3j-1} \le W^{-A}$ for sufficiently large W, since $1 - t \ge W^{-3}$. Telescoping the same convolution identity gives comparison error $C'_j(1-t)^{-j-2}\|P - \widetilde{P}_L\|_1$. Exact row mass and Ward projection always use the actual kernel. □

A.6 Geometric kernels and uniform perturbations

The following estimates separate the leading geometric mixed kernel from its finite-cycle error. Here $a \ge 1$ is an independent kernel scale, $L \ge a$, $0 < \delta \le 1$, and $\ell = \min\{L, a/\sqrt{\delta}\}$. All convolution norms use counting measure on $\mathbb{Z}$ or $\mathbb{Z}_L$, and $\Delta f(x) = f(x+1) - f(x)$.

Lemma A.8 (A periodized geometric mixed kernel). *Let $D \asymp a^2$ and*

$$P^\wedge(q) = \frac{1}{1 + 2D(1 - \cos q)}, \quad t = 1 - \delta, \quad \Xi = tP(I - tP)^{-1}.$$

On the line,

$$\Xi(x) = \frac{te^{-\lambda|x|}}{\sqrt{\delta(\delta + 4D)}}, \qquad \cosh\lambda = 1 + \frac{\delta}{2D}, \qquad \lambda \asymp \frac{\sqrt{\delta}}{a}.$$

On $\mathbb{Z}$ and after periodization to $\mathbb{Z}_L$,

$$\begin{gathered} \|\Xi\|_1 \le C\delta^{-1}, \qquad \|\Xi\|_\infty \le C(\delta\ell)^{-1}, \\ \|\Delta\Xi\|_1 \le C\lambda\delta^{-1}, \qquad \|\Delta\Xi\|_\infty \le C\lambda(\delta\ell)^{-1}, \qquad \|\Delta^2\Xi\|_1 \le C\lambda^2\delta^{-1}. \end{gathered} \tag{A.34}$$

Proof of Lemma A.8. The formula follows by solving the second-order difference equation or summing the geometric Fourier series. Each row of Ξ sums to t/δ. Periodization has supremum at most $C(\delta^{-1}\lambda + \delta^{-1}/L) \le C/(\delta\ell)$. A unit first difference is bounded in absolute value by $C\lambda$ times the sum of two translates of the original positive kernel. This proves both first-difference estimates. Away from zero, a second difference is $O(\lambda^2)$ times a translate of the kernel. The two points adjacent to the cusp contribute at most $C\lambda\Xi(0) \le C\lambda^2/\delta$ in total. Thus each absolute row sum of $\Delta^2\Xi$ is $O(\lambda^2/\delta)$. Periodization preserves the bound on the absolute row sums and commutes with differences. Notice that a pointwise second-difference bound by $\lambda^2 \|\Xi\|_\infty$ is not asserted; it fails at the cusp. □

Lemma A.9 (Small finite-kernel perturbations). *Let P_0 be the periodized kernel of Lemma A.8. Suppose P_1 is an even nonnegative probability kernel and*

$$\|P_1 - P_0\|_1 \le \varepsilon, \qquad \|P_1 - P_0\|_\infty \le \varepsilon/a.$$

If $\Xi_j = tP_j(I - tP_j)^{-1}$ and $\varepsilon \le c\delta^2$, then $E = \Xi_1 - \Xi_0$ satisfies

$$\|E\|_1 \le C\varepsilon\delta^{-2}, \qquad \|E\|_\infty \le C\varepsilon(a\delta^2)^{-1}. \tag{A.35}$$

Proof of Lemma A.9. The exact resolvent identity is

$$E = t(I - tP_1)^{-1}(P_1 - P_0)(I - tP_0)^{-1}.$$

Both inverse kernels are nonnegative and have row δ^{-1}. The row and pointwise Young inequalities give respectively $\|E\|_1 \le \varepsilon\delta^{-2}$ and $\|E\|_\infty \le \varepsilon/(a\delta^2)$. □

Lemma A.10 (Mixed kernels uniformly in the coupling). *Use the parameters and kernels of Lemma A.4. Let $t = 1 - \delta$, $0 < \delta \le 1$, and $\ell = \min\{L, g^2/\sqrt{\delta}\}$. If $L/g^2 \ge C(1 + |\log\delta|)$, then the actual mixed inverse has the decomposition*

$$(I - tP)^{-1} = I + \Xi + \mathcal{E}, \qquad \Xi = tP_0(I - tP_0)^{-1}, \qquad \varepsilon = \varepsilon_L \le c\delta^2. \tag{A.36}$$

Here ε_L is defined in (A.14), Ξ and $\mathcal{E}$ are even, and d denotes cyclic distance. The stable inverses satisfy (A.16). The actual mixed kernel satisfies

$$0 \le [P(I - tP)^{-1}](x) \le \frac{C}{\delta\ell}, \qquad \sum_x [P(I - tP)^{-1}](x) = \delta^{-1}. \tag{A.37}$$

With $\lambda := \sqrt{\delta}/g^2$, the mixed part satisfies

$$\|\Xi\|_1 \le C\delta^{-1}, \qquad \|\Xi\|_\infty \le C(\delta\ell)^{-1}, \tag{A.38}$$

$$\|\Delta\Xi\|_1 \le C\lambda\delta^{-1}, \qquad \|\Delta\Xi\|_\infty \le C\lambda(\delta\ell)^{-1}, \tag{A.39}$$

$$\left\|\Delta^2\Xi\right\|_1 \le C\lambda^2\delta^{-1}. \tag{A.40}$$

The last bound also holds for any composition of two forward or backward unit differences. The error obeys

$$\|\mathcal{E}\|_1 \le C\varepsilon\delta^{-2}, \qquad \|\mathcal{E}\|_\infty \le C\varepsilon(g^2\delta^2)^{-1}, \qquad \varepsilon \le c\delta^2. \tag{A.41}$$

In particular, $L \ge Cg^2 \log W$ suffices when $\delta \ge W^{-3}$, with no upper bound on g, L. The characteristic scales in (5.18) satisfy

$$\ell \le \ell_t \le 2\ell, \qquad \zeta_t \asymp \frac{g^2}{W\delta\ell}, \qquad A := \frac{g^2}{\delta\ell} \ge 1. \tag{A.42}$$

The actual and comparison inverses also satisfy

$$\begin{aligned} \sum_x e^{cd(x,0)/\ell}(|(I+\Xi)(x)| + |[(I-tP)^{-1}](x)|) &\le C\delta^{-1}, \\ \sum_x e^{cd(x,0)/\ell}|\mathcal{E}(x)| &\le C\varepsilon_L\delta^{-2}. \end{aligned} \tag{A.43}$$

For every fixed $r, Q > 0$, there is $C_{r,Q}$ such that, for sufficiently large W,

$$\delta \ge W^{-3}, \quad L \ge C_{r,Q}g^2 \log W \quad \Longrightarrow \quad \delta^{-r}\varepsilon_L \le W^{-Q}. \tag{A.44}$$

Finally, for any kernels satisfying (A.16) and (A.38)–(A.41), with $\varepsilon \le c\delta^2$, the local convolution bound is

$$\|\mathcal{E}\|_1 \le C, \qquad \sup_x \sum_y e^{-cd(y,0)/g^2}|K(x-y)| \le CA, \quad K \in \{I, \Xi, \mathcal{E}, \textit{stable kernels}\}. \tag{A.45}$$

Proof of Lemma A.10. Writing $H = L/g^2$, absorb $(1+H)^C$ into $Ce^{cH/2}$ in (A.14). The assumed lower bound $H \ge C(1+|\log\delta|)$ then gives $\varepsilon_L \le c\delta^2$. Use Lemma A.6 at $\gamma = g^{-2}$ and Lemmas A.8–A.9 with $a = g^2$, $D \asymp g^4$, and $\varepsilon = \varepsilon_L \le c\delta^2$. Then $(I-tP)^{-1} = I + tP(I-tP)^{-1}$ gives (A.36), and the geometric and perturbation bounds give (A.38)–(A.41), with decay rate $\lambda \asymp \sqrt{\delta}/g^2$. The stable row is (A.16).

Proof of (A.37) and (A.42). The convolution series of the probability kernel P gives positivity and row mass δ^{-1}. For $t \ge 1/2$, use $P(I-tP)^{-1} = t^{-1}(\Xi + \mathcal{E})$ and $\|\mathcal{E}\|_\infty \le C\varepsilon_L/(g^2\delta^2) \le C/g^2 \le C/(\delta\ell)$. For $t \le 1/2$, the same series and $\|P\|_\infty \le C/g^2$ give $\|P(I-tP)^{-1}\|_\infty \le C/(g^2\delta)$; here $\ell \asymp g^2$. Also (5.18) gives $\ell \le \ell_t \le 2\ell$, $\zeta_t \asymp g^2/(W\delta\ell)$, and $\delta\ell \le g^2\sqrt{\delta} \le g^2$, hence $A \ge 1$.

Proof of (A.43). In the weighted convolution norm $\|K\|_{1,c/\ell} = \sum_x e^{cd(x,0)/\ell}|K(x)|$, the explicit geometric formula in Lemma A.8 gives $\|R_0\|_{1,c/\ell} \le C\delta^{-1}$ for $R_0 = (I-tP_0)^{-1}$, by choosing c/ℓ below the decay rate when uncapped; at the cap $\ell = L$, the weight is bounded and the row sum is δ^{-1}. Since $\ell \ge g^2$, (A.14) gives $\|P - P_0\|_{1,c/\ell} \le C\varepsilon_L$. Thus

$$(I-tP)^{-1} = R_0 \sum_{j\ge 0}[t(P-P_0)R_0]^{*j}, \qquad \mathcal{E} = t(I-tP)^{-1}(P-P_0)R_0$$

converges for $\varepsilon_L \le c\delta^2$ and yields weighted inverse row $C\delta^{-1}$ and error row $C\varepsilon_L\delta^{-2}$.

Proof of (A.44) and (A.45). For fixed r, Q, choose $C_{r,Q}$ sufficiently large in $L \ge C_{r,Q}g^2 \log W$; then

$$\delta^{-r}\varepsilon_L \le CW^{3r}(1+L/g^2)^C e^{-cL/g^2} \le C'W^{3r}e^{-cL/(2g^2)} \le W^{-Q}.$$

Thus $L/(g^2 \log W) \to \infty$ permits every fixed accuracy, without an upper bound on g, L. For local convolution, the exponential summation costs Cg^2; use $g^2\|\Xi\|_\infty \le CA$ and $g^2\|\mathcal{E}\|_\infty \le C\varepsilon\delta^{-2} \le C$. The identity and stable kernels use their bounded rows and $A \ge 1$; (A.41) also gives $\|\mathcal{E}\|_1 \le C$. This argument uses only the stated abstract kernel bounds. □

A.7 Scales and deterministic propagators

Under (6.1), the following auxiliary bounds control time ratios and propagator row sums in the continuation argument. Use $1/2 \le s \le u \le t < 1$ and $1 - t \ge W^{-3}$:

$$\begin{aligned} 1 \le \ell_s \le \ell_u \le L, \qquad (W\eta_u)^{-1} &= (\Im m)^{-1}\ell_u\zeta_u, \\ c_* W^{-1} \le \zeta_s \le \zeta_u &\le \frac{1-s}{1-u}\zeta_s. \end{aligned} \tag{A.46}$$

$$\begin{aligned} \sum_b (U^{+-}_{s,u})_{ab} = \frac{1-s}{1-u}, \qquad (U^{+-}_{s,u})_{ab} \ge 0, \\ \max_a \sum_b e^{c\|a-b\|_L/\ell_0}\big(|(U^{++}_{s,u})_{ab}| + |(U^{--}_{s,u})_{ab}|\big) \le C'. \end{aligned} \tag{A.47}$$

The appendix arguments also use the following strengthened versions. Here ℓ_0 is the time-zero scale in (5.18); in the stated parameter ranges, $\ell_0 \asymp \max\{1, g^2\}$ and $1 \le \ell_0 \le \ell_s$:

$$\gamma_g \max_a \sum_b e^{c\|a-b\|_L/\ell_0}(|M_{B,ab}| + |M_{B,ab}|^2) \le C', \qquad \max_a \sum_b e^{c\|a-b\|_L/\ell_0}|[(I-uQ)^{-1}]_{ab}| \le C'. \tag{A.48}$$

The mixed kernel has row mass $(1-u)^{-1}$ and, for every fixed $A > 0$,

$$\begin{aligned} &|[P(I-uP)^{-1}]_{0,x+h} - [P(I-uP)^{-1}]_{0,x}| \\ &\quad \le \frac{C'}{(1-u)\ell_u}\min\{\ell_u^{-1}\|h\|_L, 1\}(e^{-c\|x\|_L/\ell_u} + e^{-c\|x+h\|_L/\ell_u}) + C_A\ell_u^{-1}W^{-A}. \end{aligned} \tag{A.49}$$

Let $A_{\mathbb{Z}}$ be the nearest-neighbor adjacency operator on $\mathbb{Z}$, and define the periodized line square kernel by

$$P^{\mathrm{per}}(x) = \gamma_g \sum_{j\in\mathbb{Z}} \left| \left[(\sqrt{\gamma_g} g A_{\mathbb{Z}} - E - \gamma_g m_g(E))^{-1} \right]_{0,x+jL} \right|^2 .$$

For $r_u = P(I-uP)^{-1} - P^{\mathrm{per}}(I-uP^{\mathrm{per}})^{-1}$ and every prescribed $A > 0$, uniformly for $s \le u \le t$,

$$\max_a \sum_b |(r_u)_{ab}| + \ell_u \max_{a,b} |(r_u)_{ab}| \le C_A W^{-A}. \tag{A.50}$$

The same remainder bound holds for its spatial increments and propagator version $(u-s)r_u$. Constants are independent of g, L; the width threshold may depend on the rate of divergence in the volume lower bound.

Proof of Lemma 6.1 and (A.46)–(A.50). Use (A.10) with $\gamma_g \asymp \ell_0 \asymp 1$ at bounded coupling, and Lemma A.4 together with (A.15) with $\ell_0 \asymp g^2$, $\gamma_g g^2 = 1$ at large coupling. They give $\Im m \asymp 1$, bounded entries, and

$$\gamma_g \max_a \sum_b e^{c\|a-b\|_L/\ell_0}|M_{B,ab}| \le C, \qquad |M_{B,ab}|^2 \le C|M_{B,ab}|.$$

Together with the stable rows (A.24) and (A.16), these prove (A.48) and hence (6.2). The scale definition also gives $1 \le \ell_0 \le \ell_s$. Since $U^{++}_{s,u} = I + (u-s)u^{-1}((I-uQ)^{-1} - I)$ and $u \ge 1/2$, its weighted row is bounded. Ward gives $P\mathbf{1} = \mathbf{1}$, so $U^{+-}_{s,u}\mathbf{1} = (1-s)(1-u)^{-1}\mathbf{1}$ with positive entries, proving (A.47). The mixed profile (6.3) and the increment bound (A.49) follow from Lemmas A.5 and A.7 at bounded coupling, and from the periodized geometric formula (A.28) and Lemma A.10 at large coupling; the finite remainder is checked below. For $\theta_u = (1-u)P(I-uP)^{-1}$, direct differentiation gives

$$\partial_u\theta_u = P(I-uP)^{-1}\theta_u - (1-u)^{-1}\theta_u. \tag{A.51}$$

For (A.46), the reciprocal scale

$$(\gamma_g W\zeta_u)^{-1} = \min\{L(1-u), (1-u) + g\gamma_g^{-1/2}\sqrt{1-u}\}$$

is decreasing, whereas ℓ_u is increasing. Thus $\zeta_s \le \zeta_u \le (1-s)(1-u)^{-1}\zeta_s$; $\gamma_g(1-s)\ell_s \le C$ gives $\zeta_s \ge c/W$. The identity for η_u uses $\eta_u = (1-u)\gamma_g \Im m$.

Proof of (6.4) and (6.5). Write $K_v = P(I-vP)^{-1}$ and $\mathcal{Q}_v A = A - (\operatorname{row} A)\prod_{j=2}^n \theta_v(a_1 - a_j)$. Symmetry and row mass give

$$\operatorname{row}\left(\sum_{i=1}^n K_v^{(i)} A\right) = K_v(\operatorname{row} A) + (n-1)(1-v)^{-1}\operatorname{row} A.$$

Differentiate $\mathcal{Q}_v A$ using (A.51): the normalization terms cancel the last term above, and the nonanchored generators cancel, leaving precisely the first-leg commutator (6.4). Its bracket has row $1-1=0$; bounding each product profile by $(C/\ell_v)^{n-1}$ and summing the kernel row $(1-v)^{-1}$ proves (6.5).

Proof of (A.50). At large coupling $\mathcal{E} = ur_u$, with $u \ge 1/2$; at bounded coupling use the inverse subtraction in the proof of (A.29). Thus

$$\max_a \sum_b |(r_u)_{ab}| + \ell_u \max_{a,b} |(r_u)_{ab}| \le \frac{C}{(1-u)^2}\begin{cases}(1+L)^{q+1}e^{-cL}, & \text{bounded coupling},\\ (1+\ell_u/g^2)\varepsilon_L, & \text{large coupling}.\end{cases}$$

Here $(1-u)^{-1} \le W^3$, bounded-coupling L is polynomial, and $\ell_u/g^2 \le CW^{3/2}$ at large coupling. The volume lower bounds and (A.44) absorb these costs at any fixed accuracy, chosen after the time schedule and loop and moment orders. Spatial increments are differences of translates; the propagator remainder is $(u-v)r_u$, $s \le v \le u$, so both satisfy the same bound. □

A.8 Spatial summation and weights

Proof of Lemma 6.2. The spatial budget follows from polynomial L at bounded coupling, and from $\ell_u \ge g^2$, $L/g^2 \le CW^2$ at large coupling. The normalization identity in (6.6) follows directly from the definition (5.18).

For (6.7) with $0 < \theta \le 1/2$, use the cyclic triangle inequality, $|x^\theta - y^\theta| \le |x-y|^\theta$, and the common nonnegative floor. On mesocells $n/\ell_u \le W^{-B}$, the ratio tends to one. For (6.8), retain the gain $x^\theta + y^\theta - (x+y)^\theta \ge c_\theta \min(x,y)^\theta$. Splitting the sum at the nearer endpoint leaves a summable factor of mass $C_\theta \ell_u$. The floor terms are $O(\ell_u W^{-F} + LW^{-2F})$. For the square-root version use $\sqrt{e^{-x} + W^{-F}} \le e^{-x/2} + W^{-F/2}$; its floor product costs $LW^{-F} \le \ell_u W^{-F/2}$ by the stated condition $LW^{-F/2} \le \ell_u$.

□

Lemma A.11 (Weights and time integrals for the stopped pair equation)**.** *Under (6.1), suppose $L/\ell_v \le W^B$ for a fixed B and $1/2 \le s \le v \le u \le t \le 1 - W^{-3}$. Fix $F > 0$, and set $R = (1-s)/(1-t)$ and $w_v(x) = e^{-c_{\mathrm{p}}(\|x\|_L/\ell_v)^\theta} + W^{-F}$, where $0 < \theta \le 1/2$ and $c_{\mathrm{p}} > 0$ are fixed. Then*

$$\sup_{\|x\|_L \le \ell_v (\log W)^{3/2}} w_v(x)^{-1/2} = W^{o(1)}, \tag{A.52}$$

and, for $U_{v,u} = U_{v,u}^{+-}$,

$$\sum_{a',b'} (U_{v,u})_{aa'}(U_{v,u})_{bb'} w_v(a'-b') \le C\left(\frac{1-v}{1-u}\right)^2 w_u(a-b). \tag{A.53}$$

For fixed $q \ge p \ge 0$,

$$\int_s^u \frac{\zeta_v^q}{\zeta_u^p (1-v)}\, dv \le \zeta_u^{q-p} \log R. \tag{A.54}$$

Proof of Lemma A.11. On the short halo the reciprocal weight is at most $\exp(C(\log W)^{3\theta/2}) = W^{o(1)}$, proving (A.52). Since $\ell_v \le \ell_u$, $w_v \le w_u$. The triangle inequality gives $w_u(x-y) \le w_u(x)e^{c_{\mathrm{p}}(\|y\|_L/\ell_u)^\theta}$. The exponential part of (6.3), in $U_{v,u} = I + (u-v)P(I-uP)^{-1}$, therefore has weighted row norm at most $C(1-v)/(1-u)$. Indeed, the resulting sum is bounded by $C\ell_u^{-1}\sum_y e^{-c\|y\|_L/\ell_u + c_{\mathrm{p}}(\|y\|_L/\ell_u)^\theta} \le C$. Apply this estimate successively in the two coordinates. For the remainder, use $w_v \le 2$, $w_u \ge W^{-F}$ and (A.50); choosing its accuracy after B, F absorbs the polynomial factor $(L/\ell_u)W^F$. This proves (A.53). Finally $\zeta_v \le \zeta_u$ by (A.46), so integrating $(1-v)^{-1}$ proves (A.54). □

Lemma A.12 (Scalar gains used in loop improvement)**.** *Let $R \ge 1$, $0 < \zeta_s \le \zeta_t \le R\zeta_s$, and $R^{20}\zeta_t \le W^{-c}$ for fixed $c > 0$. For every fixed C, the following quantities have a strictly negative power of W as an upper bound:*

$$(CR)^8\zeta_s \ll 1, \quad CR\sqrt{\zeta_t}\,\zeta_s^{-1/4} \ll 1, \quad CR\sqrt{\zeta_t}\,\zeta_s^{-1/8} \ll 1, \\ \zeta_t R^{10}\zeta_s^{-1/4} \ll 1, \qquad R^2\zeta_s^{1/4} \ll 1. \tag{A.55}$$

Also, for fixed $c_{\mathrm{p}}, \theta > 0$, $1 \le R \le W^3$ and $x \ge 0$, there are fixed $c', C' > 0$ such that

$$\min\{1, R^{10}e^{-c_{\mathrm{p}}x^\theta}\} \le e^{C'\sqrt{\log W}}e^{-c'x^{\theta/2}}. \tag{A.56}$$

Proof of Lemma A.12. For $b > d \ge 0$ and $a+d \le 20(b-d)$, $R^a\zeta_t^b\zeta_s^{-d} \le R^{a+d}\zeta_t^{b-d} \le W^{-(b-d)c}$. This gives the three terms with negative powers of ζ_s in (A.55); the other two follow from $\zeta_s \le \zeta_t$. Fixed powers of C preserve the strict margins. For (A.56), first suppose $x^\theta \le C(1+\log R)$. Use the bound 1 for the minimum; $x^{\theta/2} \le C'\sqrt{\log W}$ gives the desired estimate. If $x^\theta > C(1+\log R)$, choose C sufficiently large so that $R^{10}e^{-c_{\mathrm{p}}x^\theta} \le e^{-c_{\mathrm{p}}x^\theta/2} \le C'e^{-c'x^{\theta/2}}$. These two cases prove the claim. □

For covariance-weighted sums at large coupling, (6.6) gives total mass L/g^2 and halo mass ℓ_t/g^2. The restoring density $g^2\theta_t$ has size $O(g^2/\ell_t)$, so $k-1$ relative-coordinate sums cost $(L/\ell_t)^{k-1}$, a polynomial in W uniformly in g.

A.9 Deterministic comparison parameters and positive heights

Proof of Lemma 6.3(a). We construct the comparison energy and then check its regularity and scales. Set $g' = g\sqrt{s/u}$ and

$$\xi = \sqrt{s\gamma_{g'}/(u\gamma_g)}, \qquad a = u\gamma_g, \quad b = \gamma_g, \quad b' = (u/s)\gamma_g.$$

Then $2^{-1/2} \le \xi \le 1$, $a < b \le b' \le 2a$, $\xi^2 b' = \gamma_{g'}$ and $A_{g'} = \xi A_g$. Write $m(z;\beta)$ for the finite Dyson solution at fixed hopping A_g and variance β, and put $x = \Re z_u(g,E)$, $\eta_0 = \eta_u(g,E)$, $\rho(y) = \Im m(x+iy;a)$. Substitution gives $m(x+i\eta_0;a) = m(E;b)$ and $\eta_0/\rho(\eta_0) = b-a$. The Stieltjes formula $\rho(y)/y = \int((\lambda-x)^2+y^2)^{-1}\,d\nu_a(\lambda)$ shows that $y/\rho(y)$ increases strictly to infinity. Thus for every $\beta \in [b,b']$ there is a unique $\eta_\beta \ge \eta_0$ with

$$\eta_\beta/\rho(\eta_\beta) = \beta - a, \qquad e_\beta = x - (\beta-a)\Re m(x+i\eta_\beta;a). \tag{A.57}$$

Here $\eta_b = \eta_0$, $e_b = E$. The identity $e_\beta + \beta m(x+i\eta_\beta;a) = x + i\eta_\beta + am(x+i\eta_\beta;a)$ makes $m(x+i\eta_\beta;a)$ a positive-imaginary solution at (e_β,β). Ward at this covariance gives $\Re(1-\beta\langle R^2\rangle) \ge 2\beta\rho(\eta_\beta)^2 > 0$; implicit continuation into $\mathbb{C}_+$ and Herglotz uniqueness identify the root with $m(e_\beta;\beta)$. Covariance scaling now gives

$$E' = \xi e_{b'}, \qquad m_{g',L}(E') = \xi^{-1}m(e_{b'};b'), \qquad w = \xi^{-1}z_s(g',E') = x + i\eta_{b'}. \tag{A.58}$$

In particular, $\Re w = x$, $\Im w \ge \eta_0$, and $\Im w/\eta_0 = (u/s)(1-s)(1-u)^{-1}\rho(\eta_{b'})/\rho(\eta_0)$.

Bounded coupling. Here $a \geq c > 0$. Ward gives $|m(z;a)| \leq a^{-1/2}$, $|m(e_\beta;\beta)| \leq \beta^{-1/2}$ and $\eta_\beta \leq (\beta - a)\beta^{-1/2} \leq \sqrt{2}$. Since $\|A_g\| \leq 2$ and $|x| \leq C' + 1$, imaginary parts of the Dyson equation give $\rho(y) \geq cy$ on the segment. While $\rho(y) \geq c'/2$, implicit differentiation and (A.6) give $|\partial_z m(z;a)| \leq 2/(a^2(c')^2)$. A first drop from c' to $c'/2$ therefore needs a height increase at least $a^2(c')^3/4$, after which $\rho(y) \geq cy$ supplies a fixed positive lower bound. Hence $\rho \asymp 1$ throughout, and (A.58) yields

$$|E'| \leq C'', \qquad g' = g\sqrt{s/u} \leq g, \qquad \Im m_{g',L}(E') \geq (C'')^{-1}, \qquad 1 \leq \frac{\Im w}{\eta_u(g,E)} \leq C'' \frac{1-s}{1-u}. \tag{A.59}$$

The same fixed energy and density bounds hold at every e_β.

Large coupling. Now $\xi = 1$, $A_g = A_L$, $b = g^{-2}$, $b' = (g')^{-2}$. Lemma A.4 and (A.21), at coupling $\beta^{-1/2}$ and margin $\kappa/2$, give $|m(e;\beta)| \leq C$, $\Im m(e;\beta) \geq c$ and $\beta|\partial_e m(e;\beta)| \leq 1/2$ for $|e| \leq 2 - \kappa/2$, $\beta \in [b, b'] \subset [b, 2b]$. For $f_\beta(e) = e + (\beta - a)\Re m(e;\beta)$, differentiation of Dyson gives $\partial_\beta m = m\partial_e m$, hence

$$\partial_e f_\beta \in [1/2, 3/2], \qquad \partial_\beta f_\beta = \Re m + (\beta - a)\Re(m\partial_e m) = O(1).$$

Since $f_b(E) = x$, inversion gives a unique root $f_\beta(e) = x$ with $|e - E| \leq C(\beta - b)$, still in the buffered bulk for sufficiently large g. Its positive-height solution obeys (A.57), so uniqueness identifies it with e_β. Thus

$$|E' - E| \leq C(b' - b), \qquad |E'| \leq 2 - \kappa/2, \qquad 1 \leq \frac{\Im w}{\eta_u(g,E)} \leq C\frac{1-s}{1-u}. \tag{A.60}$$

Scales and the comparison segment. Use $\zeta_t(g)^{-1} = W \min\{\gamma_g L(1-t), \gamma_g(1-t) + g\sqrt{\gamma_g(1-t)}\}$. Since $s(1-s) \geq u(1-u)$ and $\gamma_{g'} \geq \gamma_g$,

$$\begin{aligned} \gamma_{g'}(1-s) + g'\sqrt{\gamma_{g'}(1-s)} &\geq \gamma_g(1-u) + g\sqrt{\gamma_g(1-u)}, \\ \gamma_{g'} L(1-s) &\geq \gamma_g L(1-u). \end{aligned} \tag{A.61}$$

Thus $\zeta_s(g') \leq \zeta_u(g)$, while $g' \asymp g$ and $\gamma_{g'} \asymp \gamma_g$ give $\zeta_s(g') \asymp \zeta_s(g)$; earlier comparison preserves both the time cutoff and scale smallness. For $\eta_0 \leq y \leq \eta_{b'}$, put $\beta = a + y/\rho(y)$, $v = a/\beta \in [s,u]$, $g_v = g\sqrt{v/u}$, $\xi_v = \sqrt{v\gamma_{g_v}/(u\gamma_g)}$ and $E_v = \xi_v e_\beta$. Then $\xi_v^{-1} z_v(g_v, E_v) = x + iy$, with the regularity bounds just proved. For the common Dyson matrix and its square kernel $Q_\beta = \beta(M^2_{\mathrm{common},ab})_{a,b}$,

$$\begin{aligned} M_{\mathrm{common}} &= (A_g - x - iy - am(x+iy;a))^{-1} \\ &= (A_g - e_\beta - \beta m(e_\beta;\beta))^{-1} = \xi_v M_B(g_v, E_v), \\ (I - (a/\beta)Q_\beta)^{-1} &= (I - vQ(g_v, E_v))^{-1}, \qquad \beta\xi_v^2 = \gamma_{g_v}. \end{aligned} \tag{A.62}$$

□

Proof of Lemma 5.11. We now pass from the deterministic comparison parameters to resolvents.

Resolvent comparisons under the Gaussian coupling. Lemma 6.3(a) gives E', w, g', ξ and their parameter bounds. Since $A_{g'} = \xi A_g$ and $s\gamma_{g'} = \xi^2 u\gamma_g$, Gaussian scaling gives, under the coupling $H_s(g') = \xi H_u(g)$,

$$\begin{aligned} \Re w = \Re z_u(g,E), \qquad 1 \leq \frac{\Im w}{\eta_u(g,E)} \leq C''R, \\ (H_u(g) - w)^{-1} = \xi G_s(g', E'). \end{aligned} \tag{A.63}$$

Write $G = G_u(g, E)$, $F = (H_u(g) - w)^{-1}$, $\eta = \eta_u(g, E)$ and $y = \Im w$. Spectral calculus and the resolvent identity give

$$\Im G \preceq \frac{y}{\eta}\Im F, \qquad G - F = i(\eta - y)GF.$$

Using $GG^* = \eta^{-1}\Im G$ and $F^*F = y^{-1}\Im F$, Cauchy–Schwarz on the intermediate block and orbital indices yields

$$\|G_{ab} - F_{ab}\|_{\mathrm{op}} \le \frac{y-\eta}{\eta}\sqrt{\|\Im F_{aa}\|_{\mathrm{op}}\|\Im F_{bb}\|_{\mathrm{op}}},$$
$$|\langle G - F\rangle_{ab}| \le \frac{y-\eta}{\eta}\sqrt{\Im\langle F\rangle_{aa}\Im\langle F\rangle_{bb}}.$$

Taking maxima and using $F = \xi G_s(g', E')$ and $y/\eta \le C''R$ proves the operator-norm and block-trace comparisons. The first spectral-calculus inequality gives the imaginary-part comparison. All three hold simultaneously under the same Gaussian coupling, which proves (5.35) in the stated distributional sense. □

A.9.1 Finite comparison chains

Proof of Lemma 6.3(b). Write g_0 for the target coupling and g for a comparison coupling. At bounded coupling, each application of part (a) above gives an energy bound and a positive lower bound on $\Im m_{g',L}(E')$, depending only on the preceding fixed parameter bounds, not on W. Since J is fixed, taking the largest energy bound and the smallest density lower bound over these finitely many stages gives uniform constants depending only on the original bounds and J; also $cg_0 \le g \le C$.

The large-coupling calculation below needs only the lower-volume condition of part (a), with no upper bound on L. Each comparison preserves ug^{-2}, and moving to an earlier time at fixed coupling decreases it. Since $u \ge 1/2$ and $g' \le g$,

$$\begin{aligned} g_0^{-2} \le g^{-2} \le u^{-1}g_0^{-2} \le 2g_0^{-2}, \\ \sum_{j=0}^{m-1}|E_{j+1} - E_j| \le C\sum_{j=0}^{m-1}(g_{j+1}^{-2} - g_j^{-2}) = C(g_m^{-2} - g_0^{-2}) \le Cg_0^{-2}. \end{aligned} \tag{A.64}$$

The second line uses (A.60) with margin $\kappa/2$, up to any first exit from that buffered bulk. Starting from $|E_0| \le 2 - 3\kappa/4$, choose C_κ so that $Cg_0^{-2} < \kappa/4$ and $g_0/\sqrt{2}$ exceeds the input thresholds; the displayed bound excludes the exit and proves (6.11). Also $L/(g^2\log W) \ge L/(g_0^2\log W) \to \infty$. The scale inequality (A.61) and monotonicity at fixed coupling show that earlier comparisons preserve both $\zeta \le W^{-c_0}$ and $1 - u \ge W^{-3}$. Finally, in the initializer range $L \le Cg_0^2W^2$, the first line of (A.64) gives $L \le 2Cg^2W^2$. The initial and sampling inputs therefore apply with a buffered bulk margin and twice the upper-volume constant.

□

A.10 Resolvent comparison and positive-height bounds

Proof of Lemma 6.4. We also prove the exponential-row refinement used below, with $\ell_0 \asymp \max\{1, g^2\}$:

$$\begin{aligned} u\gamma_g \max_a \sum_b e^{c\|a-b\|_L/\ell_0}(|M_{B,u}(z)_{ab}| + |M_{B,u}(z)_{ab}|^2) \le C', \\ \max_a \sum_b e^{c\|a-b\|_L/\ell_0}\left|\left[(I - u\gamma_g(M_{B,u}(z)^2_{ab})_{a,b})^{-1}\right]_{ab}\right| \le C'. \end{aligned} \tag{A.65}$$

This holds on the comparison segments and on the direct large-coupling domain in Lemma 6.4. On the characteristic, $M_{B,u}(z_u) = M_B(g, E)$; the real-energy row estimates used in the proof of (A.48) therefore give (6.13) also for $u < 1/2$. On a comparison segment, (A.62) identifies $M_{B,u} = \xi_v M_B(g_v, E_v)$ and its stable inverse with $(I - vQ(g_v, E_v))^{-1}$, where $u\gamma_g\xi_v^2 = v\gamma_{g_v}$, $\xi_v \asymp 1$,

and $\ell_0(g_v) \asymp \ell_0(g)$. The bounds (A.59) and (A.60) verify the same real-energy hypotheses. This proves (A.65) and (6.12), using $u \geq 1/2$ to remove u from the first row normalization. For the direct large-coupling domain, apply (A.19) and (A.20) with $u = t$, $\tau = 1$, bulk margin $\kappa/2$, and $C_0 = C$; the stated volume divergence supplies their fixed threshold.

The endpoint identity follows from (A.62) at $v = s$. For (6.14), differentiate the scalar Dyson equation at fixed $u, g, \Re z$:

$$\partial_r m_u = \frac{i\gamma_g L^{-1} \operatorname{Tr} M_{B,u}^2}{1 - u\gamma_g L^{-1} \operatorname{Tr} M_{B,u}^2}, \qquad \partial_r M_{B,u} = \frac{i\gamma_g M_{B,u}^2}{1 - u\gamma_g L^{-1} \operatorname{Tr} M_{B,u}^2}. \tag{A.66}$$

The second identity follows by substitution in the derivative of $M_{B,u}$. The scalar denominator is the zero Fourier mode of the stable inverse in (6.12), so its reciprocal is bounded. The absolute row bounds there give

$$\gamma_g \max_{a,b} |(M_{B,u}^2)_{ab}| \leq C, \qquad \gamma_g^2 \max_a \sum_b |(M_{B,u}^2)_{ab}| \leq C,$$

by convolution, proving the derivative bound. Uniqueness and the characteristic identities follow from (A.6) and substitution in (5.27), respectively. □

A.10.1 Free rows and deterministic interval interpolation

Proof of Lemma 6.5. For $w = E + m/g^2$, the assumed bounds on m give $\Im w \asymp g^{-2}$ and keep $\Re w$ in the buffered free bulk. Hence $|\rho(w)| \leq e^{-c/g^2}$ and $|f_g(w)| \leq C$. Since $L \geq 4g^2$, the denominator in (A.5) satisfies $|1 - \rho(w)^L| \geq 1 - e^{-cL/g^2} \geq c'$. Substitution in (A.5) gives the stronger pointwise estimate

$$|M_{B,bc}| \leq Ce^{-cd(b,c)/g^2}. \tag{A.67}$$

Its geometric sum proves (6.15).

We prove (6.17) through the general free resolvent $R(w) = (A_L - w)^{-1}$, where $w \notin \operatorname{Spec}(A_L)$ and $|w - E| \leq C/g^2$. With the coefficients defined in (6.16), for $a \in I$ and any b we have

$$R(w)_{ab} = \sum_\alpha c_{a\alpha} R(w)_{\alpha b} + q_{ab} + e_{ab}, \qquad |e_{ab}| \leq C\frac{|I|}{g^2}(1 + \max_\alpha |R(w)_{\alpha b}|). \tag{A.68}$$

The coefficients in (6.16) solve the free forced recurrence and are bounded because the bulk margin bounds $\cos\theta$ below. Definition 7.1, with $W = 1$, gives $T(E)^j = S_E R_E^j S_E^{-1}$, whose norm is bounded uniformly for $j \in \mathbb{Z}$. Variation of constants for its perturbation by $w - E$ on at most $|I|$ sites bounds the prefix error by $C|I||w - E|e^{C|I||w-E|} \leq C|I|/g^2$. For $a \in I$, iterate the forced recurrence from $b_0, b_0 - 1$ to a; at most one source occurs, with coefficient q_{ab}. This proves (A.68). The entry bound (6.15) then proves (6.17). For a comparison matrix write $M_{B,u}(z) = (A_L - w_u)^{-1}$ with $w_u = z + ug^{-2}m_u(z)$. Lemma 6.4 gives $|w_u - E| \leq C/g^2$ and bounded entries. The identical free recurrence therefore proves the last assertion. □

A.11 Polynomial-range derivatives and propagation scales

Lemma A.13 (Fourier inputs at polynomial coupling)**.** *Fix $B_g, B_L, A > 0$ and $\kappa \in (0, 2)$. Assume $1 \ll g < W^{B_g}$, $C_A g^2 \log W \leq L \leq W^{B_L}$, $|E| \leq 2 - \kappa$, and $0 \leq t \leq 1 - W^{-3}$, with C_A sufficiently large depending on these fixed parameters. Use the Dyson solution (5.2) and the scales (5.18). For the finite solution $m = m_{g,L}(E)$, put $f(p) = |(2\cos p - E - m/g^2)^{-1}|$ and $U_t(p) = (1 - t\widehat{P}(p))^{-1}$, with normalized finite-grid averages. Then*

$$\|f\|_\infty \leq Cg^2, \qquad \|f\|_1 \leq C(1+\log(g^2)), \qquad \sup_p \langle f(q)U_t(p-q)\rangle \leq C(1+\log(g^2)+W\zeta_t). \tag{A.69}$$

For a sufficiently small fixed $c > 0$,

$$\begin{aligned}(1-t+g^4|p|_{\mathbb{T}}^2)U_t(p) \le C \quad (|p|_{\mathbb{T}} \le c/g^2), \qquad U_t(p) \le C \quad (|p|_{\mathbb{T}} \ge c/g^2),\\ \langle \mathbf{1}_{\{|p|_{\mathbb{T}} \le c/g^2\}} U_t(p)\rangle \le CW\zeta_t/g^2, \qquad W\zeta_t \ge c.\end{aligned} \tag{A.70}$$

Lemma A.14 (Auxiliary estimates in polynomial coupling ranges)**.** *Fix $B_g, B_L > 0$ and $\kappa \in (0,2)$. Assume $1 \ll g < W^{B_g}$, $\log W \ll L \le W^{B_L}$, and $|E| \le 2-\kappa$. Use $\gamma = g^{-2}$, the diffusivity (A.27), and $\eta_t = (1-t)g^{-2}\Im m_{g,L}(E; g^{-2})$. For sufficiently large g,*

$$m_g(E;g^{-2}) = \frac{i}{\sqrt{4-E^2}} + O(g^{-2}), \qquad \Im m_g(E;g^{-2}) \asymp 1, \qquad D_g \asymp g^4.$$

Fix $A > 0$ and restrict to $0 \le t < 1$, $1 - t \ge W^{-3}$. Then $L \ge C_A g^2 \log W$ gives the same density and entry bounds for the actual solution on the finite cycle of L blocks. Its free row has pointwise bound C and weighted row sum Cg^2 at spatial scale g^2. Its mixed kernel has exact row mass $(1-t)^{-1}$, pointwise bound $O(((1-t)\ell_t)^{-1})$, and the exponential profile and increment bound (A.29), where

$$\begin{aligned}\ell_t = \min\{L, 1+g^2/\sqrt{1-t}\}, \qquad \zeta_t = \frac{g^2}{W\ell_t(1-t)} \asymp \frac{1}{W\sqrt{1-t}} + \frac{g^2}{WL(1-t)},\\ (W\eta_t)^{-1} = (\Im m)^{-1}\ell_t\zeta_t, \qquad (W/g^2)\ell_t\zeta_t = (1-t)^{-1}.\end{aligned} \tag{A.71}$$

The same-sign inverse has a bounded weighted row on scale g^2, including at $E = 0$. The finite-cycle comparisons for the Dyson function and pair kernels have error $O(W^{-A})$. For each fixed $j \ge 0$, the derivatives $\partial_E^j m_g$ are bounded and $|\partial_E^j D_g| \le C_j g^4$. For every fixed accuracy A, the finite-cycle comparisons for the Dyson solution and pair kernels have error $O(W^{-A})$ after any fixed number of E-derivatives and, for the pair kernels, t-derivatives, when the constant in $L \ge C_A g^2 \log W$ is chosen after these orders. The Dyson row and same-sign inverse bounds also hold on the comparison segments of Lemma 6.4.

Proof of Lemmas A.13 and A.14. Differentiate the line Dyson contraction established in the proof of Lemma A.4, on the buffered bulk $|E| \le 2 - \kappa/2$. Its scalar derivative is $O(g^{-2})$, giving the expansion and its differentiated form:

$$\partial_E^j m_g(E;g^{-2}) = \partial_E^j\left(\frac{i}{\sqrt{4-E^2}}\right) + O_j(g^{-2}), \qquad |\partial_E^j D_g| \le C_j g^4. \tag{A.72}$$

The Ward relation $g^{-2}|m_g|^2 = \tanh q$ and (A.27) give

$$D_g = \frac{g^4}{4|m_g(E;g^{-2})|^4} - \frac{1}{4}.$$

The lower bound on $|m_g|$ and its fixed derivative bounds therefore give $D_g \asymp g^4$ and the derivative bound in (A.72). The zero-order mixed profile and increments follow from (A.28) with $\lambda \asymp \sqrt{1-t}/g^2$; the actual finite-kernel bounds follow from Lemma A.10. The same-sign row is (A.16), since $L \ge C_A g^2 \log W$ ensures its threshold $L/g^2 \ge R_\kappa$.

The finite comparison uses the same periodic free row (A.5), now with $-\log|\rho| \asymp g^{-2}$. Its error is $O(W^C e^{-cL/g^2})$ at every prescribed derivative order. The scalar derivative is $O(g^{-2})$, so the implicit comparison gives the actual finite Dyson solution and its derivatives with that error. As in the bounded-coupling proof, compare P, Q with the periodized line square kernels. The mixed inverse costs at most $(1-t)^{-2}$; stable inverses use a convergent weighted perturbation series. Choose C_A after these polynomial costs. Row masses always use the exact finite Ward identity (A.6). On the vertical comparison segment, Lemma 6.4 gives the Dyson rows and stable inverse, with the same bulk buffer. The scale relations (A.71) follow directly from (5.18) and $\eta_t = (1-t)g^{-2}\Im m$. Since $\ell_t \le 2g^2/\sqrt{1-t}$, they also give $W\zeta_t = g^2/((1-t)\ell_t) \ge 1/2$.

Proof of (A.69) and (A.70). Write $\delta = 1 - t$. The two free poles have slopes bounded away from zero on the buffered bulk, and $\Im m/g^2 \asymp g^{-2}$. Thus $\|f\|_\infty \le Cg^2$. Summing on the finite grid in dyadic distance shells about these poles gives

$$\langle f \rangle \le C(1 + \log(g^2) + g^2/L) \le C(1 + \log(g^2)),$$

since $L \ge C_A g^2 \log W$. The line multiplier (A.28), with $D_g \asymp g^4$, and the finite-cycle comparison give

$$U_t(p) \le \frac{C}{\delta + g^4 |p|_{\mathbb{T}}^2} \quad (|p|_{\mathbb{T}} \le c/g^2), \qquad U_t(p) \le C \quad (|p|_{\mathbb{T}} \ge c/g^2).$$

Here the comparison error is chosen smaller than $c\delta^2$; all additional polynomial costs are absorbed by increasing C_A. The same symbol and normalized grid summation give, including the zero mode,

$$\langle |U_t - 1| \rangle + \langle \mathbf{1}_{\{|p|_{\mathbb{T}} \le c/g^2\}} U_t(p) \rangle \le C \left(\frac{1}{g^2 \sqrt{\delta}} + \frac{1}{L\delta} \right) \le \frac{CW\zeta_t}{g^2}.$$

This proves (A.70). Finally,

$$\sup_p \langle f(q) U_t(p - q) \rangle \le \langle f \rangle + \|f\|_\infty \langle |U_t - 1| \rangle \le C(1 + \log(g^2) + W\zeta_t),$$

which proves (A.69). □

Lemma A.15 (Early-time pair inverses)**.** *Assume (6.1) and use the time-zero scale ℓ_0 from (5.18). For some fixed $c > 0$, uniformly for $0 \le u \le 1/2$,*

$$\max_a \sum_b e^{c\|a-b\|_L/\ell_0} \left(|[(I - uP)^{-1}]_{ab}| + |[(I - uQ)^{-1}]_{ab}| \right) \le C. \tag{A.73}$$

Proof of (A.73). The exact Ward identity gives $P\mathbf{1} = \mathbf{1}$, and $|Q| \le P$. Hence the unweighted inverse rows are at most $(1 - u)^{-1} \le 2$. The exponential free-row bounds in Lemma A.3 and Lemma A.4 give $\sum_b (e^{c\|a-b\|_L/\ell_0} - 1) P_{ab} \le Cc$ for small fixed c. Choose c so that the weighted row norm of P is at most $3/2$. The Neumann series for both inverses then converge in that norm, with sum at most $(1 - 3u/2)^{-1} \le 4$. □

Corollary A.16 (Differentiated finite-cycle remainders)**.** *Under the assumptions of Lemma 6.1, (A.50) also holds after any fixed finite orders of characteristic-time differentiation. On a vertical comparison segment, use coordinates (u, r) with $z = x + i\gamma_g r$, holding g, x fixed, and put $P(u, r)_{ab} = \gamma_g |M_{B,u}(z)_{ab}|^2$. The same remainder bound holds after any fixed mixed partial derivatives in u, r, comparing this kernel with its periodized line analogue.*

Proof of Corollary A.16. On the characteristic, P is independent of time, so $\partial_u^j (I - uP)^{-1} = j! P^j (I - uP)^{-j-1}$; its row bound costs only $j!(1 - u)^{-j-1} \le j! W^{3(j+1)}$. For the segment derivatives, abbreviate $\gamma = \gamma_g$, $M = M_{B,u}(z)$, $m = L^{-1} \operatorname{Tr} M$, $S = L^{-1} \operatorname{Tr} M^2$ and $d = 1 - u\gamma S$. Differentiating the scalar Dyson equation and then $M = (A_g - z - u\gamma m)^{-1}$ gives

$$\partial_u m = \frac{\gamma m S}{d}, \qquad \partial_r m = \frac{i\gamma S}{d}, \qquad \partial_u M = \frac{\gamma m}{d} M^2, \qquad \partial_r M = \frac{i\gamma}{d} M^2.$$

The stability and weighted rows in (A.65) bound d^{-1}, all fixed-order derivatives of m, and, by repeated convolution, the entries and covariance-weighted rows of the derivatives of M. Thus all fixed-order derivatives of $P(u, r)$ have bounded weighted row norms. Put $R = (I - uP(u, r))^{-1}$. The exact row mass and the lower characteristic endpoint of the segment give

$$1 - u \sum_b P(u, r)_{ab} = \frac{r}{r + u\Im m} \ge cW^{-3}, \qquad \|R\|_{\infty \to \infty} \le CW^3.$$

Here the density is bounded above and below, and the segment has $r \geq c(1-u) \geq cW^{-3}$. The inverse derivatives are

$$\partial_u R = R(P + u\partial_u P)R, \qquad \partial_r R = R(u\partial_r P)R.$$

Every fixed mixed derivative is a finite sum of such products and therefore costs only a fixed power of W. The same identities apply to the line kernels. Subtracting the finite and periodized-line products leaves one differentiated finite-cycle error in every term; all other factors have the polynomial bounds just proved. Along $x(u) = \Re z_u(g, E)$, one has $x'(u) = -\gamma_g \Re m(g, E)$ and $\gamma_g \partial_x M = (\partial_r M)/i$, so the additional chain-rule terms for M, and hence for P by products, have the same bounds.

The error-absorption argument proving (A.50) applies with polynomial powers increased by the fixed derivative orders. Differentiating (A.5) and the finite-cycle comparisons in Lemmas A.3 and A.4 gives $C(1+L)^q e^{-cL}$ at bounded coupling and $C(1+L/g^2)^q e^{-cL/g^2}$ in row norm and in g^2 times pointwise norm at large coupling. The bounded variance interval makes $\partial_r = \gamma_g \partial_{\Im z}$ harmless. Pointwise normalization costs at most $\ell_u \leq L \leq W^{B_L}$, or $\ell_u/g^2 \leq 2W^{3/2}$, respectively. The preceding inverse-derivative bounds add only fixed powers of W, absorbed by the same volume lower bounds at any fixed accuracy. At $g = 0$, the finite and line errors vanish identically. □

Fixed derivative orders. In polynomial ranges, for every fixed derivative order the Dyson matrices and stable inverses have polynomial bounds in the spectral, coupling and time variables. For the Dyson solutions this follows by implicit differentiation in Lemma A.3 and (A.72); for the inverses differentiate $(I - uQ)^{-1}$ and use the already proved weighted row bounds. The general-variance formulas apply separately on $g \leq 1$ and $g \geq 1$.

At large coupling, these conclusions also hold at the comparison couplings $g_v \asymp g$ in (A.62): $L/(g_v^2 \log W) \to \infty$, so (A.44) and Corollary A.16 absorb the fixed polynomial costs. In particular, they include the polynomial-coupling specialization of (A.50).

A.12 Propagation of tensors with zero rows

Use the parameter ranges and deterministic scales of Theorem 5.12(i). Fix $s \leq u \leq v \leq t$ and $n \geq 2$. For an n-tensor A supported on relative diameter $H\ell_u$, $H \geq 1$, write $\mathcal{U}^{\boldsymbol{\sigma}}_{u,v}$ for the tensor product of the n external-leg propagators in (5.15). Then

$$\begin{aligned} \|(U^{+-}_{u,v})^{\otimes n} A\|_\infty &\leq C_n H^n (\zeta_v/\zeta_u)^n \|A\|_\infty, && \text{zero rows,} \\ \|\mathcal{U}^{\boldsymbol{\sigma}}_{u,v} A\|_\infty &\leq C_n H^{n-1} (\zeta_v/\zeta_u)^{n-1} \|A\|_\infty, && \text{at least one stable leg.} \end{aligned} \tag{A.74}$$

Here zero rows means $\sum_{a_2,\dots,a_n} A(a_1, \dots, a_n) = 0$ for every a_1. A same-sign neighboring pair gives a stable leg by (6.2).

Use the probability profile $\theta_v = (1-v)P(I - vP)^{-1}$ and its derivative (A.51).

Proof of (A.74). Use the one-dimensional proof of [31, Lemma 7.7(I)–(II), Appendix A.2], with (A.47) and (A.49) as the kernel inputs. Write $U = I + \Xi$, $\Xi := (v-u)P(I-vP)^{-1}$, and $\rho := (1-u)/(1-v)$. An identity or stable leg anchors one input and gives the factor $C_n(H\zeta_v/\zeta_u)^{n-1}$. For the all-$\Xi$ term, zero-row subtraction at $b_2 = \cdots = b_n = b_1$ leaves one row sum, $n-1$ pointwise kernel bounds, and the increment gain. Summing over the relative support gives

$$C_n \rho \left(\frac{\rho}{\ell_v}\right)^{n-1} (H\ell_u)^{n-1} \frac{H\ell_u}{\ell_v} \|A\|_\infty = C_n \left(\frac{\rho H \ell_u}{\ell_v}\right)^n \|A\|_\infty.$$

These prove both bounds, including $\ell_u = 1$ and $\ell_v = L$. For tails in the zero-row case, set $A_H = A\mathbf{1}_{\{\mathrm{diam} \leq H\ell_u\}}$, $H = W^\nu$. Its row is the negative of the discarded row, so restoration

against θ_v gives

$$\left\| \left(\sum_{b_2,\dots,b_n} A_H(b_1,\dots,b_n) \right) \prod_{j=2}^{n} \theta_v(b_j - b_1) \right\|_\infty \le C_n (L/\ell_v)^{n-1} \|A - A_H\|_\infty.$$

The all-Ξ subtraction for A_H leaves its nonzero row; using $n-1$ pointwise Ξ bounds and one row sum bounds this term by $((1-u)/(1-v))^n$ times the displayed right-hand side. The discarded tensor and the restoration tensor satisfy the same bound by the propagator row sums. Estimate these tensors separately: the restoration has scale ℓ_v, not support in $H\ell_u$. Since $L/\ell_v \le L/\ell_u$ and the time ratio are polynomially bounded, a tail bound W^{-A} with A exceeding these fixed polynomial costs by the desired accuracy makes these errors negligible. With a stable leg, the discarded tail is controlled directly by the propagator row sums, without restoring a row. □

A.13 A coarse stability bound in the upper half-plane

Lemma A.17 (Coarse scalar stability)**.** *Let $E \in \mathbb{R}$ and use the free transforms and Dyson solutions in (A.1)–(A.2). Take the normalized parameters $\gamma = \max\{1,g\}^{-2}$ and $z = E + i\eta$, $0 < \eta \le 1$. Suppose $u \in \mathbb{C}_+$, $|u| \le \eta^{-1}$, and*

$$u = f_g(z + \gamma u; \gamma) + r, \qquad |r| \le \eta/2.$$

Then $|u - m_g(z)| \le C\eta^{-2}|r|$, uniformly in g and E. For every $w \in \mathbb{C}_+$ with $\Im w \ge \eta$, the same parameters satisfy

$$|f_{g,L}(w;\gamma) - f_g(w;\gamma)| \le CL^{-1}\eta^{-2}.$$

Proof of Lemma A.17. Subtract the exact Dyson equation to obtain $u - m_g = q(u - m_g) + r$, where

$$q := \gamma \left\langle \frac{1}{(2\sqrt{\gamma}\, g\cos p - z - \gamma u)(2\sqrt{\gamma}\, g\cos p - z - \gamma m_g)} \right\rangle.$$

Imaginary parts and Cauchy–Schwarz give

$$|q|^2 \le \frac{\gamma(\Im u + |r|)}{\eta + \gamma \Im u} \frac{\gamma \Im m_g}{\eta + \gamma \Im m_g}.$$

Since $\gamma \le 1$, $|r| \le \eta/2$, and $\Im u \le \eta^{-1}$,

$$1 - \frac{\gamma(\Im u + |r|)}{\eta + \gamma \Im u} = \frac{\eta - \gamma|r|}{\eta + \gamma \Im u} \ge \eta^2/4.$$

The second fraction is also at most $1 - c\eta^2$, since $\Im m_g \le \eta^{-1}$. Thus $|1 - q| \ge c\eta^2$, proving the stability estimate. For the Fourier-sum estimate, the derivative of $(2\sqrt{\gamma}\, g\cos p - w)^{-1}$ has modulus at most $2\eta^{-2}$, since $\sqrt{\gamma}\, g \le 1$. The elementary Riemann-sum bound proves the claim. □

A.14 Terminal variance matching

Proof of Lemma 5.14. Set $z := E + i\eta$. The definitions (5.45) give

$$A_{g_*} = c_* A_g, \qquad t_* \gamma_{g_*} = c_*^2 \gamma_g,$$
$$E_* + \gamma_{g_*} m/c_* = c_*(z + \gamma_g m).$$

The Dyson uniqueness and Gaussian scaling calculation proving [31, Lemma 3.3, (3.11)–(3.13)], with these covariance identities, gives $m_{g_*,L}(E_*) = m/c_*$, $z_{t_*}(g_*, E_*) = c_* z$ and $H_{t_*}(g_*) \overset{d}{=} c_* H$, hence (5.47). The bulk bounds of Lemmas A.2 and A.3 give $\Im m \asymp 1$ and $|m| \le C$. Thus $1 - t_* \asymp \eta$, while $\sqrt{t_*} \le c_* \le 1$ follows directly from $\gamma_g = \max\{1,g\}^{-2}$. Together with $m_{g_*,L}(E_*) = m/c_*$ and (5.45), these facts give the parameter bounds (5.46). □

Lemma A.18 (Terminal matching at large coupling)**.** *Fix $\kappa \in (0,2)$ and assume $g \geq C_\kappa$ and $L \geq C_* g^2 \log W$, with C_κ, C_* sufficiently large. Let*

$$\tfrac{1}{2} \leq t_* < 1, \qquad g_* := \sqrt{t_*} g, \qquad \gamma_{g_*} = (t_* g^2)^{-1}.$$

The flow hopping is $A_{g_*} = \sqrt{\gamma_{g_*}} g_* A_L = A_L$, and

$$H_{t_*}(g_*) \stackrel{d}{=} H, \qquad t_* = (g_*/g)^2, \qquad t_* \gamma_{g_*} = g^{-2}. \tag{A.75}$$

More explicitly, its terminal characteristic is

$$z_{t_*}(g_*, E_0) = E_0 + (1 - t_*) \gamma_{g_*} m_{g_*, L}(E_0).$$

For sufficiently large W, its real and imaginary parts satisfy

$$\begin{gathered} \forall E \in [-2+\kappa, 2-\kappa] \quad \exists E_0 \in [-2+3\kappa/4, 2-3\kappa/4] : \\ \Re z_{t_*}(g_*, E_0) = E, \\ c g^{-2}(1-t_*) \leq \Im z_{t_*}(g_*, E_0) \leq C g^{-2}(1-t_*), \\ G(E + i \Im z_{t_*}(g_*, E_0)) \stackrel{d}{=} G_{t_*}(g_*, E_0). \end{gathered} \tag{A.76}$$

Thus (g_*, E_0) satisfies (5.40)(b).

Proof of Lemma A.18. **Proof of (A.75) and (A.76).** Here $A_{g_*} = A_g = A_L$ and $t_* \gamma_{g_*} = g^{-2}$, proving (A.75). Since $g_* \geq g/\sqrt{2}$ and $L/g_*^2 \geq L/g^2$, Lemma 5.13(b) applies at coupling g_* and $\delta = 1 - t_*$ after enlarging the fixed thresholds. It supplies E_0 in (A.76), with

$$\Im z_{t_*}(g_*, E_0) = (1-t_*)\gamma_{g_*} \Im m_{g_*, L}(E_0) \asymp \frac{1-t_*}{t_* g^2} \asymp (1-t_*) g^{-2}.$$

The real characteristic map $f(e) = e + (1-t_*)\gamma_{g_*} \Re m_{g_*, L}(e)$ has $f' \in [1/2, 3/2]$ by (A.21), so its inverse is differentiable. The resolvent identity in distribution follows from the whole-matrix equality (A.75). □

Lemma A.19 (Terminal scale and energy curve in polynomial volumes)**.** *Fix $B_g, B_L > 0$, $\kappa \in (0,2)$ and $0 < \nu < 1$, and assume $1 \ll g < W^{B_g}$ and $g^2 \log W \ll L \leq W^{B_L}$. Write $N = WL$ and $\ell_{W,g,L} = \min\{N, W^2 g^2\}$, and define*

$$t_* := 1 - \ell_{W,g,L}^{-1} g^2 W^\nu, \qquad g_* := \sqrt{t_*} g, \qquad \gamma_{g_*} = (t_* g^2)^{-1}. \tag{A.77}$$

For sufficiently large W, $1/2 \leq t_* \leq 1 - W^{-3}$ and $(1-t_*)/(g^2/N) \geq W^\nu$. For the flow coupling g_*,

$$\zeta_{t_*} \leq C \left(\sqrt{W^{-2} g^{-2} \ell_{W,g,L}}\, W^{-\nu/2} + N^{-1} \ell_{W,g,L} W^{-\nu} \right) \leq C(W^{-\nu/2} + W^{-\nu}) \leq W^{-c_0} \tag{A.78}$$

for any fixed $0 < c_0 < \nu/2$. The inverse characteristic from Lemma A.18 is differentiable, with

$$\begin{gathered} f(e) = e + (1-t_*)\gamma_{g_*} \Re m_{g_*, L}(e), \quad E_0(E) = f^{-1}(E), \quad \eta(E) = (1-t_*)\gamma_{g_*} \Im m_{g_*, L}(E_0(E)), \\ \tfrac{1}{2} \leq f' \leq \tfrac{3}{2}, \quad c \ell_{W,g,L}^{-1} W^\nu \leq \eta(E) \leq C \ell_{W,g,L}^{-1} W^\nu, \quad \eta(E)^{-1} \leq W^{C_1}, \quad |\eta'(E)| \leq W^{C_2}, \quad |E| \leq 2 - \kappa. \end{gathered} \tag{A.79}$$

Here C_1, C_2 are fixed; all constants are independent of g, L. The prescribed height $\eta_L = W^\nu / \ell_{W,g,L}$ also satisfies $\eta_L^{-1} \leq W^{2+2B_g}$.

Proof of Lemma A.19. The endpoint calculation is

$$1-t_* = \max\{W^{-1}L^{-1}g^2, W^{-2}\}W^\nu \le C((\log W)^{-1}W^{-1+\nu} + W^{-2+\nu}) = o(1),$$
$$1-t_* \ge W^{-2+\nu} \ge W^{-3}, \qquad \frac{1-t_*}{g^2/N} = \ell_{W,g,L}^{-1} N W^\nu \ge W^\nu.$$

Since $g_*^2 \asymp g^2$, substitution into (A.71) gives the first bound in (A.78). The inequalities $\ell_{W,g,L} \le W^2 g^2$ and $\ell_{W,g,L} \le N$ give the second. Lemma A.18 supplies the inverse map and the height comparison in (A.79). The derivative bounds in Lemma A.14 and $|E_0'| \le 2$ give $|\eta'| \le 2(1-t_*)\gamma_{g_*}|\partial_e m_{g_*,L}(E_0)| \le W^{C_2}$. Finally $\eta^{-1} \le C\ell_{W,g,L}W^{-\nu} \le CW^{2+2B_g-\nu} \le W^{C_1}$ for a fixed C_1; the same bound without the constant gives the assertion for η_L. □

A.15 Parameter control in polynomial volumes

Lemma A.20 (Scale and parameter control in polynomial volumes)**.** *Fix $B_g, B_L, \kappa > 0$. For $1 \ll g < W^{B_g}$, $g^2 \log W \ll L \le W^{B_L}$ and $|E| \le 2-\kappa$, the characteristic scales satisfy, for $0 \le u < 1$,*

$$\gamma_g L + L/\ell_u \le 2W^{B_L}, \qquad \gamma_g W \ell_u \zeta_u = (1-u)^{-1}, \qquad \zeta_u \ge c/W,$$
$$\ell_u \asymp g^2, \qquad \zeta_u \asymp W^{-1} \quad (0 \le u \le 1/2). \tag{A.80}$$

Starting with $|E_0| \le 2 - 3\kappa/4$, every finite comparison chain as in Lemma 6.3(b) satisfies (6.11), preserves these polynomial ranges with fixed buffered bounds, and cannot increase the scale ζ.

Proof of Lemma A.20. Here $\gamma_g = g^{-2}$ and $\ell_u = \min\{L, 1 + g^2/\sqrt{1-u}\} \ge g^2$ for large W. Thus $\gamma_g L + L/\ell_u \le 2L/g^2 \le 2W^{B_L}$. The normalization is the definition of ζ_u; its lower bound is (A.46). For $u \le 1/2$, $L \gg g^2 \log W$ leaves the scale uncapped, giving the last line of (A.80). The volume-independent calculation (A.64) and its buffered-bulk argument give (6.11); here $g_0 \to \infty$ makes the displacement smaller than the fixed buffer. The bounds $g_0/\sqrt{2} \le g \le g_0$ preserve $g < W^{B_g}$, $L \le W^{B_L}$ and $L/(g^2 \log W) \ge L/(g_0^2 \log W) \to \infty$. Equation (A.61) preserves scale smallness. □

A.16 The deformed one-block Dyson profile

Lemma A.21 (A small-norm and small-rank deformation)**.** *Fix $\kappa_0 \in (0,2)$ and $C_* > 0$. Let $B = B_{\mathrm{small}} + B_{\mathrm{bad}}$ be deterministic Hermitian, with $\|B_{\mathrm{small}}\| \le \varepsilon$ and $\operatorname{rank} B_{\mathrm{bad}} \le \varepsilon W$. For sufficiently small fixed ε, the Herglotz solution $M_B(z) = (-EI_W - B - z - m_B(z))^{-1}$, $m_B = W^{-1}\operatorname{Tr} M_B$, satisfies*

$$\Im m_B(i\eta) \asymp 1, \qquad \Im M_B(i\eta) \asymp (I_W + B^2)^{-1}, \qquad |E| \le 2-\kappa_0, \quad 0 < \eta \le C_*. \tag{A.81}$$

The constants depend only on the fixed bulk margin and height bound, not on the exceptional eigenvalues of B. The comparison for $\Im M_B$ is an inequality of quadratic forms.

Under the hypotheses of Lemma A.21, we also prove, uniformly in its energy and height ranges,

$$|m_B(i\eta)| \le 1, \qquad \left|1 - W^{-1}\operatorname{Tr} M_B(i\eta)^2\right| \ge c. \tag{A.82}$$

Proof of Lemma A.21 and (A.82). Write $M = M_B$, $m = m_B$, and $H = \max\{C_*, 2\}$. Imaginary parts and Cauchy–Schwarz give $|m| \le 1$ as in (A.6). Min–max gives $\#\{\lambda(B) \notin [-\varepsilon, \varepsilon]\} \le \operatorname{rank} B_{\mathrm{bad}} \le \varepsilon W$. On a fixed tube around the bulk semicircle branch, the imaginary gap bounds the scalar inverses and their squares. The small eigenvalues change each by $O(\varepsilon)$, and the exceptional fraction contributes $O(\varepsilon)$ regardless of its eigenvalues. Thus

$$m = -(E + i\eta + m)^{-1} + O(\varepsilon).$$

Start at $\eta = H$, where the scalar map is a contraction. The two semicircle roots remain uniformly separated for $|E| \le 2 - \kappa_0$, $0 \le \eta \le H$; the displayed residual excludes a first exit from the tube for small ε. Hence $m = m_{\mathrm{sc}}(E + i\eta) + O(\varepsilon)$ and $\Im m \asymp 1$. Applying the same eigenvalue comparison to the squared inverses yields $W^{-1} \operatorname{Tr} M^2 = m_{\mathrm{sc}}(E + i\eta)^2 + O(\varepsilon)$; the bulk semicircle stability gap proves (A.82). Finally, in each eigenvalue coordinate b of B,

$$\Im M(i\eta) = \frac{\eta + \Im m}{(E + b + \Re m)^2 + (\eta + \Im m)^2} \asymp (1 + b^2)^{-1}.$$

This proves the quadratic-form comparison uniformly in all exceptional eigenvalues and as $\varepsilon \downarrow 0$. □

Proof of Lemma 6.6: (6.18). Take $B = g^2 R$: its part on $\{|g^2 R| \le g\}$ has norm at most g, and the complementary rank is at most $W g^{1/2}$. Both fractions tend to zero, so (A.81) applies with $\varepsilon = \max\{g, g^{1/2}\}$. □

Lemma A.22 (Residual stability and vector weights for the deformed profile)**.** *Under the hypotheses of Lemma A.21, write $M = M_B(i\eta)$ and $m = m_B(i\eta)$. There are fixed $r_0, C > 0$ such that, if $|q - m| \le r_0$, then*

$$r = q - W^{-1} \operatorname{Tr}(-EI_W - B - i\eta - q)^{-1} \quad \Longrightarrow \quad |q - m| \le C|r|. \tag{A.83}$$

Uniformly in the same bulk and height ranges, for every vector v,

$$\|M\| + \|\Im M\| \le C, \qquad \eta \le C\Im m, \qquad v^* \Im M v = (\eta + \Im m)\|Mv\|_2^2 \asymp \|Mv\|_2^2. \tag{A.84}$$

Proof of Lemma A.22. The positive imaginary gap in (A.81) bounds the second derivative of the scalar Dyson map in a fixed neighborhood of m. Taylor expansion gives

$$(1 - W^{-1} \operatorname{Tr} M^2)(q - m) = r + O(|q - m|^2).$$

The stability gap in (A.82) absorbs the quadratic term when r_0 is sufficiently small, proving (A.83). For a continuous approximate solution starting close to m at the upper height, this estimate excludes a first exit from the neighborhood while the residual remains small, uniformly throughout the height interval. The same imaginary gap bounds $\|M\|$; $\Im m \asymp 1$ gives $\eta \le C\Im m$. Diagonalizing the Hermitian matrix B gives $\Im M = (\eta + \Im m) M^* M$, proving (A.84). □

B Loop identities and primitive-loop estimates

We prove the loop identities and primitive estimates stated in Section 5.2. The deterministic kernel inputs have already been proved in Appendix A; the arguments below are independent of stochastic propagation.

B.1 Exact loop algebra and mixed Ward identity

We use the flow, Dyson matrix, loops, and cut operations of Section 5. Set $\gamma = \gamma_g$, $\Lambda = A_g \otimes I_W$, and $\mathcal{S}[R] = \gamma W^{-1} \operatorname{diag}(\operatorname{Tr} R_{aa})_{a \in \mathbb{Z}_L} \otimes I_W$. The characteristic identity (A.3) gives $\Lambda - z_t - t\mathcal{S}[M] = M^{-1}$ with M independent of t. Thus [31, Assumption 4.1] holds; its general tree representation [31, Lemma 4.16, (4.15)–(4.16)] and primitive Ward identity [31, Corollary 4.21] apply. These identities require no analytic small-coupling hypothesis.

Proof of Lemma 5.1. For actual loops, cyclically place the two opposite signs next to the summed projector and use $\sum_c E_c = I/W$ and $G_t - G_t^* = 2i\eta_t G_t G_t^*$. This gives (5.11). For primitives, the symmetric coordinate covariance is $S_{ai,bj} = \gamma W^{-1} \mathbf{1}_{a=b}$. Average each external coordinate of the general primitive in [31, Corollary 4.21] against P_a/W, as in [31, Lemma 4.26, (4.38)]. The two

new normalized projectors change the coordinate cut coefficient γ/W to $(\gamma/W)W^2 = \gamma W$, so the averaged initial values and evolution are exactly (5.10). The summed Ward vertex introduces one additional W^{-1}, giving $1/(2iW\eta_t)$ in (5.11). □

Proof of (5.8) and (5.9). We start from the resolvent equation (5.4). The quadratic contraction of two differentiated factors of a loop is its cut product. A light insertion is γW times $\langle \widetilde{G}^{\sigma} \rangle_{cc}$ and the $(k+1)$-loop obtained by inserting the duplicated sign and E_c. Let $\{T_\lambda\}_{\lambda=1}^{W^2}$ be a basis of Hermitian $W \times W$ matrices with $\mathrm{Tr}(T_\lambda T_\mu) = \delta_{\lambda\mu}$, and let $T_{c,\lambda}$ be its embedding into block c. Then $d\mathcal{B}_t^{(g)} = \sqrt{\gamma/W} \sum_{c,\lambda} T_{c,\lambda}\, d\beta_{c,\lambda}$ for independent standard real Brownian motions $\beta_{c,\lambda}$. Differentiating the i-th resolvent in (5.5) and cyclically moving $d\mathcal{B}_t^{(g)}$ to the front gives $-\mathrm{Tr}(d\mathcal{B}_t^{(g)} B_i)$, with B_i defined before (5.9). Completeness of this basis gives $\sum_\lambda \mathrm{Tr}(T_{c,\lambda} B_i)\overline{\mathrm{Tr}(T_{c,\lambda} B_j')} = \mathrm{Tr}(P_c B_i P_c B_j'^{*})$, which proves (5.9). Replacing both P_c by WE_c makes this γW times joined $(2k+2)$-loops with one free block, rather than a product of unrelated loops.

Proof of Lemma 5.4: (5.24). On $[0,T]$, the auxiliary model has $\Im z_t \geq \eta_{\min} := (1-T)\gamma_g \Im m > 0$ and hence $\|G_t^{(\widetilde{W})}\| \leq \eta_{\min}^{-1}$, uniformly in $\widetilde{W}$. At fixed t, the directional derivative of a normalized trace word in a Hermitian direction V_c in block c is a sum of k terms $\widetilde{W}^{-1}\mathrm{Tr}(YV_c)$. Each Y is formed from resolvents and unnormalized block projections, compressed to block c, so $\|Y\|_{\mathrm{HS}} \leq C\sqrt{\widetilde{W}}$. Gaussian Poincare with the fixed-time covariance $t\gamma_g/\widetilde{W}$ gives

$$\sup_{0\leq t\leq T} \mathrm{Var}(\widetilde{W}^{k-1}\mathcal{L}_k^{(\widetilde{W})}(t)) \leq C_{k,g,L,E,T}\widetilde{W}^{-2}.$$

Common orbital-unitary invariance defines $R_{\widetilde{W}}(t) \in \mathbb{C}^{L\times L}$ by $\mathbb{E}(G_t^{(\widetilde{W})})_{bc} = R_{\widetilde{W}}(t)_{bc} I_{\widetilde{W}}$. By translation invariance, $\bar{m}_{\widetilde{W}}(t) := R_{\widetilde{W}}(t)_{bb} = \mathbb{E}\mathcal{L}_1^{(\widetilde{W})}(t;b,+)$ is independent of b. Gaussian integration by parts gives

$$(A_g - (z_t + t\gamma_g \bar{m}_{\widetilde{W}}(t))I_L) R_{\widetilde{W}}(t) = I_L + o(1),$$

uniformly on $[0,T]$ in $L \times L$ matrix norm. Each entry of the residual matrix is a covariance term between a normalized diagonal trace and a resolvent entry, hence is $o(1)$ by Poincare and the resolvent bound. Since $\Im \bar{m}_{\widetilde{W}}(t) \geq 0$, the inverse of $A_g - (z_t + t\gamma_g \bar{m}_{\widetilde{W}}(t))I_L$ has norm at most $\eta_{\min}^{-1}$. The uniform residual and boundedness imply that any subsequential limit μ of $\bar{m}_{\widetilde{W}}(t)$ as $\widetilde{W} \to \infty$ and $t \to t_* \in [0,T]$ satisfies

$$\mu = L^{-1}\,\mathrm{Tr}(A_g - (z_{t_*} + t_*\gamma_g \mu)I_L)^{-1}.$$

Scalar uniqueness in Lemma 6.4 and the characteristic identity (5.27) give $\mu = m$. Compactness therefore yields $\sup_{t\leq T} |\bar{m}_{\widetilde{W}}(t) - m| \to 0$. Together with Poincare, this gives

$$\sup_{0\leq t\leq T} \mathbb{E}|\mathcal{L}_1^{(\widetilde{W})}(t;b,+) - m|^2 \longrightarrow 0.$$

Take expectations in (5.8) at width $\widetilde{W}$ and normalize. The martingale has zero expectation; the centered insertion is $o(1)$ by the preceding L^2 convergence and the resolvent bound. Poincare factorizes the normalized cut products up to $o(1)$, giving

$$\frac{d}{dt}(\widetilde{W}^{k-1}\mathbb{E}\mathcal{L}_k^{(\widetilde{W})}) = \gamma_g \widetilde{W}^k \sum_{i<j,c} (\mathbb{E}\,\mathrm{Cut}_{L,i,j}^c\, \mathcal{L}_k^{(\widetilde{W})})(\mathbb{E}\,\mathrm{Cut}_{R,i,j}^c\, \mathcal{L}_k^{(\widetilde{W})}) + o(1),$$

uniformly on $[0,T]$. For cut lengths $p+q = k+2$, $\widetilde{W}^k\widetilde{W}^{1-p}\widetilde{W}^{1-q} = 1$, so the normalized equation agrees with (5.10) for $W^{k-1}\mathcal{K}_k$, up to $o(1)$; the initial data agree exactly. For $k=2$, the resolvent

bound and (5.13) bound both $\widetilde{W}\mathbb{E}\mathcal{L}_2^{(\widetilde{W})}$ and the Riccati solution $W\mathcal{K}_2$ on $[0,T]$. Subtracting their integral equations and applying Gronwall gives uniform convergence. For $k \geq 3$, adjacent cuts are linear in the length-k tensor and all other cuts use shorter words; induction and Gronwall give uniform convergence to $W^{k-1}\mathcal{K}_k$. There are finitely many label and sign tuples, proving (5.24). □

B.2 Tree representation and stable molecule definitions

For a translation-invariant k-tensor, Fourier transform the $k-1$ displacements from its first vertex; the momenta $k_1, \dots, k_k$ satisfy $\sum_i k_i = 0$. Inverse Fourier transform uses normalized averaging on each copy of $(2\pi/L)\mathbb{Z}_L$. For a sign pair define

$$V_t^{+-}(q) = V_t^{-+}(q) \coloneqq (1 - t\widehat{P}(q))^{-1}, \qquad V_t^{++}(q) \coloneqq (1 - t\widehat{Q}(q))^{-1}, \qquad V_t^{--}(q) \coloneqq \overline{V_t^{++}(q)}.$$

The mixed inverses may grow as $t \uparrow 1$; the same-sign inverses have the bounded weighted rows of Lemmas A.5 and A.4. For $k \geq 3$, define the Fourier core by

$$W^{k-1}\widehat{\mathcal{K}}_k(t;\boldsymbol{k};\boldsymbol{\sigma}) = C_t^{[k]}(\boldsymbol{k};\boldsymbol{\sigma}) \prod_{i=1}^{k} V_t^{\sigma_i\sigma_{i+1}}(k_i), \qquad \sigma_{k+1} \coloneqq \sigma_1.$$

The tree/core decomposition [31, Lemmas 4.16, 4.22, 4.25], with the block reduction [31, (4.38)–(4.42)], gives the following polygon-dual formula over noncrossing dissections:

$$C_t^{[k]} = \sum_{\mathfrak{d}} \prod_{F\in\mathfrak{d}} C_0^{[|F|]}(F) \prod_{d\in\mathfrak{d}} \gamma_g t V_t(q_d). \tag{B.1}$$

Faces have at least three sides, and each unordered nonadjacent diagonal $d = \{i,j\}$ occurs once. It carries $V_t(q_d) = V_t^{\sigma_i\sigma_j}(q_{ij})$ with $q_{ij} = k_i + \cdots + k_{j-1}$, opposite momenta in the two children, and the cut signs of (5.6). The block normalization in both coupling regimes is $\gamma_g W W^{1-k_L} W^{1-k_R}/W^{1-k} = \gamma_g$, where $k_L + k_R = k+2$.

Define a stable molecule $S_t^{[k]}$ as the spatial tensor whose Fourier transform is the sum in (B.1) restricted to dissections whose internal diagonals all carry same-sign inverses. Its external inverses are still removed. Any unrestricted dissection is obtained by joining these molecules along its mixed diagonals; the adjacency graph of the molecules is a tree. For an anchored tensor, its weighted row norm is the sum over the remaining vertices after multiplying by $e^{c\,\mathrm{diam}(0,\boldsymbol{x})}$, with periodic diameter.

B.3 Core cancellation and bounded-coupling estimates

The next two estimates are uniform under the parameter conditions (A.22).

Lemma B.1 (Spatial moments and the alternating core)**.** *Use the bounded-coupling parameters C, B_0, c', g, E of Lemma A.5, and let $S_t^{[k]}$ be the stable molecule defined after (B.1). Its norm $\|\cdot\|_{\ell_c^1}$ is the weighted row norm defined there, with a sufficiently small fixed $c > 0$. For every fixed $k \geq 3$, every external anchor, and every fixed $a \geq 1$, uniformly for $g \leq C$, $0 \leq t \leq 1$,*

$$\|S_t^{[k]}\|_{\ell_c^1} \leq C', \qquad \sum_{\boldsymbol{x}} \mathrm{diam}(0,\boldsymbol{x})^a |S_t^{[k]}(0,\boldsymbol{x})| \leq C' g^2. \tag{B.2}$$

For an alternating even molecule,

$$|\widehat{S}_t^{[k]}(\boldsymbol{k})| \leq C' \left((1-t) + g^2 \sum_i |k_i|_{\mathbb{T}}^2 \right). \tag{B.3}$$

Proof. A term with s stable diagonals has $s+1$ faces and $k+2s$ deterministic M-edges. Delete one edge in each face. The remaining edges and covariance links form a spanning tree. Deleted edges are pointwise bounded; all retained edges and stable inverses have bounded weighted rows by Lemma A.5. Leaf summation, retaining a small external exponential weight, proves the first estimate.

For small g, the free Neumann expansion and (A.32) give weighted off-diagonal rows $O(g)$ for M_B and $O(g^2)$ for each same-sign inverse, uniformly in L, t. Expand each stable dissection into diagonal and off-diagonal edges. A term with positive external diameter contains either an off-diagonal stable edge, or at least two off-diagonal M_B-edges: if all stable edges identify their endpoints, some closed face must leave a vertex and return. In the preceding tree sum, each retained marked edge keeps its weighted row gain, and each deleted marked edge keeps its pointwise gain. Thus the weighted row restricted to positive diameter is $O(g^2)$. Decreasing the weight proves every fixed positive spatial moment in (B.2); for g bounded away from zero, the first bound gives the same conclusion after enlarging C'.

For an alternating molecule, use the signed Ward-deletion and subtraction induction in [31, Corollary 4.28 and proof of Lemma 4.29, (4.45)–(4.46)]. Here the stable weighted tree bounds pure-sign rows by CW^{1-k}, each Ward deletion costs $C/[W(1-t)]$, and each normalized zero-mode mixed join costs $C/(1-t)$, since $\gamma_g, \Im m \asymp 1$. These inputs give $|\widehat{S}_t^{[k]}(0)| \le C'(1-t)$. The rows are signed; absolute summation before deletion is invalid.

Choose symmetric integer representatives for external displacements, splitting each Nyquist coordinate when L is even. The resulting smooth Fourier interpolation agrees on the finite grid. Simultaneous reflection gives zero gradient at the origin, and the second moment in (B.2) bounds its global Hessian by $C'g^2$. Taylor's theorem in any anchored momentum coordinates proves (B.3) globally on the fundamental domain. Circular distances obey $|q|_{\mathbb{T}} \le \sum_{i \ne v} |k_i|_{\mathbb{T}}$ when the charges sum to zero, so aliasing does not remove this bound. The preceding arguments apply for $t < 1$. The uniform same-sign inverse bound in (A.24) implies that $I - Q$ is invertible. These inverses and the finite stable-dissection formula extend continuously to $t = 1$; taking this limit proves (B.2) and (B.3) there. □

Proposition B.2 (Primitive-loop bounds)**.** *Under the parameter assumptions of Lemma A.5, use ℓ_t, ζ_t from (A.23) and the primitive loops (5.10). Uniformly for $0 \le t < 1$, for every fixed sign word of length $k \ge 2$ and $g > 0$,*

$$|\mathcal{K}_k(t)| \le C'\zeta_t^{k-1}.$$

The constant is uniform for $0 < g \le C$. For every fixed $B_L, \nu, A > 0$, on $1 - t \ge W^{-3}$ and $L \le W^{B_L}$, the joint tail is $O(W^{-A})$ beyond $W^{\nu}\ell_t$. The absolute pair row is at most $C'/(\gamma_g W(1-t))$, with a constant independent of g, and the exact primitive Ward identities hold.

Proof. For $k = 2$ this follows from Lemma A.5 and the explicit two-loop formulas (5.13). For $k \ge 3$, put $F_k := W^{k-1}\widehat{\mathcal{K}}_k(t)$ and use H from (A.26); every integral below is normalized finite-grid averaging. Here γ_g is bounded above and below, so its fixed-order powers are absorbed in the constants; the final estimate is stated using ζ_t. We prove the full and mixed-leg marginal estimates

$$\|F_k\|_1 \le C'H^{k-1}, \qquad A_v(q) := \int_{k_v = q} |F_k| \le C'H^{k-2}U_t(q). \tag{B.4}$$

For an alternating molecule, (B.3) and (A.25) cancel one external mixed inverse in each summand at bounded cost, since the numerator is bounded by $\sum_i((1-t) + g^2|k_i|_{\mathbb{T}}^2)$. Choose the cancelled momentum as dependent; all remaining $k-1$ independent inverse integrals cost H. For a fixed mixed leg q, use the circular triangle inequality (or anchor at that leg) to cancel a different leg; the remaining $k-2$ free integrals cost H and retain $U_t(q)$. For a nonalternating molecule there is already a stable external leg; make it dependent. At most $k-1$ mixed integrations remain, or

$k-2$ when one leg is retained. The stable core is bounded by Lemma B.1. These single-molecule estimates imply (B.4) for every $k \geq 3$.

Mixed gluing of dressed children is exactly $\gamma_g t F_L F_R / U_t(q)$; each child already contains the cut inverse. Since $\gamma_g t \leq 1$ and $k_L + k_R = k + 2$,

$$\int U_t(q)^{-1} A_L(q) A_R(-q)\, dq \leq \sup_q U_t(q)^{-1} A_L(q)\, \|F_R\|_1 \leq C' H^{k-1}.$$

For a designated retained momentum p on the left child, denote its two-leg absolute marginal by $A_L(p,q)$. Bound the right child by $\sup_q A_R(-q)/U_t(q) \leq C' H^{k_R-2}$ before integrating the left child. Since $\int A_L(p,q)\, dq = A_L(p)$, the resulting bound is $C' H^{k-2} U_t(p)$. The same argument applies with left and right interchanged. This proves the second norm in (B.4) without assuming an additional two-leg estimate. Induction along the finite molecule tree gives the bounds. Fourier inversion gives the stated pointwise bound on the primitive loops. For tails, retain an external exponential weight along a connecting tree in every dissection. Stable edges have bounded weighted rows; mixed inverses have weighted row $C'/(1-t)$ at scale ℓ_t. For some fixed $C', c > 0$ this gives $|\mathcal{K}_k(t)| \leq C' W^{1-k} (1-t)^{-C'} e^{-c\,\mathrm{diam}(\boldsymbol{a})/\ell_t}$. Since $(1-t)^{-1} \leq W^3$, the tail is arbitrarily small past $W^\nu \ell_t$. Ward and initial words are the exact algebra already verified. □

Proof of Lemma 5.2(a). Its parameter assumptions imply those of Lemma A.5, and (A.23) uses the scales (5.18). Proposition B.2 therefore proves (5.20) in case (a), its length-two bound, and the stated absolute pair-row bound. Its weighted estimate also gives the tail at $L \leq W^2$ and $1 - t \geq W^{-3}$.

B.4 The polynomial large-coupling auxiliary estimate

Proposition B.3 (Large-coupling primitive bounds)**.** *Under the assumptions of Lemma A.13, every fixed primitive word of length $k \geq 2$ satisfies*

$$|\mathcal{K}_k(t)| \leq C(1 + \log(g^2))^{6k} \zeta_t^{k-1}.$$

It has arbitrarily small polynomial joint tails beyond $W^\nu \ell_t$ for every fixed $\nu > 0$. The absolute pair row is $O(g^2/(W(1-t)))$ with no logarithmic factor.

Proof. Use the loops, primitive hierarchy and Ward identity of (5.10) and (5.11), now with cut coefficient W/g^2, pair loops $\mathcal{K}_2^{+-}(t) = (g^2/W) P (I - tP)^{-1}$ and $\mathcal{K}_2^{++}(t) = (g^2/W) Q (I - tQ)^{-1}$, and Ward denominator $2iW\eta_t$. All Fourier averages below are normalized finite-grid averages. Write $m := m_{g,L}(E)$ and $F_k := W^{k-1} \widehat{\mathcal{K}}_k(t)$. Use f, U_t and their Fourier bounds (A.69)–(A.70) from Lemma A.13. The density, free-row and stable-inverse bounds are those of Lemma A.4; the mixed row, pointwise and weighted bounds are (A.37) and (A.43). Their volume conditions follow from $L \geq C_A g^2 \log W$ and $1 - t \geq W^{-3}$ after choosing C_A sufficiently large.

The common dissection formula (B.1) applies with $\gamma_g = g^{-2}$: each internal diagonal carries tV_t/g^2. Thus the core normalization and the retained-leg convention agree with the bounded-coupling argument.

For the stable molecule cores $S^{[j]}$, repeat the weighted leaf summation in the proof of Lemma B.1 at scale g^2. A term with s diagonals has $j + s - 1$ free rows of cost Cg^2 and s links of cost C/g^2, giving row bound Cg^{2j-2}; a spatial moment of order $a = 1, 2$ costs a further g^{2a}. The signed Ward deletion and mixed-reducible subtraction in the same proof give the zero-momentum bound $O((1-t)g^{2k-2})$: a normalized deletion costs $Cg^2/(1-t)$ and a zero-mode mixed join costs $(g^2(1-t))^{-1}$, while $\sum_i j_i = k + 2(b-1)$ for a tree of b molecules. The symmetric Fourier interpolation in that proof, including the Nyquist convention, and reflection cancel the first-order term. The second-moment bound therefore gives

$$|\widehat{S}^{[k]}(\boldsymbol{k})| \leq C g^{2k-2} \left(1 - t + g^4 \sum_i |k_i|^2\right) \quad \text{on } |k_i| \leq c/g^2. \tag{B.5}$$

The momentum integration retains the factors of g^2. A stable term with s diagonals is a connected graph with k degree-two external vertices, s degree-four internal vertices, and $k+2s$ edges. An s-edge forest joins each internal vertex to an external root; its f-suprema cancel g^{-2s}. Its elimination has coefficients ± 1 on the finite grid. A spanning tree on the external roots has $k-1$ edges; outward integration uses (A.69), while the $s+1$ chords cost $\|f\|_1$. With a bounded external inverse this yields $C(1+\log(g^2))^{s+1}(1+\log(g^2)+W\zeta_t)^{k-1}$. For a retained mixed leg v with momentum q, remove its inverse and choose instead a spanning forest rooted at all other external vertices. Connectedness guarantees such a forest, with $s+1$ edges. At v, the two incident momenta satisfy

$$p_{\text{in}} - p_{\text{out}} = q.$$

Eliminating from leaves to roots therefore fixes the parent edge of v from its other incident momentum and q; at each internal vertex the zero-charge conservation law fixes its parent edge in the same way. Each solved momentum has coefficient ± 1, so this is a bijective change of the independent finite-grid variables, with no volume factor. The $s+1$ edge suprema leave $g^{2(s+1)}g^{-2s} = g^2$. After contracting the forest, the graph has $k-1$ external roots and $k+s-1$ edges: a spanning tree has $k-2$ edges and leaves $s+1$ chords. With a bounded nonretained external inverse as the final root, the same outward integration thus gives $k-2$ tree integrals and $s+1$ chord integrals, uniformly in q. For a nonalternating molecule a stable external leg is available. For an alternating one a large external momentum provides it; on the remaining region use (B.5) and the low-frequency bounds (A.70). The full and one-leg absolute Fourier bounds are thus $C(1+\log(g^2))^{2k}(W\zeta_t)^{k-1}$ and $Cg^2(1+\log(g^2))^{2k}(W\zeta_t)^{k-2}U_t(q)$, respectively. When the retained leg is mixed, its momentum is minus the sum of the other momenta, so the canceled multiplier can be chosen among the nonretained legs.

Repeat the mixed-gluing induction in the proof of Proposition B.2, now with join $tF_LF_R/(g^2U_t(q))$. The extra g^2 in the one-leg bound cancels g^{-2} at each join. For a retained leg, bound the opposite child's one-leg marginal first, as there; no two-leg supremum is needed. For a tree of b molecules of orders j_i, we have $\sum_i j_i = k+2(b-1)$ and $b \le k-2$, so the factors $(1+\log(g^2))^{2j_i}$ contribute at most $(1+\log(g^2))^{6k}$. Hence

$$|\mathcal{K}_k(t)| \le C(1+\log(g^2))^{6k}\zeta_t^{k-1}. \tag{B.6}$$

Stable edges have range g^2 and mixed edges have range ℓ_t. Retaining a small external exponential weight gives joint tails with polynomial prefactors. At diameter $W^\epsilon\ell_t$, these tails are $O_D(W^{-D})$ for every fixed $D>0$. The pair row sums are $O(g^2/(W(1-t)))$, with no logarithmic loss. Since $g^2 \le CW^{2B_g}$, every fixed $(1+\log(g^2))^{6k}$ is $W^{o(1)}$. Only the fixed-constant pair row generates the linear evolution, so a logarithm is never exponentiated. □

B.5 Uniform physical-space estimates at arbitrary coupling

The tree and Ward identities used in Appendix B.2 remain exact for arbitrarily large coupling. Their Fourier estimates can, however, lose a factor depending on $\log g$. We give a physical-space estimate that removes this loss. The pointwise norm and the absolute row norm with one external inverse removed close under the same tree gluing.

B.5.1 An explicit deterministic contract

Let $\Lambda = \mathbb{Z}/L\mathbb{Z}$ with counting measure, and let d denote cyclic distance. Fix an independent scale $a \ge 1$, $L \ge a$, and $0 < \delta \le 1$, and put

$$\ell = \min\{L, a/\sqrt{\delta}\}, \qquad A = \frac{a}{\delta\ell} \ge 1.$$

All constants below may depend on a fixed tensor order but not on a, L, δ. Convolution uses counting measure. The analytic kernel assumptions, stated in Lemmas A.4 and A.10, are (A.16)

and (A.38)–(A.41): throughout this abstract argument, replace g^2 by a in the kernel bounds, replace $\mathcal{E}$ by E, and allow any even kernels satisfying them. Different external legs may have different stable kernels; the mixed kernel $U = I + \Xi + E$ is common. No difference estimate on E is used. The molecule cores are the stable parts of the primitive dissection from Appendix B.2. Abstractly, a core $S_k : \Lambda^k \to \mathbb{C}$, $k \geq 3$, is translation invariant, invariant under simultaneous reflection, and satisfies

$$|S_k(x_1, \dots, x_k)| \leq C e^{-c \operatorname{diam}(x_1,\dots,x_k)/a}. \tag{B.7}$$

Consequently, at every anchor v,

$$\sum_{\boldsymbol{z}} |S_k(0, \boldsymbol{z})|(1 + |\boldsymbol{z}|_1^2/a^2) \leq C a^{k-1}, \tag{B.8}$$

where the other coordinates use minimal cyclic representatives. A molecule is called alternating if all its external legs are mixed. For such a molecule assume the signed row estimate

$$\left| \sum_{\boldsymbol{z}} S_k(0, \boldsymbol{z}) \right| \leq C\delta a^{k-1}. \tag{B.9}$$

Every nonalternating molecule has at least one stable external leg. Dressing means convolution separately in all external coordinates:

$$F_k(\boldsymbol{x}) = \sum_{\boldsymbol{y}} S_k(\boldsymbol{y}) \prod_{j=1}^{k} V_j(x_j - y_j), \qquad V_j \in \{U, \text{stable kernels}\}.$$

For a mixed leg v, denote by $F_k^{\langle v \rangle}$ the same expression with its factor U replaced by I. Thus this notation means removing that factor even if U has not been assumed invertible. Set

$$P(F_k) = \sup_{\boldsymbol{x}} |F_k(\boldsymbol{x})|, \quad R_v(F_k) = \sup_{\boldsymbol{x}_{\neq v}} \sum_{x_v} |F_k^{\langle v \rangle}(\boldsymbol{x})|. \tag{B.10}$$

B.5.2 Anchoring, Taylor cancellation, and the molecule estimate

Lemma B.4 (Anchoring)**.** *Suppose a tensor has the core profile (B.7). In a dressed expression, if at least one external factor has absolute row bounded by a constant, and the absolute value of each remaining factor has convolution with $e^{-cd(\cdot,0)/a}$ bounded by CA uniformly in the output coordinate, then its pointwise norm is at most CA^{k-1}. If an additional leg is undressed and summed in absolute value, the bound is CaA^{k-2}.*

Proof. The convolution hypothesis bounds the mass of each remaining factor on every ball of radius a by CA. Summing these bounds over annuli of width a gives the same convolution bound with c replaced by $c/(k-1)$, at a cost depending only on k. Anchor at a leg j with bounded row. The diameter bound implies

$$|S_k(\boldsymbol{y})| \leq C \prod_{i \neq j} e^{-cd(y_i, y_j)/((k-1)a)}.$$

For a fixed y_j, each other convolution costs at most CA. Summing the anchored kernel costs a constant. For the second assertion, the undressed leg is distinct from j; its absolute sum costs $\sum_x e^{-cd(x,y_j)/((k-1)a)} \leq C_k a$. The other $k-2$ convolutions cost CA each. The argument is uniform in all fixed output coordinates. □

The row and convolution bounds (A.16) and (A.45), with $\mathcal{E} = E$, verify the hypotheses of Lemma B.4 for I, stable kernels, and E. A nonalternating molecule is therefore controlled immediately. An alternating molecule is expanded using $U = I + \Xi + E$ on each external leg. Every term except the all-Ξ term has an anchor. For the undressed-leg row norm, expand the remaining $k-1$ factors; again only the all-Ξ term lacks an anchor.

Lemma B.5 (Discrete symmetric Taylor bound)**.** *Let B be a normed vector space on which translations in each of m cyclic coordinates are isometries. If $H(\boldsymbol{z})$ is a translated family and every mixed pair of unit differences has B-norm at most M, then*

$$\|H(\boldsymbol{z}) + H(-\boldsymbol{z}) - 2H(0)\|_B \le |\boldsymbol{z}|_1^2 M.$$

Changing the universal constant accommodates a supremum over the coordinate pairs. If $S(\boldsymbol{z}) = S(-\boldsymbol{z})$, then

$$\left\| \sum_{\boldsymbol{z}} S(\boldsymbol{z}) H(\boldsymbol{z}) \right\|_B \le \left| \sum S \right| \|H(0)\|_B + CM \sum |S(\boldsymbol{z})|\, |\boldsymbol{z}|_1^2.$$

Proof. The operator identity $T_z + T_{-z} - 2I = -(T_z - I)(T_{-z} - I)$ is exact. Write each factor $T_z - I$ as a telescoping sum of $|z|_1$ translated signed unit differences along a shortest coordinate path. Products give the first bound. Reflection pairs z with $-z$ in the second assertion. The proof works directly on the cyclic group, including its Nyquist points. □

Theorem B.6 (Single molecule)**.** *Under the kernel assumptions* (A.16) *and* (A.38)–(A.41) *and the core assumptions* (B.7)–(B.9)*,*

$$P(F_k) \le CA^{k-1}, \qquad R_v(F_k) \le CaA^{k-2}$$

for each mixed external leg v.

Proof of Theorem B.6. Only the all-Ξ terms remain. Anchor the core at its first coordinate and write $z_1 = 0$. For the pointwise norm define

$$H_{\boldsymbol{z}}(\boldsymbol{x}) = \sum_u \prod_{j=1}^{k} \Xi(x_j - u - z_j).$$

Take B to be the supremum norm in $\boldsymbol{x}$. One kernel in absolute row norm and the others in pointwise norm give

$$\|H_0\|_B \le C\delta^{-1}(\delta\ell)^{-(k-1)}.$$

A second difference in a single z_j places two differences on one kernel, so (A.40) gives

$$\|\Delta_i \Delta_j H_{\boldsymbol{z}}\|_B \le C\lambda^2\delta^{-1}(\delta\ell)^{-(k-1)}.$$

For differences on two distinct kernels use $\|\Delta\Xi\|_1 \|\Delta\Xi\|_\infty$ and $k-2$ undifferentiated pointwise factors. This gives the same bound. Applying Lemma B.5, (B.9), and (B.8) yields

$$\begin{aligned} P(F_k) &\le C\left[\delta a^{k-1}\delta^{-1}(\delta\ell)^{-(k-1)} + a^{k+1}\lambda^2\delta^{-1}(\delta\ell)^{-(k-1)}\right] \\ &\le CA^{k-1}. \end{aligned}$$

For the row norm, anchor at the undressed leg v, call its output u, and put

$$H_{\boldsymbol{z}}(u, \boldsymbol{x}_{\neq v}) = \prod_{j \neq v} \Xi(x_j - u - z_j).$$

Now take $\|H\|_B = \sup_{\boldsymbol{x}_{\neq v}} \sum_u |H(u, \boldsymbol{x}_{\neq v})|$. The same row/pointwise calculation gives

$$\|H_0\|_B \le C\delta^{-1}(\delta\ell)^{-(k-2)}, \quad \|\Delta_i\Delta_j H\|_B \le C\lambda^2\delta^{-1}(\delta\ell)^{-(k-2)}.$$

The Taylor estimate is therefore at most $Ca^{k-1}(\delta\ell)^{-(k-2)} = CaA^{k-2}$. All other terms are controlled by Lemma B.4. □

B.5.3 Exact mixed gluing and closure

Assume now that U is an even invertible convolution kernel, and that mixed gluing of children of orders k_L, k_R is

$$F(\boldsymbol{x}_L, \boldsymbol{x}_R) = \frac{t}{a} \sum_z F_L(\boldsymbol{x}_L, z) F_R^{\langle \mathrm{cut} \rangle}(z, \boldsymbol{x}_R), \qquad 0 \le t \le 1. \tag{B.11}$$

Here both cut legs are mixed and $k = k_L + k_R - 2$. The operation of undressing any retained external leg commutes with gluing. Equivalently, at core level the join is $tU(y-z)/a$ and each child has one cut-leg factor U. Formula (B.11) follows by moving one even U through the contraction and removing the other.

Theorem B.7 (Two norms close along mixed trees)**.** *If every molecule satisfies Theorem B.6, every fixed finite mixed tree satisfies*

$$P(F_k) \le C A^{k-1}, \qquad R_v(F_k) \le C a A^{k-2}.$$

The constants depend on its order and on the fixed finite collection of allowable tree topologies, but not on a, L, δ.

Proof. For a join, (B.11) gives

$$P(F) \le a^{-1} P(F_L) R_{\mathrm{cut}}(F_R) \le C A^{k_L - 1 + k_R - 2} = C A^{k-1}.$$

If a retained mixed leg v lies on the left, take its absolute row before summing the cut variable. This gives

$$R_v(F) \le a^{-1} R_v(F_L) R_{\mathrm{cut}}(F_R) \le C a A^{k_L + k_R - 4} = C a A^{k-2}.$$

The right-retained case uses the symmetric version of the gluing formula. No two-leg supremum or absolute Fourier integral occurs. Induction proves the assertions. □

For the actual kernels, the abstract assumptions with $a = g^2$ are verified by Lemmas A.4 and A.10.

B.5.4 Primitive cores and their signed sums

We connect Theorem B.7 and the finite-cycle kernel bounds of Lemma A.4 to primitive loops. The external algebra is the tree representation and Ward identity of Truong–Yang–Yin [31, Assumption 4.1, Lemma 4.16, Corollary 4.21], with core decomposition [31, Definition 4.24 and Lemma 4.25, equation (4.34)] and block reduction [31, Lemma 4.26, equation (4.38)].

For the block model, put $t = 1 - \delta$ and use $\gamma_g = g^{-2}$. Equation (A.3) verifies the external algebraic hypothesis, and Lemma A.4 supplies the density bounds. Under the block normalization, $F_k = W^{k-1} \mathcal{K}_k(t)$ has cut coefficient g^{-2}. Its mixed pair is $F_2 = g^2 P (I - tP)^{-1}$; its mixed external inverse is $U = (I - tP)^{-1}$. Core joins are exactly tU/g^2, which is the abstract formula (B.11) at $a = g^2$.

Proposition B.8 (Core hypotheses from a localized free row)**.** *Assume the free scalar row and stable inverses satisfy*

$$|M(x)| \le C e^{-cd(x,0)/a}, \qquad \sum_x e^{cd(x,0)/a} |V^{++}(x)| + \sum_x e^{cd(x,0)/a} |V^{--}(x)| \le C,$$

and that $\Im m$ is bounded above and below by positive constants. Then the stable-only molecule cores in the exact primitive tree algebra satisfy (B.7), reflection, and (B.9). Consequently, if the mixed kernel also satisfies the bounds (A.38)–(A.41), then

$$|\mathcal{K}_k(t)| \le C \left(\frac{a}{W \delta \ell} \right)^{k-1}. \tag{B.12}$$

There is no factor depending on $\log a$ in this implication.

Proof. A stable-only dissection with s internal diagonals has $k+2s$ free-resolvent edges and s stable links, each with a factor a^{-1}. Each stable link joins two internal vertices; the links are disjoint as links in the expanded graph. Contract these s links. Choose a rooted forest of s free edges connecting the resulting s internal vertices to the k external vertices, with one external root per component. Such a forest exists in every connected expanded graph: contract all external vertices to a single root, choose a spanning tree, and undo that contraction. The remaining free edges are bounded pointwise. Summing the s stable links costs C^s and summing the s retained free edges costs $(Ca)^s$, canceled by a^{-s}. Retain a smaller exponential weight before summation. Since the full graph is connected, its total edge length dominates the external diameter, so the result is (B.7). All fixed moments follow by reducing the exponential constant. Evenness of the free rows and stable kernels gives simultaneous reflection.

For the signed mass, repeat the Ward deletion and alternating-core subtraction in the proof of Lemma B.1, with normalized cut coefficient a^{-1} and $1-t=\delta$. The preceding forest bound gives pure-sign rows Ca^{k-1}; each normalized Ward deletion costs Ca/δ, since $\Im m \asymp 1$. Thus the full signed row is at most $C(a/\delta)^{k-1}$, and removing the k external zero-mode inverses gives $C\delta a^{k-1}$ for the complete alternating core. At a tree of b molecules the scale is preserved:

$$\prod_i (\delta a^{k_i-1})(a\delta)^{-(b-1)} = \delta a^{k-1}, \qquad \sum_i k_i = k+2(b-1).$$

The same subtraction induction, starting at the quadrilateral, therefore proves (B.9). Theorem B.7 then gives (B.12) for $k \geq 3$; the mixed and stable pair formulas give the $k=2$ case. □

Proof of Lemma 5.2(b). Set $\delta = 1-t$. For $L \geq C_* g^2 \log W$ and $\delta \geq W^{-3}$, Lemmas A.4 and A.10 verify the kernel hypotheses of Proposition B.8 at $a=g^2$; (A.44) supplies the required error bound after choosing C_* sufficiently large. Its conclusion (B.12), with the scale conversion (A.42), is (5.20) in case (b), including the stated bound at length two. All constants are independent of W, g, L, t.

B.6 Primitive joint tails

Proof of (5.21) in case (a). The proof of Proposition B.2 gives $|\mathcal{K}_k(t;\boldsymbol{a})| \leq CW^{1-k}(1-t)^{-C}e^{-c\,\mathrm{diam}(\boldsymbol{a})/\ell_t}$, uniformly in the signs. Since $(1-t)^{-C} \leq W^{3C}$, this is at most W^{-Q} beyond $W^{\epsilon}\ell_t$ for sufficiently large W. If $\ell_t = L$, the tail domain is empty. For arbitrarily large coupling we use the following weighted estimate, also needed in later appendices.

Corollary B.9 (Coupling-independent joint tails)**.** *Fix $0<\kappa<2$ and an integer $k \geq 2$. Assume $g \geq C_\kappa$ and $|E| \leq 2-\kappa$ as in Lemma A.4. Put $\delta = 1-t$, $0<\delta\leq 1$, and $\ell = \min\{L, g^2/\sqrt{\delta}\}$. If $\delta \geq W^{-3}$ and $L \gg g^2 \log W$, meaning $L/(g^2\log W) \to \infty$, then for every $\epsilon, Q > 0$ and sufficiently large W,*

$$\sup_{\mathrm{diam}(\boldsymbol{x})>W^{\epsilon}\ell} |\mathcal{K}_k(t;\boldsymbol{x})| \leq W^{-Q}.$$

All constants are independent of g, L; the width threshold is uniform whenever $L/(g^2\log W)$ diverges uniformly. More precisely the exact primitive satisfies the crude weighted bound

$$|W^{k-1}\mathcal{K}_k(t;\boldsymbol{x})| \leq C\delta^{-C}e^{-c\,\mathrm{diam}(\boldsymbol{x})/\ell} + O(W^{-Q})$$

with arbitrarily prescribed Q. Sharp cancellation is not needed for this tail estimate.

Proof of Corollary B.9 and (5.21) in case (b). Use the weighted mixed rows and perturbation bound (A.43), the stable rows (A.16), and the free row (A.15). Their costs are $C\delta^{-1}$, C, and Cg^2, respectively. Equation (A.44), with its fixed exponent r chosen after the number of inverse factors, makes the total finite-cycle error smaller than W^{-Q} for every prescribed Q.

In each fixed-order exact dissection, an internal covariance link contributes g^{-2}. Use the rooted forest from Proposition B.8 at $a=g^2$: one free-edge row Cg^2 for every internal link

cancels these factors. Sum every stable or mixed link and every external inverse in weighted absolute row. Their number is bounded in terms of k, so their combined cost is $C\delta^{-C}$. Bound the remaining free edges pointwise. The connected graph has total edge length at least the external diameter, so retaining a sufficiently small common exponential weight gives the displayed bound. The number of dissection topologies is finite at fixed k. If $\ell = L$, the displayed joint-tail domain is empty for large W; otherwise $\delta^{-C} \le W^{3C}$ and $e^{-cW^{\epsilon}}$ beats every fixed power of W. Finally, (A.42) gives $\ell \le \ell_t \le 2\ell$. Hence $\operatorname{diam}(\boldsymbol{x}) > W^{\epsilon}\ell_t$ implies $\operatorname{diam}(\boldsymbol{x})/\ell > W^{\epsilon}$, proving (5.21) with its stated scale. All estimates are uniform in the signs and $0 \le t \le 1 - W^{-3}$; their constants are independent of g, L, while the required error accuracy determines how large $L/(g^2 \log W)$ must be. □

B.7 Primitive inputs for initialization and stopping

The following estimates supply the primitive terms in the initial error equation, the stopped pair equation, and the polynomial-range time increments.

Lemma B.10 (Early-time primitive estimates)**.** *Fix $k \ge 2$.* (a) *Under the bounded-coupling conditions of (6.1), there are $C_k, c_k > 0$ such that*

$$\sup_{0\le t\le 1/2} \max_{\boldsymbol{a},\boldsymbol{\sigma}} e^{c_k \operatorname{diam}(\boldsymbol{a})} |\mathcal{K}_k(t;\boldsymbol{a},\boldsymbol{\sigma})| \le C_k W^{1-k}. \tag{B.13}$$

(b) *Fix $B_g, B_L > 0$ and $\kappa \in (0,2)$, and assume $C_\kappa \le g < W^{B_g}$, $g^2 \log W \ll L \le W^{B_L}$ and $|E| \le 2 - \kappa$, with C_κ as in Lemma 5.2*(b)*. Then*

$$\sup_{0\le t\le 1/2} \max_{\boldsymbol{a}_{\neq i},\boldsymbol{\sigma}} \sum_{a_i \in \mathbb{Z}_L} |\mathcal{K}_k(t;\boldsymbol{a},\boldsymbol{\sigma})| \le C_k g^2 W^{1-k}, \qquad 1 \le i \le k. \tag{B.14}$$

The constants are independent of W, g, L in their stated ranges; the width threshold in (b) *may depend on the divergence rate of $L/(g^2 \log W)$.*

***Proof of* (B.13) *and* (B.14).** For (a) with $g > 0$, use the weighted dissection estimate in the proof of Proposition B.2: $|\mathcal{K}_k| \le C_k W^{1-k}(1-t)^{-C_k} e^{-c_k \operatorname{diam}(\boldsymbol{a})/\ell_t}$. The bounded-coupling hypotheses imply (A.22); on $t \le 1/2$, both $(1-t)^{-1}$ and ℓ_t are bounded. This proves (B.13). At $g = 0$, all free rows and external inverses are diagonal with bounded entries on $[0, 1/2]$; (B.1) and (5.13) give the same bound.

For (b), (A.15) verifies the entry bound used in the proof of (7.20). Applying that estimate with $\gamma_g = g^{-2}$ and $t \le 1/2$ gives $\sum_{a_i} |\mathcal{K}_k(t)| \le C_k g^2 W^{1-k}$, uniformly in the remaining labels and signs, proving (B.14). □

Lemma B.11 (Primitive pair bounds for continuation)**.** *Assume (6.1). For $0 \le v < 1$,*

$$\max_{a,\sigma_1,\sigma_2} \sum_b |\mathcal{K}_2(v;a,b;\sigma_1,\sigma_2)| \le \frac{1}{\gamma_g W(1-v)}. \tag{B.15}$$

Fix $0 < \theta \le 1/2$, $c_{\mathrm{p}} > 0$ and $F > 0$, and put $w_v(x) = e^{-c_{\mathrm{p}}(\|x\|_L/\ell_v)^{\theta}} + W^{-F}$. For sufficiently large W, uniformly for $1/2 \le v \le 1 - W^{-3}$,

$$|\mathcal{K}_2(v;a,b;\sigma_1,\sigma_2)| \le C\zeta_v^2 w_v(a-b) \quad \text{if } \|a-b\|_L \ge \ell_v (\log W)^{3/2}. \tag{B.16}$$

***Proof of* (B.15) *and* (B.16).** Since $|Q| \le P$ and $P\mathbf{1} = \mathbf{1}$, the Neumann series give $|Q(I - vQ)^{-1}| \le P(I - vP)^{-1}$. Formula (5.13) and the mixed row mass $(1-v)^{-1}$ prove (B.15) for all signs. For (B.16), set $r = \|a-b\|_L/\ell_v$. The same pointwise comparison and (6.3) give

$$|\mathcal{K}_2(v;a,b;\sigma_1,\sigma_2)| \le C\zeta_v e^{-cr} + C_A \zeta_v W^{-A}.$$

For $r \ge (\log W)^{3/2}$, the ratio $e^{-cr + c_{\mathrm{p}} r^{\theta}}$ is $O_D(W^{-D})$ for every fixed $D > 0$. By (A.46), $\zeta_v \ge c/W$; consequently the first term is bounded by $C\zeta_v^2 w_v$. Choose $A > F + 2$ for the second term and use $w_v \ge W^{-F}$. This proves the stated weighted bound. □

C Initialization and propagation of the loop estimates

We first prove the Gaussian and Haar moment estimates used in Section 7.5, then the bounded-coupling initialization of Lemma 5.3 and the propagation theorem, Theorem 5.12. The common five-step argument follows [33, Sections 5.1–5.7] and [31, Sections 7.1–7.6].

In Appendices C.2–C.4, $\gamma = \gamma_g$ and the parameter ranges are those of Theorem 5.12, including the earlier-time pairs required in part (i). Spatial maxima and time suprema are inside probability; suprema over deterministic parameters are outside, as in that theorem. Appendix C.1 uses the hypotheses specified with its target estimates.

C.1 Gaussian integration by parts and Haar estimates

We give the Gaussian integration-by-parts and Haar calculations used in Section 7.5. The residual estimates use either parameter range of Lemma 5.9, including its comparison points, and the deterministic Dyson rows (6.13). The Haar estimate applies to an arbitrary deterministic complex matrix.

Proof of (7.26) ***and*** (5.31). Fix deterministic u, z as in Lemma 5.9, and abbreviate $M_B = M_{B,u}(z)$ and $M = M_u(z)$. Use R from (5.28) and Z from Section 7.5. We first prove (7.26) under its moment hypotheses. The assertion is immediate at $u = 0$, since $R = 0$; assume $u > 0$.

Choose a real orthonormal basis $(T_\alpha)_{\alpha=1}^{W^2}$ of Hermitian $W \times W$ matrices for the inner product $\mathrm{Tr}(AB)$, and let $T_{c,\alpha}$ denote its embedding in block c. Thus

$$\mathcal{B}_u^{(g)} = \sum_{c,\alpha} \xi_{c,\alpha} T_{c,\alpha}, \qquad \xi_{c,\alpha} \sim N(0, u\gamma_g/W) \text{ independently}, \qquad \sum_\alpha (T_\alpha)_{ij}(T_\alpha)_{k\ell} = \delta_{i\ell}\delta_{jk}.$$

Write $\lambda = (c, \alpha)$ for a Gaussian coordinate. The identities $\partial_\lambda G = -GT_{c,\alpha}G$ and $\sum_\alpha T_\alpha A T_\alpha = (\mathrm{Tr}\, A) I_W$ imply

$$R_{ab} = \sum_\lambda \left(\xi_\lambda f_\lambda - W^{-1} u\gamma_g \partial_\lambda f_\lambda \right), \qquad f_{c,\alpha} := W^{-1} M_{B,ac}\, \mathrm{Tr}(T_\alpha G_{cb}). \tag{C.1}$$

Indeed, the first term is $W^{-1}\,\mathrm{Tr}(M\mathcal{B}_u^{(g)}G)_{ab}$, and the second is

$$-W^{-1}u\gamma_g \sum_{c,\alpha} \partial_{c,\alpha} f_{c,\alpha} = W^{-2} u\gamma_g \sum_c M_{B,ac}(\mathrm{Tr}\, G_{cc})(\mathrm{Tr}\, G_{cb}) = u\gamma_g \sum_c M_{B,ac}\langle G\rangle_{cc}\langle G\rangle_{cb}.$$

In particular, $\mathbb{E}R_{ab} = 0$. More generally, for a smooth random variable F, Gaussian integration by parts gives the exact cancellation

$$\mathbb{E}[R_{ab}F] = W^{-1}u\gamma_g \sum_\lambda \mathbb{E}[f_\lambda \partial_\lambda F].$$

We apply this cancellation to $\mathbb{E}|R_{ab}|^p$. Label its p factors by $1, \dots, p$, the first $p/2$ being R_{ab} and the others $\overline{R_{ab}}$, and write $f_{j,\lambda}$ for f_λ or $\overline{f_\lambda}$ accordingly. Expanding (C.1) and integrating all Gaussian coordinates by parts gives the following finite sum. Choose q disjoint pairs of factors; their Gaussian coordinates are paired together. Each of the other $p - 2q$ coordinates from a factor i differentiates a coefficient at a factor $j \neq i$. For such a choice $\mathcal{D}$, there are $e = p - q$ freely summed Gaussian labels, and its contribution is

$$\left(W^{-1}u\gamma_g\right)^e \sum_{\boldsymbol{\lambda}\,\text{respecting the pairs}} \mathbb{E} \prod_{j=1}^p \left[\left(\prod_{i:\, i\to j} \partial_{\lambda_i} \right) f_{j,\lambda_j} \right]. \tag{C.2}$$

The sum of these contributions is exactly $\mathbb{E}|R_{ab}|^p$. To see why $j = i$ is excluded, a derivative acting on its own coefficient occurs both from integration by parts and from the negative term in

(C.1), with opposite signs. These terms cancel even when other derivatives act on that coefficient, since the derivatives commute. The number of terms depends only on p.

Block indices. An h-fold derivative of $f_{c,\alpha}$ is a sum of $h!$ terms of the form

$$W^{-1}(-1)^h M_{B,ac}\,\mathrm{Tr}(T_\alpha G_{cd_1} T_{\alpha_1} G_{d_1 d_2}\cdots T_{\alpha_h} G_{d_h b}),$$

where $(d_1,\alpha_1),\dots,(d_h,\alpha_h)$ are the incoming Gaussian labels in one of their possible orders. Conjugated coefficients give the reversed adjoint words. Thus every term has $2e$ resolvent factors and $2e$ basis-matrix occurrences. Each freely summed block label c retains the factor $M_{B,ac}$ from its original coefficient, or two such factors when its two original Gaussian coordinates are paired. No derivative acts on M_B. Consequently (6.13) gives

$$u\gamma_g \sum_c |M_{B,ac}| \le C, \qquad u\gamma_g \sum_c |M_{B,ac}|^2 \le C.$$

After applying the uniform moment bounds to the resolvent entries, these estimates sum all e block labels and cancel the factor $(u\gamma_g)^e$ in (C.2). The remaining normalization is W^{-p-e}. In particular, no free block sum is weighted only by the random array $(\langle G\rangle_{ab})_{a,b}$.

Orbital indices. Number the $2e$ basis-matrix occurrences. Let σ give the next occurrence in each original trace, and let π pair the two occurrences of each Gaussian label. If F_h is the resolvent block following occurrence h, the completeness identity for (T_α) reduces the orbital sum to

$$\sum_{i_1,\dots,i_{2e}=1}^{W} \prod_{h=1}^{2e} (F_{\pi(h)})_{i_h, i_{\sigma\pi(h)}}. \tag{C.3}$$

Each cycle of $\sigma\pi$ gives one trace of resolvent blocks. There is no cycle of length one: $\sigma\pi(h)=h$ would pair occurrence h with its predecessor in the same original trace, whereas every surviving Gaussian pair joins different original factors. Thus each resulting trace contains at least two resolvent factors.

Only now substitute $G_{cd} = \langle G\rangle_{cd} I_W + Z_{cd}$, and likewise for adjoints. If an expanded term has x factors from Z and z_0 traces containing only scalar factors, its orbital sum has at most W^{x+z_0} terms. Indeed, a trace with $x_F > 0$ error factors has at most x_F free indices; an entirely scalar trace has one. Since each entirely scalar trace contains at least two factors, $x + 2z_0 \le 2e$. Hölder's inequality uses at most $2e \le 2p$ random factors and bounds each entry product by $CK^{2e}\chi^x$. Hence every contribution is bounded by

$$CK^{2e}W^{x+z_0-p-e}\chi^x \le CK^{2p}W^{-p+x/2}\chi^x \le CK^{2p}\chi^{2p}, \tag{C.4}$$

where the last inequality uses $\chi \ge W^{-1/2}$. Summing the finitely many terms proves (7.26). The expansion was performed after every derivative, so it requires neither deterministic block traces nor independence between the block traces and Z.

For (5.31), repeat (C.1)–(C.3) and substitute $G_{cd} = M_{B,cd} I_W + \widetilde{G}_{cd}$ only after differentiation. The scalar factors are deterministic and bounded, while part (iii) controls the error entries in L^{2p}. Thus (C.4) applies with fixed K, giving $C_p\chi^{2p}$ without a W^ϵ loss. □

Proof of (7.27). Let $U \in U(W)$ be Haar and $A \in \mathbb{C}^{W\times W}$ be deterministic. For $i=j$, this is the centered quadratic-form moment bound for a uniform unit vector, obtained by normalizing a complex Gaussian vector. For $i \ne j$, condition on row i of U: row j is uniform on its orthogonal complement. Its linear-form moment bound, followed by averaging row i, gives the same estimate. □

C.2 Initialization at bounded coupling

Proof of Lemma 5.3: (5.23). Fix $g \le C$, $|E| \le B$ and $\Im m_{g,L}(E) \ge c' > 0$, with $\gamma = \gamma_g$, and take $0 \le t \le 1/2$. The estimates below are uniform in these parameters. Write

$Z_{ab} := G_{ab} - \langle G\rangle_{ab} I_W$ and use the residual of (5.28). Here $\eta_t \geq \gamma c'/2$, so $\|G_t\|_{\mathrm{op}} \leq 2/(\gamma c')$ deterministically. Exponential spatial conjugation changes only the bounded hopping; choose the weight small enough that its operator perturbation is at most $\gamma c'/4$. A resolvent Neumann series gives $\|(G_t)_{ab}\|_{\mathrm{op}} \leq C' e^{-c\|a-b\|_L}$ for every onsite realization, without an onsite norm bound. Thus every fixed actual word is $O(W^{1-k})$ with a joint exponential tail. Haar concentration gives $Z \prec W^{-1/2}$, bounded block traces, and the residual refinement gives $R \prec W^{-1}$. Starting from $\widetilde{G}_0 = 0$, (5.32) and stable inversion select the small quadratic branch by continuity: $\max_{a,b} |\langle \widetilde{G}\rangle_{ab}| \prec W^{-1}$.

For every sign pair, Cauchy–Schwarz and Ward give $\sum_b |\mathcal{L}_2(a,b)| \leq \Im\langle G\rangle_{aa}/(W\eta_t) \leq C'/W$. For the primitive loops, (B.13) gives the same W^{1-k} bound and joint exponential tail at every fixed length.

In a cut difference write $\mathcal{L}_p\mathcal{L}_q - \mathcal{K}_p\mathcal{K}_q = \mathcal{L}_p D_q + D_p\mathcal{K}_q$. Its boundary cuts define a linear error operator of norm C_k, by the pair rows and the coefficient γW; interior cuts involve shorter errors. A free block sum costs C_k, by exponential localization. Induction on k gives a drift source $O_\prec(W^{-k})$: a light insertion is $O(\gamma W \cdot W^{-1} \cdot W^{-k}) = O(W^{-k})$. The raw bracket has size $\gamma W \cdot W^{1-(2k+2)} = O(W^{-2k})$. Raw martingale BDG and pathwise Gronwall yield $D_k \prec W^{-k}$. The terminal linear operator may be random, so no anticipating stochastic integral with its terminal propagator is used. For diam $\leq C \log W$, multiplying the bound $D_k \prec W^{-k}$ by $e^{c\sqrt{\mathrm{diam}}}$ costs only $W^{o(1)}$. For diam $> C \log W$, the deterministic bound $|D_k| \leq C_k W^{1-k} e^{-c'\,\mathrm{diam}}$ absorbs the extra factor W when C is large. Thus $|D_k| \prec W^{-k}(e^{-c\sqrt{\mathrm{diam}}} + W^{-F})$ for every fixed F. The definition (5.18) gives $\ell_t \asymp 1$ and $\zeta_t \asymp W^{-1}$ on $0 \leq t \leq 1/2$.

Uniform time follows from a polynomial mesh and Brownian maximal moment estimates: with arbitrarily high probability, simultaneously for all mesh intervals,

$$\sup_{t\in[t_j,t_{j+1}]} \|\mathcal{B}_t^{(g)} - \mathcal{B}_{t_j}^{(g)}\|_{\mathrm{op}} \leq W^{1/2+\nu}(t_{j+1}-t_j)^{1/2}.$$

Resolvent identities interpolate. Spectral and coupling meshes use their deterministic polynomial derivative bounds. These meshes do not condition any Gaussian integration-by-parts calculation. □

C.3 One-step propagation

For arbitrarily large coupling, the additional verifications occur where needed: Step 1 uses transfer moments and sampling to obtain spatial and time suprema; Step 2 controls weighted sums and stopped estimates; Steps 3–4 retain the covariance factors and the cancellation in the projected hierarchy. These verifications allow the same five steps to prove (5.39) in both parameter ranges.

Proof of Theorem 5.12(i). Fix a deterministic step $[s,t]$ and a pair (g,E) satisfying (5.37) and the input and time-step assumptions of part (i). Put

$$R := \frac{1-s}{1-t}, \qquad R^{20}\zeta_t \leq W^{-c}.$$

All estimates below hold for $s \leq u \leq t$, uniformly over these pairs. In the step-goal displays, stochastic domination includes the maximum over all indicated spatial labels and signs and the supremum over $u \in [s,t]$ inside the probability; the supremum over deterministic parameters stays outside. Intermediate estimates explicitly restricted to deterministic labels or a fixed time retain that restriction. The previous-time estimates are available at (g,E) and at every earlier-time pair (g',E') specified in the input assumption of part (i). The distribution comparisons (5.35) transfer imaginary-part, block operator-norm and block-trace bounds from $G_s(g',E')$ to $G_u(g,E)$, with time ratio $(1-s)/(1-u) \leq R$.

Step 1: rough loop bounds and a weak local law. The goal of this step is to obtain, for every fixed loop length k,

$$\begin{aligned} &\sup_{s\le u\le t}\max_{a,b}\frac{\|(G_u)_{ab}\|_{\mathrm{op}}}{WR\zeta_u}\prec 1, \quad \sup_{s\le u\le t}\max_{a,b}\frac{\|(G_u)_{ab}\|_{\mathrm{HS}}^2}{W^2R\zeta_u}\prec 1,\\ &\sup_{s\le u\le t}\max_{a}\frac{|\langle\widetilde{G}_u\rangle_{aa}|}{R^2\zeta_u}\prec 1, \qquad \sup_{s\le u\le t}\max_{\boldsymbol{a},\boldsymbol{\sigma}}\frac{|\mathcal{L}_k(u;\boldsymbol{a},\boldsymbol{\sigma})|}{R^{k-1}\zeta_u^{k-1}}\prec 1. \end{aligned} \tag{C.5}$$

All maxima and time suprema are inside the probability. We then deduce

$$\sup_{s\le u\le t}\max_{a}\Im\langle G_u\rangle_{aa}\prec 1. \tag{C.6}$$

Starting bounds at a fixed time. Fix a deterministic $u\in[s,t]$ and let (g',E') be supplied by Lemma 5.11. Write $G'_s=G_s(g',E')$. The time-s positive-loop input gives, for every fixed integer $p\ge 1$,

$$W^{-2p}\|(G'_s)_{ab}\|_{\mathrm{op}}^{2p}\le W^{-2p}\operatorname{Tr}[((G'_s)_{ab}(G'_s)_{ab}^*)^p]=\operatorname{Tr}[(G'_sE_b(G'_s)^*E_a)^p]\prec\zeta_s(g')^{2p-1}. \tag{C.7}$$

The equality uses $E_a=W^{-1}P_a$. The bound is simultaneous in a,b because the positive-loop input already contains that maximum. After taking the $2p$-th root and dividing by $\zeta_s(g')$, the remaining factor is at most $CW^{1/(2p)}$, since $\zeta_s(g')\ge c/W$. For a prescribed loss W^{ϵ}, choose $p>1/\epsilon$ and use a loss $W^{\epsilon/2}$ in the root estimate. This proves $W^{-1}\max_{a,b}\|(G'_s)_{ab}\|_{\mathrm{op}}\prec\zeta_s(g')$. Together with the time-s trace input and $\zeta_s(g')\asymp\zeta_s$, (5.35) gives

$$W^{-1}\max_{a,b}\|(G_u)_{ab}\|_{\mathrm{op}}\prec R\zeta_s, \qquad \max_{a,b}|\langle G_u\rangle_{ab}|\prec R. \tag{C.8}$$

These spatial maxima are inherited from the input; they require no new union over block labels.

The characteristic row and stability bounds are (6.2) from Lemma 6.1; the corresponding comparison-segment bounds are (6.12) from Lemma 6.4.

Let w be the upper comparison point in (6.9), set $F=(H_u-w)^{-1}$, and write $G=(H_u-z)^{-1}$ for z on the vertical segment from z_u to w. Thus $1\le\Im w/\Im z\le CR$, and F has the time-s input bounds under the Gaussian coupling. In particular, $W^{-2}\|F_{ab}\|_{\mathrm{HS}}^2\prec\zeta_s$ follows from the length-two positive loop, while (C.7) gives $W^{-1}\|F_{ab}\|_{\mathrm{op}}\prec\zeta_s$. For equal real parts, spectral calculus gives $\Im G\le(\Im w/\Im z)\Im F$. To obtain the off-diagonal block traces, Cauchy–Schwarz and the two Hilbert–Schmidt Ward identities give

$$|\langle G-F\rangle_{ab}|\le|z-w|\sqrt{\frac{\Im\langle G\rangle_{aa}\Im\langle F\rangle_{bb}}{\Im z\Im w}}\le\frac{|z-w|}{\Im z}\sqrt{\Im\langle F\rangle_{aa}\Im\langle F\rangle_{bb}}\prec R.$$

Adding the time-s trace bound for F proves $\max_{a,b}|\langle G\rangle_{ab}|\prec R$ throughout the segment. The resolvent identity and the block Ward identities give

$$G-F=(z-w)GF, \qquad \|P_aG\|_{\mathrm{HS}}^2=\frac{W\Im\langle G\rangle_{aa}}{\Im z}, \qquad \|FP_b\|_{\mathrm{op}}^2=\frac{\|\Im F_{bb}\|_{\mathrm{op}}}{\Im w}.$$

Since $W^{-2}\|F_{ab}\|_{\mathrm{HS}}^2\prec\zeta_s$ and $W^{-1}\|\Im F_{bb}\|_{\mathrm{op}}\prec\zeta_s$, this yields

$$\begin{aligned} W^{-2}\|G_{ab}\|_{\mathrm{HS}}^2&\le 2W^{-2}\|F_{ab}\|_{\mathrm{HS}}^2+\frac{2|z-w|^2}{W\Im z\Im w}\Im\langle G\rangle_{aa}\|\Im F_{bb}\|_{\mathrm{op}}\\ &\prec\zeta_s(1+R\Im\langle G\rangle_{aa})\prec R^2\zeta_s. \end{aligned} \tag{C.9}$$

The displayed spatial bounds are simultaneous at each fixed (u,z); the sharper centered-trace estimate has not yet been used.

Residual estimates before Dyson stability. Write $R_{ab}(u,z)$ for the residual in (5.28). By (5.30) and (C.9), for each fixed pair a,b,

$$\max_{i,j} |G_{ai,bj} - \delta_{ij}\langle G\rangle_{ab}| \prec R\sqrt{\zeta_s}.$$

We use these pointwise entry estimates to bound residual moments under the original Gaussian law. For each fixed moment order p, any scalar field X appearing here and any event Ω^c of probability at most W^{-M}, the coarse moments in Lemma 5.10 give

$$\mathbb{E}[|X|^p \mathbf{1}_{\Omega^c}] \le (\mathbb{E}|X|^{2p})^{1/2}\mathbb{P}(\Omega^c)^{1/2} \le C_p W^{C_p - M/2}.$$

Choose M after the required moment order and accuracy. The trace and entry bounds therefore imply unconditional moment bounds with an arbitrarily small power of W. Apply the fixed-pair moment implication (7.26) with $K = CR$ and $\chi = \max\{\sqrt{\zeta_s}, W^{-1/2}\} \asymp \sqrt{\zeta_s}$, allowing that small power in K. For every fixed even $p \ge 2$ and $\delta > 0$, we obtain

$$\max_{a,b} \mathbb{E}|R_{ab}(u,z)|^p \le C_{p,\delta} W^{\delta p}(R^2\zeta_s)^p. \tag{C.10}$$

No Gaussian calculation is conditioned on a sampling or stopping event.

We next make (C.10) simultaneous in space and in the spectral heights needed for the stability argument. Fix a desired failure exponent $D > 0$ and a sufficiently large accuracy exponent Q. At each fixed (u,z), use the residual sample set from Lemma 5.10, with error W^{-Q}. Its size is at most W^{C_S}. Markov's inequality at those pairs and the sampling inequality give, for $\epsilon > \delta$,

$$\mathbb{P}\Big(\max_{a,b} |R_{ab}(u,z)| > CW^{\epsilon}R^2\zeta_s + W^{-Q}\Big) \le C_{p,\delta} W^{C_S - p(\epsilon-\delta)} + \mathbb{P}(\text{sampling fails}).$$

The sample-size exponent is fixed before choosing p. Thus the failure exponent can absorb any fixed polynomial number of time and height mesh points; this is where spatial sampling replaces a union over L^2.

For each deterministic time mesh point, use the closed comparison-height consequence (7.25). Choose its height mesh exponent $q_h > C_H + Q + 1$ before the residual moment order. It gives interpolation errors at most W^{-Q} for $G, M_u(z), R$ on the whole comparison segment, with arbitrary power probability. Only the coarse inputs of Sections 7.3 and 7.4, together with the deterministic positive-height bounds, enter this step; the random Dyson closure below has not yet been used.

We also prepare time interpolation, using the coarse increments rather than the conclusion (C.5). Fix the required loop lengths first, including a repeated positive loop of sufficiently large fixed length for the final operator estimate. For a scalar field X in Lemma 5.10(b) and a fixed normalization power $j \ge 0$, put $Y(u) = X(u)/\zeta_u^j$. Choose a fixed C_0, independent of the moment order p, for the coarse moments and increments of the finitely many fields needed below. Using $\zeta_u \ge c/W$ and $|\partial_u \log \zeta_u| \le W^3$ from the definition (5.18),

$$\begin{aligned}\|Y(v) - Y(u)\|_{L^p} &\le C_p W^{C_0+j}|v-u|^{1/2} + C_p W^{C_0+j+3}|v-u| \\ &\le C_p W^{C_1}|v-u|^{1/2}, \qquad C_1 = C_0 + j + 3.\end{aligned}$$

Request (5.34) with $w = 1$, accuracy exponent $Q + j + 2$ and failure exponent $D + 2$. Its sample set $\mathcal{S}_X$ is independent of time and has at most W^{C_S} tuples. Choose a deterministic time mesh $\mathcal{T} \subset [s,t]$ with spacing $h = W^{-q}$, including the endpoints. On each mesh interval, dyadic subdivision and Minkowski give, for $p > 2$,

$$\Big\| \sup_{u\in[v,v+h]\cap[s,t]} |Y(u) - Y(v)| \Big\|_{L^p} \le C_p W^{C_1} \sum_{m\ge 0} 2^{m/p}(h2^{-m})^{1/2} \le C_p W^{C_1} h^{1/2}.$$

Markov's inequality and the union over sampled tuples and intervals give

$$\begin{aligned}\mathbb{P}\Bigg(\max_{\boldsymbol{a}\in\mathcal{S}_X}\max_{v\in\mathcal{T}}\sup_{u\in[v,v+h]\cap[s,t]}|Y(u;\boldsymbol{a})-Y(v;\boldsymbol{a})|>W^{-Q}\Bigg)\\ \le C_pW^{C_{\mathrm{S}}+(C_1+Q)p}h^{p/2-1}.\end{aligned}\tag{C.11}$$

Choose $q>\max\{4,2(C_1+Q+1)\}$ first, and then $p(q/2-C_1-Q)>C_{\mathrm{S}}+q+D+2$. The probability is at most W^{-D-1}. The sampling error divided by ζ_u^j is $O(W^{-Q-2})$, so on the sampling and continuity events,

$$\sup_{s\le u\le t}\max_{\boldsymbol{a}}|Y(u;\boldsymbol{a})|\le C\max_{\substack{v\in\mathcal{T}\\ \boldsymbol{a}\in\mathcal{S}_X}}|Y(v;\boldsymbol{a})|+CW^{-Q}.\tag{C.12}$$

The W^2 orbital choices for entries are included in C_{S}. All mesh and sample-size exponents are fixed before the moment orders and pointwise probability accuracies.

Apply this construction first to the residual, with $j=1$. At the time and height mesh points choose the pointwise failure exponent larger than $D+q+q_h+2$; the residual sample count was already included in the Markov bound above. Height interpolation gives the second bound below, while (C.12) on the characteristic and $\zeta_s\le\zeta_u$ give the first:

$$\begin{aligned}\sup_{s\le u\le t}\max_{a,b}\frac{|R_{ab}(u,z_u)|}{R^2\zeta_u}&\prec 1,\\ \max_{v\in\mathcal{T}}\sup_{\substack{\Re z=\Re z_v\\ \eta_v\le\Im z\le\Im w(v)}}\max_{a,b}\frac{|R_{ab}(v,z)|}{R^2\zeta_s}&\prec 1.\end{aligned}\tag{C.13}$$

Here $w(v)$ is the upper comparison point at time v. The same finite union and height interpolation make the starting trace bounds simultaneous on these segments. No local law from Dyson closure has been used to obtain (C.13).

Dyson stability on the common event. Fix a mesh time v and abbreviate $G=(H_v-z)^{-1}$ on its comparison segment. Set

$$U(z)=\max_a|\langle G-M_v(z)\rangle_{aa}|,\qquad X(z)=\max_{a,b}|\langle G-M_v(z)\rangle_{ab}|.$$

The two identities in (5.32), with v in place of u and $M_v(z)$ in place of M, and the row bounds (6.12) imply

$$X(z)\le\max_{a,b}|R_{ab}(v,z)|+C\max_{a,b}|\langle G\rangle_{ab}|\,U(z),\qquad U(z)\le C\max_{a,b}|R_{ab}(v,z)|+CU(z)X(z).$$

For any sufficiently small fixed $\nu>0$, the preceding simultaneous estimates give $\max|R_{ab}(v,z)|\le W^\nu R^2\zeta_s$ and $\max|\langle G\rangle_{ab}|\le W^\nu R$ outside an event of arbitrarily small polynomial probability. Substitution and absorption of the linear term $CW^\nu R^2\zeta_sU$ yield

$$U(z)\le CW^\nu R^2\zeta_s+CW^\nu RU(z)^2,\qquad W^{2\nu}R^3\zeta_s\le W^{2\nu-c}=o(1).\tag{C.14}$$

The last inequality uses $R^{20}\zeta_t\le W^{-c}$ and $\zeta_s\le\zeta_t$. Thus the quadratic inequality excludes the interval between a small branch of size $CW^\nu R^2\zeta_s$ and a large branch of size $cW^{-\nu}R^{-1}$. At the upper point, $(H_v-w(v))^{-1}=\xi G_s(g',E')$ and $M_v(w(v))=\xi M(g',E')$ for the corresponding earlier-time pair, under the Gaussian coupling of Lemma 5.11, with the Dyson endpoint identity from Lemma 6.4. The time-s input starts $U(w(v))$ on the small branch. Continuity in $\Im z$ therefore keeps U on that branch down to z_v. Since ν can be arbitrarily small, this proves, simultaneously at all mesh times,

$$\max_{v\in\mathcal{T}}\max_a\frac{|\langle\widetilde{G}_v\rangle_{aa}|}{R^2\zeta_s}\prec 1.$$

Insert this bound into (C.9) at $z = z_v$. Since $\Im\langle G_v\rangle_{aa} = \Im m + \Im\langle \widetilde{G}_v\rangle_{aa}$ and $\Im m \asymp 1$, it gives $W^{-2}\max_{a,b}\|(G_v)_{ab}\|_{\mathrm{HS}}^2 \prec R\zeta_s$. The operator bound at mesh times is already (C.8). For $k \geq 2$, use two Hilbert–Schmidt factors and $k-2$ operator factors in the cyclic trace:

$$\max_{\boldsymbol{a},\boldsymbol{\sigma}} |\mathcal{L}_k(v)| \leq W^{-k}\left(\max_{a,b}\|(G_v)_{ab}\|_{\mathrm{HS}}\right)^2\left(\max_{a,b}\|(G_v)_{ab}\|_{\mathrm{op}}\right)^{k-2} \prec R^{k-1}\zeta_s^{k-1}.$$

For $k = 1$, use $\mathcal{L}_1(v; a, +) = m + \langle \widetilde{G}_v\rangle_{aa}$ and its conjugate. These are spatially uniform bounds at the mesh times.

To pass to all times, apply (C.12) to centered diagonal traces with $j = 1$ and to loops with $j = k-1$. The just-proved mesh bounds and $\zeta_s \leq \zeta_v$ control its right-hand side, giving the third and fourth bounds in (C.5). No new spatial union is needed at mesh times; sampling reduces the time interpolation to polynomially many tuples. Positive two-loops give the Hilbert–Schmidt bound. For the operator norm, repeated positive $2p$-loops give

$$\sup_{s\leq u\leq t}\max_{a,b}\frac{\|(G_u)_{ab}\|_{\mathrm{op}}}{WR\zeta_u} \prec \sup_{s\leq u\leq t}(R\zeta_u)^{-1/(2p)} \leq CW^{1/(2p)}.$$

For each requested domination loss $\epsilon > 0$, choose a fixed $p > 1/\epsilon$ and use loss $\epsilon/2$ in the loop estimate. This proves the first bound in (C.5); p may depend on ϵ but not on W. All deterministic (g, E) suprema remain outside the probability. This completes (C.5).

Finally, $\langle G_u\rangle_{aa} = m + \langle \widetilde{G}_u\rangle_{aa}$, $\Im m \asymp 1$ and the time-step condition (5.38), together with monotonicity of ζ, give

$$R^2\zeta_u \leq R^2\zeta_t \leq W^{-c}, \qquad s \leq u \leq t.$$

The third bound in (C.5) therefore proves (C.6).

Step 2: the sharp local law and a preliminary spatial bound. The goal of this step is to obtain the following sharp resolvent bounds and preliminary two-loop spatial bound:

$$\begin{aligned} |D_2(u; a, b; +, -)| &\prec R^{10}\zeta_u^2\left(e^{-c(\|a-b\|_L/\ell_u)^\theta} + W^{-F}\right), \\ W^{-2}\|(G_u)_{ab}\|_{\mathrm{HS}}^2 \prec \zeta_u, \qquad \|\widetilde{G}_u\|_{\max} &\prec \sqrt{\zeta_u}, \qquad \max_{a,b}|\langle \widetilde{G}_u\rangle_{ab}| \prec \zeta_u. \end{aligned} \tag{C.15}$$

Thus the resolvent bounds are sharp, while the spatial bound on D_2 still has the loss R^{10}.

We first control the free block sum using the final scale budget (6.6) of Lemma 6.2.

For the mixed pair, (5.17) becomes

$$\begin{aligned} dD_2 = &\left[P(I-uP)^{-1}D_2 + D_2P(I-uP)^{-1}\right]du \\ &+ \gamma_g W D_2^2\,du + \mathcal{W}_2\,du + d\mathcal{M}_2. \end{aligned} \tag{C.16}$$

Write $w_v(a-b) = e^{-c_{\mathrm{p}}(\|a-b\|_L/\ell_v)^\theta} + W^{-F}$, with a sufficiently small fixed $c_{\mathrm{p}} > 0$. We may increase F, since this strengthens the desired estimate; choose $F > 2C$ with C from (6.6). The convolution bounds (6.8) apply to w_u and $\sqrt{w_u}$, since (6.6) and $F > 2C$ give $LW^{-F/2} \leq \ell_u$. Fix a polynomial barrier range $1 \leq J \leq W^{C_J}$, where $C_J = 32$ is fixed before choosing the tail floor; the barrier used below has this form since $R \leq W^3$. Fix a sufficiently small $\nu > 0$. Stop at the first time τ_J when $\max_{a,b}|D_2(v; a, b; +, -)|/(\zeta_v^2 w_v(a-b))$ reaches J, or when a normalized block operator norm, block Hilbert–Schmidt norm, centered diagonal trace, or actual loop of length 3 or 6 exceeds W^ν times its Step 1 envelope. All spatial maxima of this finite list are included; these stopping tests depend only on the current and past flow. Before the stop their bounds are ordinary inequalities. Every subsequent small source loss denotes a fixed multiple of ν; choose ν so that their total is smaller than the eventual first-exit margin. Before τ_J, the quadratic term therefore satisfies

$$\gamma_g W\sum_c |D_2(u; a, c)D_2(u; c, b)| \leq C_\theta\gamma_g W\ell_u J^2\zeta_u^4 w_u(a-b) = \frac{C_\theta J^2\zeta_u^3}{1-u}w_u(a-b).$$

This is the required large-coupling normalization of the free sum. The exact contraction (5.9) retains the same factor $\gamma_g W$ in the joined six-loop bracket. The estimate needed from the stopped equation is

$$\begin{aligned} J_{\mathrm{out}} &:= \sup_{s\le v\le t\wedge\tau_J} \max_{a,b} \frac{|D_2(v;a,b;+,-)|}{\zeta_v^2 w_v(a-b)} \\ &\prec R^6 + R^2J^2\zeta_t + R^5J^{3/2}\zeta_t + R^3J^{3/2}\sqrt{\zeta_t}. \end{aligned} \tag{C.17}$$

We prove this bound by estimating the light triangle and the joined six-cycle before applying the deterministic propagators. At time u, abbreviate $w_{ab} = w_u(a-b)$ and set $h_u = \ell_u(\log W)^{3/2}$. An edge is long when its length exceeds h_u. By (B.16) and the stopped pair bound,

$$W^{-1}\|(G_u)_{ab}\|_{\mathrm{HS}} \le C\sqrt{J}\,\zeta_u\sqrt{w_{ab}} \quad \text{on long edges.}$$

On a short edge, (A.52) gives $w_{ab}^{-1/2} \le W^{\nu}$ for sufficiently large W. The stopped operator envelope therefore gives the ordinary inequality

$$W^{-1}\|(G_u)_{ab}\|_{\mathrm{op}} \le CRW^{2\nu}\zeta_u\sqrt{w_{ab}}, \qquad u \le \tau_J.$$

Here $J \ge 1$ absorbs the primitive constant. These estimates are available before any sharp local law or higher-loop tail estimate.

A distant light triangle has $d(a,b) > 2h_u$. Its direct edge ab and at least one of ac, cb are long. Choose these two as the Hilbert–Schmidt factors in its cyclic trace; bound the remaining edge by its operator norm (or its Hilbert–Schmidt norm if long). For every sign choice occurring in (5.7), this gives

$$|\mathcal{L}_3(u;a,c,b)| \prec CRJ^{3/2}\zeta_u^3\sqrt{w_{ab}w_{ac}w_{cb}}.$$

The centered insertion has size $R^2\zeta_u$ by Step 1. Multiplying by $\gamma_g W$, summing c with the square-root convolution in (6.8), and using $\gamma_g W\ell_u = [(1-u)\zeta_u]^{-1}$, we obtain

$$|\mathcal{W}_{2,ab}(u)| \prec R^3J^{3/2}(1-u)^{-1}\zeta_u^3 w_{ab}, \qquad d(a,b) > 2h_u. \tag{C.18}$$

For the martingale bracket, (5.9) joins the two pairs into the six-cycle (a,b,c,b',a',c), up to reversal and adjoints, with the same coefficient $\gamma_g W$ and one free label c. When both $d(a,b)$ and $d(a',b')$ exceed $2h_u$, the direct edges $ab, a'b'$ are long, and at least two of the four c-edges are long: c cannot be within h_u of both ends of either distant pair. Choose the direct edges as the two Hilbert–Schmidt factors. At most two remaining edges are short, so the normalized six-word is bounded by $(CR)^2J^3\zeta_u^6\prod_{\mathrm{edges}}\sqrt{w}$. Cauchy–Schwarz followed by weight convolution yields

$$\sum_c \sqrt{w_{ac}w_{bc}w_{a'c}w_{b'c}} \le \Big(\sum_c w_{ac}w_{bc}\Big)^{1/2}\Big(\sum_c w_{a'c}w_{b'c}\Big)^{1/2} \le C\ell_u\sqrt{w_{ab}w_{a'b'}}.$$

Writing $\mathcal{V}_{ab,a'b'}(u)$ for the density of $d[\mathcal{M}_{2,ab}, \overline{\mathcal{M}_{2,a'b'}}]_u$, we obtain

$$|\mathcal{V}_{ab,a'b'}(u)| \prec R^2J^3(1-u)^{-1}\zeta_u^5 w_{ab}w_{a'b'} \quad \text{when both pairs are distant.} \tag{C.19}$$

For near endpoint pairs, choose a summation halo $H_u = \ell_u(\log W)^{C_h}$ with $C_h\theta > 1$. Inside the union of the endpoint halos there are at most CH_u possible free labels. The rough triangle and six-loop estimates in (C.5) have sizes $R^2\zeta_u^2$ and $R^5\zeta_u^5$. Thus the near drift and bracket budgets are respectively

$$\begin{aligned} \gamma_g W H_u(R^2\zeta_u)(R^2\zeta_u^2) &= W^{o(1)}R^4(1-u)^{-1}\zeta_u^2, \\ \gamma_g W H_u R^5\zeta_u^5 &= W^{o(1)}R^5(1-u)^{-1}\zeta_u^4. \end{aligned}$$

Outside these halos, a c-edge is longer than H_u. Use its stopped positive-pair estimate as one Hilbert–Schmidt factor and rough bounds for the remaining factors. Its exponential is smaller

than every power of W, since $C_h\theta > 1$. The square root of its floor costs $W^{-F/2}$, multiplied only by the fixed polynomial barrier and rough powers and by the normalized sum $\gamma_g WL$. Increase F after those fixed degrees to make this contribution smaller than any prescribed W^{-A}. This uses the stopped mixed pair itself, without the joint-tail conclusion of Step 3. For $d(a,b) \le 2h_u$ one has $w_{ab}^{-1} = W^{o(1)}$, because $\theta \le 1/2$. Choose $A > 6$ in the preceding remainder estimate. Since $\zeta_u \ge c/W$, the near drift and bracket scales are at least cW^{-2} and cW^{-4}, respectively, before the near weights. Thus the W^{-A} remainder is absorbed into these near source bounds on their near endpoint support. This absorption occurs before propagation; only the resulting weighted near bounds are propagated. Restoring the near weights gives

$$\begin{aligned} |\mathcal{W}_{2,ab}(u)| &\prec R^4(1-u)^{-1}\zeta_u^2 w_{ab} \quad \text{for a near pair,} \\ |\mathcal{V}_{ab,a'b'}(u)| &\prec R^5(1-u)^{-1}\zeta_u^4 w_{ab}w_{a'b'} \quad \text{for two near pairs.} \end{aligned} \tag{C.20}$$

To include near–distant cross terms after propagation, fix a deterministic terminal pair (a,b) at time $v \ge u$ and put $\mathcal{U}_{u,v} = (U_{u,v}^{+-})^{\otimes 2}$. Its coefficients $\mathcal{U}_{u,v}(a,b;x,y) = U_{u,v}^{+-}(a,x)U_{u,v}^{+-}(b,y)$ are nonnegative. Covariance Cauchy–Schwarz gives

$$\begin{aligned} &\sum_{x,y,x',y'} \mathcal{U}_{u,v}(a,b;x,y)\mathcal{U}_{u,v}(a,b;x',y')|\mathcal{V}_{xy,x'y'}(u)| \\ &\quad \le \left(\sum_{x,y} \mathcal{U}_{u,v}(a,b;x,y)\sqrt{\mathcal{V}_{xy,xy}(u)}\right)^2. \end{aligned}$$

Split the inner sum into near and distant pairs. Their diagonal square-root bounds have coefficients $R^{5/2}(1-u)^{-1/2}\zeta_u^2$ and $RJ^{3/2}(1-u)^{-1/2}\zeta_u^{5/2}$, respectively, each times $w_u(x-y)$. By (A.53), the sum of this weight against $\mathcal{U}_{u,v}(a,b;\cdot,\cdot)$ is at most $CR^2 w_v(a-b)$. Applying $(x+y)^2 \le 2x^2+2y^2$ therefore bounds the full propagated bracket by

$$\frac{Cw_v(a-b)^2}{1-u}[R^9\zeta_u^4 + R^6J^3\zeta_u^5], \tag{C.21}$$

with the reserved small loss. The stronger floor has already absorbed all remainders into the near source bounds before this calculation.

For a deterministic terminal time v, use the pair propagator just defined in the stopped Duhamel expression

$$\begin{aligned} \widehat{D}_2(v) = \mathcal{U}_{s,v}D_2(s) &+ \int_s^v \mathbf{1}_{u\le\tau_J}\mathcal{U}_{u,v}(\gamma_g W D_2(u)^2 + \mathcal{W}_2(u))\,du \\ &+ \int_s^v \mathbf{1}_{u\le\tau_J}\mathcal{U}_{u,v}\,d\mathcal{M}_2(u). \end{aligned}$$

The stopped indicators are predictable, and $\widehat{D}_2(v) = D_2(v)$ for $v \le \tau_J$. The propagated-weight estimate (A.53) costs at most R^2 on two labels and R^4 on the four bracket labels. The initial packet contributes $O_\prec(R^2)$ after division by $\zeta_v^2 w_v(a-b)$. For the quadratic and light drifts, the preceding source bounds and (A.54) give the normalized budgets

$$\begin{aligned} R^2J^2\int_s^v \frac{\zeta_u^3}{\zeta_v^2(1-u)}\,du &\le R^2J^2\zeta_t\log R, \\ R^5J^{3/2}\int_s^v \frac{\zeta_u^3}{\zeta_v^2(1-u)}\,du &\le R^5J^{3/2}\zeta_t\log R, \\ R^6\int_s^v \frac{\zeta_u^2}{\zeta_v^2(1-u)}\,du &\le R^6\log R. \end{aligned} \tag{C.22}$$

The three lines correspond to the quadratic error, distant light insertions, and near light insertions. For the bracket, division by $\zeta_v^4 w_v(a-b)^2$ gives

$$\begin{aligned} R^6 J^3 \int_s^v \frac{\zeta_u^5}{\zeta_v^4(1-u)}\,du &\le R^6 J^3 \zeta_t \log R, \\ R^9 \int_s^v \frac{\zeta_u^4}{\zeta_v^4(1-u)}\,du &\le R^9 \log R. \end{aligned} \tag{C.23}$$

BDG takes the square root of these budgets, and therefore bounds the normalized martingale by $O_\prec(R^3 J^{3/2}\sqrt{\zeta_t} + R^{9/2})$. Since $\log R = O(\log W)$, the logarithmic and halo factors fit inside the reserved arbitrarily small domination losses. Combining the initial term, (C.22), and (C.23) proves the right-hand side of (C.17) at each deterministic pair and terminal time.

We now pass to the stopped supremum in (C.17). Choose the deterministic sample pairs from (5.34) with weight w_u, accuracy exponent $Q+4$ and failure exponent $D+3$. After division by $\zeta_u^2 \ge cW^{-2}$, its additive error is $O(W^{-Q-2})$. Apply (C.11) with $j=0$, accuracy exponent $Q+F+4$ and failure exponent $D+3$. Since $\zeta_u^2 w_u \ge cW^{-2-F}$, its raw increment error also becomes $O(W^{-Q-2})$ after normalization. Each exceptional event has probability at most W^{-D-2} for sufficiently large W. Direct differentiation gives $|\partial_u \log(\zeta_u^2 w_u)| \le C_F W^3 \log W$; on a sufficiently fine mesh, the normalizers at adjacent times therefore differ by at most a factor two. For $u \le \tau_J$ use the mesh point immediately to its left, which also precedes τ_J. On the sampling and continuity events,

$$J_{\mathrm{out}} \le 2C \max_{\substack{v\in\mathcal{T},\ v\le\tau_J \\ (a,b)\in\mathcal{S}_{D_2}}} \frac{|\widehat{D}_2(v;a,b)|}{\zeta_v^2 w_v(a-b)} + W^{-Q}. \tag{C.24}$$

Here C is the fixed norming constant. Once the deterministic pairs and times number at most W^{C_1} with fixed C_1, apply the fixed-terminal Duhamel and BDG bound with failure exponent $D+C_1+2$. The total failure probability is at most

$$W^{C_1} W^{-D-C_1-2} + 2W^{-D-2} \le W^{-D-1}.$$

This proves the supremum form of (C.17). The sampling and continuity events enter only this last comparison; they are not inserted into a stochastic integrand. Only the actual field before τ_J is interpolated.

Take $J_0 = W^\epsilon R^{10}$ with $0 < \epsilon < \min\{c/10, 1\}$. Since $R \le W^3$, this barrier satisfies $J_0 \le W^{31}$ and lies in the polynomial range fixed before the tail floor. Dividing the right-hand side of (C.17) by J_0 and using $R^{20}\zeta_t \le W^{-c}$ gives

$$\begin{aligned} &W^{-\epsilon} R^{-4} + W^\epsilon R^{12}\zeta_t + W^{\epsilon/2} R^{10}\zeta_t + W^{\epsilon/2} R^8 \sqrt{\zeta_t} \\ &\quad \le W^{-\epsilon} + W^{\epsilon-c} + W^{\epsilon/2-c} + W^{\epsilon/2-c/2} = o(1). \end{aligned}$$

Every term is polynomially small, so the stochastic-domination loss can be absorbed. The initial bound in (5.22) starts below J_0 with overwhelming probability. Continuity therefore excludes the first exit through J_0; the rough-bound exits already have arbitrarily small probability by Step 1. This proves the first line of (C.15).

For the Hilbert–Schmidt bound, the loop definition (5.5) and the mixed pair formula (5.13), together with (6.3), give

$$\begin{aligned} W^{-2}\|(G_u)_{ab}\|_{\mathrm{HS}}^2 &= \mathcal{K}_2(u;b,a;+,-) + D_2(u;b,a;+,-) \\ &\le C\zeta_u + |D_2(u;b,a;+,-)| \prec \zeta_u + R^{10}\zeta_u^2 \prec \zeta_u, \end{aligned}$$

since $R^{10}\zeta_u \le R^{20}\zeta_t \le W^{-c}$. We have proved the two-loop conclusions

$$|D_2(u;a,b;+,-)| \prec R^{10}\zeta_u^2 w_u(a-b), \qquad W^{-2}\|(G_u)_{ab}\|_{\mathrm{HS}}^2 \prec \zeta_u. \tag{C.25}$$

To make the Haar consequence simultaneous, apply (7.28) with $\Psi_u = \sqrt{\zeta_u}$ on $[s,t]$. The scale is bounded above and below by fixed powers of W, and its logarithmic derivative is at most CW^3 directly from (5.18). The fixed-pair Haar moments are thus used only at the deterministic traceless-entry samples and time mesh, including the W^2 orbital choices, before spatial norming and time interpolation. The result is

$$\sup_{s\le u\le t}\max_{ai,bj}\frac{|(G_u)_{ai,bj}-\delta_{ij}\langle G_u\rangle_{ab}|}{\sqrt{\zeta_u}}\prec 1. \tag{C.26}$$

We finish the scalar improvement in two applications of fluctuation averaging. Set $X_u = \max_{a,b}|\langle\widetilde{G}_u\rangle_{ab}|$ and $Y_u = \max_a|\langle\widetilde{G}_u\rangle_{aa}|$. The rough trace bound (C.8) is simultaneous in time by (C.12) with $j=0$, since centering subtracts the time-independent matrix M. Apply (5.29) with $K = CR$ and $\chi = \max\{\sqrt{\zeta_u}, W^{-1/2}\} \asymp \sqrt{\zeta_u}$. Its fixed-pair moment proof, residual sampling and the same time interpolation give $\sup_{u,a,b}|R_{ab}(u)|/(R^2\zeta_u)\prec 1$. The first scalar identity in (5.32) and the normalized Dyson row in (6.2) now give

$$X_u \le \max_{a,b}|R_{ab}(u)| + CY_u\max_{a,b}|\langle G_u\rangle_{ab}| \prec R^3\zeta_u \le W^{-c},$$

where $Y_u \prec R^2\zeta_u$ comes from Step 1. The bounded Dyson entries then imply $\max_{a,b}|\langle G_u\rangle_{ab}|\prec 1$. A second application of the same original-law moment implication, now with $K=C$, improves the residual to $\sup_{u,a,b}|R_{ab}(u)|/\zeta_u\prec 1$. Applying $(I-uQ)^{-1}$ and its absolute row bound in (6.2) to the second scalar identity gives

$$Y_u \le C\max_{a,b}|R_{ab}(u)| + CY_uX_u.$$

On the common event where X_u is uniformly a small power of W, absorb the last term to obtain $Y_u\prec\zeta_u$; the first scalar identity then gives $X_u\prec\zeta_u$. Finally,

$$\|\widetilde{G}_u\|_{\max}\le\max_{ai,bj}|(G_u)_{ai,bj}-\delta_{ij}\langle G_u\rangle_{ab}| + X_u \prec \sqrt{\zeta_u}.$$

All these statements have the time supremum inside probability, by the same residual sample and time-mesh construction used in Step 1. This proves every bound in (C.15).

Step 3: sharp bounds on the actual loops. The goal of this step is to obtain, for every fixed k, the sharp actual-loop bound

$$|\mathcal{L}_k(u)|\prec\zeta_u^{k-1}, \tag{C.27}$$

and the intermediate error estimate

$$|D_k(u)|\prec\zeta_s^{-1/2}\zeta_u^k. \tag{C.28}$$

We also need the joint spatial tails, for every fixed $k\ge 2$ and $\epsilon, A>0$:

$$\max_{\boldsymbol{\sigma}\in\{+,-\}^k}\max_{\boldsymbol{a}:\,\mathrm{diam}(\boldsymbol{a})>W^\epsilon\ell_u}\left(|\mathcal{L}_k(u)|+|D_k(u)|\right)\prec W^{-A}. \tag{C.29}$$

We prove the tails first, since they localize every subsequent free label. Fix the finite collection of lengths required below, and let N_0 be its largest error length. The raw brackets have length at most $M_0 = 2N_0+2$. Choose B_* so that every crude normalized block norm and $R^{10}\zeta_u^2$ is bounded by W^{B_*} on the Step 1–2 events. Let P_* exceed the fixed polynomial costs of spatial sums, row restoration and propagator bounds for these lengths. These are analytic error costs; sample and mesh cardinalities will enter only the failure probabilities. Given a final tail accuracy A, put $A_* = A+P_*+2$ and choose the stronger Step 2 floor by

$$F > 2A_* + (2M_0+1)B_* + 10. \tag{C.30}$$

A word of length $m \le M_0$ and diameter larger than $W^{\epsilon}\ell_u$ has an edge of length at least $W^{\epsilon}\ell_u/m$. The positive mixed pair on that edge is the sum of the primitive pair and its error. For large W, the edge exceeds $W^{\epsilon/2}\ell_u$, so (5.21) applies with exponent $\epsilon/2$. Together with (C.25), it gives

$$W^{-2}\|(G_u)_{ab}\|_{\mathrm{HS}}^2 \le W^{-Q} + CW^{B_*}\left(e^{-c(\|a-b\|_L/\ell_u)^{\theta}} + W^{-F}\right)$$

on the fixed Step 1–2 event; B_* includes the reserved domination loss. Choose $Q > 2A_* + 2(M_0 - 1)B_* + 4$. Use the square root of this bound as one Hilbert–Schmidt factor in the word; the remaining $m-1$ normalized factors have crude Hilbert–Schmidt bounds W^{B_*}. The primitive remainder and the error floor contribute, respectively,

$$W^{-Q/2+(m-1)B_*} \le W^{-A_*-2}, \qquad W^{B_*/2-F/2+(m-1)B_*} \le W^{-A_*-1}.$$

The stretched-exponential term is $O_D(W^{-D})$ for every fixed $D > 0$. This proves the actual-loop tail. The primitive tail in (5.21) then gives the error tail, proving (C.29). Every cut word shares its free vertex and every bracket word is a single joined cycle, so this same argument covers their forcing tensors. All summed tails keep the covariance factor, and restored rows keep their profile factors ℓ_u^{-1}; (6.6) bounds the remaining costs by $\gamma_g L$ and powers of L/ℓ_u. Thus the discarded terms remain at most $W^{P_*}W^{-A_*-1} \le W^{-A}$, including at arbitrary coupling.

The primitive size input used in this step is (5.20), together with its length-two bound. At bounded coupling its constant is uniform for $0 < g \le C$. At large coupling the constants are independent of g, L; $L/(g^2 \log W) \to \infty$ meets every required fixed volume threshold. The comparison pairs remain in these domains by Lemma 6.3.

Use the normalized random controls

$$\begin{aligned} A_j &:= \sup_{s\le v\le t} \max_{\boldsymbol{a},\boldsymbol{\sigma}} \frac{|D_j(v;\boldsymbol{a},\boldsymbol{\sigma})|}{\zeta_v^j}, \\ B_j(v) &:= \max_{\boldsymbol{a},\boldsymbol{\sigma}} \frac{|\mathcal{L}_j(v;\boldsymbol{a},\boldsymbol{\sigma})|}{\zeta_v^{j-1}}, \qquad B_j := \sup_{s\le v\le t} B_j(v). \end{aligned} \tag{C.31}$$

In particular, $A_1 \prec 1$ by Step 2, and the primitive input gives $B_j(u) \prec 1 + \zeta_u A_j$. Localize the one free block label of each source to $W^{\nu}\ell_u$ using (C.29), with arbitrarily small fixed $\nu > 0$. Using the normalization in (6.6) and $p + q = n + 2$, the preceding controls give

shorter linear cuts	$(1-u)^{-1}\zeta_u^n \max_{j<n} A_j$
quadratic errors	$(1-u)^{-1}\zeta_u^{n+1} \max_{p+q=n+2} A_pA_q$
light insertions	$(1-u)^{-1}\zeta_u^n B_{n+1}$
cross bracket	$(1-u)^{-1}\zeta_u^{2n} B_{2n+2}$.

(C.32)

Here B_j is the time supremum in (C.31); fixed-order constants and the arbitrarily small localization loss are understood. Primitive–error cuts in the first line have shorter error length; the adjacent pair terms are already in the linear generator. All terms on the right will be replaced by previously proved deterministic envelopes before the stopped estimate is applied. When a floor is summed, retain γ_g with the sum; when a row is restored, retain every factor ℓ_u^{-1} in the restoring profile. The polynomial budget is again (6.6).

We use the tensor propagator bounds (A.74) from Appendix A.12, for zero rows or at least one stable leg. A same-sign neighboring pair gives a stable leg by (6.2). For an alternating word, set $h_u(a_1) = \sum_{a_2,\dots,a_n} D_n(u;\boldsymbol{a})$ and $\theta_u = (1-u)P(I-uP)^{-1}$, a probability profile with $\|\theta_u\|_\infty \le C/\ell_u$ by (6.3). Subtract the signed row by

$$\mathcal{Q}_u D_n(\boldsymbol{a}) = D_n(\boldsymbol{a}) - h_u(a_1)\prod_{j=2}^{n}\theta_u(a_1 - a_j).$$

For every fixed $Q > 0$, the needed row and commutator bounds are

$$\begin{aligned} \|h_u\|_\infty &\prec \ell_u^{n-1}\zeta_u^n(A_{n-1} + W^{-Q}), \qquad \|D_n - \mathcal{Q}_u D_n\|_\infty \prec \zeta_u^n(A_{n-1} + W^{-Q}), \\ \|C_u(h_u)\|_\infty &\prec (1-u)^{-1}\zeta_u^n(A_{n-1} + W^{-Q}), \qquad \operatorname{row} C_u(h_u) = 0. \end{aligned} \tag{C.33}$$

We verify the signed row before taking its absolute value. By conjugation, take an alternating word beginning with $+$ and ending with $-$. Subtract the $\mathcal{L}$ and $\mathcal{K}$ identities in (5.11), and then sum the remaining $n-2$ labels. With $\boldsymbol{\sigma}^+ = (+, \sigma_2, \dots, \sigma_{n-1})$ and $\boldsymbol{\sigma}^- = (-, \sigma_2, \dots, \sigma_{n-1})$, this gives

$$h_u(a_1) = \frac{1}{2iW\eta_u} \sum_{a_2,\dots,a_{n-1}} \left[D_{n-1}(u; a_1, \dots, a_{n-1}; \boldsymbol{\sigma}^+) - D_{n-1}(u; a_1, \dots, a_{n-1}; \boldsymbol{\sigma}^-) \right].$$

The localized absolute sum has at most $C_n(W^\nu \ell_u)^{n-2}$ terms. Since $(W\eta_u)^{-1} = \ell_u \zeta_u / (\Im m)$ and $\Im m \asymp 1$,

$$|h_u(a_1)| \prec (W\eta_u)^{-1}(W^\nu \ell_u)^{n-2}\zeta_u^{n-1} A_{n-1} + \ell_u^{n-1}\zeta_u^n W^{-Q} \prec \ell_u^{n-1}\zeta_u^n(A_{n-1} + W^{-Q}). \tag{C.34}$$

The discarded row tails have only fixed polynomial costs after normalization, including powers of L/ℓ_u and ζ_u^{-1}. Choose their tail accuracy after Q and these fixed costs to leave the normalized remainder W^{-Q}. The raw random control A_{n-1} need not be bounded below, so this remainder is retained here. This calculation uses the signed Ward identity before absolute summation. Since $\|\theta_u\|_\infty \le C/\ell_u$, restoring the row costs $\ell_u^{n-1}\zeta_u^n(A_{n-1} + W^{-Q})(C/\ell_u)^{n-1} = C_n\zeta_u^n(A_{n-1} + W^{-Q})$.

The projected differential explains the remaining commutator. Write the error hierarchy (5.17), with the generator Θ from (5.14), as $dD_n = \Theta_{u,\boldsymbol{\sigma}} D_n \, du + F_n \, du + d\mathcal{M}_n$, where F_n contains the remaining cut and light sources. For the alternating word every external-leg generator is $P(I - uP)^{-1}$. Direct differentiation of the probability profile gives

$$\partial_u \theta_u = P(I - uP)^{-1}\theta_u - (1-u)^{-1}\theta_u.$$

In differentiating $\mathcal{Q}_u D_n$, the $j \ge 2$ external-leg terms cancel the corresponding profile derivatives: summing each such generator contributes $(1-u)^{-1}$ to the signed row. The difference in the first leg is exactly $C_u(h_u)$ in (6.4). Therefore

$$d(\mathcal{Q}_u D_n) = \Theta_{u,\boldsymbol{\sigma}}(\mathcal{Q}_u D_n)\, du + [\mathcal{Q}_u F_n + C_u(h_u)]\, du + \mathcal{Q}_u \, d\mathcal{M}_n. \tag{C.35}$$

The bound (6.5) with (C.34) gives $(1-u)^{-1}\zeta_u^n(A_{n-1} + W^{-Q})$ for the commutator, which proves the third line of (C.33), including its zero row. The projected bracket is $(\mathcal{Q}_u \otimes \mathcal{Q}_u)\mathcal{V}_n$, where $\mathcal{V}_n$ is the unprojected bracket tensor. It has zero rows in both sets of labels. For a source tensor A supported on relative diameter $H\ell_u$, $\|\theta_u\|_\infty \le C/\ell_u$ gives

$$\left\| (\operatorname{row} A) \prod_{j=2}^{n} \theta_u(a_1 - a_j) \right\|_\infty \le C_n(H\ell_u)^{n-1}\|A\|_\infty \ell_u^{-(n-1)} = C_n H^{n-1}\|A\|_\infty.$$

Apply this bound in each coordinate group for the bracket. With $H = W^\nu$, projection thus retains the source-table scales up to the reserved localization loss. Each restoring profile reconnects its labels to its anchor; its exponential tails follow from (6.3) and the definition of θ_u. The discarded polynomial floors are controlled next. Apply (A.74) to these zero-row tensors; nonalternating words use its stable-leg bound. This proves the projection mechanism needed for the recurrence.

For the truncation error use the tail argument in the proof of (A.74), with discarded tensor T, $\|T\|_\infty \le W^{-F}$. For $u \le v \le t$, restoration against θ_v costs at most $C_n(L/\ell_v)^{n-1}W^{-F} \le C_n(L/\ell_u)^{n-1}W^{-F}$. Propagate the discarded and restoration tensors separately; the latter has scale ℓ_v, not support in the truncated region. Since $R \le W^3$ and (6.6) bounds the remaining costs polynomially, choose F after these costs and the required accuracy. This controls the

row-restoration and covariance-weighted cut remainders without changing the main source normalizations.

The estimate supplied by the projected hierarchy is

$$A_n \prec 1 + \max_{j<n} A_j + \zeta_t \max_{2\le j\le n} A_j A_{n-j+2} + B_{n+1} + \sqrt{B_{2n+2}}. \tag{C.36}$$

We use this estimate as a bound-improvement implication: replace the controls on the right by deterministic polynomial bounds already proved at the preceding induction stages, and stop predictably at those bounds. Choose all these deterministic envelopes at least 1; they absorb the arbitrary normalized W^{-Q} remainders in the row and commutator bounds. The stopped estimate then gives the displayed deterministic bound for A_n in stochastic domination. To derive it, apply the deterministic-terminal Duhamel formula to (C.35), or to the stable-leg hierarchy for a nonalternating word. A drift source of size $(1-u)^{-1}\zeta_u^n C_u$ contributes at most

$$\zeta_v^{-n}\int_s^v (\zeta_v/\zeta_u)^n (1-u)^{-1}\zeta_u^n C_u\, du \le (\log R)\sup_{s\le u\le v} C_u.$$

The stable-leg estimate is smaller since $\zeta_v \ge \zeta_u$. The quadratic source has the additional factor $\zeta_u \le \zeta_t$. For the bracket, use order $2n$ in (A.74); its normalized integral is bounded by $(\log R)B_{2n+2}$. BDG consequently gives $(\log R)^{1/2}\sqrt{B_{2n+2}}$ for the normalized martingale. At the initial time, (5.22) bounds both $D_n(s)$ by $O_\prec(\zeta_s^n)$ and $D_{n-1}(s)$ by $O_\prec(\zeta_s^{n-1})$. The signed-row calculation therefore gives $\|\mathcal{Q}_s D_n(s)\|_\infty \prec \zeta_s^n$; the initial tails have the same floor budget. Propagation makes this a normalized contribution 1. The terminal row restoration contributes A_{n-1}. These are exactly the five terms in (C.36); the logarithms and localization powers are absorbed in the reserved small losses.

At each induction node, stop at the already proved shorter-length and preceding-round envelopes, together with a provisional barrier for the current target. Apply the projected Duhamel formula at deterministic sample tuples and terminal times. The passage to the supremum is (C.24) with D_2 replaced by D_n and normalization ζ_v^n; the proof uses the actual error only before the stop. Thus no estimate on the target at the current node is used to justify its own improvement.

We now establish the same-sign two-loop base

$$\sup_{s\le u\le t}\max_{\substack{a,b\in\mathbb{Z}_L\\ \sigma\in\{+,-\}}} \zeta_u^{-2}|D_2(u;a,b;\sigma,\sigma)| \prec R^{10}. \tag{C.37}$$

We adapt the Duhamel and martingale estimates from [33, proof of Lemma 5.11] and close the two-loop estimate by the stopping argument below. Write $D_2^{++}(u) := (D_2(u;a,b;+,+))_{a,b\in\mathbb{Z}_L}$. The exact equation is

$$\begin{aligned} dD_2^{++} = {} & [Q(I-uQ)^{-1}D_2^{++} + D_2^{++}Q(I-uQ)^{-1}]du \\ & + \gamma W(D_2^{++})^2 du + \mathcal{W}_2^{++}du + d\mathcal{M}_2^{++}. \end{aligned} \tag{C.38}$$

Here $Q_{ab} = \gamma(M_B)_{ab}^2$ as in (5.12). Its propagator has bounded absolute row and column sums by (6.2) and symmetry. The initial packet gives $\max_{a,b}|D_2^{++}(s;a,b)| \prec \zeta_s^2$. The already proved local law gives $A_1 \prec 1$, whereas the rough bounds (C.5) give $B_3 \prec R^2$ and $B_6 \prec R^5$. These are bounds on actual loops and do not require any improvement of a centered higher loop.

Fix $0<\epsilon<c/4$ and put $J_0 := W^\epsilon R^{10}$. Choose a total loss $0<\nu<\epsilon/4$, reserving smaller losses for each estimate. Stop at the first time $\max_{a,b}|D_2^{++}(u;a,b)|/\zeta_u^2$ reaches J_0, also stopping on the previously established rough, local-law and tail envelopes. All tests use only the current and past flow. Before this stop, the tails obtained above from the mixed positive pair localize the free block label in $(D_2^{++})^2$ to $W^{\nu/10}\ell_u$ sites, up to an arbitrarily small polynomial remainder. Thus

$$\gamma W\max_{a,b}|[(D_2^{++})^2]_{ab}| \le CW^{\nu/10}J_0^2(1-u)^{-1}\zeta_u^3 + W^{-A},$$

where the floor is chosen after A. The light term and the joined six-loop bracket in (C.32) are bounded, respectively, by $O_{\prec}(R^2(1-u)^{-1}\zeta_u^2)$ and $O_{\prec}(R^5(1-u)^{-1}\zeta_u^4)$.

Apply Duhamel's formula with predictable stopped sources and the deterministic propagator $U^{++}_{u,t'}$ at each deterministic terminal time t'. Its bounded rows and BDG give the stopped version of (C.36) at length two:

$$J_{\mathrm{out}} \prec 1 + R^2 + R^{5/2} + \zeta_t J_0^2. \tag{C.39}$$

Here J_{out} is the maximal normalized error up to the stop. Time integration costs at most $1+\log R$, and the localization and logarithmic losses are absorbed into an arbitrarily small domination loss. The ordinary-tuple sampling and time-increment argument in the proof of (C.24) gives the space-time supremum, also at arbitrary coupling; no random terminal propagator is inserted into a stochastic integral. With the reserved total loss W^{ν}, division by J_0 gives

$$J_0^{-1} J_{\mathrm{out}} \le CW^{\nu-\epsilon} R^{-15/2} + CW^{\nu+\epsilon} R^{10} \zeta_t = o(1)$$

with arbitrarily high probability, by $R^{20}\zeta_t \le W^{-c}$. The initial error is below J_0 with the same probability. Continuity therefore excludes the first exit. Taking ϵ arbitrarily small proves the $(+,+)$ assertion in (C.37); conjugation gives $(-,-)$. No spatial profile for the same-sign error is needed here. All constants used in this argument are the common propagation constants, so it applies to both coupling cases of Theorem 5.12.

A direct induction on n would not close, since the bracket contains an actual loop of length $2n+2$. Put $m_1 = 2\lfloor (n+1)/2 \rfloor$ and $m_2 = 2\lceil (n+1)/2 \rceil$. Trace Cauchy–Schwarz gives

$$B_{2n+2}(v) \le \zeta_v^{-1} B_{m_1}(v) B_{m_2}(v). \tag{C.40}$$

This is the four-chain Cauchy–Schwarz calculation in [31, (A.74)–(A.77)]; the present normalization contributes $\zeta_v^{m_1+m_2-2-(2n+1)} = \zeta_v^{-1}$.

For the fixed interval $[s,t]$, use double induction in the loop length n and the improvement round $j \ge 0$ to prove

$$H_{n,j} \coloneqq \zeta_s^{-1/2} + (CR)^{n-1} \zeta_s^{j/4-1}, \qquad A_n \prec H_{n,j}. \tag{C.41}$$

Here j counts improvements of the same random control, not time steps. We verify every term of (C.36) using only the already proved controls assigned to the current induction node. The rough bound (C.5) and the primitive estimate give $A_n \prec H_{n,0}$. The local law supplies $A_1 \prec 1$. The mixed and same-sign two-loop bounds give, at every round,

$$A_2 \prec R^{10} \le \zeta_s^{-1/2} \le H_{2,j},$$

since $R^{20}\zeta_s \le R^{20}\zeta_t \le W^{-c}$. For $j \ge 1$ proceed in increasing $n \ge 3$. Use shorter lengths already proved in round j for linear terms and interior quadratic products. Use round $j-1$ for B_{n+1}, the target factor A_n in $A_2 A_n$, and the two loop controls from (C.40). Their lengths satisfy $m_1, m_2 \le n+2$.

For a target length $k \ge 3$, start with rough bounds through $k + 2(k+4) = 3k+8$. At round $j = 1, \dots, k+4$, improve in increasing order through length $k + 2(k+4-j)$. The preceding round supplies every length through $n+2$ required at a current node, and the last round reaches k. Thus this construction has finitely many nodes and never uses an unknown same-round target on the right.

The scale definition (5.18) and the time-step condition (5.38) give

$$c/W \le \zeta_s \le \zeta_u \le \zeta_t, \qquad \zeta_t/\zeta_s \le R, \qquad R^{20}\zeta_t \le W^{-c}.$$

In particular $R \le \zeta_s^{-1/20}$ and $\zeta_s \le W^{-c}$. The following strict power margins absorb all fixed-order constants and the separately reserved losses at the finitely many nodes.

Use the scalar induction in [31, Lemma 7.18, Appendix A.8] with the assigned earlier envelopes. For the present time-ratio bound, substitution of (C.41) and (C.40) gives, uniformly for $s \le u \le t$,

$$\begin{aligned} B_{n+1} &\prec C\zeta_s^{1/2} H_{n,j}, \\ \sqrt{B_{2n+2}(u)} &\prec \zeta_s^{-1/2} + C\zeta_s^{3/40} H_{n,j}, \\ \zeta_t R^{10} H_{n,j-1} &\le C\zeta_s^{1/5} H_{n,j}, \\ \zeta_t H_{p,j} H_{q,j} &\le C\zeta_s^{3/20} H_{n,j} \quad (p+q = n+2,\ 3 \le p, q < n). \end{aligned}$$

These control the light, bracket, boundary quadratic and interior quadratic sources, respectively. Shorter linear terms are at most $H_{n,j}$, so every source in (C.36) has this bound. Apply its adapted stopped implication at this node and continue through the finite cone. Choosing smaller losses at its finitely many nodes preserves the requested final domination loss.

At the final round $j = k+4$ for the target k,

$$(CR)^{k-1} \zeta_s^{(k+4)/4-1} \le C_k \zeta_s^{(4k+1)/20}.$$

Thus $A_k \prec \zeta_s^{-1/2}$, proving (C.28). This argument applies to each required fixed length and the finite union of their cones.

Finally, $\zeta_u/\sqrt{\zeta_s} \le \sqrt{R\zeta_t} \le W^{-c/2}$ by the scale ratio and $R^{20}\zeta_t \le W^{-c}$. Adding the primitive bound (5.20) and the length-two estimate in Lemma 5.2 therefore yields

$$|\mathcal{L}_k(u)| \le |\mathcal{K}_k(u)| + |D_k(u)| \prec \zeta_u^{k-1}\left(1 + \frac{\zeta_u}{\sqrt{\zeta_s}}\right) \prec \zeta_u^{k-1},$$

proving (C.27); cf. [31, Section 7.4, (5.15)] and [33, Section 5.6, (2.77)].

Step 4: sharp bounds on the loop errors. The goal of this step is to remove the factor $\zeta_s^{-1/2}$ and obtain

$$|D_k(u)| \prec \zeta_u^k \tag{C.42}$$

for every fixed k. Reinsert the sharp actual loops from Step 3 and the sharp one-loop error from Step 2 into the hierarchy. The source bounds and projected propagation established in Step 3 apply to these improved inputs. In particular, the four source bounds in (C.32) give drift scale $(1-u)^{-1}\zeta_u^k$, bracket scale $(1-u)^{-1}\zeta_u^{2k}$, and an extra factor ζ_u for quadratic errors, also at arbitrarily large coupling.

With $B_j \prec 1$ at every required length, the stopped implication (C.36) reduces to

$$A_n \prec 1 + \max_{j<n} A_j + \zeta_t \max_{2\le j\le n} A_j A_{n-j+2}.$$

Use the length induction of [31, Section 7.5, (7.66)–(7.67)] on this recurrence. With $A_1 \prec 1$, the successive input envelopes $A_2 \prec \zeta_s^{-1/2}$ from Step 3 and then $A_2 \prec R$ give

$$A_2 \prec 1 + \frac{\zeta_t}{\zeta_s} \lesssim R, \qquad A_2 \prec 1 + \zeta_t R^2 \lesssim 1. \tag{C.43}$$

Here $\zeta_t/\zeta_s \le R$ and $\zeta_t R^2 \le W^{-c}$. Each application stops at its already proved input envelope and then uses the deterministic-sample and time-interpolation argument following (C.36); the new output is not used in its own integrand.

For $n \ge 3$, use shorter current-stage bounds $A_m \prec 1$ in the linear and interior quadratic terms, and the Step 3 envelope $A_n \prec \zeta_s^{-1/2}$ in the boundary products $A_2 A_n$. Then

$$A_n \prec 1 + \zeta_t + \zeta_t \zeta_s^{-1/2} \prec 1,$$

since $\zeta_t/\sqrt{\zeta_s} \le \sqrt{R\zeta_t} \le W^{-c/2}$. All input controls precede this induction node. The same joint-tail budget (C.30) covers the localized sums and stopped brackets. The finitely many loss and failure reserves therefore give the spatial and time supremum in (C.42) for every fixed length, uniformly in the deterministic parameters.

Step 5: restoring the spatial profile and the next initial data. The goal of this step is to remove the spatial loss R^{10}:

$$|D_2(u;a,b;+,-)| \prec \zeta_u^2 \left(e^{-c(\|a-b\|_L/\ell_u)^{\theta/2}} + W^{-F} \right). \tag{C.44}$$

Together with the joint tails (C.29) and the sharp bounds from Steps 2–4, this restores every estimate in (5.39).

Put $x = \|a-b\|_L/\ell_u$ and apply Step 2 with a stronger floor exponent $F' > F + 32$. Because $R \le W^3$, its floor satisfies $R^{10}W^{-F'} \le W^{-F-2}$. Taking the minimum with Step 4 gives

$$\frac{|D_2(u;a,b;+,-)|}{\zeta_u^2} \prec \min\{1, R^{10}e^{-c_{\mathrm{p}}x^\theta}\} + W^{-F-2}.$$

The inequality $\min\{1, a+b\} \le \min\{1,a\}+b$ separates the floor. Apply (A.56) with $x = \|a-b\|_L/\ell_u$ and $1 \le R \le W^3$. Its prefactor $e^{C'\sqrt{\log W}} = W^{o(1)}$ is absorbed in the reserved domination loss. Together with the W^{-F-2} remainder, this proves (C.44). The constants are uniform in a, b, u and the allowed parameters. A fixed finite number of steps leaves a fixed positive profile exponent.

For the operator input at the next time, the sharp actual loops include

$$\mathrm{Tr}[(G_tE_bG_t^*E_a)^p] = W^{-2p}\,\mathrm{Tr}[((G_t)_{ab}(G_t)_{ab}^*)^p] \prec \zeta_t^{2p-1}. \tag{C.45}$$

Consequently,

$$W^{-1}\max_{a,b}\|(G_t)_{ab}\|_{\mathrm{op}} \prec \zeta_t. \tag{C.46}$$

Repeat the Schatten-to-operator argument following (C.7), now using (C.45) at time t. For each $\epsilon, D > 0$, choose a fixed integer $p > 1/\epsilon$ before the loop-length cone, use loss $\epsilon/2$, and invoke $\zeta_t \ge c/W$. The spatial maximum is inherited from the loop estimate, so no L^2 union is needed. The same positive words supply the fixed Schatten moments, including for off-diagonal blocks. Together with the averaged law and the comparison (5.35), all inputs for the next step are now available. All these estimates already include the spatial and time suprema by (C.11) and (C.24). For each prescribed $\epsilon, D > 0$, reserve smaller losses at the finitely many uses above and failure W^{-D-2} for each step; their union has probability at most $5W^{-D-2} \le W^{-D}$. Constants are uniform in deterministic (g, E), so their supremum remains outside the probability as required in (5.39). This proves part (i). □

C.4 Finite time iteration

Proof of Theorem 5.12(ii). We prove (5.42) with all required finite orders and probability estimates fixed before the forward iteration. Write g_0 for the coupling in the desired conclusion. For the fixed $c_0 > 0$ and target $1/2 \le t_* \le 1 - W^{-3}$ with $\zeta_{t_*} \le W^{-c_0}$, set $J = \lceil 240c_0^{-1} \rceil + 1$ and

$$s_j := 1 - \max\{1 - t_*, \tfrac{1}{2}W^{-jc_0/80}\}, \qquad 0 \le j \le J. \tag{C.47}$$

Since $Jc_0/80 > 3$, these times run from $s_0 = 1/2$ to $s_J = t_*$; omit repeated endpoints. The definition and monotonicity of ζ give $R_j := (1-s_{j-1})/(1-s_j) \le W^{c_0/80}$ and $R_j^{20}\zeta_{s_j} \le W^{-3c_0/4}$, while $1 - s_j \ge 1 - t_* \ge W^{-3}$. Thus part (i) applies with $c = c_0/2$; J is fixed independently of loop orders, losses, floors and failure exponents.

We must supply the starting estimates at every deterministic comparison pair used within a step. Lemma 6.3(b) applies to each backward comparison chain of length at most J. At bounded coupling it gives $0 < c_J g_0 \le g \le C$, bounded energies, and a fixed positive lower bound on $\Im m$, preserving the input range and the original $L \gg \log W$ assumption. At large coupling, (6.11) gives $g_0/\sqrt{2} \le g \le g_0$ and $|E| \le 2 - \kappa/2$. Consequently $L \gg g_0^2 \log W$ implies $L \gg g^2 \log W$, while $L \le Cg_0^2W^2$ implies $L \le 2Cg^2W^2$. The preserved domain (6.10) and its equality $s\gamma_{g'} = u\gamma_g$ give precisely the earlier-time admissibility needed in Step 1. The same

lemma preserves $1-u \geq W^{-3}$ and the endpoint scale bound $\zeta_u(g) \leq \zeta_{t_*}(g_0) \leq W^{-c_0}$. For an earlier comparison pair required at time s, every preceding time $v \leq s$ satisfies

$$v\gamma_{g'} \leq s\gamma_{g'} = u\gamma_g \leq g_0^{-2}.$$

Propagate that pair only through preceding intervals ending at s, using the same schedule with deterministic admissibility truncation when needed; propagation up to t_* at the smaller g' is not required. These are deterministic parameter bounds; uniformity in them is outside the probability and requires no union over an energy or coupling continuum.

Fix the desired final loop lengths, domination loss, spatial-tail accuracy and failure exponent. For every operator loss W^{δ} needed at the next step, first select the fixed integer $p > 1/\delta$ in (C.45) and include its positive loop of length $2p$. For an actual-loop output of length n, the finite cone in Step 3 requires rough lengths through $n + 2(n+4) = 3n+8$. For an error output of length k, Step 4 needs actual loops through $2k+2$, hence rough lengths through $6k+14$ suffice. Take the finite union of these requirements and reserve smaller losses at its nodes so that products and square roots leave the prescribed output loss. The starting data at this step must supply the resulting finite list at the original and the deterministic comparison pairs.

Now fix the analytic tail accuracies for this finite list. If N_0 is the largest required error length, its raw joined bracket has length at most $M_0 = 2N_0 + 2$. Such a bracket uses the long-edge tail argument of Step 3 and crude bounds on the remaining factors, not sharp errors of length M_0. Fix their crude exponent B_* and the polynomial spatial-sum, row-restoration and propagator cost P_*. Given tail accuracy A, choose $A_* = A + P_* + 2$, the primitive remainder accuracy $Q > 2A_* + 2(M_0 - 1)B_* + 4$ from the long-edge calculation, and then the floor in (C.30). For an output profile floor W^{-F}, choose the stronger input floor $F' > F + 32$ used in Step 5 as well. Increase the finitely many input floor exponents to meet both requirements. The discarded analytic errors are at most $W^{P_*}W^{-A_*-1} \leq W^{-A}$.

Only after these floor choices, fix each global norming and interpolation event's failure probability at W^{-D-2} for the desired final exponent D. Invoke (5.34) at this fixed failure exponent and the required raw accuracy, and choose polynomial time meshes using (5.33). For example, a weighted n-loop normalization costs at most $C_n W^{F+n}$ because $\zeta_u \geq c/W$; a raw error $W^{-Q-F-n-2}$ is therefore sufficient for normalized accuracy W^{-Q-1}. Vertical comparisons use the closed (7.25); its coarse envelope exponent is fixed before its height mesh and moments. The Duhamel integrals are exact, and interpolation compares a point with its nearest or left mesh point and takes a supremum. No mesh cardinality multiplies a deterministic tail error. Mesh sizes enter only the probability union.

Let the combined spatial, orbital and time or height meshes at all nodes have at most $W^{Q_{\mathrm{mesh}}}$ points. Their cardinality may depend on the already fixed global failure exponent $D+2$. The union of those global exceptional events has probability $O(W^{-D-2})$. At the resulting fixed points, require probability $W^{-D-Q_{\mathrm{mesh}}-2}$ only for the pointwise Gaussian, Haar and stopped-martingale estimates. Their union costs another $O(W^{-D-2})$, so the combined failure is at most W^{-D} for large W. Do not request the global sampling or interpolation input again with this larger pointwise failure exponent. Choose the BDG and Markov moment orders for these pointwise estimates last: a threshold gap W^{ρ} is handled by any fixed order q with $q\rho > D + Q_{\mathrm{mesh}} + 2$. Increasing this moment order does not enlarge the sharp-loop cone. For a target of length n, the joined bracket word has the fixed length $2n+2$; its deterministic envelope is imposed by an adapted stop before BDG is applied. Higher moments are then taken of that stopped martingale with its already bounded bracket. Only the positive-loop order p selected for the operator loss changes the required loop lengths, and it was fixed before constructing the cone. Brownian interpolation events are used after these adapted estimates; no moment estimate is conditioned on a future interpolation event.

Assign this finite list of starting lengths, floors, losses and failure accuracies to the preceding time step, and repeat backwards at most J times. Each individual step has finitely many

improvement nodes, and each comparison remains in the domains already verified. This backward construction never requires one finite list to be closed under repeated enlargement at the same time level. Its constants and orders do not alter the deterministic schedule. Finally choose the accuracy-dependent deterministic constants; the divergent volume assumptions accommodate every one for sufficiently large W. The bounded initial estimate (5.23) in Lemma 5.3, or its large-coupling version in Lemma 5.6, supplies all these fixed-order inputs at $s_0 = 1/2$, uniformly at the buffered pairs.

Apply part (i) forward over the chosen distinct intervals. The preceding step provides the next starting packet and the operator/Schatten inputs proved in Step 5, also at every comparison pair. The profile exponent starts at $1/2$ and is halved at most J times, so one may take $\theta_* = 2^{-J-1} > 0$, with the minimum of the finitely many positive profile constants. The time suprema are already inside the probability on each interval; their finite union gives the full interval $[1/2, t_*]$. For the original coupling $g = g_0$, the domain $u\gamma_{g_0} \le g_0^{-2}$ imposes no additional restriction. This proves (5.42) with its stated uniformity. □

D Orbital spreading and the rough density law

The rough density law at the end of this appendix is used in both parts of Theorem 1.4.

D.1 Common unitary conjugation

Proof of (2.3) ***and*** (2.2). Let U be independent Haar unitary. The GUE law and scalar hopping give $(I_L \otimes U)H(I_L \otimes U^*) \overset{\text{law}}{=} H$. Conditional on H, rotate all eigenvectors by $I_L \otimes U$; each normalized nonzero block is uniform on the unit sphere of $\mathbb{C}^W$. For a uniform unit vector $v \in \mathbb{C}^W$, the coordinate tail $\mathbb{P}(|v_j|^2 > s) = (1-s)_+^{W-1}$ for $s \ge 0$, and a union bound over the N^2 eigenvector-coordinate pairs, give with probability at least $1 - N^{-D}$

$$|\psi_k(i)|^2 \le C_D W^{-1} \log N \, \|\psi_{k,a}\|_2^2, \qquad i \in \mathcal{I}_a, \quad a \in \mathbb{Z}_L, \quad 1 \le k \le N. \tag{D.1}$$

The half-mass definition immediately gives (2.3).

For (2.2), fix a normalized eigenvector and let $r = \lceil c_0 W \rceil < W$ with c_0 sufficiently small. A coordinate interval of length r meets at most two blocks. If it carries half the mass, it contains mass at least $1/4$ from one of the at most four blocks whose total mass is at least $1/4$. For a fixed set I of r coordinates, the $\mathrm{Beta}(r, W-r)$ tail gives

$$\mathbb{P}\left(\sum_{j \in I} |v_j|^2 \ge 1/4 \right) \le e^{-cW}.$$

A union bound over at most W cyclic consecutive coordinate sets in each of the four blocks bounds the conditional probability of $\ell(\psi_k) < c_0 W$ by $4We^{-cW}$. Averaging over the WL eigenvectors and then over H proves (2.2). □

D.2 The rough density law

Proof of (1.15). Write $\gamma = \gamma_g$, so $H = A_g \otimes I_W + \sqrt{\gamma} V$, with $0 < \gamma \le 1$ and $\|A_g\| \le 2$. For $z = E + i\eta$, $0 < \eta \le 1$, translation invariance gives $\bar{m} := \mathbb{E}\langle G(z)\rangle_{aa} = N^{-1}\mathbb{E}\,\mathrm{Tr}\, G(z)$. Gaussian Poincaré and integration by parts yield

$$\mathbb{E}|\langle G\rangle_{aa} - \bar{m}|^2 \le C\gamma W^{-2}\eta^{-4}, \tag{D.2}$$

and

$$(A_g \otimes I_W - z - \gamma\bar{m})\mathbb{E}G = I + \gamma\mathbb{E}[\mathrm{diag}((\langle G\rangle_{aa} - \bar{m})I_W)G]. \tag{D.3}$$

Cauchy–Schwarz bounds the error on the right in operator norm by $CW^{-1}\eta^{-3}$, independently of L, g. After inversion and normalized trace, the Fourier-sum estimate in Lemma A.17 gives

$$\bar{m} = f_g(z + \gamma\bar{m}; \gamma) + O\left(W^{-1}\eta^{-4} + L^{-1}\eta^{-2}\right). \tag{D.4}$$

For fixed η and sufficiently large W, L, the error is at most $\eta/2$, so the same lemma gives

$$|\bar{m} - m_g(z)| \le C\left(W^{-1}\eta^{-6} + L^{-1}\eta^{-4}\right). \tag{D.5}$$

Gaussian concentration gives

$$\mathbb{P}\left(\left|N^{-1}\operatorname{Tr} G(E + i\eta) - \bar{m}\right| > u\right) \le 4\exp(-cW^2L\eta^4u^2). \tag{D.6}$$

The Gaussian second-moment bound $\mathbb{P}(N^{-1}\operatorname{Tr} H^2 > 4) \le e^{-cW^2L}$ also gives empirical mass at most $4/R^2$ outside $[-R, R]$, with the same uniformity in g.

Fix $\delta > 0$. Choose $R > 4$ large, then $h > 0$ small and $\eta > 0$ with η/h small, so that

$$R^{-2} + \eta/h + \sup_{g,x} \nu_g([x - 2h, x + 2h]) \ll \delta,$$

using (1.13). On the second-moment event, Poisson smoothing reduces the distribution-function comparison to resolvent control at height η on $[-R-1, R+1]$. Apply (D.5) and (D.6) on a finite energy mesh; the resolvent derivative bound $2\eta^{-2}$ fills the gaps. Choose the mesh and accuracy in terms of δ, R, h, η, then take W, L sufficiently large. Removing the smoothing costs at most $C\eta/h$, the uniform interval mass above, and the tails. Thus the distribution functions differ by at most δ, with failure probability $C_\delta e^{-c_\delta W^2 L}$. All choices depend only on δ, which proves (1.15) uniformly in g. □

E Fractional moments and eigenvector concentration

We prove the fractional-moment conversion in Lemma 8.1 and the finite-volume criterion in Proposition 8.2.

E.1 From fractional moments to concentration

***Proof of Lemma 8.1*(i).** Define the block correlator $\mathcal{Q}_{ab}(J) := \sum_{\lambda_k \in J} \|\psi_{k,a}\|_2 \|\psi_{k,b}\|_2$. Since $g \le W^{C_0}$, the conditional densities and qth moments of the diagonal GUE entries of H are bounded by powers of W. The finite-volume spectral-averaging argument in [3, Appendix A, proof of Theorem A.1, (A.10)–(A.12)], applied to the integral over J of (8.2) and summed over the W^2 coordinate pairs in blocks a, b, gives

$$\mathbb{E}\mathcal{Q}_{ab}(J) \le W^C e^{-c\|a-b\|_L/\mathfrak{b}}, \qquad a \ne b. \tag{E.1}$$

Choose a maximal-mass block a_k for each eigenvector, and let $B_D(R)$ be the number counted in (3.6). Since $\|\psi_{k,b}\|_2^2 \le \|\psi_{k,a_k}\|_2 \|\psi_{k,b}\|_2$, summing (E.1) gives

$$B_D(R) \le W^D \sum_a \sum_{\|b-a\|_L > R} \mathcal{Q}_{ab}(J), \qquad \frac{\mathbb{E}B_D(R)}{WL} \le W^{D+C-1}\mathfrak{b}\, e^{-cR/\mathfrak{b}}. \tag{E.2}$$

As $\mathfrak{b} \le W^{C_0}$, taking $R = \lceil C_D \mathfrak{b} \log W \rceil$ with sufficiently large C_D proves (3.6). □

***Proof of Lemma 8.1*(ii).** To obtain (3.6), we first control the mass correlations between distant cells and then sum over their positions. Use the assumptions of part (ii). Since $g \ge W$, for sufficiently large W Lemma 7.2 applies on J with window length at most $2g^2$. Set $n = \lfloor g^2 \rfloor \asymp g^2$;

the constants and width thresholds below are uniform in g, L. For $L \geq 3n$, partition the cycle into consecutive cells of lengths in $[n, 2n]$. Write $n_A = |A|$ for each cell and let $d(A, B)$ be the minimum block distance between two cells. Put

$$m_A(k) = \|P_A \psi_k\|^2, \qquad Q(A, B) = \sum_{\lambda_k \in J} \sqrt{m_A(k) m_B(k)},$$

where P_A projects onto all coordinates in A. Our target is

$$\mathbb{E}Q(A, B) \leq nW^C e^{-cd(A,B)/(Wg^2)}. \tag{E.3}$$

For $A = B$, this follows from $Q(A, A) \leq n_A W \leq 2nW$. For $A \neq B$, set $t = q/4$ and

$$Q_t(A, B) = \sum_{\lambda_k \in J} m_A(k)^{1-t} m_B(k)^t.$$

Cauchy–Schwarz over eigenstates gives $Q(A, B)^2 \leq Q_t(A, B) Q_t(B, A)$, so it suffices to prove the same decay bound for $\mathbb{E}Q_t(A, B)$.

To express the mass ratio m_B/m_A through the resolvent on A^c, we first compare the boundary values of an eigenvector with its cell mass. For the transfer products in a cell A, number its blocks by $1, \ldots, n_A$ in their natural order, and put

$$F_j(E) = T_j(E) \cdots T_1(E), \qquad K_A(E) = \max_{0 \leq j \leq n_A} \max\{1, \|F_j(E)\|, \|F_j(E)^{-1}\|\}.$$

Here $T_j(E)$ is defined in Definition 7.1 and $F_0 = I$. Since $n_A \leq 2g^2$, Lemma 7.2 gives $\mathbb{E}K_A(E)^p \leq C_p$ for every fixed $p > 0$, uniformly on J. Fix a normalized eigenpair (λ, ψ), suppress its index in m_A, m_B, and evaluate K_A, K_B at λ. Let $\psi_{A,\text{left}}, \psi_{A,\text{right}}$ be the endpoint values on A, and x_-, x_+ their respective neighbors outside A, the endpoints of the Dirichlet path $\Lambda = A^c$.

The inverse-prefix bound, summed over the $n_A - 1$ interior pairs, controls the initial pair $(\psi_{A,\text{left}}, \psi_{x_-})$ by $CK_A\sqrt{m_A/n_A}$. Forward propagation on A, B then gives, with $b_0, b_0 + 1$ the first two blocks of B,

$$\begin{aligned} \|\psi_{A,\text{left}}\| + \|\psi_{A,\text{right}}\| &\leq CK_A^2 \sqrt{m_A/n_A}, \\ m_B &\leq Cn_B K_B^4 (\|\psi_{b_0}\|^2 + \|\psi_{b_0+1}\|^2). \end{aligned}$$

Almost surely, H and H_Λ have no common eigenvalue: condition on the exterior and all Gaussian directions in A except the common scalar shift. Every eigenvalue is strictly increasing in this shift, since a solution vanishing on two consecutive blocks is zero. Thus

$$\psi_b = -G_\Lambda(b, x_-; \lambda)\psi_{A,\text{left}} - G_\Lambda(b, x_+; \lambda)\psi_{A,\text{right}}.$$

Since $n_B/n_A \leq 2$,

$$\frac{m_B}{m_A} \leq CK_A^4 K_B^4 \sum_{b \in \{b_0, b_0+1\},\ x \in \{x_-, x_+\}} \|G_\Lambda(b, x; \lambda)\|^2. \tag{E.4}$$

For fixed A, B, define the exterior-measurable function

$$f(E, H_{A^c}) = K_B(E)^{4t} \sum_{b \in \{b_0, b_0+1\},\ x \in \{x_-, x_+\}} \|G_\Lambda(b, x; E)\|^{2t}.$$

Since $4t = q < 1$ and $K_A \geq 1$, (E.4) gives

$$Q_t(A, B) \leq C \sum_{\lambda_k \in J} m_A(k) K_A(\lambda_k) f(\lambda_k, H_{A^c}). \tag{E.5}$$

The input (8.3) controls an energy integral, whereas (E.5) is evaluated at the random eigenvalues. We therefore need the following averaging estimate, which retains the transfer weight and is uniform in the exterior configuration.

Claim (Weighted averaging over one cell). For each cell A and every nonnegative measurable $f(E, H_{A^c})$,

$$\mathbb{E}_A \sum_{\lambda_k \in J} m_A(k) K_A(\lambda_k) f(\lambda_k, H_{A^c}) \le C n_A W^2 \int_J f(E, H_{A^c})\, dE. \tag{E.6}$$

Here $\mathbb{E}_A$ averages over the blocks in A, with the exterior fixed. We use classical spectral averaging and monotone boundary phases; see [3, Appendix A] and [26, Section 4, Theorem 5 and Lemma 2]. The proof below keeps the transfer weight and the uniform constants needed on cells of length comparable to g^2.

Proof of the claim. Fix the exterior configuration. Since $\operatorname{Var}(\operatorname{Tr} V_j) = 1$, Gaussian orthogonal decomposition gives

$$u = n_A^{-1/2} \sum_{j \in A} \operatorname{Tr} V_j \sim N(0,1), \qquad V_j = V_j^\circ + \frac{u}{W\sqrt{n_A}} I_W,$$

where u is independent of the residual blocks $(V_j^\circ)_{j \in A}$. Write $\mathbb{E}_u$ for expectation over u at fixed residual blocks and $\mathbb{E}_\circ$ for their expectation. Their joint average $\mathbb{E}_A = \mathbb{E}_\circ \mathbb{E}_u$ restores the original independent GUE law on A. We have $H(u) = H^\circ + \alpha u P_A$, with $\alpha = (gW\sqrt{n_A})^{-1}$; $K_A(E,u)$ denotes the same transfer weight for these shifted blocks. By the two-block uniqueness used above, the compression of P_A to every eigenspace is positive definite. The Rellich branches satisfy $\lambda_k'(u) = \alpha m_A(k) > 0$, with this compression diagonalized at crossings. The scalar area formula, with standard Gaussian density φ, yields

$$\mathbb{E}_u \sum_{\lambda_k \in J} m_A(k) K_A(\lambda_k, u) f(\lambda_k, H_{A^c}) = \alpha^{-1} \int_J f(E, H_{A^c}) \sum_{u : E \in \sigma(H(u))} \varphi(u) K_A(E,u)\, dE. \tag{E.7}$$

In part (i), the conditional densities arising in this conversion from eigenvalue sums to energy integrals are bounded by powers of W because $g \le W^{C_0}$. Here cell averaging and normalization remove this coupling dependence. Here and below, each root u is counted with multiplicity $\dim\ker(H(u) - E)$. It suffices to prove, for almost every $E \in J$,

$$\mathbb{E}_\circ \sum_{u : E \in \sigma(H(u))} \varphi(u) K_A(E,u) \le CW. \tag{E.8}$$

Fix $E \notin \sigma(H_{A^c})$. For a solution of the cell recurrence put

$$v = (\psi_1, \psi_{n_A}), \quad w = (\psi_0, \psi_{n_A+1}), \qquad a = 2^{-1/2}(v + iw), \quad b = 2^{-1/2}(v - iw).$$

Discrete Green's identity gives $v^* \widetilde{w} - w^* \widetilde{v} = 0$ for any two cell solutions. Thus the boundary data form a Lagrangian space. The map from the $2W$ initial data to $a \in \mathbb{C}^{2W}$ is injective: $a = 0$ implies $b = 0$, hence $v = w = 0$ and then the entire recurrence vanishes. It is therefore invertible. We obtain a smooth unitary $b = C_A(E,u)a$ for every real u.

The solution depends linearly on a; let $\Psi_A : \mathbb{C}^{2W} \to \mathbb{C}^{n_A W}$ map a to $(\psi_1, \dots, \psi_{n_A})$. Since the initial pair (ψ_1, ψ_0) is part of (v, w) and $\|v\|^2 + \|w\|^2 = 2\|a\|^2$,

$$\|\Psi_A\|^2 \le 2 n_A K_A^2. \tag{E.9}$$

Differentiating the solution at fixed a and summing by parts gives

$$v^* \partial_u w - w^* \partial_u v = -\alpha \|\Psi_A a\|^2 .$$

Together with $a^*da - b^*db = i(v^*dw - w^*dv)$, this quadratic-form identity yields

$$-iC_A^*\partial_u C_A = \alpha\Psi_A^*\Psi_A > 0, \qquad \|-iC_A^*\partial_u C_A\| \le 2\alpha n_A K_A^2. \tag{E.10}$$

Strict positivity follows from the same two-block uniqueness property.

Eliminating the fixed exterior gives a unitary $C_{\mathrm{ext}}(E)$, independent of u, in the same boundary coordinates. The full spectral equation is $1 \in \sigma(C_{\mathrm{ext}}^* C_A(E,u))$. The positive generator in (E.10) makes all lifted eigenphases strictly increasing, also at multiple phases by compression to their eigenspaces. Let $N_I(E)$ count the roots in a unit interval I, with multiplicity. The sum of the eigenphase increments equals the integral of the trace of the generator, so

$$\begin{aligned} N_I(E) &\le 2W + \frac{1}{2\pi}\int_I \mathrm{Tr}(-iC_A^*\partial_u C_A)\,du \\ &\le 2W + C\alpha n_A W \sup_I K_A^2 \le CW(1+\sup_I K_A^2). \end{aligned} \tag{E.11}$$

Shared endpoints of adjacent intervals only overcount roots; the finite set of excluded energies does not affect (E.7).

It remains to average weighted suprema. Product differentiation gives

$$\partial_u F_j = \sum_{k=1}^{j} F_j F_k^{-1}(\partial_u T_k)F_{k-1}, \qquad \|\partial_u T_k\| = \alpha.$$

Using also the derivative of the inverse, the finite maximum K_A is locally Lipschitz and satisfies almost everywhere

$$|\partial_u K_A| \le C\alpha n_A K_A^5.$$

Since $\alpha n_A \le C/W$, applying $\sup_I z \le \int_I z + \int_I |z'|$ to $z = K_A^3$ on $I_\ell = [\ell, \ell+1]$ gives

$$\sup_{I_\ell} K_A^3 \le C\int_{I_\ell} K_A^7\,du. \tag{E.12}$$

For $u \in I_\ell$, $\sup_{I_\ell}\varphi/\varphi(u) \le e^{|u|+C}$. Using (E.12), Cauchy–Schwarz, and Lemma 7.2,

$$\begin{aligned} \mathbb{E}_\circ \sum_\ell (\sup_{I_\ell}\varphi)(\sup_{I_\ell} K_A^3) &\le C\mathbb{E}_A[e^{|u|}K_A^7] \\ &\le C(\mathbb{E}_A e^{2|u|})^{1/2}(\mathbb{E}_A K_A^{14})^{1/2} \le C. \end{aligned} \tag{E.13}$$

Since $K_A \ge 1$, (E.11) and (E.13) give (E.8). Finally, $\alpha^{-1} = gW\sqrt{n_A} \le \sqrt{2}n_A W$ proves (E.6). □

The averaging cost before using $n_A \asymp g^2$ is $CgW^2\sqrt{n_A}$. After normalization by the cell dimension it is

$$\frac{CgW^2\sqrt{n_A}}{n_A W} = \frac{CgW}{\sqrt{n_A}} \asymp CW.$$

We now apply (E.6) to the function f in (E.5). Averaging the exterior and using Cauchy–Schwarz jointly in energy and the exterior variables gives

$$\begin{aligned} \mathbb{E}Q_t(A,B) &\le CnW^2 \sum_{b,x}\int_J \mathbb{E}[K_B(E)^{4t}\,\|G_\Lambda(b,x;E)\|^{2t}]\,dE \\ &\le nW^C e^{-cd(A,B)/(Wg^2)}. \end{aligned}$$

The second inequality uses the $8t$th transfer moment from Lemma 7.2, the $4t = q$ moment in (8.3), and that the relevant path distances are at least $d(A,B) - 1$. Interchanging A, B and applying Cauchy–Schwarz in probability to $Q(A,B)^2 \le Q_t(A,B)Q_t(B,A)$ proves (E.3).

Exceptional proportions. If the proposed block radius fills the cycle, the tail is empty; otherwise $L \geq 3n$ and we use the cell partition above. For each eigenvector choose a maximal-mass cell A_k. Since $m_B(k) \leq \sqrt{m_{A_k}(k) m_B(k)}$, the number of eigenvectors in J with $\sum_{B:d(A_k,B)>r} m_B(k) > W^{-D}$ is at most $W^D \sum_A \sum_{B:d(A,B)>r} \sum_{\lambda_k \in J} \sqrt{m_A(k) m_B(k)}$. Using $O(L/n)$ cells, (E.3), and the geometric tail sum gives a normalized exceptional proportion at most

$$CW^D \frac{L/n}{WL}\, nW^C \frac{\mathfrak{b}}{n} e^{-c'r/\mathfrak{b}} \leq W^{D+C} e^{-c'r/\mathfrak{b}},$$

since $\mathfrak{b} = Wg^2$ and $\mathfrak{b}/n \leq CW$. Choose $r = C_D \mathfrak{b} \log W$ with $c' C_D > 2D + C + 1$. Every cell at distance at most r from the maximal-mass cell lies in a block neighborhood of radius at most $r + 4n \leq C_D W g^2 \log W$, after increasing C_D, around a block of the maximal-mass cell. This proves (3.6) and the half-mass bound with the cap WL. □

E.2 Finite-volume criterion for fractional-moment decay

Proof of Proposition 8.2. Fix $E \in J$ and the length n from (8.6). For a path $\Gamma = [l, b]$, $x < y$ with $y - x > n$, put $\Lambda = [x, r]$, $r = x + n$. Conditional on the exterior of Λ, the Schur complement gives

$$G_\Gamma(x, y; E) = -\frac{g}{\max\{1, g\}} G_\Lambda^{L_0, R_0}(x, r; E) G_{[r+1,b]}(r + 1, y; E),$$

where L_0, R_0 are Hermitian shifts determined by the exterior. The last factor is fixed under this conditional average, so (8.6) yields

$$\mathbb{E}[\|G_\Gamma(x, y; E)\|^s \mid (V_i)_{i \notin \Lambda}] \leq W^{-2} \|G_{[r+1,b]}(r + 1, y; E)\|^s.$$

No independence between R_0 and the last factor is needed. Iterating as in [3, proof of Theorem 1.1, (2.23)–(2.24)], using (8.7) on the remainder and $2 \log W/(n + 1) \geq c/\mathfrak{b}$, gives

$$\mathbb{E}\|G_\Gamma(x, y; E)\|^s \leq W^C e^{-c|x-y|/\mathfrak{b}}, \qquad E \in J, \quad x, y \in \Gamma. \tag{E.14}$$

The constants are independent of the interval length; adjunction handles $x > y$, and (8.7) handles $x = y$.

On a cycle, deleting block a leaves a Dirichlet path with resolvent $G^{(a)} = (H_{\mathbb{Z}_L \setminus \{a\}} - E)^{-1}$. The Schur identity is

$$G_{ab} = -\frac{g}{\max\{1, g\}} G_{aa} (G^{(a)}_{a+1,b} + G^{(a)}_{a-1,b}), \qquad a \neq b. \tag{E.15}$$

Both path distances are at least $\|a - b\|_L - 1$. Cauchy–Schwarz, (8.7) and (E.14), with $q = s/2$, give

$$\mathbb{E}\|G_{ab}(E)\|^q \leq W^C e^{-c\|a-b\|_L/\mathfrak{b}}, \qquad a \neq b, \quad E \in J. \tag{E.16}$$

Jensen's inequality lowers the exponent in (E.14) to $s/2$; (8.7) handles the diagonal. This proves (8.2) on paths and cycles. □

F Circle identities and operator estimates

We use the form J_{sig} and group $\mathcal{G}$ from Definition 8.7, and the volume ratio c from (8.17). The operator iteration is given in Section 8.3.

F.1 The quotient circle and its exact Jacobian

Lemma F.1 (Signature quotient)**.** *Let $U \subset \mathbb{C}^{2W}$ be a deterministic subspace of dimension $W-1$ on which the form $x^* J_{\mathrm{sig}} y$ vanishes. Then*

$$H_U := (U + J_{\mathrm{sig}} U)^{\perp}$$

is J_{sig}-invariant, has complex dimension two, and has signature $(1,1)$. Choose unit vectors $e, f \in H_U$ with $J_{\mathrm{sig}} e = e$ and $J_{\mathrm{sig}} f = -f$. Then the extensions are $Y = U \oplus \mathbb{C} v_\theta$, where

$$v_\theta := (e + e^{i\theta} f)/\sqrt{2}, \qquad dm_U = d\theta/(2\pi). \tag{F.1}$$

For $\mathcal{T} \in \mathcal{G}$, set

$$A_{\mathcal{T}} : H_U \longrightarrow H_{\mathcal{T}U}, \qquad A_{\mathcal{T}} v := \mathrm{proj}_{H_{\mathcal{T}U}} \mathcal{T} v.$$

This is the induced J_{sig}-unitary quotient isomorphism. For the volume ratio $c(\mathcal{T}; U, Y)$ in (8.17), the angular Jacobian is (8.36); equivalently, $e^{-2c(\mathcal{T};U,Y)} = \|A_{\mathcal{T}} v_\theta\|^{-2}$. The function c is a cocycle, and

$$|c(\mathcal{T}; U, Y)| \le \log \|\mathcal{T}\|. \tag{F.2}$$

Proof of (8.36). Isotropy makes $U \perp J_{\mathrm{sig}} U$. Their sum is J_{sig}-invariant and has signature $(W-1, W-1)$, which proves the claims concerning H_U. Moreover $U^{\perp_{J_{\mathrm{sig}}}} = U \oplus H_U$. If $v \in H_U$, then $\mathcal{T} v \in (\mathcal{T}U)^{\perp_{J_{\mathrm{sig}}}} = \mathcal{T}U \oplus H_{\mathcal{T}U}$. The removed component lies in the isotropic space $\mathcal{T}U$ and is J_{sig}-orthogonal to all of $(\mathcal{T}U)^{\perp_{J_{\mathrm{sig}}}}$. Thus the projection preserves the quotient form. The quotient map for $\mathcal{T}^{-1}$ is its inverse; their norms are bounded by $\|\mathcal{T}\|$ and $\|\mathcal{T}^{-1}\|$, respectively. Since $\mathcal{T}^{-1} = J_{\mathrm{sig}} \mathcal{T}^* J_{\mathrm{sig}}$, we have $\|\mathcal{T}^{-1}\| = \|\mathcal{T}\|$, hence $\|\mathcal{T}\|^{-1} \le \|A_{\mathcal{T}} v_\theta\| \le \|\mathcal{T}\|$.

Choose unit J_{sig}-eigenvectors e', f' of $H_{\mathcal{T}U}$ with eigenvalues $+1, -1$. In the bases e, f of H_U and e', f' of $H_{\mathcal{T}U}$, write $A_{\mathcal{T}} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. On $|z| = 1$ its action is $z' = (c + dz)/(a + bz)$. Since $A_{\mathcal{T}}$ preserves $\mathrm{diag}(1, -1)$, the two output components of $A_{\mathcal{T}}(1, z)^t/\sqrt{2}$ have equal modulus and $\|A_{\mathcal{T}} v_\theta\|^2 = |a + bz|^2$. Differentiating the fraction, and using the form-preserving identities and $|\det A_{\mathcal{T}}| = 1$, gives $d\theta'/d\theta = |a + bz|^{-2} > 0$. Finally Gram–Schmidt on $\mathcal{T}Y$, beginning with an orthonormal basis of $\mathcal{T}U$, identifies $\|A_{\mathcal{T}} v_\theta\|$ with the last volume ratio. The cocycle identity is the multiplicativity of restricted volumes, and the norm bounds on the quotient and its inverse give (F.2). □

The reciprocal convention is useful to record:

$$\frac{d(\mathcal{T}_* m_U)}{dm_{\mathcal{T}U}}(\mathcal{T}Y) = e^{2c(\mathcal{T};U,Y)}.$$

This circle has real dimension one. The exponent two in (8.36) must not be doubled because the ambient vector spaces are complex.

F.2 The inverse matrix and the last-volume ratios

Lemma F.2 (Omitted-column identity)**.** *Let $\mathcal{T} \in \mathcal{G}$ for the group (8.24), fix $0 < s < 1$, and let $\Phi \in \mathcal{F}_W$ from (8.8), $Y := \mathrm{ran}\, \Phi$, and U_j the span of all its columns except the j-th. Set $D := \sqrt{2} \begin{pmatrix} I_W & 0 \end{pmatrix} \mathcal{T} \Phi$. Then*

$$e^{-c(\mathcal{T};U_j,Y)} = \sqrt{[(D^* D)^{-1}]_{jj}} = \|e_j^* D^{-1}\|. \tag{F.3}$$

Consequently

$$\|D^{-1}\|^s \le \sum_{j=1}^{W} e^{-sc(\mathcal{T};U_j,Y)}. \tag{F.4}$$

***Proof of Lemma F.2 and* (8.23).** The identity $(\mathcal{T}\Phi)^* J_{\mathrm{sig}}(\mathcal{T}\Phi) = 0$ equates the Gram matrices of the upper and lower blocks, including cross terms. Thus $(\mathcal{T}\Phi)^*(\mathcal{T}\Phi) = D^*D > 0$, which explains the factor $\sqrt{2}$ and proves that D is invertible. The squared restricted W-volume is $\det(D^*D)$, and the squared $(W-1)$-volume is its j-th principal minor. Their quotient is the reciprocal of the j-th diagonal entry of $(D^*D)^{-1}$. Also $(D^*D)^{-1} = D^{-1}D^{-*}$. Finally, $\|D^{-1}\|^s \le \|D^{-1}\|_{\mathrm{HS}}^s \le \sum_j \|e_j^* D^{-1}\|^s$, by concavity for $s/2 < 1$. □

F.3 Bounds for the good and bad operators

The following estimates apply to any smooth good component with the stated support and derivative bounds. Use the Lie algebra (10.1). For a smooth density on $\mathcal{G}$, use the right derivative [10, (3)]

$$X_M f(\mathcal{T}) := \frac{d}{dt} f(\mathcal{T} e^{tM})\Big|_{t=0}, \qquad \|X_M f\|_1 := \int |X_M f(\mathcal{T})|\, d\mathcal{T}, \qquad M \in \mathfrak{g}. \tag{F.5}$$

Here $d\mathcal{T}$ is Haar measure on $\mathcal{G}$. We use the real Hilbert–Schmidt inner product $\langle A, B\rangle = \Re \operatorname{Tr}(A^*B)$ on $\mathfrak{g}$.

Lemma F.3 (Bounds for the good and bad operators)**.** *Use the group and operators defined in Section 8.3, the Lie algebra in (10.1), and the right derivatives in (F.5). Let $\mu = \widetilde{\pi} f\, d\mathcal{T} + \mu_{\mathrm{bad}}$ be a probability measure, where $0 < \widetilde{\pi} \le 1$, f is a smooth compactly supported probability density, and μ_{bad} is a positive measure satisfying $\int \|\mathcal{T}\|^{1/2}\, d\mu_{\mathrm{bad}}(\mathcal{T}) \le 1/2$. Set*

$$K := \sup_{\mathcal{T} \in \operatorname{supp} f} \max\{\|\mathcal{T}\|, \|\mathcal{T}^{-1}\|\}.$$

There is an absolute constant $C < \infty$ such that every bounded measurable h satisfies

$$\|G_{1/2} h\|_\infty \le CK \left(1 + \sup_{\substack{M \in \mathfrak{g} \\ \|M\|_{\mathrm{HS}} = 1}} \|X_M f\|_1\right)^{1/4} \times \sup_U \|h(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)}, \tag{F.6}$$

$$\|B_{1/2} h\|_\infty \le \tfrac{1}{2} \|h\|_\infty. \tag{F.7}$$

Proof of Lemma F.3. From angular L^4 to a fixed flag. Fix U, Y and take the compact Lie direction K_U which is $\frac{1}{2}\operatorname{diag}(-i, i)$ in an orthonormal J_{sig}-eigenbasis of H_U, ordered $+1, -1$, and zero on $U + J_{\mathrm{sig}} U$. It is skew-Hermitian, has bounded Hilbert–Schmidt norm, and rotates the source circle with unit angular speed while fixing its base. For a function ψ, measurable in the target base and smooth on each circle,

$$X_{K_U} \psi(\mathcal{T}U, \mathcal{T}Y) = e^{-2c(\mathcal{T};U,Y)} \partial_\theta \psi(\mathcal{T}U, \mathcal{T}Y).$$

On the support of f, Lemma F.1 gives

$$e^{2c(\mathcal{T};U,Y)} \le K^2, \qquad |X_{K_U}(e^{2c(\mathcal{T};U,Y)})| \le CK^2.$$

Indeed, $e^{2c(\mathcal{T};U,Y)} = \|A_{\mathcal{T}} v_\theta\|^2$, and differentiation of the source unit vector gives the second bound from $\|A_{\mathcal{T}}\| \le K$. Haar integration by parts therefore yields

$$\begin{aligned} \left| \int \partial_\theta \psi(\mathcal{T}U, \mathcal{T}Y) f(\mathcal{T})\, d\mathcal{T} \right| &\le \|\psi\|_\infty \|X_{K_U}(e^{2c(\mathcal{T};U,Y)} f(\mathcal{T}))\|_1 \\ &\le CK^2 \left(1 + \sup_{\substack{M \in \mathfrak{g} \\ \|M\|_{\mathrm{HS}} = 1}} \|X_M f\|_1\right) \|\psi\|_\infty. \end{aligned} \tag{F.8}$$

This step uses the derivative bound over all Lie directions, since K_U is skew-Hermitian.

Disintegrate the output law of $(\mathcal{T}U, \mathcal{T}Y)$ under $f(\mathcal{T})\,d\mathcal{T}$ over its base marginal $\bar{\nu}(dV)$. Equation (F.8) bounds the vertical distributional derivative. Choose a countable dense family of smooth circle tests with supremum norm at most one. In (F.8), choose measurably at each base the largest signed value among the first finitely many tests, then pass to the limit. This bounds the integral of the conditional total variations, so the conditional circle laws have BV densities k_V with

$$\int \|\partial_\theta k_V\|_{\mathrm{TV}}\, d\bar{\nu}(V) \le CK^2 \left(1 + \sup_{\substack{M\in\mathfrak{g}\\ \|M\|_{\mathrm{HS}}=1}} \|X_M f\|_1\right).$$

For clarity, a circle measure with a finite distributional derivative has a BV density: convolution by circle mollifiers gives densities with uniformly bounded variation, and BV compactness gives the claim on almost every conditional circle. Since $\int k_V\, dm_V = 1$,

$$\int \|k_V\|_\infty\, d\bar{\nu}(V) \le CK^2 \left(1 + \sup_{\substack{M\in\mathfrak{g}\\ \|M\|_{\mathrm{HS}}=1}} \|X_M f\|_1\right).$$

The weight $e^{-c(\mathcal{T};U,Y)/2}$ is at most $K^{1/2}$. Hölder's inequality on each circle and then concavity give

$$\begin{aligned}
|(G_{1/2}h)(U,Y)| &\le K^{1/2} \iint |h(V,Z)| k_V(Z)\, dm_V(Z)\, d\bar{\nu}(V)\\
&\le K^{1/2} \sup_V \|h(V,\cdot)\|_{L^4(\mathcal{C}_V, dm_V)} \int \|k_V\|_{4/3}\, d\bar{\nu}(V)\\
&\le K^{1/2} \sup_V \|h(V,\cdot)\|_{L^4(\mathcal{C}_V, dm_V)} \left(\int \|k_V\|_\infty\, d\bar{\nu}(V)\right)^{1/4}.
\end{aligned}$$

Here $\widetilde{\pi} \le 1$ was discarded and $\|k_V\|_{4/3} \le \|k_V\|_\infty^{1/4}$. This proves (F.6), uniformly for the fixed source flag. No density estimate on the base marginal is required.

The bad component. Finally, $|c(\mathcal{T};U,Y)| \le \log\|\mathcal{T}\|$ gives

$$|(B_{1/2}h)(U,Y)| \le \|h\|_\infty \int \|\mathcal{T}\|^{1/2}\, d\mu_{\mathrm{bad}}(\mathcal{T}) \le \tfrac{1}{2}\|h\|_\infty,$$

which is (F.7). □

G GUE estimates and the common density

The relative local law, preparation estimates and Riccati product bounds are used in Section 9.4. The deterministic Dyson profile is given by Lemma A.21 in Appendix A.16. Structured Gaussian coercivity then supplies the cutoff for the common density, whose construction proves Lemma 8.9. The proof is completed by the $G_{1/2}, B_{1/2}$ estimates of Appendix F.3.

G.1 The relative local law and preparation of the incoming flag

Lemma G.1 (Local law for a deformed GUE)**.** *Fix $\kappa_0 \in (0,2)$, $C_*, K > 0$, and sufficiently small fixed $\varepsilon > 0$. Let $B = B_{\mathrm{small}} + B_{\mathrm{bad}}$ be Hermitian with $\|B_{\mathrm{small}}\| \le \varepsilon$, $\operatorname{rank} B_{\mathrm{bad}} \le \varepsilon W$ and $\|B\| \le W^K$. Let V be normalized complex GUE, and set*

$$G := (V - EI_W - B - i\eta)^{-1}, \qquad M := M_B(i\eta), \qquad m := W^{-1}\operatorname{Tr} M,$$

where M_B is the Herglotz solution of Lemma A.21 with $\Im m > 0$. For every fixed $\delta > 0$, uniformly for $|E| \le 2 - \kappa_0$ and $W^{-1+\delta} \le \eta \le C_*$,

$$\left|W^{-1} \operatorname{Tr} G - m\right| \prec W^{-1}\eta^{-1}, \qquad |u^*(G-M)v| \prec (W\eta)^{-1/2} \|Mu\|_2 \|Mv\|_2 \tag{G.1}$$

for deterministic vectors u, v. In particular, for each deterministic $v \neq 0$,

$$\left|\frac{\Im(v^*Gv)}{v^*\Im Mv} - 1\right| \prec (W\eta)^{-1/2}, \qquad \Im(v^*Gv) \asymp v^*(I_W + B^2)^{-1}v \tag{G.2}$$

with the comparison holding with arbitrarily high polynomial probability. The constants and probability bounds are uniform over the deterministic parameters just specified; the event need not be common to all vectors.

For the isotropic and anisotropic local-law framework see [19] and [20, Section 12, Theorem 12.2].

Proof of (G.1) ***and*** (G.2). We use the averaged estimate in [18, Theorem 1.9], with deformation $-EI_W - B$ and spectral parameter $i\eta$. Its bounded-deformation assumption can be relaxed as explained in [18, Remark 1.11]: Theorem 1.5 there has no such assumption. The deterministic inputs to its Section 4 proof are (A.81), (A.82), and (A.84). The local stability bound (A.83) verifies [18, Definition 4.9], uniformly along the height interval. The norm and density bounds in (A.84) replace the uses of a bounded deformation in its Lemmas 4.1–4.2. In its Lemma 4.6, $\|B\| \le W^K$ only requires a finer polynomial mesh for the resolvent-continuity estimates. The proof of its Theorem 1.9 therefore gives the first estimate in (G.1), with uniform constants. The complex case is covered by [18, Appendix B].

We next retain the vector weights, using the averaged estimate in (G.1) and the Gaussian structure. By unitary invariance we may diagonalize B. Let $G^{(i)}$ be the resolvent with row and column i deleted, and let $h_i := (V_{ki})_{k\neq i}$. Schur's formula gives

$$G_{ii}^{-1} = -E - B_{ii} - i\eta + V_{ii} - h_i^* G^{(i)} h_i.$$

Conditional Gaussian quadratic-form estimates and Ward's identity give

$$\left|h_i^* G^{(i)} h_i - W^{-1} \operatorname{Tr} G^{(i)}\right| \prec \left(W^{-2}\eta^{-1}\Im \operatorname{Tr} G^{(i)}\right)^{1/2} \prec (W\eta)^{-1/2}.$$

Here the averaged estimate in (G.1) and the interlacing bound $|\operatorname{Tr} G - \operatorname{Tr} G^{(i)}| \le C\eta^{-1}$ control the trace on the right. Consequently,

$$G_{ii}^{-1} = M_{ii}^{-1} + O_\prec((W\eta)^{-1/2}), \qquad |G_{ii} - M_{ii}| \prec (W\eta)^{-1/2}|M_{ii}|^2. \tag{G.3}$$

Since $|M_{ii}| \le C$, inversion is justified uniformly, even when $|M_{ii}|$ is small. The same calculation for $G_{jj}^{(i)}$, using the resolvent with i, j deleted, gives $\Im G_{jj}^{(i)} \prec |M_{jj}|^2$; deleting two indices changes $W^{-1}\operatorname{Tr} G$ by $O((W\eta)^{-1})$. For $i \neq j$, the off-diagonal Schur formula and conditional Gaussian linear-form estimate now yield

$$G_{ij} = -G_{ii}\sum_{k\neq i} V_{ik} G_{kj}^{(i)}, \qquad |G_{ij}| \prec |M_{ii}|\left(W^{-1}\eta^{-1}\Im G_{jj}^{(i)}\right)^{1/2} \prec (W\eta)^{-1/2}|M_{ii}M_{jj}|. \tag{G.4}$$

To pass to deterministic vectors, introduce independent uniform unit complex numbers ω_i. With $\Omega := \operatorname{diag}(\omega_1, \dots, \omega_W)$, the pair (G, M) has the same distribution as $(\Omega G \Omega^*, M)$. For every fixed integer $p \ge 1$ and every array (c_{ij}), the usual decoupling and Khintchine inequalities give

$$\mathbb{E}_\omega \left|\sum_{i\neq j} c_{ij}\omega_i\overline{\omega_j}\right|^{2p} \le C_p \left(\sum_{i\neq j}|c_{ij}|^2\right)^p.$$

Apply this conditionally on G, with $c_{ij} = \overline{u_i} G_{ij} v_j$. Equation (G.4) bounds the sum of squares by $(W\eta)^{-1} \|Mu\|_2^2 \|Mv\|_2^2$, in stochastic domination. By (G.3), the diagonal contribution is bounded by $(W\eta)^{-1/2} \sum_i |u_i v_i| |M_{ii}|^2$, again in stochastic domination. Cauchy–Schwarz and arbitrarily high moments prove the second estimate in (G.1). Finally, (A.84) converts the weighted estimate to (G.2). □

G.1.1 Preparation of the incoming flag

Proof of Lemma 9.1. To justify chart entry for every incoming flag, write

$$SY = \operatorname{ran} \begin{pmatrix} A \\ B \end{pmatrix}, \qquad A^*B = B^*A, \qquad \operatorname{rank} \begin{pmatrix} A \\ B \end{pmatrix} = W.$$

Isotropy and full rank imply that $iA - B$ has trivial kernel, so $\det(zA - B)$ is not the zero polynomial. The first upper block is therefore invertible almost surely under the fresh GUE perturbation. The second block gives an inverse Riccati state, followed by two further steps; in particular,

$$R_4 = (V_4 - EI - g^2 R_3)^{-1}.$$

The line spanned by $R_4^{-1} v_4$ is fixed before V_4 is drawn, and v_4 is determined before V_5. The one-block bounds (9.58) give the inverse-norm and eigenvalue-count bounds. In particular, taking $\delta = 2$ there and applying Markov's inequality to the count on $[-g, g]$ gives regularity (9.3) with failure $O(W^{-2} + g^{1/2})$ for each outgoing state, uniformly in the past. In regime (i), their total failure is $O(W^{-2} + W^{-\alpha/2} + W^{-\varepsilon/800})$. In regime (ii), the inverse-norm failure is $O(W^{-\theta})$, and the count has mean at most $CWg^2W^{\theta} \le CW^{2\theta}$, so Markov's inequality gives failure $O(W^{-\theta})$. The last bound in (9.7) follows from $g^2 \le W^{-1+\theta}$.

It remains to compare $\|Pv_4\|^2$ with b_4 in regime (i). On a regular past the deformation has polynomial norm, while its small part and exceptional rank fraction tend to zero uniformly. Since $Wg^2 \ge W^{1-2\beta}$, Lemma G.1 applies at cg^2, g^2, Cg^2 for fixed positive c, C, with common constants and any prescribed failure power. The tested line is fixed by conditioning on the past; no common event over all vectors is needed.

Set

$$u := R_4^{-1} v_4, \qquad b_4 = g^{-2} \Im \left[u^* (R_4^{-1} - ig^2)^{-1} u \right].$$

Its line is fixed before V_4, so the homogeneous local law applies despite its random length. By (6.18), the Poisson integrals at the three scales are comparable to $u^*(I + g^4 R_3^2)^{-1} u$. In the integral at g^2, the contribution from $|\lambda| \le cg^2$ is at most $2c$ times the integral at cg^2, and that from $|\lambda| \ge Cg^2$ is at most $2/C$ times the integral at Cg^2. Taking c small and C large leaves a fixed fraction on the annulus between them, which is retained by P for large W. Comparing the inverse-square and Poisson kernels there gives

$$\|Pv_4\|^2 \ge c' g^{-2} u^* (I + g^4 R_3^2)^{-1} u \ge c'' b_4.$$

This proves $b_4 \le C\|Pv_4\|^2$. On the regular past, $(I + g^4 R_3^2)^{-1} \ge cW^{-6} I_W$, and $\|R_4^{-1} v_4\| \ge W^{-3} \|v_4\|$ since $\|R_4\| \le W^3$. Thus $\|Pv_4\|^2 \ge cW^{-12} \|v_4\|^2$, which gives the weaker stated $\|Pv_4\| \ge W^{-10} \|v_4\|$ at a common threshold.

□

G.2 Gaussian and spectral inputs for inverse images

G.2.1 Gaussian estimates for the dependent projection

We verify (9.45)–(9.47) under the assumptions of Lemma 9.6. Use the notation from its proof,

$$Y_\perp = (I - P_F) R Y, \qquad E_> = \operatorname{ran} \mathbf{1}_{\{|R| \ge \eta^{-1}\}}.$$

Proof of (9.45). The compression of RY to $E_> \cap F^\perp$ has row covariance at least $(W\eta^2)^{-1}I$ and at least $C_0 d - d$ rows. Factoring its covariance reduces (9.45) to the rectangular Gaussian least-singular-value bound. For a Gaussian matrix with n rows and r columns, the rectangular bound here only needs n to exceed a sufficiently large constant times r. For a fixed unit column vector, the squared Gaussian image norm is a scaled sum of independent exponential variables, with lower and upper tails e^{-cn}. A fixed fine net in the column sphere has $e^{O(r)}$ points. Its upper bound controls the operator norm, and then its lower bound extends to the sphere. With $n \geq Cr$ this proves the asserted least singular value and exponential failure. A general row covariance bounded below is factored out before this argument.

Proof of (9.46). The bounds in (9.46) follow from a constant net: the expected norm in a fixed column direction is at most $\|R_<\|_{\mathrm{HS}}/\sqrt{W}$, and the Gaussian Lipschitz deviation is $\|R_<\|/\sqrt{W}$; use a deviation of order $\sqrt{d}$. The unweighted projection bounds are the same calculation.

Proof of (9.47). Set

$$M := (I - P_F)R_<.$$

The spectral assumptions give $|\operatorname{Tr} M| \leq CW \log W + d/\eta$, $\|M\|_{\mathrm{HS}} \leq C\sqrt{W/\eta}$, and $\|M\| \leq \eta^{-1}$. The centered Gaussian quadratic-form estimate and a constant net in the column space give

$$\|Y^* M Y\| \leq |\operatorname{Tr} M|/W + C\sqrt{d}\,\|M\|_{\mathrm{HS}}/W + Cd\,\|M\|/W \leq C \log W.$$

For non-Hermitian M, apply this to its Hermitian real and imaginary parts. The scalar tail estimate follows by diagonalization and the exponential moment of centered independent exponential variables; the net has $\exp(O(d))$ points. All failures are Ce^{-cd}.

G.2.2 Spectral hypotheses for the complementary inverse

Proof of (9.48). Apply the averaged local law (G.1) to the complementary GUE with its $o(1)$-norm deformation and bulk energy. Set $\eta = Ad/W$ and compare the imaginary parts of the normalized resolvent traces at heights $c\eta$ and η. Outside $[-\eta, \eta]$, the first Poisson kernel is at most $2c$ times the second. Choose c small to obtain at least $c'W\eta$ eigenvalues in this interval, then choose A large. The upper count follows from the kernel at η. The upper count is $\operatorname{Tr} \mathbf{1}_{\{|\Gamma_c| \leq r\}} \leq CWr$ for $\eta \leq r \leq 1$. Thus, for $j = 1, 2$,

$$\operatorname{Tr} |R_<|^j \leq W + C \sum_{0 \leq l \leq \lceil \log_2(1/\eta) \rceil} (2^l \eta)^{-j} W 2^l \eta \leq \begin{cases} CW \log W, & j = 1, \\ CW/\eta, & j = 2. \end{cases}$$

The remaining norm bound $\|R_<\| \leq \eta^{-1}$ is the definition of the truncation. These are precisely the two truncated trace bounds in (9.43). All these scales are in the local-law domain, since $W\eta \geq AW^\delta$. No bound on the full real inverse is needed in Lemma 9.6. For a complement of dimension $W - d$, its entry variance is $1/W$, so divide the matrix, energy, and spectral parameter by $\sqrt{(W-d)/W}$ before applying Lemma G.1 in dimension $W - d$. Since $d = o(W)$, this changes only fixed constants and keeps the energy in a fixed bulk interval.

G.3 Moment and lower bounds for Riccati products

For fixed $b > 0$ assume

$$W^{-b} \leq g = o(1), \qquad |E| \leq 2 - \kappa_0. \tag{G.5}$$

Lemma G.2 (Killed product moment). *Assume (G.5). Fix an integer $h \geq 1$ and a deterministic Hermitian R_4 satisfying (9.3). Let $V_5, \ldots, V_{h+4}$ be independent normalized complex GUE matrices, and define*

$$R_j := (V_j - EI - g^2 R_{j-1})^{-1}, \qquad 5 \leq j \leq h + 4.$$

Let $0 < p < 2/(h+1)$. For every $e > 0$ there is a scalar spectral good event of probability $1 - o(1)$ on which

$$\operatorname{Tr} |R_{h+4} \cdots R_5|^p \le W^{1+e}. \tag{G.6}$$

This event is invariant under simultaneous conjugation of $R_4, V_5, \dots, V_{h+4}$. The probability statement is uniform over the deterministic E, g, R_4 just specified.

Proof of (G.6). Choose a small $0 < \delta < 1$. Kill at any step where the inverse norm exceeds $W^{1+\delta}$, the outgoing state is not regular, or an inverse dyadic count exceeds $CW^{1+\delta}2^{-l}$. Equation (9.58) and a dyadic union bound make the total failure $O_h((\log W)W^{-\delta} + g^{1/2})$. On regular pasts, Lemma G.1 gives, by testing every fixed vector and then taking expectation,

$$\mathbb{E}[\mathbf{1}_{\{|\Gamma| \le t\}} \mid \text{past}] \le CW^{2\delta} tI, \qquad W^{-1-\delta} \le t \le 1. \tag{G.7}$$

For $t \ge W^{-1+\delta}$ use $\mathbf{1}_{|x| \le t} \le 2t\Im(x - it)^{-1}$ and the local law; below that scale use monotonicity at $W^{-1+\delta}$. The exceptional probability can have an arbitrarily large power, and the projection is bounded by I.

For $5 \le j \le h+4$, split a killed R_j into bands $2^{l_j} A_{j,l_j}$, where $\left\|A_{j,l_j}\right\| \le 1$ and the $l_j = 0$ band contains the part of norm at most one. Each killing indicator records survival through its current step. In successive conditioning, earlier indicators are measurable with respect to the past and the current indicator is bounded by one; no future survival event is conditioned on. For $Y := A_{h+4,l_{h+4}} \cdots A_{5,l_5}$, including these indicators, (G.7) gives

$$\mathbb{E} \|Y\|_{\mathrm{HS}}^2 \le C_h W^{1+2h\delta} 2^{-\sum l_j}, \qquad \operatorname{rank} Y \le CW^{1+\delta} 2^{-\max_j l_j}.$$

The power-mean inequality for singular values and concavity imply

$$\mathbb{E} \operatorname{Tr} |2^{\sum l_j} Y|^p \le C_h W^{1+\delta(1-p/2+hp)} 2^{-[1-p(h+1)/2]\max_j l_j}.$$

The sum over multi-indices converges up to its harmless polynomial multiplicity in $\max_j l_j$. To sum matrix bands, for M_* matrices use the rank approximation inequality $N_{\sum Y_i}(t) \le \sum_i N_{Y_i}(t/M_*)$ and integrate it, obtaining $\operatorname{Tr} |\sum Y_i|^p \le M_*^p \sum_i \operatorname{Tr} |Y_i|^p$. Here $M_* = O((\log W)^h)$, absorbed by an arbitrarily small power. Choose δ small and apply Markov with remaining power slack. On the all-good event the killed product is the original product. □

For the application to Lemma 8.12(i), take $h = q - 6$ and $e = \varepsilon/3200$, with q, p from (9.54). One uniform choice of the small exponent in the proof is

$$\delta := \min\left\{ \frac{\varepsilon}{204800(q-4)}, \frac{1-2\beta}{64}, \frac{\alpha}{64}, \frac{\varepsilon}{51200} \right\}. \tag{G.8}$$

Apply Lemma G.1 with domain exponent $\delta/4$ and stochastic-domination slack $\delta/8$. The scales $W^{-1+\delta}$ and cg^2, g^2, Cg^2 lie in this fixed domain. The geometric summation margin is $1 - p(q-5)/2 = \varepsilon/6400$, and $1 - p/2 + (q-6)p < 3$. Thus the powers of W^δ and the logarithmic factors consume less than half the available $\varepsilon/3200$ slack. The remaining Markov failure is at most $W^{-\varepsilon/6400}$, while the killing failure is at most $C_{q-5}((\log W)W^{-\delta} + W^{-\alpha/2})$. All parameters were fixed before $a \in [\alpha, \beta]$ and W, which proves the asserted uniformity.

G.3.1 The output in the critical window

Lemma G.3 (Buffered product lower bound)**.** *Under the assumptions of Lemma 8.12(ii), use its θ, κ_0 and define*

$$\upsilon := W^{-1+2\theta}.$$

The matrices $V_5, \dots, V_{q-1}$ are independent normalized complex GUE blocks. Define

$$R_j := (V_j - EI - g^2 R_{j-1})^{-1}, \qquad 5 \le j \le q-1,$$

with deterministic initial R_4 satisfying $\|R_4\| \le W^{1+\theta}$ and regularity (9.3). Fix a nonzero vector v_4 and $|E| \le 2-\kappa_0$. For fixed q and every fixed $D' > 0$, the probability that the chain is regular and (9.69) fails for this v_4 is at most $W^{-D'}$.

Proof of (9.69). Set $F_j^{(\upsilon)} = (I + \upsilon^2 R_j^2)^{-1}$ for $4 \le j \le q-1$, with the empty-product convention $R_{[4,5]} = I_W$. At each step $5 \le j \le q-1$, a vector u measurable before V_j is fixed after conditioning on the past. The exact spectral identity and Lemma G.1 give

$$u^* R_j F_j^{(\upsilon)} R_j u = \upsilon^{-1} \Im u^* (R_j^{-1} - i\upsilon)^{-1} u \ge c\upsilon^{-1} u^* (I + g^4 R_{j-1}^2)^{-1} u \ge c\upsilon^{-1} u^* F_{j-1}^{(\upsilon)} u. \tag{G.9}$$

The first comparison in (G.9) has failure at most $W^{-D'}$ for any prescribed D', uniformly on regular pasts: $W\upsilon = W^{2\theta}$ and the relative Dyson profile in (6.18) is retained. The second is a deterministic comparison of two functions of the same Hermitian matrix, using $g^2 \le \upsilon$. Apply this with $u = R_{[j-1,5]} v_4$, normalized before each local-law application. A finite conditional union bound, intersected with chain regularity, gives the iterated lower bound. Finally $\upsilon \|R_4\| \le W^{3\theta}$, so $v_4^* F_4^{(\upsilon)} v_4 \ge \frac{1}{2} W^{-6\theta} \|v_4\|^2$, and $F_{q-1}^{(\upsilon)} \le I$. This proves (9.69). □

G.4 Coercivity of a structured map of GUE matrices

We adapt the proof of [10, Lemma A.2] to complex GUE matrices. The complex map has the additional kernel direction iI_W, which is removed by the restriction $\Im \operatorname{Tr} P = 0$ below. After this adjustment, we follow the same decoupling, norm-convergence and Gaussian concentration argument.

Lemma G.4 (Structured Gaussian coercivity)**.** *Let $m := 128$ and let $\widehat{V} = (\widehat{V}_1, \dots, \widehat{V}_m)$ consist of independent complex GUE matrices with $\mathbb{E}(\widehat{V}_j)_{ab}(\widehat{V}_k)_{cd} = W^{-1}\delta_{jk}\delta_{ad}\delta_{bc}$. On the real Hilbert space*

$$\mathcal{Q}_W := \{P \in \mathbb{C}^{W \times W} : \Im \operatorname{Tr} P = 0\}$$

define

$$\mathcal{L}_{\widehat{V}}(P) := (\widehat{V}_j P + P^* \widehat{V}_j)_{j=1}^m. \tag{G.10}$$

There exist absolute $\tau, c, C > 0$ such that, for sufficiently large W,

$$\mathbb{P}\{s_{\min}(\mathcal{L}_{\widehat{V}}|_{\mathcal{Q}_W}) < \tau\} \le Ce^{-cW}. \tag{G.11}$$

For any fixed finite additional family of normalized GUE matrices (possibly including matrices from which the $\widehat{V}_j$ are formed as disjoint normalized differences), a sufficiently large fixed operator-norm cutoff has the same exponentially small failure bound. There is a C^∞ cutoff $0 \le \chi_P \le 1$, compactly supported in the input Hermitian variables, equal to one with probability $1 - Ce^{-cW}$, supported where the singular value and norm conditions hold with fixed margins, and

$$\|\nabla \chi_P\| \le C. \tag{G.12}$$

Here the gradient is taken in all input Hermitian variables with their real Hilbert–Schmidt product inner product. The number of input matrices is fixed before W varies.

Proof of (10.5) ***and*** (10.14). We follow [10, Appendix A.1, proof of Lemma A.2], recording the complex-valued adjustment and the cutoff construction. Put

$$S := \sum_j \widehat{V}_j^2, \qquad \Phi(P) := \sum_j \widehat{V}_j P^* \widehat{V}_j, \qquad S_W := s_{\min}(\mathcal{L}_{\widehat{V}}|_{\mathcal{Q}_W}).$$

The unrestricted map annihilates iI_W. On the full real matrix space, $\mathcal{L}^*_{\widehat{V}}\mathcal{L}_{\widehat{V}}(P) = 2SP + 2\Phi(P)$, and GUE contraction gives

$$\mathbb{E}\Phi(P) = W^{-1}m\overline{\operatorname{Tr} P}\, I_W, \qquad \langle P, \mathbb{E}\Phi(P)\rangle = W^{-1}m\Re(\operatorname{Tr} P)^2 \geq 0 \quad (P \in \mathcal{Q}_W).$$

Thus the restriction $\Im \operatorname{Tr} P = 0$ removes the kernel direction and gives the analogue of [10, (42)]:

$$S_W^2 \geq 2\lambda_{\min}(S) - 2\|\Phi - \mathbb{E}\Phi\|. \tag{G.13}$$

Let $\widehat{V}'$ be an independent copy of $\widehat{V}$. Jensen's inequality bounds $\mathbb{E}\|\Phi_{\widehat{V}} - \mathbb{E}\Phi_{\widehat{V}}\|$ by $\mathbb{E}\|\Phi_{\widehat{V}} - \Phi_{\widehat{V}'}\|$. Under the Gaussian rotation $\widehat{V} = (B + C)/\sqrt{2}$, $\widehat{V}' = (B - C)/\sqrt{2}$,

$$(\Phi_{\widehat{V}} - \Phi_{\widehat{V}'})(P) = \sum_j (B_j P^* C_j + C_j P^* B_j).$$

The two cross operators have the same distribution. The triangle inequality and the real isometry $P \mapsto P^*$ give, as in [10, Appendix A.1],

$$\mathbb{E}\|\Phi - \mathbb{E}\Phi\| \leq 2\mathbb{E}\left\|\sum_j B_j \otimes C_j^t\right\|. \tag{G.14}$$

Use [6, Corollary 1.2 and Lemma 9.6] for ordinary GUE norm convergence and for the two independent tensor factors, respectively. The normalization is exactly that of [6, Definition 2.8(a)]; transpose preserves the GUE law. For variance-one semicircular families s_j, t_j on the two factors, the free bounds used in [10, Appendix A.1] imply

$$\sum_j s_j^2 \geq (m - 2\sqrt{m})I, \qquad \left\|\sum_j s_j \otimes t_j\right\| \leq 2(\sqrt{m} + 1). \tag{G.15}$$

The lower spectral bound follows from norm convergence applied to $cI - S$ for a sufficiently large fixed c. Fixed GUE norm moments give uniform integrability in both limits. Consequently, (G.13)–(G.15) yield

$$\liminf_{W\to\infty} \mathbb{E}S_W^2 \geq 2(m - 2\sqrt{m}) - 8(\sqrt{m} + 1) > 0 \quad (m = 128). \tag{G.16}$$

Also $\mathbb{E}S_W^4 \leq C$, since $S_W \leq 2(\sum_j \|\widehat{V}_j\|^2)^{1/2}$. Paley–Zygmund gives $\mathbb{E}S_W \geq c_0 > 0$ for large W. The map $\widehat{V} \mapsto S_W$ is 2-Lipschitz in the product Hilbert–Schmidt norm, so Gaussian concentration with coordinate covariance I/W proves (G.11). Standard GUE operator-norm tails give the same failure bound for the fixed additional family.

For the cutoff, disjoint normalized differences are an orthogonal Gaussian transformation, so the same Lipschitz bound holds in the original inputs. Multiply Lipschitz cutoffs of S_W and the input norms, with fixed nested margins, and mollify in the real Hermitian input space inside a sufficiently small fixed Hilbert–Schmidt ball. This preserves the support margins and the uniform gradient bound, and leaves $\chi_P = 1$ on a stronger event of probability $1 - Ce^{-cW}$. The norm cutoff makes its support compact. This proves (G.12) and the lemma. □

G.5 Construction of the common density

Lemma 8.9 asks for a smooth good component of the batch law, the polynomial estimates (8.27), and the two operator bounds (8.29). We first split the law by a cutoff in the GUE inputs. To show that the retained measure has a smooth density, we prove that the product map has surjective differential on the cutoff support. A controlled right inverse of this differential then gives an auxiliary density-derivative bound by Gaussian integration by parts. The operator bounds follow from Lemma F.3. This construction follows [10, Proposition 4.1, Lemma 4.2 and Section 4.3]; the GUE coercivity input is Lemma G.4 in Appendix G.4.

***Proof of* (8.26), (8.27), *and* (8.29).** Fix B, E_0 and deterministic $W^{-B} \le g \le W^B$, $|E| \le E_0$. Let q be the fixed multiple of 1000 in the lemma. In particular, q is even and independent of W; all constants below may depend on B, E_0, q.

Step 1: split the batch law and control the discarded part. The goal of this step is to obtain (8.27) from the cutoff conditions (10.5). Use T_i from (8.9) and the fixed unitary matrix S from Definition 7.1. Set $F := T_1 \cdots T_q$. By reversal of the independent inputs, S^*FS has the batch law μ. Use C_i from (10.11) and put $h_g := \max\{1, g^{-1}\}$. For a cutoff $0 \le \chi_P \le 1$ to be chosen, put $\widetilde{\pi} := \mathbb{E}\chi_P$ and define the positive measure μ_{bad} by

$$\int u(\mathcal{T})\, d\mu_{\mathrm{bad}}(\mathcal{T}) := \mathbb{E}[(1-\chi_P)u(S^*FS)]$$

for bounded measurable u. Then $\mu - \mu_{\mathrm{bad}}$ is positive, has mass $\widetilde{\pi}$, and $\mu_{\mathrm{bad}}(\mathcal{G}) = 1 - \widetilde{\pi}$. We will prove in Step 2 that $\mu - \mu_{\mathrm{bad}}$ has a smooth density $\widetilde{\pi} f$.

For the mass and support estimates, we require

$$\chi_P \in C_c^\infty, \qquad 0 \le \chi_P \le 1, \qquad \mathbb{P}(\chi_P = 1) \ge 1 - Ce^{-cW},$$
$$\operatorname{supp}\chi_P \subset \{\max_i \|V_i\| \le C\}.$$

Step 2 will construct such a cutoff. We first check the consequences of these requirements. The probability bound gives $\widetilde{\pi} \ge 1/2$; its stronger exponential form will also control the weighted bad mass.

On $\operatorname{supp}\chi_P$, each $\|C_i\| \le Ch_g$, and hence $\|F\| = \|F^{-1}\| \le Ch_g^q \le W^C$. Fixed GUE norm moments give

$$\mathbb{E}\|F\| \le Ch_g^q.$$

Consequently,

$$\int \|\mathcal{T}\|^{1/2}\, d\mu_{\mathrm{bad}}(\mathcal{T}) \le \mathbb{P}(\chi_P \ne 1)^{1/2}(\mathbb{E}\|F\|)^{1/2} \le Ce^{-cW}h_g^{q/2} \le \tfrac{1}{2}$$

for large W. Thus all estimates in (8.27) are available once the density is shown to exist.

Step 2: construct the cutoff and obtain a smooth density. The goal of this step is to obtain (8.26) by proving (10.10). **Proof of (10.10).** We choose χ_P so that $V \mapsto F(V)$ has surjective differential on a neighborhood of $\operatorname{supp}\chi_P$; this will give the required smooth density. Conjugation by S preserves this property.

For a Hermitian input variation $H = (H_i)$, differentiate the product as in [10, Claim 4.5]:

$$\mathcal{A}(H) := F^{-1}(H \cdot \nabla_V)F = g^{-1}\sum_{i=1}^{q} Q_i^{-1}\begin{pmatrix} 0 & 0 \\ H_i & 0 \end{pmatrix} Q_i, \qquad Q_i := T_{i+1}\cdots T_q, \tag{G.17}$$

with $Q_q = I$. We must show that $\mathcal{A}$ is onto. For a target Z, use the block form (10.3), with (A, B_1, B_2) denoted by (P, U, R). We do so by recovering every Z from $\mathcal{A}^*Z$; the adjoint of this reconstruction will also provide the input perturbation needed in Step 3.

Write $Z_i := Q_i^{-*}ZQ_i^*$ and denote its blocks as in (10.3) by (P_i, U_i, R_i), so $(P_q, U_q, R_q) = (P, U, R)$. The trace pairing and transfer recursion, as in [10, Proposition 4.4, pp. 34–35], give

$$\mathcal{A}^*Z = g^{-1}(R_i)_{i=1}^q, \qquad \begin{aligned} P_{i-1} &= -R_iC_i - P_i^*, \qquad U_{i-1} = -R_i, \\ R_{i-1} &= P_iC_i + C_iP_i^* + C_iR_iC_i - U_i, \end{aligned} \quad 2 \le i \le q. \tag{G.18}$$

Thus the data are $\mathbf{R} = (R_1, \dots, R_q)$, and we need to recover P, U, R. For any Hermitian data array define $E_q := 0$ and $E_{i-1} := -R_iC_i - E_i^*$ for $2 \le i \le q$. When the data come from Z as above, the first recursion gives

$$P_i = \begin{cases} P + E_i, & i \text{ even}, \\ -P^* + E_i, & i \text{ odd}. \end{cases}$$

At an interior odd index, substituting this into the remaining recursions yields

$$C_iP + P^*C_i = F_i(\mathbf{R}), \qquad F_i(\mathbf{R}) := E_iC_i + C_iE_i^* + C_iR_iC_i + R_{i+1} - R_{i-1}. \tag{G.19}$$

To recover P from these equations, subtract them at disjoint pairs of interior odd indices. Set

$$(i_j, k_j) := (4j+1, 4j-1), \qquad 1 \le j \le 128, \qquad \widehat{V}_j := 2^{-1/2} g(C_{i_j} - C_{k_j}) = 2^{-1/2}(V_{k_j} - V_{i_j}).$$

The largest selected index is 513, and its neighbor 514 is still within the batch since $q \ge 1000$. The common energy term cancels, and the disjoint pairs make the $\widehat{V}_j$ independent normalized GUE matrices. Since $\Im \operatorname{Tr} P = 0$, use $\mathcal{L} = \mathcal{L}_{\widehat{V}}$ from (G.10) for these $\widehat{V}_j$. With

$$\mathcal{F}\mathbf{R} := (F_{i_j}(\mathbf{R}) - F_{k_j}(\mathbf{R}))_{j=1}^{128},$$

we obtain $\mathcal{L}P = (g/\sqrt{2})\mathcal{F}\mathbf{R}$. Only these odd-index pairs are needed to recover P; the other blocks remain in the product and in the data $\mathbf{R}$. Each F_i is linear in $\mathbf{R}$ and has degree at most two in C.

Thus it suffices to arrange $s_{\min}(\mathcal{L}) \ge c$ on the cutoff support. Lemma G.4, applied also to the norms of all q input blocks, now supplies a cutoff satisfying the requirements of Step 1 and, in addition,

$$\operatorname{supp} \chi_P \subset \{s_{\min}(\mathcal{L}) \ge c\}, \qquad \|\nabla_V \chi_P\| \le C.$$

The lower singular-value bound and the input norm bounds hold on a neighborhood of $\operatorname{supp} \chi_P$, after adjusting the fixed constants. This choice of χ_P depends on the input blocks. On this neighborhood, $\mathcal{L}$ has a bounded left inverse. Define, on the cutoff neighborhood,

$$\mathcal{J} := (\mathcal{L}^*\mathcal{L})^{-1}\mathcal{L}^*, \qquad \mathcal{K}_P := 2^{-1/2} g \mathcal{J}\mathcal{F}.$$

Then $P = \mathcal{K}_P\mathbf{R}$. The terminal recursion recovers U from $P, \mathbf{R}$, while $R = R_q$ is already observed. Accordingly, for arbitrary Hermitian data define

$$\mathcal{K}_0\mathbf{R} := \begin{pmatrix} P & U \\ R_q & -P^* \end{pmatrix}, \qquad P := \mathcal{K}_P\mathbf{R}, \qquad U := PC_q + C_qP^* + C_qR_qC_q - R_{q-1}. \tag{G.20}$$

This matrix belongs to the Lie algebra. For the data associated with Z we have just proved $\mathcal{K}_0\mathbf{R} = Z$. Since $\mathcal{A}^*Z = g^{-1}\mathbf{R}$, taking adjoints gives

$$g\mathcal{K}_0\mathcal{A}^* = I, \qquad \mathcal{A}(g\mathcal{K}_0^*) = I. \tag{G.21}$$

In particular $\mathcal{A}$ is onto on the cutoff neighborhood.

Cover the compact cutoff support by finitely many submersion charts and choose a partition of unity. In each chart, integration of the smooth compactly supported input density along the fibers gives a smooth target density. Their sum is a smooth compactly supported Haar density for $\mu - \mu_{\mathrm{bad}}$; no global submersion or properness assumption is needed. Its normalization defines f by

$$\begin{aligned} \widetilde{\pi} \int u(\mathcal{T}) f(\mathcal{T})\, d\mathcal{T} &:= \mathbb{E}[\chi_P\, u(S^*FS)], \\ \int u(\mathcal{T})\, d\mu_{\mathrm{bad}}(\mathcal{T}) &= \mathbb{E}[(1-\chi_P) u(S^*FS)]. \end{aligned} \tag{G.22}$$

Therefore $f \ge 0$, $\int f = 1$, $f \in C_c^\infty(\mathcal{G})$, and $\mu = \widetilde{\pi} f\, d\mathcal{T} + \mu_{\mathrm{bad}}$, proving (8.26). This density and cutoff were chosen before any flag or differentiation direction.

Step 3: bound the directional derivatives of this density. The goal of this step is to obtain the following auxiliary bound for the smoothing estimate in Step 4:

$$\sup_{\substack{M \in \mathfrak{g} \\ \|M\|_{\mathrm{HS}} = 1}} \|X_M f\|_1 \le W^C. \tag{G.23}$$

The directions and derivatives are defined in (10.1)–(F.5). For M with $S^*MS \in \mathfrak{g}$ and $\|M\|_{\mathrm{HS}} = 1$, take the field

$$X := \sqrt{2}\, g\mathcal{K}_0^* M, \qquad \mathcal{A}(X) = \sqrt{2}\, M, \qquad D_X F = \sqrt{2}\, FM.$$

The last two identities follow from (G.21). Extend $\chi_P X$ by zero and let γ be the Gaussian law of the original inputs V_i, with covariance $W^{-1}I$ in real trace-orthonormal Hermitian coordinates. The definition (G.22), the chain rule, and Gaussian integration by parts [10, Lemma 4.2] give, for $u \in C_c^\infty(\mathcal{G})$,

$$\begin{aligned}\sqrt{2}\,\widetilde{\pi} \int (X_{S^*MS}u)(\mathcal{T}) f(\mathcal{T})\, d\mathcal{T} &= \mathbb{E}[\chi_P D_X[u(S^*FS)]] \\ &= \mathbb{E}[u(S^*FS)\, \delta_\gamma(\chi_P X)].\end{aligned}$$

Here $D_X = X \cdot \nabla_V$, and $\mathbb{E} D_X \varphi = \mathbb{E}[\varphi \delta_\gamma X]$ defines the Gaussian divergence. For completeness, we give the second-moment calculation. **Proof of (10.4).** In real trace-orthonormal coordinates v_i of variance $1/W$, put $Y = \chi_P X$ and $\delta_\gamma Y = \sum_i (W v_i Y_i - \partial_i Y_i)$. Two Gaussian integrations by parts give

$$\mathbb{E}|\delta_\gamma Y|^2 = W\mathbb{E}\sum_i Y_i^2 + \mathbb{E}\sum_{i,j}(\partial_i Y_j)(\partial_j Y_i) \le W\mathbb{E}\|Y\|^2 + \mathbb{E}\|\nabla Y\|_{\mathrm{HS}}^2.$$

The field is real on the Hermitian input space, and Y has compact support, so no boundary term occurs. This proves (10.4).

The reconstruction formulas use a fixed number of matrix products and inverses, with $\|C_i\| \le C h_g$ and $s_{\min}(\mathcal{L}) \ge c$ on the cutoff neighborhood. Differentiating them, and summing over the qW^2 input coordinates, therefore gives

$$\sup_{\|M\|_{\mathrm{HS}}=1} (\|X\| + \|\nabla_V X\|_{\mathrm{HS}}) + \|\nabla_V \chi_P\| \le W^C.$$

Haar integration by parts and duality over $\|u\|_\infty \le 1$ now yield

$$\|X_{S^*MS} f\|_1 \le 2^{-1/2}\widetilde{\pi}^{-1} \|\delta_\gamma(\chi_P X)\|_2 \le W^C. \tag{G.24}$$

Since S is unitary and $\widetilde{\pi} \ge 1/2$, this proves (G.23) uniformly over unit Lie directions.

Step 4: deduce the two operator bounds. The goal of this step is to obtain (8.29). The support bound in (8.27) and the derivative bound (G.23), both proved above, are the inputs to the smoothing estimate (F.6) in Lemma F.3; it gives

$$\|G_{1/2} h\|_\infty \le W^C \sup_U \|h(U,\cdot)\|_{L^4(\mathcal{C}_U, dm_U)}.$$

For the bad operator, $e^{-c(\mathcal{T};U,Y)/2} \le \|\mathcal{T}\|^{1/2}$ and Step 1 give

$$\|B_{1/2} h\|_\infty \le \|h\|_\infty \int \|\mathcal{T}\|^{1/2}\, d\mu_{\mathrm{bad}}(\mathcal{T}) \le \tfrac{1}{2}\|h\|_\infty.$$

The decomposition of μ gives $P_{1/2} = G_{1/2} + B_{1/2}$, so (8.29) follows.

All constants and width thresholds depend only on B, E_0, q, as asserted in Lemma 8.9. □

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